

Siu-Kui Au

Operational Modal Analysis

Modeling, Bayesian Inference,
Uncertainty Laws

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Dedicated to my wife and mother

Preface

Operational modal analysis (OMA) aims at identifying the modal properties (natural frequencies, damping ratios, mode shapes, etc.) of a structure using vibration response measurements only. The possibility of doing so without knowing the input forces is exciting from both a theoretical and practical point of view. It significantly enhances the feasibility and reduces the cost of obtaining in-situ information about structures. Together with the advent of sensing, data acquisition and computer technology, the bar to enter the community of vibration testing or related research is disappearing. Activities are booming in all directions, e.g., theory development, experimentation and applications; and spanning across civil, mechanical, aerospace and electrical power engineering.

I started exploring OMA in an industrial project in 2008 when I was looking for a conventional method to identify the modal properties of several tall buildings from acceleration time histories during typhoons. At that time the literature was already quite mature, with many smart ideas of coming up with an estimate for modal properties from apparently random vibration time histories that one can hardly make any sense of. After experimenting with different methods and data under various situations (some similar and some different) I soon realized that there was a significant scatter in the estimates, despite the fact that the fit between the theoretical and empirical statistics (from which the estimates were obtained) appeared quite reasonable in each case. Damping ratio was the quantity of most interest in the project, but its scatter was quite large, which inevitably undermined the value of the work.

The main issue with the scatter was perhaps not so much with its existence but rather with the lack of understanding, which motivated subsequent research in OMA. The objective is not so much to come up with another estimate for the modal properties (there are already many methods) but rather to have a scientifically sound method that tells how large their ‘uncertainty’ is; and, if possible, how it depends on the factors that can be quantified.

Uncertainty is as a lack of information. Using probability for quantification, Bayes’ theorem provides the fundamental principle for processing information and hence uncertainty. It yields the probability distribution of modal properties

conditional on available data and modeling assumptions, naturally addressing both estimation and uncertainty quantification in OMA. The concept is metaphysical but very useful for decision making. The mathematics is not trivial, but it can be worked out and made efficient in the form of computer algorithms. After some years of development, I believe the algorithms for Bayesian OMA are now ready for industrial applications and the theory allows us to understand how identification uncertainty depends on test configuration. Of course, there are still many uncertainty-related problems in OMA and applications that have yet to be addressed. It will require concerted efforts in the future.

Adopting a Bayesian perspective means that we need to be clear about what assumptions we make on OMA data, because our conclusions are all conditional on them. Results are as good as assumptions. This is generally true but is particularly important to bear in mind when making Bayesian inference. It is tempting to believe that the calculated uncertainties are universal numbers; they are not. Throughout this book, the assumptions on OMA data are invariably the same. They are the conventional ones used in structural dynamic analysis and design, e.g., linear classically damped dynamics and stochastic stationary response.

This is intended to be a book for learning the basic assumptions and probabilistic modeling of ambient vibration data, how to make Bayesian inference effectively, and identification uncertainty in OMA. The primary audience are graduate students and researchers, although practitioners should also find the application part of the book a useful reference. Even if Bayesian OMA algorithms may not be adopted, I hope the book can still arouse interests in field testing and provide materials for understanding ambient vibration data and uncertainties in OMA.

Liverpool, UK
November 2016

Siu-Kui Au

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I have been influenced by several people in developing the research that led to this book. Prof. Costas Papadimitriou introduced me to engineering reliability research, along which I learnt the asymptotics of Laplace integrals, which is closely related to the Gaussian approximation of posterior distribution in globally identifiable problems. I took a series of engineering mathematics courses (AM125a, b, c) from Prof. James Beck (also my Ph.D. advisor), which equipped me with the theoretical background in linear space theory for developing the Bayesian OMA algorithms. His Bayesian philosophy shapes my perspectives on research related to system identification and uncertainty in general. The series of courses on stochastic processes (AMA153a, b, c) by Dr. Uri Keich made me not afraid of stochastic processes and their mathematics. I started developing a serious interest in field vibration testing and learning how to do it from Prof. James Brownjohn through a week-long field test for the Humber Bridge (UK). I am grateful to these wonderful teachers. Of course, any mistakes or misconceptions in this book are my responsibility.

I started seriously playing with field test equipment in the Structural Vibration Laboratory at the City University of Hong Kong. Establishing the laboratory would not have been possible without the friendship, collaboration and help from Dr. Heung-Fai Lam; and generous administrative and funding help from Prof. Yee-Tak Leung and Prof. Sritawat Kitipornchai (successive department heads). Chi-Kin Lai provided unfailing and dedicated support on developing innovative in-house field test equipment. Collaboration with Dr. Ping To (Arup HK) keeps motivating me on developing fast Bayesian OMA algorithms and exploring industrial applications. He also provided excellent logistics supports for obtaining field test data from many interesting structures in Hong Kong.

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Performing field tests requires teamwork. In my case it cannot be realized without the enthusiastic participation from students and postdocs, including (but not limited to) Ir Michael Sien, Ir Grace Chua, Ir Hing-Lun Yip, Ir Heung-Ming Chow, Dr. Fengliang Zhang, Dr. Yanchun Ni, Dr. Jiahua Yang, Dr. Qing Hu, Dr. Ching-Tai Ng, Yichen Zhu and Yanlong Xie.

This book is dedicated to my wife, Janice, and my mother, 阿媽, for their love, patience and support of my education, career and life goals. I am also indebted to my uncles, 大舅父, 謙舅父, 大細舅父, 中細舅父, 孝記舅父 and 烈記舅父, for their encouragement and the precious moments they deliberated; and my cousin 阿峰 for the wonderful times we share. These all shaped my values and added colors to my life since childhood.

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Abbreviations

c.o.v.	Coefficient of variation = ratio of standard deviation to mean
CRB	Cramér Rao Bound
DAQ	Data acquisition
DFT	Discrete Fourier Transform
DOF	Degree of freedom
DTFT	Discrete-time Fourier Transform
e.g.	For example
FFT	Fast Fourier Transform
FIM	Fisher information matrix
FT	Fourier Transform
Hz	Hertz (cycles per second)
i.e.	That is
i.i.d.	Independent and identically distributed
LHS	Left hand side
MAC	Modal assurance criterion
MDOF	Multi degree of freedom
MPV	Most probable value
NLLF	Negative log-likelihood function
ODE	Ordinary differential equation
OMA	Operational modal analysis
PDF	Probability density function
PSD	Power spectral density
RHS	Right hand side
RMS	Root mean square
s/n	Signal-to-noise
SDOF	Single degree of freedom
SV	Singular value
w.r.t.	With respect to

Nomenclature

$\ln x$	Natural logarithm of x , i.e., to base e ($\approx 2.71828\dots$)
$\log_p x$	Logarithm of x to base p
$\mathbf{i}$	Purely imaginary number, $\mathbf{i}^2 = -1$
$\bar{x}$	Complex conjugate of x
$\mathbf{I}_n, \mathbf{I}$	$n \times n$ identity matrix, identity matrix (dimension not indicated)
$\mathbf{0}_n, \mathbf{0}_{n \times m}$	$n \times n$ zero matrix, $n \times m$ zero matrix
$[a_1, a_2, a_3]$	1×3 row vector with entries a_1, a_2 and a_3
$[a_1; a_2; a_3]$	3×1 column vector with entries a_1, a_2 and a_3
$\text{diag}[\mathbf{x}]$	Diagonal matrix formed by the entries of vector $\mathbf{x}$
$\text{diag}\{x_i\}_{i=1}^n$	Diagonal matrix formed by $x_1, \dots, x_n$
$\text{vec}(\mathbf{A})$	Vectorization of matrix $\mathbf{A}$
$\mathbf{A} \otimes \mathbf{B}$	Kronecker product of matrices $\mathbf{A}$ and $\mathbf{B}$
$ x $	Absolute value of scalar x
$ \mathbf{A} $	Determinant of matrix $\mathbf{A}$
$\mathbf{A}^T$	Transpose of matrix $\mathbf{A}$
$\mathbf{A}^*$	Conjugate transpose of matrix $\mathbf{A}$, i.e., $\overline{\mathbf{A}}^T$
$\text{tr}(\mathbf{A})$	Trace of matrix $\mathbf{A}$, i.e., sum of diagonal entries
∇	Gradient of a function
∇^2	Hessian matrix of a function
$x \sim y$	x is asymptotic to y , i.e., $x/y \rightarrow 1$
$x \approx y$	x is approximately equal to y
$x = O(y)$	x is of the order of y , i.e., $x/y \rightarrow$ non-zero finite constant
$\ \mathbf{x}\ $	Euclidean norm of vector $\mathbf{x}$, i.e., the square root sum of squared entries
$\{\mathbf{x}_j\}$	Collection of variables indexed by j
$\text{var}[X]$	Variance of scalar random variable X
$\text{cov}[\mathbf{X}]$	(Auto) Covariance matrix of random vector $\mathbf{X}$
$\text{cov}[\mathbf{X}, \mathbf{Y}]$	(Cross) Covariance matrix between random vectors $\mathbf{X}$ and $\mathbf{Y}$

Part I

Modeling

Chapter 1

Introduction

Abstract This chapter introduces modal testing and identification of structures, covering free vibration test, forced vibration test and ambient vibration test (operational modal analysis). The uncertainties in operational modal analysis are discussed. Regarding uncertainty as a lack of information, sources due to variability and identification error are distinguished. The unknown nature of excitation is an issue that must be tackled in any operational modal analysis method. Existing strategies attempt to eliminate the effect of this uncertainty or model it. After a brief tour on conventional (non-Bayesian) methods, Bayesian approach is introduced as a fundamental means to address uncertainties using probability. An overview of the book is given with specific advice for readers from different sectors.

Keywords Free vibration test • Forced vibration test • Ambient vibration test • Modal identification

The ‘modal properties’ of a structure include primarily its natural frequencies, damping ratios and mode shapes. They are the characteristics governing the response of a structure under dynamic loads, playing an important role in analysis and design. Dynamic response is significantly larger if the structure is in resonance with the loading; and this is governed by natural frequencies. Damping is associated with energy dissipation; the higher the damping, the lower the response. Mode shape affects the nature and spatial distribution of vibration.

Forward and Backward Problems

In a conventional dynamics problem, structural properties are specified, from which modal properties can be determined. The response of different modes, distinguished by mode shapes, are then calculated and summed to give a prediction of the structural response. These are referred as ‘forward problems’, depicted by the two right-pointing arrows in Fig. 1.1.

Modal Identification

‘Modal identification’ aims at identifying the modal properties of a structure, typically using measured vibration time history data (e.g., acceleration). ‘Structural

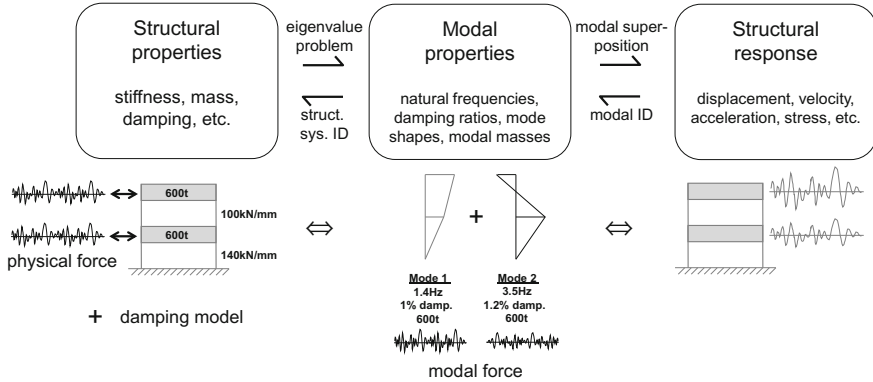


Fig. 1.1 Forward (right-arrow) and backward (left-arrow) problems in structural dynamics

system identification’ aims at identifying the structural properties. These are ‘backward’, or ‘inverse’, problems, depicted by the left-pointing arrows in Fig. 1.1. Although not indicated in the figure, structural system identification can also be based directly on the data. Modal and structural system identification are inter-related objectives. The general aim is to bridge the gap between structural models and their constructed counterparts, improve the quality of response predictions and provide updated structure-specific information for decision making. It involves a range of expertise such as modeling, experimentation, data processing and system identification theory. See Catbas et al. (2011) for a state-of-the-art report; Friswell and Mottershead (1995) and Doebling et al. (1998) on structural system identification.

In-situ Versus Modeled Modal Properties

Modal identification provides information about the ‘in-situ’ modal properties of a constructed structure, which can differ significantly from model predictions. Discrepancies in the order of 30% in predicting natural frequencies are not uncommon. They can be due to deviations/variations in stiffness and mass distribution. Boundary conditions and contributions from non-structural components are difficult to model. Natural frequencies can change with vibration amplitude, temperature and humidity; and with aging throughout the lifetime of a structure. Damping mechanism is difficult to model at the structural level and there is currently no accepted method for prediction from first principles. A further complication is its amplitude-dependent nature. Representative values or empirical formulas used in design are based on databases of identified values (with significant scatter). See, e.g., EC1 Annex F (2005), ESDU 83009 (2012), ISO 4354:2009 Annex E and ISO 10137:2007 Annex B, Bachmann et al. (1995).

A Practical Means for Updating

Modal identification is a practical means for updating the state of a structure and response prediction in a future dynamic loading event. By virtue of modal superposition, the modal properties of contributing modes are already sufficient for a

good estimation of the dynamic response, without knowing the structural properties. The modal properties of the modes that significantly contribute to the measured data can be identified uniquely. The same is not necessarily true for structural properties, however. It depends on the complexity of the structural model used, how it is parameterized, and the sensitivity of modeled data to parameters.

Demand for Modal Identification

New design concepts, tightened performance criteria and targets for cost-effectiveness have led to an increasing demand for modal identification. With new architectural forms and the use of advanced materials, modern structures tend to be lighter, more slender and more sensitive to dynamic loads. Serviceability performance assumes increasing importance, besides ultimate load capacity and resilience. In super-tall buildings, for example, designing and maintaining a weathertight envelope and controlling vibrations for human comfort more frequently becomes the governing challenge. Controlling vibrations below serviceability limits is also relevant for long-span slabs, pedestrian bridges and roofs. Design decisions outside or at the boundary of normal experience demand higher-than-normal quality in modeling information and assumptions in capturing reality. This leads to a demand for the knowledge of in-situ properties of constructed facilities in general. Databases of identified modal properties allow modeling assumptions to be calibrated for structures of similar configurations, in the long run improving the fidelity of models at the design stage. In a vibration assessment, control or retrofit project of an existing structure, modal identification is among the first few tasks to establish baseline dynamic properties. In ‘structural health monitoring’ (SHM) applications, modal properties are almost the default quantities of investigation and affect many decisions downstream; see Brownjohn (2007), Farrar and Worden (2012) and Xu and Xia (2012).

1.1 Vibration Tests

Whether and how the modal properties of a structure can be identified from measured vibration data depend critically on the available information about the input excitation during test. Generally, the measured vibration data can be viewed as

$$\begin{aligned} \text{measured vibration data} = & \text{free vibration from unknown initial condition} \\ & + \text{forced vibration from known excitation} \\ & + \text{forced vibration from unknown excitation} \\ & + \text{measurement noise (unknown)} \end{aligned} \quad (1.1)$$

The first three terms on the right hand side (RHS) depend on modal properties and hence contain information for identifying them. Transforming this principle into a

modal identification method is not trivial, however. Different terms have different sensitivities to modal properties. They may depend on other unknown properties that may need to be identified together. The second term is completely determined by modal properties and the known excitation. In addition to the modal properties, the first term also depends on initial conditions, i.e., displacement and velocity of the structure, which are generally unknown. For the third term, if the unknown excitation is totally arbitrary then the resulting forced vibration response will involve too many unknowns that are impossible to identify. Possibility exists when the unknown excitation is assumed to have some statistical properties so that the resulting forced vibration response can be described statistically. Such information can be used for modal identification, although the theory is much more involved and the results are only as good as the statistical assumptions.

Developing a modal identification method that accounts for all the terms on the RHS of (1.1) can be theoretically and computationally involved. It has not been done so far; and it need not be cost-effective to do so in the future. Existing methods mostly consider the special cases where only one of the first three terms dominates. Accounting for measurement noise (the last term) does not create much complication in method development. Dominance by the first or second term can be created artificially through experimental design, leading to ‘free vibration test’ and ‘forced vibration test’, respectively. There are many naturally occurring scenarios where the third term is dominant with reasonably tractable and justifiable statistical properties, leading to ‘ambient vibration test’.

The basic principles of the three types of tests are next described in Sects. 1.1.1–1.1.3. This book is about ‘operational modal analysis’ (OMA), which aims at identifying the modal properties using the data from ambient vibration test. Methods for free and forced vibration tests are not formally covered but their concepts are introduced in Chap. 3 as an application of structural dynamics. See also McConnell (1995), Maia and Silva (1997) and Ewins (2000).

1.1.1 Free Vibration Test

In a free vibration test, the structure is initially set into motion by some artificial means and then left to vibrate on its own. The response time histories during the free vibration phase at selected locations are measured and used for modal identification. To take advantage of free vibration test, the initiation excitation should be large enough so that the induced dynamic response is significantly larger than the response due to other unmodeled (e.g., ambient) excitations. Figure 1.2 shows an example where the first vertical mode of a footbridge is excited by human jumping near resonance. This of course is not feasible for massive structures. For example, the horizontal response due to a few participants pushing on the wall (even at the natural frequency) of a tall building is unlikely to be noticeable in the measured data, which is dominated by ambient vibrations.

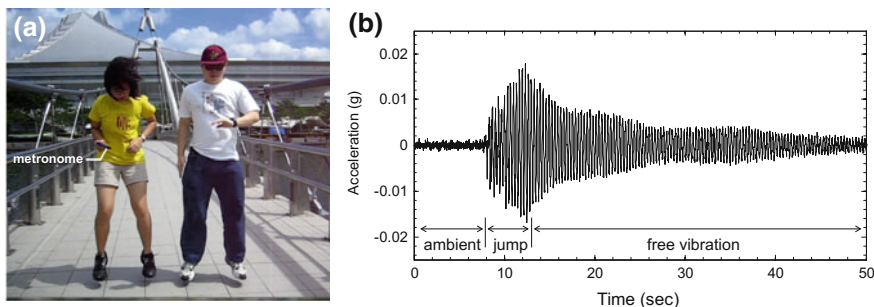


Fig. 1.2 Free vibration test with human excitation. **a** To excite the first vertical mode, participants tried to jump at the natural frequency, following the beeps of the metronome held in the lady's right hand; **b** acceleration data measured by sensor near the midspan

Modal identification using free vibration data dominated by a single mode is conceptually straightforward. In this case the data is oscillatory with a single frequency and exponentially decaying amplitude. For good quality data, the estimation procedure can be as simple as counting cycles for the natural frequency and calculating the rate of amplitude decay for the damping ratio (Sect. 3.1.4).

In reality, it can be difficult to excite only a single mode of vibration. The data may be dominated by more than one mode, as in Fig. 1.2b. Numerically filtering out the contribution from other unwanted modes is one option, as long as it does not significantly distort the contribution from the target mode. More generally, one can exploit the full time history of data, rather than just their amplitudes. The theoretical free vibration response can be written explicitly in terms of the natural frequencies, damping ratios, mode shapes and modal initial conditions. These properties affect the free vibration response in different characteristic ways. In principle, they can be identified uniquely by curve-fitting in either the time or frequency domain. Alternatively, one can approximate the free vibration equation or response in discrete-time and estimate the coefficient matrices, whose eigenvalue properties can be used for back-calculating the modal properties. See Sect. 1.4.2 later.

1.1.2 Forced Vibration Test

In a forced vibration test (known input) the structure is excited by some artificial means, e.g., a shaker or a sledge hammer. In addition to the vibration response, the force applied to the structure is also measured, and hence the term 'known input'. The (input) force and (output) response time histories are used together for modal identification. To take advantage of forced vibration test, the response due to the known force should dominate those from other unknown excitation sources. Forced vibration test could be expensive to perform due to the equipment and logistics required. Safety and risk considerations also increase the overheads or discourage its use.

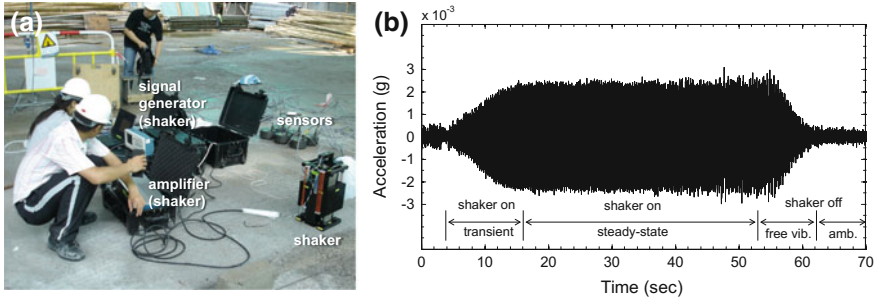


Fig. 1.3 Forced vibration test for the vertical modes of a floor slab with a shaker. **a** Equipment and **b** vertical acceleration measured near the midspan of the floor slab

However, when suitably applied with sufficient excitation power, forced vibration test allows the modal properties to be identified with good precision and at a desired vibration level, e.g., near serviceability limits (important for damping ratios). The data in a forced vibration test naturally contains a free vibration phase after the artificial force is removed. Figure 1.3 shows an example with an electro-magnetic shaker. With a payload of a few hundreds of Newton, such shaker has been conventionally applied to investigate the vibrations of floor slabs and footbridges. See Sect. 3.4.

1.1.3 Ambient Vibration Test

If the excitation is not measured and no assumption is imposed, it is generally not possible to determine the modal properties based on the (output) vibration response only. This is because the latter reflects the dynamic characteristics of the structure as well as the excitation. Possibility exists if one is willing to make some assumptions on the ‘statistical properties’ of the excitation. One conventional assumption is that the excitation is ‘broadband random’, which can be justified physically and is found to give reasonable results. The specific assumptions and their validity vary from one method to another, and so is the reliability of the identified modal properties. Results are as good as assumptions. Typically, the structure can be under its ‘operating condition’, i.e., no initial excitation or artificial excitation required during the time span of recorded data; and hence the term ‘operational modal analysis’. The structure is subjected to a variety of excitations which are not measured, and almost invariably cannot be measured.

Cost-Effectiveness

High implementation economy is one primary advantage of ambient vibration test. This is particularly attractive for civil engineering structures, where it can be expensive, disruptive or risky to carry out free vibration or forced vibration tests. Of course, ambient vibration test is not always the best solution. It is a matter of feasibility

and cost-effectiveness. For example, mechanical systems or components, for their small size, can often be tested with forced vibration under controlled laboratory conditions with affordable effort, achieving a much higher identification precision than ambient vibration test. For its importance, one would not expect to identify the modal properties of a satellite prototype primarily by ambient vibration test in the assembly plant. A series of forced vibration tests is more appropriate. However, when the satellite is in operation, forced vibration test can be difficult to implement and ambient vibration data can be useful for updating dynamic characteristics.

Drawbacks

The following are some limitations or disadvantages of ambient vibration test, primarily due to the fact that the ambient excitation is not measured:

- (1) It is not possible to identify the modal mass directly;
- (2) Identified modal properties only reflect their values at ambient vibration level, which is usually at least an order of magnitude lower than the serviceability level or other design levels of interest. This is especially relevant for the damping ratio, which can be amplitude-dependent;
- (3) Identified modal properties often have much lower precision than their counterparts from well-controlled free or forced vibration data. The results are only as good as the statistical assumptions about the unknown excitation.
- (4) Identification methods are more sophisticated than those for free or forced vibration data; and ‘closed’ in some sense like a ‘black-box’. They involve statistical concepts in the modeling or treatment of the unknown excitation. The results require more technical background to interpret correctly. These reduce confidence in the results.

1.2 Uncertainties

In OMA, information about the input excitation is absent and the vibration response can be so small that measurement noise is no longer negligible. As a result, the identified modal properties often have significantly higher uncertainty than their counterparts from free or forced vibration data.

1.2.1 Variability and Identification Uncertainty

The meaning of ‘uncertainty’ has yet to be defined, but intuitively it is associated with the experience that one obtains different answers from different data sets. Two sources can be distinguished here, namely, ‘variability’ and ‘identification uncertainty’. Variability refers to whether the modal properties of the structure have changed from one data set to another. If there is such a change then it is reasonable to expect different answers. Identification uncertainty is associated with the precision

with which one can estimate the modal properties using a given data set. It stems from incomplete knowledge in modeling (imperfect) and data (limited and noisy). Ideally, if variability and identification uncertainty were both absent, the answers from two data sets must be identical. In the presence of identification uncertainty, the answers will differ even if there is no variability. In reality, both variability and identification uncertainty are present. It is not trivial to distinguish them, especially when they have a similar order of magnitude. In terms of methodology, this book focuses on identification uncertainty, although variability can be naturally observed in field test examples with multiple setups (Chap. 14).

1.2.2 Sources of Identification Uncertainty

Regardless of the method used, one should not expect to be able to identify the modal properties with perfect precision from a given set of ambient vibration data, simply because there is not enough information. The lack of information arises primarily from the following sources:

- (1) Limited data—the amount of data is finite in terms of the number of measured degrees of freedom (DOFs), sampling rate and duration;
- (2) Measurement error—the recorded data is contaminated by noise due to the sensor, transmission medium (e.g., cable) and digitizing hardware;
- (3) Unknown excitation—the excitation is not measured but is modeled statistically with unknown parameters, which can be part of the properties to be identified;
- (4) Modeling error—theory only provides a way to view the world but nature does not care. The measured data need not obey modeling assumptions, e.g., the excitation need not be broadband; the structure need not be viscously or classically damped.

In principle, Source 1 can be reduced using a sufficiently large amount of data. Source 2 can be reduced or practically eliminated with high quality equipment. Source 3 cannot be eliminated and its effect will always stay. Source 4 can lead to bias in the identification results. To a certain extent, it can be reduced with advances in modeling but the progress can be slow; or for practical purposes one would still choose to use a conventional model and live with the modeling error. Depending on the source, it may be possible to verify, control or at least reveal the potential model error in the interpretation of results.

1.3 OMA Methods

OMA is a research subject of a few decades old. For its importance, a variety of methods have been developed. A quick tour of the scope and the ways where OMA has been approached is given below. The measured vibration in OMA stems from

Table 1.1 Approach and scope of OMA methods

Aspect	Attribute	Treatment
Approach	Vibration due to unknown excitation	Eliminate or model
	Governing equation	Physical or state-space
	Analysis perspective	Time domain or frequency domain
	Identification philosophy	Frequentist or Bayesian
Scope	Measured DOFs	Single → multiple
	Mode proximity	Well-separated → close
	Setup	Single → multiple
	Data synchronization	Synchronous → asynchronous
	Identification uncertainty	Not calculated → calculated → insight

‘→’ indicates increasing sophistication

unknown excitations, which must be addressed upfront in method development. The common strategy is either to *eliminate* or *model* the vibration due to unknown excitations. Methods can differ in the way they approach the governing dynamic equation. A method can operate in the time or frequency domain, or take advantage of both. It can operate in terms of physical or state-space coordinates. Methods can also differ in the philosophy about the inverse problem. The two main schools of thoughts are ‘frequentist’ (non-Bayesian) and ‘Bayesian’ perspective. The former constructs an estimate (a proxy) for the modal properties and views uncertainty as the variability of the estimate calculated using data from repeated experiments. The Bayesian perspective models the modal properties as random variables whose probability distribution (and hence uncertainty) depends on the available data and modeling assumptions.

As methods populate, they expand in scope and become more sophisticated. Early methods can only handle data from a single measured DOF, but now most methods can handle data from multiple DOFs, allowing the mode shapes at the measured DOFs to be identified. Conventional methods assume well-separated modes but there are now algorithms dealing with close modes. Synchronous data in a single setup is conventionally assumed, but now multiple setup data are also considered. There are also algorithms that can handle asynchronous data (e.g., those from different smart phones). Conventional focus is on constructing an estimate for modal properties but contemporary methods also aim at calculating their uncertainty and building insights. Table 1.1 summarizes the aspects discussed above. Some notable classes of methods are next discussed in Sects. 1.4 and 1.5.

1.4 Non-Bayesian Approach

Non-Bayesian methods construct an estimate as a proxy for the modal properties, leveraging on the developers’ insights about the physical nature of ambient vibration and theoretical origin. Notable methods along the idea of eliminating

random response or exploiting their statistics are discussed below. The treatment of uncertainty follows.

1.4.1 Eliminating Random Response

‘Random decrement method’ exploits the idea that averaging a sufficiently large number of ambient vibration sample histories with some non-canceling initial conditions (e.g., of the same sign) will result in a time history that mimics the free vibration time history. The idea stems from the fact that dynamic response generally comprises a free vibration (due to initial conditions) and forced vibration (due to excitations) component. Under random excitations the latter will cancel out statistically in the average, following an inverse square root law, i.e., magnitude $\propto 1/\sqrt{\text{no. of samples}}$. The averaged time history is called a ‘random decrement signature’ (RDS). It is used for modal identification as if it were a free vibration time history. This idea is conceptually simple and generally applicable for both linear and nonlinear dynamic response. Care should be taken not to extrapolate the meaning of RDS, however. RDS is a derived quantity rather than the actual free vibration history in a real experiment. The physical response of a structure may exhibit amplitude-dependent damping behavior, but it need not be reflected in the same way as is detected in the RDS. There is generally a (non-vanishing) bias in the modal property estimate because RDS is a statistical average conditional on some specified initial conditions, biasing it away from the free vibration history. See Cole (1973) (original work), Cole (2014) (recent review) and Aquino and Tamura (2016) (recent discussion).

1.4.2 Exploiting Statistics

Instead of eliminating the random component of ambient vibration in the measured data, one can model it using probability theory and exploit its theoretical statistical properties for modal identification. For the theoretical properties to be mathematically tractable, it is often assumed that the structure is time-invariant, the measured data is stochastic stationary (i.e., statistics do not change with time) and the excitation is broadband random. Exploiting statistical properties in the time or frequency domain leads to different methods with different implied assumptions, limitations, or computational effort.

Time Domain Statistics

In the time domain, assuming white noise excitation, the correlation function of ambient data can be shown to satisfy the same free vibration equation of motion of the structure, and so it can be treated as free vibration data for modal identification. In the general case, the correlation function contains dynamics arising from the

non-white nature of excitation. Although derived for the same purpose, the correlation function should not be confused with the RDS in random decrement method. They are derived quantities used mathematically as free vibration data for modal identification, but their meaning should not be extrapolated to be associated with the data physically measured in a free vibration test.

The theoretical correlation function is not known and so it must be estimated from data. The sample correlation function has a statistical error, following an inverse square root law. Direct fitting of the sample correlation function with the theoretical expression of free vibration response results in a high-dimensional nonlinear optimization problem involving the modal properties. To resolve this, one computational strategy is to work with the dynamic equation or response in the discrete-time domain. Simplification results by taking approximations for small time interval, justified when the interval is small compared to the natural periods of interest. An output ‘state vector’ is defined to comprise the values of the correlation function at a finite number of consecutive time steps. The latter is related to the order of the modeled dynamics. The output state vector at the next time step can be written recursively as a linear function of that at the current step, related by a coefficient matrix that depends on the modal properties. This matrix can be estimated efficiently by least squares method or singular value decomposition. The modal properties are estimated from the estimated coefficient matrix.

Notable developments based on the above idea or its variants include the Ibrahim Time Domain Method (Ibrahim and Milkulcik 1976), Poly Reference Method (Vold et al. 1982) and Eigen Realization Algorithm (Juang and Pappa 1985). Stochastic Subspace Identification (SSI, Overschee and De Moor 1996) is a popular method of this class. ‘Covariance-driven’ SSI operates on the sample correlation function. ‘Data-driven’ SSI operates directly on the measured data. See Peeters and De Roeck (2001) for a review.

Frequency Domain Statistics

In the frequency domain, the power spectral density (PSD) matrix of ambient data can be theoretically written in terms of the transfer functions and mode shapes of the contributing modes. For a well-separated mode, the PSD matrix is dominated by the contribution of the mode near the natural frequency. This allows the modal properties of the mode to be directly estimated from the eigenvalue properties of the PSD matrix. The eigenvalue can be used to estimate the natural frequency and damping ratio; the eigenvector gives the mode shape. Estimation is non-trivial for close modes, whose contributions are ‘entangled’ because generally modal excitations are correlated and mode shapes (confined to measured DOFs only) are not orthogonal. The eigenvectors of the PSD matrix do not necessarily give the mode shapes for close modes. They are orthogonal vectors spanning the subspace of the mode shapes, and are commonly called ‘operational deflection shapes’. The same issue and terminology also apply to time domain methods when the mode shapes are directly estimated from matrix decomposition.

In implementation, the sample PSD matrix estimated from data is used as a proxy for the theoretical PSD matrix. For a given frequency resolution, the

estimation error follows an inverse square root law. ‘Peak picking’ is a simple procedure that uses the eigenvalue properties at the natural frequency for estimating the modal properties. ‘Frequency domain decomposition’ (FDD, Brincker et al. 2001) with enhanced variants is a notable method exploiting the band near the natural frequency. See also Pintelon and Schoukens (2001).

1.4.3 Identification Uncertainty

The methods described in Sects. 1.4.1 and 1.4.2 focus on developing an algorithm that makes use of the measured data to calculate an estimate that will be used as a proxy for the modal properties. Using a particular method with a given data set, suppose the natural frequency estimate is calculated to be 1.2 Hz. For practical use (e.g., to feed into design calculations) this 1.2 Hz will be taken as the natural frequency of the structure, i.e., a proxy. This is despite the fact that the estimate based on another data set is unlikely to be exactly equal to 1.2 Hz. This motivates one to define uncertainty to reflect the variability of the estimate over different data sets. Implementing this idea quantitatively via statistics, identification uncertainty is defined as the variance of the estimate over hypothetical ‘identically controlled experiments’. Identically controlled experiments do not imply identical data, because there are still factors that cannot be controlled, e.g., unknown ambient excitation and measurement noise. Uncertainty here is defined in the frequentist sense via the relative frequency of occurrence of events, even though identically controlled experiments are neither possible nor actually performed.

Inherent Uncertainty

Calculating the variance of modal property estimate requires a separate algorithm from that for obtaining the estimate. Mathematically, the variance of the estimate is defined as the expectation (ensemble average) of squared deviation from the mean, taken over random data assumed to arise from the ‘unknown true’ modal properties. It will therefore depend on the unknown true modal properties but not on a particular data set. This is a ‘frequentist’ definition of uncertainty and is ‘inherent’ in nature, in that there is an unknown but fixed value regardless of the data at hand. Clearly, the expression of the variance, if it can be derived, will depend on the unknown true values of the modal properties. In implementation, the variance is estimated by substituting the true values by their estimates.

Perturbation Expressions

Developing the expression of variance is not trivial and it can be challenging because the estimate depends in a complicated (possibly implicit) manner on the data. One strategy to simplify derivation is to consider the first order sensitivity of the modal property estimate due to variability of data from the measured set. The resulting derivation and expression are still algebraically involved but at least mathematically tractable as an algorithm. The resulting expression is approximate but sufficient for practical purposes. See Pintelon et al. (2007) and Reynders et al. (2008)

for uncertainty in SSI with single setup data; and Döhler et al. (2013) with multiple setup data.

Empirical Estimates

Besides analytical expressions, the uncertainty of modal properties in a frequentist sense can also be calculated as the sample variance of the estimate using data from multiple experiments. The motivation and idea are straightforward, but one should note that the sample variance generally comprises identification error and variability stemming from the tested structure and environment. The sample variance need not agree with the value calculated from the theoretical expression of variance, because the latter is derived based on some assumed probability distribution of data that nature does not care about.

1.5 Bayesian Approach

The methods described in Sect. 1.4 aim at a mathematically ‘smart’ and tractable way of exploiting the information in the data to produce an estimate for the modal properties. It should be smart so that the estimate is around the actual value to be estimated and has small variance. It should be tractable so that it can be implemented in an efficient algorithm. This strategy is a natural step when one transforms intuitions about the forward problem into a solution for the backward problem. It need not be the best way, however. The algorithm and hence the quality of the estimate depends on the insights and mathematical facility of the developer. For example, why correlation function? Why least squares method? If one is not able to identify a given mode, is it because the method has not made full use of the information contained in the data, or such information is not there anyway (so no method can do)? How does one come up with the best estimate? The methods inevitably make use of some theoretical statistical properties of data, which are unknown and must be substituted by their sample counterparts. Downstream calculations based on the sample counterparts produce quantities that need not obey the modeling assumptions in structural dynamic analysis or design. For example, the coefficient matrices estimated in a time domain approach need not correspond to one realizable from a structure satisfying the Newton’s equation of motion. The theoretical variance of the estimates for modal properties generally involves their unknown true values, whose existence is questionable.

1.5.1 Philosophy

This book focuses on Bayesian methods for OMA, which differ at least philosophically from non-Bayesian ones. Probability is still used to treat uncertainties. Instead of constructing a proxy estimate, however, one models the modal properties

as random variables with a probability distribution that depends on the available information. One still makes assumptions about structural dynamics, the unknown ambient excitation and measurement noise. One still solves the forward problem to obtain the theoretical statistical properties of the measured data for given modal properties. In fact, the forward problem is addressed in full detail: given the modal properties, what is the distribution of data? Bayes' Theorem is invoked to swap the roles of the modal properties and available data, so that one can address the backward problem, also in full detail: given the data, what is the distribution of the modal properties? Mathematically, Bayes' Theorem is a just statement of conditional probability, but its philosophical meaning is far reaching. It leads to a method that fundamentally transforms one's available information into the required information that addresses the question being asked.

1.5.2 Posterior Distribution and Statistics

In a Bayesian approach, all identification results about the modal properties are encapsulated in the 'posterior' (i.e., given data) distribution, which is a joint probability density function conditional on modeling assumptions and available data. In modal identification, 'prior' information (i.e., in the absence of data) is typically negligible compared to that provided by data. In this case, the posterior distribution is directly proportional to the 'likelihood function', i.e., the (modeled) distribution of data for given modal properties. This is a function that depends on both the data and modal properties. It was the answer in the forward problem where the modal properties were fixed. Fixing the data (as measured) now and viewing the likelihood as a function of modal properties gives the posterior distribution.

For OMA, the likelihood function depends in a non-trivial manner on the modal properties. As a result, the posterior distribution does not belong to any standard distribution (e.g., Gaussian, exponential), but it still has a centralized shape. With sufficient data, it can be well-approximated by a Gaussian distribution. The peak location of the distribution represents the posterior 'most probable value' (MPV) of the modal properties. The covariance matrix of the modal properties implied by the distribution, i.e., 'posterior covariance matrix', quantifies their identification uncertainty.

In contrast to the frequentist approach, no true parameter value or concept of repeated experiments is involved in the quantification of identification uncertainty. Up to the same modeling assumptions and data, the likelihood function is unique, and so are the posterior distribution and its descriptive statistics (e.g., MPV). The identification uncertainty depends on, and can be calculated from, a given data set. There is no inherent uncertainty; one can get a bigger uncertainty with one data set but a smaller one with another. Different data sets provide different information about the modal properties and hence lead to different levels of identification uncertainty.

1.5.3 Computing Posterior Statistics

Modeling assumptions define the likelihood function and hence a Bayesian ‘formulation’ of the OMA inference problem. Deriving the likelihood function involves solving the forward problem, for which techniques are well-established, e.g., stochastic processes and stochastic dynamics. The likelihood function should be derived in explicit form so that its value can be calculated efficiently. The remaining tasks are computational in nature, i.e., to determine the posterior MPV and covariance matrix. Determining the MPV involves solving an optimization problem with respect to (w.r.t.) the modal properties, whose number can be large. Determining the covariance matrix involves calculating the Hessian matrix of the negative logarithm of likelihood function (NLLF) w.r.t. the modal properties. Brute-force numerical optimization for MPV is prohibitive and unlikely to converge reliably. Finite difference method for Hessian is time-consuming and inaccurate. Proper algorithms are required for accurate and efficient computation.

Existing strategies for determining the posterior MPV exploit the mathematical structure of the NLLF to derive semi-analytical solutions for the MPV of different groups of modal properties. This allows the MPV to be determined in an iterative manner, updating different groups until convergence. This is especially critical for mode shape values because their number can be large in applications. The semi-analytical solution for the MPV of mode shape values effectively suppresses the growth of computational effort with the number of DOFs. On the other hand, analytical expressions can be derived for the Hessian of NLLF to allow accurate and efficient calculation of the posterior covariance matrix. One major issue is how to systematically account for the constraints among the modal properties (e.g., mode shape under scaling constraint). This can be handled through Lagrange multiplier concepts.

1.5.4 Formulations and Algorithms

Bayesian OMA formulations can differ in their assumptions and the form of data used. For synchronous data acquired in a single setup (the nominal setting in OMA), formulations have been developed chronologically based on time domain data (Yuen and Katafygiotis 2001a), sample power spectral density (Yuen and Katafygiotis 2001b) and Fast Fourier Transform (FFT) of data (Yuen and Katafygiotis 2003). They all originated from Yuen’s MPhil thesis (Yuen 1999). Time domain formulation is too restrictive in the modeling assumptions on ambient excitation and measurement noise. Sample PSD formulation requires averaging concept (as in non-Bayesian methods) and is less fundamental than FFT formulation. It gives no computational advantage either (Yan and Katafygiotis 2014; Au 2016).

This book focuses on Bayesian formulations based on the FFT of ambient vibration data, which is found to be mathematically tractable with a fairly robust set of assumptions. Computational algorithms for single setup data have been developed in Au (2011) for well-separated modes (see also Zhang and Au 2013) and Au (2012a, b) for multiple modes. Formulation for multiple setup data and computational algorithms are developed in Au and Zhang (2012a) and Zhang et al. (2015), which originated from Zhang's PhD thesis (Zhang 2011). Field applications of Bayesian OMA are growing but reports in the literature are still limited. See, Au and To (2012), Au and Zhang (2012b), Au et al. (2012a, b), Kuok and Yuen (2012), Ni et al. (2015), Liu et al. (2016), Zhang et al. (2016a, b, c), Ni and Zhang (2016), Ni et al. (2016a, b, 2017) and Lam et al. (2017).

1.5.5 Maximum Likelihood Estimation

Bayesian methods are sometimes confused with 'maximum likelihood estimation' (MLE). MLE is a non-Bayesian method where the modal property estimate is determined as the one that maximizes the likelihood function for a given data set. When the same likelihood function in the Bayesian method is used and no prior information on the modal properties is used, the posterior MPV is equal to the estimate by MLE. This is perhaps the reason for the confusion. Identification uncertainty in MLE, however, is still defined as the variance of the estimate over repeated experiments, as in a non-Bayesian method.

MLE and Bayesian method share some similarities in their mathematical structure (not philosophy) of the problem. For a sufficiently large data size and assuming that the data is indeed distributed as the likelihood function for some 'true' properties (assumed to exist), the MLE estimator has the smallest variance among all unbiased estimators. The smallest variance is given by the tightest Cramér Rao Bound, mathematically equal to the inverse of the full Fisher information matrix. It coincides with the leading order of the posterior variance in the Bayesian method under the same set of (frequentist) assumptions just described. See Sect. 9.6 for details. Matarazzo and Pakzad (2016a, b) proposed a MLE method for OMA based on a state-space formulation.

1.5.6 Drawbacks and Limitations

Bayesian OMA methods are implicit, requiring numerical optimization to determine the posterior MPV. This is typical for Bayesian methods in general, because of the nonlinear relationship between the likelihood function and the model parameters to identify. In contrast, most frequentist OMA methods are explicit in nature, requiring no numerical optimization or iterations. Existing Bayesian OMA methods are capable of reducing the dimension of the numerical optimization problem, but not

eliminating the need for solving it. The methods are sustainable with the number of measured DOFs but not the number of close modes that must be identified together. Generally, well-separated modes can be identified in a matter of seconds on a laptop regardless of the number of measured DOFs. Close modes require significantly longer time to converge, from a few seconds to a few minutes on a laptop depending on the signal-to-noise ratio of data and collinearity of mode shapes; but relatively insensitive to the number of measured DOFs. It is possible that the solution of MPV does not converge when the data does not contain enough information for identification. In typical situations where the number of close modes does not exceed three, the problem is still manageable.

Existing Bayesian OMA methods mostly address identification uncertainty but not variability. The posterior variance tells how precise a modal property can be identified based on the given data and modeling assumptions, but it does not tell what can be expected from another data set without additional assumptions. This limitation applies to OMA in general. Recognizing this aspect is important for proper interpretation of results and further advance. Hopefully, a scientifically sound methodology can be established in the near future for distinguishing variability and identification uncertainty. This is important for SHM, because systematic change in the monitored structure cannot be efficiently detected if it cannot be distinguished from identification error.

1.6 Overview of This Book

As the title tells, this book is about the modeling of ambient vibration data and making inference about modal properties following a Bayesian approach. It has taken a different perspective to OMA from the current conventional approaches, which are mostly non-Bayesian. Insights about dynamics are still discussed, but they no longer drive the development of OMA method. Rather, Bayes' theorem is invoked to give the answer to modal identification. Non-Bayesian OMA methods are not covered. They can be found in review papers and monographs; see, e.g., Wenzel and Pichler (2005), Reynders (2012) and Brincker and Ventura (2015). Although only Bayesian methods are discussed formally for modal identification, Chaps. 2–7 on modeling are relevant to OMA in general. The same is also true for Chap. 15 on managing identification uncertainty. Chapters 8 and 9 discuss Bayesian and non-Bayesian approaches in a general context, not just for OMA.

Basic Assumptions

Bayesian inference in this book makes use of the FFT of ambient vibration data. This is especially suitable for OMA because the information of different vibration modes is naturally partitioned in the frequency domain by way of resonance. Time-invariant linear classically damped dynamics is assumed. This is the conventional assumption in structural dynamic analysis and design; and so the identification results are directly transferrable. The excitation is assumed to be

broadband random in the sense that the modal forces of the subject modes have a constant PSD near their natural frequencies. This assumption turns out to be fairly robust (Sect. 10.2).

Application Context

Although the application context of this book is on civil engineering structures (e.g., buildings, bridges), modal identification can also be applied and is demanded for mechanical and aerospace structures, e.g., ground testing of aircrafts. It is also demanded in electrical power networks, i.e., ‘wide-area monitoring’, where electro-mechanical oscillations obey exactly the same dynamic equation of motion with analogous definition (e.g., voltage $\leftrightarrow$ displacement). See Pierre et al. (1997), Rogers (2000) and Seppanen et al. (2016). The theories and algorithms developed in this book are applicable regardless of discipline, as long as the governing equations and basic assumptions fit in.

The chapters in this book can be grouped in terms of ‘modeling’, ‘inference’, ‘algorithms’ and ‘uncertainty laws’. They are outlined in the following.

1.6.1 Modeling

Chapters 2–7 contribute to the theoretical modeling of ambient vibration data. Chapter 2 introduces the Fourier analysis of a deterministic process, providing the basics for frequency domain perspective and techniques in a non-probabilistic context. Chapter 3 discusses the dynamic response of a structure subjected to known excitations. Principles of modal testing are introduced to strengthen dynamics concepts and give an idea of how modal properties can be estimated. Chapters 4 and 5 are probabilistic counterparts of Chaps. 2 and 3, respectively, involving unknown excitations modeled by a stationary stochastic process. They allow one to model ambient vibration response in a probabilistic manner. They do not yet fully explain the measured ambient vibration data, which is the output of sensors and data acquisition hardware and is contaminated by measurement noise. These aspects are discussed in Chap. 6. Chapter 7 introduces the basic tools for analyzing ambient vibration data and explains the results based on the theories in the previous chapters.

1.6.2 Inference

Chapter 8 introduces Bayesian inference in a general context, covering different cases of identifiability where the posterior distribution has different topologies and requires different computational strategies for characterization. Chapter 9 discusses the frequentist approach of system identification. Similarities and differences with

the Bayesian approach are highlighted. The mathematical connection between the frequentist and Bayesian notion of identification uncertainty leads to an effective means for investigating the achievable precision of OMA in Chap. 16 later. Chapter 10 formulates Bayesian OMA problems for different situations, covering data from a single setup, multiple setups and asynchronous channels. Chapter 11 discusses the general computational strategies for determining the posterior MPV and covariance matrix. Computing the Hessian of the NLLF incorporating constraints (e.g., from mode shape scaling) is a major topic there. Systematic and efficient strategies are presented.

1.6.3 Algorithms

Chapters 12–14 develop and provide the recipe for Bayesian OMA algorithms in different settings within the computational framework in Chap. 11. Chapters 12 and 13 consider respectively well-separated modes and multiple (possibly close) modes, both assuming data from a single setup. Chapter 14 considers well-separated modes with data from multiple setups. As a basic structure of these chapters, the likelihood function is analyzed in order to develop an efficient computational procedure for the posterior MPV. For the posterior covariance matrix, analytical formulas are presented within the framework of Chap. 11. Examples based on synthetic data are presented to illustrate the problem, method or interpretation of results. Application is illustrated using laboratory and field test data.

1.6.4 Uncertainty Laws

In addition to modeling and Bayesian inference, this book also covers the ‘uncertainty laws’ of OMA, which are closed form analytical expressions giving the leading order term of the posterior variance of modal properties. This is one of the latest advances in OMA, providing insights into the achievable precision of OMA and a scientific basis for planning ambient vibration tests. Key results are discussed in Chap. 15, requiring minimal mathematical facility. Examples are also given to illustrate how the test configuration in an ambient vibration test can be decided based on the uncertainty laws. Detailed theory and derivations are provided in Chap. 16.

Three appendix chapters provide supplementary materials to the main chapters. Appendix A on complex Gaussian distribution supplements the theory on the statistical properties of FFT in Chap. 4. Appendix B on Hessian under constraints supplements the theory for calculating the posterior covariance matrix in Chap. 11. Appendix C supplements the general mathematical tools referred in different chapters.

1.7 How to Use This Book

This book is written with the primary aim to provide a one-stop reference for learning the theoretical, computational and practical aspects of OMA and its Bayesian solution. Each chapter is written in a self-contained manner while connections with other chapters are highlighted in the beginning. The chapters have been arranged in a logical manner where pre-requisites needed in the current chapter can be found in the previous chapters. Following this order is natural in presentation and writing but not necessarily the best way to learn OMA. In my experience, a problem-driven approach, i.e., setting a target problem upfront, trying to solve it, looking for and learning the required skills along the way, is a better way to grasp a subject.

This book grows out of my personal need to advise undergraduate and master degree projects, doctoral and postdoctoral research, and disseminate research findings to academics and industry. This spectrum of audience presents conflicting demands on the contents and the style of writing. For example, undergraduate and master degree projects have a short time span, e.g., a year. It is vital for them to have a quick start, to get interested and be on the right track. Proofs are of secondary importance, at least on first reading. A quick start is also good for PhD students and postdocs, but they have a longer term agenda and need a strong theoretical foundation for original research and further developments. Proofs are needed for scientific rigor and provide insights into the mathematical structure of problems. They may also involve important techniques, mastering which allows one to explore and solve new problems. Practitioners are interested in applications, and in some cases, implementation (algorithms).

To achieve my primary aim while balancing interests for different groups of audience, I have tried to present materials from a utility point of view. Theories are motivated from use. Results are developed from mathematical reasoning when it is not difficult to understand. Proofs that may disturb the flow of reading are postponed until the end of discussion. Proofs that are too technical are omitted. Examples are provided to illustrate theory, method or application; or to provide details relevant to specific contexts. Proofs and examples are supplementary; skipping them should not affect understanding the main text.

Some specific advice is given below for different groups of audience. For this purpose (only) I would stereotype their background and objectives.

1.7.1 *Student*

By a ‘student’ I suppose you are a senior undergraduate or master degree student pursuing a project related to modal identification or its applications. You have learnt basic probability and statistics in junior years. You may not have taken a course in

structural dynamics; or you are taking it concurrently with the project; or you have taken it before but are rusty on it.

Structural Dynamics

You may first take a look at Chap. 3. There is no need to finish the whole chapter at this stage. Target to grasp the concepts about natural frequency, damping ratio, dynamic amplification factor and resonance in Sect. 3.1; mode shape and modal superposition in Sect. 3.2. The derivations in this chapter should not be difficult to follow; otherwise skip them for the time being and revisit later. Build insights into the results by plotting the theoretical dynamic response and changing some parameters to see the effects. The response to any arbitrary loading can be computed using the Newmark method (Sect. 3.8), or simply the function `lsim` if you are using Matlab. Compute and plot the response for some simple load cases and see if the results make sense.

Generating and Analyzing Synthetic Data

When you are comfortable with structural dynamics you may take a jump start on analyzing ambient data. Although experimental data is always the target, it is easier to start with synthetic data because you can have good control over it. Generate a time history of white noise (Sect. 7.7) and take it as an excitation to a single-DOF structure. Adding white measurement noise to the response gives the synthetic data. Plot the data with time. Get the units right and indicate them. You may not find anything interesting in the plot but looking at the original data is always the first thing an analyst should do. The interesting things are in the PSD and singular value (SV) spectra. Follow the procedures in Sects. 7.2 and 7.3 to plot and interpret these spectra. Get the units right and indicate them. There are terminology and concepts in Chap. 7 that you may not understand, e.g., PSD, SV, FFT. These are to do with viewing a time series as a sum of sines and cosines, i.e., ‘Fourier analysis’, and applying probability concepts for analysis. Chapter 2 introduces Fourier analysis for a given time series that does not involve any probability concept. You have learnt probability but the level required for PSD and SV is higher than that. You will also need Chaps. 4 and 5. Sections 4.8–4.10 and 5.3–5.4 may be skipped at the moment. Skip difficult proofs on first reading and revisit later if that can improve understanding. Study the theories in Chaps. 2, 4 and 5 as far as you need them in Chap. 7.

Analyzing Real Data

Once you are comfortable with analyzing synthetic data, it is time you meet reality. Obtain a set of ambient vibration data from a simple structure in the laboratory as soon as possible. Get the units right. It is time to take a look at Chap. 6 to know something about the process that gives you the data, so that you are mindful of the possible distortions and noise. Repeat experiments to build confidence with the measured data. Change settings, one at a time, and see if you get what you expected. Once you are confident with laboratory data, plan and perform your field test. You may need a different type of sensor (e.g., servo-accelerometer) to pick up the small ambient vibrations in civil engineering structures. Build confidence with your sensors in the laboratory first. Make necessary preparations (e.g., software) so

that you can analyze field data with your laptop on site (and you are sure your day is worth before you leave).

Bayesian OMA

By now hopefully you are confident with acquiring and analyzing ambient data, and estimating modal properties empirically from the SV spectrum. You may then explore modal identification more formally. For non-Bayesian methods, you may consult the references mentioned in Sect. 1.4 or the beginning of Sect. 1.6. For Bayesian methods, the algorithm for the posterior MPV of modal properties for well-separated modes in Chap. 12 should be simple to program. The one for posterior covariance matrix takes more time, involving analytical expressions for the second derivatives of the NLLF. Program and check them individually before assembling into a big program. If this sounds tedious, finite difference approximation may be resorted but do check its convergence with different step sizes. Another alternative is the uncertainty laws in Chap. 15, which gives the ballpark figure of identification uncertainty and may be taken as a good approximation. They are also quite insightful. If you are more ambitious and time permits, you are encouraged to try multiple setups (Chap. 14), which is quite useful and allows you to produce a (more interesting) detailed mode shape with a small number of sensors. The algorithm for multiple (possibly close) modes (Chap. 13) can take some time to program, be it for the posterior MPV or covariance matrix. When time does not permit, it may be better to use an existing program (see Sect. 1.7.4) to explore interesting applications. Other chapters may be consulted depending on interest and background needed. Perhaps by this time you may have new ideas to pursue.

1.7.2 Researcher

By a ‘researcher’ I suppose you are not only interested in using OMA methods but also mastering the underlying principles and mathematical techniques so that you can develop original research. You may be a PhD student, postdoc, an academic, engineer or scientist in a research laboratory. You may be familiar with Fourier analysis, structural dynamics and their probabilistic form, though at varying degree. Chapters 2–7 are reference materials that may be consulted if needed. In particular, the long-data asymptotic behavior of FFT is an important fundamental result that is worth pursuing; see Sects. 4.8 and 4.9. If you want to see what Bayesian OMA methods can offer, a tour at the examples in Chaps. 12–14 and the uncertainty laws in Chap. 15 will give you an idea. To have a better appreciation of results, Chap. 10 discusses the underlying assumptions and formulations in different cases. To see how Bayesian identification results may be interpreted differently from those by conventional frequentist methods, Chaps. 8 and 9 may be consulted.

Chapters 12–14 provide the recipe for programing the posterior MPV and covariance matrix of modal properties in different settings. Programming for the MPV is generally easier than the covariance matrix. The algorithms for the MPV

for well-separated modes are straightforward to program, be it for single or multiple setups. The algorithm for the MPV for close modes takes more time. The same applies to the covariance matrix using analytical formulas. Finite difference approximation may provide a temporary solution and a reference for checking. So are the uncertainty laws (Chap. 15) for well-separated modes.

The uncertainty laws are one of the latest advances in OMA. Take a look at Chap. 15 to see if they make sense. Chapter 16 outlines how they can be derived, with details provided to illustrate the techniques in the derivations. The connection between frequentist and Bayesian uncertainty in Sect. 9.6 provides an effective systematic means for deriving the uncertainty laws. As a result, the Cramér Rao Bound (Sect. 9.3) and Fisher information matrix (Sect. 9.4) are relevant, despite their non-Bayesian origin. Appendix A on complex Gaussian distribution and Appendix B on Hessian under constraints are general tools that you may find applications in other research areas.

1.7.3 Practitioner

By a ‘practitioner’ I suppose you are an engineer primarily interested in applying OMA to assist engineering solutions in projects. You may take a look at the field data examples in Chaps. 12–14 to see what Bayesian OMA methods can offer. You will see that, in addition to the posterior MPV that is analogous to the modal property estimate in a non-Bayesian method, the methods can also determine the identification uncertainty. How the uncertainty depends on test configuration is generally complicated, but the relationship reduces to remarkably simple form for sufficiently long data and small damping, both typically encountered in applications. That is presented in a set of formulas called ‘uncertainty laws’ in Chap. 15. Using such formulas, the required data length to achieve a desired identification precision can be calculated. The chapter shall convince you that, as far as natural frequencies and damping ratios are concerned, getting a good sensor is more cost-effective than getting more sensors. For algorithms, see the advice for researchers in Sect. 1.7.2. See the advice for students in Sect. 1.7.1 if you are interested in building some in-house experience with analyzing ambient data. Other chapters may be consulted when being referred to. Proofs may be skipped.

1.7.4 Supporting Resources

Resources are being developed to disseminate Bayesian OMA research. They may include computer codes, examples, tutorial notes, database for OMA data and modal identification results, method developments and applications. They may be searched on the internet with keywords such as ‘Bayesian operational modal analysis’, ‘Bayesian OMA’ or ‘bayoma’.

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Chapter 2

Spectral Analysis of Deterministic Process

Abstract This chapter analyzes a deterministic process in the frequency domain via Fourier series (for period process) and Fourier Transform (for non-period process). Estimators based on a sample time history of the process are introduced and their possible distortions arising from finite sampling rate (aliasing) and duration (leakage) are discussed. The Fast Fourier Transform provides an efficient computational tool for digital implementation. The chapter contains a section that connects the presented mathematical theory to implementation in Matlab, which is a convenient platform for scientific computing.

Keywords Fourier series • Fourier Transform • Fast Fourier Transform • Matlab • Aliasing • Leakage

A time series, or ‘process’, describes how a quantity varies with time. Viewing as a function of time allows one to see, e.g., when it is zero, how fast it changes, the minimum and maximum values. This ‘time domain’ view is not the only perspective. One useful alternative is the ‘frequency domain’ view, as a sum of ‘harmonics’ (sines and cosines) at different frequencies and studying how their amplitude varies with frequency. This view is especially relevant for a variety of processes that exhibit variations at different time scales, revealing the characteristics of contributing activities. It also offers a powerful means for studying the oscillatory response of systems with resonance behavior.

In this chapter we introduce the theory for analyzing a process in the frequency domain, namely, ‘Fourier analysis’. The basic result is that a periodic process can be written as a ‘Fourier series’ (FS), which is a sum of harmonics at discrete frequencies. An analogous result holds for a non-periodic process with finite energy, where the sum becomes an integral over a continuum of frequencies and is called ‘Fourier Transform’ (FT). In digital computations, the integrals involved in Fourier analysis can be approximated in discrete time and computed efficiently via the ‘Fast Fourier Transform’ (FFT) algorithm. This approximation leads to distortions, namely, limited scope by ‘Nyquist frequency’, ‘aliasing’ and ‘leakage’. These must be in check so that the calculated results reflect well their targets and are

correctly interpreted. Fourier theory is often introduced in undergraduate texts of differential equations, e.g., Boyce and DiPrima (2005). See Champeney (1987) for Fourier theorems and Sundararajan (2001) for FFT. Digital signal processing is a closely related subject, see, e.g., Lathi (2000). Smith (2002) gives a non-technical coverage.

As the title tells, this chapter is about ‘deterministic process’, where the subject time series is taken as ‘fixed’ or ‘given’. No probability concept is involved. While all data (including the ones in operational modal analysis) are by definition deterministic when they are obtained, their downstream effects can be understood much better when they are modeled using probabilistic concepts. This will be taken up in Chap. 4, where Fourier analysis is applied in a probabilistic context.

2.1 Periodic Process (Fourier Series)

A periodic process repeats itself at a fixed time interval. We say that a function $x(t)$ of time t (s) is ‘periodic with period T ’ if T is the smallest value such that

$$x(t+T) = x(t) \quad \text{for any } t \quad (2.1)$$

The proviso ‘the smallest value’ is necessary to remove ambiguity, because if (2.1) holds then for any integer m :

$$x(t+mT) = x(t+(m-1)T+T) = x(t+(m-1)T) = \cdots = x(t+T) = x(t) \quad (2.2)$$

According to the Fourier theorem, a periodic process $x(t)$ can be written as a ‘Fourier series’ (FS):

$$x(t) = a_0 + \sum_{k=1}^{\infty} a_k \cos \bar{\omega}_k t + \sum_{k=1}^{\infty} b_k \sin \bar{\omega}_k t \quad \bar{\omega}_k = \frac{2\pi k}{T} \quad (2.3)$$

Here, a_k and b_k are called the (real) ‘Fourier series coefficients’, associated with harmonic oscillations with period T/k (s), i.e., frequency $\bar{\omega}_k = 2\pi k/T$ (rad/s). This can be seen by noting that as t goes from 0 to T/k the argument $\bar{\omega}_k t$ in the cosine and sine terms goes from 0 to 2π , hence completing one cycle. The term a_0 accounts for the constant ‘static’ level of the process. Generally, a periodic process need not be just a finite sum of cosines and sines, but the Fourier theorem says that by including an *infinite* number of them with systematically increasing frequencies it is possible to represent *any* periodic process.

Amplitude and Phase

By writing

$$a_k \cos \bar{\omega}_k t + b_k \sin \bar{\omega}_k t = \sqrt{a_k^2 + b_k^2} \left[\underbrace{\frac{a_k}{\sqrt{a_k^2 + b_k^2}}}_{\cos \phi_k} \cos \bar{\omega}_k t + \underbrace{\frac{b_k}{\sqrt{a_k^2 + b_k^2}}}_{\sin \phi_k} \sin \bar{\omega}_k t \right] \quad (2.4)$$

and using the compound angle formula $\cos(\theta_1 - \theta_2) = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2$, (2.3) can be written as

$$x(t) = a_0 + \sum_{k=1}^{\infty} \underbrace{\sqrt{a_k^2 + b_k^2}}_{\text{amplitude}} \cos(\underbrace{\bar{\omega}_k}_{\text{frequency}} t - \underbrace{\phi_k}_{\text{phase}}) \quad \tan \phi_k = \frac{b_k}{a_k} \quad (2.5)$$

Expressions of Fourier Series Coefficients

Clearly, a_k and b_k depend on $x(t)$. They are given by

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \quad (2.6)$$

$$\begin{aligned} a_k &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos \bar{\omega}_k t dt \\ b_k &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin \bar{\omega}_k t dt \end{aligned} \quad k = 1, 2, \dots \quad (2.7)$$

In these expressions, the integration domain can be any continuous interval of length T because $x(t)$ has period T .

Proof of (2.6) and (2.7) (Fourier Series Coefficients)

The expression of a_0 in (2.6) can be shown by integrating both sides of (2.3) w.r.t. t from $-T/2$ to $T/2$ and noting that the integrals of sines and cosines on the RHS are all equal to zero. The derivation of a_k and b_k ($k \geq 1$) makes use of the following results (j and k are non-zero integers):

$$\int_{-T/2}^{T/2} \cos \bar{\omega}_j t \cos \bar{\omega}_k t dt = \begin{cases} 0 & j \neq k \\ T/2 & j = k \end{cases} \quad (2.8)$$

$$\int_{-T/2}^{T/2} \sin \bar{\omega}_j t \sin \bar{\omega}_k t dt = \begin{cases} 0 & j \neq k \\ T/2 & j = k \end{cases} \quad (2.9)$$

$$\int_{-T/2}^{T/2} \cos \bar{\omega}_j t \sin \bar{\omega}_k t dt = 0 \quad \text{any } j, k \quad (2.10)$$

In particular, multiplying both sides of $x(t) = a_0 + \sum_{j=1}^{\infty} a_j \cos \bar{\omega}_j t + \sum_{j=1}^{\infty} b_j \sin \bar{\omega}_j t$ with $\cos \bar{\omega}_k t$ and integrating from $-T/2$ to $T/2$,

$$\begin{aligned} \int_{-T/2}^{T/2} x(t) \cos \bar{\omega}_k t dt &= a_0 \underbrace{\int_{-T/2}^{T/2} \cos \bar{\omega}_k t dt}_0 + \sum_{j=1}^{\infty} a_j \underbrace{\int_{-T/2}^{T/2} \cos \bar{\omega}_j t \cos \bar{\omega}_k t dt}_{T/2 \text{ if } j=k; \text{ else } 0} \\ &+ \sum_{j=1}^{\infty} b_j \underbrace{\int_{-T/2}^{T/2} \sin \bar{\omega}_j t \cos \bar{\omega}_k t dt}_0 = \frac{a_k T}{2} \end{aligned} \quad (2.11)$$

Rearranging gives the expression of a_k in (2.7). Multiplying $x(t) = a_0 + \sum_{j=1}^{\infty} a_j \cos \bar{\omega}_j t + \sum_{j=1}^{\infty} b_j \sin \bar{\omega}_j t$ with $\sin \bar{\omega}_k t$ and following similar steps gives the expression of b_k in (2.7). ■

2.1.1 Complex Exponential Form

The FS in (2.3) contains for each frequency a sine and cosine term. It is possible to combine the two terms into a single term using the ‘Euler formula’:

$$e^{i\theta} = \cos \theta + \mathbf{i} \sin \theta \quad \mathbf{i}^2 = -1 \quad (2.12)$$

for any real number θ . Conversely, cosine and sine can be written in terms of $e^{\pm i\theta}$ as

$$\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta}) \quad \sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta}) \quad (2.13)$$

Applying these identities to (2.3) and writing $1/2i = -i/2$ gives

$$\begin{aligned} x(t) &= a_0 + \sum_{k=1}^{\infty} \frac{a_k}{2} (e^{i\bar{\omega}_k t} + e^{-i\bar{\omega}_k t}) + \sum_{k=1}^{\infty} -\frac{ib_k}{2} (e^{i\bar{\omega}_k t} - e^{-i\bar{\omega}_k t}) \\ &= \underbrace{a_0}_{c_0} + \sum_{k=1}^{\infty} \underbrace{\frac{1}{2}(a_k - ib_k)}_{c_k} e^{i\bar{\omega}_k t} + \sum_{k=1}^{\infty} \underbrace{\frac{1}{2}(a_k + ib_k)}_{c_{-k}} e^{-i\bar{\omega}_k t} \end{aligned} \quad (2.14)$$

The FS can then be written in a compact manner as

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{i\bar{\omega}_k t} \quad \bar{\omega}_k = \frac{2\pi k}{T} \quad (2.15)$$

where $\{c_k\}_{k=-\infty}^{\infty}$ are the ‘complex Fourier series coefficients’, related to the real ones by

$$c_0 = a_0 \quad \begin{aligned} c_k &= \frac{1}{2}(a_k - ib_k) \\ c_{-k} &= \frac{1}{2}(a_k + ib_k) \end{aligned} \quad k = 1, 2, \dots \quad (2.16)$$

Substituting (2.7) into (2.16) gives c_k in terms of $x(t)$:

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-i\bar{\omega}_k t} dt \quad \bar{\omega}_k = \frac{2\pi k}{T}, \quad k = 0, \pm 1, \pm 2, \dots \quad (2.17)$$

It can be verified that c_k has the ‘conjugate mirror property’:

$$c_{-k} = \overline{c_k} \quad (2.18)$$

where a bar on top denotes the complex conjugate, i.e., $\overline{a + bi} = a - bi$ for real numbers a and b .

The complex form of FS significantly simplifies algebra in Fourier analysis and is widely used. It takes some time to master algebraic skills with complex numbers but is worthwhile to do so.

2.1.2 Parseval Equality

The ‘energy’ of a process when viewed in the time or frequency domain can be equated via the ‘Parseval equality’, which appears in different forms depending on the context. For a periodic process with finite energy $\int_{-T/2}^{T/2} x(t)^2 dt < \infty$,

$$\underbrace{\frac{1}{T} \int_{-T/2}^{T/2} x(t)^2 dt}_{\text{Time domain}} = \underbrace{a_0^2 + \frac{1}{2} \sum_{k=1}^{\infty} (a_k^2 + b_k^2)}_{\text{Freq. domain real coeff.}} = \underbrace{\sum_{k=-\infty}^{\infty} |c_k|^2}_{\text{Freq. domain complex coeff.}} \quad (2.19)$$

The leftmost expression may be interpreted as the average energy per unit time. The Parseval equality says that this energy can be viewed as the sum of contributions from harmonics at different frequencies. If one substitutes the FS of $x(t)$ in (2.3) into the leftmost expression, after squaring, one will obtain an integral of a double sum of cosine and sine products. The non-trivial (and beautiful) result is that only the integrals of $\cos \times \cos$ and $\sin \times \sin$ terms with the same frequency are non-zero, giving the neat frequency domain expressions in the middle and rightmost that contain no cross terms.

Proof of (2.19) (Parseval Equality, Fourier Series)

We prove the Parseval equality (2.19) using the complex FS in (2.15). Writing $x^2 = x\bar{x}$,

$$\begin{aligned} \frac{1}{T} \int_{-T/2}^{T/2} x(t)^2 dt &= \frac{1}{T} \int_{-T/2}^{T/2} \overbrace{\left(\sum_{k=-\infty}^{\infty} c_k e^{i\bar{\omega}_k t} \right)}^{x(t)} \overbrace{\left(\sum_{j=-\infty}^{\infty} \bar{c}_j e^{-i\bar{\omega}_j t} \right)}^{\bar{x}(t)} dt \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} c_k \bar{c}_j \underbrace{\int_{-T/2}^{T/2} e^{i(\bar{\omega}_k - \bar{\omega}_j)t} dt}_{\substack{T \text{ if } j=k; \text{ else } 0}} \\ &= \sum_{k=-\infty}^{\infty} |c_k|^2 \end{aligned} \quad (2.20)$$

In the second equality, the order of infinite sum and integration has been swapped, which is legitimate when the process has finite energy. The result of the integral can be reasoned as follow. Clearly, it is equal to T when $j = k$. Otherwise ($j \neq k$),

$$\int_{-T/2}^{T/2} e^{i(\bar{\omega}_k - \bar{\omega}_j)t} dt = \frac{e^{i(\bar{\omega}_k - \bar{\omega}_j)T/2} - e^{-i(\bar{\omega}_k - \bar{\omega}_j)T/2}}{i(\bar{\omega}_k - \bar{\omega}_j)} = \frac{2i \sin \pi(k-j)/T}{2\pi i(k-j)/T} = 0 \quad (2.21)$$

■

2.2 Non-periodic Process (Fourier Transform)

If a process is not periodic then it cannot be written as a FS. In this case, if it has finite energy then it can still be represented as a sum of harmonics, although now there is a continuum of contributing frequencies, each with infinitesimal contribution. Specifically, a process $x(t)$ defined for $-\infty < t < \infty$ with finite energy $\int_{-\infty}^{\infty} x(t)^2 dt < \infty$ can be written as

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega \quad (2.22)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt \quad (2.23)$$

Here, $X(\omega)$ is called the ‘Fourier Transform’ (FT) of $x(t)$; and $x(t)$ is the ‘inverse Fourier Transform’ of $X(\omega)$. In FT, the frequency ω is continuous-valued. This is in contrast to FS, where the frequencies $\{\bar{\omega}_k\}$ are discrete-valued. The factor $1/2\pi$ in (2.22) may appear peculiar but it can be explained by consideration of units; see Sect. 2.7.1 later.

2.2.1 From Fourier Series to Fourier Transform

FT can be reasoned from FS as follow. Consider approximating $x(t)$ by a periodic function $x_p(t)$ with period T , where $x_p(t) = x(t)$ for $-T/2 < t < T/2$ but simply repeats itself elsewhere. This is illustrated in Fig. 2.1.

Intuitively, $x_p(t)$ (as a function) converges to $x(t)$ as $T \rightarrow \infty$. For a given T , let c_k be the complex FS coefficients of $x_p(t)$ at frequency $\bar{\omega}_k = 2\pi k/T$. Then

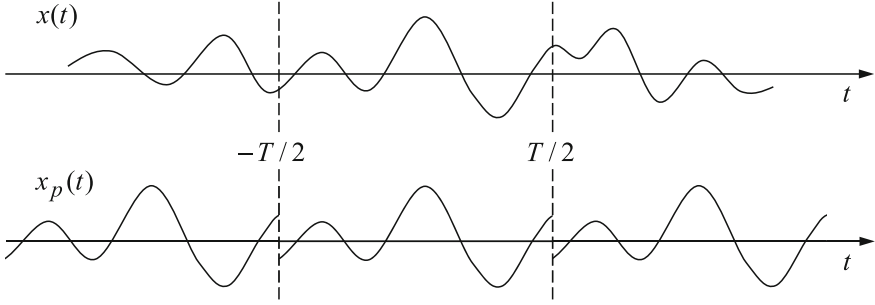


Fig. 2.1 Original process $x(t)$ and periodic proxy $x_p(t)$

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} x_p(t) e^{-i\bar{\omega}_k t} dt = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-i\bar{\omega}_k t} dt \quad (2.24)$$

since $x_p(t) = x(t)$ for $-T/2 < t < T/2$. Using Cauchy-Schwartz inequality (Sect. C.5.6),

$$|c_k|^2 \leq \underbrace{\frac{1}{T^2} \int_{-T/2}^{T/2} x(t)^2 dt}_{\leq \int_{-\infty}^{\infty} x(t)^2 dt} \times \underbrace{\int_{-T/2}^{T/2} \underbrace{|e^{-i\bar{\omega}_k t}|^2}_1 dt}_T \leq \frac{1}{T} \int_{-\infty}^{\infty} x(t)^2 dt \quad (2.25)$$

Since $\int_{-\infty}^{\infty} x(t)^2 dt < \infty$, the above implies $|c_k| \rightarrow 0$ as $T \rightarrow \infty$. This indicates that FS coefficients are not legitimate quantities for studying the frequency characteristics of a non-period process, because they all diminish trivially as $T \rightarrow \infty$, no matter what the process is. The factor $1/T$ is the source of diminishing magnitude.

The following function is motivated by taking out the factor $1/T$ in (2.24) and replacing $\bar{\omega}_k$ by the continuous-valued frequency variable ω :

$$X_p(\omega) = \int_{-T/2}^{T/2} x(t) e^{-i\omega t} dt \quad (2.26)$$

By construction, $c_k = X_p(\bar{\omega}_k)/T$ and so

$$x_p(t) = \sum_{k=-\infty}^{\infty} c_k e^{i\bar{\omega}_k t} = \sum_{k=-\infty}^{\infty} X_p(\bar{\omega}_k) e^{i\bar{\omega}_k t} \frac{1}{T} \quad (2.27)$$

As $T \rightarrow \infty$, $x_p(t) \rightarrow x(t)$ as a function. Also, the frequency interval $\Delta\omega = \bar{\omega}_{k+1} - \bar{\omega}_k = 2\pi/T$ diminishes. The infinite sum on the RHS of (2.27) tends to an integral. Thus,

$$x(t) = \lim_{T \rightarrow \infty} x_p(t) = \lim_{T \rightarrow \infty} \sum_{k=-\infty}^{\infty} X_p(\bar{\omega}_k) e^{i\bar{\omega}_k t} \frac{\Delta\omega}{2\pi} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega \quad (2.28)$$

where

$$X(\omega) = \lim_{T \rightarrow \infty} X_p(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt \quad (2.29)$$

In the above reasoning, we have swapped the order of limit and integration. This is legitimate when the process has finite energy.

2.2.2 Properties of Fourier Transform

Some properties of FT are listed in Table 2.1. They can be shown directly from definition. The symbol $\mathcal{F}\{x\}$ denotes the FT of $x(t)$. It is a function of frequency ω but this is omitted for simplicity.

Table 2.1 Some properties of Fourier Transform

Property	Description
Conjugate mirror	Let $X(\omega)$ be the FT of $x(t)$. Then $X(-\omega) = \overline{X(\omega)}$
Linearity	For any scalars a and b , $\mathcal{F}\{ax + by\} = a\mathcal{F}\{x\} + b\mathcal{F}\{y\}$
Differentiation	$\mathcal{F}\{\dot{x}\} = i\omega\mathcal{F}\{x\}$
Time shift	For any s , let $y(t) = x(t + s)$. Then $\mathcal{F}\{y\} = e^{i\omega s} \mathcal{F}\{x\}$
Convolution	Let $z(t)$ be the ‘convolution’ between $x(t)$ and $y(t)$, defined as $z(t) = \int_{-\infty}^{\infty} x(t - \tau)y(\tau)d\tau$ The FT of convolution is equal to the product of FTs, i.e., $\mathcal{F}\{z\} = \mathcal{F}\{x\}\mathcal{F}\{y\}$

2.2.3 Dirac Delta Function

The ‘Dirac Delta function’ $\delta(t)$, or Delta function in short, is an idealized unit impulse of arbitrarily short duration centered at $t = 0$. It does not exist physically but is frequently used in analysis and modeling. It has the property that

$$\int_{-\varepsilon}^{\varepsilon} \delta(t)x(t) dt = x(0) \quad (2.30)$$

for any $\varepsilon > 0$ and function $x(t)$. The FT of the Delta function is simply the constant 1 because

$$\int_{-\infty}^{\infty} \delta(t)e^{-i\omega t} dt = e^{-i\omega(0)} = 1 \quad \text{for any } \omega \quad (2.31)$$

The inverse FT of the constant 1 gives the FT representation of the Delta function:

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega t} d\omega \quad (2.32)$$

This is frequently used in Fourier analysis.

2.2.4 Parseval Equality

For a non-period process $x(t)$ that has FT $X(\omega)$, the Parseval equality reads

$$\int_{-\infty}^{\infty} x(t)^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega \quad (2.33)$$

Proof of (2.33) (Parseval Equality, Fourier Transform)

The proof for (2.33) has a similar structure as the one in (2.20) for FS, except that the infinite sums now become integrals:

$$\begin{aligned}
\int_{-\infty}^{\infty} x(t)^2 dt &= \int_{-\infty}^{\infty} \overbrace{\left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega \right]}^{x(t)} \overbrace{\left[\frac{1}{2\pi} \int_{-\infty}^{\infty} \overline{X(\omega')} e^{-i\omega' t} d\omega' \right]}^{\overline{x(t)}} dt \\
&= \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(\omega) \overline{X(\omega')} e^{i(\omega - \omega')t} d\omega' d\omega dt \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} X(\omega) \overline{X(\omega')} \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i(\omega - \omega')t} dt}_{\delta(\omega - \omega')} d\omega' d\omega \quad (2.34) \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \underbrace{\int_{-\infty}^{\infty} \overline{X(\omega')} \delta(\omega - \omega') d\omega'}_{\overline{X(\omega)}} d\omega \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega
\end{aligned}$$

In arriving at the third equality, the order of integration w.r.t. t and (ω, ω') has been swapped. This is legitimate when the process has finite energy. ■

2.3 Discrete-Time Approximation with FFT

In digital computations, a process is sampled at discrete time instants. The integrals in FS and FT can be approximated by a Riemann sum on the grid of sampled time instants. Let $\{x_j = x(j\Delta t)\}_{j=0}^{N-1}$ be the N sample values of $x(t)$ at equal time interval Δt (s). A discrete-time approximation to the FT of $x(t)$ is constructed by replacing the integral in (2.23) with a Riemann sum:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt \approx \hat{X}(\omega) = \sum_{j=0}^{N-1} x_j e^{-i\omega j\Delta t} \Delta t \quad (2.35)$$

Here, $\hat{X}(\omega)$ is called the ‘discrete-time Fourier Transform’ (DTFT) of $x(t)$. Evaluating $\hat{X}(\omega)$ for a given value of ω involves a summation in the time domain. In practice it is only calculated at the following uniformly spaced frequencies:

$$\omega_k = \frac{2\pi k}{N\Delta t} \text{ (rad/s)} \quad k = 0, \dots, N-1 \text{ (FFT frequencies)} \quad (2.36)$$

It is because the values of $\hat{X}(\omega)$ at these frequencies can be evaluated very efficiently. This is discussed next.

2.3.1 Fast Fourier Transform

The ‘Fast Fourier Transform’ (FFT) algorithm (Cooley and Tukey 1965) provides an efficient means for calculating the values of DTFT at a specific set of frequencies as in (2.36). It is commonly coded in commercial software or programming packages; see Sect. 2.9 later for an introduction. Here we focus on the definition and properties of FFT.

The FFT of $\{x_j\}_{j=0}^{N-1}$ is the sequence $\{y_k\}_{k=0}^{N-1}$ defined by

$$y_k = \sum_{j=0}^{N-1} x_j e^{-2\pi i j k / N} \quad k = 0, \dots, N-1 \quad (\text{FFT}) \quad (2.37)$$

The ‘inverse FFT’ of $\{y_k\}_{k=0}^{N-1}$ is defined as the sequence $\{z_j\}_{j=0}^{N-1}$ where

$$z_j = \frac{1}{N} \sum_{k=0}^{N-1} y_k e^{2\pi i j k / N} \quad j = 0, \dots, N-1 \quad (\text{inverse FFT}) \quad (2.38)$$

Note that $\{y_k\}_{k=0}^{N-1}$ is generally complex-valued, even though $\{x_j\}_{j=0}^{N-1}$ is real-valued. In the literature, $\{y_k\}_{k=0}^{N-1}$ is referred as the ‘discrete Fourier Transform’ (DFT, not to be confused with DTFT) of $\{x_j\}_{j=0}^{N-1}$. In this book, we simply refer it as FFT, because DFT is almost always evaluated via FFT and there is little distinction between the two terms.

Inverse FFT Recovers the Original Sequence

The following shows that inverse FFT indeed recovers the original sequence that produces the FFT, i.e., $x_j = z_j, j = 0, \dots, N-1$. Substituting $y_k = \sum_{r=0}^{N-1} x_r e^{-2\pi i r k / N}$ into (2.38),

$$z_j = \frac{1}{N} \sum_{k=0}^{N-1} \underbrace{\sum_{r=0}^{N-1} x_r e^{-2\pi i r k / N}}_{y_k} e^{2\pi i j k / N} = \frac{1}{N} \sum_{r=0}^{N-1} x_r \underbrace{\sum_{k=0}^{N-1} e^{2\pi i (j-r) k / N}}_{N \text{ if } r=j; \text{ else } 0} = x_j \quad (2.39)$$

The result of the sum over k used above is discussed next.

Exponential Sum Formula

Sums of exponentials are frequently encountered in Fourier analysis and it is worth to get familiarized with their analytical formulas. One basic result is that for any integer k ,

$$\sum_{j=0}^{N-1} e^{2\pi i j k / N} = \begin{cases} N & k/N = 0, \pm 1, \pm 2, \dots \\ 0 & \text{otherwise} \end{cases} \quad (2.40)$$

The first case when k is an integer multiple of N is trivial because $e^{2\pi i p} = 1$ for any integer p . Otherwise, using the geometric series summation formula $\sum_{j=0}^{N-1} a^j = (1 - a^N)/(1 - a)$ with $a = e^{2\pi i k / N}$,

$$\sum_{j=0}^{N-1} e^{2\pi i j k / N} = \frac{1 - \overbrace{e^{2\pi i k}}^1}{1 - \underbrace{e^{2\pi i k / N}}_{\neq 1}} = 0 \quad \left(\frac{k}{N} \text{ not integer}\right) \quad (2.41)$$

Taking complex conjugate on (2.40) shows that the result is the same if $2\pi i j k / N$ is replaced by $-2\pi i j k / N$.

Conjugate Mirror Property

The FFT $\{y_k\}_{k=0}^{N-1}$ of a real-valued sequence $\{x_j\}_{j=0}^{N-1}$ has the conjugate mirror property that

$$y_{N-k} = \overline{y_k} \quad (2.42)$$

This is because

$$y_{N-k} = \sum_{j=0}^{N-1} x_j e^{-2\pi i j (N-k) / N} = \sum_{j=0}^{N-1} x_j \underbrace{e^{-2\pi i j}}_1 e^{2\pi i j k / N} = \overline{\sum_{j=0}^{N-1} x_j e^{-2\pi i j k / N}} = \overline{y_k} \quad (2.43)$$

The conjugate mirror property is illustrated in Fig. 2.2. About half of the FFT sequence carries redundant information, in the sense that it can be produced as the complex conjugate of the other half.

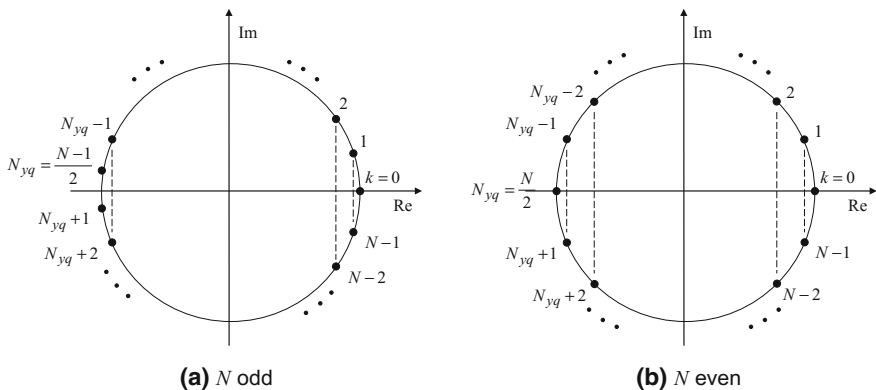


Fig. 2.2 Conjugate mirror property of the FFT of a real-valued sequence. **a** N odd; **b** N even. N_{yq} = integer part of $N/2$, is the index at or just below the ‘Nyquist frequency’ (Sect. 2.4.1)

2.3.2 Approximating Fourier Transform and Fourier Series

Back to the problem of approximating the FT $X(\omega_k)$ with the FFT of $\{x_j = x(j\Delta t)\}_{j=0}^{N-1}$ as in (2.35). Denoting $\hat{X}_k = \hat{X}(\omega_k)$ and noting $\omega_k j \Delta t = 2\pi jk/N$,

$$\hat{X}_k = \Delta t \underbrace{\sum_{j=0}^{N-1} x_j e^{-2\pi i j k / N}}_{y_k(\text{FFT})} \quad (2.44)$$

Thus,

$$X(\omega_k) \approx \hat{X}_k = y_k \Delta t \quad (2.45)$$

For FS coefficients, let $\{x_j = x(j\Delta t)\}_{j=0}^{N-1}$ be the sampled sequence of a periodic process $x(t)$ with period T . Assume that $N\Delta t = T$ so that the FFT frequency $\omega_k = 2\pi k/N\Delta t$ coincides with the FS frequency $\bar{\omega}_k = 2\pi k/T$. Approximating the integral in (2.17) by a Riemann sum,

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-i\bar{\omega}_k t} dt \approx \hat{c}_k = \frac{1}{N\Delta t} \underbrace{\sum_{j=0}^{N-1} x_j e^{-2\pi i j k / N}}_{y_k} \Delta t \quad (2.46)$$

Thus

$$c_k \approx \hat{c}_k = \frac{y_k}{N} \quad (2.47)$$

The use of $\hat{c}_k$ in (2.46) can be generalized to allow $N\Delta t \neq T$, which may provide convenience in practice, e.g., when the period is not known, data duration is not equal to the period, or it is desirable to estimate the FS of measured data using more than one period to average out noise.

2.3.3 Parseval Equality

For a real-valued sequence $\{x_j\}_{j=0}^{N-1}$ with FFT $\{y_k\}_{k=0}^{N-1}$, Parseval equality reads

$$\sum_{j=0}^{N-1} x_j^2 = \frac{1}{N} \sum_{k=0}^{N-1} |y_k|^2 \quad (2.48)$$

Proof of (2.48) (Parseval Equality, FFT)

The structure of the proof is similar to that for FS in (2.20) or FT in (2.34):

$$\begin{aligned} \sum_{j=0}^{N-1} x_j^2 &= \sum_{j=0}^{N-1} \left[\frac{1}{N} \sum_{k=0}^{N-1} \overbrace{x_j}^{x_j} e^{2\pi i j k / N} \right] \left[\frac{1}{N} \sum_{r=0}^{N-1} \overbrace{\bar{x}_j}^{\bar{x}_j} e^{-2\pi i j r / N} \right] \\ &= \frac{1}{N^2} \sum_{j=0}^{N-1} \sum_{k=0}^{N-1} \sum_{r=0}^{N-1} y_k \bar{y}_r e^{2\pi i j (k-r) / N} \\ &= \frac{1}{N^2} \sum_{k=0}^{N-1} \sum_{r=0}^{N-1} y_k \bar{y}_r \underbrace{\sum_{j=0}^{N-1} e^{2\pi i j (k-r) / N}}_{N \text{ if } r=k; \text{ else } 0} = \frac{1}{N} \sum_{k=0}^{N-1} |y_k|^2 \end{aligned} \quad (2.49)$$

■

2.4 Distortions in Fourier Series

The discrete-time approximation of FS and FT leads to errors of a characteristic nature. In the first place, the approximation is valid only up to the ‘Nyquist frequency’ ($1/2\Delta t$ Hz). It is also contaminated with harmonics in the original process

beyond the Nyquist frequency. This is known as ‘aliasing’, where high frequency variations are mistaken as low frequency ones. It is the same mechanism that our eyes see the rotor blade of a flying helicopter as slowly rotating (sometimes reversing direction). There is further contamination from harmonics that do not have an integer multiple of cycles within the measured time span, in a $O(1/N\Delta t)$ neighborhood of the subject frequency (and aliased counterparts). This is known as ‘leakage’. Aliasing is not repairable after discrete-time sampling and so harmonics beyond the Nyquist frequency should be filtered out beforehand. Leakage can be suppressed by increasing data duration. We first discuss the distortions in the FFT approximation of FS. The discussion for FT follows in the next section.

Recall the context in (2.46), where $x(t)$ is a periodic process with complex FS coefficients $\{c_k\}_{k=-\infty}^{\infty}$; $\{x_j = x(j\Delta t)\}_{j=0}^{N-1}$ is a discrete-time sample sequence of $x(t)$; $\{y_k\}_{k=0}^{N-1}$ is the FFT of $\{x_j\}_{j=0}^{N-1}$; and $\hat{c}_k = y_k/N$ is a discrete-time approximation of c_k .

2.4.1 Nyquist Frequency

Although $\{\hat{c}_k\}_{k=0}^{N-1}$ is a sequence with N terms, only the first half is informative. This stems from the conjugate mirror property:

$$\hat{c}_{N-k} = \overline{\hat{c}_k} \quad (2.50)$$

It implies that $\hat{c}_k$ is conjugate symmetric about the index $N/2$, i.e., frequency $f_{N/2} = (N/2)/N\Delta t = 1/2\Delta t$ (Hz), which is called the ‘Nyquist frequency’. As a result, $\hat{c}_k$ can only give a proper estimation of c_k up to the Nyquist frequency.

2.4.2 Aliasing

Aliasing occurs when the original process $x(t)$ contains harmonics at frequencies beyond the Nyquist frequency. Suppose the data duration is equal to the period of $x(t)$, i.e., $N\Delta t = T$. Then it can be shown that (see the end)

$$\hat{c}_k = \sum_{m=-\infty}^{\infty} c_{mN+k} \quad (2.51)$$

In addition to c_k (the $m = 0$ term), $\hat{c}_k$ also contains other terms, $c_{k\pm N}$, $c_{k\pm 2N}$ and so on. To see the contributing (positive) frequencies, separate the sum into positive and negative m , and use the conjugate mirror property of c_k :

$$\begin{aligned}
\hat{c}_k &= c_k + \sum_{m=1}^{\infty} c_{mN+k} + \sum_{m=-\infty}^{-1} c_{mN+k} \\
&= c_k + (c_{N+k} + c_{2N+k} + \cdots) + (c_{-N+k} + c_{-2N+k} + \cdots) \\
&= c_k + (c_{N+k} + c_{2N+k} + \cdots) + (\overline{c_{N-k}} + \overline{c_{2N-k}} + \cdots)
\end{aligned} \tag{2.52}$$

Combining the two infinite sums,

$$\underbrace{\hat{c}_k}_{\text{freq. } f_k} = \underbrace{c_k}_{\text{freq. } f_k} + \underbrace{\sum_{m=1}^{\infty} \left(\underbrace{c_{mN+k}}_{\text{freq. } mf_s + f_k} + \underbrace{\overline{c_{mN-k}}}_{\text{freq. } mf_s - f_k} \right)}_{\text{aliasing}} \tag{2.53}$$

That is, $\hat{c}_k$ is contaminated with contributions from frequencies (in Hz) $f_s \pm f_k$, $2f_s \pm f_k$, ..., where $f_s = 1/\Delta t$ is the sampling frequency and $f_k = k/N\Delta t$ is the subject FFT frequency. Aliasing occurs by the same mechanism when the data duration is an integer multiple of the period. When the data duration is not even an integer multiple of the period, there will also be ‘leakage’; see the next section.

Example 2.1 (Aliasing with a single harmonic) Consider $x(t) = 2 \cos 2\pi f t$, which is a single harmonic with frequency f (Hz). Its real FS coefficients $\{a_k, b_k\}$ are zero except $a_1 = 2$. Its complex FS coefficients $\{c_k\}$ are zero except $c_1 = 1$ and $c_{-1} = 1$. Suppose we obtain the samples $\{x_j = x(j\Delta t)\}_{j=0}^{N-1}$ at $\Delta t = 0.1$ s and $N = 10$, i.e., for a duration of $N\Delta t = 1$ s. Using $\{x_j\}_{j=0}^{N-1}$, we estimate the complex FS coefficients by the FFT approximation $\hat{c}_k = N^{-1} \sum_{j=0}^{N-1} x_j e^{-2\pi i j k / N}$ in (2.46).

Figure 2.3 illustrates the possible distortions in $\hat{c}_k$, depending on the source frequency f . The plots on the left column show $x(t)$ (dashed line) and the sample points $\{x_j\}_{j=0}^{N-1}$ (dots). The plots on the right column show $|\hat{c}_k|$ (dot with stick) versus the FFT frequency $f_k = k/N\Delta t$ (Hz) for $k = 0, \dots, N-1$. The shaded part from 5 Hz (Nyquist frequency) to 10 Hz (sampling frequency) is just the mirror image of that from 0 to 5 Hz. It is usually not plotted but is shown here for illustration. When $f = 1$ or 4 Hz, $|\hat{c}_k|$ is correctly estimated up to the Nyquist frequency. When $f = 6$ Hz or 11 Hz, which are beyond the Nyquist frequency, the harmonic is mistaken (aliased) to be 4 Hz and 1 Hz, respectively. ■

Proof of (2.51) (Aliasing, Fourier Series)

Since $N\Delta t = T$, $\bar{\omega}_r = 2\pi r/T = 2\pi r/N\Delta t$ and so $\bar{\omega}_r j\Delta t = 2\pi j r/N$. Substituting $x_j = \sum_{r=-\infty}^{\infty} c_r e^{2\pi i j r/N}$ into $\hat{c}_k$ in (2.46),

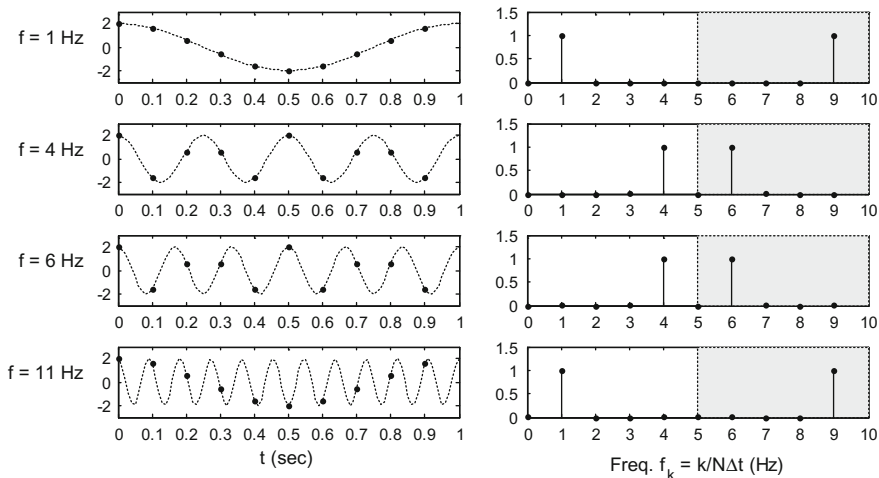


Fig. 2.3 FS amplitude of $x(t) = 2 \cos 2\pi ft$ estimated via the FFT of $\{x_j = x(j\Delta t)\}_{j=0}^{N-1}$ with $\Delta t = 0.1$ s and $N = 10$, for $f = 1, 4, 6$ and 11 Hz. *Left column:* dashed line = $x(t)$, dot = x_j . *Right column:* dot with stick = $|\hat{c}_k|$

$$\begin{aligned}
 \hat{c}_k &= \frac{1}{N} \sum_{j=0}^{N-1} \overbrace{\sum_{r=-\infty}^{\infty} c_r e^{2\pi i j r / N}}^{x_j} e^{-2\pi i j k / N} \\
 &= \frac{1}{N} \sum_{r=-\infty}^{\infty} c_r \times \underbrace{\sum_{j=0}^{N-1} e^{2\pi i j (r-k) / N}}_{N \text{ if } (r-k)/N=0, \pm 1, \pm 2, \dots; \text{ else } 0} = \sum_{m=-\infty}^{\infty} c_{mN+k}
 \end{aligned} \tag{2.54}$$

■

2.4.3 Leakage

Leakage occurs in the FFT approximation of FS when the data duration $N\Delta t$ is not an integer multiple of the period T . In this case, a given FS frequency $\bar{\omega}_r = 2\pi r/T$ need not be matched by a FFT frequency $\omega_k = 2\pi k/N\Delta t$. When this occurs, the FS harmonic will ‘leak out’ to other FFT frequencies. Leakage (and aliasing) can be explained by the following general formula that expresses $\hat{c}_k$ as the convolution of $\{c_r\}_{r=-\infty}^{\infty}$ with a ‘kernel function’ $K_1(\omega)$:

$$\hat{c}_k = \sum_{r=-\infty}^{\infty} c_r K_1(\omega_k - \bar{\omega}_r) \quad (2.55)$$

$$K_1(\omega) = \frac{1}{N} \sum_{j=0}^{N-1} e^{-i\omega j \Delta t} = \frac{\sin(N\omega \Delta t/2)}{N \sin(\omega \Delta t/2)} e^{-i(N-1)\omega \Delta t/2} \quad (2.56)$$

See the end for proof. In terms of the dimensionless variable $u = \omega \Delta t/2\pi$,

$$K_1(\omega) = D_N(u) e^{-i\pi(N-1)u} \quad u = \frac{\omega \Delta t}{2\pi} \quad (2.57)$$

where

$$D_N(u) = \frac{\sin N\pi u}{N \sin \pi u} \quad (2.58)$$

is known as the ‘Dirichlet kernel’, which plays an important role in Fourier theory.

Equation (2.55) indicates that $\hat{c}_k$ contains contributions from all FS frequencies. The sum need not even contain a term at the subject frequency ω_k . The contribution from frequency $\bar{\omega}_r$ is not directly c_r , but is ‘attenuated’ by $K_1(\omega_k - \bar{\omega}_r)$, which depends on how far $\bar{\omega}_r$ is from ω_k .

Figure 2.4 shows a schematic plot of $|D_N(u)|$. It has a period of 1, with a symmetric basic branch on $(-1/2, 1/2)$. In this branch it has a global maximum of 1 at $u = 0$; and a series of zeros at $u = \pm 1/N, \pm 2/N, \dots$, up to $\pm 1/2$. For u ranging between $\pm 1/2$, $\omega = 2\pi u/\Delta t$ ranges between $\pm \pi/\Delta t$ (rad/s), i.e., $\pm$ Nyquist frequency. Due to convolution effect, $\hat{c}_k$ comprises the harmonics in $x(t)$ at frequencies

- (1) near the subject frequency $f_k = k/N\Delta t$ Hz;
- (2) in a $O(1/N\Delta t)$ (Hz) neighborhood around f_k (leakage); and
- (3) in the $O(1/N\Delta t)$ (Hz) neighborhoods around $f_s \pm f_k, 2f_s \pm f_k, \dots$ (aliased counterparts of leakage), where $f_s = 1/\Delta t$ Hz is the sampling frequency.

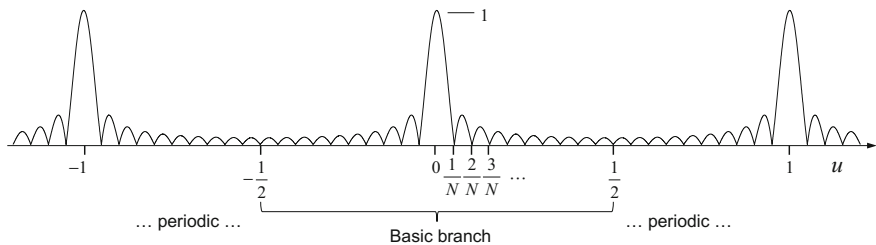


Fig. 2.4 Schematic plot of $|D_N(u)|$, the modulus of the Dirichlet kernel

Example 2.2 (Leakage from a single harmonic) Revisit Example 2.1. Everything else being the same, the sampling duration is now slightly extended to $N\Delta t = 1.2$ s, i.e., with $N = 1.2/0.1 = 12$ points ($\Delta t = 0.1$ s). Results analogous to Fig. 2.3 are shown in Fig. 2.5. For $f = 1, 4, 6$ and 11 Hz, the number of cycles within the data duration is $fN\Delta t = 1.2, 4.8, 7.2$ and 13.2 . None of these are integer and so leakage occurs, in addition to aliasing (when $f = 6$ and 11 Hz). The FFT frequencies are now $f_k = k/N\Delta t = 0, 0.833, 1.667, \dots, 9.167$ Hz, instead of $0, 1, \dots, 9$ Hz in Fig. 2.3. ■

Proof of (2.55) (Leakage, Fourier Series)

Substituting the FS $x_j = \sum_{r=-\infty}^{\infty} c_r e^{i\bar{\omega}_r j\Delta t}$ into $\hat{c}_k$ in (2.46),

$$\hat{c}_k = \frac{1}{N} \sum_{j=0}^{N-1} \underbrace{\sum_{r=-\infty}^{\infty} c_r e^{i\bar{\omega}_r j\Delta t}}_{x_j} e^{-i\omega_k j\Delta t} = \sum_{r=-\infty}^{\infty} c_r \underbrace{\frac{1}{N} \sum_{j=0}^{N-1} e^{-i(\omega_k - \bar{\omega}_r)j\Delta t}}_{K_1(\omega_k - \bar{\omega}_r)} \quad (2.59)$$

where

$$K_1(\omega) = \frac{1}{N} \sum_{j=0}^{N-1} e^{-i\omega j\Delta t} \quad (2.60)$$

as defined in (2.56). Using $\sum_{j=0}^{N-1} a^j = (1 - a^N)/(1 - a)$ with $a = e^{-i\omega\Delta t}$,

$$K_1(\omega) = \frac{1 - e^{-iN\omega\Delta t}}{N(1 - e^{-i\omega\Delta t})} \quad (2.61)$$

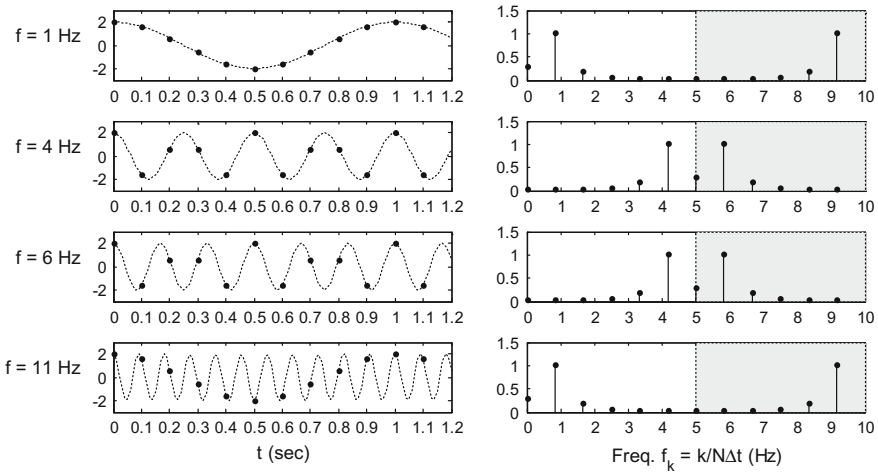


Fig. 2.5 FS amplitude of $x(t) = 2 \cos 2\pi ft$ estimated via the FFT of $\{x_j = x(j\Delta t)\}_{j=0}^{N-1}$ with $\Delta t = 0.1$ s and $N = 12$, for $f = 1, 4, 6$ and 11 Hz. Same legend as in Fig. 2.3

Note that for any real θ ,

$$1 - e^{-i\theta} = e^{-i\theta/2} \underbrace{(e^{i\theta/2} - e^{-i\theta/2})}_{2i \sin(\theta/2)} = 2ie^{-i\theta/2} \sin(\theta/2) \quad (2.62)$$

Using this identity,

$$K_1(\omega) = \frac{2ie^{-iN\omega\Delta t/2} \sin(N\omega\Delta t/2)}{2ie^{-i\omega\Delta t/2} N \sin(\omega\Delta t/2)} = \frac{\sin(N\omega\Delta t/2)}{N \sin(\omega\Delta t/2)} e^{-i(N-1)\omega\Delta t/2} \quad (2.63)$$

which is the rightmost expression in (2.56). ■

2.5 Distortions in Fourier Transform

Nyquist frequency limit, aliasing and leakage in the FFT approximation of FT occurs by a similar mechanism as in FS. Analogous to (2.55), the DTFT approximation $\hat{X}(\omega)$ in (2.35) is related to the target FT $X(\omega)$ by a convolution integral:

$$\hat{X}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega') K_2(\omega - \omega') d\omega' \quad (2.64)$$

$$K_2(\omega) = N\Delta t K_1(\omega) = N\Delta t D_N(u) e^{-i\pi(N-1)u} \quad u = \frac{\omega\Delta t}{2\pi} \quad (2.65)$$

See the end for proof.

Aliasing and leakage can be explained based on (2.64). To see aliasing, note that $|D_N(u)|$ has local maxima 1 at $u = 0, \pm 1, \pm 2, \dots$; see Fig. 2.4. Correspondingly, $|K_2(\omega - \omega')|$ as a function of ω' has local maxima at $\omega' = \omega \pm 2\pi r/\Delta t$ for $r = 0, 1, 2, \dots$. Using the conjugate mirror property of FT, $X(\omega - 2\pi r/\Delta t) = \overline{X(2\pi r/\Delta t - \omega)}$, and so $\hat{X}(\omega)$ receives significant contributions from the frequencies $2\pi r/\Delta t \pm \omega$ ($r = 0, 1, 2, \dots$). This is aliasing.

To see leakage, note that $\hat{X}(\omega)$ at frequency ω receives contribution from the FT $X(\omega')$ of the original process at frequency ω' . The contribution is attenuated by $K_2(\omega - \omega')$, which depends on how far ω' is from ω . For the contribution to be non-zero, ω' need not be the subject frequency ω or aliased counterparts. This is leakage. Since a non-periodic process generally contains harmonics at a continuum of frequencies, leakage exists regardless of the data duration. Nevertheless, the effect diminishes as the data duration increases, because $|K_2(\omega - \omega')|$ is negligible for $|\omega - \omega'| \gg 2\pi/N\Delta t$.

Proof of (2.64) (Distortion, Fourier Transform)

Substituting the inverse FT $x_j = (2\pi)^{-1} \int_{-\infty}^{\infty} X(\omega') e^{i\omega' j \Delta t} d\omega'$ into $\hat{X}(\omega)$ in (2.35),

$$\begin{aligned}
 \hat{X}(\omega) &= \sum_{j=0}^{N-1} \overbrace{\left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega') e^{i\omega' j \Delta t} d\omega' \right]}^{x_j} e^{-i\omega j \Delta t} \Delta t \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega') N \Delta t \times \underbrace{\left[\frac{1}{N} \sum_{j=0}^{N-1} e^{-i(\omega - \omega') j \Delta t} \right]}_{K_2(\omega - \omega')} d\omega'
 \end{aligned} \tag{2.66}$$

which gives (2.64). ■

2.6 Summary of FFT Approximations

Table 2.2 summarizes the Fourier formulas and their FFT approximations. They are generally related by a convolution (sum or integral). Relevant sections are indicated. The last row for power spectral density applies to a stationary stochastic process, which is discussed in Chap. 4.

2.7 Summary of Fourier Formulas, Units and Conventions

Table 2.3 summarizes the Fourier formulas and their Parseval equalities. Unit matters, and is indicated. The table assumes that $x(t)$ has a unit of volt (V) and time is measured in second (s). Relevant sections are indicated.

2.7.1 Multiplier in Fourier Transform

One common confusion in Fourier analysis is the definition of the inverse FT, in particular, the multiplier $1/2\pi$ in $x(t) = (2\pi)^{-1} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega$. Different authors may use a different multiplier. It may appear arbitrary but in fact has a direct implication on the unit of the FT $X(\omega)$. As seen in Table 2.3, if $x(t)$ has unit V,

Table 2.2 Summary of FFT approximations to Fourier series, Fourier Transform and power spectral density; $y_k = \sum_{j=0}^{N-1} x_j e^{-2\pi i j k / N}$ is the FFT of $\{x_j = x(j\Delta t)\}_{j=0}^{N-1}$; $D_N(u) = \sin(N\pi u) / N \sin \pi u$ is the Dirichlet kernel

Process	Theoretical		FFT approximation at $\omega_k = 2\pi k / N\Delta t$ (rad/s)	Relationship between theory and approximation
	Time domain	Frequency domain		
Periodic (period T)	$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{i\omega_k t}$ $\bar{\omega}_k = 2\pi k / T$ (rad/s) (Sect. 2.1.1)	$c_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-i\omega_k t} dt$ (Sect. 2.1.1)	$\hat{c}_k = \frac{1}{N} y_k$ (Sect. 2.3.2)	$\hat{c}_k = \sum_{r=-\infty}^{\infty} c_r K_1(\omega_k - \bar{\omega}_r)$ $K_1(\omega) = D_N(u) e^{-i\pi(N-1)u}$, $u = \omega\Delta t / 2\pi$ (Sect. 2.4.3)
Non-periodic	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega$ (Sect. 2.2)	$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt$ (Sect. 2.2)	$\hat{X}_k = \Delta t y_k$ (Sect. 2.3.2)	$\hat{X}_k = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) K_2(\omega_k - \omega) d\omega$ $K_2(\omega) = N\Delta t D_N(u) e^{-i\pi(N-1)u}$, $u = \omega\Delta t / 2\pi$ (Sect. 2.5)
Stochastic stationary	$\mathbf{x}(t)$ = vector process	$\mathbf{X}_T(\omega) = \frac{1}{\sqrt{T}} \int_0^T \mathbf{x}(t) e^{-i\omega t} dt$	$\hat{\mathbf{X}}_k = \sqrt{\Delta t / N} \mathbf{y}_k$ $\{\mathbf{y}_k\}_{k=0}^{N-1}$ is the FFT of $\{\mathbf{x}(j\Delta t)\}_{j=0}^{N-1}$	
	$\mathbf{R}(\tau) = E[\mathbf{x}(t+\tau)\mathbf{x}(t)^T]$ (Sect. 4.1)	$\mathbf{S}(\omega) = \lim_{T \rightarrow \infty} E[\mathbf{X}_T(\omega)\mathbf{X}_T(\omega)^*]$ (Sect. 4.2)	$\hat{\mathbf{S}}_k = \hat{\mathbf{X}}_k \hat{\mathbf{X}}_k^*$ (Sect. 4.5.2)	$E[\hat{\mathbf{S}}_k] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega) F_N(\omega_k - \omega) d\omega$
	$\mathbf{R}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega) e^{i\omega\tau} d\omega$ (Sect. 4.4.3)	$\mathbf{S}(\omega) = \int_{-\infty}^{\infty} \mathbf{R}(\tau) e^{-i\omega\tau} d\tau$ (Sect. 4.4.3)		$F_N(\omega) = N\Delta t D_N^2(u)$, $u = \omega\Delta t / 2\pi$ (Sect. 4.7.2)

Table 2.3 Summary of Fourier formulas and units

Quantity	Time domain	Frequency domain	Parseval equality
Fourier series (period T)	$\underbrace{x(t)}_{\text{V}} = \underbrace{\sum_{k=-\infty}^{\infty} c_k e^{j\omega_k t}}_{\text{V}}$	$\underbrace{c_k}_{\text{V}} = \underbrace{\frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-j\omega_k t} dt}_{1/\text{s}}$	$\underbrace{\int_{-T/2}^{T/2} x(t)^2}_{\text{V}^2} dt = \underbrace{\frac{T}{s} \sum_{k=-\infty}^{\infty} c_k ^2}_{\text{V}^2}$ (Sect. 2.1.2)
Fourier Transform	$\underbrace{x(t)}_{\text{V}} = \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega}_{\text{V/Hz}}$	$\underbrace{X(\omega)}_{\text{V/Hz}} = \underbrace{\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt}_{\text{V}}$	$\underbrace{\int_{-\infty}^{\infty} x(t)^2}_{\text{V}^2} dt = \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) ^2 d\omega}_{\text{V}^2/\text{Hz}^2}$ (Sect. 2.2.4)
Fast Fourier Transform	$\underbrace{x_j}_{\text{V}} = \underbrace{\frac{1}{N} \sum_{k=0}^{N-1} y_k e^{2\pi j k/N}}_{\text{V}}$	$\underbrace{y_k}_{\text{V}} = \underbrace{\sum_{j=0}^{N-1} x_j e^{-2\pi j k/N}}_{\text{V}}$	$\sum_{j=0}^{N-1} x_j^2 = \frac{1}{N} \sum_{k=0}^{N-1} y_k ^2$ (Sect. 2.3.3)

$X(\omega)$ has unit V/Hz rather than V/(rad/s), despite the fact that the integral in $x(t) = (2\pi)^{-1} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega$ is w.r.t. ω (rad/s). The factor $1/2\pi$ (with unit 1/rad) makes up for this; $\omega/2\pi$ is the frequency in Hz. If one omits 2π and writes $x(t) = \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega$, then $X(\omega)$ must have a unit of V/(rad/s), and vice versa. The same rule applies to the FFT approximation $\hat{X}(\omega)$. This issue is not relevant to the FS coefficient c_k because the multiplier $e^{i\tilde{\omega}_k t}$ in $x(t) = \sum_{k=-\infty}^{\infty} c_k e^{i\tilde{\omega}_k t}$ is dimensionless.

2.8 Connecting Theory with Matlab

Matlab provides a convenient platform for signal processing and scientific/engineering computing in general. This section presents the connection of some theoretical results with the functions in Matlab. The focus is on FFT and related functions. In the following, quantities in Matlab are in typewriter font.

In Matlab, the index of an array starts from 1. In this book, the index of a discrete-time sequence starts from 0, which is found to simplify presentation. Thus, $\{x_j\}_{j=0}^{N-1}$ in this book is an array `x` of length `N` in Matlab with

$$\mathbf{x}(1) = x_0, \quad \mathbf{x}(2) = x_1, \dots, \mathbf{x}(N) = x_{N-1}.$$

Let $\{y_k\}_{k=0}^{N-1}$ be the FFT of $\{x_j\}_{j=0}^{N-1}$. In Matlab, FFT can be performed by the built-in function `fft`. The call `y = fft(x)` returns an array `y` of length `N`. By definition,

$$y_k = \sum_{j=0}^{N-1} x_j e^{-2\pi i j k / N} \quad k = 0, \dots, N-1 \quad (2.67)$$

$$y(k) = \sum_{j=1}^N x(j) e^{-2\pi i (j-1)(k-1)/N} \quad k = 1, \dots, N \quad (2.68)$$

Correspondingly,

$$\begin{aligned} y(1) &= y_0 && \text{(zero frequency)} \\ y(2) &= y_1 && \text{(frequency } f_1 = 1/N\Delta t \text{ Hz)} \\ y(3) &= y_2 && \text{(frequency } f_2 = 2/N\Delta t \text{ Hz)} \end{aligned}$$

and so on, up to

$$y(Nq+1) = y_{N_yq}$$

where $N_q = \text{floor}(N/2)$ (integer part of $N/2$). The remaining entries in y are just the conjugate mirror image of those below the Nyquist frequency.

Table 2.4 shows the connection of some theoretical properties with Matlab. The symbol ‘ $\equiv$ ’ denotes that the quantities on both sides are numerically the same when evaluated in Matlab. Unless otherwise stated, x and y are assumed to be real N by 1 array. See Chap. 4 for the last three properties regarding correlation function and power spectral density.

2.9 FFT Algorithm

To supplement Sect. 2.3.1, the FFT algorithm is briefly introduced here. Originally, evaluating the DTFT in (2.35) at N frequencies requires a computational effort of the order of N^2 , i.e., $O(N^2)$. Using the FFT algorithm, it is reduced to $O(N \log_2 N)$, although the values are evaluated at a specific set of equally spaced frequencies. The key lies in the discovery that, for N being some power of 2, i.e., $N = 2^m$ for some integer m , a FFT sequence of length N can be obtained from two FFT sequences of length $N/2$; and similarly each FFT sequence of length $N/2$ can be obtained from another two FFT sequences of length $N/4$; and so on. The general algorithm has provisions for other cases of N , but here we confine our discussion to $N = 2^m$.

Table 2.4 Connection between theory and Matlab

Property	Matlab
First entry of FFT	$y = \text{fft}(x);$ $y(1) \equiv \text{sum}(x)$
Conjugate mirror	$y = \text{fft}(x);$ $y(2:\text{end}) \equiv \text{conj}(y(\text{end}:-1:2))$
Inverse FFT	$\text{ifft}(\text{fft}(x)) \equiv x$
Parseval equality	$\text{sum}(\text{abs}(\text{fft}(x)).^2) \equiv N * \text{sum}(x.^2)$
Symmetry of convolution	$\text{conv}(x, y) \equiv \text{conv}(y, x)$
Convolution theorem	$\text{fft}(\text{conv}(x, y)) \equiv \text{fft}([x; \text{zeros}(N-1, 1)]) .* \text{fft}([y; \text{zeros}(N-1, 1)])$
Wiener-Khinchin theorem	$s = \text{fft}([0; \text{xcorr}(x, y)]);$ % (2N, 1) array $s(1:2:\text{end}-1) \equiv \text{fft}(x) .* \text{conj}(\text{fft}(y))$
Asymmetry of xcorr	$\text{xcorr}(x, y) \equiv \text{xcorr}(y(\text{end}:-1:1), x(\text{end}:-1:1))$
xcorr and conv	$\text{xcorr}(x, y) \equiv \text{conv}(x, y(\text{end}:-1:1))$

2.9.1 Basic Idea

Given $\{x_j\}_{j=0}^{N-1}$, let $\{y_k\}_{k=0}^{N-1}$ be the FFT sequence to be computed, i.e.,

$$y_k = \sum_{j=0}^{N-1} x_j e^{-2\pi i j k / N} \quad k = 0, \dots, N-1 \quad (2.69)$$

Let

$$w_N = e^{-2\pi i / N} \quad (2.70)$$

so that $(w_N)^N = 1$. Then y_k can be written in terms of w_N :

$$y_k = \sum_{j=0}^{N-1} x_j w_N^{jk} \quad (2.71)$$

Separating the sum into even and odd terms of j ,

$$\begin{aligned} y_k &= x_0 w_N^{0 \cdot k} + x_2 w_N^{2 \cdot k} + \dots + x_{N-2} w_N^{(N-2) \cdot k} & (j \text{ even}) \\ &\quad + x_1 w_N^{1 \cdot k} + x_3 w_N^{3 \cdot k} + \dots + x_{N-1} w_N^{(N-1) \cdot k} & (j \text{ odd}) \\ &= \sum_{r=0}^{(N/2)-1} x_{2r} w_N^{2rk} + \sum_{r=0}^{(N/2)-1} x_{2r+1} w_N^{(2r+1)k} \end{aligned} \quad (2.72)$$

Since $w_N^2 = w_{N/2}$, we can write $w_N^{2rk} = w_{N/2}^{rk}$ and $w_N^{(2r+1)k} = w_{N/2}^{rk} w_N^k$. Then

$$y_k = \sum_{r=0}^{(N/2)-1} x_{2r} w_{N/2}^{rk} + w_N^k \sum_{r=0}^{(N/2)-1} x_{2r+1} w_{N/2}^{rk} \quad (2.73)$$

On the other hand, suppose we separate $\{x_j\}_{j=0}^{N-1}$ into two sequences of length $N/2$, one containing the even j terms and the other containing the odd j terms, i.e., $\{x_{2r}\}_{r=0}^{(N/2)-1}$ and $\{x_{2r+1}\}_{r=0}^{(N/2)-1}$, respectively. Their FFTs are respectively given by

$$\begin{aligned} y'_k &= \sum_{r=0}^{(N/2)-1} x_{2r} e^{-2\pi i r k / (N/2)} = \sum_{r=0}^{(N/2)-1} x_{2r} w_{N/2}^{rk} \\ y''_k &= \sum_{r=0}^{(N/2)-1} x_{2r+1} e^{-2\pi i r k / (N/2)} = \sum_{r=0}^{(N/2)-1} x_{2r+1} w_{N/2}^{rk} \end{aligned} \quad k = 0, \dots, \frac{N}{2} - 1 \quad (2.74)$$

Comparing (2.73) and (2.74), we see that

$$y_k = y'_k + w_N^k y''_k \quad k = 0, \dots, \frac{N}{2} - 1 \quad (2.75)$$

The first half of $\{y_k\}_{k=0}^{N-1}$ can thus be obtained from $\{y'_k\}_{k=0}^{(N/2)-1}$ and $\{y''_k\}_{k=0}^{(N/2)-1}$. The remaining half can be produced using the conjugate mirror property of FFT, i.e., $y_k = \bar{y}_{N-k}$.

The above shows that the FFT of a length N sequence can be obtained from the FFTs of two length $N/2$ sequences. The FFT of each length $N/2$ sequence can be further obtained from the FFTs of another two length $N/4$ sequences, and so on. Carrying this on recursively, eventually it involves the FFT of a length 1 sequence, i.e., a number, which is the sequence itself.

2.9.2 Computational Effort

To assess the computational effort of the FFT algorithm, let C_N denote the number of multiplications to produce a FFT sequence of length N . The effort for additions, subtractions and complex conjugate can be considered negligible. Also, assume that $\{w_N^k\}_{k=0}^{N-1}$ has been computed upfront. Its computational effort is not counted in C_N . Based on (2.75),

$$\underbrace{C_N}_{\text{FFT of length } N} = 2 \underbrace{C_{N/2}}_{\text{FFT of length } N/2} + \underbrace{N/2}_{\text{Multiply } w_N^k \text{ for } N/2 \text{ times}} \quad (2.76)$$

By sequential substitution or induction, and recalling $N = 2^m$, it can be shown that

$$C_N = C_1 N + \frac{N}{2} \log_2 N \quad (2.77)$$

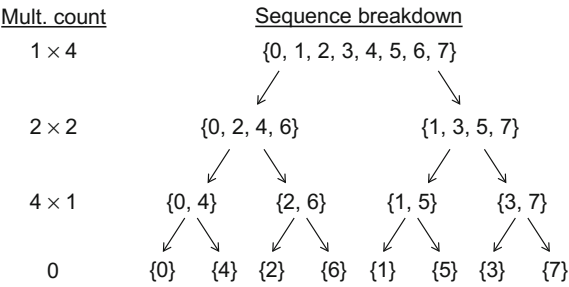
Since the FFT of a length 1 sequence is just the sequence itself, $C_1 = 0$. Consequently,

$$C_N = \frac{N}{2} \log_2 N \quad (2.78)$$

The computational effort for the FFT of a length N sequence is therefore $O(N \log_2 N)$ instead of $O(N^2)$.

Example 2.3 (FFT algorithm) Figure 2.6 illustrates the recursive breakdown of the calculations for a FFT sequence of length $N = 2^3 = 8$. On the right, each brace

Fig. 2.6 Illustration of FFT algorithm for a sequence of length $N = 8$



contains a sequence. The FFT of the original sequence $\{0, 1, \dots, 7\}$ is obtained from the FFTs of two shorter sequences $\{0, 2, 4, 6\}$ and $\{1, 3, 5, 7\}$. The same applies to other sequences.

The left side of the figure counts the number of multiplications involved in producing the longer sequences from the shorter ones. Starting from the bottom, no multiplication is needed to obtain the FFTs of the sequences of length 1. To produce the FFT of $\{0, 4\}$ from the FFTs of $\{0\}$ and $\{4\}$, it requires 1 multiplication because the first FFT entry ($k = 0$) involves a multiplication with w_2^k ; the second can be produced as the complex conjugate of the first. The same applies to other sequences $\{2, 6\}$, $\{1, 5\}$ and $\{3, 7\}$. The number of multiplications to produce the four sequences of length 2 from 8 sequences of length 1 is therefore 4×1 . Similarly, to produce the FFT of $\{0, 2, 4, 6\}$ from the FFTs of $\{0, 4\}$ and $\{2, 6\}$, it involves 2 multiplications for the first two FFT entries ($k = 0, 1$); the other two entries ($k = 2, 3$) do not involve any multiplication as they are produced from the complex conjugate of the first two. The number of multiplications to produce the FFT of two sequences of length 4 from four sequences of length 2 is therefore 2×2 . Finally, it involves 1×4 multiplications to obtain the FFT of $\{0, 1, \dots, 7\}$ from the FFTs of $\{0, 2, 4, 6\}$ and $\{1, 3, 5, 7\}$. The total number of multiplications is $4 \times 1 + 2 \times 2 + 1 \times 4 = 12$. This checks with (2.78), which gives $C_8 = (8/2) \log_2 8 = 12$. ■

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Chapter 3

Structural Dynamics and Modal Testing

Abstract This chapter analyzes the response of a structure subjected to deterministic excitations. Conventional topics in single-degree-of-freedom and multi-degree-of-freedom structures are covered, including free vibration, forced vibration due to harmonic excitation, periodic excitation, impulsive excitation and arbitrary excitation, and modal superposition. Basic concepts in experimental modal testing are discussed as a simple application of structural dynamics, including logarithmic decrement, half-power bandwidth, harmonic load test and impact hammer test. The state-space approach is introduced for analyzing general dynamical systems. The basic principles of numerical solution and Newmark integration schemes are introduced, which allow one to compute the structural response for a given time history of excitation.

Keywords Structural dynamics • Modal testing • Logarithmic decrement • Half-power bandwidth • Newmark scheme • Impact hammer test

In this chapter we discuss the subject of ‘structural dynamics’, which studies the response (e.g., displacement, velocity) of a structure obeying Newton’s law of motion. The focus is on oscillatory behavior, commonly known as ‘vibration’. ‘Resonance’ plays an important role, where an excitation of oscillatory nature can generate significantly larger response if it varies at a pace near the ‘natural frequency’ of the structure. Just as a musical instrument can produce sounds at different pitches, a structure can have more than one natural frequency. Associated with each frequency there is a specific spatial vibration pattern, called ‘mode shape’. Natural frequencies and mode shapes are determined by an ‘eigenvalue equation’ that depends on the stiffness and mass of the structure. Together with damping characteristics they completely determine the structural dynamic response under applied loads. For textbooks on structural dynamics, see, e.g., Meirovitch (1986), Clough and Penzien (1993) and Beards (1996).

We first discuss the vibration response of a single-degree-freedom (SDOF) structure, i.e., with only one variable in the equation of motion. Different types of excitations are considered. Multi-degree-of-freedom (MDOF) structures are next

Table 3.1 Modal testing concepts introduced in this chapter

Dynamics	Concept	Section, example
Free vibration	Logarithmic decrement	Section 3.1.4
Forced vibration	Half-power bandwidth	Section 3.1.8, Example 3.2
	Harmonic load	Section 3.4, Example 3.7
	Impact hammer	Section 3.5

discussed. Their response can be obtained as the sum of contributions from different modes, distinguished by mode shapes. Under conventional assumptions on damping, the response contribution from each mode can be obtained via the solution of a SDOF equation, and so the techniques and insights gained from SDOF dynamics are generally useful. We also discuss ‘state-space approach’, which is a powerful means for analyzing general dynamical systems. Time integration method is then introduced. It allows one to compute structural dynamic response based on a given discrete-time history of the excitation. This is an indispensable tool in applications.

‘Modal testing’ concepts are introduced as an application of structural dynamics. They illustrate the basic principles where the modal properties (i.e., natural frequencies, damping ratios, etc.) of a structure can be estimated from conceptually simple vibration tests. They also provide an interesting way of understanding structural dynamics and building intuitions. The relevant sections are listed in Table 3.1 for easy reference, skipping which will not affect understanding the main theory. See also McConnell (1995), Maia and Silva (1997) and Ewins (2000).

3.1 SDOF Dynamics

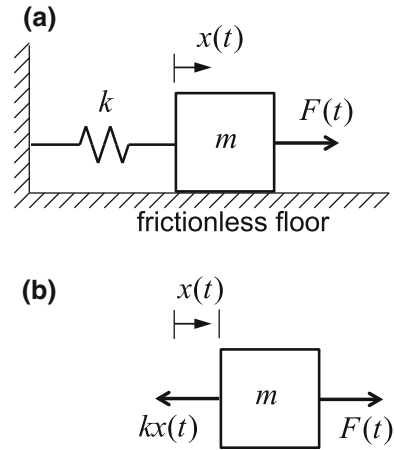
Consider a mass m (kg) on a frictionless floor and connected with a spring of stiffness k (N/m) to a fixed wall, as shown in Fig. 3.1a. It is subjected to a force $F(t)$ (N) that can vary with time t (s). The position of the mass from its static equilibrium position is measured by $x(t)$ (m).

As shown in Fig. 3.1b, at a particular time t , the mass is at $x(t)$ and it is subjected to a restoring force $kx(t)$ in the opposite direction, in addition to the applied force $F(t)$. Applying Newton’s second law (mass $\times$ acceleration = net force) on the mass gives $m\ddot{x}(t) = -kx(t) + F(t)$. This is conventionally written in the form

$$\underbrace{m\ddot{x}(t)}_{\text{inertia}} + \underbrace{kx(t)}_{\text{stiffness}} = \underbrace{F(t)}_{\text{applied force}} \quad (3.1)$$

The stiffness and inertia terms refer to those proportional to displacement and acceleration, respectively. This equation does not have ‘dissipative’ effect, which is not realistic. According to the equation, if the mass is given an initial kick, it will oscillate forever even if there is no applied force afterwards. In reality, it will

Fig. 3.1 SDOF structure;
a schematic diagram; **b** free
 body diagram at time t



eventually stop, due to irreversible mechanisms (e.g., friction, forming of micro-cracks) converting kinetic/strain energy to other forms (e.g., heat, sound). This is collectively referred as ‘damping’. To match reality, a conventional way is to introduce a damping term:

$$\underbrace{m\ddot{x}(t)}_{\text{inertia}} + \underbrace{c\dot{x}(t)}_{\text{damping}} + \underbrace{kx(t)}_{\text{stiffness}} = \underbrace{F(t)}_{\text{applied force}} \quad (3.2)$$

where c (N/(m/s)) is the ‘damping constant’. The damping term here is ‘viscous’ in nature, as it is proportional to velocity. This need not be the case in reality, e.g., dissipation due to friction is not. Nevertheless, it is still the conventional assumption in analysis, design and identification of lightly damped structures. See Sect. 3.3 for further remarks.

3.1.1 Natural Frequency

For instructional purpose we first consider ‘undamped free vibration’, where the structure has no damping and the applied force is absent, i.e., $c = 0$ and $F(t) \equiv 0$. In this case the response is purely due to initial conditions. The equation of motion in (3.2) reduces to

$$m\ddot{x}(t) + kx(t) = 0 \quad (3.3)$$

Suppose the initial conditions are

$$x(0) = u \quad \dot{x}(0) = v \quad (3.4)$$

Equation (3.3) is a second order ordinary differential equation (ODE). Its solution has two integration constants, which can be determined by the two initial conditions in (3.4). By inspection, it can be satisfied by a simple harmonic motion, e.g., $x(t) = a \cos \omega_1 t$, where ω_1 and a are constants to be determined. Substituting $x(t) = a \cos \omega_1 t$ and $\ddot{x}(t) = -a\omega_1^2 \cos \omega_1 t$ into (3.3) and collecting terms,

$$(-m\omega_1^2 + k)a \cos \omega_1 t = 0 \quad (3.5)$$

This equation must hold for all t . Since $a \neq 0$ (otherwise $x(t) \equiv 0$) and $\cos \omega_1 t$ cannot be zero for all t , the only possibility is $-m\omega_1^2 + k = 0$. Solving this equation gives two roots for ω_1 , i.e., $\pm \sqrt{k/m}$, but they lead to the same solution form for $x(t)$. It is therefore sufficient to consider only one of the roots. Taking the positive root gives

$$\omega_1 = \sqrt{\frac{k}{m}} \quad (\text{Natural frequency in rad/s}) \quad (3.6)$$

This is called the ‘natural frequency’ of the structure. The unit of ω_1 is radian per second, which can be reasoned by noting that the argument $\omega_1 t$ in $\cos \omega_1 t$ has a unit of radian and t has a unit of second. In vibration applications, it is more intuitive to refer frequency in Hz (cycles per second), i.e.,

$$f_1 = \frac{\omega_1}{2\pi} \quad (\text{Natural frequency in Hz}) \quad (3.7)$$

The ‘natural period’ is the time it takes to complete one cycle of oscillation:

$$T_1 = \frac{1}{f_1} = \frac{2\pi}{\omega_1} \quad (\text{Natural period in sec}) \quad (3.8)$$

Repeating the above argument by substituting $x(t) = b \sin \omega_1 t$ for some constants ω_1 and b leads to the same conclusion that $\omega_1 = \sqrt{k/m}$. Since (3.3) is a linear equation and there are two possible solution forms, the response is generally of the form

$$x(t) = a \cos \omega_1 t + b \sin \omega_1 t \quad (3.9)$$

Check that it satisfies (3.3) for arbitrary a and b . The constants a and b can be determined from the two initial conditions in (3.4). According to (3.9), $x(0) = a$ and $\dot{x}(0) = b\omega_1$. Enforcing the initial conditions in (3.4) gives $a = u$ and $b = v/\omega_1$. As a result,

$$x(t) = u \cos \omega_1 t + \frac{v}{\omega_1} \sin \omega_1 t \quad (\text{undamped free vibration}) \quad (3.10)$$

Using the compound angle formula $\cos(\theta_1 - \theta_2) = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2$, $x(t)$ can be written in a form that shows explicitly the amplitude and phase:

$$\begin{aligned}
 x(t) &= \sqrt{u^2 + v^2/\omega_1^2} \left[\underbrace{\frac{u}{\sqrt{u^2 + v^2/\omega_1^2}}}_{\cos \phi} \cos \omega_1 t + \underbrace{\frac{v/\omega_1}{\sqrt{u^2 + v^2/\omega_1^2}}}_{\sin \phi} \sin \omega_1 t \right] \\
 &= \underbrace{\sqrt{u^2 + v^2/\omega_1^2}}_{\text{amplitude}} \cos(\underbrace{\omega_1}_{\text{frequency}} t - \underbrace{\phi}_{\text{phase}})
 \end{aligned} \tag{3.11}$$

where $\tan \phi = v/\omega_1 u$.

3.1.2 Damping Ratio

Equation (3.11) indicates that undamped free vibration never dies down, because the cosine term does not. This is not realistic. The general case of a SDOF structure with damping, i.e., (3.2), will be considered in the remaining of this chapter. It is mathematically more convenient to work with the following form

$$\ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) = p(t) \tag{3.12}$$

where we have divided (3.2) by m and replaced k/m by ω_1^2 ;

$$p(t) = \frac{F(t)}{m} \tag{3.13}$$

is the applied force per unit structural mass; and

$$\zeta = \frac{c}{2\sqrt{mk}} \tag{3.14}$$

is the (dimensionless) ‘damping ratio’. In most structural vibration applications, it is ζ rather c that is referred to. Typically, $0 < \zeta < 1$ with ζ taking small values of a few percent. Our discussion shall focus on the standard form (3.12) for different cases of $p(t)$.

3.1.3 Damped Free Vibration

Consider again free vibration ($p(t) \equiv 0$) but now with damping ($\zeta > 0$). Equation (3.12) reduces to

$$\ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) = 0 \quad (3.15)$$

As before let the initial conditions be

$$x(0) = u \quad \dot{x}(0) = v \quad (3.16)$$

Consider solution of the form $x(t) = ae^{\alpha t}$ for some constants a and α to be determined. Then $\dot{x}(t) = \alpha ae^{\alpha t}$ and $\ddot{x}(t) = \alpha^2 ae^{\alpha t}$. Substituting into (3.15) and collecting terms,

$$(\alpha^2 + 2\zeta\omega_1\alpha + \omega_1^2)ae^{\alpha t} = 0 \quad (3.17)$$

This equation must hold for all t . Since $a \neq 0$ and it is not possible for $e^{\alpha t}$ to be zero for all t , the only possibility is $\alpha^2 + 2\zeta\omega_1\alpha + \omega_1^2 = 0$. Solving this equation for α gives two roots

$$\alpha_1 = -\zeta\omega_1 + \omega_1\sqrt{\zeta^2 - 1} \quad \alpha_2 = -\zeta\omega_1 - \omega_1\sqrt{\zeta^2 - 1} \quad (3.18)$$

Since there are two possibilities and the equation of motion is linear, the solution is a linear combination of the two possibilities, i.e.,

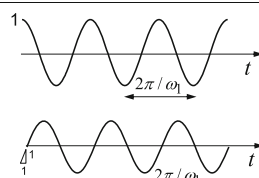
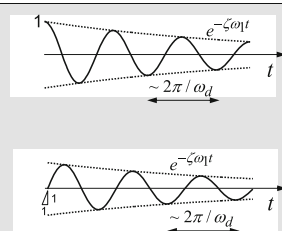
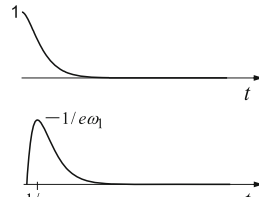
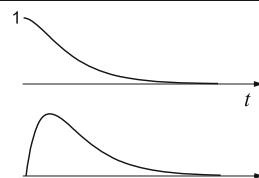
$$x(t) = a_1e^{\alpha_1 t} + a_2e^{\alpha_2 t} \quad (3.19)$$

The constants a_1 and a_2 can be determined from the initial conditions in (3.16), which give two algebraic equations. The solutions of a_1 and a_2 can be complex-valued, even though $x(t)$ is always real-valued. Substituting the solution of a_1 and a_2 into (3.19), the response can be written as

$$x(t) = ug_1(t) + vg_2(t) \quad (3.20)$$

where $g_1(t)$ and $g_2(t)$ are given in Table 3.2 for different cases of ζ ; see the end for derivation. The under damped case ($0 < \zeta < 1$) is highlighted as it is typical in applications. Clearly, $x(t)$ is a linear function of u and v . Setting $u = 1$ and $v = 0$ shows that $g_1(t)$ is the response with unit initial displacement and zero velocity. Setting $u = 0$ and $v = 1$ shows that $g_2(t)$ is the response with zero initial displacement and unit velocity. For their nature, $g_1(t)$ and $g_2(t)$ are referred as the ‘complementary free vibration response’. These functions and hence the response $x(t)$ have characteristic behavior depending on whether $0 \leq \zeta < 1$ (oscillatory) or $\zeta \geq 1$ (decay without rebound). In the former, the frequency of oscillation (ignoring exponential decay effect) is $\omega_d = \omega_1\sqrt{1 - \zeta^2}$ (rad/s), which is called the ‘damped natural frequency’. In typical applications where ζ is small, $\omega_d \approx \omega_1$ and so the

Table 3.2 Expressions for complementary free vibration responses $g_1(t)$ and $g_2(t)$

Case	$g_1(t)$ and $g_2(t)$	Behavior
Undamped $\zeta = 0$	$g_1(t) = \cos \omega_1 t$ $g_2(t) = \frac{\sin \omega_1 t}{\omega_1}$	
Under damped $0 < \zeta < 1$ (typical in structural vibrations)	$g_1(t) = e^{-\zeta \omega_1 t} \left(\cos \omega_d t + \frac{\zeta \omega_1}{\omega_d} \sin \omega_d t \right)$ $\omega_d = \omega_1 \sqrt{1 - \zeta^2}$ $g_2(t) = \frac{e^{-\zeta \omega_1 t}}{\omega_d} \sin \omega_d t$	
Critically damped $\zeta = 1$	$g_1(t) = e^{-\omega_1 t} (1 + \omega_1 t)$ $g_2(t) = t e^{-\omega_1 t}$	
Over damped $\zeta > 1$	$g_1(t) = e^{-\zeta \omega_1 t} \left(\cosh \omega_D t + \frac{\zeta \omega_1}{\omega_D} \sinh \omega_D t \right)$ $\omega_D = \omega_1 \sqrt{\zeta^2 - 1}$ $g_2(t) = \frac{e^{-\zeta \omega_1 t}}{\omega_D} \sinh \omega_D t$	

damped natural frequency ω_d is seldom distinguished from the (undamped) natural frequency ω_1 .

The expressions in Table 3.2 match at the boundaries of ζ . The expressions for $\zeta = 0$ agree with the limit of those for $0 < \zeta < 1$ as $\zeta \rightarrow 0$. Similarly, taking limit $\zeta \rightarrow 1$ on the expressions for $0 < \zeta < 1$ or $\zeta > 1$ gives the ones for $\zeta = 1$. Interestingly, the expressions for $\zeta > 1$ can be obtained from those for $0 < \zeta < 1$ by replacing ω_d with $\omega_D = \omega_1 \sqrt{\zeta^2 - 1}$, and cosines/sines with hyperbolic cosines/sines.

Derivation of $g_1(t)$ and $g_2(t)$ in Table 3.2

We first solve for a_1 and a_2 in (3.19) from the initial conditions in (3.16). From (3.19), $x(t) = a_1 e^{\alpha_1 t} + a_2 e^{\alpha_2 t}$. Setting $x(0) = u$ and $\dot{x}(0) = v$ gives

$$a_1 + a_2 = u \quad \alpha_1 a_1 + \alpha_2 a_2 = v \quad (3.21)$$

Solving for a_1 and a_2 ,

$$a_1 = \frac{-\alpha_2 u + v}{\alpha_1 - \alpha_2} \quad a_2 = \frac{\alpha_1 u - v}{\alpha_1 - \alpha_2} \quad (3.22)$$

Substituting into $x(t) = a_1 e^{\alpha_1 t} + a_2 e^{\alpha_2 t}$ and collecting terms of u and v gives $x(t) = u g_1(t) + v g_2(t)$ in (3.20) where

$$g_1(t) = \frac{\alpha_1 e^{\alpha_2 t} - \alpha_2 e^{\alpha_1 t}}{\alpha_1 - \alpha_2} \quad g_2(t) = \frac{e^{\alpha_1 t} - e^{\alpha_2 t}}{\alpha_1 - \alpha_2} \quad (3.23)$$

Note that $g_1(t)$ and $g_2(t)$ are always real-valued even though α_1 and α_2 in (3.18) can be complex-valued. To write them explicitly in terms of real quantities, we express $e^{\alpha_1 t}$ and $e^{\alpha_2 t}$ in terms of cosines and sines when $\zeta < 1$; or hyperbolic cosines and sines when $\zeta > 1$. Details are as follow.

Under Damped

When $0 < \zeta < 1$, α_1 and α_2 are complex. Using (3.18), writing $\sqrt{\zeta^2 - 1} = \mathbf{i}\sqrt{1 - \zeta^2}$ ($\mathbf{i}^2 = -1$) and using the Euler formula $e^{\mathbf{i}\theta} = \cos \theta + \mathbf{i} \sin \theta$ (for any real θ),

$$\left. \begin{matrix} e^{\alpha_1 t} \\ e^{\alpha_2 t} \end{matrix} \right\} = e^{-\zeta \omega_1 t} \times \underbrace{e^{\pm \mathbf{i} \omega_1 \sqrt{1 - \zeta^2} t}}_{\cos \omega_d t \pm \mathbf{i} \sin \omega_d t} = e^{-\zeta \omega_1 t} (\cos \omega_d t \pm \mathbf{i} \sin \omega_d t) \quad (3.24)$$

Substituting into (3.23) and simplifying gives $g_1(t)$ and $g_2(t)$ in Table 3.2 for $0 < \zeta < 1$.

Over Damped

When $\zeta > 1$, α_1 and α_2 are both real. The terms $e^{\alpha_1 t}$ and $e^{\alpha_2 t}$ can be expressed in terms of hyperbolic cosines and sines by noting the following identities for any real θ :

$$\cosh \theta = \frac{e^{\theta} + e^{-\theta}}{2} \quad \sinh \theta = \frac{e^{\theta} - e^{-\theta}}{2} \quad e^{\theta} = \cosh \theta + \sinh \theta \quad (3.25)$$

Using (3.18),

$$\left. \begin{matrix} e^{x_1 t} \\ e^{x_2 t} \end{matrix} \right\} = e^{-\zeta \omega_1 t} \times \underbrace{e^{\pm \omega_1 \sqrt{\zeta^2 - 1} t}}_{\cosh \omega_D t \pm \sinh \omega_D t} = e^{-\zeta \omega_1 t} (\cosh \omega_D t \pm \sinh \omega_D t) \quad (3.26)$$

Substituting into (3.23) gives $g_1(t)$ and $g_2(t)$ in Table 3.2 for $\zeta > 1$. The algebra involved is similar to the case for $0 < \zeta < 1$, where the sine and cosine terms are replaced by their hyperbolic counterparts.

Matching at Boundaries

The expressions for $g_1(t)$ and $g_2(t)$ for the boundary cases of $\zeta = 0$ and $\zeta = 1$ can be derived from first principle by substituting the value of ζ into (3.18) and then obtaining the coefficients a_1 and a_2 . Alternatively, they can be obtained by taking limits of ζ on the expressions in Table 3.2. The case for $0 < \zeta < 1$ as $\zeta \rightarrow 0$ is straightforward. For $0 < \zeta < 1$ as $\zeta \rightarrow 1$, note that $\omega_d = \omega_1 \sqrt{1 - \zeta^2} \rightarrow 0$ as $\zeta \rightarrow 1$. Then

$$\lim_{\zeta \rightarrow 1} g_1(t) = \lim_{\zeta \rightarrow 1} \underbrace{e^{-\zeta \omega_1 t}}_{\rightarrow e^{-\omega_1 t}} \left(\underbrace{\cos \omega_d t}_{\rightarrow 1} + \underbrace{\zeta \omega_1}_{\rightarrow \omega_1} \underbrace{\frac{\sin \omega_d t}{\omega_d}}_{\rightarrow t} \right) = e^{-\omega_1 t} (1 + \omega_1 t) \quad (3.27)$$

$$\lim_{\zeta \rightarrow 1} g_2(t) = \lim_{\zeta \rightarrow 1} \underbrace{e^{-\zeta \omega_1 t}}_{\rightarrow e^{-\omega_1 t}} \underbrace{\frac{\sin \omega_d t}{\omega_d}}_{\rightarrow t} = t e^{-\omega_1 t} \quad (3.28)$$

The case for $\zeta > 1$ as $\zeta \rightarrow 1$ is similar. ■

3.1.4 Logarithmic Decrement Method

For lightly damped structures ($\zeta \ll 1$), the oscillatory nature of free vibration with a decaying amplitude provides an intuitive way for estimating the natural frequency and damping ratio. By combining the sine and cosine terms in $g_1(t)$ and $g_2(t)$, one can reason from (3.20) that the free vibration displacement starting from some initial amplitude x_0 can be written in the form

$$x(t) = \underbrace{x_0}_{\text{initial amplitude}} \underbrace{e^{-\zeta \omega_1 t}}_{\text{exponential decay}} \underbrace{\cos \omega_d t}_{\text{oscillatory}} \quad (3.29)$$

This is illustrated in Fig. 3.2.

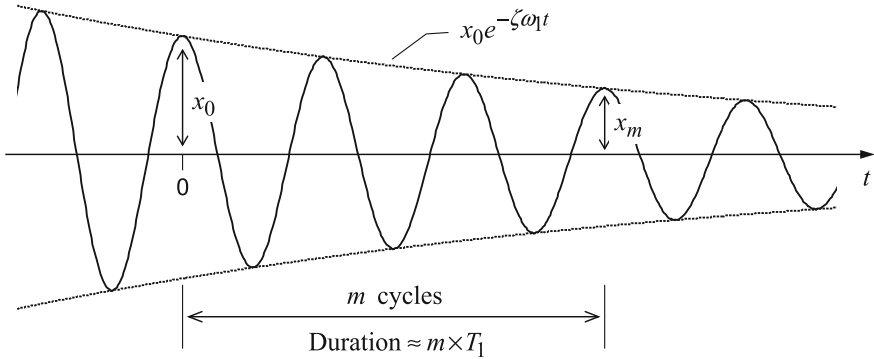


Fig. 3.2 Cycle counting and logarithmic decrement method ($m = 3$)

Since $\zeta \ll 1$, $\omega_d \approx \omega_1$. The natural period $T_1 = 2\pi/\omega_1$ (sec) can be estimated by ‘cycle counting’ as the average time it takes to complete one cycle, i.e.,

$$\text{Natural period, } T_1 \approx \frac{\text{Time for } m \text{ cycles}}{m} \quad (\text{cycle-counting}) \quad (3.30)$$

Estimating using multiple cycles helps reduce error from measurement noise. On the other hand, the damping ratio is related to the rate of amplitude decay. By noting that the cosine term in (3.29) is approximately equal to 1 at the peaks,

$$\frac{x_0}{x_m} \approx \frac{x_0}{x_0 e^{-\zeta \omega_1 m T_1}} = e^{2\pi m \zeta} \quad (3.31)$$

since $\omega_1 T_1 = 2\pi$. Rearranging gives an estimation of the damping ratio:

$$\zeta \approx \frac{\ln(x_0/x_m)}{2\pi m} \quad (\text{log-decrement}) \quad (3.32)$$

This is known as the ‘logarithmic decrement method’. The same formula can be applied to velocity or acceleration response data, as it can be reasoned that they can also be written in the form of (3.29). In reality, vibration data contains measurement noise and contribution from ambient excitations. Their spurious effects may be reduced by averaging the data over repeated trials.

3.1.5 Harmonic Excitation

From now on we limit our scope to the under damped case ($0 < \zeta < 1$), which is the primary case of interest in structural vibrations. Suppose the excitation is ‘harmonic’, say,

$$p(t) = p_0 \cos \omega t \quad (3.33)$$

where ω (rad/s) is the ‘excitation frequency’ and p_0 is the ‘excitation amplitude’ (per unit structural mass). As before, let the initial conditions be

$$x(0) = u \quad \dot{x}(0) = v \quad (3.34)$$

The response to harmonic excitation comprises a ‘transient response’ and a ‘steady-state response’:

$$x(t) = \underbrace{C_1 g_1(t) + C_2 g_2(t)}_{\substack{\text{transient} \\ \rightarrow 0 \text{ as } t \rightarrow \infty}} + \underbrace{x_s(t)}_{\text{steady-state}} \quad (3.35)$$

where C_1 and C_2 are constants (often not of concern) that can be determined from initial conditions; $g_1(t)$ and $g_2(t)$ are the complementary free vibration responses in Table 3.2.

Different Roles

In ODE theory, the transient response and steady-state response are respectively the ‘homogeneous solution’ and ‘particular solution’. They have different roles. Substituting the transient response into the equation of motion (3.12) gives a zero LHS. It is therefore the steady-state response that maintains a balance of dynamic forces with the excitation. On the other hand, the steady-state response alone need not satisfy initial conditions, i.e., generally $x_s(0) \neq u$ and $\dot{x}_s(0) \neq v$. The transient response makes this up with the right coefficients C_1 and C_2 so that $x(t)$ satisfies the initial conditions. As time goes on ($t \rightarrow \infty$) the transient response diminishes, leaving only the steady-state response, and hence the name.

Steady-State Response

The steady-state response only needs to satisfy the equation of motion. The functional form of the excitation in (3.33) suggests the form

$$x_s(t) = a \cos \omega t + b \sin \omega t \quad (3.36)$$

for some constants a and b to be determined. Direct differentiation gives

$$\begin{aligned} \dot{x}_s(t) &= -a\omega \sin \omega t + b\omega \cos \omega t \\ \ddot{x}_s(t) &= -a\omega^2 \cos \omega t - b\omega^2 \sin \omega t \end{aligned} \quad (3.37)$$

Substituting into $\ddot{x}_s + 2\zeta\omega_1\dot{x}_s + \omega_1^2 x_s = p_0 \cos \omega t$ and collecting the cosine and sine terms gives

$$[(\omega_1^2 - \omega^2)a + (2\zeta\omega_1\omega)b - p_0] \cos \omega t + [-(2\zeta\omega_1\omega)a + (\omega_1^2 - \omega^2)b] \sin \omega t = 0 \quad (3.38)$$

The bracketed terms must be zero because it is not possible for $\cos \omega t$, $\sin \omega t$ or their linear combination to be zero for all t , unless their coefficients are zero. Setting the bracketed terms to zero and solving for a and b gives

$$a = \frac{p_0}{\omega_1^2} \left[\frac{1 - \beta^2}{(1 - \beta^2)^2 + (2\zeta\beta)^2} \right] \quad b = \frac{p_0}{\omega_1^2} \left[\frac{2\zeta\beta}{(1 - \beta^2)^2 + (2\zeta\beta)^2} \right] \quad (3.39)$$

where

$$\beta = \frac{\omega}{\omega_1} \quad (3.40)$$

is the ratio of the excitation frequency to the natural frequency.

Amplitude and Phase

Using the compound angle formula $\cos(\theta_1 - \theta_2) = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2$, the steady-state response in (3.36) can be written explicitly in terms of amplitude and phase:

$$\begin{aligned} x_s(t) &= a \cos \omega t + b \sin \omega t \\ &= \underbrace{\sqrt{a^2 + b^2}}_{X_s} \left(\underbrace{\frac{a}{\sqrt{a^2 + b^2}}}_{\cos \phi} \cos \omega t + \underbrace{\frac{b}{\sqrt{a^2 + b^2}}}_{\sin \phi} \sin \omega t \right) \\ &= \underbrace{X_s}_{\text{amplitude}} \cos \left(\underbrace{\omega}_{\text{frequency}} t - \underbrace{\phi}_{\text{phase}} \right) \end{aligned} \quad (3.41)$$

where $\tan \phi = b/a$. Substituting (3.39) into $X_s = \sqrt{a^2 + b^2}$ and $\tan \phi = b/a$ gives

$$X_s = \underbrace{\frac{p_0}{\omega_1^2}}_{\text{static response}} \times \underbrace{\frac{1}{\sqrt{(1 - \beta^2)^2 + (2\zeta\beta)^2}}}_{\substack{\text{dynamic amplification} \\ A(\beta)}} \quad \tan \phi = \frac{2\zeta\beta}{1 - \beta^2} \quad (0 \leq \phi < \pi) \quad (3.42)$$

The term p_0/ω_1^2 is the static displacement if p_0 were applied in a static manner. The constraint $0 \leq \phi < \pi$ arises from the consideration that $\sin \phi = 2\zeta\beta A(\beta) \geq 0$ and so ϕ must lie in the first or second quadrant, depending on whether $\cos \phi = (1 - \beta^2)A(\beta)$ is positive or negative, i.e., $\beta < 1$ or $\beta > 1$.

Sinusoidal Excitation

The steady-state response when the excitation is $p(t) = p_0 \sin \omega t$ has the same form as (3.41):

$$x_s(t) = X_s \sin(\omega t - \phi) \quad (3.43)$$

where X_s and ϕ are still given by (3.42). This can be reasoned by noting that $p_0 \sin \omega t = p_0 \cos(\omega t - \pi/2)$ and so has the steady-state response $X_s \cos(\omega t - \pi/2 - \phi)$, which is equal to $X_s \sin(\omega t - \phi)$.

Transient Response

The constants C_1 and C_2 in the transient response in (3.35) are determined from initial conditions. Setting $x(0) = u$ and $\dot{x}(0) = v$, and noting $g_1(0) = 1$, $\dot{g}_1(0) = 0$, $g_2(0) = 0$ and $\dot{g}_2(0) = 1$ gives

$$\begin{aligned} C_1 + x_s(0) &= u \\ C_2 + \dot{x}_s(0) &= v \end{aligned} \quad \Rightarrow \quad \begin{aligned} C_1 &= u - x_s(0) \\ C_2 &= v - \dot{x}_s(0) \end{aligned} \quad (3.44)$$

Thus, C_1 makes up for the difference in initial displacement between $x(t)$ and $x_s(t)$; C_2 makes up for the difference in initial velocity.

3.1.6 Simplifying Algebra with Complex Number

As in Fourier analysis (Sect. 2.1.1), complex number provides a convenient means for obtaining the steady-state response. Suppose we want to determine the steady-state response of $x(t)$ and $y(t)$, driven respectively by cosine and sine excitation:

$$\begin{aligned} \ddot{x} + 2\zeta\omega_1\dot{x} + \omega_1^2x &= p_0 \cos \omega t \\ \ddot{y} + 2\zeta\omega_1\dot{y} + \omega_1^2y &= p_0 \sin \omega t \end{aligned} \quad (3.45)$$

Instead of repeating the same procedure twice, we can determine the steady-state response of

$$z(t) = x(t) + \mathbf{i}y(t) \quad (3.46)$$

After that, $x(t)$ and $y(t)$ can be recovered by taking the real and imaginary part of $z(t)$, respectively. It turns out that $z(t)$ can be obtained with simpler algebra, as seen in the following.

Multiplying the second equation in (3.45) by $\mathbf{i}$ and adding with the first, we obtain the equation for $z(t)$:

$$\begin{aligned}
 \underbrace{(\ddot{x} + \mathbf{i}\ddot{y})}_{\ddot{z}} + 2\zeta\omega_1 \underbrace{(\dot{x} + \mathbf{i}\dot{y})}_{\dot{z}} + \omega_1^2 \underbrace{(x + \mathbf{i}y)}_z &= p_0 \underbrace{(\cos \omega t + \mathbf{i} \sin \omega t)}_{e^{\mathbf{i}\omega t}} \\
 \Rightarrow \ddot{z} + 2\zeta\omega_1 \dot{z} + \omega_1^2 z &= p_0 e^{\mathbf{i}\omega t}
 \end{aligned} \tag{3.47}$$

Assume steady-state solution of the form

$$z_s(t) = Z_s e^{\mathbf{i}\omega t} \tag{3.48}$$

for some constant Z_s to be determined. Then $\dot{z}_s = \mathbf{i}\omega Z_s e^{\mathbf{i}\omega t}$ and $\ddot{z}_s = -\omega^2 Z_s e^{\mathbf{i}\omega t}$. Substituting into (3.47) and collecting terms,

$$(\omega_1^2 - \omega^2 + 2\zeta\omega_1\omega\mathbf{i})Z_s e^{\mathbf{i}\omega t} = p_0 e^{\mathbf{i}\omega t} \tag{3.49}$$

This implies that

$$Z_s = \frac{p_0}{\omega_1^2 - \omega^2 + 2\zeta\omega_1\omega\mathbf{i}} = \underbrace{\frac{p_0}{\omega_1^2}}_{\text{static}} \times \underbrace{\frac{1}{1 - \beta^2 + 2\zeta\beta\mathbf{i}}}_{\text{dynamic}} \tag{3.50}$$

where $\beta = \omega/\omega_1$ as defined in (3.40). Substituting Z_s into (3.48) and taking the real/imaginary part gives the same steady-state response for cosine/sine excitations, but now with simpler algebra.

3.1.7 Dynamic Amplification

According to (3.42), the amplitude of the steady-state displacement to harmonic excitation is equal to the static displacement p_0/ω_1^2 multiplied by the ‘dynamic amplification factor’:

$$A(\beta) = \frac{1}{\sqrt{(1 - \beta^2)^2 + (2\zeta\beta)^2}} \quad \beta = \frac{\omega}{\omega_1} \tag{3.51}$$

Figure 3.3 shows $A(\beta)$ versus β for different values of ζ . It has a peak value of

$$A_{\max} = \frac{1}{2\zeta\sqrt{1 - \zeta^2}} \quad \text{at} \quad \beta = \sqrt{1 - 2\zeta^2} \tag{3.52}$$

For small $\zeta \ll 1$, $A_{\max} \sim 1/2\zeta$ at $\beta \sim 1$, i.e., $\omega \sim \omega_1$. This is commonly referred as ‘resonance’, which is one key aspect of dynamics. For $\zeta = 1\%$, $A_{\max} \sim 50$, i.e., a

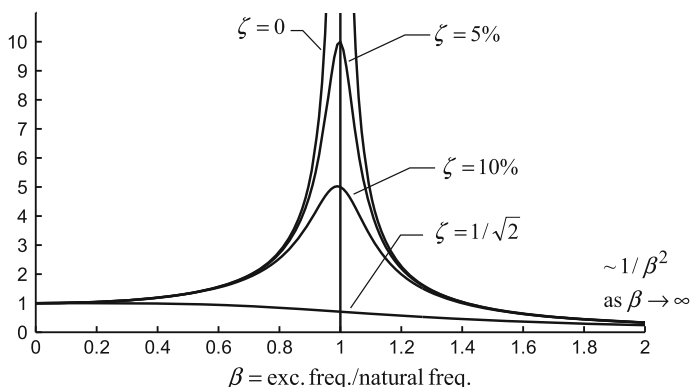


Fig. 3.3 Dynamic amplification factor $A(\beta)$

harmonic load applied at resonance can generate 50 times the displacement than if it is applied in a static manner.

As $\beta \rightarrow 0$, $A(\beta) \rightarrow 1$ regardless of the value of ζ . That is, when the excitation frequency is low (i.e., slowly varying) compared to the natural frequency, there is no amplification; the response is ‘pseudo-static’. At the other extreme, $A(\beta) \sim 1/\beta^2$ as $\beta \rightarrow \infty$. In this case the excitation is changing so fast that the structure cannot ‘catch up’. The faster the change, the smaller the response. Finally, $A(\beta)$ decreases monotonically with β for $\zeta \geq 1/\sqrt{2}$, i.e., no resonance when the damping is too high (yet still under damped).

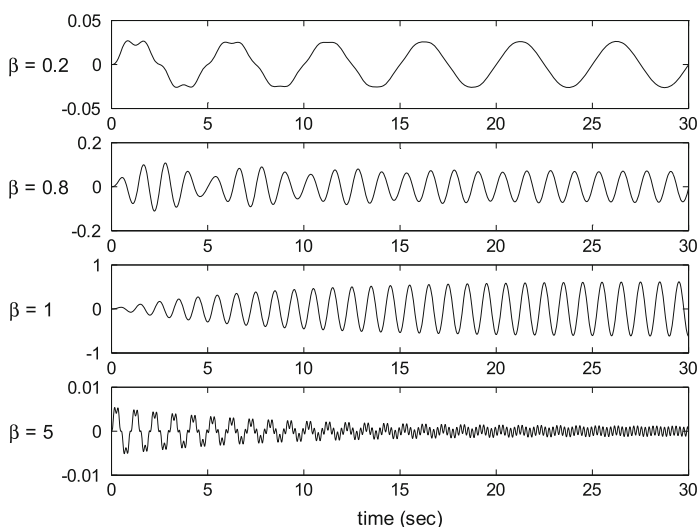


Fig. 3.4 SDOF response to sinusoidal excitations; β = excitation frequency/natural frequency

Example 3.1 (Transient versus steady-state response) Consider a SDOF structure with natural frequency $\omega_1 = 2\pi$ (rad/s), i.e., 1 Hz, and damping ratio $\zeta = 2\%$. Figure 3.4 shows the response $x(t)$ to sinusoidal excitation $p(t) = \sin \omega t$ for different frequencies indicated by the frequency ratio $\beta = \omega/\omega_1$. Generally, the response is dominated by the steady-state component for time beyond 15 s (say). For $\beta = 0.2$, the excitation varies so slowly that the response is able to catch up closely, except for the small transience in the beginning. At the other extreme, when $\beta = 5$ the response cannot catch up. The small fast varying ripples are due to the excitation while the large slow variations are due to transience. At resonance ($\beta = 1$), the response amplitude grows towards the steady-state value. ■

3.1.8 Half-Power Bandwidth Method

The resonance nature of the dynamic amplification factor provides an intuitive way for estimating the natural frequency and damping ratio. Suppose we observe the steady-state displacement amplitude of the structure when it is subjected to harmonic excitations of different frequencies. If the excitation amplitude does not change with frequency, a plot of the displacement amplitude with frequency will be proportional to the dynamic amplification factor. The shape of this plot depends on the natural frequency and damping ratio, which allows one to estimate them.

For lightly damped structures ($\zeta \ll 1$), one conceptually simple way is to make use of the ‘half-power frequencies’, which are the two points on either side of the natural frequency such that the dynamic amplification is equal to $1/\sqrt{2}$ of the peak value; it is $1/\sqrt{2}$ instead of $1/2$ because power is proportional to the square of amplitude. Using (3.51) and (3.52), and solving $A(\beta) = A_{\max}/\sqrt{2}$ for β gives the frequency ratios at the half-power points:

$$\beta = \sqrt{1 - 2\zeta^2 \pm 2\zeta\sqrt{1 - \zeta^2}} \quad (3.53)$$

For small damping ratio $\zeta \ll 1$,

$$\beta \sim 1 \pm \zeta \quad (3.54)$$

That is, the half-power frequencies are located symmetrically on either side of the natural frequency, differing by a fraction of ζ . These are illustrated in Fig. 3.5.

In implementation, the half-power frequencies can be located using the plot of any quantity that is proportional to the dynamic amplification factor. Based on (3.54), the natural frequency can be estimated as the *average* of the

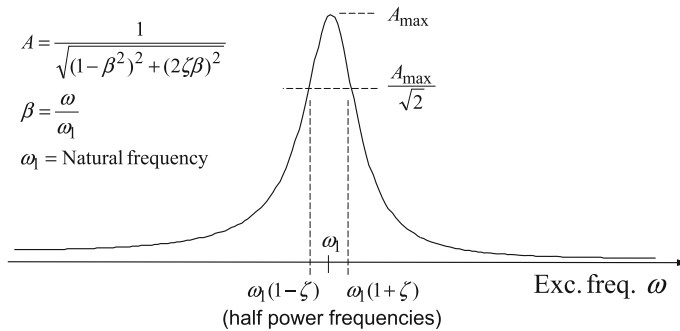


Fig. 3.5 Dynamic amplification factor and half-power frequencies for small damping

half-power frequencies. The *ratio of the difference to the sum* of the half-power frequencies gives an estimate for the damping ratio. This is known as the ‘half-power bandwidth method’.

Generalization

Equation (3.54) is derived based on the dynamic amplification factor in (3.51), which is associated with the displacement amplitude. When applying the half-power bandwidth method with velocity or acceleration amplitudes, they should be first converted to displacement amplitudes. Dividing the velocity amplitude (for example) by the excitation frequency (in rad/s) gives the displacement amplitude. Modification is also needed to remove the effect from any frequency-dependence of the excitation amplitude. For example, if the excitation amplitude is proportional to the square of frequency (e.g., in eccentric motors), the displacement amplitude should be divided by the square of frequency.

Example 3.2 (Shaker test on floor slab) Consider a floor slab subjected to a harmonic vertical force near the midspan generated by an electro-magnetic shaker, as shown in Fig. 3.6a. The magnitude of vertical force exerted by the shaker on the floor is equal to the product of the (moving) shaker mass and its acceleration. The latter is measured by an accelerometer (circled by dashed line in the figure). The vertical acceleration of the structure near the same location is measured by another accelerometer.

With more than one DOF, the floor slab has more than one natural frequency. The complete theoretical relationship between the acceleration amplitude of the shaker mass and the structure requires multi-DOF dynamics concepts to obtain (see Sect. 3.2 later). It can be reasoned that when the excitation frequency is near a particular natural frequency the structural acceleration amplitude (A_x) is still approximately related by the dynamic identification factor to the shaker force amplitude, and hence shaker mass acceleration amplitude (A_p). Thus, we still have, for the dynamic amplification factor in (3.51),

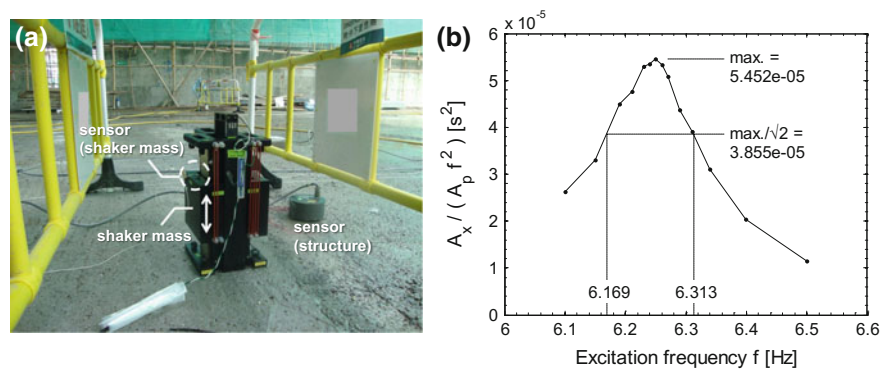


Fig. 3.6 Floor slab subjected to harmonic vertical force by a shaker near the midspan. **a** Setup; **b** dynamic amplification and half-power frequencies

$$A \propto \frac{A_x}{A_p f^2} \tag{3.55}$$

where $f = \omega/2\pi$ is the excitation frequency in Hz; the factor f^2 in the denominator accounts for acceleration (rather than displacement) measurement. Note that A , A_x and A_p depend on f but this has been omitted to simplify notation. The natural frequency and damping ratio can be obtained by half-power bandwidth method using a plot of the RHS of (3.55) for different values of f .

Table 3.3 shows the measured acceleration amplitudes of the shaker mass and the structure for different excitation frequencies. The frequencies are not evenly

Table 3.3 Acceleration amplitudes of shaker mass and structure at different frequencies

No.	Excitation frequency f (Hz)	Acc. amplitude shaker mass A_p (milli-g)	Acc. amplitude structure A_x (milli-g)	$A_x/A_p f^2$ ($\times 10^{-6} s^2$)
1	6.10	1171	1.146	26.30
2	6.15	1170	1.458	32.94
3	6.19	1170	2.016	44.96
4	6.21	1170	2.151	47.67
5	6.23	1170	2.409	53.03
6	6.24	1171	2.439	53.50
7	6.25	1174	2.500	54.52
8	6.26	1173	2.455	53.39
9	6.27	1173	2.347	50.91
10	6.29	1172	2.029	43.75
11	6.31	1171	1.825	39.14
12	6.34	1170	1.459	31.03
13	6.40	1169	0.974	20.35
14	6.50	1167	0.563	11.42

spaced as they were adjusted manually on a signal generator during the test to populate around the natural frequency. Plotting the last column versus the excitation frequency gives Fig. 3.6b. Dividing the peak value ($5.452\text{e-}5$) by $\sqrt{2}$ gives the half-power value ($3.855\text{e-}5$), from which the half-power frequencies can be located as 6.169 and 6.313 Hz. Using the half-power bandwidth method, the natural frequency and damping ratio are estimated as

$$\begin{aligned}\text{Natural frequency} &\approx \frac{6.313 + 6.169}{2} = 6.24 \text{ Hz} \\ \text{Damping ratio} &\approx \frac{6.313 - 6.169}{6.313 + 6.169} = 1.15\%\end{aligned}$$

See Example 3.7 for the complete relationship between A_x and A_p , which also allows one to estimate the effective (modal) mass of the structure associated with the natural frequency. Modal testing with harmonic loads is discussed in more detail in Sect. 3.4 later. ■

3.1.9 Principle of Superposition

To prepare for the study on periodic and arbitrary excitations in Sects. 3.1.10 to 3.1.12, we detour to discuss the ‘principle of superposition’. It generally holds for linear elastic structures, or mathematically, linear equations, be it algebraic or differential in nature. An equation $L(x) = 0$ is ‘linear’ if $L(ax + by) = aL(x) + bL(y)$ for any scalars a and b . In our case, the equation of motion and initial conditions are linear functions of x and its derivatives, and so the problem is linear. The principle of superposition then holds, which says that the response to the sum of multiple effects is simply the sum of the responses to the individual effects. Specifically, let the response (with zero initial conditions) to excitation $p_i(t)$ ($i = 1, \dots, q$) be $x_i(t)$, satisfying

$$\ddot{x}_i(t) + 2\zeta\omega_1\dot{x}_i(t) + \omega_1^2x_i(t) = p_i(t) \quad x_i(0) = 0 \quad \dot{x}_i(0) = 0 \quad (3.56)$$

Also, let $x_0(t)$ be the free vibration response satisfying the initial conditions, i.e.,

$$\ddot{x}_0(t) + 2\zeta\omega_1\dot{x}_0(t) + \omega_1^2x_0(t) = 0 \quad x_0(0) = u \quad \dot{x}_0(0) = v \quad (3.57)$$

When the structure is subjected to a linear combination of the excitations, say, $p(t) = \sum_{i=1}^q c_i p_i(t)$ for some constants $\{c_i\}_{i=1}^q$, and under initial conditions $x(0) = u$ and $\dot{x}(0) = v$, the response is simply the same linear combination of the individual responses, i.e.,

$$\left. \begin{aligned} \ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) &= \sum_{i=1}^q c_i p_i(t) \\ x(0) = u, \dot{x}(0) &= v \end{aligned} \right\} \Rightarrow x(t) = x_0(t) + \sum_{i=1}^q c_i x_i(t) \quad (3.58)$$

This can be shown by multiplying (3.56) with c_i , summing over $i = 1, \dots, q$, and then adding with (3.57):

$$\begin{aligned} (\ddot{x}_0 + 2\zeta\omega_1\dot{x}_0 + \omega_1^2x_0) + \sum_{i=1}^q c_i (\ddot{x}_i + 2\zeta\omega_1\dot{x}_i + \omega_1^2x_i) &= \sum_{i=1}^q c_i p_i \\ \Rightarrow \underbrace{\left(\ddot{x}_0 + \sum_{i=1}^q c_i \ddot{x}_i \right)}_{\ddot{x}} + 2\zeta\omega_1 \underbrace{\left(\dot{x}_0 + \sum_{i=1}^q c_i \dot{x}_i \right)}_{\dot{x}} + \omega_1^2 \underbrace{\left(x_0 + \sum_{i=1}^q c_i x_i \right)}_x &= \underbrace{\left(\sum_{i=1}^q c_i p_i \right)}_p \\ \Rightarrow \ddot{x} + 2\zeta\omega_1\dot{x} + \omega_1^2x &= p \end{aligned} \quad (3.59)$$

That is, $x(t)$ satisfies the equation of motion. It also satisfies the initial conditions because

$$x(0) = \underbrace{x_0(0)}_u + \sum_{i=1}^q c_i \underbrace{x_i(0)}_0 = u \quad \dot{x}(0) = \underbrace{\dot{x}_0(0)}_v + \sum_{i=1}^q c_i \underbrace{\dot{x}_i(0)}_0 = v \quad (3.60)$$

The principle of superposition allows one to assess the response due to a potentially complicated excitation by looking at the effects of its constituents. For example, the response due to a periodic loading (Sect. 3.1.10) can be obtained by simply summing the responses due to the individual harmonics; the response due to an arbitrary excitation (Sect. 3.1.12) can be obtained as the sum of contributions from individual impulses at different time instants in the past.

3.1.10 Periodic Excitation

When the excitation $p(t)$ is periodic, it can be written as a Fourier series (FS, Sect. 2.1):

$$p(t) = a_0 + \sum_{r=1}^{\infty} a_r \cos \bar{\omega}_r t + \sum_{r=1}^{\infty} b_r \sin \bar{\omega}_r t \quad \bar{\omega}_r = \frac{2\pi r}{T} \quad (3.61)$$

where T (s) is the period, i.e., the smallest value such that $p(t+T) = p(t)$ for any t ; $\bar{\omega}_r = 2\pi r/T$ (rad/s) is the FS frequency of the r th harmonic;

$$a_0 = \frac{1}{T} \int_0^T p(t) dt \quad a_r = \frac{2}{T} \int_0^T p(t) \cos \bar{\omega}_r t dt \quad b_r = \frac{2}{T} \int_0^T p(t) \sin \bar{\omega}_r t dt \quad r \geq 1 \quad (3.62)$$

are the real FS coefficients. Applying the principle of superposition, the response $x(t)$ is the sum of those due to different harmonics and initial conditions. The response due to each harmonic comprises a transient and a steady-state component. Combining the transient components, one can write

$$x(t) = \underbrace{e^{-\zeta \omega_d t} (C_1 \cos \omega_d t + C_2 \sin \omega_d t)}_{\substack{\text{transient} \\ \rightarrow 0 \text{ as } t \rightarrow \infty}} + \underbrace{x_s(t)}_{\text{steady-state}} \quad (3.63)$$

where C_1 and C_2 are constants (often not of concern) which can be determined from initial conditions; and (see (3.41))

$$x_s(t) = \underbrace{\frac{a_0}{\omega_1^2}}_{\text{static}} + \sum_{r=1}^{\infty} \underbrace{\frac{a_r}{\omega_1^2}}_{\text{static}} \underbrace{A(\beta_r)}_{\substack{\text{dyn.} \\ \text{amp.}}} \underbrace{\cos(\bar{\omega}_r t - \phi_r)}_{\substack{\text{lagged-time} \\ \text{harmonic}}} + \sum_{r=1}^{\infty} \underbrace{\frac{b_r}{\omega_1^2}}_{\text{static}} \underbrace{A(\beta_r)}_{\substack{\text{dyn.} \\ \text{amp.}}} \underbrace{\sin(\bar{\omega}_r t - \phi_r)}_{\substack{\text{lagged-time} \\ \text{harmonic}}} \quad (3.64)$$

$$A(\beta_r) = \frac{1}{\sqrt{(1 - \beta_r^2)^2 + (2\zeta\beta_r)^2}} \quad \beta_r = \frac{\bar{\omega}_r}{\omega_1} \quad \tan \phi_r = \frac{2\zeta\beta_r}{1 - \beta_r^2} \quad (3.65)$$

Alternatively, the FS of excitation can be written in complex exponential form:

$$p(t) = \sum_{r=-\infty}^{\infty} c_r e^{i\bar{\omega}_r t} \quad (3.66)$$

$$c_r = \frac{1}{T} \int_0^T p(t) e^{-i\bar{\omega}_r t} dt \quad (3.67)$$

The steady-state response is then given by (see (3.48))

$$x_s(t) = \sum_{r=-\infty}^{\infty} \underbrace{\frac{c_r}{\omega_1^2}}_{\text{static}} \underbrace{\frac{1}{1 - \beta_r^2 + 2\zeta\beta_r i}}_{\text{dynamic}} \underbrace{e^{i\bar{\omega}_r t}}_{\substack{\text{excitation} \\ \text{harmonic}}} \quad \beta_r = \frac{\bar{\omega}_r}{\omega_1} \quad (3.68)$$

Although the individual terms in the sum of (3.66) are generally complex-valued, $p(t)$ is still real-valued because the r th and $-r$ th term are complex conjugate of each other. The same is also true for $x_s(t)$ in (3.68).

3.1.11 Ideal Impulse Excitation

An ‘impulsive excitation’ generally refers to one that lasts for a very short time compared to the natural period of the structure. An ‘ideal impulse’ is a mathematical idealization whose duration is arbitrarily small. Consider a unit rectangular impulse, equal to $1/\varepsilon$ for a duration of ε . As $\varepsilon \rightarrow 0$, the height of the impulse grows to infinity but the area under the impulse is still $\varepsilon \times 1/\varepsilon = 1$ (hence the name ‘unit’). The ideal unit impulse can be considered as the limit of the unit rectangular impulse as $\varepsilon \rightarrow 0$, although this representation is not unique. It does not exist in reality but it can provide a good convenient approximation if the actual impulse duration is very short compared to the natural period of the structure.

Consider a SDOF structure initially at rest and subjected to an ideal impulse at $t = 0^+$:

$$\ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) = \delta(t - 0^+) \quad x(0) = \dot{x}(0) = 0 \quad (3.69)$$

Here, 0^+ denotes symbolically the time ‘slightly’ (in a limiting sense) to the right of the origin $t = 0$; $\delta(t - 0^+)$ denotes the Dirac Delta function (Sect. 2.2.3) slightly shifted so that $\int_0^\varepsilon \delta(t - 0^+)dt = 1$ (instead of 1/2) for any $\varepsilon > 0$.

The response to the ideal unit impulse with zero initial conditions ($x(0) = \dot{x}(0) = 0$) is simply

$$x(t) = g_2(t) = \frac{e^{-\zeta\omega_1 t}}{\omega_d} \sin \omega_d t \quad \omega_d = \omega_1 \sqrt{1 - \zeta^2} \quad (3.70)$$

where $g_2(t)$ is the free vibration response with zero initial displacement and unit initial velocity, as given in Table 3.2. This result can be reasoned as follow. Integrating (3.69) from 0 to $\varepsilon > 0$,

$$\dot{x}(\varepsilon) + 2\zeta\omega_1x(\varepsilon) + \omega_1^2 \int_0^\varepsilon x(t)dt = \int_0^\varepsilon \delta(t - 0^+)dt = 1 \quad (3.71)$$

As $\varepsilon \rightarrow 0$, $x(\varepsilon) \rightarrow x(0) = 0$ and so is $\int_0^\varepsilon x(t)dt \rightarrow 0$. The equation then implies $\dot{x}(0^+) = \lim_{\varepsilon \rightarrow 0} \dot{x}(\varepsilon) = 1$. Since the excitation is zero after the impulse, the response thereafter is a free vibration with zero initial displacement and unit initial velocity, i.e., $g_2(t)$.

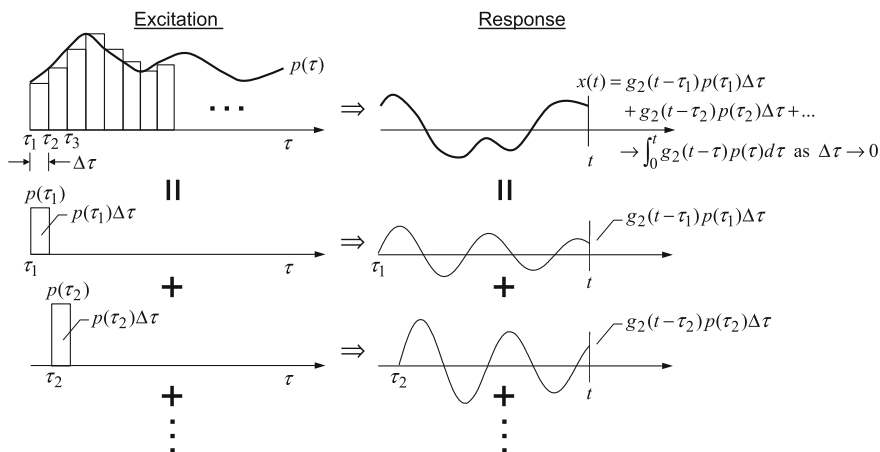


Fig. 3.7 Illustration of Duhamel's integral

3.1.12 Arbitrary Excitation

For a general excitation $p(t)$, the response with zero initial conditions is given by

$$x(t) = \int_0^t g_2(t-\tau)p(\tau)d\tau \quad (3.72)$$

This is a 'convolution integral', often called the "Duhamel's integral" in structural dynamics. It can be reasoned as the sum of responses due to the impulses at different time instants up to t , as illustrated in Fig. 3.7.

Clearly $x(t)$ depends on the excitation from time 0 up to t . Consider approximating the excitation on $[0, t)$ by a sequence of rectangular impulses of duration $\Delta\tau$ at time instants $\tau_1 = 0$, $\tau_2 = \Delta\tau$, $\tau_3 = 2\Delta\tau$, ..., up to time t . The impulse at time τ has area $p(\tau)\Delta\tau$. At time t , which is $t - \tau$ after the impulse, the induced response is approximately $g_2(t - \tau) \times p(\tau)\Delta\tau$. Summing the contributions from the impulses occurring from time 0 to t gives

$$x(t) \approx \sum_{0 \leq \tau < t} g_2(t-\tau)p(\tau)\Delta\tau \quad (3.73)$$

As $\Delta\tau \rightarrow 0$, the sum becomes an integral and the approximation becomes exact:

$$x(t) = \lim_{\Delta\tau \rightarrow 0} \sum_{0 \leq \tau < t} g_2(t-\tau)p(\tau)\Delta\tau = \int_0^t g_2(t-\tau)p(\tau)d\tau \quad (3.74)$$

In the general case when the structure starts from non-zero initial conditions, the response is given by

$$x(t) = \underbrace{ug_1(t) + vg_2(t)}_{\text{initial condition effect}} + \underbrace{\int_0^t g_2(t-\tau)p(\tau)d\tau}_{\text{excitation effect}} \quad (x(0) = u, \dot{x}(0) = v) \quad (3.75)$$

Check that it satisfies $x(0) = u$ and $\dot{x}(0) = v$. It also satisfies $\ddot{x} + 2\zeta\omega_1\dot{x} + \omega_1^2x = p(t)$ because $ug_1(t) + vg_2(t)$ gives a zero LHS and the Duhamel's integral gives a LHS equal to $p(t)$ (see proof below).

Proof of (3.72) (Duhamel's integral satisfying the equation of motion)

Here we show that $x(t) = \int_0^t g_2(t-\tau)p(\tau)d\tau$ in (3.72) satisfies $\ddot{x} + 2\zeta\omega_1\dot{x} + \omega_1^2x = p(t)$. To obtain $\dot{x}$ and $\ddot{x}$, use the 'Leibniz rule of differentiation' under integral sign, i.e., for functions $w(\tau, t)$, $a(t)$ and $b(t)$,

$$\frac{d}{dt} \left[\int_{a(t)}^{b(t)} w(\tau, t) d\tau \right] = w(b(t), t) \dot{b}(t) - w(a(t), t) \dot{a}(t) + \int_{a(t)}^{b(t)} \frac{\partial w}{\partial t} d\tau \quad (3.76)$$

Applying this rule,

$$\begin{aligned} \dot{x}(t) &= \underbrace{g_2(t-t)}_0 p(t) + \int_0^t \dot{g}_2(t-\tau)p(\tau)d\tau = \int_0^t \dot{g}_2(t-\tau)p(\tau)d\tau \\ \ddot{x}(t) &= \underbrace{\dot{g}_2(t-t)}_1 p(t) + \int_0^t \ddot{g}_2(t-\tau)p(\tau)d\tau = p(t) + \int_0^t \ddot{g}_2(t-\tau)p(\tau)d\tau \end{aligned} \quad (3.77)$$

Using these expressions,

$$\begin{aligned} \ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) &= p(t) + \int_0^t \ddot{g}_2(t-\tau)p(\tau)d\tau + 2\zeta\omega_1 \int_0^t \dot{g}_2(t-\tau)p(\tau)d\tau + \omega_1^2 \int_0^t g_2(t-\tau)p(\tau)d\tau \\ &= p(t) + \int_0^t \underbrace{[\ddot{g}_2(t-\tau) + 2\zeta\omega_1\dot{g}_2(t-\tau) + \omega_1^2g_2(t-\tau)]}_{=0 \text{ since } g_2 \text{ satisfies free vibration equation}} p(\tau)d\tau = p(t) \end{aligned} \quad (3.78)$$

■

3.1.13 Summary of SDOF Response

The SDOF response $x(t)$ satisfying

$$\ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) = p(t) \quad x(0) = u, \dot{x}(0) = v \quad (3.79)$$

is summarized in Table 3.4. Relevant sections are indicated.

3.2 MDOF Dynamics

The dynamics of a structure is generally governed by multiple (possibly infinitely many) DOFs. For instructional purpose, consider the two-DOF structure in Fig. 3.8a.

The DOFs here comprise the displacements $x_1(t)$ and $x_2(t)$ of the two masses. Two equations of motion can be derived by applying Newton's second law on each mass. As shown in Fig. 3.8b, at a particular time t , the mass m_1 is displaced to the right by $x_1(t)$ and m_2 by $x_2(t)$. In addition to the applied force $F_1(t)$, m_1 is subjected to a spring force of $k_1x_1(t)$ to the left and another $k_2[x_2(t) - x_1(t)]$ to the right. Applying Newton's second law to m_1 gives

$$m_1\ddot{x}_1(t) = -k_1x_1(t) + k_2[x_2(t) - x_1(t)] + F_1(t) \quad (3.80)$$

For m_2 , in addition to $F_2(t)$, it is subjected to a spring force of $k_2[x_2(t) - x_1(t)]$ to the left. Newton's second law gives

$$m_2\ddot{x}_2(t) = -k_2[x_2(t) - x_1(t)] + F_2(t) \quad (3.81)$$

Writing the unknowns on the LHS and applied forces on the RHS, (3.80) and (3.81) can be written as

$$\begin{array}{rcccl} m_1\ddot{x}_1(t) & + & (k_1 + k_2)x_1(t) & - & k_2x_2(t) & = & F_1(t) \\ m_2\ddot{x}_2(t) & - & k_2x_1(t) & + & k_2x_2(t) & = & F_2(t) \end{array} \quad (3.82)$$

These can be further assembled in matrix form:

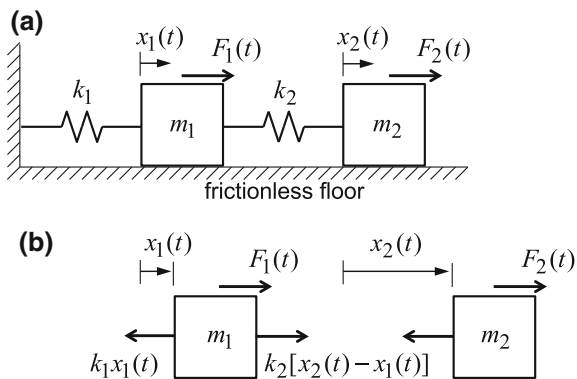
$$\underbrace{\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}}_{\substack{\mathbf{M} \\ \text{mass} \\ \text{matrix}}} \underbrace{\begin{bmatrix} \ddot{x}_1(t) \\ \ddot{x}_2(t) \end{bmatrix}}_{\substack{\ddot{\mathbf{x}}(t) \\ \text{acc.} \\ \text{vector}}} + \underbrace{\begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}}_{\substack{\mathbf{K} \\ \text{stiffness} \\ \text{matrix}}} \underbrace{\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}}_{\substack{\mathbf{x}(t) \\ \text{disp.} \\ \text{vector}}} = \underbrace{\begin{bmatrix} F_1(t) \\ F_2(t) \end{bmatrix}}_{\substack{\mathbf{F}(t) \\ \text{load} \\ \text{vector}}} \quad (3.83)$$

Table 3.4 Summary of SDOF response to different excitations ($0 < \zeta < 1$)

Excitation $p(t)$	Response $x(t)$
Free vibration (3.1.3) ($p(t) \equiv 0$)	$x(t) = u_g(t) + v g_2(t)$
Harmonic (3.1.5) $p_0 \cos \omega t$	$\underbrace{C_1 g_1(t) + C_2 g_2(t)}_{\text{transient} \rightarrow 0 \text{ as } t \rightarrow \infty} + \underbrace{\frac{p_0}{\omega_1^2} A(\beta) \cos(\omega t - \phi)}_{\text{steady-state } x_s(t)}$ $C_1 = u - x_s(0), C_2 = v - \dot{x}_s(0)$ $A(\beta) = [(1 - \beta^2)^2 + (2\zeta\beta)^2]^{-1/2}, \beta = \omega/\omega_1, \tan \phi = 2\zeta\beta/(1 - \beta^2), 0 \leq \phi < \pi$
Periodic (real Fourier series) (3.1.10) $a_0 + \sum_{r=1}^{\infty} (a_r \cos \bar{\omega}_r t + b_r \sin \bar{\omega}_r t)$ $\bar{\omega}_r = 2\pi r/T, T = \text{period}$	$\underbrace{C_1 g_1(t) + C_2 g_2(t)}_{\text{transient} \rightarrow 0 \text{ as } t \rightarrow \infty} + \underbrace{\sum_{r=1}^{\infty} \frac{A(\beta_r)}{\omega_1^2} [a_r \cos(\bar{\omega}_r t - \phi_r) + b_r \sin(\bar{\omega}_r t - \phi_r)]}_{\text{steady-state } x_s(t)}$ $C_1 = u - x_s(0), C_2 = v - \dot{x}_s(0)$ $A(\beta_r) = [(1 - \beta_r^2)^2 + (2\zeta\beta_r)^2]^{-1/2}, \beta_r = \bar{\omega}_r/\omega_1, \tan \phi_r = 2\zeta\beta_r/(1 - \beta_r^2), 0 \leq \phi_r < \pi$
Periodic (complex Fourier series) (3.1.10) $\sum_{r=-\infty}^{\infty} c_r e^{i\bar{\omega}_r t}$ $\bar{\omega}_r = 2\pi r/T, T = \text{period}$	$\underbrace{C_1 g_1(t) + C_2 g_2(t)}_{\text{transient} \rightarrow 0 \text{ as } t \rightarrow \infty} + \underbrace{\sum_{r=-\infty}^{\infty} \frac{h(\beta_r)}{\omega_1^2} c_r e^{i\bar{\omega}_r t}}_{\text{steady-state } x_s(t)}$ $C_1 = u - x_s(0), C_2 = v - \dot{x}_s(0)$ $h(\beta_r) = [1 - \beta_r^2 + 2\zeta\beta_r i]^{-1}, \beta_r = \bar{\omega}_r/\omega_1$
Arbitrary (3.1.12)	$\underbrace{u g_1(t) + v g_2(t)}_{\text{initial condition effect}} + \underbrace{\int_0^t g_2(t - \tau) p(\tau) d\tau}_{\text{excitation effect}}$

 $\omega_1 = \text{natural frequency (rad/s)}, \zeta = \text{damping ratio}, \omega_d = \omega_1 \sqrt{1 - \zeta^2} = \text{damped natural frequency}$
 $u = \text{initial displacement}, v = \text{initial velocity}, g_1(t) = e^{-\zeta\omega_1 t} (\cos \omega_d t + \frac{\zeta\omega_1}{\omega_d} \sin \omega_d t), g_2(t) = \frac{e^{-\zeta\omega_1 t}}{\omega_d} \sin \omega_d t$

Fig. 3.8 Two-DOF structure;
a schematic diagram; **b** free
 body diagram at time t



This is a second order vector ODE in $\mathbf{x}(t) = [x_1(t), x_2(t)]^T$. It does not have any damping. For general problems where there are n DOFs and damping is modeled by a viscous term, the equation of motion reads

$$\underset{n \times n}{\mathbf{M}} \underset{n \times 1}{\ddot{\mathbf{x}}(t)} + \underset{n \times n}{\mathbf{C}} \underset{n \times 1}{\dot{\mathbf{x}}(t)} + \underset{n \times n}{\mathbf{K}} \underset{n \times 1}{\mathbf{x}(t)} = \underset{n \times 1}{\mathbf{F}(t)} \quad (3.84)$$

where $\mathbf{C}$ is called the ‘damping matrix’. Finite element method (Hughes 1987; Bathe 1982) is a conventional tool for constructing $\mathbf{M}$, $\mathbf{K}$ and $\mathbf{F}$ from physical specifications and has been implemented in a user-friendly fashion in commercial software. In structural vibration applications, the damping matrix $\mathbf{C}$ is seldom constructed from first principles. Rather, damping is accounted empirically through ‘modal damping ratios’, in which case it may not even be necessary to construct $\mathbf{C}$ in order to determine the response; see Sect. 3.2.3 later.

Positive Definiteness and Energy

Typically, $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ are real symmetric and positive definite. That is, for $\mathbf{M}$, all entries are real-valued, $\mathbf{M}^T = \mathbf{M}$ (symmetric) and $\mathbf{u}^T \mathbf{M} \mathbf{u} > 0$ for any non-zero $n \times 1$ real vector $\mathbf{u}$ (positive definite). The same properties apply to $\mathbf{C}$ and $\mathbf{K}$. Positive definiteness of $\mathbf{M}$ reflects that the kinetic energy $\dot{\mathbf{x}}^T \mathbf{M} \dot{\mathbf{x}}/2$ is always positive for non-zero $\dot{\mathbf{x}}$. Of $\mathbf{C}$ it reflects positive energy dissipation rate $\dot{\mathbf{x}}^T \mathbf{C} \dot{\mathbf{x}} > 0$. Of $\mathbf{K}$ it reflects that the structure is stable, where displacement always involves positive strain energy $\mathbf{x}^T \mathbf{K} \mathbf{x}/2$.

The rate of change of energies from kinetics, dissipation and strain is in balance with the rate of work done (i.e., power) by the external force. This can be established by pre-multiplying (3.84) with $\dot{\mathbf{x}}^T$, and noting that $\dot{\mathbf{x}}^T \mathbf{M} \ddot{\mathbf{x}} = d(\dot{\mathbf{x}}^T \mathbf{M} \dot{\mathbf{x}}/2)/dt$ and $\dot{\mathbf{x}}^T \mathbf{K} \mathbf{x} = d(\dot{\mathbf{x}}^T \mathbf{K} \mathbf{x}/2)/dt$:

$$\underbrace{\frac{d}{dt} \left(\frac{1}{2} \dot{\mathbf{x}}^T \mathbf{M} \dot{\mathbf{x}} \right)}_{\text{kinetic energy}} + \underbrace{\dot{\mathbf{x}}^T \mathbf{C} \dot{\mathbf{x}}}_{\substack{\text{energy} \\ \text{dissipation} \\ \text{rate}}} + \underbrace{\frac{d}{dt} (\mathbf{x}^T \mathbf{K} \mathbf{x})}_{\substack{\text{strain} \\ \text{energy}}} = \underbrace{\dot{\mathbf{x}}^T \mathbf{F}}_{\substack{\text{external rate} \\ \text{of work done} \\ \text{i.e., power}}} \quad (3.85)$$

where the dependence on t has been omitted for simplicity.

3.2.1 Natural Frequencies and Mode Shapes

We first consider the undamped free vibration of a MDOF structure to illustrate the basic characteristics, i.e., the existence of more than one natural frequency, each associated with a spatial vibration pattern called ‘mode shape’. When there is no damping ($\mathbf{C} = \mathbf{0}$) and the applied force is absent ($\mathbf{F}(t) \equiv \mathbf{0}$), (3.84) reduces to

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) = \mathbf{0} \quad (3.86)$$

Assume solution of the form $\mathbf{x}(t) = \boldsymbol{\psi}\eta(t)$ where $\boldsymbol{\psi}$ ($n \times 1$) is a constant vector and $\eta(t)$ is a scalar function of t . This assumes that spatial and temporal variations are ‘separable’. It need not be true but we will see later what needs to be revised to arrive at the correct solution. Substituting $\mathbf{x}(t) = \boldsymbol{\psi}\eta(t)$ and $\ddot{\mathbf{x}}(t) = \boldsymbol{\psi}\ddot{\eta}(t)$ into (3.86) gives, after rearranging,

$$\underbrace{\mathbf{K}\boldsymbol{\psi}}_{n \times 1} = \underbrace{-\frac{\ddot{\eta}(t)}{\eta(t)}}_{\text{scalar}} \underbrace{\mathbf{M}\boldsymbol{\psi}}_{n \times 1} \quad (3.87)$$

This equation must hold for all t . Since $\mathbf{K}\boldsymbol{\psi}$ and $\mathbf{M}\boldsymbol{\psi}$ do not depend on t , $\ddot{\eta}(t)/\eta(t)$ must be a constant. This constant must be negative, which can be seen by pre-multiplying both sides of (3.87) with $\boldsymbol{\psi}^T$ and noting that $\mathbf{K}$ and $\mathbf{M}$ are positive definite:

$$\underbrace{\boldsymbol{\psi}^T \mathbf{K} \boldsymbol{\psi}}_{>0} = \underbrace{-\frac{\ddot{\eta}(t)}{\eta(t)}}_{\text{must be } >0} \underbrace{\boldsymbol{\psi}^T \mathbf{M} \boldsymbol{\psi}}_{>0} \quad (3.88)$$

Thus, let $\ddot{\eta}(t)/\eta(t) = -\omega^2$ where ω is a real positive number to be determined; the possibility of a negative value for ω can be eliminated as it gives the same solution as the one by the positive value. We now have two equations, one in the time domain and the other in the spatial domain:

$$\ddot{\eta}(t) + \omega^2 \eta(t) = 0 \quad (\text{time}) \quad (3.89)$$

$$\mathbf{K}\boldsymbol{\Psi} = \omega^2 \mathbf{M}\boldsymbol{\Psi} \quad (\text{spatial}) \quad (3.90)$$

The first equation in $\eta(t)$ describes the dynamics of a SDOF undamped free vibration with natural frequency ω (rad/s). The second corresponds to the ‘generalized eigenvalue problem’, which looks for the ‘eigenvalue’ $\lambda = \omega^2$ and ‘eigenvector’ $\boldsymbol{\Psi} (\neq \mathbf{0})$ so that the equation holds. The *square root* of the eigenvalue gives the natural frequency. The eigenvector gives the ‘mode shape’. To supplement, the ‘standard eigenvalue problem’ corresponds to the case when $\mathbf{M}$ is the identity matrix. See Sect. C.2.4 for more details.

A discussion of the eigenvalue problem is postponed until the next section. It can be shown that there are n different possible solutions, $\{\omega_i, \boldsymbol{\Psi}_i\}_{i=1}^n$, and correspondingly n ‘modal responses’ $\{\eta_i(t)\}_{i=1}^n$. The response $\mathbf{x}(t)$ is then given by

$$\mathbf{x}(t) = \sum_{i=1}^n \boldsymbol{\Psi}_i \eta_i(t) \quad (3.91)$$

where each $\eta_i(t)$ satisfies (3.89) with $\omega = \omega_i$:

$$\ddot{\eta}_i(t) + \omega_i^2 \eta_i(t) = 0 \quad (3.92)$$

The solution to this SDOF equation is obtained by applying (3.10):

$$\eta_i(t) = \eta_i(0) \cos \omega_i t + \frac{\dot{\eta}_i(0)}{\omega_i} \sin \omega_i t \quad (3.93)$$

The initial conditions of $\{\eta_i(t)\}_{i=1}^n$ can be determined from those of $\mathbf{x}(t)$. Let

$$\underset{n \times 1}{\boldsymbol{\eta}(t)} = \begin{bmatrix} \eta_1(t) \\ \vdots \\ \eta_n(t) \end{bmatrix} \quad \underset{n \times n}{\boldsymbol{\Psi}} = [\boldsymbol{\Psi}_1, \dots, \boldsymbol{\Psi}_n] \quad (3.94)$$

Then $\mathbf{x}(t) = \boldsymbol{\Psi} \boldsymbol{\eta}(t)$ and $\dot{\mathbf{x}}(t) = \boldsymbol{\Psi} \dot{\boldsymbol{\eta}}(t)$. Evaluating at $t = 0$ and inverting gives

$$\begin{bmatrix} \eta_1(0) \\ \vdots \\ \eta_n(0) \end{bmatrix} = \boldsymbol{\eta}(0) = \boldsymbol{\Psi}^{-1} \mathbf{x}(0) \quad \begin{bmatrix} \dot{\eta}_1(0) \\ \vdots \\ \dot{\eta}_n(0) \end{bmatrix} = \dot{\boldsymbol{\eta}}(0) = \boldsymbol{\Psi}^{-1} \dot{\mathbf{x}}(0) \quad (3.95)$$

3.2.2 Eigenvalue Problem

Here we detour to discuss the natural frequencies and mode shapes of a structure, which are solutions to the eigenvalue problem in (3.90). This prepares us to obtain the response for the general case with damping and applied force in Sect. 3.2.3 later. For the discussion here, we write the eigenvalue problem explicitly in terms of the eigenvalue $\lambda = \omega^2$:

$$\mathbf{K}\boldsymbol{\psi} = \lambda\mathbf{M}\boldsymbol{\psi} \quad (3.96)$$

To see the basic nature of the eigenvalue problem, if we take an arbitrary vector $\boldsymbol{\psi}$, the vector $\mathbf{K}\boldsymbol{\psi}$ need not be a scalar multiple of the vector $\mathbf{M}\boldsymbol{\psi}$. The ‘eigenvector’ $\boldsymbol{\psi}$ is the special vector that makes this true. When this is true, the proportionality constant is not arbitrary either. The special constant is the ‘eigenvalue’ λ .

Characteristic Equation

As another way of looking at the eigenvalue equation, rewrite it as

$$(\mathbf{K} - \lambda\mathbf{M})\boldsymbol{\psi} = \mathbf{0} \quad (3.97)$$

For non-trivial solution of $\boldsymbol{\psi}$ to exist, i.e., $\boldsymbol{\psi} \neq \mathbf{0}$, the determinant of $\mathbf{K} - \lambda\mathbf{M}$ must be zero. This gives the ‘characteristic equation’ for the eigenvalue λ :

$$|\mathbf{K} - \lambda\mathbf{M}| = 0 \quad (3.98)$$

Evaluating the determinant gives a polynomial in λ of order n , and so there are n (generally complex) eigenvalues. Once an eigenvalue is found, it can be substituted into (3.97), from which the corresponding eigenvector $\boldsymbol{\psi}$ can be determined.

There are efficient algorithms for solving the eigenvalue problem. Standard functions are available in commercial software, e.g., `eig` in Matlab. These will not be discussed here. Rather, we mention some important properties relevant to structural dynamics.

Mode Shape Scaling

Eigenvectors are only determined up to a scaling factor, i.e., if $\boldsymbol{\psi}$ is an eigenvector then so is $a\boldsymbol{\psi}$ for any scalar $a \neq 0$. This follows because $\mathbf{K}\boldsymbol{\psi} = \lambda\mathbf{M}\boldsymbol{\psi}$ implies $\mathbf{K}(a\boldsymbol{\psi}) = \lambda\mathbf{M}(a\boldsymbol{\psi})$. In computations, a scaling constraint is applied to uniquely determine the eigenvector. One common way is to scale it to have unit ‘Euclidean norm’, i.e., $\sqrt{\boldsymbol{\psi}^T \boldsymbol{\psi}} = 1$, which is equivalent to having the sum of the squared values of the vector to be equal to 1. Another way is to scale it to be 1 at a particular DOF. Scaling affects the modal response $\eta_i(t)$ but not the physical response $x(t)$; nature does not care about what scaling we use. It is a good practice to indicate the scaling applied when presenting results.

Repeated Eigenvalues

Eigenvalues need not be distinct. Two different eigenvectors can have the same eigenvalue. For example, the natural frequencies of the translational modes along two principal horizontal directions of a building are theoretically the same when the two directions have the same stiffness and mass properties.

For a given eigenvalue, the number of times it is repeated is called ‘algebraic multiplicity’. The number of ‘different’, or technically, linearly independent, eigenvectors associated with the eigenvalue is called ‘geometric multiplicity’. It can be reasoned that geometric $\leq$ algebraic multiplicity. If $\mathbf{M}$ and $\mathbf{K}$ are arbitrary matrices, then it is possible that geometric $<$ algebraic multiplicity. For example, an eigenvalue can be repeated three times but one can only find two linearly independent eigenvectors; all other eigenvectors with the same eigenvalue are linear combination of these two eigenvectors.

For structural dynamics problems where $\mathbf{M}$ and $\mathbf{K}$ are both real symmetric, it can be shown that geometric = algebraic multiplicity. This is an important property, implying that there are always as many linearly independent mode shapes as the number of DOFs, regardless of the presence of repeated eigenvalues. It is then always possible to write the structural response at any particular time instant as a linear combination of the mode shapes, although the coefficients may change with time. This fact will be used in (3.104) later to obtain the response in the general case.

Mode Shape Orthogonality

As just mentioned, the real symmetric property of $\mathbf{M}$ and $\mathbf{K}$ guarantees n linearly independent eigenvectors $\{\psi_i\}_{i=1}^n$ with eigenvalues $\{\lambda_i\}_{i=1}^n$. Not only are $\{\psi_i\}_{i=1}^n$ linearly independent, they are also ‘orthogonal’ w.r.t. $\mathbf{K}$ and $\mathbf{M}$, i.e.,

$$\psi_i^T \mathbf{K} \psi_j = 0 \quad \psi_i^T \mathbf{M} \psi_j = 0 \quad i \neq j \quad (3.99)$$

Recall $\Psi = [\psi_1, \dots, \psi_n]$ ($n \times n$). Then $\Psi^T \mathbf{K} \Psi$ is a $n \times n$ diagonal matrix because its (i, j) -entry is $\psi_i^T \mathbf{K} \psi_j$. The same is also true for $\Psi^T \mathbf{M} \Psi$. In this sense, $\mathbf{K}$ and $\mathbf{M}$ are ‘diagonalizable’ by the eigenvectors.

Example 3.3 (Two-DOF structure) Here we illustrate how to obtain analytically the natural frequencies and mode shapes of the two-DOF structure in Fig. 3.8. For simplicity, assume $k_1 = k_2 = k_0$ and $m_1 = m_2 = m_0$. Then

$$\mathbf{K} - \lambda \mathbf{M} = k_0 \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} - \lambda m_0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = m_0 \begin{bmatrix} 2\lambda_0 - \lambda & -\lambda_0 \\ -\lambda_0 & \lambda_0 - \lambda \end{bmatrix}$$

where $\lambda_0 = k_0/m_0$. Setting $|\mathbf{K} - \lambda\mathbf{M}| = 0$ gives

$$(2\lambda_0 - \lambda)(\lambda_0 - \lambda) - \lambda_0^2 = 0 \Rightarrow \lambda^2 - 3\lambda_0\lambda + \lambda_0^2 = 0 \Rightarrow \lambda = \lambda_0 \left(\frac{3}{2} \pm \frac{\sqrt{5}}{2} \right)$$

$$\Rightarrow \lambda_1 = 0.382\lambda_0 \quad \lambda_2 = 2.618\lambda_0$$

$$\Rightarrow \omega_1 = \sqrt{\lambda_1} = 0.618\sqrt{\lambda_0} \quad \omega_2 = \sqrt{\lambda_2} = 1.618\sqrt{\lambda_0}$$

Mode Shapes

Let the two mode shapes be

$$\boldsymbol{\psi}_1 = \begin{bmatrix} \phi_{11} \\ \phi_{21} \end{bmatrix} \quad \boldsymbol{\psi}_2 = \begin{bmatrix} \phi_{12} \\ \phi_{22} \end{bmatrix}$$

Substituting $\lambda_1 = 0.382\lambda_0$ into $(\mathbf{K} - \lambda_1\mathbf{M})\boldsymbol{\psi}_1 = \mathbf{0}$ gives

$$m_0\lambda_0 \begin{bmatrix} 1.618 & -1 \\ -1 & 0.618 \end{bmatrix} \begin{bmatrix} \phi_{11} \\ \phi_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The first row gives $1.618\phi_{11} - \phi_{21} = 0$ and so $\phi_{21} = 1.618\phi_{11}$. The second row gives $-\phi_{11} + 0.618\phi_{21} = 0$. Dividing it by 0.618 gives the same conclusion that $\phi_{21} = 1.618\phi_{11}$. We see that one equation is redundant. This agrees with the fact that the determinant of the matrix on the LHS is zero. The mode shape can only be determined up to a scaling constant. If we scale it to be 1 at DOF 1, i.e., $\phi_{11} = 1$, then $\boldsymbol{\psi}_1 = [1 \ 1.618]^T$. If we scale it to have unit norm, $\boldsymbol{\psi}_1 = [1 \ 1.618]^T / \sqrt{1^2 + 1.618^2} = [0.526 \ 0.851]^T$. This mode corresponds to two masses moving in the same direction.

The procedure for the second mode is similar. Substituting $\lambda_2 = 2.618\lambda_0$ into $(\mathbf{K} - \lambda_2\mathbf{M})\boldsymbol{\psi}_2 = \mathbf{0}$ gives

$$m\lambda_0 \begin{bmatrix} -0.618 & -1 \\ -1 & -1.618 \end{bmatrix} \begin{bmatrix} \phi_{12} \\ \phi_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The first row gives $\phi_{22} = -0.618\phi_{12}$. The second row gives the same result. If we scale the mode shape to be 1 at DOF 1, $\boldsymbol{\psi}_2 = [1 \ -0.618]^T$. If we scale it to have unit norm, $\boldsymbol{\psi}_2 = [1 \ -0.618]^T / \sqrt{1^2 + (-0.618)^2} = [0.851 \ -0.526]^T$. This mode corresponds to two masses moving in opposite directions.

Mode Shape Orthogonality

Using the mode shapes scaled to be 1 at DOF 1,

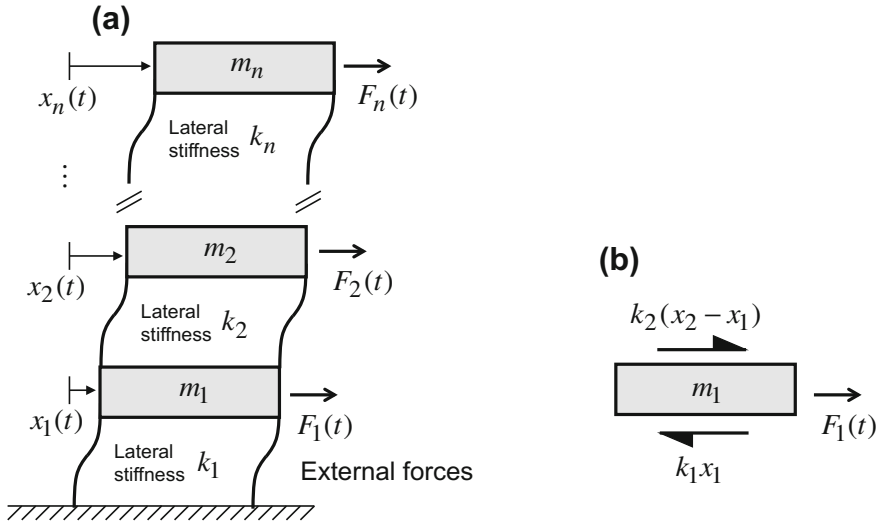


Fig. 3.9 Shear building; **a** schematic diagram; **b** free body diagram of the first floor mass

$$\begin{aligned}\psi_1^T \mathbf{K} \psi_2 &= \begin{bmatrix} 1 \\ 1.618 \end{bmatrix}^T k_0 \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -0.618 \end{bmatrix} = 7.6 \times 10^{-5} k_0 \\ \psi_1^T \mathbf{K} \psi_1 &= \begin{bmatrix} 1 \\ 1.618 \end{bmatrix}^T k_0 \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1.618 \end{bmatrix} = 1.38 k_0 \\ \psi_2^T \mathbf{K} \psi_2 &= \begin{bmatrix} 1 \\ -0.618 \end{bmatrix}^T k_0 \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -0.618 \end{bmatrix} = 3.62 k_0\end{aligned}$$

Due to round off errors, the calculated value of $\psi_1^T \mathbf{K} \psi_2$ is not zero. However, it is small compared to $\psi_1^T \mathbf{K} \psi_1$ and $\psi_2^T \mathbf{K} \psi_2$, which can be used as a criterion for checking. Checks w.r.t. the mass matrix give similar conclusions, where $\psi_1^T \mathbf{M} \psi_2 = 7.6 \times 10^{-5} m_0$, $\psi_1^T \mathbf{M} \psi_1 = 3.62 m_0$ and $\psi_2^T \mathbf{M} \psi_2 = 1.38 m_0$. ■

Example 3.4 (Shear building) In a ‘shear building’, the restoring force between two adjacent floors is proportional to the difference of their displacements. This is illustrated in Fig. 3.9. Applying Newton’s second law to the floor masses gives a matrix equation of motion where the stiffness and mass matrices have exactly the same form as those for the mass-spring system in Fig. 3.8.

As an example, for a six-storied shear building ($n = 6$), the stiffness matrix $\mathbf{K}$ and mass matrix $\mathbf{M}$ are given by

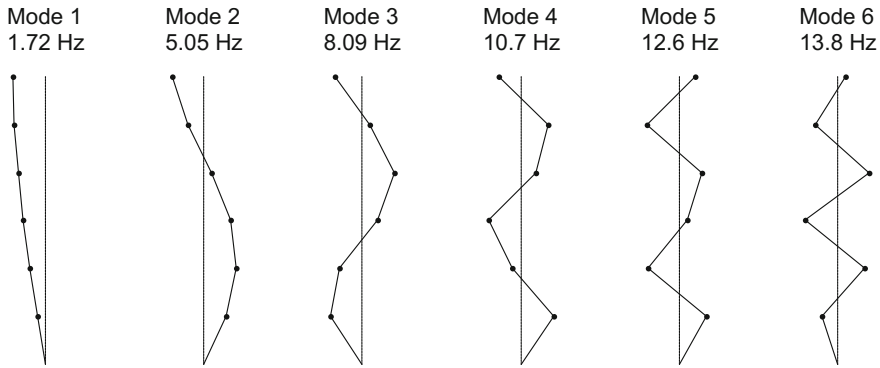


Fig. 3.10 Natural frequencies and mode shapes of a six-storied shear building

$$\mathbf{K} = \begin{bmatrix} k_1 + k_2 & -k_2 & & & & \\ -k_2 & k_2 + k_3 & -k_3 & & & \\ & -k_3 & k_3 + k_4 & -k_4 & & \\ & & -k_4 & k_4 + k_5 & -k_5 & \\ & & & -k_5 & k_5 + k_6 & -k_6 \\ & & & & -k_6 & k_6 \end{bmatrix} \quad (3.100)$$

$$\mathbf{M} = \begin{bmatrix} m_1 & & & & & \\ & m_2 & & & & \\ & & m_3 & & & \\ & & & m_4 & & \\ & & & & m_5 & \\ & & & & & m_6 \end{bmatrix} \quad (3.101)$$

In the present case, it is prohibitive to obtain the natural frequencies and mode shapes analytically. Instead, they are calculated numerically, as is usually done in applications. For uniform stiffness $k_i = 200$ kN/mm and floor mass $m_i = 100$ tons ($i = 1, \dots, 6$), the results calculated using the Matlab function `eig` are shown in Fig. 3.10. ■

3.2.3 Modal Superposition and Classical Damping

Section 3.2.1 shows that the undamped free vibration response of a structure with n DOFs is simply the sum of contributions from n modes, where each modal response satisfies its own SDOF equation of motion ‘uncoupled’ from others. This effectively reduces the original $n \times 1$ vector ODE to n scalar ODEs, which are much easier to analyze. In this section, we show that when damping and applied forces are present, the response can still be written as the sum of modal

contributions. However, the equations governing the modal responses are generally coupled, unless the damping matrix has a special algebraic structure, known as ‘classical damping’.

Consider the equation of motion (3.84) in the general case:

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) = \mathbf{F}(t) \quad (3.102)$$

Recall that the natural frequencies $\{\omega_i\}_{i=1}^n$ and mode shapes $\{\boldsymbol{\psi}_i\}_{i=1}^n$ satisfy the eigenvalue equation:

$$\mathbf{K}\boldsymbol{\psi}_i = \omega_i^2 \mathbf{M}\boldsymbol{\psi}_i \quad (3.103)$$

Since $\{\boldsymbol{\psi}_i\}_{i=1}^n$ are n linearly independent vectors, they ‘span’ the n -dimensional space and hence form a ‘basis’. That is, any $n \times 1$ vector can always be written as a linear combination of $\{\boldsymbol{\psi}_i\}_{i=1}^n$. This implies that at any given time t the response $\mathbf{x}(t)$ can be written as

$$\mathbf{x}(t) = \sum_{i=1}^n \boldsymbol{\psi}_i \eta_i(t) \quad (3.104)$$

for some $\{\eta_i(t)\}_{i=1}^n$ that depend on t . Substituting $\mathbf{x}(t) = \sum_{j=1}^n \boldsymbol{\psi}_j \eta_j(t)$ into (3.102) and pre-multiplying by $\boldsymbol{\psi}_i^T$ for a particular mode i ,

$$\sum_{j=1}^n (\boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_j) \ddot{\eta}_j(t) + \sum_{j=1}^n (\boldsymbol{\psi}_i^T \mathbf{C} \boldsymbol{\psi}_j) \dot{\eta}_j(t) + \sum_{j=1}^n (\boldsymbol{\psi}_i^T \mathbf{K} \boldsymbol{\psi}_j) \eta_j(t) = \boldsymbol{\psi}_i^T \mathbf{F}(t) \quad (3.105)$$

Using the orthogonality property in (3.99), only the $j = i$ term in the first and last sum on the LHS is non-zero, giving

$$(\boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i) \ddot{\eta}_i(t) + \sum_{j=1}^n (\boldsymbol{\psi}_i^T \mathbf{C} \boldsymbol{\psi}_j) \dot{\eta}_j(t) + (\boldsymbol{\psi}_i^T \mathbf{K} \boldsymbol{\psi}_i) \eta_i(t) = \boldsymbol{\psi}_i^T \mathbf{F}(t) \quad i = 1, \dots, n \quad (3.106)$$

Classical Damping

Equation (3.106) contains not only η_i (and its derivatives) but also other $\dot{\eta}_j$ s ($j \neq i$), through the cross terms $\{\boldsymbol{\psi}_i^T \mathbf{C} \boldsymbol{\psi}_j\}_{j \neq i}$. One conventional *assumption* to simplify analysis is ‘classical damping’, i.e., $\mathbf{C}$ is diagonalizable by the mode shapes as well:

$$\boldsymbol{\psi}_i^T \mathbf{C} \boldsymbol{\psi}_j = 0 \quad i \neq j \quad (3.107)$$

Equation (3.106) is then uncoupled from other $\dot{\eta}_j$ s ($j \neq i$). Dividing by $\boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i$ and noting $\boldsymbol{\psi}_i^T \mathbf{K} \boldsymbol{\psi}_i / \boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i = \omega_i^2$ gives

$$\ddot{\eta}_i(t) + 2\zeta_i\omega_i\dot{\eta}_i(t) + \omega_i^2\eta_i(t) = p_i(t) \quad (i = 1, \dots, n) \quad (3.108)$$

where ζ_i and $p_i(t)$ are respectively the ‘modal damping ratio’ and ‘modal force’ (per unit modal mass) of the i th mode, defined through

$$2\zeta_i\omega_i = \frac{\boldsymbol{\psi}_i^T \mathbf{C} \boldsymbol{\psi}_i}{\boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i} \quad p_i(t) = \frac{\boldsymbol{\psi}_i^T \mathbf{F}(t)}{\boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i} \quad (3.109)$$

so that (3.108) has the same form as the SDOF equation in (3.12). The initial conditions of $\{\eta_i(t)\}_{i=1}^n$ can be determined from those of $\mathbf{x}(t)$, as in (3.95).

To summarize, for a classically damped structure, the response $\mathbf{x}(t)$ with initial displacement $\mathbf{x}(0)$ and velocity $\dot{\mathbf{x}}(0)$ can be obtained as follow.

Modal superposition

1. Determine the natural frequencies $\{\omega_i\}_{i=1}^n$ and mode shapes $\{\boldsymbol{\psi}_i\}_{i=1}^n$ by solving the eigenvalue problem $\mathbf{K}\boldsymbol{\psi} = \omega^2\mathbf{M}\boldsymbol{\psi}$.
2. Determine the initial conditions of modal response, $\boldsymbol{\eta}(0) = \boldsymbol{\Psi}^{-1}\mathbf{x}(0)$ and $\dot{\boldsymbol{\eta}}(0) = \boldsymbol{\Psi}^{-1}\dot{\mathbf{x}}(0)$, where $\boldsymbol{\Psi} = [\boldsymbol{\psi}_1, \dots, \boldsymbol{\psi}_n]$.
3. For $i = 1, \dots, n$, determine the modal response $\eta_i(t)$ by solving the SDOF equation (3.108) with initial conditions $\eta_i(0)$ and $\dot{\eta}_i(0)$.
4. Obtain $\mathbf{x}(t) = \sum_{i=1}^n \boldsymbol{\psi}_i \eta_i(t)$.

3.2.4 Rayleigh Quotient

Given the stiffness matrix $\mathbf{K}$ and mass matrix $\mathbf{M}$, consider the following scalar function of $n \times 1$ real vector $\mathbf{u}$:

$$R(\mathbf{u}) = \frac{\mathbf{u}^T \mathbf{K} \mathbf{u}}{\mathbf{u}^T \mathbf{M} \mathbf{u}} \quad (3.110)$$

This is called the ‘Rayleigh quotient’. It is invariant to scaling, i.e., $R(a\mathbf{u}) = R(\mathbf{u})$ for any non-zero scalar a . It gives the eigenvalue λ_i of the i th mode when $\mathbf{u} = \boldsymbol{\psi}_i$, i.e.,

$$R(\boldsymbol{\psi}_i) = \frac{\boldsymbol{\psi}_i^T \mathbf{K} \boldsymbol{\psi}_i}{\boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i} = \lambda_i \quad (3.111)$$

This follows directly by pre-multiplying both sides of the eigenvalue equation $\mathbf{K}\boldsymbol{\psi}_i = \lambda_i \mathbf{M}\boldsymbol{\psi}_i$ by $\boldsymbol{\psi}_i^T$ and rearranging. Clearly, when $\mathbf{u}$ is not an eigenvector, $R(\mathbf{u})$ is not equal to the eigenvalue. However, it turns out that $R(\mathbf{u})$ can give a good approximation of the eigenvalue if $\mathbf{u}$ does not differ significantly from the eigenvector. For this reason, the Rayleigh quotient is often used as a quick means for estimating the natural frequency based on a guess of the mode shape.

Second Order Accuracy

The approximation capability of the Rayleigh quotient stems from the fact that its gradient is zero at the eigenvectors, i.e., the latter are ‘stationary points’:

$$\nabla R(\mathbf{u})|_{\mathbf{u}=\boldsymbol{\psi}_i} = \left[\frac{\partial R}{\partial u_1}, \dots, \frac{\partial R}{\partial u_n} \right] \bigg|_{\mathbf{u}=\boldsymbol{\psi}_i} = \mathbf{0} \quad i = 1, \dots, n \quad (3.112)$$

A first order deviation of $\mathbf{u}$ from $\boldsymbol{\psi}_i$ then leads to only a second order (rather than first order) difference of $R(\mathbf{u})$ from λ_i . Denote the difference between $\mathbf{u}$ and $\boldsymbol{\psi}_i$ by $\boldsymbol{\varepsilon}_i = \mathbf{u} - \boldsymbol{\psi}_i$ and assume that it is small. By noting that $R(\boldsymbol{\psi}_i) = \lambda_i$, the Taylor series of $R(\mathbf{u})$ about $\boldsymbol{\psi}_i$ reads

$$R(\mathbf{u}) = \lambda_i + \underbrace{\frac{1}{2} \boldsymbol{\varepsilon}_i^T \mathbf{H}_i \boldsymbol{\varepsilon}_i}_{\text{2nd order small}} + \text{higher order terms of } \boldsymbol{\varepsilon}_i \quad (3.113)$$

where the first order term vanishes because the gradient is zero; $\mathbf{H}_i$ is the Hessian matrix of $R(\mathbf{u})$ at $\mathbf{u} = \boldsymbol{\psi}_i$, which is a $n \times n$ matrix with the (r, s) -entry equal to $\partial^2 R / \partial u_r \partial u_s |_{\mathbf{u}=\boldsymbol{\psi}_i}$. It can be shown that

$$\mathbf{H}_i = 2(\boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i)^{-1} (\mathbf{K} - \lambda_i \mathbf{M}) \quad (3.114)$$

The gradient and Hessian of the Rayleigh quotient can be derived by repeated application of chain rule and differentiation rule for linear and quadratic forms. See Sects. C.4 and C.5.4 for details.

Lower and Upper Bounds

The Rayleigh quotient is bounded between the smallest and largest eigenvalues, i.e.,

$$\lambda_1 \leq R(\mathbf{u}) \leq \lambda_n \quad (3.115)$$

Clearly, the lower and upper bounds are attainable at $\mathbf{u} = \boldsymbol{\psi}_1$ and $\mathbf{u} = \boldsymbol{\psi}_n$, respectively. The Rayleigh quotient therefore has a global minimum at the eigenvector of the lowest mode and a global maximum at the highest mode. The lower bound is particularly useful for estimating the natural frequency of the lowest mode, which often contributes significantly to the dynamic response of a structure. It

implies that when estimating the lowest mode natural frequency using the Rayleigh quotient with different ‘trial mode shapes’ (i.e., not exact), the smallest estimate of natural frequency should be taken. This rule only applies to the lowest mode, however. Higher modes do not have this lower bound property.

Proof of (3.115) (Bounds on Rayleigh quotient)

For any arbitrary vector $\mathbf{u}$, let it be written as $\mathbf{u} = \sum_{i=1}^n a_i \boldsymbol{\psi}_i$ for some coefficients $\{a_i\}_{i=1}^n$. This is always possible since $\{\boldsymbol{\psi}_i\}_{i=1}^n$ forms a basis in the n -dimensional space. Then

$$\begin{aligned} \mathbf{u}^T \mathbf{K} \mathbf{u} &= \left(\sum_{i=1}^n a_i \boldsymbol{\psi}_i \right)^T \mathbf{K} \left(\sum_{j=1}^n a_j \boldsymbol{\psi}_j \right) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \underbrace{\boldsymbol{\psi}_i^T \mathbf{K} \boldsymbol{\psi}_j}_{\substack{\lambda_i \boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i \text{ if } i=j; \\ \text{zero otherwise}}} = \sum_{i=1}^n \lambda_i a_i^2 \boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i \end{aligned} \quad (3.116)$$

A similar expression can be obtained for $\mathbf{u}^T \mathbf{M} \mathbf{u}$, giving

$$\mathbf{u}^T \mathbf{K} \mathbf{u} = \sum_{i=1}^n \lambda_i a_i^2 \boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i \quad \mathbf{u}^T \mathbf{M} \mathbf{u} = \sum_{i=1}^n a_i^2 \boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i \quad (3.117)$$

Using these expressions and noting $\lambda_1 \leq \lambda_i \leq \lambda_n$ ($i = 1, \dots, n$),

$$\lambda_1 \mathbf{u}^T \mathbf{M} \mathbf{u} = \sum_{i=1}^n \underbrace{\lambda_1}_{\leq \lambda_i} a_i^2 \boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i \leq \sum_{i=1}^n \underbrace{\lambda_i}_{\leq \lambda_n} a_i^2 \boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i \leq \lambda_n \mathbf{u}^T \mathbf{M} \mathbf{u} \quad (3.118)$$

$\mathbf{u}^T \mathbf{K} \mathbf{u}$

Dividing throughout by $\mathbf{u}^T \mathbf{M} \mathbf{u}$ and noting $R(\mathbf{u}) = (\mathbf{u}^T \mathbf{K} \mathbf{u})/(\mathbf{u}^T \mathbf{M} \mathbf{u})$ gives (3.115). ■

Example 3.5 (Shear building, Rayleigh quotient) For a shear building structure with stiffness and mass matrices of the form respectively in (3.100) and (3.101), the numerator and denominator of the Rayleigh quotient simplify to

$$\mathbf{u}^T \mathbf{K} \mathbf{u} = \sum_{i=1}^n k_i (u_i - u_{i-1})^2 \quad \mathbf{u}^T \mathbf{M} \mathbf{u} = \sum_{i=1}^n m_i u_i^2 \quad (\text{shear building}) \quad (3.119)$$

where $\mathbf{u} = [u_1, \dots, u_n]^T$ and $u_0 = 0$.

Recall the six-storied shear building in Example 3.4, with uniform stiffness $k_i = 200$ kN/mm and floor mass $m_i = 100$ tons ($i = 1, \dots, 6$). Here we estimate the

natural frequencies of the first two modes using the Rayleigh quotient. The mode shapes in Fig. 3.10 are typical and they provide intuitions for suggesting trial mode shapes in general situations.

First Mode

For the first mode, assume the trial mode shape

$$\mathbf{u} = [1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6]^T$$

whose linearly increasing nature with height is simple and intuitive. Substituting into the Rayleigh quotient (3.110) and using (3.119) gives

$$\begin{aligned} R(\mathbf{u}) &= \frac{200 \times (1-0)^2 + 200 \times (2-1)^2 + \cdots + 200 \times (6-5)^2}{100 \times (1)^2 + 100 \times (2)^2 + \cdots + 100 \times (6)^2} \times \frac{10^6 \text{ N/m}}{10^3 \text{ kg}} \\ &= 131.8 \text{ s}^{-2} \end{aligned}$$

Note the unit conversion factor, missing which the result will err by orders of magnitude. An estimate for the first mode natural frequency is given by

$$f_1 \approx \frac{\sqrt{131.8 \text{ s}^{-2}}}{2\pi} = 1.83 \text{ Hz}$$

This estimate is larger than the exact value of 1.72 Hz (see Fig. 3.10), as guaranteed by the lower bound in (3.115). The error is 6%, which is sufficient for typical use (e.g., initial design).

Second Mode

For the second mode, assume the trial mode shape

$$\mathbf{u} = [1 \quad 1.5 \quad 1 \quad 0 \quad -1 \quad -1.5]^T$$

whose variation with positive and negative sides is intuitive. Then

$$\begin{aligned} R(\mathbf{u}) &= \frac{200 \times (1-0)^2 + 200 \times (1.5-1)^2 + \cdots + 200 \times [-1.5 - (-1)]^2}{100 \times (1)^2 + 100 \times (1.5)^2 + \cdots + 100 \times (-1.5)^2} \\ &\quad \times \frac{10^6 \text{ N/m}}{10^3 \text{ kg}} \\ &= 1000 \text{ s}^{-2} \end{aligned}$$

$$f_2 \approx \frac{\sqrt{1000 \text{ s}^{-2}}}{2\pi} = 5.03 \text{ Hz}$$

This estimate is lower than the exact value of 5.05 Hz (see Fig. 3.10), demonstrating that the natural frequency estimates of higher modes do not have the lower bound property. The error is still small, less than 1%. ■

3.3 Remarks on Damping

Unlike the stiffness or mass matrix that can be modeled by basic mechanics, there is still no accepted method for modeling structural damping, except for contributions from major discrete damping elements (e.g., dampers). Classical damping is conventionally assumed. Modal damping ratios are specified directly rather than calculated from the damping matrix. The latter is seldom constructed from first principles. Mathematical convenience and lack of knowledge in modeling damping are the main reasons for this practice.

Necessary and Sufficient Condition

Theoretically, a necessary and sufficient condition for classical damping is

$$\mathbf{KM}^{-1}\mathbf{C} = \mathbf{CM}^{-1}\mathbf{K} \quad (3.120)$$

The proof can be found at the end. See also Caughey and O’Kelly (1965); and Adhikari (2014) for general damping models. In structural dynamic applications, however, this condition is rarely checked to justify the use of classical damping; $\mathbf{C}$ is not available in the first place.

Repeatability

Measured vibrations from full-scale structures show ‘point-wise’ agreement with the oscillatory behavior of response assuming viscous damping. That is, the measured vibration (e.g., acceleration) time history often agrees with the theoretical prediction for some value of damping; but the same value of damping need not predict well another set of measured time history. ‘Repeatability’ is the issue.

Amplitude Dependence

‘Amplitude-dependent damping’ is one important characteristic that is not accounted by the viscous model with constant damping matrix. The identified values of damping ratio from vibration measurements exhibit a general increase (though with significant scatter) with the ‘amplitude’ of vibration, the latter defined in some empirical way depending on the nature of vibration (e.g., steady-state or stochastic). Even in this case the common treatment is still based on viscous damping with empirical modifications, e.g., using different values of damping ratio at different target amplitudes of vibration (see Example 3.6). In most structural vibration applications, damping is more of an empirical parameter to match theory with reality, rather than one that can be derived fundamentally from physical modeling.

Damping is one important challenging subject in wind engineering of tall buildings and long-span bridges. In addition to structural damping, it can also be attributed to aerodynamic effects (relative motion between structure and wind) and soil-structure interaction. Early discussion of structural damping mechanisms can be found in Jeary (1986, 1997). Beards (1996) has a chapter devoted to damping

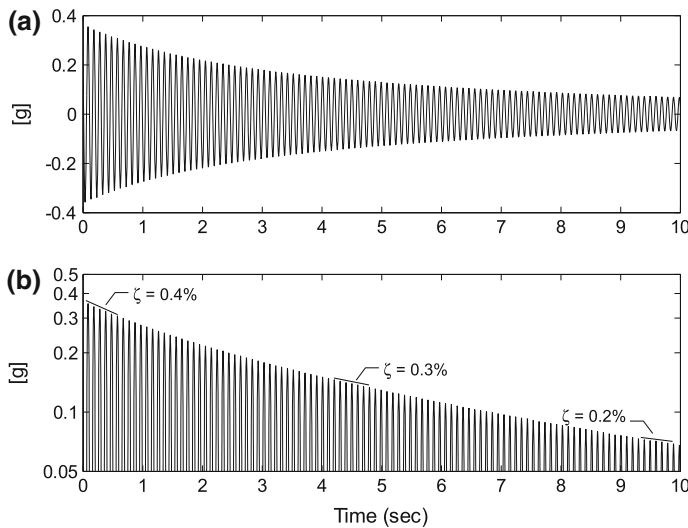


Fig. 3.11 Free vibration acceleration time history of a SDOF laboratory structure exhibiting amplitude dependence. Data is plotted on linear scale in (a) and $\log_{10}$ -scale in (b)

with introductory coverage on different sources. See recent reports in Tamura and Kareem (2013) and Kwok et al. (2015). There have been many attempts on identifying and predicting amplitude-dependent damping from full-scale vibration measurements but results are mixed with large scatter. See, e.g., Fukuwa et al. (1996), Li et al. (2000), Satake et al. (2003) and Kijewski-Correa et al. (2006). Identified values form the basis of guidelines currently used in design. See, e.g., EC1 Annex F (2005), ESDU 83009 (2012), ISO 4354:2009 Annex E, and ISO 10137:2007 Annex B, Bachmann et al. (1995).

Example 3.6 (Amplitude-dependent damping) Figure 3.11a shows the free vibration time history of acceleration measured from a SDOF structure under laboratory condition. Figure 3.11b shows the same plot but on a $\log_{10}$ -scale. If the damping does not depend on amplitude, the envelope of the peaks in (b) should form a straight line, because the envelope is $A \propto e^{-\zeta\omega_1 t}$, i.e.,

$$\log_{10} A = \underbrace{(-\zeta\omega_1 \log_{10} e)}_{\text{slope}} \times t + \text{constant} \quad (3.121)$$

where ω_1 (rad/s) is the natural frequency and ζ is the damping ratio. In Fig. 3.11b, the slope of the envelope decreases in magnitude as the amplitude decreases with time, which is an evidence of amplitude-dependent damping. The damping ratios indicated in the figure were estimated using logarithmic decrement method (Sect. 3.1.4) with five cycles. ■

Proof of (3.120) (Necessary and sufficient condition for classical damping)

A conventional way to prove is to write the equation of motion in (3.84) as (after pre-multiplying by $\mathbf{M}^{-1/2}$)

$$\underbrace{\mathbf{M}^{1/2}\ddot{\mathbf{x}}(t)}_{\ddot{\mathbf{q}}(t)} + \underbrace{\mathbf{M}^{-1/2}\mathbf{C}\mathbf{M}^{-1/2}}_{\tilde{\mathbf{C}}} \underbrace{\mathbf{M}^{1/2}\dot{\mathbf{x}}(t)}_{\dot{\mathbf{q}}(t)} + \underbrace{\mathbf{M}^{-1/2}\mathbf{K}\mathbf{M}^{-1/2}}_{\tilde{\mathbf{K}}} \underbrace{\mathbf{M}^{1/2}\mathbf{x}(t)}_{\mathbf{q}(t)} = \mathbf{M}^{-1/2}\mathbf{F}(t) \quad (3.122)$$

and use the result in linear algebra that the real symmetric (hence diagonalizable) matrices $\tilde{\mathbf{C}}$ and $\tilde{\mathbf{K}}$ are diagonalizable by the same set of eigenvectors if and only if they commute, i.e., $\tilde{\mathbf{C}}\tilde{\mathbf{K}} = \tilde{\mathbf{K}}\tilde{\mathbf{C}}$, which is equivalent to (3.120). Here we use an alternative approach to illustrate the algebraic structure of the problem.

Let $\Psi = [\psi_1, \dots, \psi_n]$, where ψ_i ($i = 1, \dots, n$) is the eigenvector associated with eigenvalue λ_i , satisfying $\mathbf{K}\psi_i = \lambda_i\mathbf{M}\psi_i$. We show that $D = \Psi^T\mathbf{C}\Psi$ is a diagonal matrix if and only if (3.120) holds, i.e., $\mathbf{K}\mathbf{M}^{-1}\mathbf{C} = \mathbf{C}\mathbf{M}^{-1}\mathbf{K}$.

By orthogonality, $A = \Psi^T\mathbf{M}\Psi$ and $B = \Psi^T\mathbf{K}\Psi$ are diagonal matrices. Pre-multiplying these by Ψ^{-T} and post-multiplying by Ψ^{-1} ,

$$\mathbf{M} = \Psi^{-T}A\Psi^{-1} \quad \mathbf{K} = \Psi^{-T}B\Psi^{-1} \quad (3.123)$$

Using $(XYZ)^{-1} = Z^{-1}Y^{-1}X^{-1}$ for invertible matrices X , Y and Z ,

$$\mathbf{M}^{-1} = \Psi A^{-1}\Psi^T \quad (3.124)$$

Necessary Condition

We first assume that $D = \Psi^T\mathbf{C}\Psi$ is a diagonal matrix and show that (3.120) holds. Similar to (3.123),

$$\mathbf{C} = \Psi^{-T}D\Psi^{-1} \quad (3.125)$$

Substituting (3.123) to (3.125) into the LHS of (3.120),

$$\mathbf{K}\mathbf{M}^{-1}\mathbf{C} = (\underbrace{\Psi^{-T}B\Psi^{-1}}_{\mathbf{I}_n})(\underbrace{\Psi A^{-1}\Psi^T}_{\mathbf{I}_n})(\Psi^{-T}D\Psi^{-1}) = \Psi^{-T}B A^{-1}D\Psi^{-1} \quad (3.126)$$

Since A^{-1} , B and D are diagonal matrices, $BA^{-1}D = DA^{-1}B$. Then (3.120) follows because

$$\mathbf{K}\mathbf{M}^{-1}\mathbf{C} = \Psi^{-T}DA^{-1}B\Psi^{-1} = \underbrace{(\Psi^{-T}D\Psi^{-1})}_{\mathbf{C}} \underbrace{(\Psi A^{-1}\Psi^T)}_{\mathbf{M}^{-1}} \underbrace{(\Psi^{-T}B\Psi^{-1})}_{\mathbf{K}} \quad (3.127)$$

Sufficient Condition

Conversely, assume that (3.120) holds, i.e., $\mathbf{KM}^{-1}\mathbf{C} = \mathbf{CM}^{-1}\mathbf{K}$. We will show that $\mathbf{\Psi}^T\mathbf{C}\mathbf{\Psi}$ is a diagonal matrix. Substituting (3.123) and (3.124) into this equation gives

$$\begin{aligned} (\underbrace{\mathbf{\Psi}^{-T}\mathbf{B}\mathbf{\Psi}^{-1}}_{\mathbf{I}_n})(\underbrace{\mathbf{\Psi}\mathbf{A}^{-1}\mathbf{\Psi}^T}_{\mathbf{I}_n})\mathbf{C} &= \mathbf{C}(\underbrace{\mathbf{\Psi}\mathbf{A}^{-1}}_{\mathbf{I}_n})(\underbrace{\mathbf{\Psi}^T}_{\mathbf{I}_n})(\underbrace{\mathbf{\Psi}^{-T}\mathbf{B}\mathbf{\Psi}^{-1}}_{\mathbf{I}_n}) \\ \Rightarrow \mathbf{\Psi}^{-T}\mathbf{B}\mathbf{A}^{-1}\mathbf{\Psi}^T\mathbf{C} &= \mathbf{C}\mathbf{\Psi}\mathbf{A}^{-1}\mathbf{B}\mathbf{\Psi}^{-1} \end{aligned} \quad (3.128)$$

Pre-multiplying by $\mathbf{\Psi}^T$ and post-multiplying by $\mathbf{\Psi}$,

$$\mathbf{B}\mathbf{A}^{-1}(\mathbf{\Psi}^T\mathbf{C}\mathbf{\Psi}) = (\mathbf{\Psi}^T\mathbf{C}\mathbf{\Psi})\mathbf{A}^{-1}\mathbf{B} \quad (3.129)$$

Note that $\mathbf{B}\mathbf{A}^{-1}$ and $\mathbf{A}^{-1}\mathbf{B}$ are both equal to a diagonal matrix with entries $\{\omega_i^2\}_{i=1}^n$. Let D_{ij} be the (i,j) -entry of $\mathbf{\Psi}^T\mathbf{C}\mathbf{\Psi}$. Reading the (i,j) -entry on both sides of (3.129) and rearranging gives

$$(\omega_i^2 - \omega_j^2)D_{ij} = 0 \quad i, j = 1, \dots, n \quad (3.130)$$

This equation must hold for all $i, j = 1, \dots, n$. For $i = j$, it holds trivially. For $i \neq j$, if $\omega_i \neq \omega_j$ then $D_{ij} = 0$. This implies that the off-diagonal entries of $\mathbf{\Psi}^T\mathbf{C}\mathbf{\Psi}$ are zero for any two modes with distinct eigenvalues. The case for modes with the same (repeated) eigenvalue requires more linear algebra arguments, whose details are omitted here. ■

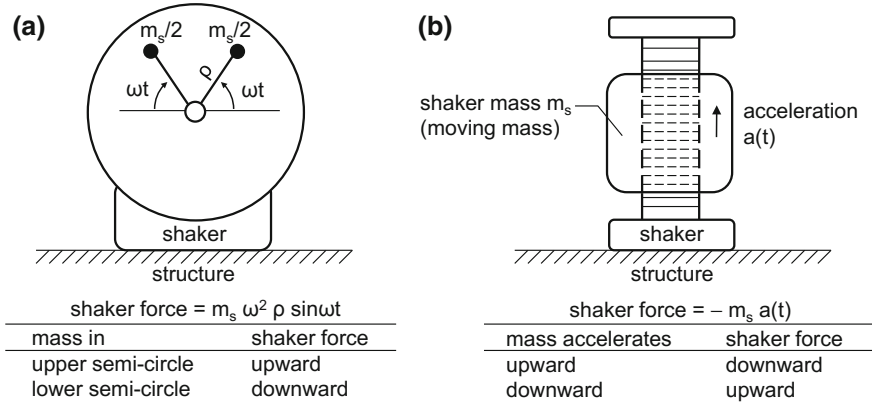


Fig. 3.12 Two examples of shaker. **a** Reciprocating motor; **b** electro-magnetic shaker. Expressions of shaker force assumes upward positive

3.4 Harmonic Load Test

Structural vibration generally comprises contributions from different modes. When a structure is excited by a harmonic force near the natural frequency of a particular mode, however, its response is dominated by the mode due to resonance; dynamic amplification of the mode is much greater than those of other modes. This observation suggests a conceptually simple method for identifying the modal properties of a MDOF structure using harmonic excitations, often artificially generated by ‘shakers’. A shaker derives its payload through the motion of an electro-mechanically driven moving mass. One simple mechanism is a ‘reciprocating motor’, as shown in Fig. 3.12a. The two eccentric masses ($m_s/2$ each) rotating at a constant angular speed ω (rad/s) in opposite sense generates a harmonic vertical force with amplitude $m_s \rho \omega^2$ but ideally no horizontal force due to cancellation.

Figure 3.12b is a more general mechanism where the motion of the moving mass is driven by an electro-magnetic force that can be assigned arbitrarily (within physical limits). Neglecting the inertia of other parts, the ‘shaker force’, i.e., force exerted by the shaker on the structure, is simply $-m_s a(t)$. The negative sign accounts for the action-reaction pairs (Newton’s third law) of interactive forces between the moving mass, shaker (excluding the moving mass) and the structure. For convenience, the moving mass is referred as the ‘shaker mass’, which should not be confused with the total mass of the shaker. For the shaker mass to accelerate upward (say), it must receive an upward force from the shaker (e.g., magnetic force by solenoid). By action-reaction pair, there is an equal but opposite (i.e., downward) force on the shaker. This downward force is further transmitted from the shaker to the structure by another action-reaction pair. The shaker force is often measured indirectly by an accelerometer attached to the shaker mass.

3.4.1 Collocated Setup

Consider a simple setup where the shaker in Fig. 3.12b exerts a vertical harmonic force on the structure. There are two accelerometers measuring vertical motion, one on the shaker mass and the other on the structure at the same location of the shaker, i.e., ‘collocated’. ‘Vertical direction’ here is only for instructional purpose and in general it can be any DOF of the structure. For a particular mode, if we scale the mode shape such that its value at this DOF is 1, the amplitude of modal force (per unit modal mass) due to the shaker will be $m_s A_p / M$, where A_p is the acceleration amplitude of the shaker mass (m_s), and M is the modal mass of the structure. If the excitation frequency ω (rad/s) is near the natural frequency ω_0 (rad/s) of a lightly damped mode well-separated from others, the steady-state acceleration amplitude

A_x of the structure measured at the same location can be approximated by the contribution from the mode only:

$$\underbrace{A_x}_{\substack{\text{struct.} \\ \text{acc.} \\ \text{amp.}}} \approx \underbrace{\frac{m_s A_p}{M \omega_0^2}}_{\text{static disp.}} \times \underbrace{A}_{\substack{\text{dyn. amp. between} \\ \text{modal force and disp.}}} \times \underbrace{\omega^2}_{\substack{\text{convert disp.} \\ \text{to acc. amp.}}} \quad (\omega \approx \omega_0) \quad (3.131)$$

where $A = 1/\sqrt{(1 - \beta^2)^2 + (2\zeta\beta)^2}$ is the dynamic amplification factor in (3.51); $\beta = \omega/\omega_0$ and ζ is the damping ratio of the mode. Note that A_x , A_p and A depend on ω but this has been omitted to simplify notation. Equation (3.131) can be rearranged to give $A \propto A_x/A_p\omega^2$, plotting which allows the half-power bandwidth method in Sect. 3.1.8 to be applied for estimating the natural frequency and damping ratio. See Example 3.2 in that section.

Modal Mass

The modal mass can also be further estimated, which is generally not possible unless the input force is measured. When the excitation frequency is equal to the natural frequency, i.e., $\omega = \omega_0$, $A \approx 1/2\zeta$ for small $\zeta \ll 1$. Substituting into (3.131) and rearranging gives

$$M \approx \frac{A_p m_s}{A_x 2\zeta} \quad (\omega = \omega_0) \quad (3.132)$$

3.4.2 Least Squares Approach

The last section illustrates the basic principle of how modal properties can be estimated from steady-state response amplitudes. Better (and often more sophisticated) methods can be adopted that make use of the information at different excitation frequencies and address the presence of measurement noise. For example, the natural frequency ω_0 , damping ratio ζ and shaker mass ratio $r = m_s/M$ can be estimated as the ones that minimize the following ‘objective function’, defined as the sum of squared difference between the LHS and RHS of (3.131):

$$J(\omega_0, \zeta, r) = \sum_{\omega} \left[\underbrace{A_x(\omega)}_{\text{measured}} - \frac{r\omega^2}{\omega_0^2} \underbrace{A_p(\omega)}_{\text{measured}} \underbrace{A(\omega)}_{\substack{\text{depends} \\ \text{on } \omega_0 \text{ and } \zeta}} \right]^2 \quad (3.133)$$

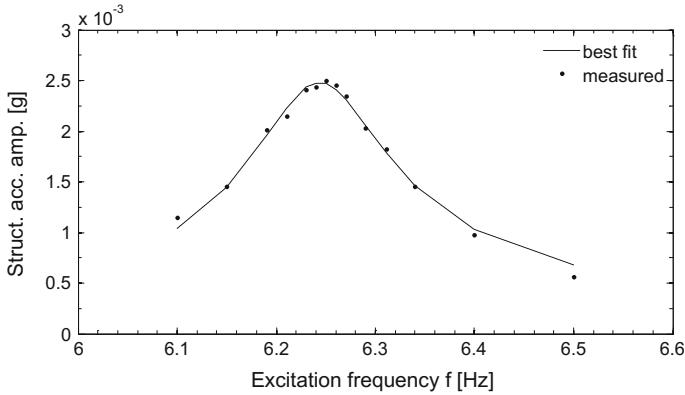


Fig. 3.13 Measured and best least squares fit of structural acceleration amplitude

where the sum is over all tested values of the excitation frequency ω ; the dependence of A_x , A_p and A on ω has been emphasized. As indicated, $A = \sqrt{(1 - \beta^2)^2 + (2\zeta\beta)^2}$ ($\beta = \omega/\omega_0$) also depends on ω_0 and ζ . The three-dimensional minimization problem can be solved numerically using standard algorithms, e.g., `fminsearch` in Matlab. Once r has been estimated, the modal mass can be recovered from $M = m_s/r$. The above is only one of the many possibilities where the modal properties can be estimated.

Example 3.7 (Shaker test on floor slab, modal mass) Consider estimating the modal mass of the floor slab in Example 3.2. The shaker mass is known to be $m_s = 13.2$ kg. The shaker and accelerometer on the structure are close to each other compared to the dimension of the slab (about $35 \text{ m} \times 35 \text{ m}$) and so they may be assumed to be collocated. Equation (3.132) then applies. It was estimated in Example 3.2 that the natural frequency was 6.24 Hz and damping ratio was 1.15%. From Table 3.3, at $f = 6.24$ Hz, $A_p = 1171$ milli-g and $A_x = 2.439$ milli-g. Substituting into (3.132) gives

$$M \approx \frac{1171}{2.439} \left[\frac{\text{milli-g}}{\text{milli-g}} \right] \times \frac{13.2 [\text{kg}]}{2(1.15\%)} = 276 \times 10^3 \text{ kg}$$

The value of modal mass depends on the scaling of mode shape, which has been assumed to be 1 vertically at the shaker location. If, relative to this scaling, the mode shape is multiplied by a factor, the modal mass should be multiplied by the square of that factor.

Least Squares Method

Alternatively, using the values of A_x and A_p in Table 3.3, the modal properties that minimize the objective function in (3.133) are found to be $f_0 = \omega_0/2\pi \approx 6.24$ Hz,

$\zeta \approx 1.11\%$ and $r \approx 46.78 \times 10^{-6}$. The corresponding estimate for the modal mass is then $M = m_s/r \approx 13.2/(46.78 \times 10^{-6}) = 282 \times 10^3$ kg. These estimates are similar to those found earlier, differing by less than 5%. Figure 3.13 shows the measured and best fitted structural acceleration amplitudes. Further details about modal testing of the structure in this example can be found in Au et al. (2012). ■

3.5 Impact Hammer Test

The input-output relationship between an impulsive applied force and vibration response in the frequency domain can be exploited for estimating the modal properties of a structure. The basic idea is as follow. Consider a structure subjected to an ideal impulsive force $Q\delta(t)$ at DOF r , where Q is the amount of impulse (area of force with time) and $\delta(t)$ is the Dirac Delta function (Sect. 3.1.11). Applying (3.109), the modal force (per unit modal mass) is given by

$$p(t) = \frac{\phi_r Q}{M} \delta(t) \quad (3.134)$$

where $M = \Psi^T \mathbf{M} \Psi$ is the modal mass and $\Psi = [\phi_1, \dots, \phi_n]^T$ is the mode shape. Note that $p(t)$, M and Ψ depend on the mode number but this has been omitted to simplify notation.

3.5.1 Frequency Response

In the frequency domain, the Fourier Transform (FT, Sect. 2.2) of the applied force $Q\delta(t)$ is simply the constant Q because the FT of $\delta(t)$ is 1. Correspondingly, the FT of the modal force $p(t)$ is the constant $\phi_r Q/M$. This turns out to provide convenience for estimating the modal properties. The FT of structural response generally comprises contributions from different modes. However, near the natural frequency of a lightly damped mode well-separated from others, the FT is dominated by the contribution from the mode, due to dynamic amplification. Near the natural frequency ω_0 (say), the FT of displacement at DOF j is then approximately given by

$$X_j(\omega) \approx \phi_j \eta(\omega) \quad (\omega \approx \omega_0) \quad (3.135)$$

where ω (rad/s) is the frequency variable of FT; ϕ_j is the mode shape value at DOF j ; $\eta(\omega)$ is the FT of modal displacement whose time-domain counterpart satisfies the modal equation

$$\ddot{\eta}(t) + 2\zeta\omega_0\dot{\eta}(t) + \omega_0^2\eta(t) = \frac{\phi_r Q}{M} \delta(t) \quad (3.136)$$

and ζ is the damping ratio of the mode. Taking FT on both sides of (3.136) and noting that the FTs of $\dot{\eta}(t)$ and $\ddot{\eta}(t)$ are respectively $i\omega\eta(\omega)$ and $-\omega^2\eta(\omega)$, one obtains, after rearranging,

$$\eta(\omega) = G(\omega) \frac{\phi_r Q}{M} \quad (3.137)$$

where

$$G(\omega) = \omega_0^{-2} (1 - \beta^2 + 2\zeta\beta i)^{-1} \quad \beta = \frac{\omega}{\omega_0} \quad (3.138)$$

is the ‘transfer function’ of the mode. Note that

$$|G(\omega)| = \omega_0^{-2} A(\beta) \quad \beta = \frac{\omega}{\omega_0} \quad (3.139)$$

where $A(\beta)$ is the dynamic amplification factor in (3.51). Substituting (3.137) into (3.135) gives

$$\underbrace{X_j(\omega)}_{\text{FT of response at DOF } j} \approx \underbrace{\phi_j}_{\text{mode shape value at DOF } j} \underbrace{G(\omega)}_{\text{modal transfer function}} \underbrace{\frac{\phi_r Q}{M}}_{\text{FT of modal force}} \quad (\omega \approx \omega_0) \quad (3.140)$$

Taking modulus and using (3.139),

$$|X_j(\omega)| \approx \left| \frac{\phi_j \phi_r Q}{M} \right| |G(\omega)| \propto A(\beta) \quad \beta = \frac{\omega}{\omega_0} \quad (3.141)$$

A plot of $|X_j(\omega)|$ versus ω can then be used for estimating the natural frequency and damping ratio, e.g., using the half-power bandwidth method. The estimates based on different measured DOFs are generally different. They may be empirically averaged to give a representative value.

Mode Shape

For two DOFs j and k , considering the ratio $X_j(\omega)/X_k(\omega)$ using (3.140) gives

$$\frac{\phi_j}{\phi_k} \approx \frac{X_j(\omega)}{X_k(\omega)} \quad (\omega \approx \omega_0) \quad (3.142)$$

This implies that the ratio of mode shape values can be estimated as the ratio of the FTs of response near the natural frequency. Together with a chosen scaling, this determines the mode shape values at the measured DOFs.

Velocity or Acceleration Measurement

In the above, the modal identification principle has been explained in terms of displacement measurement. It can also be applied with velocity or acceleration measurement, provided that the transfer function $G(\omega)$ is modified accordingly. Multiply $G(\omega)$ by $i\omega$ for velocity and $(i\omega)^2$ for acceleration measurement. In implementation with digital data, the FT is replaced by its discrete-time counterpart (Sect. 2.3).

Impact Source

Practically, the impulsive input force can be provided by, e.g., a drop weight or an impact hammer. In the minimum setting, the input force need not be measured but doing so offers more information (e.g., modal mass, see Sect. 3.5.2 later) and flexibility in testing (see Sect. 3.5.3 later). In reality, the input force is not an ideal impulse. Its FT is only approximately constant with a finite bandwidth limited by non-zero impulse duration. Correspondingly, the value of Q in the foregoing discussion should be a representative FT value of the input force near the natural frequency. The accuracy of modal property estimates is not significantly affected by the limited bandwidth of the input impulse as long as its FT is reasonably constant near the natural frequency and the mode is adequately excited. Measurement noise can be reduced by averaging the FTs of the measured response over multiple trials.

3.5.2 Least Squares Approach

The modal properties can be estimated in a more sophisticated manner using a least squares approach, making use of the FT of measured response near the natural frequency. Suppose the mode shape is scaled so that $\phi_r = 1$. Let

$$a_j = \frac{\phi_j Q}{M} \quad (3.143)$$

Then ω_0 , ζ and a_j at the measured DOFs can be estimated as the ones that minimize the following objective function, defined as the squared modulus of the difference between the LHS and RHS of (3.140) summed over ω near the natural frequency and all measured DOFs:

$$J(\omega_0, \zeta, \{a_j\}) = \sum_j \sum_{\omega} \left| \underbrace{X_j(\omega)}_{\text{measured}} - a_j \underbrace{G(\omega)}_{\substack{\text{depends} \\ \text{on } \omega_0 \text{ and } \zeta}} \right|^2 \quad (3.144)$$

where $\{a_j\}$ denotes the collection of a_j s at the measured DOFs. Note that J depends on ω_0 and ζ through $G(\omega)$. Since a_j affects J in a quadratic manner, its optimal value, denoted by $\hat{a}_j$, can be obtained analytically as:

$$\hat{a}_j = \frac{\sum_{\omega} \operatorname{Re}[X_j(\omega) \overline{G(\omega)}]}{\sum_{\omega} |G(\omega)|^2} \quad (3.145)$$

where $\overline{G(\omega)}$ denotes the complex conjugate of $G(\omega)$. Substituting $a_j = \hat{a}_j$ into (3.144) gives an expression that depends on ω_0 and ζ only. Numerically minimizing such expression gives the optimal values for ω_0 and ζ . Substituting these optimal values into $G(\omega)$ and then into (3.145) gives the optimal value for a_j .

Mode Shape and Modal Mass

If the vibration at the force DOF r is measured, one can use $\phi_r = 1$ to recover the ratio $Q/M = a_r$, which follows directly from (3.143). The mode shape values at other measured DOFs can be determined from $\phi_j = a_j/a_r$. If the input force is also measured so that Q is known, the modal mass can be recovered from $M = Q/a_r$. On the other hand, if the vibration at the force DOF r is not measured, one can still use $\phi_j \propto a_j$ and apply a scaling constraint to uniquely determine the mode shape values at the measured DOFs. The potential conflict of the new scaling with $\phi_r = 1$ is irrelevant because DOF r is not measured. Since a_r is not known (DOF r is not measured), it is not possible to recover the modal mass from $M = Q/a_r$, unless a_r is estimated by other means, e.g., interpolation using the mode shape values at the measured DOFs.

3.5.3 Covering DOFs in Multiple Setups

To obtain mode shape values at the desired DOFs, a direct way is to deploy multiple vibration sensors (e.g., accelerometers) to cover them all in a single setup. When this is not possible (e.g., not enough sensors), one can cover the DOFs in multiple setups, assuming that the overall mode shape remains the same in different setups. The possibilities are summarized in Table 3.5. When the input force is not measured (Case 1), the setups need to share some common ‘reference DOFs’ so that their ‘local mode shapes’ (i.e., with DOFs in a given setup only) can

Table 3.5 Cases where global mode shape can be determined from multiple setups

Case	Force measured?	Force DOF	Sensor DOF(s)	Remark
1	No	Fixed/Roved	Roved	Global mode shape covering <i>sensor DOFs</i> can be determined if the setups share some reference DOF(s)
2	Yes	Fixed	Roved	Global mode shape covering <i>sensor DOFs</i> can be determined; reference sensor DOFs not needed
3	Yes	Roved	Fixed	Global mode shape covering <i>force DOFs</i> can be determined

be assembled to form the ‘global mode shape’ containing all DOFs. This belongs to the general problem of ‘mode shape assembly’. See Sects. 14.1 and 14.2 for least squares methods. When the input force is measured, one can either fix the input force and ‘rove’ the sensor(s) (Case 2); or, fix the sensor(s) and rove the input force (Case 3). These two cases are described next.

Roving Sensor DOF(s)

The principle in this case is similar to the single-setup setting, except that the input force must now be measured because it need not be the same in different setups. Consider two setups, say, 1 and 2, where the input force is always applied at DOF r . Let the vibrations at DOFs j and k be measured in Setups 1 and 2, respectively. Then (3.140) reads

$$\begin{aligned} X_j(\omega) &\approx \phi_j G(\omega) \frac{\phi_r Q_1}{M} & \omega \approx \omega_0 \\ & & j \text{ in Setup 1} \\ X_k(\omega) &\approx \phi_k G(\omega) \frac{\phi_r Q_2}{M} & k \text{ in Setup 2} \end{aligned} \quad (3.146)$$

where Q_1 and Q_2 are the FTs of the input force in Setups 1 and 2, respectively. Considering the ratio $X_j(\omega)/X_k(\omega)$ and rearranging gives

$$\frac{\phi_j}{\phi_k} \approx \frac{X_j(\omega) Q_2}{X_k(\omega) Q_1} \quad \begin{aligned} &\omega \approx \omega_0 \\ &j \text{ in Setup 1} \\ &k \text{ in Setup 2} \end{aligned} \quad (3.147)$$

This allows the mode shape values of the DOFs from different setups to be combined to form the global mode shape, even when the setups do not share any sensor DOFs in common. The term Q_2/Q_1 in the expression explains why it is necessary to measure the input force.

Roving Force DOF

Instead of roving vibration sensors, it is possible to obtain the global mode shape by roving the input force over the DOFs in different setups. In the simplest setting, suppose the vibration at only one DOF is measured, always at DOF j in all setups. Let $X_{jr}(\omega)$ denote the FT of the response at DOF j when the input force is applied at DOF r with FT Q_r near the natural frequency. Consider the ratio of the measured response FTs of two tests, one with input force at DOF r and the other at DOF s . Equation (3.140) now reads

$$\begin{aligned} X_{jr}(\omega) &\approx \phi_j G(\omega) \frac{\phi_r Q_r}{M} \\ X_{js}(\omega) &\approx \phi_j G(\omega) \frac{\phi_s Q_s}{M} \end{aligned} \quad (\omega \approx \omega_0) \quad (3.148)$$

Considering the ratio $X_{js}(\omega)/X_{jr}(\omega)$ and rearranging gives

$$\frac{\phi_s}{\phi_r} \approx \frac{X_{js}(\omega)Q_r}{X_{jr}(\omega)Q_s} \quad (\omega \approx \omega_0) \quad (3.149)$$

Thus, by applying the input force at different DOFs (s) in different setups, the mode shape value (ϕ_s) of these DOFs can be determined (together with a chosen scaling). A simple choice for the sensor DOF j and the reference force DOF r is to take them collocated at a DOF expected to have large mode shape value ϕ_r .

3.6 State-Space Approach

In Sects. 3.1 and 3.2, the response of a structure is obtained by direct analysis of the equation of motion, which is a second order ODE. In this section we introduce the ‘state-space’ approach, where dynamics is studied through an equivalent first order ODE. Elegant theory for the latter has been developed, providing a means for analyzing general dynamical systems. State-space approach is applied in Sects. 3.7 and 3.8 later for studying the numerical stability of time integration schemes; and in Sects. 5.3 and 5.4 for studying the transient response statistics of structures subjected to stochastic excitations.

State-Space Equation of Structural Dynamics

Recall the equation of motion (3.84) of a structure with n DOFs:

$$\underset{n \times n}{\mathbf{M}} \underset{n \times 1}{\ddot{\mathbf{x}}(t)} + \underset{n \times n}{\mathbf{C}} \underset{n \times 1}{\dot{\mathbf{x}}(t)} + \underset{n \times n}{\mathbf{K}} \underset{n \times 1}{\mathbf{x}(t)} = \underset{n \times 1}{\mathbf{F}(t)} \quad (3.150)$$

This is a second order vector ODE. In a state-space approach, it is converted to a first order vector ODE. Define the ‘state-vector’ $\mathbf{y}(t)$ by

$$\underset{2n \times 1}{\mathbf{y}(t)} = \begin{bmatrix} \underset{n \times 1}{\mathbf{y}_1(t)} \\ \underset{n \times 1}{\mathbf{y}_2(t)} \end{bmatrix} = \begin{bmatrix} \underset{n \times 1}{\mathbf{x}(t)} \\ \underset{n \times 1}{\dot{\mathbf{x}}(t)} \end{bmatrix} \quad (3.151)$$

Then $\mathbf{y}_1(t)$ and $\mathbf{y}_2(t)$ satisfy

$$\begin{aligned} \dot{\mathbf{y}}_1(t) &= \mathbf{y}_2(t) \\ \dot{\mathbf{y}}_2(t) &= -\mathbf{M}^{-1}\mathbf{K}\mathbf{y}_1(t) - \mathbf{M}^{-1}\mathbf{C}\mathbf{y}_2(t) + \mathbf{M}^{-1}\mathbf{F}(t) \end{aligned} \quad (3.152)$$

The first equation follows from the definition in (3.151). The second is just the equation of motion in (3.150) but now written in terms of $\mathbf{y}_1$ and $\mathbf{y}_2$. The two equations in (3.152) can be written as a first order ODE in $\mathbf{y}(t)$:

$$\dot{\mathbf{y}}(t) = \underset{2n \times 1}{\mathbf{A}} \underset{2n \times 2n}{\mathbf{y}}(t) + \underset{2n \times n}{\mathbf{B}} \underset{n \times 1}{\mathbf{F}}(t) \quad (3.153)$$

$$\mathbf{A} = \begin{bmatrix} \mathbf{0}_n & \mathbf{I}_n \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} \mathbf{0}_n \\ \mathbf{M}^{-1} \end{bmatrix} \quad (3.154)$$

where $\mathbf{0}_n$ and $\mathbf{I}_n$ denote the $n \times n$ zero and identity matrix, respectively. The matrices $\mathbf{A}$ and $\mathbf{B}$ are often called the ‘system matrices’ of state-space dynamics.

Generalization

The above can be generalized. One can always convert a n -dimensional vector ODE of order p to a np -dimensional first order vector ODE:

$$\begin{aligned} & \mathbf{A}_p \mathbf{x}^{(p)} + \mathbf{A}_{p-1} \mathbf{x}^{(p-1)} + \cdots + \mathbf{A}_1 \mathbf{x}^{(1)} + \mathbf{A}_0 \mathbf{x} = \mathbf{F}(t) \\ \Leftrightarrow & \frac{d}{dt} \underbrace{\begin{bmatrix} \mathbf{x} \\ \mathbf{x}^{(1)} \\ \vdots \\ \mathbf{x}^{(p-1)} \end{bmatrix}}_{\substack{\dot{\mathbf{y}}(t) \\ np \times 1}} = \underbrace{\begin{bmatrix} \mathbf{0}_n & \mathbf{I}_n & & \\ \vdots & & \ddots & \\ \mathbf{0}_n & & & \mathbf{I}_n \\ -\mathbf{A}_p^{-1}\mathbf{A}_0 & -\mathbf{A}_p^{-1}\mathbf{A}_1 & \cdots & -\mathbf{A}_p^{-1}\mathbf{A}_{p-1} \end{bmatrix}}_{\substack{\mathbf{A} \\ np \times np}} \underbrace{\begin{bmatrix} \mathbf{x} \\ \mathbf{x}^{(1)} \\ \vdots \\ \mathbf{x}^{(p-1)} \end{bmatrix}}_{\substack{\mathbf{y}(t) \\ np \times 1}} + \underbrace{\begin{bmatrix} \mathbf{0}_n \\ \vdots \\ \mathbf{0}_n \\ \mathbf{A}_p^{-1} \end{bmatrix}}_{\substack{\mathbf{B} \\ np \times n}} \underbrace{\mathbf{F}(t)}_{n \times 1} \\ \Leftrightarrow & \dot{\mathbf{y}}(t) = \underset{np \times 1}{\mathbf{A}} \underset{np \times np}{\mathbf{y}}(t) + \underset{np \times n}{\mathbf{B}} \underset{n \times 1}{\mathbf{F}}(t) \end{aligned} \quad (3.155)$$

where $\mathbf{x}^{(r)}$ denotes the r th derivative of $\mathbf{x}(t)$ w.r.t. t ; the dependence on t has been omitted for simplicity.

Analytical Solution

The solution of the first order ODE $\dot{\mathbf{y}}(t) = \mathbf{A}\mathbf{y}(t) + \mathbf{B}\mathbf{F}(t)$ given the initial condition $\mathbf{y}(0)$ has a simple analytical form (Duhamel’s integral):

$$\mathbf{y}(t) = e^{\mathbf{A}t} \mathbf{y}(0) + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{F}(\tau) d\tau \quad (3.156)$$

The terms $e^{\mathbf{A}t}$ and $e^{\mathbf{A}(t-\tau)}$ are ‘matrix exponentials’, to be discussed next in Sect. 3.6.1. It can be shown that $e^{\mathbf{A}t}$ becomes the identity matrix when $t = 0$. Then it is clear that (3.156) satisfies the initial condition because the integral is zero when $t = 0$. It can also be shown that $de^{\mathbf{A}t}/dt = \mathbf{A}e^{\mathbf{A}t}$, which is analogous to the differentiation rule when $\mathbf{A}$ is a scalar. Using these results and the Leibniz rule of differentiation in (3.76), the following shows that (3.156) satisfies the state-space equation:

$$\begin{aligned}
\dot{\mathbf{y}}(t) &= \mathbf{A}e^{\mathbf{A}t}\mathbf{y}(0) + \underbrace{\mathbf{e}^{\mathbf{A}(t-t)}}_{\mathbf{I}}\mathbf{B}\mathbf{F}(t) + \int_0^t \mathbf{A}e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{F}(\tau)d\tau \\
&= \mathbf{A} \underbrace{\left[\mathbf{e}^{\mathbf{A}t}\mathbf{y}(0) + \int_0^t \mathbf{e}^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{F}(\tau)d\tau \right]}_{\mathbf{y}(t)} + \mathbf{B}\mathbf{F}(t) = \mathbf{A}\mathbf{y}(t) + \mathbf{B}\mathbf{F}(t)
\end{aligned} \tag{3.157}$$

where $\mathbf{I}$ denotes the identity matrix.

3.6.1 Matrix Exponential

Let $\mathbf{A}$ be a $q \times q$ real-valued matrix with eigenvalues $\{\lambda_i\}_{i=1}^q$ and linearly independent eigenvectors $\{\xi_i\}_{i=1}^q$, which are generally complex-valued. Assembling the eigenvalue equation $\mathbf{A}\xi_i = \lambda_i\xi_i$ row-wise for $i = 1, \dots, q$ gives

$$\mathbf{A} \underbrace{\begin{bmatrix} \xi_1 & \cdots & \xi_q \end{bmatrix}}_{\mathbf{Q}} = \underbrace{\begin{bmatrix} \xi_1 & \cdots & \xi_q \end{bmatrix}}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_q \end{bmatrix}}_{\Lambda} \tag{3.158}$$

$$\Rightarrow \mathbf{A}\mathbf{Q} = \mathbf{Q}\Lambda \quad \Lambda_{q \times q} = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_q \end{bmatrix} \quad \mathbf{Q}_{q \times q} = \begin{bmatrix} \xi_1 & \cdots & \xi_q \end{bmatrix} \tag{3.159}$$

Post-multiplying by $\mathbf{Q}^{-1}$ gives the ‘eigenmatrix representation’ of $\mathbf{A}$:

$$\mathbf{A} = \mathbf{Q}\Lambda\mathbf{Q}^{-1} \tag{3.160}$$

The matrix exponential $e^{\mathbf{A}t}$, or generally an arbitrary function of $\mathbf{A}$, is defined through eigenvalues and eigenvectors. Let $g(u)$ be a scalar-valued function of scalar variable u . Then $g(\mathbf{A})$ is defined as a matrix with eigenvalues $\{g(\lambda_i)\}_{i=1}^q$ and the same eigenvectors as $\mathbf{A}$:

$$g(\mathbf{A}) = \mathbf{Q}g(\Lambda)\mathbf{Q}^{-1} \tag{3.161}$$

where $g(\Lambda)$ for a diagonal matrix Λ is simply

$$g(\Lambda) = \begin{bmatrix} g(\lambda_1) & & \\ & \ddots & \\ & & g(\lambda_q) \end{bmatrix} \quad (3.162)$$

Taking $g(u) = e^{ut}$ gives

$$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_q t} \end{bmatrix} \quad e^{\mathbf{A}t} = \mathbf{Q} e^{\Lambda t} \mathbf{Q}^{-1} \quad (3.163)$$

Since $\mathbf{Q}$ does not depend on t ,

$$\begin{aligned} \frac{d}{dt} e^{\mathbf{A}t} &= \mathbf{Q} \left(\frac{d}{dt} e^{\Lambda t} \right) \mathbf{Q}^{-1} = \mathbf{Q} \begin{bmatrix} \lambda_1 e^{\lambda_1 t} & & \\ & \ddots & \\ & & \lambda_q e^{\lambda_q t} \end{bmatrix} \mathbf{Q}^{-1} \\ &= \underbrace{\mathbf{Q} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_q \end{bmatrix}}_{\mathbf{A}} \underbrace{\mathbf{Q}^{-1} \mathbf{Q} \begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_q t} \end{bmatrix} \mathbf{Q}^{-1}}_{e^{\mathbf{A}t}} = \mathbf{A} e^{\mathbf{A}t} \end{aligned} \quad (3.164)$$

This is analogous to the ordinary differentiation rule when $\mathbf{A}$ is a scalar. Similar analogies apply to many other functions or operations as well, although they should be verified case by case. For example, by similar steps one can show that

$$\sin^2 \mathbf{A} + \cos^2 \mathbf{A} = \mathbf{I} \quad \int_0^t e^{\mathbf{A}t} dt = \mathbf{A}^{-1} (e^{\mathbf{A}t} - \mathbf{I}) \quad (3.165)$$

where $\mathbf{I}$ denotes the identity matrix.

3.6.2 Eigenvalue Properties of System Matrix

When the system matrix $\mathbf{A}$ is given by (3.154) and the structure is classically damped, its eigenvalues and eigenvectors can be expressed analytically in terms of modal properties. As in Sect. 3.2.3, let $\{\omega_i\}_{i=1}^n$ and $\{\Psi_i\}_{i=1}^n$ be the natural frequencies and mode shapes of the structure, satisfying the eigenvalue equation

$\mathbf{K}\boldsymbol{\psi}_i = \omega_i^2 \mathbf{M}\boldsymbol{\psi}_i$ ($i = 1, \dots, n$). Also, let $\{\zeta_i\}_{i=1}^n$ be the modal damping ratios, i.e., $2\zeta_i\omega_i = (\boldsymbol{\psi}_i^T \mathbf{C}\boldsymbol{\psi}_i) / (\boldsymbol{\psi}_i^T \mathbf{M}\boldsymbol{\psi}_i)$. Then it can be shown that the eigenvalues of $\mathbf{A}$ appear in complex conjugate pairs given by

$$\begin{aligned}\lambda_i &= -\zeta_i\omega_i + \mathbf{i}\omega_i\sqrt{1-\zeta_i^2} \\ \lambda_{n+i} &= \overline{\lambda_i} = -\zeta_i\omega_i - \mathbf{i}\omega_i\sqrt{1-\zeta_i^2} \quad i = 1, \dots, n\end{aligned}\quad (3.166)$$

The eigenvector corresponding to λ_i ($i = 1, \dots, 2n$) is of the form

$$\boldsymbol{\xi}_i = \begin{bmatrix} \boldsymbol{\psi}_i \\ \lambda_i \boldsymbol{\psi}_i \end{bmatrix} \quad (\text{replace } \boldsymbol{\psi}_i \text{ by } \boldsymbol{\psi}_{i-n} \text{ for } i > n) \quad (3.167)$$

See the end for proof.

Example 3.8 (SDOF structure, state-space) Consider the SDOF equation $\ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) = p(t)$. In state-space form, the state vector is $\mathbf{y}(t) = [x(t), \dot{x}(t)]^T$ and the system matrices are

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\omega_1^2 & -2\zeta\omega_1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (3.168)$$

Here we evaluate the matrix exponential $e^{\mathbf{A}t}$ in terms of the modal properties ω_1 and ζ . For this purpose we first evaluate the eigenvalues and eigenmatrix of $\mathbf{A}$, as well as the inverse of the eigenmatrix.

Using (3.166), the eigenvalues of $\mathbf{A}$ are given by

$$\lambda_1 = -\zeta\omega_1 + \mathbf{i}\omega_d \quad \lambda_2 = -\zeta\omega_1 - \mathbf{i}\omega_d \quad (3.169)$$

where $\omega_d = \omega_1\sqrt{1-\zeta^2}$ is the damped natural frequency. Since there is only one DOF, the mode shape is just a scalar. We can simply take $\boldsymbol{\psi}_1 = 1$. Using (3.167), the eigenmatrix of $\mathbf{A}$ is

$$\mathbf{Q} = [\boldsymbol{\xi}_1, \boldsymbol{\xi}_2] = \begin{bmatrix} \boldsymbol{\psi}_1 & \boldsymbol{\psi}_1 \\ \lambda_1\boldsymbol{\psi}_1 & \lambda_2\boldsymbol{\psi}_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -\zeta\omega_1 + \mathbf{i}\omega_d & -\zeta\omega_1 - \mathbf{i}\omega_d \end{bmatrix} \quad (3.170)$$

The determinant and inverse of $\mathbf{Q}$ are given by

$$|\mathbf{Q}| = -2\mathbf{i}\omega_d \quad \mathbf{Q}^{-1} = \frac{1}{-2\mathbf{i}\omega_d} \begin{bmatrix} -\zeta\omega_1 - \mathbf{i}\omega_d & -1 \\ \zeta\omega_1 - \mathbf{i}\omega_d & 1 \end{bmatrix} \quad (3.171)$$

Substituting λ_1 , λ_2 , $\mathbf{Q}$ and $\mathbf{Q}^{-1}$ into (3.163) gives, after algebra,

$$\mathbf{e}^{\mathbf{A}t} = \begin{bmatrix} e^{-\zeta\omega_d t} \left(\cos \omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t \right) & \frac{e^{-\zeta\omega_d t}}{\omega_d} \sin \omega_d t \\ -\frac{\omega_1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_1 t} \sin \omega_d t & e^{-\zeta\omega_1 t} \left(\cos \omega_d t - \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t \right) \end{bmatrix} \quad (3.172)$$

The entries of $\mathbf{e}^{\mathbf{A}t}$ can be identified with the complementary free vibration responses in (3.20):

$$\mathbf{e}^{\mathbf{A}t} = \begin{bmatrix} g_1(t) & g_2(t) \\ \dot{g}_1(t) & \dot{g}_2(t) \end{bmatrix} \quad (3.173)$$

where $g_1(t)$ is the free vibration displacement due to unit initial displacement and zero velocity; and $g_2(t)$ is the free vibration displacement due to unit initial velocity and zero displacement. This can be reasoned by noting that the free vibration response of $\mathbf{y}(t) = [x(t), \dot{x}(t)]^T$ starting from initial condition $\mathbf{y}(0) = [x(0), \dot{x}(0)]^T$ is $\mathbf{y}(t) = \mathbf{e}^{\mathbf{A}t} \mathbf{y}(0)$. The expressions of $g_1(t)$ and $g_2(t)$, respectively in the (1,1)- and (1,2)- entry in (3.172), are identical to those in Table 3.2 of Sect. 3.1.3. ■

Derivation of eigenvalue properties in (3.166) and (3.167)

Here we show that if the structure is classically damped then the eigenvalues and eigenvectors of $\mathbf{A}$ in (3.154) are given by (3.166) and (3.167), respectively. Let λ be an eigenvalue and $\xi = [\mathbf{u}; \mathbf{v}]$ ($\mathbf{u}$ and $\mathbf{v}$ both $n \times 1$) be the corresponding eigenvector of $\mathbf{A}$. Then the eigenvalue equation $\mathbf{A}\xi = \lambda\xi$ reads

$$\begin{bmatrix} \mathbf{0}_n & \mathbf{I}_n \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \end{bmatrix} = \lambda \begin{bmatrix} \mathbf{u} \\ \mathbf{v} \end{bmatrix} \quad (3.174)$$

Writing out the upper and lower partitions,

$$\mathbf{v} = \lambda \mathbf{u} \quad -\mathbf{M}^{-1}\mathbf{K}\mathbf{u} - \mathbf{M}^{-1}\mathbf{C}\mathbf{v} = \lambda \mathbf{v} \quad (3.175)$$

Substituting the first equation into the second and pre-multiplying by $\mathbf{M}$ gives, after rearranging,

$$(\lambda^2 \mathbf{M} + \lambda \mathbf{C} + \mathbf{K})\mathbf{u} = \mathbf{0} \quad (3.176)$$

The eigenvalue λ is a solution of the characteristic equation

$$|\lambda^2 \mathbf{M} + \lambda \mathbf{C} + \mathbf{K}| = 0 \quad (3.177)$$

Note that $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ can be expressed as

$$\begin{aligned}\mathbf{M} &= \mathbf{M}\Psi \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \Psi^{-1} \quad \mathbf{C} = \mathbf{M}\Psi \begin{bmatrix} 2\zeta_1\omega_1 & & \\ & \ddots & \\ & & 2\zeta_n\omega_n \end{bmatrix} \Psi^{-1} \\ \mathbf{K} &= \mathbf{M}\Psi \begin{bmatrix} \omega_1^2 & & \\ & \ddots & \\ & & \omega_n^2 \end{bmatrix} \Psi^{-1}\end{aligned}\quad (3.178)$$

where $\Psi = [\psi_1, \dots, \psi_n]$. The expression of $\mathbf{M}$ is trivial. The expression of $\mathbf{K}$ follows from $\mathbf{K}\Psi = \mathbf{M}\Psi\Lambda$ ($\Lambda = \text{diag}(\{\omega_i^2\}_{i=1}^n)$), which is the assembled form of $\mathbf{K}\psi_i = \omega_i^2\mathbf{M}\psi_i$ ($i = 1, \dots, n$). The expression of $\mathbf{C}$ follows from classical damping assumption. Using (3.178),

$$\lambda^2\mathbf{M} + \lambda\mathbf{C} + \mathbf{K} = \mathbf{M}\Psi \begin{bmatrix} \lambda^2 + 2\zeta_1\omega_1\lambda + \omega_1^2 & & \\ & \ddots & \\ & & \lambda^2 + 2\zeta_n\omega_n\lambda + \omega_n^2 \end{bmatrix} \Psi^{-1}\quad (3.179)$$

Using the fact that the determinant of the product of square matrices is equal to the product of their determinants, and noting $|\Psi||\Psi^{-1}| = 1$, we have

$$|\lambda^2\mathbf{M} + \lambda\mathbf{C} + \mathbf{K}| = |\mathbf{M}| \prod_{j=1}^n (\lambda^2 + 2\zeta_j\omega_j\lambda + \omega_j^2)\quad (3.180)$$

Since $|\mathbf{M}| \neq 0$, λ must make at least one of the terms in the product on the RHS to be zero. First consider the case when it makes exactly one term to be zero, say, the i th term, i.e., $\lambda^2 + 2\zeta_i\omega_i\lambda + \omega_i^2 = 0$. Solving this equation for λ gives the two roots in (3.166). To determine the eigenvector, substitute (3.179) with $\lambda = \lambda_i$ into (3.176) and pre-multiply by $(\mathbf{M}\Psi)^{-1}$. Writing $\mathbf{u} = \Psi\mathbf{z}$, where $\mathbf{z} = [z_1, \dots, z_n]^T$, gives

$$\begin{bmatrix} \lambda_i^2 + 2\zeta_1\omega_1\lambda_i + \omega_1^2 & & \\ & \ddots & \\ & & \lambda_i^2 + 2\zeta_n\omega_n\lambda_i + \omega_n^2 \end{bmatrix} \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} = \mathbf{0}\quad (3.181)$$

The j th row reads

$$(\lambda_i^2 + 2\zeta_j\omega_j\lambda_i + \omega_j^2)z_j = 0 \quad j = 1, \dots, n\quad (3.182)$$

When $j = i$ the parenthesis on the LHS is zero and the equality is satisfied. When $j \neq i$ the parenthesis is not zero and so $z_j = 0$. Thus, only z_i is non-zero and $\mathbf{u} = z_i \boldsymbol{\psi}_i$, which is proportional to $\boldsymbol{\psi}_i$. Together with $\mathbf{v} = \lambda_i \mathbf{u}$, the eigenvector corresponding to λ_i is of the form in (3.167).

In the general case when λ_i makes m_i terms (say) in the product to be zero, the same argument shows that $\mathbf{u}$ lies in the m_i -dimensional subspace spanned by the corresponding $\boldsymbol{\psi}_i$ s but is otherwise arbitrary. We can then set the eigenvectors of these m_i repeated eigenvalues as the $\boldsymbol{\psi}_i$ s so that the eigenvector $\boldsymbol{\xi}$ is still of the form in (3.167). ■

3.7 Time Integration Scheme

In the previous sections, the response of a structure was determined analytically in terms of the excitation time history. As a common situation in applications, one needs to compute the response based on a given discrete-time excitation history. The latter may come from real measurements (e.g., ground acceleration during an earthquake) or it may be synthetically generated (e.g., in a parametric study). A ‘time integration scheme’ sequentially calculates a discrete-time approximation of the response at the future time instant based on information up to the present.

In this section we introduce the basic concepts of time integration. This is followed by the ‘Newmark method’ in Sect. 3.8, which is a popular time integration scheme in structural dynamic applications. The scope is confined to linear time-invariant dynamics, although the concepts can be applied in a more general context to time-variant nonlinear dynamics. For general discussions on time integration methods, see Hildebrand (1987), Hughes (1987), Subbaraj and Dokainish (1989 a, b) and Hulbert (2004).

3.7.1 Numerical Stability and Accuracy

A time integration scheme is developed based on a discrete-time approximation of the governing equation. As a basic requirement, it should be ‘consistent’ (error converges to zero as time interval decreases) and ‘numerically stable’ (error does not grow unbounded with time). A further consideration is accuracy, or ‘convergence rate’, i.e., the order at which error scales with the time interval. Depending on the approximation applied, different schemes can have different stability and accuracy characteristics.

To introduce these concepts, consider the first order equation

$$\dot{x}(t) + ax(t) = w(t) \quad x(0) = x_0 \quad (3.183)$$

Assume $a > 0$ so that the system is stable. Given the values of $w_j = w(t_j)$ at time instants $t_j = j\Delta t$ ($j = 0, 1, 2, \dots$) of interval Δt , we want to determine $x(t)$ at these time instants. Let the computed response at t_j be $\tilde{x}_j$. The tilde ‘ $\sim$ ’ reminds that $\tilde{x}_j$ is only an approximation to $x(t_j)$.

Discrete-Time Approximation

To develop a time integration scheme, suppose we make the ‘forward-difference’ approximation:

$$\dot{x}(t_j) \approx \frac{x(t_{j+1}) - x(t_j)}{\Delta t} \quad (3.184)$$

Considering (3.183) at t_j and replacing $\dot{x}(t_j)$ by $(\tilde{x}_{j+1} - \tilde{x}_j)/\Delta t$ gives $(\tilde{x}_{j+1} - \tilde{x}_j)/\Delta t + a\tilde{x}_j = w_j$. Rearranging,

$$\begin{aligned} \tilde{x}_{j+1} &= A\tilde{x}_j + w_j\Delta t \\ A &= 1 - a\Delta t \end{aligned} \quad j = 1, 2, \dots \quad (3.185)$$

This recurrence relationship allows one to compute $\tilde{x}_j$ sequentially. Starting from $\tilde{x}_0 = x_0$ (initial condition), $\tilde{x}_1$ can be determined from $\tilde{x}_0$, then $\tilde{x}_2$ from $\tilde{x}_1$, and so on.

Numerical Stability

The scheme (3.185) is supposed to give an approximation $\tilde{x}_j$ of $x(t_j)$ but it need not even do a ‘proper’ job. By sequential substitution,

$$\begin{aligned} \tilde{x}_{j+1} &= A \overbrace{(A\tilde{x}_{j-1} + w_{j-1}\Delta t)}^{\tilde{x}_j} + w_j\Delta t = A^2\tilde{x}_{j-1} + (w_j + w_{j-1}A)\Delta t = \dots \\ &= A^{j+1}\tilde{x}_0 + \sum_{r=0}^j w_{j-r}A^r\Delta t \end{aligned} \quad (3.186)$$

If $|A| > 1$, the first term on the RHS and hence $\tilde{x}_{j+1}$ will grow unbounded with the time index j ; the sum will be dominated by terms with large r , i.e., by excitations that are in the far past. This is the case even when the original system is stable. Such a scheme is ‘numerically unstable’ and is clearly not doing a proper job. A proper scheme must be ‘numerically stable’, returning bounded output for bounded input when the original system is stable. Here it requires $|A| \leq 1$, i.e., $0 \leq \Delta t \leq 2/a$. This effectively puts an upper bound on Δt , besides accuracy requirement.

Error Propagation and Convergence Rate

The error in $\tilde{x}_j$ propagates with the same dynamic characteristics as itself. Let

$$\varepsilon_j = \tilde{x}_j - x(t_j) \quad (3.187)$$

be the error at t_j . As a refined statement of (3.184), by Taylor series,

$$x(t_{j+1}) = x(t_j) + \dot{x}(t_j)\Delta t + R_j \quad (3.188)$$

where R_j is the remainder term. Generally, it is $O(\Delta t^2)$, i.e., $\lim_{\Delta t \rightarrow 0} R_j/\Delta t^2$ is non-zero and finite. Substituting $\dot{x}(t_j) = -ax(t_j) + w_j$ gives

$$x(t_{j+1}) = Ax(t_j) + w_j\Delta t + R_j \quad A = 1 - a\Delta t \quad (3.189)$$

Subtracting this equation from (3.185) and using the definition of ε_j gives

$$\varepsilon_{j+1} = A\varepsilon_j + R_j \quad (3.190)$$

Compared with (3.185), ε_j evolves with time by the same mechanism as $\tilde{x}_j$ but now it is driven by R_j . If the scheme is numerically unstable, i.e., $|A| > 1$, then ε_j will grow unbounded as time goes on. Similar to (3.186),

$$\varepsilon_{j+1} = A^{j+1}\varepsilon_0 + \underbrace{\sum_{r=0}^j \overbrace{R_{j-r}}^{O(\Delta t^2)} A^r}_{O(\Delta t)} \quad (3.191)$$

The sum on the RHS can be reasoned to be $O(\Delta t)$ by writing it as a Riemann sum. This shows that the error is $O(\Delta t)$ and the convergence rate is first order.

Example 3.9 (First order process) Consider the first order equation in (3.183) with $a = 1$, $w(t) \equiv 0$ and $x(0) = 1$. Numerical stability requires $\Delta t \leq 2$ s. Figure 3.14 shows the computed response $\tilde{x}_j$ (line with dots) for $\Delta t = 0.5$ s (stable), 2 s (marginally stable) and 2.1 s (unstable). These pictures also indicate how an initial numerical error of 1 unit is propagated. In (a) it diminishes with time; in (b) it stays; in (c) it grows.

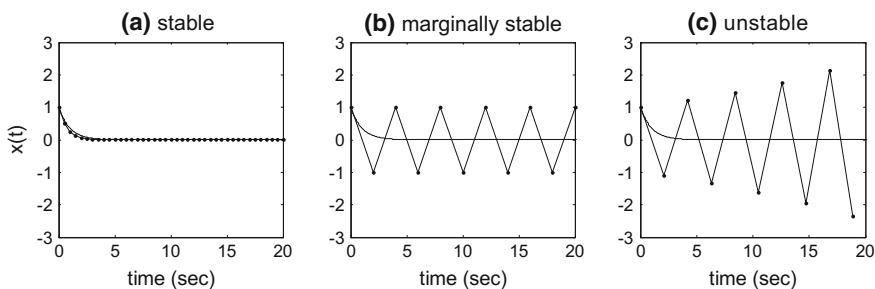


Fig. 3.14 Numerical stability of a first order equation. *Dashed line exact solution, line with dots numerical solution.* **a** $\Delta t = 0.5$ s (stable); **b** $\Delta t = 2$ s (marginally stable); **c** $\Delta t = 2.1$ s (unstable)

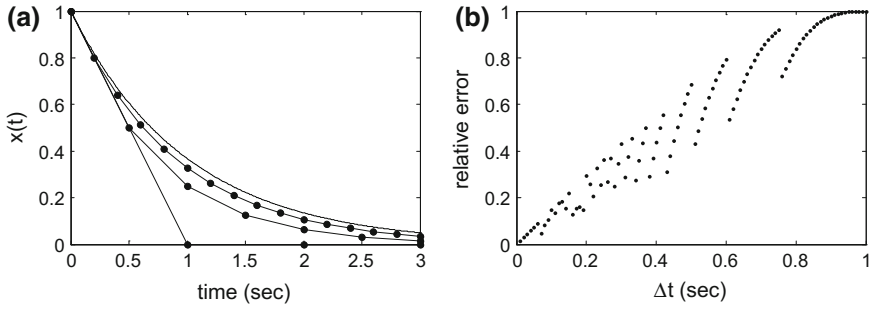


Fig. 3.15 Numerical solution of first order equation. In **a**, from top to bottom: exact, $\Delta t = 0.2, 0.5$ and 1 s; **b** relative error $|\tilde{x}_j - x(t_j)|/|x(t_j)|$ at $t_j = 3$ s

A numerically stable solution need not be accurate. Figure 3.15a shows $\tilde{x}_j$ (line with dots) for $\Delta t = 0.2, 0.5$ and 1 s, all numerically stable. In (b), the relative error $|\tilde{x}_j - x(t_j)|/|x(t_j)|$ at $t_j = 3$ s is plotted versus Δt . Here, the exact solution is $x(t) = e^{-at}$. Viewing macroscopically, for small Δt the error grows linearly. This agrees with the fact that the convergence rate is first order. ■

3.7.2 Discrete-Time State-Space Analysis

The forgoing concepts can be extended to general time integration schemes. One can write the recurrence relationship in a first order discrete-time state-space form:

$$\tilde{\mathbf{y}}_{j+1} = \mathbf{A}\tilde{\mathbf{y}}_j + \mathbf{B}_j\mathbf{w}_j \quad (3.192)$$

where $\tilde{\mathbf{y}}_j$ denotes a vector of quantities being updated from one time step to another; $\mathbf{A}$ and $\mathbf{B}_j$ are matrices that depend on the scheme; $\mathbf{w}_j$ denotes a vector containing the excitations. The scheme is stable if and only if all eigenvalues of $\mathbf{A}$ have a modulus bounded above by 1. To see this, analogous to (3.186),

$$\tilde{\mathbf{y}}_{j+1} = \mathbf{A}^{j+1}\tilde{\mathbf{y}}_0 + \sum_{r=0}^j \mathbf{A}^r \mathbf{B}_{j-r} \mathbf{w}_{j-r} \quad (3.193)$$

Assume that the eigenvectors of $\mathbf{A}$ are linearly independent. One can write $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}$ where $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues and $\mathbf{Q}$ contains in its columns the eigenvectors (Sect. 3.6). Then

$$\mathbf{A}^j = \underbrace{(\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1})(\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}) \cdots (\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1})}_{j \text{ times}} = \mathbf{Q}\mathbf{\Lambda}^j\mathbf{Q}^{-1} \quad (3.194)$$

The diagonal entries of Λ^j are simply the j th power of the eigenvalues. If there is an eigenvalue with a modulus greater than 1, the corresponding entry in Λ^j will grow unbounded with j ; and so will $\tilde{\mathbf{y}}_j$.

For a time integration scheme, stability criteria are established to guide the choice of algorithmic parameters (e.g., Δt). In theory, they can be derived directly from the analytical expression of eigenvalues. This is difficult when the latter is mathematically intractable. Methods have been developed to determine stability based on the coefficients in the characteristic equation, bypassing the determination of eigenvalues. The Ruth-Hurwitz test is a classical method often discussed in control theory for determining whether the roots of a polynomial equation (hence eigenvalues) have a negative real part, which is relevant for stability analysis of continuous-time systems. By making a transformation of the variable in the equation, the method can be adapted to investigate whether the roots have a modulus less than 1. This is relevant for discrete-time time systems and hence stability analysis of time integration schemes. Other methods are available, e.g., the Bistritz test (Bistritz 1984, 2002).

Example 3.10 (Central difference, numerical stability) Consider the SDOF equation $\ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) = p(t)$. Suppose we develop a scheme using the ‘central difference’ approximation of $\dot{x}$ and $\ddot{x}$ (Fig. 3.16):

$$\tilde{x}_j = \frac{\tilde{x}_{j+1} - \tilde{x}_{j-1}}{2\Delta t} \quad \tilde{\ddot{x}}_j = \frac{1}{\Delta t} \underbrace{\left[\frac{(\tilde{x}_{j+1} - \tilde{x}_j)}{\Delta t} - \frac{(\tilde{x}_j - \tilde{x}_{j-1}))}{\Delta t} \right]}_{\text{approx. change of velocity during } \Delta t} = \frac{\tilde{x}_{j+1} - 2\tilde{x}_j + \tilde{x}_{j-1}}{\Delta t^2} \quad (3.195)$$

Substituting (3.195) into $\tilde{\ddot{x}}_j + 2\zeta\omega_1\tilde{\dot{x}}_j + \omega_1^2\tilde{x}_j = p_j$ and solving for $\tilde{x}_{j+1}$ gives

$$\begin{aligned} \tilde{x}_{j+1} &= a_1\tilde{x}_j - a_2\tilde{x}_{j-1} + a_3p_j \\ a_1 &= \frac{2 - \omega_1^2\Delta t^2}{1 + \zeta\omega_1\Delta t} \quad a_2 = \frac{1 - \zeta\omega_1\Delta t}{1 + \zeta\omega_1\Delta t} \quad a_3 = \frac{\Delta t^2}{1 + \zeta\omega_1\Delta t} \end{aligned} \quad (3.196)$$

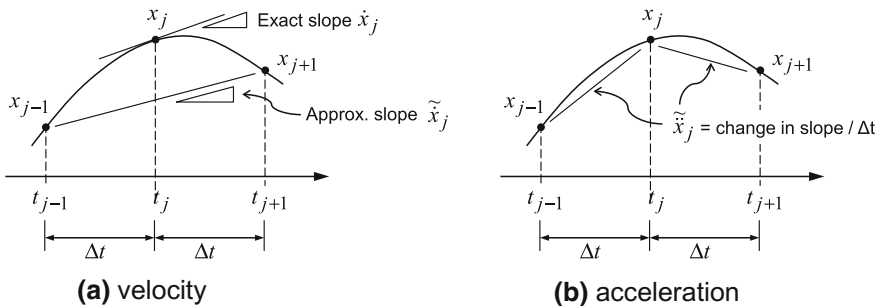


Fig. 3.16 Central difference approximation, **a** for velocity; **b** for acceleration

This scheme allows one to determine the future ($\tilde{x}_{j+1}$) based on the present ($\tilde{x}_j$) and the nearest past ($\tilde{x}_{j-1}$). To investigate numerical stability, define the state vector $\tilde{\mathbf{y}}_j = [\tilde{x}_j, \tilde{x}_{j-1}]^T$. It changes according to

$$\underbrace{\begin{bmatrix} \tilde{x}_{j+1} \\ \tilde{x}_j \end{bmatrix}}_{\tilde{\mathbf{y}}_{j+1}} = \underbrace{\begin{bmatrix} a_1 & -a_2 \\ 1 & 0 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \tilde{x}_j \\ \tilde{x}_{j-1} \end{bmatrix}}_{\tilde{\mathbf{y}}_j} + \underbrace{\begin{bmatrix} a_3 \\ 0 \end{bmatrix}}_{\mathbf{B}_j} p_j \quad (3.197)$$

which is in the form of (3.192). The first row comes from the first equation in (3.196). The second row is just $\tilde{x}_j = \tilde{x}_j$, which is trivial but necessary to make up the state-space equation. The scheme is stable if and only if all eigenvalues of $\mathbf{A}$ have a modulus bounded above by 1. The eigenvalues of $\mathbf{A}$ are solutions of the characteristic equation:

$$|\mathbf{A} - \lambda \mathbf{I}| = 0 \quad \Rightarrow \quad \lambda^2 - a_1 \lambda + a_2 = 0 \quad (3.198)$$

Solving the equation gives two roots, $\lambda = (a_1 \pm \sqrt{a_1^2 - 4a_2})/2$.

A simple stability criterion can be derived by considering (conservatively) $\zeta = 0$. In this case, $a_1 = 2 - \omega_1^2 \Delta t^2$, $a_2 = 1$ and $\lambda = (a_1 \pm \sqrt{a_1^2 - 4})/2$. If $a_1^2 - 4 > 0$, there are two distinct real roots. Since their product is equal to 1 ($= a_2$), one of them must have an absolute value greater than 1 and so the scheme is unstable. If $a_1^2 - 4 \leq 0$, the two roots are complex conjugate of each other and so they have the same modulus. Since their product is equal to 1, their modulus must both be equal to 1, and so the scheme is stable. In conclusion, when $\zeta = 0$, the scheme is stable if and only if $a_1^2 - 4 \leq 0$, i.e., $0 \leq \Delta t \leq T/\pi$ where $T = 2\pi/\omega_1$ is the natural period. In the general case when $\zeta > 0$, this is only a sufficient condition (i.e., possibly more stringent than it needs to be) but is simple enough to apply regardless of ζ . ■

3.8 Newmark Scheme

We now introduce the Newmark method for time integration, which is a general and popular scheme in structural dynamics. Consider first the SDOF equation:

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = F(t) \quad (3.199)$$

Newmark scheme is developed based on the approximation of $\ddot{x}(t)$.

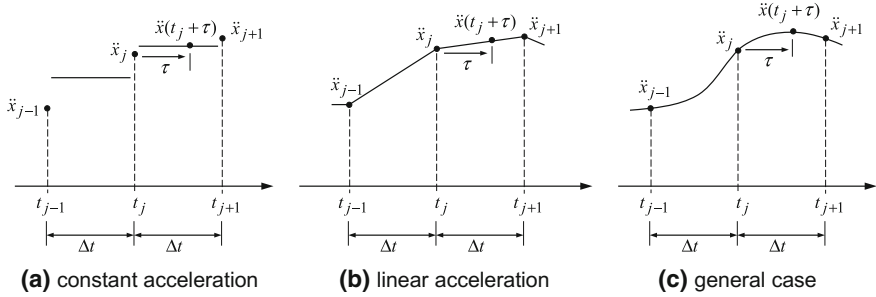


Fig. 3.17 Approximation in Newmark scheme. **a** Constant acceleration; **b** linear acceleration; **c** general case

3.8.1 SDOF Linear Acceleration

Suppose $\ddot{x}(t)$ varies piecewise linearly from one time step to another, as in Fig. 3.17b. Then for $0 \leq \tau \leq \Delta t$,

$$\ddot{x}(t_j + \tau) = \ddot{x}(t_j) + [\ddot{x}(t_{j+1}) - \ddot{x}(t_j)] \frac{\tau}{\Delta t} \quad (3.200)$$

Integrating successively w.r.t. τ from $\tau = 0$ gives, after rearranging,

$$\begin{aligned} \dot{x}(t_j + \tau) &= \dot{x}(t_j) + \tau \ddot{x}(t_j) + \frac{\tau^2}{2\Delta t} [\ddot{x}(t_{j+1}) - \ddot{x}(t_j)] \\ x(t_j + \tau) &= x(t_j) + \tau \dot{x}(t_j) + \frac{1}{2} \tau^2 \ddot{x}(t_j) + \frac{\tau^3}{6\Delta t} [\ddot{x}(t_{j+1}) - \ddot{x}(t_j)] \end{aligned} \quad (3.201)$$

Evaluating at $\tau = \Delta t$, noting $\dot{x}(t_j + \Delta t) = \dot{x}(t_{j+1})$ and $x(t_j + \Delta t) = x(t_{j+1})$, and collecting terms,

$$\begin{aligned} \dot{x}(t_{j+1}) &= \underbrace{\dot{x}(t_j) + \frac{\Delta t}{2} \ddot{x}(t_j)}_{\text{present}} + \underbrace{\frac{\Delta t}{2} \ddot{x}(t_{j+1})}_{\text{future}} \\ x(t_{j+1}) &= \underbrace{x(t_j) + \Delta t \dot{x}(t_j) + \frac{\Delta t^2}{3} \ddot{x}(t_j)}_{\text{present}} + \underbrace{\frac{\Delta t^2}{6} \ddot{x}(t_{j+1})}_{\text{future}} \end{aligned} \quad (3.202)$$

Suppose $\{x(t_j), \dot{x}(t_j), \ddot{x}(t_j)\}$ at the present time step t_j are known. The above equations indicate that, in order to determine $x(t_{j+1})$ and $\dot{x}(t_{j+1})$ at the future time step t_{j+1} , one only needs to know the future acceleration $\ddot{x}(t_{j+1})$. Substituting them into (3.199) at $t = t_{j+1}$ and solving gives $\ddot{x}(t_{j+1})$ in terms of the present responses $\{x(t_j), \dot{x}(t_j), \ddot{x}(t_j)\}$ and the future excitation $F(t_{j+1})$. Here, the computed response is exact when $\ddot{x}(t)$ is piecewise linear within the time interval. Otherwise it is only approximate.

3.8.2 SDOF General Scheme

Linear acceleration approximation is a special case in the Newmark scheme. In the general case (Fig. 3.17c), one can write

$$\ddot{x}(t_j + \tau) = \ddot{x}(t_j) + [\ddot{x}(t_{j+1}) - \ddot{x}(t_j)]r_j(\tau) \quad 0 \leq \tau \leq \Delta t \quad (3.203)$$

for some (unknown) function $r_j(\tau)$ with the property that $r_j(0) = 0$ and $r_j(\Delta t) = 1$. Following similar steps as before, one obtains

$$\begin{aligned} \dot{x}(t_{j+1}) &= \underbrace{\dot{x}(t_j) + (1 - \gamma_j)\Delta t \ddot{x}(t_j)}_{\text{present}} + \underbrace{\gamma_j \Delta t \ddot{x}(t_{j+1})}_{\text{future}} \\ x(t_{j+1}) &= \underbrace{x(t_j) + \Delta t \dot{x}(t_j) + \left(\frac{1}{2} - \beta_j\right)\Delta t^2 \ddot{x}(t_j)}_{\text{present}} + \underbrace{\beta_j \Delta t^2 \ddot{x}(t_{j+1})}_{\text{future}} \end{aligned} \quad (3.204)$$

where

$$\gamma_j = \frac{1}{\Delta t} \int_0^{\Delta t} r_j(\tau) d\tau \quad \beta_j = \frac{1}{\Delta t^2} \int_0^{\Delta t} \int_0^u r_j(s) ds du \quad (3.205)$$

Of course, γ_j and β_j are generally unknown and they may depend on j . Taking them (heuristically) to be constant, say, γ and β , leads to a general approximation in the Newmark scheme:

$$\begin{aligned} \tilde{\ddot{x}}_{j+1} &= \tilde{\ddot{x}}_j + (1 - \gamma)\Delta t \tilde{\ddot{x}}_j + \gamma \Delta t \tilde{\ddot{x}}_{j+1} \\ \tilde{x}_{j+1} &= \tilde{x}_j + \Delta t \tilde{\dot{x}}_j + \left(\frac{1}{2} - \beta\right)\Delta t^2 \tilde{\ddot{x}}_j + \beta \Delta t^2 \tilde{\ddot{x}}_{j+1} \end{aligned} \quad (3.206)$$

Again, the tilde ‘ $\sim$ ’ reminds that the quantity is only an approximation. Substituting into (3.199) at $t = t_{j+1}$ and rearranging gives

$$\begin{aligned} \tilde{\ddot{x}}_{j+1} &= D^{-1} P_{j+1} \\ D &= m + \gamma \Delta t c + \beta \Delta t^2 k \\ P_{j+1} &= F_{j+1} - c [\tilde{x}_j + (1 - \gamma)\Delta t \tilde{\dot{x}}_j] - k \left[\tilde{x}_j + \Delta t \tilde{\dot{x}}_j + \left(\frac{1}{2} - \beta\right)\Delta t^2 \tilde{\ddot{x}}_j \right] \end{aligned} \quad (3.207)$$

where $F_{j+1} = F(t_{j+1})$.

Given the initial conditions $x(0) = u$ and $\dot{x}(0) = v$, the value of $\ddot{x}_0$ can be obtained from (3.199) at $t = 0$ to be $\ddot{x}_0 = m^{-1}(F_0 - cv - ku)$. The Newmark scheme is summarized as follow.

Newmark scheme for SDOF Eq. (3.199)

Given $x(0) = u$, $\dot{x}(0) = v$ and $\{F_j = F(j\Delta t) : j = 0, 1, 2, \dots\}$

Step 1. Set $\tilde{x}_0 = u$, $\dot{\tilde{x}}_0 = v$ and $\ddot{\tilde{x}}_0 = m^{-1}(F_0 - cv - ku)$

Step 2. For $j = 0, 1, 2, \dots$

2.1. Calculate $\ddot{\tilde{x}}_{j+1}$ from (3.207).

2.2. Calculate $\dot{\tilde{x}}_{j+1}$ and $\tilde{x}_{j+1}$ from (3.206).

End for j

3.8.3 General MDOF Scheme

The SDOF Newmark scheme can be directly generalized for solving the MDOF equation:

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) = \mathbf{F}(t) \quad (3.208)$$

In this case, (3.206) and (3.207) read

$$\begin{aligned} \ddot{\tilde{\mathbf{x}}}_{j+1} &= \ddot{\tilde{\mathbf{x}}}_j + (1 - \gamma)\Delta t \ddot{\tilde{\mathbf{x}}}_j + \gamma\Delta t \ddot{\tilde{\mathbf{x}}}_{j+1} \\ \tilde{\mathbf{x}}_{j+1} &= \tilde{\mathbf{x}}_j + \Delta t \dot{\tilde{\mathbf{x}}}_j + \left(\frac{1}{2} - \beta\right)\Delta t^2 \ddot{\tilde{\mathbf{x}}}_j + \beta\Delta t^2 \ddot{\tilde{\mathbf{x}}}_{j+1} \end{aligned} \quad (3.209)$$

$$\begin{aligned} \ddot{\tilde{\mathbf{x}}}_{j+1} &= \mathbf{D}^{-1}\mathbf{P}_{j+1} \\ \mathbf{D} &= \mathbf{M} + \gamma\Delta t \mathbf{C} + \beta\Delta t^2 \mathbf{K} \\ \mathbf{P}_{j+1} &= \mathbf{F}_{j+1} - \mathbf{C}[\dot{\tilde{\mathbf{x}}}_j + (1 - \gamma)\Delta t \ddot{\tilde{\mathbf{x}}}_j] - \mathbf{K}\left[\tilde{\mathbf{x}}_j + \Delta t \dot{\tilde{\mathbf{x}}}_j + \left(\frac{1}{2} - \beta\right)\Delta t^2 \ddot{\tilde{\mathbf{x}}}_j\right] \end{aligned} \quad (3.210)$$

Note that $\mathbf{D}$ does not depend on the time step j and so its inverse only needs to be computed once in the beginning.

Newmark scheme for MDOF Eq. (3.208)

Given $\mathbf{x}(0) = \mathbf{u}$, $\dot{\mathbf{x}}(0) = \mathbf{v}$ and $\{\mathbf{F}_j = \mathbf{F}(j\Delta t) : j = 0, 1, 2, \dots\}$

Step 1. Set $\tilde{\mathbf{x}}_0 = \mathbf{u}$, $\dot{\tilde{\mathbf{x}}}_0 = \mathbf{v}$ and $\ddot{\tilde{\mathbf{x}}}_0 = \mathbf{M}^{-1}(\mathbf{F}_0 - \mathbf{C}\mathbf{v} - \mathbf{K}\mathbf{u})$

Step 2. For $j = 0, 1, 2, \dots$

2.1. Calculate $\ddot{\tilde{\mathbf{x}}}_{j+1}$ from (3.210).

2.2. Calculate $\tilde{\mathbf{x}}_{j+1}$ and $\dot{\tilde{\mathbf{x}}}_{j+1}$ from (3.209).

End for j

3.8.4 Parameters and Numerical Stability

The numerical properties of the Newmark scheme depend on the choice of γ and β . The linear acceleration scheme corresponds to $\gamma = 1/2$ and $\beta = 1/6$. Setting $\gamma = 1/2$ and $\beta = 1/4$ gives the ‘constant acceleration scheme’, which is exact if the acceleration is piecewise constant. See Fig. 3.17a, for which $r_j(\tau) \equiv 1/2$ in (3.203). Setting $\gamma = 1/2$ and $\beta = 0$ gives the central difference scheme; see Example 3.10 and Example 3.11. Using $\gamma > 1/2$ introduces artificial positive damping and $\gamma < 1/2$ introduces artificial negative damping. In most applications, $\gamma = 1/2$ is used, for which there is no artificial damping and the accuracy is second order.

Assume (conservatively) zero structural damping. Then the Newmark scheme is stable if $\gamma \geq 1/2$ and $\beta \geq \gamma/2$, regardless of Δt . If $\gamma \geq 1/2$ but $\beta < \gamma/2$, it is only conditionally stable, requiring $\Delta t/T \leq 1/2\pi\sqrt{\gamma/2 - \beta}$, where T is the shortest period of the structure. The constant acceleration scheme is therefore unconditionally stable. The linear acceleration and central difference scheme are only conditionally stable. Substituting $\gamma = 1/2$ and $\beta = 1/6$ gives the stability criterion $\Delta t/T \leq \sqrt{3}/\pi$ for the linear acceleration scheme. Substituting $\gamma = 1/2$ and $\beta = 0$ gives $\Delta t/T \leq 1/\pi$ for the central difference scheme, which agrees with the finding in Example 3.10. For structures with a large number of DOFs, the highest mode can have a very short period. This can impose a requirement on Δt that is much more stringent than is necessary for acceptable accuracy. In this sense the constant acceleration scheme is a more robust choice. Figure 3.18 summarizes the stability criteria for the Newmark scheme for zero damping.

Example 3.11 (Central difference as special case of Newmark scheme) This example shows that setting $\gamma = 1/2$ and $\beta = 0$ in the Newmark scheme gives the central difference scheme in Example 3.10, which may not be obvious from the recurrence equations. Substituting these values into (3.206) gives

$$\tilde{\mathbf{x}}_{j+1} = \tilde{\mathbf{x}}_j + \Delta t \dot{\tilde{\mathbf{x}}}_j + \frac{\Delta t^2}{2} \ddot{\tilde{\mathbf{x}}}_j \quad (3.211)$$

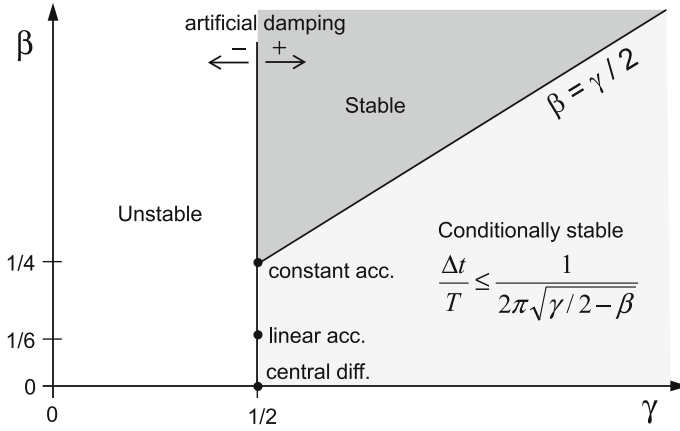


Fig. 3.18 Stability diagram of the Newmark scheme for zero damping

$$\tilde{x}_{j+1} = \tilde{x}_j + \frac{\Delta t}{2} \tilde{\ddot{x}}_j + \frac{\Delta t}{2} \tilde{\ddot{x}}_{j+1} \quad (3.212)$$

For the central difference scheme, from the first equation of (3.195), $\tilde{x}_{j-1} = \tilde{x}_{j+1} - 2\Delta t \tilde{\ddot{x}}_j$. Substituting into the expression of $\tilde{\ddot{x}}_j$ in (3.195) and solving for $\tilde{x}_{j+1}$ gives (3.211). On the other hand, replacing j by $j+1$ in (3.195) gives

$$\tilde{\ddot{x}}_{j+1} = \frac{\tilde{x}_{j+2} - \tilde{x}_j}{2\Delta t} \quad (3.213)$$

Replacing j by $j+1$ in (3.211), which has just been proven,

$$\tilde{x}_{j+2} = \tilde{x}_{j+1} + \Delta t \tilde{\ddot{x}}_{j+1} + \frac{\Delta t^2}{2} \tilde{\ddot{x}}_{j+1} = \tilde{x}_j + \underbrace{\Delta t \tilde{\ddot{x}}_j + \frac{\Delta t^2}{2} \tilde{\ddot{x}}_j}_{\tilde{x}_{j+1}} + \Delta t \tilde{\ddot{x}}_{j+1} + \frac{\Delta t^2}{2} \tilde{\ddot{x}}_{j+1} \quad (3.214)$$

Substituting into (3.213) and solving for $\tilde{x}_{j+1}$ gives (3.212). ■

3.8.5 Derivation of Stability Criterion

The stability criterion for the Newmark scheme can be derived using the concepts in Sect. 3.7.2. For simplicity, assume zero structural damping, which also leads to a conservative criterion. Consider first the SDOF equation in (3.199). For convenience, rewrite it as

$$\ddot{x}(t) + \omega_1^2 x(t) = p(t) \quad (3.215)$$

where $\omega_1 = \sqrt{k/m}$ is the natural frequency (rad/s) and $p(t) = F(t)/m$. Substitute $\tilde{\tilde{x}}_j = p_j - \omega_1^2 \tilde{x}_j$ and $\tilde{\tilde{x}}_{j+1} = p_{j+1} - \omega_1^2 \tilde{x}_{j+1}$ into (3.206). Writing in matrix form,

$$\underbrace{\begin{bmatrix} \tilde{\tilde{x}}_{j+1} \\ \tilde{\tilde{x}}_{j+1} \end{bmatrix}}_{\tilde{\mathbf{y}}_{j+1}} = \underbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} \tilde{\tilde{x}}_j \\ \tilde{\tilde{x}}_j \end{bmatrix}}_{\tilde{\mathbf{y}}_j} + \underbrace{\mathbf{B}}_{\substack{2 \times 2 \\ \text{irrelevant}}} \underbrace{\begin{bmatrix} p_j \\ p_{j+1} \end{bmatrix}}_{\mathbf{w}_{j+1}} \quad (3.216)$$

$$\begin{aligned} a_{11} &= 1 - \frac{\alpha^2}{2(1 + \beta\alpha^2)} & a_{12} &= \frac{\Delta t}{1 + \beta\alpha^2} \\ a_{21} &= \frac{\alpha^2}{\Delta t} \left[\frac{\gamma\alpha^2}{2(1 + \beta\alpha^2)} - 1 \right] & a_{22} &= 1 - \frac{\gamma\alpha^2}{1 + \beta\alpha^2} \\ \alpha &= \omega_1 \Delta t \end{aligned} \quad (3.217)$$

The eigenvalue of $\mathbf{A}$ satisfies the characteristic equation:

$$|\mathbf{A} - \lambda \mathbf{I}| = 0 \quad \Rightarrow \quad \lambda^2 - b\lambda + c = 0 \quad (3.218)$$

where

$$\begin{aligned} b &= a_{11} + a_{22} & c &= a_{11}a_{22} - a_{12}a_{21} \\ &= 2 - \frac{\alpha^2(1 + 2\gamma)}{2(1 + \beta\alpha^2)} & &= 1 - \frac{\alpha^2(\gamma - 1/2)}{1 + \beta\alpha^2} \end{aligned} \quad (3.219)$$

Solving (3.218) gives two eigenvalues:

$$\lambda_1 = \frac{1}{2} \left(b + \sqrt{b^2 - 4c} \right) \quad \lambda_2 = \frac{1}{2} \left(b - \sqrt{b^2 - 4c} \right) \quad (3.220)$$

There are two cases to consider:

- (1) If $b^2 - 4c \leq 0$, the eigenvalues are complex conjugates (including the case of repeated roots). Since their product is c , and $c \geq b^2/4 \geq 0$, the modulus of the eigenvalues are both equal to $\sqrt{c}$. Stability then requires $c \leq 1$.
- (2) If $b^2 - 4c > 0$, the eigenvalues are real and distinct. If $b \geq 0$ then $\lambda_1 > 0$ and $|\lambda_1| \geq |\lambda_2|$. Stability then requires $\lambda_1 \leq 1$. Solving this inequality gives $b - c \leq 1$. Similarly, if $b < 0$ then $\lambda_2 < 0$ and $|\lambda_1| < |\lambda_2|$. Stability then requires $\lambda_2 \geq -1$. Solving this inequality gives $b + c \geq -1$.

Combining the above considerations, the value of (b, c) when the scheme is stable is shaded in Fig. 3.19. The eigenvalues are complex conjugates, real-repeated or real-distinct when (b, c) is above, on, or below the parabola, respectively.

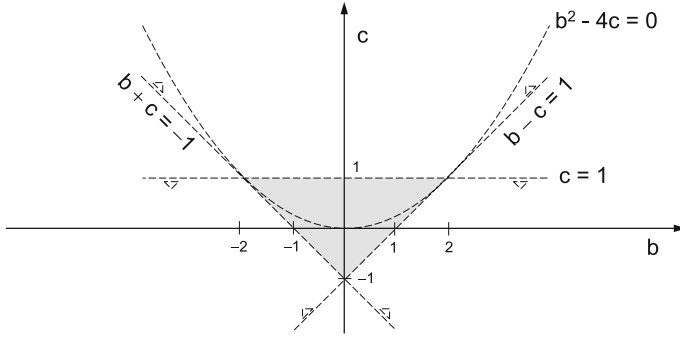


Fig. 3.19 Stability region (*shaded* and including boundaries)

The figure shows that the scheme is stable if and only if all the following conditions are met:

$$c \leq 1 \quad b - c \leq 1 \quad b + c \geq -1 \quad (3.221)$$

Using (3.219),

$$c = 1 - \frac{\alpha^2(\gamma - 1/2)}{1 + \beta\alpha^2} \quad b - c = 1 - \frac{\alpha^2}{1 + \beta\alpha^2} \quad b + c = 3 - \frac{2\gamma\alpha^2}{1 + \beta\alpha^2} \quad (3.222)$$

The first inequality in (3.221) is equivalent to $\gamma \geq 1/2$. The second is always satisfied. The third is equivalent to $\omega_1^2 \Delta t^2 (\gamma/2 - \beta) \leq 1$. If $\beta \geq \gamma/2$ it is always satisfied and hence imposes no constraint on Δt . Otherwise ($\beta < \gamma/2$) it is equivalent to $\omega_1 \Delta t \leq 1/\sqrt{\gamma/2 - \beta}$, i.e., $\Delta t/T \leq 1/2\pi\sqrt{\gamma/2 - \beta}$ where $T = 2\pi/\omega_1$ is the natural period.

MDOF Equation

The proof for the case of MDOF equation has a similar logical structure, which is now applied to the modal response. In this case, write

$$\tilde{\mathbf{x}}_j = \sum_{i=1}^n \boldsymbol{\psi}_i \tilde{\eta}_{ij} \quad \tilde{\mathbf{x}}_j = \sum_{i=1}^n \boldsymbol{\psi}_i \tilde{\eta}_{ij} \quad \tilde{\mathbf{x}}_j = \sum_{i=1}^n \boldsymbol{\psi}_i \tilde{\eta}_{ij} \quad (3.223)$$

where $\{\boldsymbol{\psi}_i\}_{i=1}^n$ are the mode shapes (with unit norm) corresponding to natural frequencies $\{\omega_i\}_{i=1}^n$, satisfying $\mathbf{K}\boldsymbol{\psi}_i = \omega_i^2 \mathbf{M}\boldsymbol{\psi}_i$; $\{\tilde{\eta}_{ij}, \tilde{\eta}_{ij}, \tilde{\eta}_{ij}\}$ are the modal counterparts of the computed response. This is always possible since $\{\boldsymbol{\psi}_i\}_{i=1}^n$ is a basis in the n -dimensional space. Substituting (3.223) into (3.208) to (3.210), pre-multiplying by $\boldsymbol{\psi}_r^T$ ($r = 1, \dots, n$) and following similar steps in the SDOF case shows that the state vector $\mathbf{y} = [\tilde{\eta}_{rj}, \dot{\tilde{\eta}}_{rj}]^T$ satisfies a similar equation as (3.216), but now ω_1 is replaced by ω_r . A similar stability criterion then follows by noting that

$\{\tilde{\mathbf{x}}_r, \tilde{\dot{\mathbf{x}}}_r\}$ are bounded if and only if $\{\tilde{\eta}_{rj}, \tilde{\dot{\eta}}_{rj}\}$ are bounded for all modes $r = 1, \dots, n$. When $\gamma \geq 1/2$ and $\beta \geq \gamma/2$, all modes are stable and so is the scheme. When $\gamma \geq 1/2$ but $\beta < \gamma/2$, numerical stability requires $\omega_r \Delta t \leq 1/\sqrt{\gamma/2 - \beta}$ for all modes. The mode with the highest frequency then governs the upper bound of Δt .

As a remark, (3.221) gives the key inequalities governing stability. In the present case it has been reasoned based on the specific properties of quadratic equation. The inequalities could have been arrived more directly via the Ruth-Hurwitz or Bistritz test (Bistritz 1984, 2002), which are also applicable to polynomial equations of general order.

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Chapter 4

Spectral Analysis of Stationary Stochastic Process

Abstract This chapter analyzes an unknown time series modeled by a stationary stochastic process. It is a probabilistic version of Chap. 2, and allows one to model the statistical properties of ambient vibration data. Assuming pre-requisites in undergraduate probability and statistics, the chapter introduces the characterization of a stochastic process by correlation function in the time domain and power spectral density in the frequency domain. Estimators based on a sample time history with finite duration and finite sampling rate are introduced; and their statistical properties are discussed. For long data duration, the statistical properties have remarkably simple form, which is conventionally adopted in applications and is one of the pillars in Bayesian operational modal analysis in the frequency domain.

Keywords Stationary stochastic process • Fourier analysis • Correlation function • Power spectral density

In Chap. 2 we studied the frequency characteristics of a deterministic process, where analysis results were expressed directly in terms of its values. This chapter has a similar objective but now the process is ‘unknown’ and modeled as a ‘stochastic process’ using probability theory. The analysis results are defined and expressed in terms of statistical properties of the process.

A ‘stochastic process’ is a sequence of random variables, conventionally indexed w.r.t. time. Just as a random variable is characterized by its probability distribution, a stochastic process is characterized by the joint distribution among its values at different time instants. The distribution can be complicated for a general process. Modeling assumptions allow it to be simplified, often in terms of parameters in the mechanism driving the process or statistical parameters that can be estimated from observations. A ‘stationary stochastic process’ refers to one whose statistical properties do not change with time. It is much easier to analyze or model than a non-stationary process.

Supported by theoretical arguments and field observations, over suitable time scales, it is reasonable to model ambient vibration data as a stochastic stationary process. Whether this is justified depends on factors such as the time scale under

question and the nature of excitations that drive the process. During a typhoon that lasts for a day, for example, the vibration of a building is non-stationary over the whole event but it may be considered stationary within a thirty-minute time window. The free vibration response of a footbridge shortly after the footfall of a single pedestrian is clearly non-stationary but the response in a time span of ten minutes over which many pedestrians walk by in an unorganized fashion may be considered stationary for vibration assessment purposes.

We start with continuous-time process and characterize it in the time domain through the ‘correlation function’ and in the frequency domain through the ‘power spectral density’ (PSD). Estimators for these statistics based on a sample history of the process are then introduced. The ‘Wiener-Khinchin formula’ shows that the correlation function and PSD are Fourier Transform pairs. The remaining sections focus on discrete-time sample process, which is what one deals with in digital computations. The Fast Fourier Transform (FFT) provides an efficient tool. The distribution of the scaled FFT of stochastic stationary data for long duration is pivotal to Bayesian operational modal analysis (OMA) theory.

This chapter assumes acquaintance with basic probability concepts, e.g., random variable, probability distribution, expectation, variance and correlation. One may refer to textbooks, e.g., Ross (2011), Ang and Tang (2007). We omit mathematical details when they are too technical or save to ignore in applications. For example, details on the existence of derivatives/integrals or exchanging their order are omitted. They are legitimate for ambient vibration data. Stochastic process is a broad subject with applications in many disciplines. See, e.g., Cramer and Leadbetter (1962), Papoulis (1991) and Ross (1996) for general theory. Further theoretical details on statistical spectral analysis can be found in Brillinger (1981), Bendat and Piersol (1993) and Kay (1993).

4.1 Correlation Function

Let $\mathbf{x}(t) = [x_1(t), \dots, x_n(t)]^T$ be a real-valued stochastic vector process w.r.t. time t . Without loss of generality, it is assumed to have zero mean:

$$E[\mathbf{x}(t)] = \mathbf{0} \quad (4.1)$$

where $E[\cdot]$ denotes the ‘expectation’ (ensemble average). In the time domain, one basic statistical characterization is the ‘correlation function’:

$$\mathbf{R}(s, t) = E[\underbrace{\mathbf{x}(s)}_{n \times 1} \underbrace{\mathbf{x}(t)^T}_{1 \times n}] \quad (4.2)$$

This is a matrix function of two arguments, s and t . The (i, j) -entry of $\mathbf{R}(s, t)$ is $E[x_i(s)x_j(t)]$. The ‘covariance matrix’ of $\mathbf{x}(t)$ is

$$\mathbf{R}(t, t) = E[\mathbf{x}(t)\mathbf{x}(t)^T] \quad (4.3)$$

This is a $n \times n$ real symmetric and positive semi-definite matrix.

A stochastic process is ‘strict-sense stationary’ if the joint distribution of its values at different time instants is invariant to a time shift. That is, for any time lag τ , the joint distribution of $\mathbf{x}(t + \tau)$ and $\mathbf{x}(t)$ does not depend on the time shift t . In practice this can be difficult to model or verify. ‘Wide-sense stationary’ process is a more common notion, where only the correlation function is required to be invariant to a time shift. In this case, the correlation function depends only on the time lag τ :

$$\mathbf{R}(\tau) = E[\mathbf{x}(t + \tau)\mathbf{x}(t)^T] \quad (4.4)$$

Generally, $t + \tau$ and t in the expectation cannot be swapped. That is, $\mathbf{R}(\tau) \neq E[\mathbf{x}(t)\mathbf{x}(t + \tau)^T]$. The correlation function need not be a symmetric matrix. Instead, for a stationary process it has the ‘transpose mirror property’ that

$$\mathbf{R}(-\tau) = \mathbf{R}(\tau)^T \quad (4.5)$$

This holds because

$$\mathbf{R}(-\tau) = E[\mathbf{x}(t - \tau)\mathbf{x}(t)^T] \underset{\substack{\text{replace } t \\ \text{by } t + \tau}}{=} E[\mathbf{x}(t)\mathbf{x}(t + \tau)^T] = \underbrace{E[\mathbf{x}(t + \tau)\mathbf{x}(t)^T]}_{\mathbf{R}(\tau)}^T \quad (4.6)$$

The second equality follows because the expectation is invariant to t (stationary).

Cross Correlation Function

For two scalar processes $x(t)$ and $y(t)$, their ‘cross correlation function’ is defined as

$$R_{xy}(s, t) = E[x(s)y(t)] \quad (4.7)$$

If they are ‘jointly stationary’, their cross correlation function only depends on time lag but not on time shift, giving

$$R_{xy}(\tau) = E[x(t + \tau)y(t)] \quad (4.8)$$

Note that $R_{xy}(-\tau) = R_{yx}(\tau)$. The cross correlation function is often used for studying the statistical relationship between two processes in the time domain. By definition, the (i, j) -entry ($i \neq j$) of the correlation function of a vector process is the cross correlation function between its i th and j th component. Although one can define the cross correlation function between two vector processes in an analogous manner, this will not be pursued here because in principle one can combine two vector processes into a single one comprising the two and study the correlation function of the resulting vector process.

4.2 Power Spectral Density

In the frequency domain, a stationary vector process can be characterized by the ‘power spectral density’ (PSD) matrix:

$$\mathbf{S}(\omega) = \lim_{n \times n} E[\underbrace{\mathbf{X}_T(\omega)}_{n \times 1} \underbrace{\mathbf{X}_T(\omega)^*}_{1 \times n}] \quad (4.9)$$

where ‘*’ denotes the conjugate transpose; and

$$\mathbf{X}_T(\omega) = \frac{1}{\sqrt{T}} \int_0^T \mathbf{x}(t) e^{-i\omega t} dt \quad (4.10)$$

is the (finite-time) Fourier Transform (FT) of $\mathbf{x}(t)$ scaled by $1/\sqrt{T}$. Taking conjugate transpose on (4.9) shows that $\mathbf{S}(\omega)$ is ‘Hermitian’:

$$\mathbf{S}(\omega)^* = \mathbf{S}(\omega) \quad (4.11)$$

It also has the transpose mirror property that

$$\mathbf{S}(-\omega) = \mathbf{S}(\omega)^T \quad (4.12)$$

This follows because $\mathbf{X}_T(-\omega) = \overline{\mathbf{X}_T(\omega)}$ and so

$$\mathbf{S}(-\omega) = \lim_{T \rightarrow \infty} E[\underbrace{\mathbf{X}_T(-\omega)}_{\overline{\mathbf{X}_T(\omega)}} \underbrace{\mathbf{X}_T(-\omega)^*}_{\mathbf{X}_T(\omega)^T}] = \lim_{T \rightarrow \infty} E[\underbrace{\mathbf{X}_T(\omega) \mathbf{X}_T(\omega)^*}_{\mathbf{S}(\omega)}]^T \quad (4.13)$$

Finally, $\mathbf{S}(\omega)$ is positive semi-definite, because for any fixed $n \times 1$ complex vector $\mathbf{u}$,

$$\mathbf{u}^* \mathbf{S}(\omega) \mathbf{u} = \lim_{T \rightarrow \infty} E[\underbrace{\mathbf{u}^* \mathbf{X}_T(\omega)}_{Z} \underbrace{\mathbf{X}_T(\omega)^* \mathbf{u}}_{\bar{Z}}] = \lim_{T \rightarrow \infty} E[|Z|^2] \geq 0 \quad (4.14)$$

Operationally, it is more convenient to interpret the definition of $\mathbf{S}(\omega)$ as

$$\mathbf{S}(\omega) = E[\mathbf{X}(\omega) \mathbf{X}(\omega)^*] \quad (4.15)$$

where

$$\mathbf{X}(\omega) = \lim_{T \rightarrow \infty} \mathbf{X}_T(\omega) \quad (4.16)$$

denotes a random vector with the same distribution as $\mathbf{X}_T(\omega)$ in the limit $T \rightarrow \infty$. This definition is only ‘symbolic’ because the limit of a random vector has yet to be defined (omitted here). It appears as resulting from carrying the limit in (4.9) inside the expectation and then the product. Omitting technical details, the interpretation in (4.15) is legitimate for ‘well-behaved’ processes, justified for OMA. It provides convenience for deriving the relationship between the PSDs of the input and output of a linear system. When there is no confusion, $\mathbf{X}(\omega)$ is also referred as the scaled FT of $\mathbf{x}(t)$ for convenience.

Cross PSD and Coherence

For two jointly stationary scalar processes $x(t)$ and $y(t)$ with corresponding scaled FTs $X(\omega)$ and $Y(\omega)$, their ‘cross power spectral density’ (cross PSD) is defined as

$$S_{xy}(\omega) = E[X(\omega)Y(\omega)^*] \quad (4.17)$$

Note that $S_{xy}(\omega)^* = S_{yx}(\omega)$. By definition, the (i, j) -entry ($i \neq j$) of the PSD matrix of a vector process is the cross PSD between its i th and j th component.

Analogous to cross correlation function, the cross PSD is often used for studying the statistical relationship between two processes in the frequency domain. When t is measured in seconds, the unit of the cross PSD between $x(t)$ and $y(t)$ is equal to the product of their units divided by Hz. The ‘coherence’ is a dimensionless quantity that is conveniently used in analysis:

$$\chi_{xy}(\omega) = \frac{S_{xy}(\omega)}{\sqrt{S_{xx}(\omega)S_{yy}(\omega)}} \quad (4.18)$$

where (consistent with notation) $S_{xx}(\omega)$ and $S_{yy}(\omega)$ are respectively the PSD of $x(t)$ and $y(t)$. Note that coherence is complex-valued and its modulus is bounded above by 1.

4.3 Fourier Series, Fourier Transform and PSD

The definition of PSD may look peculiar but it turns out to offer a very powerful tool for analyzing and modeling stationary stochastic processes. The following motivates the definition, through a line of query on the legitimate quantity for characterizing processes of different nature.

Consider first a periodic process $\mathbf{x}(t) = [x_1(t), \dots, x_n(t)]^T$ with period T . Then it can be written as a Fourier series (Sect. 2.1.1):

$$\mathbf{x}(t) = \sum_{k=-\infty}^{\infty} \mathbf{c}_k e^{i\bar{\omega}_k t} \quad (4.19)$$

where

$$\mathbf{c}_k = \frac{1}{T} \int_0^T \mathbf{x}(t) e^{-i\bar{\omega}_k t} dt \quad (4.20)$$

is the (complex) Fourier series (FS) coefficient associated with frequency $\bar{\omega}_k = 2\pi k/T$. A periodic process is characterized by its FS coefficients at a discrete set of frequencies.

Next, suppose $\mathbf{x}(t)$ is a non-periodic process with finite energy, i.e., $\int_0^\infty x_i(t)^2 dt < \infty$ ($i = 1, \dots, n$). Conceptually, it can be considered periodic with an infinitely long period. Taking $T \rightarrow \infty$, the sum in (4.19) becomes an integral, giving (Sect. 2.2.1)

$$\mathbf{x}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{X}(\omega) e^{i\omega t} d\omega \quad (4.21)$$

where

$$\mathbf{X}(\omega) = \int_0^\infty \mathbf{x}(t) e^{-i\omega t} dt \quad (4.22)$$

is the FT of $\mathbf{x}(t)$. A non-periodic process with finite energy is characterized by its FT over a continuum of frequencies.

Consider now when $\mathbf{x}(t)$ is a zero-mean stationary stochastic process, the focus in this chapter. It does not have a FS because it is not periodic. Its FT does not exist either because it extends over $(0, \infty)$ without decaying; $\int_0^T x_i(t)^2 dt$ grows without bound as T increases. As the value of $\mathbf{x}(t)$ at different t s are not fully correlated, the integral $\int_0^T \mathbf{x}(t) e^{-i\omega t} dt$ is a random quantity whose magnitude grows with $\sqrt{T}$ (not T). Both the FS coefficient and FT involve this integral but they have different multipliers, $1/T$ and 1 , respectively. As $T \rightarrow \infty$, the FS coefficient is $O(1/\sqrt{T})$ and it tends to zero. The FT is $O(\sqrt{T})$ and it tends to infinity. Neither one is a legitimate quantity to characterize the process in the frequency domain. The proper one has the multiplier $1/\sqrt{T}$, i.e., $\mathbf{X}_T(\omega) = T^{-1/2} \int_0^T \mathbf{x}(t) e^{-i\omega t} dt$, so that it has converging and bounded statistics. As $\mathbf{X}_T(\omega)$ is a random complex vector with zero mean, the PSD in (4.9) is simply its covariance matrix (generalized for complex vectors) in the limit.

Section 4.4.3 later shows that the correlation function and PSD are Fourier Transform pairs, therefore carrying the same piece of information in the time and frequency domain, respectively. Figure 4.1 summarizes the characterization of different processes.

Proper Scaling

The above reveals the importance of using a transform that scales properly with the data duration, depending on the nature of time series. Failing to do so gives trivially diminishing or diverging results. Table 4.1 shows what happens when a time series of a particular nature is analyzed by different tools as the data duration T increases. For example, calculating the FS coefficients of the sample history of a stationary process, one will find that the magnitude of the FS coefficients reduces systematically with the data duration, in an inverse square root manner.

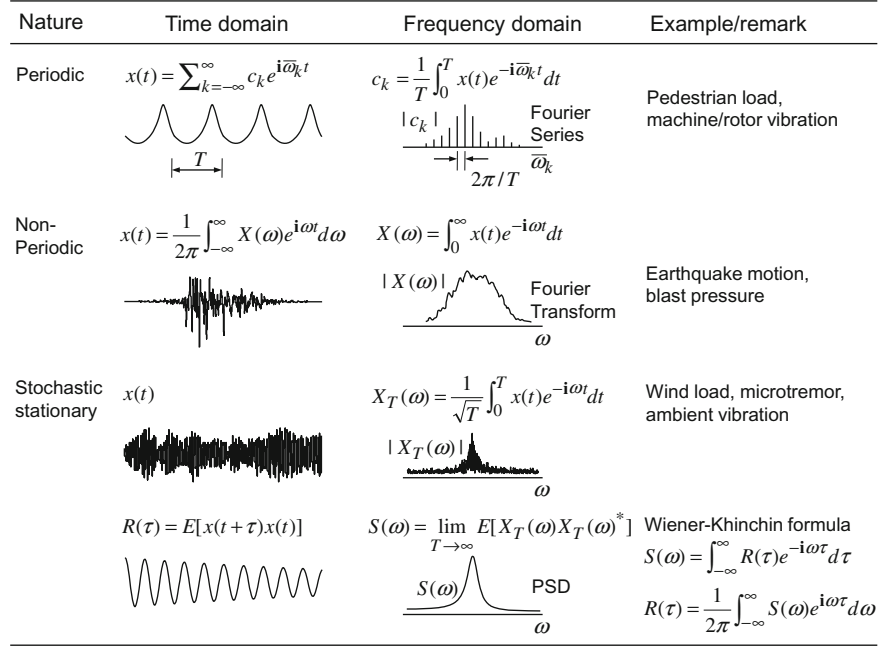


Fig. 4.1 Characterization of different (scalar) processes (showing positive frequencies only)

Table 4.1 Analyzing time series of duration T with different tools

Nature of $\mathbf{x}(t)$	Analysis tool		
	Fourier series $\mathbf{c}_k$	PSD $\mathbf{S}(\omega)$	Fourier Transform $\mathbf{X}(\omega)$
Periodic	Proper and converging	$O(\sqrt{T}), \rightarrow \infty$	$O(T), \rightarrow \infty$
Stochastic stationary	$O(1/\sqrt{T}), \rightarrow 0$	Proper and converging	$O(\sqrt{T}), \rightarrow \infty$
Non-periodic, finite energy	$O(1/T), \rightarrow 0$	$O(1/\sqrt{T}), \rightarrow 0$	Proper and converging

4.4 Continuous-Time Sample Process

The correlation function and PSD are theoretical characterization of a stationary process in the time and frequency domain, respectively. In applications we need to work with a given time history which is modeled as a stationary process. For example, we may use the time history to estimate the statistical properties of the process. In analysis we may generate random samples of the process and study its effects on system response. In this section, we consider estimating the correlation function and PSD using a continuous-time sample history. The next section extends to discrete-time sample history, the typical scenario in modern applications with digital computers.

4.4.1 Sample Correlation Function

The correlation function $\mathbf{R}(\tau)$ in (4.4) was defined in terms of the expectation of $\mathbf{x}(t+\tau)\mathbf{x}(t)^T$ for a given t . Estimating it in this spirit requires multiple samples of $\mathbf{x}(t)$ for a given t . This could imply a very tedious data collection process; imagine measuring the temperature at the same place and same time everyday. For a stationary process, it is possible to estimate using only a single rather than multiple time histories.

Definition from First Principle

Using a sample history $\{\mathbf{x}(t) : 0 \leq t \leq T\}$ of duration T , the correlation function at time lag τ ($|\tau| < T$) can be estimated by the ‘sample correlation function’, defined as

$$\hat{\mathbf{R}}_T(\tau) = \begin{cases} \frac{1}{T} \int_0^{T-\tau} \mathbf{x}(t+\tau)\mathbf{x}(t)^T dt & 0 \leq \tau < T \\ \frac{1}{T} \int_{|\tau|}^T \mathbf{x}(t+\tau)\mathbf{x}(t)^T dt & -T < \tau < 0 \end{cases} \quad (4.23)$$

This definition is motivated from $E[\mathbf{x}(t+\tau)\mathbf{x}(t)^T]$ and makes use of its invariance w.r.t. t for a stationary process. For a given τ , the term $\mathbf{x}(t+\tau)\mathbf{x}(t)^T$ is averaged over different t s. This works when the process is stationary and ‘ergodic’, i.e., the (temporal) average over time of a single sample history converges to the (ensemble) average over different histories as the data duration increases. Ergodicity is an advanced topic which will not be further discussed here. Interested readers may refer to Yaglom (1989) and Papoulis (1991). Typically, ambient vibration data are assumed to be ergodic.

Some peculiar features are in place in the definition (4.23) for technical reasons. Although the integral covers a duration of $T - |\tau|$, the averaging factor is T rather than $T - |\tau|$. This is so that $\hat{\mathbf{R}}_T(\tau)$ satisfies the Wiener-Khinchin formula (see Sect. 4.4.3 later), at the expense of a bias for finite T . The definition differs in the

integration limit for $\tau > 0$ or $\tau < 0$, so that only the value of $\mathbf{x}(t)$ within $[0, T]$ is involved.

Transpose Mirror Property

The value of $\hat{\mathbf{R}}_T(\tau)$ for negative time lag can be produced from that for positive lag because

$$\hat{\mathbf{R}}_T(-\tau) = \hat{\mathbf{R}}_T(\tau)^T \quad -T < \tau < T \quad (4.24)$$

This is analogous to (4.5). It can be verified by noting that, for $\tau > 0$, changing integration variable from t to $s = t - \tau$,

$$\begin{aligned} \hat{\mathbf{R}}_T(\underbrace{-\tau}_{<0}) &= \frac{1}{T} \int_{\tau}^T \mathbf{x}(t - \tau) \mathbf{x}(t)^T dt \\ &= \frac{1}{T} \int_0^{T-\tau} \mathbf{x}(s) \mathbf{x}(s + \tau)^T ds = \underbrace{\left[\frac{1}{T} \int_0^{T-\tau} \mathbf{x}(s + \tau) \mathbf{x}(s)^T ds \right]^T}_{\hat{\mathbf{R}}_T(\tau)} \end{aligned} \quad (4.25)$$

The case for $\tau < 0$ is similar. Equation (4.24) implies that only the values for non-negative time lag are needed; the ones for negative time lag are redundant.

Definition Based on Windowed Process

The sample correlation function in (4.23) can be written in a compact manner as

$$\hat{\mathbf{R}}_T(\tau) = \frac{1}{T} \int_{-\infty}^{\infty} \mathbf{x}_T(t + \tau) \mathbf{x}_T(t)^T dt \quad (4.26)$$

where

$$\mathbf{x}_T(t) = \begin{cases} \mathbf{x}(t) & 0 \leq t \leq T \\ \mathbf{0} & \text{otherwise} \end{cases} \quad (4.27)$$

is a ‘windowed’ version of $\mathbf{x}(t)$. Check that this gives the same expression in (4.23) for $-T \leq \tau \leq T$; and it is zero otherwise:

$$\hat{\mathbf{R}}_T(\tau) = \mathbf{0} \quad |\tau| > T \quad (4.28)$$

Essentially, the integrand in (4.26) is zero whenever $t + \tau$ or t is outside $[0, T]$. This allows $\hat{\mathbf{R}}_T(\tau)$ to be represented by a single expression regardless of the sign of τ . The integration domain can also extend over $(-\infty, \infty)$ independent of T or τ . Equation (4.28) extends the definition of $\hat{\mathbf{R}}_T(\tau)$ for $|\tau| > T$. These features provide some convenience in analysis.

4.4.2 Sample Power Spectral Density

Using the sample history $\{\mathbf{x}(t) : 0 \leq t \leq T\}$, the ‘sample PSD’ is defined as

$$\hat{\mathbf{S}}_T(\omega) = \mathbf{X}_T(\omega) \mathbf{X}_T(\omega)^* \quad (4.29)$$

where $\mathbf{X}_T(\omega) = T^{-1/2} \int_0^T \mathbf{x}(t) e^{-i\omega t} dt$ as in (4.10). Analogous to $\mathbf{S}(\omega)$, it can be verified that $\hat{\mathbf{S}}_T(\omega)$ is Hermitian and it has the transpose mirror property:

$$\hat{\mathbf{S}}_T(\omega)^* = \hat{\mathbf{S}}_T(\omega) \quad \hat{\mathbf{S}}_T(-\omega) = \hat{\mathbf{S}}_T(\omega)^T \quad (4.30)$$

It is also positive semi-definite.

4.4.3 Wiener-Khinchin Formula

One important result in the theory of stationary process is that the correlation function and PSD are Fourier Transform pairs:

$$\mathbf{S}(\omega) = \int_{-\infty}^{\infty} \mathbf{R}(\tau) e^{-i\omega\tau} d\tau \quad (4.31)$$

$$\mathbf{R}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega) e^{i\omega\tau} d\omega \quad (4.32)$$

These are known as the ‘Wiener-Khinchin formula’. For the sample counterparts,

$$\hat{\mathbf{S}}_T(\omega) = \int_{-\infty}^{\infty} \hat{\mathbf{R}}_T(\tau) e^{-i\omega\tau} d\tau \quad (4.33)$$

$$\hat{\mathbf{R}}_T(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\mathbf{S}}_T(\omega) e^{i\omega\tau} d\omega \quad (4.34)$$

with the convention that $\hat{\mathbf{R}}_T(\tau) = \mathbf{0}$ for $|\tau| > T$. As the correlation function and PSD can be converted interchangeably, they are equivalent characterization of a stationary process.

Proof of (4.31) to (4.34) (Wiener-Khinchin Formula, Continuous-Time)

Here we show (4.33), i.e., $\hat{\mathbf{S}}_T(\omega)$ is the FT of $\hat{\mathbf{R}}_T(\tau)$. Then $\hat{\mathbf{R}}_T(\tau)$ is the inverse FT of $\hat{\mathbf{S}}_T$, i.e., (4.34). Taking limit $T \rightarrow \infty$ of (4.33) and (4.34) gives (4.31) and (4.32), respectively.

Starting with the RHS of (4.33) and using (4.26),

$$\int_{-\infty}^{\infty} \hat{\mathbf{R}}_T(\tau) e^{-i\omega\tau} d\tau = \frac{1}{T} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{x}_T(t+\tau) \mathbf{x}_T(t)^T e^{-i\omega\tau} dt d\tau \quad (4.35)$$

Let $s = t + \tau$ and change integration variable from (τ, t) to (s, t) . The determinant of Jacobian of this transformation is 1. The new integration domain is still $(-\infty, \infty) \times (-\infty, \infty)$. Then

$$\begin{aligned} \int_{-\infty}^{\infty} \hat{\mathbf{R}}_T(\tau) e^{-i\omega\tau} d\tau &= \frac{1}{T} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{x}_T(s) \mathbf{x}_T(t)^T e^{-i\omega(s-t)} dt ds \\ &= \frac{1}{T} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{x}_T(s) e^{-i\omega s} \mathbf{x}_T(t)^T e^{i\omega t} ds dt \\ &= \underbrace{\left[\frac{1}{\sqrt{T}} \int_{-\infty}^{\infty} \mathbf{x}_T(s) e^{-i\omega s} ds \right]}_{\mathbf{X}_T(\omega)} \underbrace{\left[\frac{1}{\sqrt{T}} \int_{-\infty}^{\infty} \mathbf{x}_T(t) e^{-i\omega t} dt \right]^*}_{\mathbf{X}_T(\omega)^*} = \hat{\mathbf{S}}_T(\omega) \end{aligned} \quad (4.36)$$

■

4.4.4 Parseval Equality

For a stationary process, the Parseval equality explains the variance as a sum of PSD contributions from different frequencies. It results directly from the Wiener-Khinchin formula at zero time lag. Setting $\tau = 0$ in (4.32) gives

$$\underbrace{E[\mathbf{x}(t) \mathbf{x}(t)^T]}_{\mathbf{R}(0)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega) d\omega \quad (4.37)$$

For the sample process $\{\mathbf{x}(t) : 0 \leq t \leq T\}$, the Parseval equality explains the mean square value as a sum of sample PSD contributions. Setting $\tau = 0$ in (4.34) gives

$$\underbrace{\frac{1}{T} \int_0^T \mathbf{x}(t) \mathbf{x}(t)^T dt}_{\hat{\mathbf{R}}_T(0)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{\mathbf{S}}_T(\omega) d\omega \quad (4.38)$$

For a scalar process, the Parseval equality implies that the area under the PSD in the frequency domain (w.r.t. $\omega/2\pi$, in Hz) gives the variance. This allows one to calculate the variance contributed only by the activities of interest. For example, to calculate the variance due to a particular vibration mode, one integrates the PSD over the resonance band of the mode only.

4.4.5 White Noise

Consider a scalar process $w(t)$ with a constant PSD given by

$$S(\omega) \equiv S_w \quad (4.39)$$

Using the Wiener-Khinchin formula, the correlation function is

$$R(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_w e^{i\omega\tau} d\omega = S_w \delta(\tau) \quad (4.40)$$

where $\delta(\tau) = (2\pi)^{-1} \int_{-\infty}^{\infty} e^{i\omega\tau} d\omega$ is the Dirac Delta function (Sect. 2.2.3). The process $w(t)$ is called ‘white noise’. It is used widely in stochastic modeling and analysis because of the simple constant PSD. It does not exist in reality, however. Some unrealistic features are: the variance $\text{var}[w(t)] = R(0)$ is unbounded; the values of $w(t)$ at any two distinct time instants are uncorrelated, no matter how close they are. The PSD $S(\omega)$ is non-zero for all frequencies, no matter how high. These features may not affect the result of a particular analysis but should be kept in mind to ascertain their effects. In reality, the PSD of a physical process decays to zero for sufficiently high frequency. Conceptually, a process may be modeled as a ‘band-limited white noise’. That is, the PSD is constant only within some frequency band of relevance.

Example 4.1 (First order process driven by white noise) Consider a scalar process $x(t)$ governed by

$$\dot{x}(t) + ax(t) = w(t) \quad (4.41)$$

where $a > 0$ and $w(t)$ is white noise with PSD S_w . Here we determine the correlation function and PSD of $x(t)$ using their original definitions. After that we verify the Wiener-Khinchin formula and Parseval equality.

Correlation Function from Definition

The correlation function $R(\tau) = E[x(t+\tau)x(t)]$ can be obtained by a standard procedure in stochastic dynamics as follow. See Sect. 5.4 for a general treatment. For $\tau > 0$, write (4.41) at $t + \tau$ and multiply by $x(t)$. Taking expectation,

$$\underbrace{E[\dot{x}(t+\tau)x(t)]}_{\dot{R}(\tau)} + a \underbrace{E[x(t+\tau)x(t)]}_{R(\tau)} = \underbrace{E[w(t+\tau)x(t)]}_0 \quad (4.42)$$

On the RHS above, $E[w(t+\tau)x(t)] = 0$ because $x(t)$ depends only on the past $\{w(s) : s \leq t\}$, which is independent of the future $w(t+\tau)$ for $\tau > 0$ (white noise). Solving $\dot{R}(\tau) + aR(\tau) = 0$ yields

$$R(\tau) = R(0)e^{-a\tau} \quad \tau > 0 \quad (4.43)$$

It remains to determine $R(0)$. Multiplying (4.41) by $x(t)$ and taking expectation,

$$E[\dot{x}(t)x(t)] + aE[x(t)^2] = E[w(t)x(t)] \quad (4.44)$$

Note that $E[x(t)^2] = R(0)$ and

$$E[\dot{x}(t)x(t)] = E\left\{\frac{d}{dt}\left[\frac{x(t)^2}{2}\right]\right\} = \frac{d}{dt}E\left[\frac{x(t)^2}{2}\right] = 0 \quad (4.45)$$

since we are considering $x(t)$ to be in a stationary state. Substituting into (4.44) gives

$$R(0) = \frac{1}{a}E[w(t)x(t)] \quad (4.46)$$

To evaluate $E[w(t)x(t)]$, express $x(t)$ in terms of $w(t)$ as the Duhamel's integral solution to (4.41) (Sect. 3.1.12):

$$x(t) = e^{-at}x(0) + \int_0^t e^{-a(t-s)}w(s)ds \quad (4.47)$$

Multiplying by $w(t)$ and taking expectation,

$$E[w(t)x(t)] = e^{-at}x(0)\underbrace{E[w(t)]}_0 + \int_0^t e^{-a(t-s)}\underbrace{E[w(t)w(s)]}_{S_w\delta(t-s)}ds = \frac{S_w}{2} \quad (4.48)$$

where the factor $1/2$ results because the integral only covers the right half of the Delta function. Substituting into (4.46) gives $R(0) = S_w/2a$. Equation (4.43) then gives $R(\tau) = S_w e^{-a\tau}/2a$ for $\tau > 0$. Since $R(-\tau) = R(\tau)$ for a scalar process, we conclude

$$R(\tau) = \frac{S_w}{2a} e^{-a|\tau|} \quad -\infty < \tau < \infty \quad (4.49)$$

PSD from Definition

The PSD of $x(t)$ is given by $S(\omega) = E[X(\omega)X(\omega)^*]$ where $X(\omega)$ is the scaled FT of $x(t)$. Taking scaled FT on (4.41) gives $i\omega X(\omega) + aX(\omega) = W(\omega)$, where $W(\omega)$ is the scaled FT of $w(t)$. Rearranging gives $X(\omega) = W(\omega)/(a + i\omega)$. Thus,

$$S(\omega) = E[X(\omega)X(\omega)^*] = \frac{E[W(\omega)W(\omega)^*]}{(a + i\omega)(a - i\omega)} = \frac{S_w}{a^2 + \omega^2} \quad (4.50)$$

Figure 4.2 shows a schematic plot of the correlation function and PSD of $\mathbf{x}(t)$.

Wiener-Khinchin Formula

Using (4.49),

$$\int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau = \int_{-\infty}^{\infty} \frac{S_w}{2a} e^{-a|\tau|} e^{-i\omega\tau} d\tau \quad (4.51)$$

Separate the integral into two, one for $\tau \geq 0$ and the other for $\tau < 0$. For the latter, change integration variable from τ to $s = -\tau$. This gives

$$\begin{aligned} \int_{-\infty}^{\infty} R(\tau) e^{-i\omega\tau} d\tau &= \int_0^{\infty} \frac{S_w}{2a} e^{-a\tau} e^{-i\omega\tau} d\tau + \int_0^{\infty} \frac{S_w}{2a} e^{-as} e^{i\omega s} ds \\ &= \frac{S_w}{2a} \left[\frac{1}{a + i\omega} + \frac{1}{a - i\omega} \right] = \frac{S_w}{a^2 + \omega^2} \end{aligned} \quad (4.52)$$

This is the same expression in (4.50), verifying the Wiener-Khinchin formula.

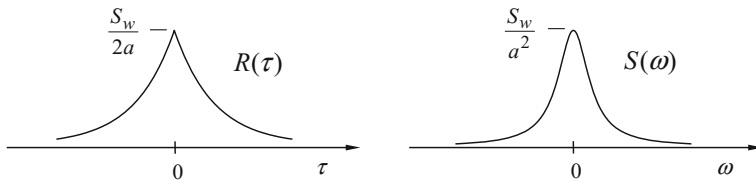


Fig. 4.2 Correlation function $R(\tau)$ and PSD $S(\omega)$ of a first order process driven by white noise

Parseval Equality

Using (4.50),

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} S(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{S_w}{a^2 + \omega^2} d\omega = \frac{S_w}{2a} = R(0) \quad (4.53)$$

where the integral can be obtained by letting $\omega = a \tan \phi$ and changing integration variable from ω to ϕ . This verifies the Parseval equality. ■

4.5 Discrete-Time Sample Process

When a continuous-time process is recorded on a digital computer, it is sampled in discrete time. The estimators for the correlation function and PSD based on such sample history are discussed in this section. This connects the theoretical properties of the process to its sample counterparts that can be calculated and analyzed on a digital computer.

As before, let $\mathbf{x}(t)$ be a stationary process with zero mean, correlation function $\mathbf{R}(\tau)$ and PSD $\mathbf{S}(\omega)$. The discrete-time sample process is the sequence $\{\mathbf{x}_j = \mathbf{x}(j\Delta t)\}_{j=0}^{N-1}$ where Δt (s) is the sampling interval.

4.5.1 Sample Correlation Function

Analogous to $\hat{\mathbf{R}}_T(\tau)$ in (4.23), the value of the correlation function $\mathbf{R}(\tau)$ at time lag $\tau = r\Delta t$ can be estimated by averaging $\mathbf{x}_{j+r}\mathbf{x}_j^T$ over different j s. This leads to the following estimator for $\mathbf{R}(r\Delta t)$:

$$\hat{\mathbf{R}}_r = \begin{cases} \frac{1}{N} \sum_{j=0}^{N-1-r} \mathbf{x}_{j+r} \mathbf{x}_j^T & 0 \leq r \leq N-1 \\ \frac{1}{N} \sum_{j=|r|}^{N-1} \mathbf{x}_{j+r} \mathbf{x}_j^T & -(N-1) \leq r \leq -1 \end{cases} \quad (4.54)$$

Based on $\{\mathbf{x}_j\}_{j=0}^{N-1}$, the maximum time lag at which the correlation can be estimated is $(N-1)\Delta t$ and so $\hat{\mathbf{R}}_r$ is only defined for $-(N-1) \leq r \leq (N-1)$. The averaging factor is N instead of $N-|r|$, for the same reason that the averaging factor for

$\hat{\mathbf{R}}_T(\tau)$ in (4.23) is T instead of $T - |\tau|$, i.e., so that $\hat{\mathbf{R}}_r$ satisfies the Wiener-Khinchin formula (Sect. 4.5.3).

The expressions for positive and negative r differ in their summation limit, as constrained by the values available in the sequence $\{\mathbf{x}_j\}_{j=0}^{N-1}$. Analogous to (4.24), the transpose mirror property reads

$$\hat{\mathbf{R}}_{-r} = \hat{\mathbf{R}}_r^T \quad (4.55)$$

This can be verified by noting that for $r > 0$,

$$\underbrace{\hat{\mathbf{R}}_{-r}}_{<0} = \frac{1}{N} \sum_{j=r}^{N-1} \mathbf{x}_{j-r} \mathbf{x}_j^T \stackrel{\text{change index from } j \text{ to } i=j-r}{=} \frac{1}{N} \sum_{i=0}^{N-1-r} \mathbf{x}_i \mathbf{x}_{i+r}^T = \left[\frac{1}{N} \sum_{i=0}^{N-1-r} \mathbf{x}_{i+r} \mathbf{x}_i^T \right]^T = \hat{\mathbf{R}}_r^T \quad (4.56)$$

Equation (4.55) implies that only the values for $r \geq 0$ are needed; the ones for $r < 0$ are redundant.

Example 4.2 (Structure of sample correlation function) Here we illustrate the structure of the sum in the sample correlation function and its transpose mirror property. Consider a discrete-time process with $N = 4$ samples:

$$\{\mathbf{x}_j\}_{j=0}^{N-1} = \{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$$

For $0 \leq r \leq N - 1$, i.e., $0 \leq r \leq 3$, using the first expression in (4.54),

$$r = 0, \quad \hat{\mathbf{R}}_0 = \frac{1}{4} \sum_{j=0}^3 \mathbf{x}_j \mathbf{x}_j^T = \frac{1}{4} (\mathbf{x}_0 \mathbf{x}_0^T + \mathbf{x}_1 \mathbf{x}_1^T + \mathbf{x}_2 \mathbf{x}_2^T + \mathbf{x}_3 \mathbf{x}_3^T)$$

$$r = 1, \quad \hat{\mathbf{R}}_1 = \frac{1}{4} \sum_{j=0}^2 \mathbf{x}_{j+1} \mathbf{x}_j^T = \frac{1}{4} (\mathbf{x}_1 \mathbf{x}_0^T + \mathbf{x}_2 \mathbf{x}_1^T + \mathbf{x}_3 \mathbf{x}_2^T)$$

$$r = 2, \quad \hat{\mathbf{R}}_2 = \frac{1}{4} \sum_{j=0}^1 \mathbf{x}_{j+2} \mathbf{x}_j^T = \frac{1}{4} (\mathbf{x}_2 \mathbf{x}_0^T + \mathbf{x}_3 \mathbf{x}_1^T)$$

$$r = 3, \quad \hat{\mathbf{R}}_3 = \frac{1}{4} \sum_{j=0}^0 \mathbf{x}_{j+3} \mathbf{x}_j^T = \frac{1}{4} (\mathbf{x}_3 \mathbf{x}_0^T)$$

For $-(N - 1) \leq r \leq -1$, i.e., $-3 \leq r \leq -1$, using the second expression in (4.54),

$$\begin{aligned}
r = -1, \quad \hat{\mathbf{R}}_{-1} &= \frac{1}{4} \sum_{j=1}^3 \mathbf{x}_{j-1} \mathbf{x}_j^T = \frac{1}{4} (\mathbf{x}_0 \mathbf{x}_1^T + \mathbf{x}_1 \mathbf{x}_2^T + \mathbf{x}_2 \mathbf{x}_3^T) = \hat{\mathbf{R}}_1^T \\
r = -2, \quad \hat{\mathbf{R}}_{-2} &= \frac{1}{4} \sum_{j=2}^3 \mathbf{x}_{j-2} \mathbf{x}_j^T = \frac{1}{4} (\mathbf{x}_0 \mathbf{x}_2^T + \mathbf{x}_1 \mathbf{x}_3^T) = \hat{\mathbf{R}}_2^T \\
r = -3, \quad \hat{\mathbf{R}}_{-3} &= \frac{1}{4} \sum_{j=3}^3 \mathbf{x}_{j-3} \mathbf{x}_j^T = \frac{1}{4} (\mathbf{x}_0 \mathbf{x}_3^T) = \hat{\mathbf{R}}_3^T
\end{aligned}$$

■

4.5.2 Sample Power Spectral Density

Analogous to $\hat{\mathbf{S}}_T(\omega) = \mathbf{X}_T(\omega) \mathbf{X}_T(\omega)^*$ in (4.29), the sample PSD using the discrete-time history $\{\mathbf{x}_j\}_{j=0}^{N-1}$ is defined as

$$\hat{\mathbf{S}}_N(\omega) = \hat{\mathbf{X}}(\omega) \hat{\mathbf{X}}(\omega)^* \quad (4.57)$$

where

$$\hat{\mathbf{X}}(\omega) = \sqrt{\frac{\Delta t}{N}} \sum_{j=0}^{N-1} \mathbf{x}_j e^{-i\omega j \Delta t} \quad (4.58)$$

is a discrete-time approximation to $\mathbf{X}_T(\omega) = T^{-1/2} \int_0^T \mathbf{x}(t) e^{-i\omega t} dt$ in (4.10). It can be verified that $\hat{\mathbf{S}}_N(\omega)$ is Hermitian and it has the transpose mirror property:

$$\hat{\mathbf{S}}_N(\omega)^* = \hat{\mathbf{S}}_N(\omega) \quad \hat{\mathbf{S}}_N(-\omega) = \hat{\mathbf{S}}_N(\omega)^T \quad (4.59)$$

It is also positive semi-definite.

Sample PSD at FFT Frequencies

Evaluating $\hat{\mathbf{S}}_N(\omega)$ for different values of ω involves a separate summation. In practice it is calculated at the following frequencies:

$$\omega_k = \frac{2\pi k}{N\Delta t} \text{ (rad/s)} \quad k = 0, \dots, N-1 \quad (4.60)$$

This is because the values of $\hat{\mathbf{X}}(\omega)$ at these frequencies can be evaluated efficiently by the Fast Fourier Transform (FFT) algorithm (Sect. 2.3). As these values play a central role in applications, they are denoted specifically by

$$\hat{\mathbf{S}}_k = \hat{\mathbf{X}}_k \hat{\mathbf{X}}_k^* \quad (4.61)$$

where, noting $\omega_k j \Delta t = 2\pi j k / N$,

$$\hat{\mathbf{X}}_k = \hat{\mathbf{X}}(\omega_k) = \underbrace{\sqrt{\frac{\Delta t}{N}} \sum_{j=0}^{N-1} \mathbf{x}_j e^{-2\pi i j k / N}}_{\text{FFT of } \{\mathbf{x}_j\}_{j=0}^{N-1}} \quad (4.62)$$

The sequence $\{\hat{\mathbf{X}}_k\}_{k=0}^{N-1}$ is referred as the ‘scaled FFT’ of $\{\mathbf{x}_j\}_{j=0}^{N-1}$; the scaling factor is $\sqrt{\Delta t / N}$.

4.5.3 Wiener-Khinchin Formula

Analogous to (4.33) and (4.34), the Wiener-Khinchin formula for a discrete-time sample process reads

$$\hat{\mathbf{S}}_N(\omega) = \Delta t \sum_{r=-(N-1)}^{N-1} \hat{\mathbf{R}}_r e^{-i\omega r \Delta t} \quad (4.63)$$

$$\hat{\mathbf{R}}_r = \frac{1}{2\pi} \int_{-\pi/\Delta t}^{\pi/\Delta t} \hat{\mathbf{S}}_N(\omega) e^{i\omega r \Delta t} d\omega \quad (4.64)$$

Equation (4.63) can be viewed as a Fourier series for $\hat{\mathbf{S}}_N(\omega)$, where the roles of time and frequency are swapped. This agrees with the fact that $\hat{\mathbf{S}}_N(\omega) = \hat{\mathbf{X}}(\omega) \hat{\mathbf{X}}(\omega)^*$ is a periodic function of ω with period $2\pi/\Delta t$, because $\hat{\mathbf{X}}(\omega)$ is. Equation (4.64) results from the expression of the FS coefficients.

Evaluating (4.63) at the FFT frequency $\omega_k = 2\pi k / N \Delta t$ and noting $\hat{\mathbf{S}}_N(\omega_k) = \hat{\mathbf{S}}_k$,

$$\hat{\mathbf{S}}_k = \Delta t \sum_{r=-(N-1)}^{N-1} \hat{\mathbf{R}}_r e^{-2\pi i r k / N} \quad (4.65)$$

Despite the appearance in (4.65), the sequence $\{\hat{\mathbf{S}}_k\}_{k=0}^{N-1}$ is *not* the FFT of $\{\hat{\mathbf{R}}_r \Delta t\}_{r=-(N-1)}^{N-1}$. Rather, $\hat{\mathbf{S}}_k$ is equal to the $2k$ th entry of the FFT of the $2N$ -sequence formed by padding a zero in the beginning of $\{\hat{\mathbf{R}}_r \Delta t\}_{r=-(N-1)}^{N-1}$. See Table 2.4 in Sect. 2.8. The inverse FFT of $\{\hat{\mathbf{S}}_k\}_{k=0}^{N-1}$ does not return $\hat{\mathbf{R}}_r \Delta t$ either.

Proof of (4.63) and (4.64) (Wiener-Khinchin formula, discrete-time)

From definition,

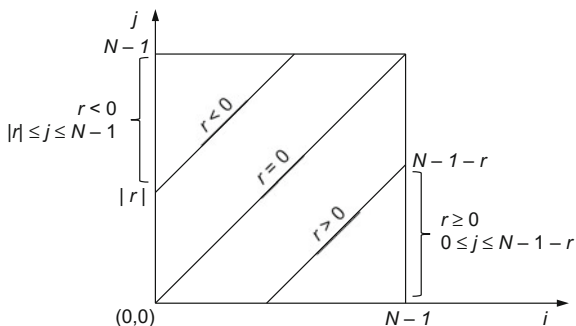
$$\hat{\mathbf{S}}_N(\omega) = \underbrace{\sqrt{\frac{\Delta t}{N}} \sum_{i=0}^{N-1} \mathbf{x}_i e^{-i\omega i \Delta t}}_{\hat{\mathbf{X}}(\omega)} \underbrace{\sqrt{\frac{\Delta t}{N}} \sum_{j=0}^{N-1} \mathbf{x}_j^T e^{i\omega j \Delta t}}_{\hat{\mathbf{X}}(\omega)^*} = \frac{\Delta t}{N} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} \mathbf{x}_i \mathbf{x}_j^T e^{-i\omega(i-j)\Delta t} \quad (4.66)$$

Change index from (i, j) to (r, j) where $r = i - j$ ranges between $-(N-1)$ and $N-1$. For every r , the summation limit for j can be determined from Fig. 4.3. For $r \geq 0$, j ranges from 0 to $N-1-r$. For $r < 0$, j ranges from $|r|$ to $N-1$. Equation (4.66) can then be written as

$$\begin{aligned} \hat{\mathbf{S}}_N(\omega) &= \Delta t \sum_{r=0}^{N-1} \underbrace{\frac{1}{N} \sum_{j=0}^{N-1-r} \mathbf{x}_{j+r} \mathbf{x}_j^T}_{\hat{\mathbf{R}}_r \text{ for } r \geq 0} e^{-i\omega r \Delta t} + \Delta t \sum_{r=-(N-1)}^{-1} \underbrace{\frac{1}{N} \sum_{j=|r|}^{N-1} \mathbf{x}_{j+r} \mathbf{x}_j^T}_{\hat{\mathbf{R}}_r \text{ for } r < 0} e^{-i\omega r \Delta t} \\ &= \Delta t \sum_{r=-(N-1)}^{N-1} \hat{\mathbf{R}}_r e^{-i\omega r \Delta t} \end{aligned} \quad (4.67)$$

This proves (4.63).

Fig. 4.3 Summation limit of j



To show (4.64), starting from the RHS and using (4.63),

$$\begin{aligned}
 \frac{1}{2\pi} \int_{-\pi/\Delta t}^{\pi/\Delta t} \hat{\mathbf{S}}_N(\omega) e^{i\omega r \Delta t} d\omega &= \frac{1}{2\pi} \int_{-\pi/\Delta t}^{\pi/\Delta t} \Delta t \overbrace{\sum_{j=-(N-1)}^{N-1} \hat{\mathbf{R}}_j e^{-i\omega j \Delta t}}^{\hat{\mathbf{S}}_N(\omega)} e^{i\omega r \Delta t} d\omega \\
 &= \frac{\Delta t}{2\pi} \sum_{j=-(N-1)}^{N-1} \hat{\mathbf{R}}_j \underbrace{\int_{-\pi/\Delta t}^{\pi/\Delta t} e^{i\omega(r-j)\Delta t} d\omega}_{2\pi/\Delta t \text{ when } j=r; \text{ else } 0} = \hat{\mathbf{R}}_r
 \end{aligned} \tag{4.68}$$

■

4.5.4 Parseval Equality

Setting $r = 0$ in (4.64) gives the Parseval equality for the discrete-time sample process $\{\mathbf{x}_j\}_{j=0}^{N-1}$:

$$\underbrace{\frac{1}{N} \sum_{j=0}^{N-1} \mathbf{x}_j \mathbf{x}_j^T}_{\hat{\mathbf{R}}_0} = \frac{1}{2\pi} \int_{-\pi/\Delta t}^{\pi/\Delta t} \hat{\mathbf{S}}_N(\omega) d\omega \tag{4.69}$$

On the other hand, summing $\hat{\mathbf{S}}_k$ in (4.65) over $k = 0, \dots, N-1$ and multiplying by the FFT frequency interval $\Delta f = 1/N\Delta t$ (Hz) gives

$$\Delta f \sum_{k=0}^{N-1} \hat{\mathbf{S}}_k = \frac{\Delta t}{N\Delta t} \sum_{k=0}^{N-1} \sum_{r=-(N-1)}^{N-1} \hat{\mathbf{R}}_r e^{-2\pi i r k / N} = \frac{1}{N} \sum_{r=-(N-1)}^{N-1} \hat{\mathbf{R}}_r \underbrace{\sum_{k=0}^{N-1} e^{-2\pi i r k / N}}_{N \text{ for } r=0; \text{ else } 0} = \hat{\mathbf{R}}_0 \tag{4.70}$$

where the exponential sum has been evaluated in Sect. 2.3.1. Thus,

$$\underbrace{\frac{1}{N} \sum_{j=0}^{N-1} \mathbf{x}_j \mathbf{x}_j^T}_{\hat{\mathbf{R}}_0} = \Delta f \sum_{k=0}^{N-1} \hat{\mathbf{S}}_k \tag{4.71}$$

Equations (4.69) and (4.71) express the sample covariance matrix as a sum of PSD contributions from different frequencies.

Example 4.3 (Simulation of discrete-time white noise) A sample history $\{w_j\}_{j=0}^{N-1}$ of a scalar Gaussian white noise process with PSD S_w band-limited between $\pm 1/2\Delta t$ (Hz) can be generated by

$$w_j = \sqrt{\frac{S_w}{\Delta t}} Z_j \quad j = 0, \dots, N-1 \quad (4.72)$$

where Δt (s) is the sampling interval and $\{Z_j\}_{j=0}^{N-1}$ are i.i.d. standard Gaussian (zero mean and unit variance). Here we verify that the expectation of the sample PSD is indeed S_w . The scaled FFT of $\{w_j\}_{j=0}^{N-1}$ at frequency $f_k = k/N\Delta t$ (Hz) is

$$W_k = \sqrt{\frac{\Delta t}{N}} \sum_{j=0}^{N-1} w_j e^{-2\pi i j k / N} \quad (4.73)$$

The expectation of the sample PSD at f_k is then

$$E[\hat{S}_k] = E[W_k W_k^*] = \frac{\Delta t}{N} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} \underbrace{E[w_i w_j]}_{\substack{S_w/\Delta t \text{ for } j=i; \\ \text{else } 0}} e^{-2\pi i (i-j)k/N} = \frac{\Delta t}{N} \sum_{i=0}^{N-1} \frac{S_w}{\Delta t} = S_w \quad (4.74)$$

Alternatively, the simulation formula (4.72) can be reasoned as follow. To be white noise the w_j s must be i.i.d. Gaussian and so $w_j = c Z_j$ where $\{Z_j\}_{j=0}^{N-1}$ are i.i.d. standard Gaussian and c is some constant. To determine c , note that $\text{var}[w_j] = c^2$. From the Parseval equality, variance is equal to the area of PSD in the frequency domain (in Hz). Since the PSD is equal to S_w over the band from $-1/2\Delta t$ to $1/2\Delta t$ (Hz), the area is $S_w/\Delta t$. This implies that $c^2 = S_w/\Delta t$ and so $c = \sqrt{S_w/\Delta t}$. ■

4.6 Averaging Sample PSD

The sample PSD $\hat{S}_N(\omega)$ in (4.57) was motivated from its theoretical counterpart $S(\omega)$. In application it is not used directly for estimating $S(\omega)$ because its statistical error does not diminish as data length increases. We first consider an example to illustrate this point. After that a converging estimator shall be introduced.

Example 4.4 (Variability of raw sample PSD) Consider synthetic acceleration data $\{\hat{x}_j\}_{j=0}^{N-1}$ generated at sampling rate 100 Hz ($\Delta t = 0.01$ s) according to

$$\hat{x}_j = \ddot{x}(t_j) + \varepsilon_j \quad j = 0, \dots, N-1 \quad (4.75)$$

where $\{\varepsilon_j\}_{j=0}^{N-1}$ is white noise with PSD $S_e = 1(\mu\text{g})^2/\text{Hz}$; $x(t)$ satisfies

$$\ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) = p(t) \quad (4.76)$$

with natural frequency $\omega_1 = 2\pi$ (rad/s) and damping ratio $\zeta = 1\%$; $p(t)$ is white noise with PSD $S_p = 0.04(\mu\text{g})^2/\text{Hz}$. It can be shown that (Sect. 5.1.2) the theoretical PSD of $\hat{x}(t)$ is $S(\omega) = S_p D(\omega) + S_e$, where $D(\omega) = \left[(1 - \beta^2)^2 + (2\zeta\beta)^2 \right]^{-1}$, $\beta = \omega_1/\omega$.

Figure 4.4 shows the sample PSD calculated using data of different durations $N\Delta t$. As the duration increases, the frequency resolution increases (horizontal separation between dots narrows). There are more points in the spectrum but they show no continuity. There is no sign of convergence with increasing duration. The values at different frequencies appear uncorrelated. ■

Simple Average

As demonstrated in Example 4.4, the sample PSD in (4.57) does not converge to the theoretical PSD as the data duration increases. Essentially, the additional data goes into improving the frequency resolution rather than reducing the statistical variability at a given frequency. The ‘raw’ spectra in Fig. 4.4 comprising scattered points around the theoretical PSD are difficult to interpret or infer about the

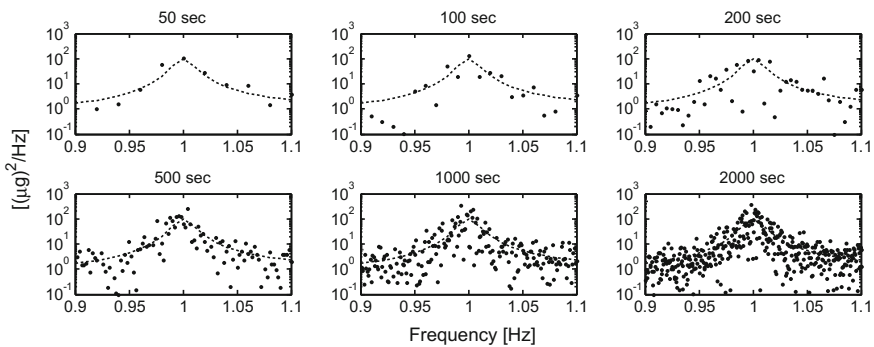


Fig. 4.4 Sample PSD (dots at FFT frequencies) calculated from data of different durations. Dashed line theoretical PSD

theoretical spectrum. A converging estimator for the PSD can be obtained by averaging the sample PSDs calculated from multiple sample histories.

Let $\{\mathbf{x}_j^{(r)}\}_{j=0}^{N-1}$ be the r th set of sample history ($r = 1, \dots, M$) of $\mathbf{x}(t)$. The (raw) sample PSD at the FFT frequency $\omega_k = 2\pi k/N\Delta t$ (rad/s) based on this sample is

$$\hat{\mathbf{S}}_k^{(r)} = \hat{\mathbf{X}}_k^{(r)} \hat{\mathbf{X}}_k^{(r)*} \quad (4.77)$$

$$\hat{\mathbf{X}}_k^{(r)} = \sqrt{\frac{\Delta t}{N}} \sum_{j=0}^{N-1} \mathbf{x}_j^{(r)} e^{-2\pi i j k / N} \quad (4.78)$$

The ‘averaged sample PSD’, or simply sample PSD when understood, is defined as

$$\hat{\mathbf{E}}_k = \frac{1}{M} \sum_{r=1}^M \hat{\mathbf{S}}_k^{(r)} = \frac{1}{M} \sum_{r=1}^M \hat{\mathbf{X}}_k^{(r)} \hat{\mathbf{X}}_k^{(r)*} \quad (4.79)$$

As a result of averaging, if the segments $\{\mathbf{x}_j^{(r)}\}_{j=0}^{N-1}$ are i.i.d. for different r , then the variance of $\hat{\mathbf{E}}_k$ will be $1/M$ of the variance of $\hat{\mathbf{S}}_k^{(r)}$, thereby producing a converging estimator. The statistical properties of $\hat{\mathbf{E}}_k$ are discussed in Sect. 4.10.

In typical situations there is only one time history available, say $\{\mathbf{x}_j\}_{j=0}^{N-1}$. It may be divided into M non-overlapping time histories to give M segments. That is, assuming N/M is an integer,

$$\mathbf{x}_j^{(r)} = \mathbf{x}_{(r-1)N/M + j} \quad j = 0, \dots, \frac{N}{M} - 1 \quad (4.80)$$

In this case, each segment has only N/M points and hence shorter duration $N\Delta t/M$. The frequency resolution of the averaged spectrum is M times lower, with a frequency interval of $M/N\Delta t$ (Hz). The choice of M is a trade-off between frequency resolution and statistical accuracy in the spectrum. See Sect. 7.2.

The averaged spectrum here is based on non-overlapping segments and is conceptually simple. There are more sophisticated means for averaging. It is possible to make use of segments that overlap or modified by a window function (Welch 1967). The choice of window function allows one to improve certain statistical features of the spectrum. Interested readers can refer to signal processing texts.

Example 4.5 (Sample PSD, averaged) Consider the 2000 s data in Example 4.4. Dividing it into M segments, we calculate the averaged sample PSD. The results are

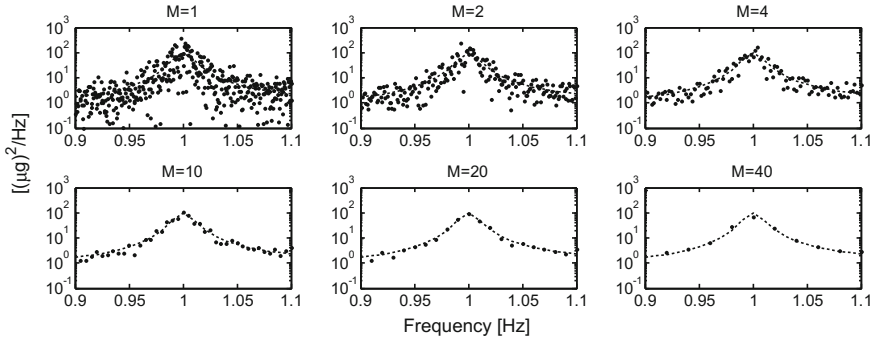


Fig. 4.5 Sample PSD (dot) averaged over different number of segments M from 2000 s data. Dashed line theoretical PSD

shown in Fig. 4.5. As M increases, the spectrum looks smoother, as a result of averaging. This is at the expense of reducing frequency resolution and increasing bias in the estimation; see Sect. 4.7 next. ■

4.7 Distortions in Sample Estimators

The sample correlation function and sample PSD are ‘statistical estimators’ for the theoretical quantity they estimate. Clearly, their values depend on the particular data set. Variability in data induces variability in the sample estimator. The quality of a sample estimator is often quantified in terms of its bias (mean deviation from the target value) and variance (mean squared deviation from the target value).

In this section we derive the expectation and discuss the bias of the sample correlation function and sample PSD. Of particular attention is the sample PSD, whose bias is of a characteristic nature stemming from the properties of discrete-time Fourier Transform (DTFT). Finite data duration causes leakage and limits frequency resolution. Finite sampling rate limits the highest frequency that can be properly estimated and causes aliasing. Discussion on the variance of the sample PSD is postponed until Sect. 4.10.

4.7.1 Sample Correlation Function

Based on a continuous time sample $\{\mathbf{x}(t) : 0 \leq t \leq T\}$, the sample correlation function $\hat{\mathbf{R}}_T(\tau)$ in (4.23) is a biased estimator for $\mathbf{R}(\tau)$ for finite T but is asymptotically unbiased as $T \rightarrow \infty$. To see this, take expectation on the first expression in (4.23) ($0 \leq \tau < T$) and swap the order of expectation and integration:

$$E[\hat{\mathbf{R}}_T(\tau)] = \frac{1}{T} \int_0^{T-\tau} \underbrace{E[\mathbf{x}(t+\tau)\mathbf{x}(t)^T]}_{\mathbf{R}(\tau)} dt = \left(1 - \frac{\tau}{T}\right) \mathbf{R}(\tau) \quad (4.81)$$

For a given τ , as $T \rightarrow \infty$, $E[\hat{\mathbf{R}}_T(\tau)] \rightarrow \mathbf{R}(\tau)$, i.e., asymptotically unbiased. The case for $-T < \tau < 0$ follows similarly by using the transpose mirror property.

An analogous result holds in discrete time. Using $\{\mathbf{x}_j = \mathbf{x}(j\Delta t)\}_{j=0}^{N-1}$, the sample correlation function $\hat{\mathbf{R}}_r$ in (4.54) is an asymptotically unbiased estimator for $\mathbf{R}(r\Delta t)$ as $N \rightarrow \infty$. Taking expectation of the first expression ($0 \leq r \leq N-1$),

$$E[\hat{\mathbf{R}}_r] = \frac{1}{N} \sum_{j=0}^{N-1-r} \underbrace{E[\mathbf{x}_{j+r}\mathbf{x}_j^T]}_{\mathbf{R}(r\Delta t)} = \left(1 - \frac{r}{N}\right) \mathbf{R}(r\Delta t) \quad (4.82)$$

and so for a given r we have $E[\hat{\mathbf{R}}_r] \rightarrow \mathbf{R}(r\Delta t)$ as $N \rightarrow \infty$. The case for $-(N-1) \leq r \leq -1$ follows similarly from the transpose mirror property. The above argument assumes that Δt is fixed, which implies $T = N\Delta t \rightarrow \infty$ as $N \rightarrow \infty$. Otherwise it need not hold. For example, suppose T is fixed. Then $\Delta t = T/N \rightarrow 0$ as $N \rightarrow \infty$. If we estimate $\mathbf{R}(\tau)$ for fixed $\tau = r\Delta t$, then $r = \tau/\Delta t = N\tau/T$ has to increase with N . In this case, in (4.82), $1 - r/N = 1 - \tau/T$ will not tend to 1 and $E[\hat{\mathbf{R}}_r]$ will not tend to $\mathbf{R}(r\Delta t)$ even when $N \rightarrow \infty$. Intuitively, increasing the sampling rate alone does not eliminate bias. The data duration has to increase as well.

4.7.2 Sample PSD

Similar to the sample correlation function, the sample PSD is also asymptotically unbiased. While the bias in the sample correlation function is simply by a factor, the bias in the sample PSD is of a more characteristic nature. The expectation of the sample PSD is equal to the convolution of the theoretical PSD with a kernel function.

Continuous-Time Sample PSD

The expectation of the sample PSD in (4.29) is given by

$$E[\hat{\mathbf{S}}_T(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega') F_T(\omega - \omega') d\omega' \quad (4.83)$$

$$F_T(\omega) = \frac{4 \sin^2(\omega T/2)}{\omega^2 T} \quad (4.84)$$

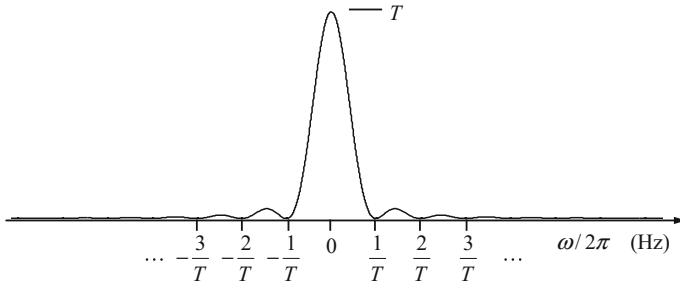


Fig. 4.6 Schematic plot of $F_T(\omega)$

See the end for proof. Figure 4.6 shows a schematic plot of the kernel function $F_T(\omega)$. It is concentrated in a neighborhood of $O(2\pi/T)$ around the origin. This implies that the value of $E[\hat{\mathbf{S}}_T(\omega)]$ is affected by the values of $\mathbf{S}(\omega)$ in a neighborhood of $O(2\pi/T)$ around the subject frequency ω . This is leakage, as encountered in Sect. 2.4.3. It implies a frequency resolution of $O(2\pi/T)$ (rad/s), or $O(1/T)$ (Hz), because one is not able to distinguish harmonics in the neighborhood. It can be shown that the area of $F_T(\omega)/2\pi$ is always equal to 1, i.e., $\int_{-\infty}^{\infty} F_T(\omega) d\omega/2\pi = 1$. As $T \rightarrow \infty$, $F_T(\omega)/2\pi$ tends to the Dirac Delta function $\delta(\omega)$. This implies that $E[\hat{\mathbf{S}}_T(\omega)] \rightarrow \mathbf{S}(\omega)$, i.e., asymptotically unbiased.

Discrete-Time Sample PSD

Analogous to (4.83), the expectation of the sample PSD in (4.57) is given by

$$E[\hat{\mathbf{S}}_N(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega') F_N(\omega - \omega') d\omega' \quad (4.85)$$

$$F_N(\omega) = \frac{\Delta t \sin^2(N\omega\Delta t/2)}{N \sin^2(\omega\Delta t/2)} \quad (4.86)$$

See the end for proof. It is more insightful to write $F_N(\omega)$ in terms of the dimensionless variable $u = \omega\Delta t/2\pi$:

$$F_N(\omega) = N\Delta t D_N^2(u) \quad u = \frac{\omega\Delta t}{2\pi} \quad (4.87)$$

where

$$D_N(u) = \frac{\sin N\pi u}{N \sin \pi u} \quad (4.88)$$

is the Dirichlet kernel that appeared in Sects. 2.4.3. Figure 4.7 shows a schematic plot of $F_N(\omega)$, whose properties stem from those of $D_N(u)$. It is periodic with a basic branch in $(-\pi/\Delta t, \pi/\Delta t)$, i.e., $\pm$ Nyquist frequency. Within the branch, $F_N(\omega)$ is similar to $F_T(\omega)$ in Fig. 4.6. It can be shown that the area of $F_N(\omega)/2\pi$ in the branch is always equal to 1, i.e., $\int_{-\pi/\Delta t}^{\pi/\Delta t} F_N(\omega) d\omega / 2\pi = 1$; see Sect. 4.8.2 later. For fixed data duration $T = N\Delta t$, taking $\Delta t \rightarrow 0$ ($N = T/\Delta t \rightarrow \infty$) yields $F_N(\omega) \rightarrow F_T(\omega)$ and so $E[\hat{S}_N(\omega)] \rightarrow E[\hat{S}_T(\omega)]$. That is, the discrete-time and continuous-time estimator coincide as the sampling interval diminishes, which can be intuitively expected. For $\hat{S}_N(\omega)$ to be asymptotically unbiased, it is required that $\Delta t \rightarrow 0$ and $T = N\Delta t \rightarrow \infty$.

It can be reasoned that $\hat{S}_N(\omega)$ is periodic with the same period as $F_N(\omega)$. As a result, $\hat{S}_N(\omega)$ can at best estimate $S(\omega)$ for frequencies up to the Nyquist frequency, i.e., $\pi/\Delta t$ (rad/s) or $1/2\Delta t$ (Hz). Not only is the frequency band of estimation limited, it is also contaminated by frequency components outside the band. Since $F_N(\omega)$ is periodic with a basic branch on $(-\pi/\Delta t, \pi/\Delta t)$, the integral in (4.85) includes the contribution of $S(\omega)$ from the frequency neighborhoods $\omega \pm 2\pi k/\Delta t$ ($k = 1, 2, \dots$) recurring outside the band. This is aliasing, as encountered in Sect. 2.4.2. Considering positive frequencies only and taking into account the conjugate mirror property of PSD, the aliased neighborhoods are around $2\pi k/\Delta t \pm \omega$ ($k = 1, 2, \dots$). Figure 4.8 illustrates the distortion of the sample PSD due to finite duration and finite sampling rate.

Example 4.6 (Leakage) Consider the sample PSD in Example 4.4 computed from synthetic ambient acceleration data. Figure 4.9 shows a plot of the theoretical PSD (solid line) and the expectation of sample PSD (dots) for different N_c = data duration/natural period. The expectation is calculated based on (4.85), where the

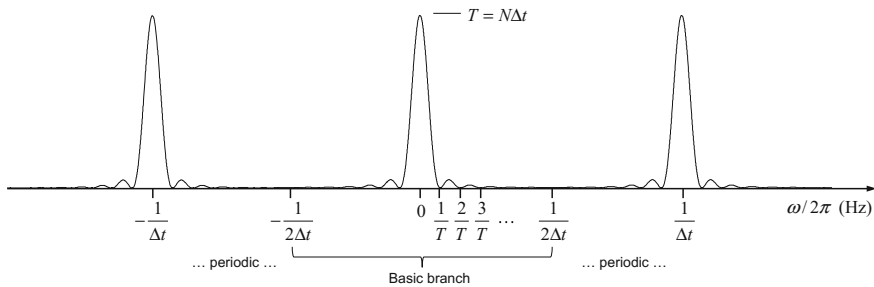


Fig. 4.7 Schematic plot of $F_N(\omega)$

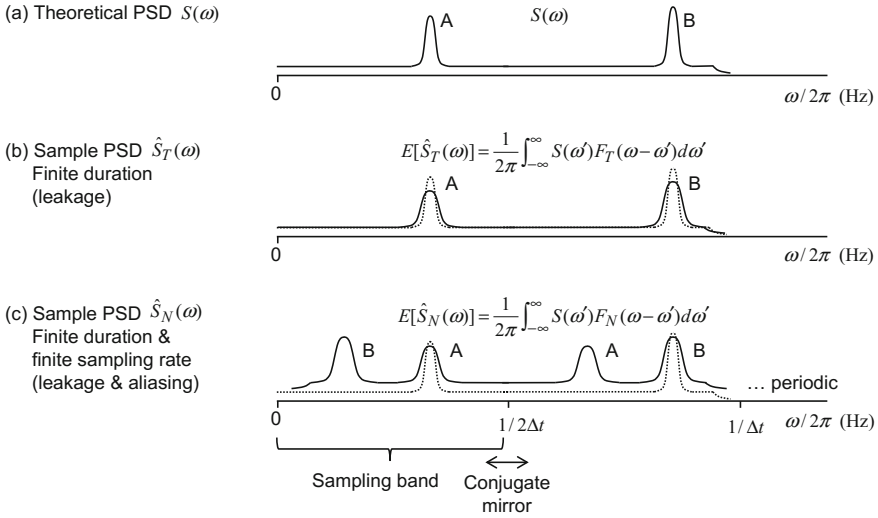


Fig. 4.8 Distortion in the sample PSD. **a** Theoretical; **b** continuous-time finite duration sample; **c** discrete-time finite duration sample

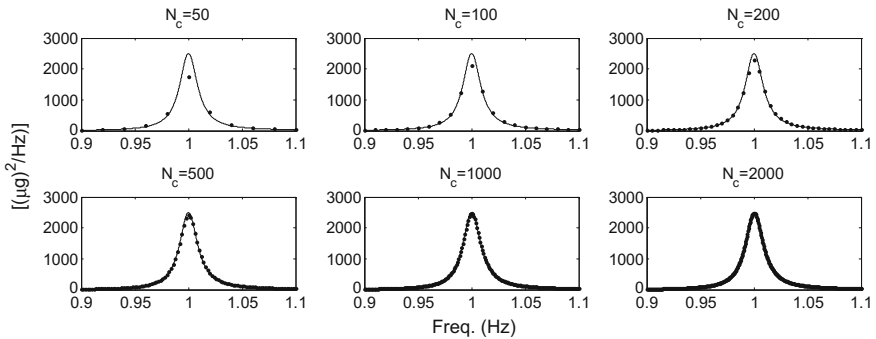


Fig. 4.9 Theoretical PSD (solid line) and expectation of sample PSD (dots) for different data durations, N_c = data duration/natural period

convolution is performed numerically. As N_c increases, the expectation of the sample PSD tends to the theoretical PSD. When $N_c = 50$ the difference is quite significant. When $N_c \geq 500$ the two PSDs visually coincide.

Figure 4.10 shows the ratio of the two PSDs at the natural frequency versus N_c . This ratio is less than 1, indicating that the sample PSD is biased low at the natural frequency. Here, $N_c = 500$ is required to have a percentage bias less than 5%. ■

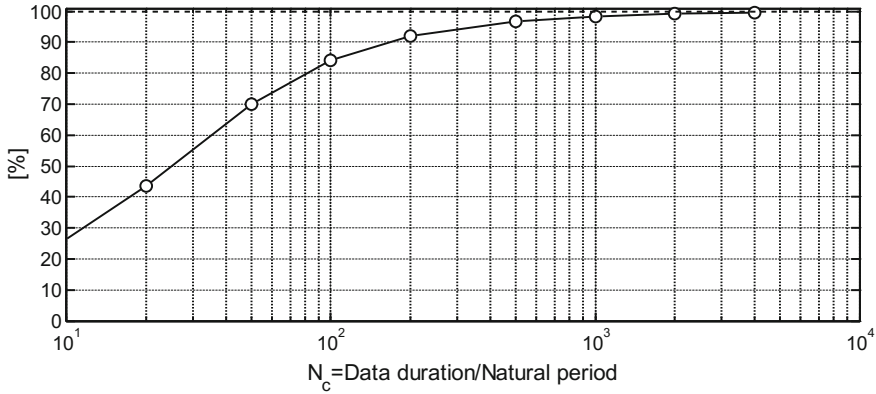


Fig. 4.10 Ratio of the expectation of sample PSD to theoretical PSD at the natural frequency

Proof of (4.83) (Expectation of Sample PSD, Continuous-Time)

From definition,

$$\begin{aligned}
 E[\hat{\mathbf{S}}_T(\omega)] &= E\left[\underbrace{\frac{1}{\sqrt{T}} \int_0^T \mathbf{x}(t) e^{-i\omega t} dt}_{\mathbf{X}_T(\omega)} \underbrace{\frac{1}{\sqrt{T}} \int_0^T \mathbf{x}(s)^T e^{i\omega s} ds}_{\mathbf{X}_T(\omega)^*}\right] \\
 &= \frac{1}{T} \int_0^T \int_0^T \underbrace{E[\mathbf{x}(t)\mathbf{x}(s)^T]}_{\mathbf{R}(t-s)} e^{-i\omega(t-s)} dt ds
 \end{aligned} \tag{4.89}$$

Substituting the Wiener-Khinchin formula

$$\mathbf{R}(t-s) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega') e^{i\omega'(t-s)} d\omega' \tag{4.90}$$

and carrying the integral w.r.t. ω' outside,

$$\begin{aligned}
 E[\hat{\mathbf{S}}_T(\omega)] &= \frac{1}{2\pi T} \int_{-\infty}^{\infty} \mathbf{S}(\omega') \left[\int_0^T \int_0^T e^{-i(\omega-\omega')(t-s)} dt ds \right] d\omega' \\
 &= \frac{T}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega') \underbrace{\left[\frac{1}{T} \int_0^T e^{-i(\omega-\omega')t} dt \right]}_{K(\omega-\omega')} \underbrace{\left[\frac{1}{T} \int_0^T e^{-i(\omega-\omega')s} ds \right]^*}_{K(\omega-\omega')^*} d\omega' \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega') F_T(\omega - \omega') d\omega'
 \end{aligned} \tag{4.91}$$

where

$$F_T(\omega) = T|K(\omega)|^2 \quad (4.92)$$

Using the identity $1 - e^{-i\theta} = 2ie^{-i\theta/2} \sin(\theta/2)$ for any real θ ,

$$K(\omega) = \frac{1}{T} \int_0^T e^{-i\omega t} dt = \frac{1 - e^{-i\omega T}}{i\omega T} = \frac{2}{\omega T} e^{-i\omega T/2} \sin(\omega T/2) \quad (4.93)$$

$$|K(\omega)|^2 = \frac{4 \sin^2(\omega T/2)}{\omega^2 T^2} \quad (4.94)$$

Substituting into (4.92) gives the expression of $F_T(\omega)$ in (4.84).

The following shows that the area of $F_T(\omega)/2\pi$ is equal to 1:

$$\begin{aligned} \frac{1}{2\pi} \int_{-\infty}^{\infty} F_T(\omega) d\omega &= \frac{T}{2\pi} \int_{-\infty}^{\infty} \overbrace{\left[\frac{1}{T} \int_0^T e^{-i\omega t} dt \right]}^{K(\omega)} \overbrace{\left[\frac{1}{T} \int_0^T e^{i\omega s} ds \right]}^{\overline{K(\omega)}} d\omega \\ &= \frac{1}{2\pi T} \int_{-\infty}^{\infty} \int_0^T \int_0^T e^{-i\omega(t-s)} dt ds d\omega \\ &= \frac{1}{T} \int_0^T \int_0^T \underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega(t-s)} d\omega}_{\delta(t-s)} dt ds = \frac{1}{T} \int_0^T ds = 1 \end{aligned} \quad (4.95)$$

■

Proof of (4.85) (Expectation of Sample PSD, Discrete-Time)

The structure of the proof is similar to that for (4.83). From definition,

$$\begin{aligned} E[\hat{\mathbf{S}}_N(\omega)] &= E \left[\underbrace{\sqrt{\frac{\Delta t}{N}} \sum_{r=0}^{N-1} \mathbf{x}_r e^{-i\omega r \Delta t}}_{\hat{\mathbf{X}}(\omega)} \underbrace{\sqrt{\frac{\Delta t}{N}} \sum_{s=0}^{N-1} \mathbf{x}_s^T e^{i\omega s \Delta t}}_{\hat{\mathbf{X}}(\omega)^*} \right] \\ &= \frac{\Delta t}{N} \sum_{r=0}^{N-1} \sum_{s=0}^{N-1} \underbrace{E[\mathbf{x}_r \mathbf{x}_s^T]}_{\mathbf{R}((r-s)\Delta t)} e^{-i\omega(r-s)\Delta t} \end{aligned} \quad (4.96)$$

Substituting the Wiener-Khinchin formula

$$\mathbf{R}((r-s)\Delta t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega') e^{i\omega'(r-s)\Delta t} d\omega' \quad (4.97)$$

and carrying the integral outside,

$$\begin{aligned} E[\hat{\mathbf{S}}_N(\omega)] &= \frac{\Delta t}{2\pi N} \int_{-\infty}^{\infty} \mathbf{S}(\omega') \sum_{r=0}^{N-1} \sum_{s=0}^{N-1} e^{-i(\omega-\omega')(r-s)\Delta t} d\omega' \\ &= \frac{N\Delta t}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega') \underbrace{\left[\frac{1}{N} \sum_{r=0}^{N-1} e^{-i(\omega-\omega')r\Delta t} \right]}_{K_1(\omega-\omega')} \underbrace{\left[\frac{1}{N} \sum_{s=0}^{N-1} e^{-i(\omega-\omega')s\Delta t} \right]^*}_{K_1(\omega-\omega')^*} d\omega' \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega') F_N(\omega - \omega') d\omega' \end{aligned} \quad (4.98)$$

where

$$F_N(\omega) = N\Delta t |K_1(\omega)|^2 \quad (4.99)$$

$K_1(\omega)$ appeared in Sect. 2.4.3 and was found to be

$$K_1(\omega) = \frac{\sin(N\omega\Delta t/2)}{N \sin(\omega\Delta t/2)} e^{-i(N-1)\omega\Delta t/2} \quad (4.100)$$

This gives

$$|K_1(\omega)|^2 = \frac{\sin^2(N\omega\Delta t/2)}{N^2 \sin^2(\omega\Delta t/2)} \quad (4.101)$$

Substituting into (4.99) gives the expression of $F_N(\omega)$ in (4.86). ■

4.8 Second Order Statistics of Scaled DTFT

In this section we study the second order statistics of the scaled DTFTs $\{\hat{\mathbf{X}}(\omega)\}$ in (4.58):

$$\hat{\mathbf{X}}(\omega) = \sqrt{\frac{\Delta t}{N}} \sum_{j=0}^{N-1} \mathbf{x}_j e^{-i\omega j \Delta t} \quad (4.102)$$

As before, $\{\mathbf{x}_j = \mathbf{x}(j\Delta t)\}_{j=0}^{N-1}$ are sample values of a zero-mean stationary process $\mathbf{x}(t)$ with PSD matrix $\mathbf{S}(\omega)$. The statistical properties of $\hat{\mathbf{X}}(\omega)$ at the FFT frequencies are of fundamental importance to Bayesian OMA theory.

4.8.1 Complex Covariance and Pseudo-covariance Matrix

Since $\hat{\mathbf{X}}(\omega)$ is a complex vector, its second order statistics comprise those of the real and imaginary parts. For two frequencies α and β , the second order statistics between $\hat{\mathbf{X}}(\alpha)$ and $\hat{\mathbf{X}}(\beta)$ comprise the following real-valued cross covariance matrices:

$$\begin{aligned} E[\operatorname{Re} \hat{\mathbf{X}}(\alpha) \operatorname{Re} \hat{\mathbf{X}}(\beta)^T] & \quad E[\operatorname{Re} \hat{\mathbf{X}}(\alpha) \operatorname{Im} \hat{\mathbf{X}}(\beta)^T] \\ E[\operatorname{Im} \hat{\mathbf{X}}(\alpha) \operatorname{Re} \hat{\mathbf{X}}(\beta)^T] & \quad E[\operatorname{Im} \hat{\mathbf{X}}(\alpha) \operatorname{Im} \hat{\mathbf{X}}(\beta)^T] \end{aligned} \quad (4.103)$$

Working with these matrices can be cumbersome for analysis or computation, involving algebraic operations that are similar or repetitive. An equivalent but more elegant way is to work with the complex-valued generalization of the cross covariance matrix and the cross ‘pseudo-covariance matrix’, defined respectively as

$$\mathbf{E} = E[\hat{\mathbf{X}}(\alpha) \hat{\mathbf{X}}(\beta)^*] \quad \mathbf{Q} = E[\hat{\mathbf{X}}(\alpha) \hat{\mathbf{X}}(\beta)^T] \quad (4.104)$$

Both $\mathbf{E}$ and $\mathbf{Q}$ are $n \times n$ complex matrix. They can be determined from the matrices in (4.103), and vice versa (Section A.2):

$$\begin{aligned} E[\operatorname{Re} \hat{\mathbf{X}}(\alpha) \operatorname{Re} \hat{\mathbf{X}}(\beta)^T] &= \frac{\operatorname{Re} \mathbf{E} + \operatorname{Re} \mathbf{Q}}{2} & E[\operatorname{Re} \hat{\mathbf{X}}(\alpha) \operatorname{Im} \hat{\mathbf{X}}(\beta)^T] &= \frac{-\operatorname{Im} \mathbf{E} + \operatorname{Im} \mathbf{Q}}{2} \\ E[\operatorname{Im} \hat{\mathbf{X}}(\alpha) \operatorname{Re} \hat{\mathbf{X}}(\beta)^T] &= \frac{\operatorname{Im} \mathbf{E} + \operatorname{Im} \mathbf{Q}}{2} & E[\operatorname{Im} \hat{\mathbf{X}}(\alpha) \operatorname{Im} \hat{\mathbf{X}}(\beta)^T] &= \frac{\operatorname{Re} \mathbf{E} - \operatorname{Re} \mathbf{Q}}{2} \end{aligned} \quad (4.105)$$

Two complex vectors are uncorrelated if their cross covariance and pseudo-covariance matrices are zero.

4.8.2 Convolution Formula

The cross covariance and pseudo-covariance matrices of scaled DTFTs can be expressed in terms of the PSD matrix $\mathbf{S}(\omega)$. For any two frequencies α and β (rad/s),

$$E[\hat{\mathbf{X}}(\alpha)\hat{\mathbf{X}}(\beta)^*] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega) F_{\alpha\beta}(\omega) d\omega \quad (4.106)$$

$$F_{\alpha\beta}(\omega) = N\Delta t K_1(\alpha - \omega) K_1(\beta - \omega)^* \quad (4.107)$$

where

$$K_1(\omega) = \frac{1}{N} \sum_{j=0}^{N-1} e^{-i\omega j\Delta t} = \frac{\sin(N\omega\Delta t/2)}{N \sin(\omega\Delta t/2)} e^{-i(N-1)\omega\Delta t/2} \quad (4.108)$$

appeared in Sect. 2.4.3. See the end for proof.

PSD Matrix

Setting $\beta = \alpha$ in (4.106) gives the expectation of sample PSD matrix $E[\hat{\mathbf{S}}_N(\alpha)] = E[\hat{\mathbf{X}}(\alpha)\hat{\mathbf{X}}(\alpha)^*]$. The resulting expression agrees with (4.85) because

$$F_{\alpha\alpha}(\omega) = N\Delta t |K_1(\alpha - \omega)|^2 = F_N(\alpha - \omega) \quad (4.109)$$

Cross Pseudo-covariance Matrix

Although presented like a covariance matrix, (4.106) can be used for obtaining the pseudo-covariance by replacing β with $-\beta$. This is because $\hat{\mathbf{X}}(\beta)^T = \hat{\mathbf{X}}(-\beta)^*$ and so $E[\hat{\mathbf{X}}(\alpha)\hat{\mathbf{X}}(\beta)^T] = E[\hat{\mathbf{X}}(\alpha)\hat{\mathbf{X}}(-\beta)^*]$. Then

$$E[\hat{\mathbf{X}}(\alpha)\hat{\mathbf{X}}(\beta)^T] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega) F_{\alpha,-\beta}(\omega) d\omega \quad (4.110)$$

Kernel Function

As mentioned in Sect. 2.4.3, in terms of the dimensionless variable $u = \omega\Delta t/2\pi$,

$$K_1(\omega) = D_N(u) e^{-i\pi(N-1)u} \quad u = \frac{\omega\Delta t}{2\pi} \quad (4.111)$$

where $D_N(u) = \sin(N\pi u)/N \sin \pi u$ is the Dirichlet kernel. Substituting into (4.107) gives

$$F_{\alpha\beta}(\omega) = G_{\alpha\beta}(u) e^{-i\pi(N-1)(u_\alpha - u_\beta)} \quad (4.112)$$

$$G_{\alpha\beta}(u) = N\Delta t D_N(u_\alpha - u) D_N(u_\beta - u) \quad (4.113)$$

$$u = \frac{\omega \Delta t}{2\pi} \quad u_\alpha = \frac{\alpha \Delta t}{2\pi} \quad u_\beta = \frac{\beta \Delta t}{2\pi} \quad (4.114)$$

It can be shown that (see the end)

$$\int_{-\pi/\Delta t}^{\pi/\Delta t} K_1(\alpha - \omega) K_1(\beta - \omega)^* d\omega = \frac{2\pi}{N\Delta t} K_1(\alpha - \beta) \quad (4.115)$$

and so

$$\frac{1}{2\pi} \int_{-\pi/\Delta t}^{\pi/\Delta t} F_{\alpha\beta}(\omega) d\omega = K_1(\alpha - \beta) \quad (4.116)$$

Setting $\beta = \alpha$ gives $\int_{-\pi/\Delta t}^{\pi/\Delta t} F_{\alpha\alpha}(\omega) d\omega / 2\pi = K_1(0) = 1$. This proves the claim in Sect. 4.7.2 that $\int_{-\pi/\Delta t}^{\pi/\Delta t} F_N(\omega) d\omega / 2\pi = 1$, because $F_{\alpha\alpha}(\omega) = F_N(\alpha - \omega)$ and its integral over one period is invariant to the shift α .

Proof of (4.106) (Cross Covariance Matrix of Scaled DTFT)

The structure of the proof is similar to that for (4.85). From definition,

$$\begin{aligned} E[\hat{\mathbf{X}}(\alpha) \hat{\mathbf{X}}(\beta)^*] &= E \left[\underbrace{\sqrt{\frac{\Delta t}{N}} \sum_{r=0}^{N-1} \mathbf{x}_r e^{-i\alpha r \Delta t}}_{\hat{\mathbf{X}}(\alpha)} \underbrace{\sqrt{\frac{\Delta t}{N}} \sum_{s=0}^{N-1} \mathbf{x}_s^T e^{i\beta s \Delta t}}_{\hat{\mathbf{X}}(\beta)^*} \right] \\ &= \frac{\Delta t}{N} \sum_{r=0}^{N-1} \sum_{s=0}^{N-1} \underbrace{E[\mathbf{x}_r \mathbf{x}_s^T]}_{\mathbf{R}((r-s)\Delta t)} e^{-i(\alpha r - \beta s)\Delta t} \end{aligned} \quad (4.117)$$

Using the Wiener-Khinchin formula,

$$\mathbf{R}((r-s)\Delta t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega) e^{i\omega(r-s)\Delta t} d\omega \quad (4.118)$$

Substituting into (4.117) and carrying the integral outside,

$$\begin{aligned}
E[\hat{\mathbf{X}}(\alpha)\hat{\mathbf{X}}(\beta)^*] &= \frac{\Delta t}{2\pi N} \int_{-\infty}^{\infty} \mathbf{S}(\omega) \left[\sum_{r=0}^{N-1} \sum_{s=0}^{N-1} e^{\mathbf{i}\omega(r-s)\Delta t} e^{-\mathbf{i}(\alpha r - \beta s)\Delta t} \right] d\omega \\
&= \frac{\Delta t}{2\pi N} \int_{-\infty}^{\infty} \mathbf{S}(\omega) \left[\sum_{r=0}^{N-1} \sum_{s=0}^{N-1} e^{-\mathbf{i}(\alpha - \omega)r\Delta t} e^{\mathbf{i}(\beta - \omega)s\Delta t} \right] d\omega \\
&= \frac{N\Delta t}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}(\omega) \underbrace{\left[\frac{1}{N} \sum_{r=0}^{N-1} e^{-\mathbf{i}(\alpha - \omega)r\Delta t} \right]}_{K_1(\alpha - \omega)} \underbrace{\left[\frac{1}{N} \sum_{s=0}^{N-1} e^{-\mathbf{i}(\beta - \omega)s\Delta t} \right]^*}_{K_1(\beta - \omega)^*} d\omega
\end{aligned} \tag{4.119}$$

■

Proof of (4.115) (Integral of Kernel Product)

$$\begin{aligned}
K_1(\alpha - \omega)K_1(\beta - \omega)^* &= \left[\frac{1}{N} \sum_{r=0}^{N-1} e^{-\mathbf{i}(\alpha - \omega)r\Delta t} \right] \left[\frac{1}{N} \sum_{s=0}^{N-1} e^{\mathbf{i}(\beta - \omega)s\Delta t} \right] \\
&= \frac{1}{N^2} \sum_{r=0}^{N-1} \sum_{s=0}^{N-1} e^{-\mathbf{i}(\alpha r - \beta s)\Delta t} e^{\mathbf{i}\omega(r-s)\Delta t}
\end{aligned} \tag{4.120}$$

Integrating w.r.t. ω ,

$$\begin{aligned}
\int_{-\pi/\Delta t}^{\pi/\Delta t} K_1(\alpha - \omega)K_1(\beta - \omega)^* d\omega &= \frac{1}{N^2} \sum_{r=0}^{N-1} \sum_{s=0}^{N-1} e^{-\mathbf{i}(\alpha r - \beta s)\Delta t} \underbrace{\int_{-\pi/\Delta t}^{\pi/\Delta t} e^{\mathbf{i}\omega(r-s)\Delta t} d\omega}_{2\pi/\Delta t \text{ if } s=r; \text{ else } 0} \\
&= \frac{2\pi}{N\Delta t} \times \underbrace{\frac{1}{N} \sum_{r=0}^{N-1} e^{-\mathbf{i}(\alpha - \beta)r\Delta t}}_{K_1(\alpha - \beta)}
\end{aligned} \tag{4.121}$$

■

4.8.3 Long-Data Asymptotics of Scaled FFT

The second order statistics of scaled FFTs can be obtained by evaluating (4.106) at the FFT frequencies. The resulting expression simplifies significantly when the data duration is ‘long’. Consider the scaled FFTs

$$\hat{\mathbf{X}}_p = \hat{\mathbf{X}}(\alpha) \quad \hat{\mathbf{X}}_q = \hat{\mathbf{X}}(\beta) \quad (4.122)$$

at the FFT frequencies

$$\alpha = \frac{2\pi p}{N\Delta t} \quad \beta = \frac{2\pi q}{N\Delta t} \quad (\text{rad/s}) \quad (4.123)$$

For fixed Δt , it can be shown that as $N \rightarrow \infty$ (and hence data duration $N\Delta t \rightarrow \infty$):

For positive integers p and q below the Nyquist index (integer part of $N/2$),

A1. Cross-covariance $E[\hat{\mathbf{X}}_p \hat{\mathbf{X}}_q^*] \rightarrow \mathbf{0}$ for any $p \neq q$

A2. Cross pseudo-covariance $E[\hat{\mathbf{X}}_p \hat{\mathbf{X}}_q^T] \rightarrow \mathbf{0}$ for any p, q

In addition, if there is no aliasing (e.g., $\Delta t \rightarrow 0$), then

A3. Auto-covariance $E[\hat{\mathbf{X}}_p \hat{\mathbf{X}}_p^*] \rightarrow \mathbf{S}(\omega_p)$

These asymptotic results are *remarkably clean*. Omitting technical details, they can be reasoned based on (4.106) and the behavior of $G_{\alpha\beta}(u)$ in (4.113). The latter is illustrated in Fig. 4.11 ($\alpha \neq \beta$) and Fig. 4.12 ($\alpha = \beta$). The function $G_{\alpha\beta}(u)$ has a

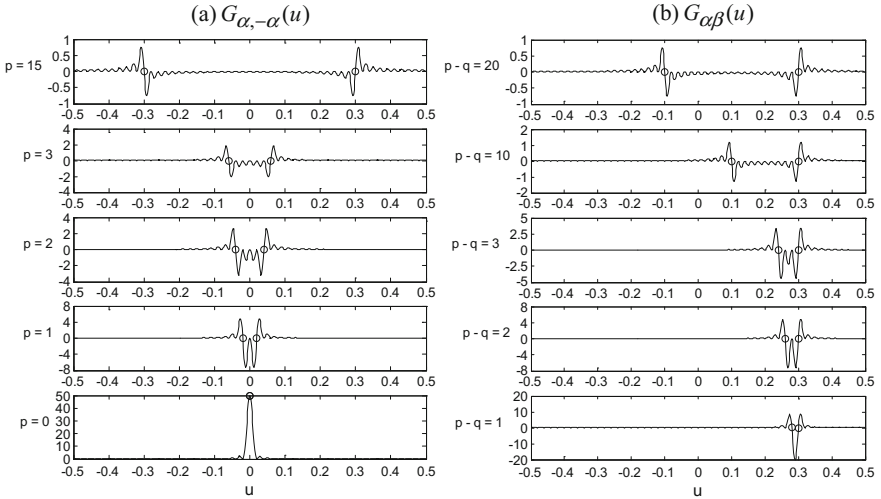


Fig. 4.11 Plot of $G_{\alpha\beta}(u)$ for $\alpha = 2\pi p/N\Delta t$, $\beta = 2\pi q/N\Delta t$ and $N = 50$. In **a** $q = -p$, i.e., pseudo-covariance at ω_p ; the function is symmetric about the origin; for large N it is concentrated near $u = \pm p/N$; when $p = 0$ it tends to the Delta function as $N \rightarrow \infty$. In **b** $p/N = 0.3$ and $q \neq p$, i.e., cross covariance or pseudo-covariance between two different frequencies

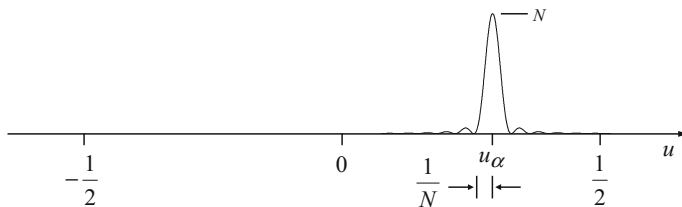


Fig. 4.12 Plot of $G_{xx}(u)$. It becomes the Delta function centered at $u_{\alpha} = \alpha\Delta t/2\pi$ as $N \rightarrow \infty$

period of 1 and so the pattern in the figures is repeating outside the range $(-1/2, 1/2)$. It is fast varying over a scale of $O(1/N)$. For $\alpha \neq \beta$, the positive and negative part of $G_{\alpha\beta}$ tend to cancel out each other, giving an integral that approaches zero as $N \rightarrow \infty$. This leads to A1 and A2. In Fig. 4.12, $G_{xx}(u)$ tends to the Dirac Delta function as $N \rightarrow \infty$ and so $E[\hat{\mathbf{X}}_p \hat{\mathbf{X}}_p^*] \rightarrow \mathbf{S}(\omega_p) +$ aliased contributions from other branches. A3 then follows if there is no aliasing.

White noise

When $\mathbf{S}(\omega) = \mathbf{S}_w$ is constant, i.e., white noise, A1 and A2 holds for *any duration*, i.e., even when $N\Delta t < \infty$. To see this, note that if the process is band-limited white between $\pm\pi/\Delta t$ (rad/s) then, using (4.116),

$$E[\hat{\mathbf{X}}(\alpha)\hat{\mathbf{X}}(\beta)^*] = \mathbf{S}_w K_1(\alpha - \beta) \quad (4.124)$$

This is zero when α and β are distinct FFT frequencies because then $\alpha - \beta$ is an integer multiple of $2\pi/N\Delta t$, at which $K_1(\alpha - \beta) = 0$ (see Fig. 4.7). For (ideal) white noise over frequency $\pm\infty$, the integral in (4.106) is still zero because the integrand has a period of $2\pi/\Delta t$.

For A3 to hold, it is still required that $N \rightarrow \infty$ and there is no aliasing. Thus, for general non-white process and finite data duration in practice, A1 and A2 are more robust approximations than A3.

4.8.4 How Long Is Long?

For finite data duration, the quality of long-data asymptotic approximation depends on the shape of the PSD $\mathbf{S}(\omega)$. As discussed in the last section, when $\mathbf{S}(\omega)$ is constant, i.e., white noise, A1 and A2 hold even for finite duration. In the general case, the approximation is good when $F_{\alpha\beta}(\omega)$ is ‘fast varying’ compared to $\mathbf{S}(\omega)$. Translating this in the context of OMA,

In the resonance band of a vibration mode, the number of FFT points should be large compared to 1

This can be reasoned as follow.

Consider the resonance band of a mode where $\mathbf{S}(\omega)$ varies significantly. The frequency ω ranges between $2\pi f(1 \pm \zeta\kappa)$ (rad/s) where f is the natural frequency (Hz), ζ is the damping ratio and κ is a dimensionless bandwidth factor; e.g., $\kappa = 1$ corresponds to the half-power band. The resonance bandwidth is then $4\pi\zeta\kappa f$ (rad/s). The function $F_{\alpha\beta}(\omega)$ can be considered fast varying compared to $\mathbf{S}(\omega)$ when it is concentrated in a small region compared to the resonance bandwidth. Note that $F_{\alpha\beta}(\omega)$ is proportional to $G_{\alpha\beta}(u)$ ($u = \omega\Delta t/2\pi$). The latter is concentrated in a neighborhood of $O(1/N)$ w.r.t. u . This implies that $F_{\alpha\beta}(\omega)$ is concentrated in a neighborhood of $O(2\pi/T)$ w.r.t. ω , where $T = N\Delta t$ is the data duration. Thus, long-data asymptotic approximation is good when $2\pi/T \ll 4\pi\zeta\kappa f$. That is,

$$N_f = 2\zeta\kappa fT \gg 1 \quad (4.125)$$

Note that N_f is the number of FFT points in the resonance band because the bandwidth is $2\zeta\kappa f$ (Hz) and the FFT frequencies are equally spaced at $\Delta f = 1/T$ (Hz). This is an important scale in the study of ‘uncertainty laws’ in Chaps. 15 and 16.

Example 4.7 (Correlation of scaled FFT) This example illustrates the cross correlation of scaled FFTs at different frequencies of the synthetic acceleration data in Example 4.4. Section 4.8.3 says that $E[\hat{X}_p\hat{X}_q^*]$ ($p \neq q$) is asymptotically zero for long data. Consider the cross correlation coefficient between two FFT frequencies $f_p = p/N\Delta t$ and $f_q = q/N\Delta t$ (Hz), defined as $|E[\hat{X}_p\hat{X}_q^*]|/\sqrt{S_p S_q}$, where $S_p = E[|\hat{X}_p|^2]$ and $S_q = E[|\hat{X}_q|^2]$. It is dimensionless and ranges between 0 and 1. When $p = q$ it is equal to 1. Figure 4.13 shows the cross correlation coefficients between

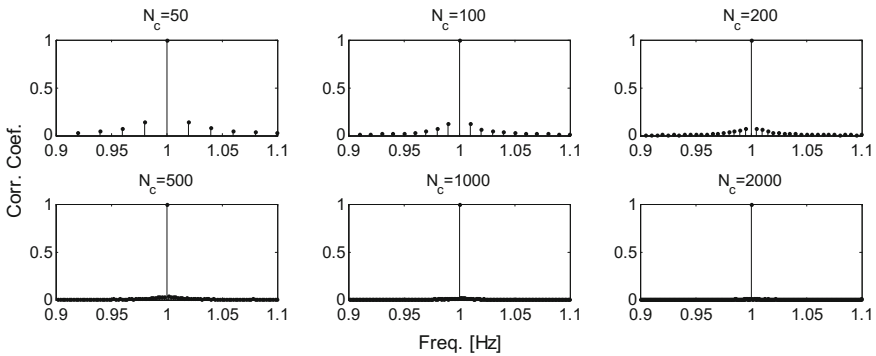


Fig. 4.13 Cross correlation coefficient of scaled FFTs between the natural frequency and other frequencies; N_c = data duration/natural period

the natural frequency and other frequencies for different data durations. They are estimated from 100,000 i.i.d. time histories. As the data duration increases, the cross correlation tends to zero. ■

4.9 Asymptotic Distribution of Scaled FFT

As $\{\hat{\mathbf{X}}_k\}$ are complex vectors, their joint distribution is given by the joint distribution of their real and imaginary parts. For long data, they asymptotically have a ‘complex Gaussian distribution’, i.e., their real and imaginary parts are jointly Gaussian. This is by virtue of the Central Limit Theorem (details omitted), as each $\hat{\mathbf{X}}_k$ is a sum of N correlated random complex vectors. When $\mathbf{x}(t)$ is a Gaussian process, $\{\hat{\mathbf{X}}_k\}$ are complex Gaussian regardless of the data duration because Gaussian vectors remain to be so under a linear transformation (FFT is linear). For long data, since $\{\hat{\mathbf{X}}_k\}$ are asymptotically Gaussian and uncorrelated (Sect. 4.8.3), they are also independent, i.e., their joint PDF is equal to the product of their marginal PDFs.

The marginal PDF of the $n \times 1$ complex vector $\hat{\mathbf{X}}_k$ is the PDF of the $2n \times 1$ real vector $[\text{Re } \hat{\mathbf{X}}_k; \text{Im } \hat{\mathbf{X}}_k]$. For long data, it is asymptotically given by

$$p_{\hat{\mathbf{X}}_k}(\mathbf{x}_k) = (2\pi)^{-n} |\mathbf{C}_k|^{-1/2} \left\{ -\frac{1}{2} \begin{bmatrix} \text{Re } \mathbf{x}_k \\ \text{Im } \mathbf{x}_k \end{bmatrix}^T \mathbf{C}_k^{-1} \begin{bmatrix} \text{Re } \mathbf{x}_k \\ \text{Im } \mathbf{x}_k \end{bmatrix} \right\} \quad \mathbf{x}_k \text{ } n \times 1 \text{ complex} \quad (4.126)$$

where

$$\mathbf{C}_k = \begin{bmatrix} E \left[\text{Re } \hat{\mathbf{X}}_k \text{Re } \hat{\mathbf{X}}_k^T \right] & E \left[\text{Re } \hat{\mathbf{X}}_k \text{Im } \hat{\mathbf{X}}_k^T \right] \\ E \left[\text{Im } \hat{\mathbf{X}}_k \text{Re } \hat{\mathbf{X}}_k^T \right] & E \left[\text{Im } \hat{\mathbf{X}}_k \text{Im } \hat{\mathbf{X}}_k^T \right] \end{bmatrix} \quad (4.127)$$

Equation (4.126) can be written in a compact form in terms of the complex covariance matrix $\mathbf{E}_k = E[\hat{\mathbf{X}}_k \hat{\mathbf{X}}_k^*]$ and pseudo-covariance matrix $\mathbf{Q}_k = E[\hat{\mathbf{X}}_k \hat{\mathbf{X}}_k^T]$:

$$p_{\hat{\mathbf{X}}_k}(\mathbf{x}_k) = \pi^{-n} |\mathbf{E}_k|^{-1/2} |\mathbf{P}_k|^{-1/2} \exp \left\{ -\frac{1}{2} \begin{bmatrix} \mathbf{x}_k \\ \overline{\mathbf{x}}_k \end{bmatrix}^* \begin{bmatrix} \mathbf{E}_k & \mathbf{Q}_k \\ \mathbf{Q}_k^* & \overline{\mathbf{E}}_k \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{x}_k \\ \overline{\mathbf{x}}_k \end{bmatrix} \right\} \quad (4.128)$$

where

$$\mathbf{P}_k = \overline{\mathbf{E}}_k - \mathbf{Q}_k^* \mathbf{E}_k^{-1} \mathbf{Q}_k \quad (4.129)$$

See Sect. A.3 for proof and Chap. A for further details on complex Gaussian distribution.

Further simplification results by noting that for long data the pseudo-covariance matrix $\mathbf{Q}_k$ is asymptotically zero. The marginal PDF of $\hat{\mathbf{X}}_k$ in (4.128) then simplifies to

$$p_{\hat{\mathbf{X}}_k}(\mathbf{x}_k) = \pi^{-n} |\mathbf{E}_k|^{-1} \exp(-\mathbf{x}_k^* \mathbf{E}_k^{-1} \mathbf{x}_k) \quad (\text{long data}) \quad (4.130)$$

If there is no aliasing (e.g., $\Delta t \rightarrow 0$), $\mathbf{E}_k$ can be further replaced by the theoretical PSD matrix $\mathbf{S}(\omega_k)$.

4.10 Asymptotic Distribution of Sample PSD

To supplement Sect. 4.6, this section presents the distribution of the averaged sample PSD based on M i.i.d. data segments of a zero-mean stationary process:

$$\hat{\mathbf{E}}_k = \frac{1}{M} \sum_{r=1}^M \hat{\mathbf{X}}_k^{(r)} \hat{\mathbf{X}}_k^{(r)*} \quad (4.131)$$

Assume long data so that $\hat{\mathbf{X}}_k^{(r)}$ s are complex Gaussian. Then it can be shown that the sum $\sum_{r=1}^M \hat{\mathbf{X}}_k^{(r)} \hat{\mathbf{X}}_k^{(r)*}$ has a ‘complex Wishart distribution’ with M degrees of freedom. The PDF of $\hat{\mathbf{E}}_k$ is given by

$$p_{\hat{\mathbf{E}}_k}(\mathbf{E}) = M^{nM} \frac{|\mathbf{E}|^{M-n}}{I(\mathbf{E}_k)} e^{-M \text{tr}(\mathbf{E}_k^{-1} \mathbf{E})} \quad (M \geq n, \mathbf{E} \text{ Hermitian}) \quad (4.132)$$

$$I(\mathbf{E}_k) = \pi^{n(n-1)/2} |\mathbf{E}_k|^M \prod_{i=1}^n (M-i)! \quad (4.133)$$

$$\mathbf{E}_k = E[\hat{\mathbf{E}}_k] \quad (4.134)$$

This PDF is in fact the joint PDF of the (real) diagonal entries and the real and imaginary parts of the lower (or upper) off-diagonal entries of $\hat{\mathbf{E}}_k$. It is valid for $M \geq n$, which ensures that $\hat{\mathbf{E}}_k$ is non-singular. The sample PSDs at different frequencies are asymptotically independent because the scaled FFTs are. Their joint

PDF is equal to the product of their marginal PDFs. See Sect. A.5 for further details on complex Wishart distribution.

4.10.1 Scalar Process

Applying (4.132) to a scalar process ($n = 1$) gives the PDF of the averaged sample PSD, which is often plotted for visualization. For $M = 1$ (no averaging) the PDF reduces to the exponential PDF:

$$p_{\hat{E}_k}(S) = \frac{1}{E_k} \exp\left(-\frac{S}{E_k}\right) \quad (S \geq 0) \quad (4.135)$$

where E_k is the expectation of $\hat{E}_k$. Note that the standard deviation of an exponential random variable is equal to its mean. This implies that the standard deviation of the sample PSD without averaging is as large as its mean, no matter how high the sample rate or how long the data is. Intuitively, the additional information from data goes into exploring higher frequencies (former) or improving frequency resolution (latter) rather than improving the statistical accuracy of the sample PSD at a particular frequency.

For the general case $M \geq 1$, the PDF is given by

$$p_{\hat{E}_k}(S) = \frac{M^M S^{M-1}}{(M-1)! E_k^M} \exp\left(-\frac{MS}{E_k}\right) \quad (S \geq 0) \quad (4.136)$$

Using this expression, it can be reasoned by PDF transformation that the random variable $2M\hat{E}_k/E_k$ has a Chi-square distribution with $2M$ degrees of freedom and so it has mean $2M$ and variance $4M$. This implies that $\text{var}[\hat{E}_k] = E_k^2/M$, which agrees with the fact that $\hat{E}_k$ is the average of M i.i.d. random variables with mean E_k and variance E_k^2 .

Consider the dimensionless random variable $Y = \hat{E}_k/E_k$. Its PDF is, by PDF transformation,

$$p_Y(y) = \frac{M^M y^{M-1}}{(M-1)!} \exp(-My) \quad (y \geq 0) \quad (4.137)$$

Clearly, $E[Y] = 1$ and $\text{var}[Y] = 1/M$. Figure 4.14 shows $p_Y(y)$ for different M . When $M = 1$ it is simply the exponential PDF with mean 1. As M increases it tends to a Gaussian PDF centered at 1 with diminishing variance $1/M$. This agrees with the Central Limit Theorem, as Y is in fact the average of M i.i.d. variables, each with mean 1 and variance 1.

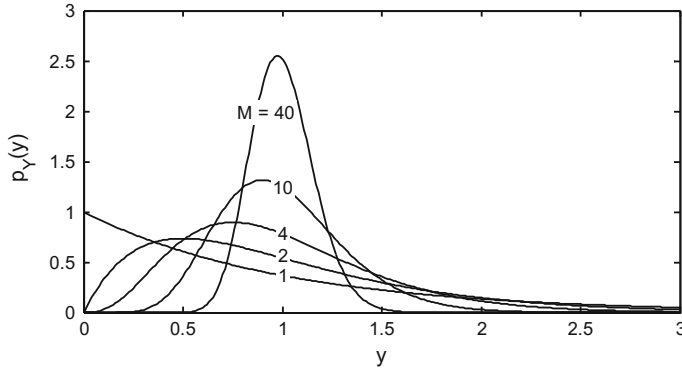


Fig. 4.14 PDF of $Y = \hat{E}_k / E_k$ for different number of segments M

4.11 Summary of Fourier Formulas, Units and Conventions

Table 4.2 summarizes the expectation of sample estimators for the correlation function and PSD; and their relationship with the theoretical quantity they estimate. The units indicated assume that $\mathbf{x}(t)$ has a unit of volt (V) and time is measured in second (s). Relevant sections are indicated.

4.11.1 Multiplier in Wiener-Khinchin Formula

As in the case of deterministic process (Sect. 2.7.1), the multiplier $1/2\pi$ in the Wiener-Khinchin formula $\mathbf{R}(\tau) = (2\pi)^{-1} \int_{-\infty}^{\infty} \mathbf{S}(\omega) e^{i\omega\tau} d\omega$ has a direct implication on the units of $\mathbf{S}(\omega)$ and the scaled FT $\mathbf{X}_T(\omega)$. The PSD $\mathbf{S}(\omega)$ here has unit V^2/Hz , not $V^2/(\text{rad/s})$, despite the fact that the integral is over ω (rad/s). Correspondingly, $\mathbf{X}_T(\omega)$ has unit $V/\sqrt{\text{Hz}}$, not $V/\sqrt{\text{rad/s}}$. If one uses $\mathbf{R}(\tau) = \int_{-\infty}^{\infty} \mathbf{S}(\omega) e^{i\omega\tau} d\omega$ then $\mathbf{S}(\omega)$ must have unit $V^2/(\text{rad/s})$ and $\mathbf{X}_T(\omega)$ has unit $V/\sqrt{\text{rad/s}}$; and vice versa. The same rule applies to the sample counterparts.

4.11.2 One-Sided Versus Two-Sided Spectrum

The PSDs in this chapter have been defined in a ‘two-sided’ manner, where the frequency ω takes on both positive and negative values. In the Parseval equality and the inverse FT for the correlation function (Wiener-Khinchin formula), the integration domain covers both positive and negative frequencies. The two-sided

Table 4.2 Summary of sample estimators, their units and expectation; $D_N(u) = \sin(N\pi u)/N \sin \pi u$ is the Dirichlet kernel

Type	Statistic	Expectation
Theoretical	$\underbrace{\mathbf{R}(\tau)}_{V^2} = E[\underbrace{\mathbf{x}(t+\tau)}_V \underbrace{\mathbf{x}(t)^T}_V]$ (Sect. 4.1) $\underbrace{\mathbf{S}(\omega)}_{V^2/\text{Hz}} = \lim_{T \rightarrow \infty} E[\underbrace{\mathbf{X}_T(\omega)}_{V/\sqrt{\text{Hz}}} \underbrace{\mathbf{X}_T(\omega)^*}_{V/\sqrt{\text{Hz}}}]$ $\underbrace{\mathbf{X}_T(\omega)}_{V/\sqrt{\text{Hz}}} = \frac{1}{\sqrt{T}} \underbrace{\int_0^T \mathbf{x}(t)}_V \underbrace{e^{-i\omega t}}_{1/\sqrt{s}} \underbrace{dt}_s$ (Sect. 4.2)	
Sample $\{\mathbf{x}(t) : 0 \leq t \leq T\}$ continuous time of duration T	$\underbrace{\hat{\mathbf{R}}_T(\tau)}_{V^2} = \underbrace{\frac{1}{T}}_{1/s} \underbrace{\int_0^T \mathbf{x}(t+\tau)}_V \underbrace{\mathbf{x}(t)^T}_V \underbrace{dt}_s$ $(0 \leq \tau < T)$ (Sect. 4.4.1) $\underbrace{\hat{\mathbf{S}}_T(\omega)}_{V^2/\text{Hz}} = \underbrace{\mathbf{X}_T(\omega)}_{V/\sqrt{\text{Hz}}} \underbrace{\mathbf{X}_T(\omega)^*}_{V/\sqrt{\text{Hz}}}$ (Sect. 4.4.2)	$E[\hat{\mathbf{R}}_T(\tau)] = (1 - \tau/T) \mathbf{R}(\tau)$ (Sect. 4.7.1) $E[\hat{\mathbf{S}}_T(\omega)] = \underbrace{\frac{1}{2\pi}}_{1/\text{rad}} \underbrace{\int_{-\infty}^{\infty} \mathbf{S}(\omega')}_{V^2/\text{Hz}} \underbrace{F_T(\omega - \omega')}_s \underbrace{d\omega'}_{\text{rad/s}}$ $F_T(\omega) = 4 \sin^2(\omega T/2)/\omega^2 T$ (Sect. 4.7.2)

(continued)

Table 4.2 (continued)

Type	Statistic	Expectation
Sample $\{\mathbf{x}_j = \mathbf{x}(j\Delta t)\}_{j=0}^{N-1}$ discrete-time at interval Δt and duration $N\Delta t$		
	$\underbrace{\hat{\mathbf{R}}_r}_{V^2} = \frac{1}{N} \sum_{j=0}^{N-1-r} \underbrace{\mathbf{x}_{j+r}}_V \underbrace{\mathbf{x}_j^T}_V \quad (0 \leq r \leq N-1)$ (Sect. 4.5.1)	$E[\hat{\mathbf{R}}_r] = (1 - r/N) \mathbf{R}(r\Delta t)$ (Sect. 4.7.1)
	$\underbrace{\hat{\mathbf{S}}_N(\omega)}_{V^2/\text{Hz}} = \underbrace{\hat{\mathbf{X}}(\omega)}_{V/\sqrt{\text{Hz}}} \underbrace{\hat{\mathbf{X}}(\omega)^*}_{V/\sqrt{\text{Hz}}} \underbrace{\frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \mathbf{x}_j}_{\sqrt{s}} \underbrace{e^{-i\omega j\Delta t}}_V$ $\underbrace{\hat{\mathbf{X}}(\omega)}_{V/\sqrt{\text{Hz}}} = \underbrace{\sqrt{\frac{\Delta t}{N}}}_{\sqrt{s}} \underbrace{\sum_{j=0}^{N-1} \mathbf{x}_j}_V e^{-i\omega j\Delta t}$ (Sect. 4.5.2)	$E[\hat{\mathbf{S}}_N(\omega)] = \underbrace{\frac{1}{2\pi}}_{V^2/\text{Hz}} \underbrace{\int_{-\infty}^{\infty} \mathbf{S}(\omega')}_{V^2/\text{Hz}} \underbrace{F_N(\omega - \omega')}_s d\omega'_{\text{rad/s}}$ $F_N(\omega) = N\Delta t D_N^2(u)$ $u = \omega\Delta t/2\pi$ (Sect. 4.7.2)
	$\underbrace{\hat{\mathbf{S}}_N(\alpha, \beta)}_{V^2/\text{Hz}} = \underbrace{\hat{\mathbf{X}}(\alpha)}_{V/\sqrt{\text{Hz}}} \underbrace{\hat{\mathbf{X}}(\beta)^*}_{V/\sqrt{\text{Hz}}}$ (Sect. 4.8.1)	$E[\hat{\mathbf{S}}_N(\alpha, \beta)] = \underbrace{\frac{1}{2\pi}}_{V^2/\text{Hz}} \underbrace{\int_{-\infty}^{\infty} \mathbf{S}(\omega)}_{V^2/\text{Hz}} \underbrace{F_{\alpha\beta}(\omega)}_s d\omega_{\text{rad/s}}$ $F_{\alpha\beta}(\omega) = N\Delta t D_N(u_\alpha - u) D_N(u_\beta - u) e^{-i\pi(N-1)(u_\alpha - u_\beta)}$ $u = \omega\Delta t/2\pi, u_\alpha = \alpha\Delta t/2\pi, u_\beta = \beta\Delta t/2\pi$ (Sect. 4.8.2)

definition stems mathematically from the FTs. For a scalar process, since the PSD for negative frequencies is just the mirror image of the PSD for positive frequencies, it is sufficient in applications to consider positive frequencies only. In terms of this ‘one-sided’ PSD, the integration domain of the Parseval equality and the inverse FT for the correlation function only covers positive frequencies. To maintain correctness of these theoretical relationships, the one-sided PSD is defined as *twice* of the two-sided PSD. In the study of input-output relationship of a system, whether the PSD is defined in a one-sided or two-sided manner does not affect the derived results, as long as their inputs and outputs are consistently defined. That is, when the input is one-sided, so is the output; and vice versa.

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Chapter 5

Stochastic Structural Dynamics

Abstract This chapter analyzes the statistical response of a linear-elastic structure subjected to unknown excitations modeled by a stationary stochastic process. It is a probabilistic version of Chap. 3. Both stationary and transient response statistics are analyzed. For stationary statistics, a modal superposition approach in the frequency domain is adopted. For transient statistics, a state-space approach in the time domain is adopted. Although transient statistics are of less relevance in operational modal analysis, they are discussed for better appreciation of the meaning of stationary statistics. Their large time limits are the stationary statistics.

Keywords Response variance • Transient statistics • Lyapunov equation

If we know the properties of a structure, its initial conditions and excitation time history at all DOFs, then theoretically by solving the equation of motion we can predict the response as a function of time. This is a (deterministic) structural dynamics problem, as discussed in Chap. 3. In that context there is no uncertainty associated with the response, even when its time history may look erratic.

If the excitation is unknown then logically we cannot predict the response. Nevertheless, if we are willing to make some ‘probabilistic assumptions’ about the excitation then it is possible to predict the response in a probabilistic manner. Instead of predicting what its value will be (which is not possible), we aim at predicting the statistical properties (e.g., mean, variance) or even the probability distribution of the response (generally a more difficult task). This is a rational approach for studying the ambient vibration of structures because the excitation sources (e.g., wind, microtremor and cultural activities) are so abundant that it is impractical to measure them all or to make assumptions about what their time histories are in a precise manner.

In this chapter we discuss the statistical properties of the response of a linear-elastic time-invariant structure subjected to unknown excitations modeled as a stationary stochastic process. This belongs to the subject of ‘random vibration analysis’ or ‘stochastic structural dynamics’. Comprehensive references are available, e.g., Lin (1967), Soong and Grigoriu (1993) and Lutes and Sarkani (1996).

In a general context, this subject can be technically involved, e.g., in the existence of limits of integrals and derivatives of stochastic processes. These issues are treated formally in the subject of ‘stochastic calculus’, which has wide applications in science and engineering; see, e.g., Kloeden and Platen (1992) and Grigoriu (2002). We assume that the excitation is ‘well-behaved’ so that the mathematical operations involved are valid. This has traded off simplicity with mathematical rigor but is sufficient for practical purposes.

Analogous to the deterministic case, for a structure under stochastic stationary excitation, the response statistics generally comprise a transient and a stationary component. We first study the stationary response statistics in Sects. 5.1 and 5.2. They are the main target in applications and are easier to analyze. Transient statistics are studied later in Sects. 5.3 and 5.4. They are of secondary importance in operational modal analysis (OMA) but their discussion helps clarify the meaning of stationary statistics.

The response statistics can be obtained in different ways, e.g., by modal superposition or state-space approach, in the time or frequency domain. When studying stationary response statistics we adopt modal superposition in the frequency domain. It yields insightful expressions often used in applications. A time-domain state-space approach is adopted when studying transient response statistics. See Sect. 5.5 for a quick summary of the theories in this chapter and their connections.

5.1 Stationary SDOF Response

Consider a single-degree-of freedom (SDOF) structure with natural frequency ω_1 (rad/s), damping ratio ζ and subjected to excitation $w(t)$:

$$\ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) = w(t) \quad (5.1)$$

Generally, the response $x(t)$ is a sum of contributions from initial conditions and excitation. When $w(t)$ is harmonic (i.e., sine or cosine), we learnt in Sect. 3.1.5 that as time goes on the response will approach a ‘steady state’ where the transience due to initial conditions has died out, leaving the excitation effect in the steady-state response. An analogous result holds for the response statistics when $w(t)$ is a stationary stochastic process. Initially, they change with time but as time goes on they converge to a constant (stationary) value. At the stationary state, the correlation function $R(\tau) = E[x(t+\tau)x(t)]$ depends only on the time lag τ but not the time shift t . By the Wiener-Khinchin formula (Sect. 4.4.3), the Fourier Transform (FT) of this correlation function gives the power spectral density (PSD). Instead of going through the correlation function, we will obtain the PSD directly from the scaled FT of the equation of motion, which is algebraically easier.

5.1.1 Scaled Fourier Transform

By definition, the PSD of $x(t)$ is $S_x(\omega) = E[\mathcal{F}_x(\omega)\mathcal{F}_x(\omega)^*]$ where

$$\mathcal{F}_x(\omega) = \lim_{T \rightarrow \infty} \frac{1}{\sqrt{T}} \int_0^T x(t) e^{-i\omega t} dt \quad (5.2)$$

is the scaled FT of $x(t)$. It can be expressed in terms of the scaled FT of $w(t)$. Taking scaled FT on both sides of (5.1),

$$\mathcal{F}_{\ddot{x}}(\omega) + 2\zeta\omega_1\mathcal{F}_{\dot{x}}(\omega) + \omega_1^2\mathcal{F}_x(\omega) = \mathcal{F}_w(\omega) \quad (5.3)$$

where $\mathcal{F}_{\dot{x}}(\omega)$, $\mathcal{F}_{\ddot{x}}(\omega)$ and $\mathcal{F}_w(\omega)$ denote the scaled FTs of $\dot{x}(t)$, $\ddot{x}(t)$ and $w(t)$, respectively. The scaled FTs $\mathcal{F}_{\dot{x}}(\omega)$ and $\mathcal{F}_{\ddot{x}}(\omega)$ can be expressed in terms of $\mathcal{F}_x(\omega)$:

$$\mathcal{F}_{\dot{x}}(\omega) = i\omega\mathcal{F}_x(\omega) \quad \mathcal{F}_{\ddot{x}}(\omega) = i\omega\mathcal{F}_{\dot{x}}(\omega) = -\omega^2\mathcal{F}_x(\omega) \quad (5.4)$$

Substituting into (5.3) and rearranging gives

$$\mathcal{F}_x(\omega) = g(\omega)\mathcal{F}_w(\omega) \quad (5.5)$$

where

$$g(\omega) = (\omega_1^2 - \omega^2 + 2\zeta\omega\omega_1 i)^{-1} = \omega_1^{-2}(1 - \beta^2 + 2\zeta\beta i)^{-1} \quad \beta = \frac{\omega}{\omega_1} \quad (5.6)$$

is the ‘transfer function’ between $x(t)$ and $w(t)$. Substituting (5.5) into (5.4),

$$\mathcal{F}_{\dot{x}}(\omega) = i\omega g(\omega)\mathcal{F}_w(\omega) \quad \mathcal{F}_{\ddot{x}}(\omega) = -\omega^2 g(\omega)\mathcal{F}_w(\omega) \quad (5.7)$$

Transfer Function of Response Acceleration

The scaled FTs $\mathcal{F}_w(\omega)$ and $\mathcal{F}_{\ddot{x}}(\omega)$ have the same unit and so $g(\omega)$ has the same unit as ω^{-2} . When working with acceleration (often measured in OMA) it is more convenient to express $\mathcal{F}_{\ddot{x}}(\omega)$ as

$$\mathcal{F}_{\ddot{x}}(\omega) = h(\omega)\mathcal{F}_w(\omega) \quad (5.8)$$

where

$$h(\omega) = (1 - \beta^2 - 2\zeta\beta i)^{-1} \quad \beta = \frac{\omega}{\omega_1} \quad (5.9)$$

is the (dimensionless) transfer function between $\ddot{x}(t)$ and $w(t)$. Note that in the expression of $h(\omega)$ the frequency ratio β is defined as ω_1/ω rather than ω/ω_1 . In terms of $h(\omega)$,

$$\mathcal{F}_x(\omega) = -\omega^{-2}h(\omega)\mathcal{F}_w(\omega) \quad \mathcal{F}_{\dot{x}}(\omega) = -i\omega^{-1}h(\omega)\mathcal{F}_w(\omega) \quad (5.10)$$

5.1.2 Power Spectral Density

Using the scaled FTs of $x(t)$, $\dot{x}(t)$ and $\ddot{x}(t)$ in (5.8) and (5.10), their PSDs are given by

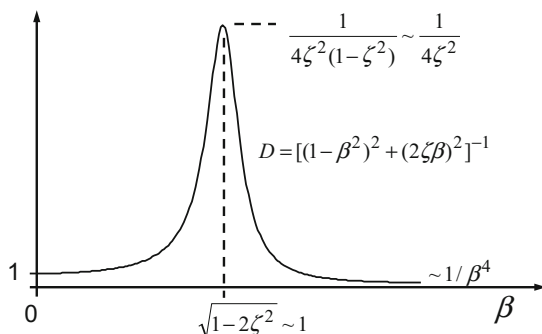
$$\begin{aligned} S_x(\omega) &= E[\mathcal{F}_x \mathcal{F}_x^*] = \omega^{-4}D(\omega)S_w(\omega) \\ S_{\dot{x}}(\omega) &= E[\mathcal{F}_{\dot{x}} \mathcal{F}_{\dot{x}}^*] = \omega^{-2}D(\omega)S_w(\omega) \\ S_{\ddot{x}}(\omega) &= E[\mathcal{F}_{\ddot{x}} \mathcal{F}_{\ddot{x}}^*] = D(\omega)S_w(\omega) \end{aligned} \quad (5.11)$$

where $S_w(\omega) = E[\mathcal{F}_w \mathcal{F}_w^*]$ is the PSD of $w(t)$;

$$D(\omega) = |h(\omega)|^2 = \left[(1 - \beta^2)^2 + (2\zeta\beta)^2 \right]^{-1} \quad \beta = \frac{\omega_1}{\omega} \quad (5.12)$$

is the dynamic amplification factor between the PSDs of $\ddot{x}(t)$ and $w(t)$. Figure 5.1 shows a schematic plot of D versus β for $0 < \zeta < 1$. The square root of D in terms of β has the same expression as the dynamic amplification factor $A(\beta)$ in harmonic response (Sect. 3.1.7) but now β is defined differently.

Fig. 5.1 Dynamic amplification factor D between the PSDs of $\ddot{x}(t)$ and $w(t)$. Asymptotic expressions (after ‘ $\sim$ ’) are indicated for small damping $\zeta \ll 1$



5.1.3 Response Variance

The intensity of a stochastic vibration response can be measured in terms of its variance. At the stationary state this does not depend on time. Without loss of generality, assume that $w(t)$ has zero mean. Taking expectation on the equation of motion (5.1) shows that $x(t)$, $\dot{x}(t)$ and $\ddot{x}(t)$ also have zero mean. Their variance is then equal to the expectation of their square. Using the Parseval equality (Sect. 4.4.4),

$$\begin{aligned}\text{var}[x] &= E[x^2] = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_x(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \omega^{-4} D(\omega) S_w(\omega) d\omega \\ \text{var}[\dot{x}] &= E[\dot{x}^2] = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{\dot{x}}(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \omega^{-2} D(\omega) S_w(\omega) d\omega \\ \text{var}[\ddot{x}] &= E[\ddot{x}^2] = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{\ddot{x}}(\omega) d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} D(\omega) S_w(\omega) d\omega\end{aligned}\quad (5.13)$$

The covariance between any pair of $x(t)$, $\dot{x}(t)$ and $\ddot{x}(t)$ can also be obtained through the Parseval equality. For example,

$$E[x\dot{x}] = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{x\dot{x}}(\omega) d\omega \quad (5.14)$$

where

$$S_{x\dot{x}}(\omega) = E[\mathcal{F}_x(\omega) \mathcal{F}_{\dot{x}}(\omega)^*] \quad (5.15)$$

is the cross PSD between $x(t)$ and $\dot{x}(t)$. Using $\mathcal{F}_{\dot{x}}(\omega) = i\omega \mathcal{F}_x(\omega)$,

$$S_{x\dot{x}}(\omega) = -i\omega E[\mathcal{F}_x(\omega) \mathcal{F}_x(\omega)^*] = -i\omega S_x(\omega) \quad (5.16)$$

Substituting into (5.14) and noting $S_x(-\omega) = S_x(\omega)$ gives

$$E[x\dot{x}] = 0 \quad (5.17)$$

That is, at the stationary state, displacement and velocity response are uncorrelated. The same applies to velocity and acceleration response.

5.1.4 Response to White Noise

Simplification results when the excitation is white noise, where $S_w(\omega)$ is a constant. In this case, the integrals for $\text{var}[x]$ and $\text{var}[\dot{x}]$ in (5.13) can be determined analytically (e.g., using ‘residue theorem’ in complex analysis, out of scope here), giving

$$\text{var}[x] = \frac{S_w}{4\zeta\omega_1^3} = \frac{S_w}{32\pi^3\zeta f_1^3} \quad \text{var}[\dot{x}] = \frac{S_w}{4\zeta\omega_1} = \frac{S_w}{8\pi\zeta f_1} \quad (5.18)$$

where $f_1 = \omega_1/2\pi$ is the natural frequency in Hz. Note that $\text{var}[\ddot{x}] = \omega_1^2 \text{var}[\dot{x}]$, which is intuitive but not trivial from the integrals.

Theoretically, $\text{var}[\ddot{x}]$ is unbounded for white noise excitation because $D(\omega) \rightarrow 1$ as $\omega \rightarrow \infty$. This can also be reasoned from (5.1), where $\text{var}[w]$ is infinite but $\text{var}[x]$ and $\text{var}[\dot{x}]$ are bounded, forcing $\text{var}[\ddot{x}]$ to be infinite. The unbounded nature of $\text{var}[\ddot{x}]$ stems from the fact that white noise has a non-zero PSD no matter how high the frequency is. In reality the PSD of a physical process decays to zero when the frequency is sufficiently high, for which $\text{var}[\ddot{x}]$ is bounded. One simple approximation, which is justified for small damping, is to take

$$\text{var}[\ddot{x}] \approx \omega_1^2 \text{var}[\dot{x}] = \frac{S_w\omega_1}{4\zeta} = \frac{\pi S_w f_1}{2\zeta} \quad (5.19)$$

Example 5.1 (SDOF variance to white noise) Consider the SDOF structure in (5.1) with natural frequency $\omega_1 = 2\pi$ rad/s, i.e., $f_1 = 1$ Hz, damping ratio $\zeta = 1\%$, and subjected to white noise excitation with PSD $S_w = 0.04 (\mu\text{g})^2/\text{Hz}$ ($\mu = 10^{-6}$, $g = 9.81 \text{ m/s}^2$). The stationary variance of displacement is given by (5.18):

$$\text{var}[x] = \frac{(0.04)}{32\pi^3(0.01)(1)^3} \left[\frac{(\mu\text{g})^2/\text{Hz}}{\text{Hz}^3} \right] = 4.031 \times 10^{-3} \left[\frac{(\mu\text{g})^2}{\text{Hz}^4} \right] = 0.3879 (\mu\text{m})^2$$

Taking square root gives a standard deviation of $0.623 \mu\text{m}$. Similarly, the standard deviation of velocity is calculated to be $3.91 \mu\text{m/s}$. Using (5.19), the standard deviation of acceleration is approximately $24.6 \mu\text{m/s}^2$, i.e., $2.51 \mu\text{g}$. ■

5.2 Stationary MDOF Response

Consider a classically damped multi-degree-of-freedom (MDOF) structure satisfying the equation of motion

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) = \mathbf{w}(t) \quad (5.20)$$

where $\mathbf{x}(t)$, $\dot{\mathbf{x}}(t)$ and $\ddot{\mathbf{x}}(t)$ are respectively the $n \times 1$ displacement, velocity and acceleration response vector; $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ are respectively the $n \times n$ mass, damping and stiffness matrix, all real-symmetric and positive-definite; n is the number of DOFs. Recall from Sect. 3.2.1 that the natural frequencies $\{\omega_i\}_{i=1}^n$ and mode shapes $\{\boldsymbol{\psi}_i\}_{i=1}^n$ ($n \times 1$) of the structure are solutions of the eigenvalue problem:

$$\mathbf{K}\boldsymbol{\psi}_i = \omega_i^2 \mathbf{M}\boldsymbol{\psi}_i \quad i = 1, \dots, n \quad (5.21)$$

The response is given by the sum of modal contributions:

$$\mathbf{x}(t) = \sum_{i=1}^n \boldsymbol{\psi}_i \eta_i(t) \quad (5.22)$$

where $\{\eta_i(t)\}_{i=1}^n$ are the modal responses, each satisfying its own SDOF equation of motion:

$$\ddot{\eta}_i(t) + 2\zeta_i \omega_i \dot{\eta}_i(t) + \omega_i^2 \eta_i(t) = p_i(t) \quad i = 1, \dots, n \quad (5.23)$$

ζ_i and $p_i(t)$ are respectively the modal damping ratio and modal force of the i th mode, defined such that

$$2\zeta_i \omega_i = \frac{\boldsymbol{\psi}_i^T \mathbf{C} \boldsymbol{\psi}_i}{\boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i} \quad p_i(t) = \frac{\boldsymbol{\psi}_i^T \mathbf{w}(t)}{\boldsymbol{\psi}_i^T \mathbf{M} \boldsymbol{\psi}_i} \quad (5.24)$$

Equation (5.22) can be written in matrix form:

$$\mathbf{x}(t) = \underset{n \times 1}{\boldsymbol{\Psi}} \underset{n \times n}{\boldsymbol{\eta}(t)} \underset{n \times 1}{\boldsymbol{\eta}(t)} \quad (5.25)$$

where

$$\boldsymbol{\Psi} = [\boldsymbol{\psi}_1, \dots, \boldsymbol{\psi}_n] \quad \boldsymbol{\eta}(t) = \begin{bmatrix} \eta_1(t) \\ \vdots \\ \eta_n(t) \end{bmatrix} \quad (5.26)$$

5.2.1 Scaled Fourier Transform

Using (5.22), the scaled FT of $\mathbf{x}(t)$ is

$$\mathcal{F}_{\mathbf{x}}(\omega) = \sum_{i=1}^n \boldsymbol{\Psi}_i \mathcal{F}_{\eta_i}(\omega) \quad (5.27)$$

where $\mathcal{F}_{\eta_i}(\omega)$ is the scaled FT of $\eta_i(t)$. Since each $\eta_i(t)$ satisfies the SDOF equation in (5.23), (5.10) can be applied to give:

$$\mathcal{F}_{\eta_i}(\omega) = -\omega^{-2} h_i(\omega) \mathcal{F}_{p_i}(\omega) \quad (5.28)$$

where $\mathcal{F}_{p_i}(\omega)$ is the scaled FT of $p_i(t)$; and

$$h_i(\omega) = (1 - \beta_i^2 - 2\zeta_i \beta_i \mathbf{i})^{-1} \quad \beta_i = \frac{\omega_i}{\omega} \quad (5.29)$$

Substituting (5.28) into (5.27),

$$\mathcal{F}_{\mathbf{x}}(\omega) = -\omega^{-2} \sum_{i=1}^n \boldsymbol{\Psi}_i h_i(\omega) \mathcal{F}_{p_i}(\omega) \quad (5.30)$$

In matrix form,

$$\mathcal{F}_{\mathbf{x}}(\omega) = -\omega^{-2} \boldsymbol{\Psi} \overbrace{\text{diag}[\mathbf{h}(\omega)]}^{\mathcal{F}_{\mathbf{\ddot{\eta}}}(\omega)} \mathcal{F}_{\mathbf{p}}(\omega) \quad (5.31)$$

where

$$\mathcal{F}_{\mathbf{p}}(\omega) = \begin{bmatrix} \mathcal{F}_{p_1}(\omega) \\ \vdots \\ \mathcal{F}_{p_n}(\omega) \end{bmatrix} = (\boldsymbol{\Psi}^T \mathbf{M} \boldsymbol{\Psi})^{-1} \boldsymbol{\Psi}^T \mathcal{F}_{\mathbf{w}}(\omega) \quad (5.32)$$

$$\mathcal{F}_{\mathbf{\ddot{\eta}}}(\omega) = \begin{bmatrix} \mathcal{F}_{\ddot{\eta}_1}(\omega) \\ \vdots \\ \mathcal{F}_{\ddot{\eta}_n}(\omega) \end{bmatrix} = \underbrace{\begin{bmatrix} h_1(\omega) & & \\ & \ddots & \\ & & h_n(\omega) \end{bmatrix}}_{\text{diag}[\mathbf{h}(\omega)]} \underbrace{\begin{bmatrix} \mathcal{F}_{p_1}(\omega) \\ \vdots \\ \mathcal{F}_{p_n}(\omega) \end{bmatrix}}_{\mathcal{F}_{\mathbf{p}}(\omega)} \quad (5.33)$$

Using $\mathcal{F}_{\dot{\mathbf{x}}}(\omega) = i\omega\mathcal{F}_{\mathbf{x}}(\omega)$ and $\mathcal{F}_{\ddot{\mathbf{x}}}(\omega) = -\omega^2\mathcal{F}_{\mathbf{x}}(\omega)$,

$$\mathcal{F}_{\dot{\mathbf{x}}}(\omega) = -i\omega^{-1}\Psi\text{diag}[\mathbf{h}(\omega)]\mathcal{F}_{\mathbf{p}}(\omega) \quad \mathcal{F}_{\ddot{\mathbf{x}}}(\omega) = \Psi\text{diag}[\mathbf{h}(\omega)]\mathcal{F}_{\mathbf{p}}(\omega) \quad (5.34)$$

5.2.2 Power Spectral Density

Using (5.31) and (5.34), the PSD matrices of $\mathbf{x}(t)$, $\dot{\mathbf{x}}(t)$ and $\ddot{\mathbf{x}}(t)$ are given by

$$\begin{aligned} \mathbf{S}_{\mathbf{x}}(\omega) &= E[\mathcal{F}_{\mathbf{x}}\mathcal{F}_{\mathbf{x}}^*] = \omega^{-4}\Psi\mathbf{H}(\omega)\Psi^T & \mathbf{S}_{\dot{\mathbf{x}}}(\omega) &= E[\mathcal{F}_{\dot{\mathbf{x}}}\mathcal{F}_{\dot{\mathbf{x}}}^*] = \omega^{-2}\Psi\mathbf{H}(\omega)\Psi^T \\ \mathbf{S}_{\ddot{\mathbf{x}}}(\omega) &= E[\mathcal{F}_{\ddot{\mathbf{x}}}\mathcal{F}_{\ddot{\mathbf{x}}}^*] = \Psi\mathbf{H}(\omega)\Psi^T \end{aligned} \quad (5.35)$$

where

$$\mathbf{H}(\omega) = E[\mathcal{F}_{\ddot{\mathbf{q}}}(\omega)\mathcal{F}_{\ddot{\mathbf{q}}}(\omega)^*] = \text{diag}[\mathbf{h}(\omega)]\mathbf{S}_{\mathbf{p}}(\omega)\text{diag}[\mathbf{h}(\omega)]^* \quad (5.36)$$

and

$$\mathbf{S}_{\mathbf{p}}(\omega) = E[\mathcal{F}_{\mathbf{p}}(\omega)\mathcal{F}_{\mathbf{p}}(\omega)^*] = (\Psi^T\mathbf{M}\Psi)^{-1}\Psi^T\mathbf{S}_{\mathbf{w}}(\omega)\Psi(\Psi^T\mathbf{M}\Psi)^{-1} \quad (5.37)$$

is the PSD matrix of modal forces; $\mathbf{S}_{\mathbf{w}}(\omega) = E[\mathcal{F}_{\mathbf{w}}(\omega)\mathcal{F}_{\mathbf{w}}(\omega)^*]$ is the PSD matrix of $\mathbf{w}(t)$. The (i,j) -entry of $\mathbf{H}(\omega)$ and $\mathbf{S}_{\mathbf{p}}(\omega)$ are (omitting dependence on ω)

$$\mathbf{H}(i,j) = E[\mathcal{F}_{\ddot{\eta}_i}\mathcal{F}_{\ddot{\eta}_j}^*] = h_i h_j^* \mathbf{S}_{\mathbf{p}}(i,j) \quad (5.38)$$

$$\mathbf{S}_{\mathbf{p}}(i,j) = E[\mathcal{F}_{p_i}\mathcal{F}_{p_j}^*] = \frac{\psi_i^T \mathbf{S}_{\mathbf{w}} \psi_j}{(\psi_i^T \mathbf{M} \psi_i)(\psi_j^T \mathbf{M} \psi_j)} \quad (5.39)$$

Correlation Between Modal Forces

By orthogonality, $\Psi^T\mathbf{M}\Psi$ and hence $(\Psi^T\mathbf{M}\Psi)^{-1}$ are diagonal matrices. However, the same need not be true for $\mathbf{S}_{\mathbf{p}}(\omega)$ because the modal forces of different modes can be dependent, or equivalently, because $\Psi^T\mathbf{S}_{\mathbf{w}}\Psi$ in (5.37) need not be a diagonal matrix. Independence or uncorrelation among the excitations at different DOFs does not carry over to the modal forces, and vice versa. The modal forces of different modes can be correlated even if the excitations at different DOFs are uncorrelated. Conversely they can be uncorrelated even if the excitations are correlated.

5.2.3 Response Variance

Similar to the SDOF case, the stationary response covariance matrix of a MDOF structure can be obtained by integrating the PSD matrix via the Parseval equality. For displacement response, the covariance matrix is given by

$$E[\mathbf{x}\mathbf{x}^T] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}_{\mathbf{x}}(\omega) d\omega \quad (5.40)$$

The (i, j) -entry of $E[\mathbf{x}\mathbf{x}^T]$ is $E[x_i x_j]$. The covariance matrix between other quantity pairs can also be obtained via the Parseval equality. For example,

$$E[\mathbf{x}\dot{\mathbf{x}}^T] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathbf{S}_{\mathbf{x}\dot{\mathbf{x}}}(\omega) d\omega \quad (5.41)$$

where

$$\mathbf{S}_{\mathbf{x}\dot{\mathbf{x}}}(\omega) = E[\mathcal{F}_{\mathbf{x}}(\omega) \mathcal{F}_{\dot{\mathbf{x}}}(\omega)^*] \quad (5.42)$$

is the cross PSD between $\mathbf{x}(t)$ and $\dot{\mathbf{x}}(t)$.

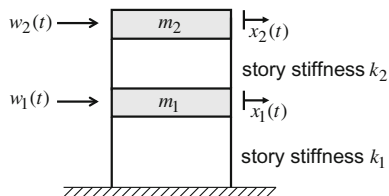
5.2.4 Mode Shape Scaling

The scaling of mode shape Ψ_i affects the scaling of the modal response $\eta_i(t)$ and modal force $p_i(t)$. If Ψ_i is multiplied by a factor, $\eta_i(t)$ and $p_i(t)$ will be divided by the same factor. The modal force PSD S_{p_i} will be divided by the square of that factor. These can be easily reasoned by noting that the physical response $\mathbf{x}(t) = \sum_{i=1}^n \Psi_i \eta_i(t)$ is invariant to mode shape scaling. Let $\{c_i\}_{i=1}^n$ be the factors applied to $\{\Psi_i\}_{i=1}^n$. Then the modal force PSD matrix $\mathbf{S}_{\mathbf{p}}(\omega)$ under the newly scaled mode shapes $\{\Psi'_i = c_i \Psi_i\}_{i=1}^n$ is given by

$$\mathbf{S}_{\mathbf{p}'}(\omega) = \begin{bmatrix} c_1^{-1} & & \\ & \ddots & \\ & & c_n^{-1} \end{bmatrix} \mathbf{S}_{\mathbf{p}}(\omega) \begin{bmatrix} c_1^{-1} & & \\ & \ddots & \\ & & c_n^{-1} \end{bmatrix} \quad (5.43)$$

If we take the mode shapes $\{\Psi_i\}_{i=1}^n$ to be dimensionless then the modal responses $\{\eta_i(t)\}_{i=1}^n$ will have the same unit as the physical displacement $\mathbf{x}(t)$. If the mass matrix $\mathbf{M}$ has a unit of kg and the excitation $\mathbf{w}(t)$ has a unit of N, then the

Fig. 5.2 Two-DOF shear building



modal forces $\{p_i(t)\}_{i=1}^n$ will have a unit of N/kg and their PSD matrix $\mathbf{S}_p(\omega)$ has a unit of $(\text{N/kg})^2/\text{Hz}$.

Example 5.2 (Modal force PSD, two-DOF shear building) Consider the two-storied shear building in Fig. 5.2 with uniform floor mass $m_1 = m_2 = 1000$ tons and interstory stiffness $k_1 = k_2 = 1000$ kN/mm. The first and second floor are subjected to independent white noise excitations with PSDs S_1 and S_2 , respectively. Here we illustrate how the modal force PSD matrix depends on the scaling of mode shapes.

The stiffness and mass matrix are given by

$$\mathbf{K} = \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} = \begin{bmatrix} 2000 & -1000 \\ -1000 & 1000 \end{bmatrix} \times 10^6 \text{ [N/m]}$$

$$\mathbf{M} = \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} = \begin{bmatrix} 1000 & 0 \\ 0 & 1000 \end{bmatrix} \times 10^3 \text{ [kg]}$$

Solving the eigenvalue problem $\mathbf{K}\boldsymbol{\psi} = \omega^2\mathbf{M}\boldsymbol{\psi}$ gives the natural frequencies and mode shapes:

$$\begin{aligned} \omega_1 &= 19.5 \text{ rad/s (3.11 Hz)} & \boldsymbol{\psi}_1 &= \begin{bmatrix} 0.5257 \\ 0.8507 \end{bmatrix} \\ \omega_2 &= 51.2 \text{ rad/s (8.14 Hz)} & \boldsymbol{\psi}_2 &= \begin{bmatrix} -0.8507 \\ 0.5257 \end{bmatrix} \end{aligned}$$

These mode shapes have been scaled to have unit norm, i.e., $\sqrt{\boldsymbol{\psi}_i^T \boldsymbol{\psi}_i} = 1$. Under this scaling, the modal masses are calculated to be the same:

$$\mu_1 = \boldsymbol{\psi}_1^T \mathbf{M} \boldsymbol{\psi}_1 = 10^6 \text{ kg} \quad \mu_2 = \boldsymbol{\psi}_2^T \mathbf{M} \boldsymbol{\psi}_2 = 10^6 \text{ kg}$$

Since the excitations at the two floors are independent, their PSD matrix is diagonal:

$$\mathbf{S}_w = \begin{bmatrix} S_1 & \\ & S_2 \end{bmatrix}$$

Coherence of Modal Forces

Suppose $S_1 = 0.2(\text{kN})^2/\text{Hz}$ and $S_2 = 0.1(\text{kN})^2/\text{Hz}$. Using (5.37), the modal force PSD matrix is calculated to be

$$\mathbf{S}_p = \begin{bmatrix} 0.1276 & -0.0447 \\ -0.0447 & 0.1724 \end{bmatrix} \times 10^{-6} \quad [(\text{N/kg})^2/\text{Hz}]$$

which is not a diagonal matrix. If instead $S_1 = 0.1(\text{kN})^2/\text{Hz}$ and $S_2 = 0.1(\text{kN})^2/\text{Hz}$,

$$\mathbf{S}_p = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} \times 10^{-6} \quad [(\text{N/kg})^2/\text{Hz}]$$

which is now a diagonal matrix. Here, the modal force PSD matrix can be non-diagonal simply because the excitations at different locations do not have the same PSD.

Scaling of Modal Forces

Instead of scaling the mode shapes to have unit norm, suppose we scale them so that the value at the second DOF is 1. The mode shapes under this scaling are

$$\boldsymbol{\psi}'_1 = \begin{bmatrix} 0.618 \\ 1 \end{bmatrix} \quad \boldsymbol{\psi}'_2 = \begin{bmatrix} -1.618 \\ 1 \end{bmatrix}$$

Check that $\boldsymbol{\psi}'_1 = c_1 \boldsymbol{\psi}_1$ and $\boldsymbol{\psi}'_2 = c_2 \boldsymbol{\psi}_2$ where $c_1 = 1/0.8507$ and $c_2 = 1/0.5257$. The modal masses look very different now:

$$\mu'_1 = \boldsymbol{\psi}_1'^T \mathbf{M} \boldsymbol{\psi}'_1 = 1.382 \times 10^6 \text{ kg} \quad \mu'_2 = \boldsymbol{\psi}_2'^T \mathbf{M} \boldsymbol{\psi}'_2 = 3.618 \times 10^6 \text{ kg}$$

Check that $\mu'_1 = c_1^2 \mu_1$ and $\mu'_2 = c_2^2 \mu_2$. For $S_1 = 0.2(\text{kN})^2/\text{Hz}$ and $S_2 = 0.1(\text{kN})^2/\text{Hz}$, the modal force PSD matrix is

$$\mathbf{S}_{p'} = \begin{bmatrix} 0.0923 & -0.02 \\ -0.02 & 0.0476 \end{bmatrix} \times 10^{-6} \quad [(\text{N/kg})^2/\text{Hz}]$$

Check that

$$\mathbf{S}_{p'} = \begin{bmatrix} c_1^{-1} & \\ & c_2^{-1} \end{bmatrix} \mathbf{S}_p \begin{bmatrix} c_1^{-1} & \\ & c_2^{-1} \end{bmatrix}$$

Ambient Vibration Level

The modal force PSDs in this example may look small but in fact they are sufficient to create ambient vibrations large enough for OMA. Assuming a damping ratio of $\zeta = 1\%$, the PSD of the acceleration at the roof at the first mode frequency is about

$$\begin{aligned}
\frac{\psi_{21}^2 S_{p_1}}{4\zeta^2} &= \frac{(0.8507)^2 (0.1276 \times 10^{-6}) (\text{N/kg})^2 / \text{Hz}}{(4)(0.01)^2} \\
&= 0.232 \times 10^{-3} (\text{m/s}^2)^2 / \text{Hz} \\
&\approx 2.4 \times 10^6 (\mu\text{g})^2 / \text{Hz}
\end{aligned}$$

since $1 \text{ m/s}^2 = (10^6/9.81) \mu\text{g}$. In terms of root PSD, this is about $\sqrt{2.4 \times 10^6} \approx 155 \mu\text{g}/\sqrt{\text{Hz}}$, which is well above the noise level of many servo-accelerometers (in the order of $1 \mu\text{g}/\sqrt{\text{Hz}}$). ■

5.3 Transient Response Variance

The response variance derived in Sects. 5.1 and 5.2 applies when the response is in a statistically stationary state where transient effects have died out. In this section we consider the transient variation of the response variance when the structure is subjected to stochastic stationary excitations. The next section considers the transience in the correlation. Generally, transience in response statistics occurs when they are not equal to their stationary value. The latter can be seen as the limit for large time. This and the next section may be skipped for readers primarily interested in OMA, which is concerned with stationary response statistics.

A state-space approach (Sect. 3.6) is adopted for analysis, as it elegantly produces the response statistics for all quantities. Recall the MDOF equation of motion in (5.20):

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) = \mathbf{w}(t) \quad (5.44)$$

It is first converted to a state-space form:

$$\begin{aligned}
\dot{\mathbf{y}}(t) &= \mathbf{A}\mathbf{y}(t) + \mathbf{B}\mathbf{w}(t) \\
\mathbf{y}(t) &= \begin{bmatrix} \mathbf{x}(t) \\ \dot{\mathbf{x}}(t) \end{bmatrix}_{2n \times 1} \quad \mathbf{A}_{2n \times 2n} = \begin{bmatrix} \mathbf{0}_n & \mathbf{I}_n \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{C} \end{bmatrix} \quad \mathbf{B}_{2n \times n} = \begin{bmatrix} \mathbf{0}_n \\ \mathbf{M}^{-1} \end{bmatrix} \quad (5.45)
\end{aligned}$$

The solution is

$$\mathbf{y}(t) = e^{\mathbf{A}t} \mathbf{y}(0) + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{w}(\tau) d\tau \quad (5.46)$$

As in Sect. 5.2, assume that $\mathbf{w}(t)$ is a stationary stochastic process with zero mean, PSD matrix $\mathbf{S}_w(\omega)$ and correlation function $\mathbf{R}_w(\tau)$. The initial condition $\mathbf{y}(0)$ can be either known or random, but is otherwise independent of $\mathbf{w}(t)$. Using (5.46) and noting $E[\mathbf{w}(\tau)] = 0$, the expectation of $\mathbf{y}(t)$ is

$$\boldsymbol{\mu}(t) = E[\mathbf{y}(t)] = e^{\mathbf{A}t} E[\mathbf{y}(0)] \quad (5.47)$$

The covariance matrix of $\mathbf{y}(t)$ is given by

$$E\{[\mathbf{y}(t) - \boldsymbol{\mu}(t)][\mathbf{y}(t) - \boldsymbol{\mu}(t)]^T\} = E[\mathbf{y}(t)\mathbf{y}(t)^T] - \boldsymbol{\mu}(t)\boldsymbol{\mu}(t)^T \quad (5.48)$$

where the expression on the RHS can be obtained by expanding the product on the LHS and evaluating the expression of the individual terms. It remains to determine

$$\mathbf{V}(t) = E[\mathbf{y}(t)\mathbf{y}(t)^T] \quad (5.49)$$

Clearly,

$$\mathbf{V}(0) = E[\mathbf{y}(0)\mathbf{y}(0)^T] \quad (5.50)$$

This is assumed to be given as initial condition. We shall derive the equation governing the change of $\mathbf{V}(t)$ and then solve it together with this initial condition.

5.3.1 Governing Equation

Differentiating (5.49) w.r.t. t ,

$$\dot{\mathbf{V}}(t) = E[\dot{\mathbf{y}}(t)\mathbf{y}(t)^T] + E[\mathbf{y}(t)\dot{\mathbf{y}}(t)^T] \quad (5.51)$$

Post-multiplying the state-space equation in (5.45) by $\mathbf{y}(t)^T$ and taking expectation,

$$E[\dot{\mathbf{y}}(t)\mathbf{y}(t)^T] = \mathbf{A}E[\mathbf{y}(t)\mathbf{y}(t)^T] + \mathbf{B}E[\mathbf{w}(t)\mathbf{y}(t)^T] = \mathbf{A}\mathbf{V}(t) + \mathbf{B}E[\mathbf{w}(t)\mathbf{y}(t)^T] \quad (5.52)$$

Substituting into (5.51) and noting $\mathbf{V}(t)^T = \mathbf{V}(t)$,

$$\dot{\mathbf{V}}(t) = \mathbf{A}\mathbf{V}(t) + \mathbf{V}(t)\mathbf{A}^T + E[\mathbf{y}(t)\mathbf{w}(t)^T]\mathbf{B}^T + \{E[\mathbf{y}(t)\mathbf{w}(t)^T]\mathbf{B}^T\}^T \quad (5.53)$$

To obtain $E[\mathbf{y}(t)\mathbf{w}(t)^T]$, post-multiply (5.46) with $\mathbf{w}(t)^T$, take expectation and note that $\mathbf{y}(0)$ and $\mathbf{w}(t)$ are independent. This gives

$$E[\mathbf{y}(t)\mathbf{w}(t)^T] = e^{\mathbf{A}t} E[\mathbf{y}(0)] \underbrace{E[\mathbf{w}(t)^T]}_{\mathbf{0}} + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \underbrace{E[\mathbf{w}(\tau)\mathbf{w}(t)^T]}_{\mathbf{R}_w(t-\tau)^T} d\tau \quad (5.54)$$

Changing integration variable from τ to $u = t - \tau$ gives

$$E[\mathbf{y}(t)\mathbf{w}(t)^T] = \int_0^t e^{\mathbf{A}u} \mathbf{B} \mathbf{R}_w(u)^T du \quad (5.55)$$

Substituting into (5.53) yields

$$\dot{\mathbf{V}}(t) = \mathbf{A} \mathbf{V}(t) + \mathbf{V}(t) \mathbf{A}^T + \mathbf{P}(t) \quad (5.56)$$

$$\mathbf{P}(t) = \int_0^t e^{\mathbf{A}u} \mathbf{B} \mathbf{R}_w(u)^T \mathbf{B}^T du + \left[\int_0^t e^{\mathbf{A}u} \mathbf{B} \mathbf{R}_w(u)^T \mathbf{B}^T du \right]^T \quad (5.57)$$

5.3.2 Solution Procedure

Equation (5.56) is a first order $2n \times 2n$ matrix ODE in $\mathbf{V}(t)$. Taking the symmetry of $\mathbf{V}(t)$ into account, in principle the equation can be converted to a first order vector ODE for the $n(2n+1)$ lower (or upper) and diagonal entries of $\mathbf{V}(t)$. Obtaining the solution from such equation is not preferred because the dimension of the system matrix ($n(2n+1)$ by $n(2n+1)$) is unnecessarily large. A more accurate and efficient strategy (Hammarling 1982) is to first determine

$$\mathbf{Z}(t) = \mathbf{Q}^{-1} \mathbf{V}(t) \mathbf{Q}^{-T} \quad (5.58)$$

($\mathbf{Q}^{-T} = (\mathbf{Q}^{-1})^T$) and then recover $\mathbf{V}(t)$ from

$$\mathbf{V}(t) = \mathbf{Q} \mathbf{Z}(t) \mathbf{Q}^T \quad (5.59)$$

where $\mathbf{Q}$ is the eigenmatrix of $\mathbf{A}$ such that $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^{-1}$; $\mathbf{\Lambda}$ is the diagonal matrix of eigenvalues of $\mathbf{A}$. This works because each entry of $\mathbf{Z}(t)$ satisfies its own ODE, as seen in the following.

Uncoupled ODEs for $\mathbf{Z}(t)$

Substituting $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^{-1}$ into (5.56),

$$\dot{\mathbf{V}}(t) = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^{-1} \mathbf{V}(t) + \mathbf{V}(t) \mathbf{Q}^{-T} \mathbf{\Lambda} \mathbf{Q}^T + \mathbf{P}(t) \quad (5.60)$$

Pre-multiplying by $\mathbf{Q}^{-1}$ and post-multiplying by $\mathbf{Q}^{-T}$, and noting that $\mathbf{Q}^{-1} \dot{\mathbf{V}}(t) \mathbf{Q}^{-T} = \dot{\mathbf{Z}}(t)$,

$$\dot{\mathbf{Z}}(t) = \Lambda \mathbf{Z}(t) + \mathbf{Z}(t) \Lambda + \mathbf{L}(t) \quad (5.61)$$

$$\mathbf{L}(t) = \mathbf{Q}^{-1} \mathbf{P}(t) \mathbf{Q}^{-T} \quad (5.62)$$

Equation (5.61) is still a matrix ODE but the entries of $\mathbf{Z}(t)$ now satisfy their own ODE uncoupled from each other, because Λ is a diagonal matrix. The (i, j) -entry of the equation reads:

$$\dot{Z}_{ij}(t) = (\lambda_i + \lambda_j) Z_{ij}(t) + L_{ij}(t) \quad (5.63)$$

where $Z_{ij}(t)$ and $L_{ij}(t)$ denote the (i, j) -entry of $\mathbf{Z}(t)$ and $\mathbf{L}(t)$, respectively. The initial condition for $\mathbf{Z}(t)$ is given by $\mathbf{Z}(0) = \mathbf{Q}^{-1} \mathbf{V}(0) \mathbf{Q}^{-T}$. Note that $\mathbf{P}(t)$ is symmetric and so is $\mathbf{L}(t)$, i.e., $L_{ij}(t) = L_{ji}(t)$. This implies that $Z_{ij}(t)$ and $Z_{ji}(t)$ satisfy the same equation. This agrees with the fact that $\mathbf{Z}(t)$ is symmetric, as evidenced from (5.58).

In summary, to determine $\mathbf{V}(t) = E[\mathbf{y}(t)\mathbf{y}(t)^T]$ where $\mathbf{y}(t)$ satisfies (5.45) with initial condition $\mathbf{V}(0) = E[\mathbf{y}(0)\mathbf{y}(0)^T]$,

1. Determine the eigenvalues $\{\lambda_i\}_{i=1}^{2n}$ and eigenmatrix $\mathbf{Q}$ of $\mathbf{A}$.
2. Determine the initial condition $\mathbf{Z}(0) = \mathbf{Q}^{-1} \mathbf{V}(0) \mathbf{Q}^{-T}$.
3. Solve $Z_{ij}(t)$ from (5.63) for $i = 1, \dots, n; j \leq i$, with initial condition $Z_{ij}(0)$ from the (i, j) -entry of $\mathbf{Z}(0)$. Set $Z_{ji}(t) = Z_{ij}(t)$.
4. Obtain $\mathbf{V}(t) = \mathbf{Q} \mathbf{Z}(t) \mathbf{Q}^T$.

5.3.3 Limiting Stationary Value

According to Sect. 3.6.2, if the structure is stable (all $\omega_i > 0$) with positive damping (all $\zeta_i > 0$), the eigenvalues of $\mathbf{A}$ will all have a negative real part. Then $\mathbf{V}(t)$ will converge to a bounded value as $t \rightarrow \infty$. This limiting value should coincide with the stationary response variance derived in Sect. 5.2. Let $\mathbf{V}_\infty = \lim_{t \rightarrow \infty} \mathbf{V}(t)$. Setting $\dot{\mathbf{V}}(t) = 0$ (stationary) in (5.56) gives

$$\mathbf{A} \mathbf{V}_\infty + \mathbf{V}_\infty \mathbf{A}^T + \mathbf{P}_\infty = \mathbf{0} \quad (5.64)$$

$$\mathbf{P}_\infty = \int_0^\infty \mathbf{e}^{\mathbf{A}u} \mathbf{B} \mathbf{R}_w(u)^T \mathbf{B}^T du + \left[\int_0^\infty \mathbf{e}^{\mathbf{A}u} \mathbf{B} \mathbf{R}_w(u)^T \mathbf{B}^T du \right]^T \quad (5.65)$$

Equation (5.64) is known as the ‘Lyapunov equation’. The solution is

$$\mathbf{V}_\infty = \mathbf{Q}\mathbf{Z}_\infty\mathbf{Q}^T \quad (5.66)$$

where $\mathbf{Z}_\infty = \lim_{t \rightarrow \infty} \mathbf{Z}(t)$. Setting $\dot{Z}_{ij}(t) = 0$ and solving (5.63) gives the (i, j) -entry of $\mathbf{Z}_\infty$:

$$Z_\infty(i, j) = -\frac{\mathbf{L}_\infty(i, j)}{\lambda_i + \lambda_j} \quad \mathbf{L}_\infty = \mathbf{Q}^{-1}\mathbf{P}_\infty\mathbf{Q}^{-T} \quad (5.67)$$

5.3.4 Response to White Noise

Consider the case when $\mathbf{w}(t)$ is white noise with covariance matrix

$$\mathbf{R}_w(\tau) = \mathbf{S}_w\delta(\tau) \quad (5.68)$$

where $\delta(\tau)$ is the Dirac Delta function and $\mathbf{S}_w$ is the (constant) PSD matrix of $\mathbf{w}(t)$. Check that the FT of $\mathbf{R}_w(\tau)$ is $\mathbf{S}_w$. Also, $\mathbf{S}_w$ is real symmetric because $\mathbf{R}_w(0)$ is. Substituting (5.68) into (5.57) and noting that the integrals only cover the right half of the Delta function,

$$\mathbf{P}(t) = \mathbf{B}\mathbf{S}_w\mathbf{B}^T \quad (5.69)$$

Substituting into (5.62),

$$\mathbf{L}(t) = \mathbf{Q}^{-1}\mathbf{B}\mathbf{S}_w\mathbf{B}^T\mathbf{Q}^{-T} \quad (5.70)$$

Here, $\mathbf{P}(t)$ and $\mathbf{L}(t)$ do not depend on t and so they are equal to their limiting values, $\mathbf{P}_\infty$ and $\mathbf{L}_\infty$, respectively. Equation (5.63) can be solved analytically, giving

$$Z_{ij}(t) = e^{(\lambda_i + \lambda_j)t} Z_{ij}(0) - \frac{\mathbf{L}_\infty(i, j)}{\lambda_i + \lambda_j} \left[1 - e^{(\lambda_i + \lambda_j)t} \right] \quad (5.71)$$

As a check, setting $t = 0$ on the RHS gives $Z_{ij}(0)$. Differentiating the RHS w.r.t. t gives the RHS of (5.63). Taking $t \rightarrow \infty$ gives $\mathbf{Z}_\infty(i, j)$ in (5.67), as the exponential terms tend to zero.

Example 5.3 (First order process under white noise, transient variance)
Consider a scalar process $y(t)$ satisfying

$$\dot{y}(t) + ay(t) = w(t) \quad y(0) = 0 \quad (5.72)$$

where $a > 0$ and $w(t)$ is white noise with PSD S_w and correlation function $R_w(\tau) = S_w\delta(\tau)$. The PSD and correlation function of $y(t)$ at the stationary state

was obtained in Example 4.1 of Sect. 4.4.5. Here we obtain the transient variance $V(t) = E[y(t)^2]$. In the context of the state-space equation in (5.45), $\mathbf{A} = -a$ and $\mathbf{B} = 1$. Then (5.69) gives $\mathbf{P}(t) = S_w$ and (5.56) reads

$$\dot{V}(t) = -2aV(t) + S_w \quad (5.73)$$

Solving this equation with initial condition $V(0) = 0$ gives

$$V(t) = \frac{S_w}{2a}(1 - e^{-2at}) \quad (5.74)$$

This is a monotonic increasing function of t . It converges to $S_w/2a$ as $t \rightarrow \infty$, which agrees with the finding in Example 4.1. ■

Example 5.4 (SDOF under white noise, transient variance) Consider the SDOF response $x(t)$ satisfying $\ddot{x}(t) + 2\zeta\omega_1\dot{x}(t) + \omega_1^2x(t) = w(t)$, starting from rest ($x(0) = \dot{x}(0) = 0$) and subjected to white noise excitation $w(t)$ with PSD S_w . In the context of the state-space equation in (5.45),

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\omega_1^2 & -2\zeta\omega_1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (5.75)$$

The eigenvalues and eigenmatrix of $\mathbf{A}$ were obtained in Example 3.8 of Sect. 3.6.2:

$$\lambda_1 = -\zeta\omega_1 + \mathbf{i}\omega_d \quad \lambda_2 = -\zeta\omega_1 - \mathbf{i}\omega_d \quad (5.76)$$

$$\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ -\zeta\omega_1 + \mathbf{i}\omega_d & -\zeta\omega_1 - \mathbf{i}\omega_d \end{bmatrix} \quad \mathbf{Q}^{-1} = \frac{1}{-2\mathbf{i}\omega_d} \begin{bmatrix} -\zeta\omega_1 - \mathbf{i}\omega_d & -1 \\ \zeta\omega_1 - \mathbf{i}\omega_d & 1 \end{bmatrix} \quad (5.77)$$

where $\omega_d = \omega_1\sqrt{1 - \zeta^2}$. Equations (5.69) and (5.70) give

$$\mathbf{P}(t) = S_w \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{L}(t) = \frac{-S_w}{4\omega_d^2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad (5.78)$$

Using (5.71) with $Z_{ij}(0) = 0$ (since $\mathbf{V}(0) = \mathbf{0}$) to obtain $\mathbf{Z}(t)$ and then substituting into $\mathbf{V}(t) = \mathbf{Q}\mathbf{Z}(t)\mathbf{Q}^T$ gives, after algebra,

$$V_{11}(t) = \frac{S_w}{4\zeta\omega_1^3} \left\{ 1 - \frac{e^{-2\zeta\omega_1 t}}{1 - \zeta^2} [1 + \zeta \sin(2\omega_d t - \phi)] \right\} \quad \tan \phi = \frac{\zeta}{\sqrt{1 - \zeta^2}} \quad (5.79)$$

$$V_{22}(t) = \frac{S_w}{4\zeta\omega_1} \left\{ 1 - \frac{e^{-2\zeta\omega_1 t}}{1 - \zeta^2} [1 - \zeta \sin(2\omega_d t + \phi)] \right\} \quad (5.80)$$

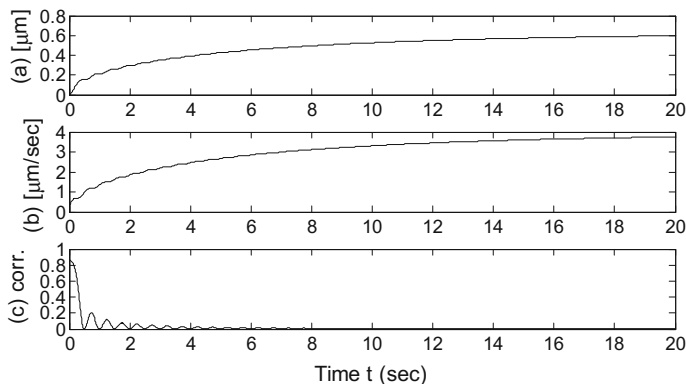


Fig. 5.3 Transient response statistics of SDOF structure under white noise. **a** $\sqrt{E[x(t)^2]}$, **b** $\sqrt{E[\dot{x}(t)^2]}$, **c** correlation coefficient $= E[x(t)\dot{x}(t)]/\sqrt{E[x(t)^2]E[\dot{x}(t)^2]}$

$$V_{12}(t) = \frac{S_w e^{-2\zeta\omega_1 t}}{2\omega_1^2(1 - \zeta^2)} \sin^2 \omega_d t \quad (5.81)$$

Note that $V_{21}(t) = V_{12}(t)$. Taking $t \rightarrow \infty$ gives the stationary covariance matrix:

$$\begin{bmatrix} E[x^2] & E[x\dot{x}] \\ E[\dot{x}x] & E[\dot{x}^2] \end{bmatrix} = \lim_{t \rightarrow \infty} \underbrace{\begin{bmatrix} V_{11}(t) & V_{12}(t) \\ V_{21}(t) & V_{22}(t) \end{bmatrix}}_{\mathbf{V}_\infty} = \begin{bmatrix} \frac{S_w}{4\zeta\omega_1^3} & 0 \\ 0 & \frac{S_w}{4\zeta\omega_1} \end{bmatrix} \quad (5.82)$$

This result could have been obtained directly from the solution of the Lyapunov equation in (5.66). It also agrees with (5.18), which was derived in the frequency domain using the Parseval equality.

Figure 5.3 shows the response statistics with time for $\omega_1 = 2\pi$ rad/s (1 Hz), $\zeta = 1\%$ and $S_w = 0.04(\mu\text{g})^2/\text{Hz}$ as in Example 5.1. For large t they tend to a limiting stationary value. In (5.79)–(5.81), since transient effect decays through the term $\exp(-2\zeta\omega_1 t)$, the time it takes to approach near the stationary state is $O(1/\zeta\omega_1)$. Check from the figure that the limiting values of $\sqrt{E[x^2]}$ and $\sqrt{E[\dot{x}^2]}$ agree with those found in Example 5.1, i.e., $0.623 \mu\text{m}$ and $3.91 \mu\text{m/s}$, respectively. ■

5.4 Transient Response Correlation

We next consider the transience in the correlation of response. The context is the same as the last section and analysis technique is similar. Analogous to (5.48), the correlation function of $\mathbf{y}(t)$ at time lag τ can be expressed as

$$E\{[\mathbf{y}(t+\tau) - \boldsymbol{\mu}(t+\tau)][\mathbf{y}(t) - \boldsymbol{\mu}(t)]^T\} = E[\mathbf{y}(t+\tau)\mathbf{y}(t)^T] - \boldsymbol{\mu}(t+\tau)\boldsymbol{\mu}(t)^T \quad (5.83)$$

where $\boldsymbol{\mu}(t) = E[\mathbf{y}(t)]$ as in (5.47). The main task is therefore to determine

$$\mathbf{R}_t(\tau) = E[\mathbf{y}(t+\tau)\mathbf{y}(t)^T] \quad (5.84)$$

This function depends on both the time shift t and time lag τ because the response is non-stationary.

Transpose Mirror Property

For finite $t < \infty$, the transpose mirror property does not hold for $\mathbf{R}_t(\tau)$, i.e., generally $\mathbf{R}_t(-\tau) \neq \mathbf{R}_t(\tau)^T$. Rather, for any t and τ ,

$$\mathbf{R}_{t+\tau}(-\tau) = \mathbf{R}_t(\tau)^T \quad (5.85)$$

This holds because

$$\mathbf{R}_{t+\tau}(-\tau) = E[\mathbf{y}(t+\tau-\tau)\mathbf{y}(t+\tau)^T] = E[\mathbf{y}(t+\tau)\mathbf{y}(t)^T]^T = \mathbf{R}_t(\tau)^T \quad (5.86)$$

Equation (5.85) reduces to the transpose mirror property as $t \rightarrow \infty$, because then $\mathbf{R}_t$ and $\mathbf{R}_{t+\tau}$ both become $\mathbf{R}_\infty$.

5.4.1 Governing Equation

We first derive the equation governing the change of $\mathbf{R}_t(\tau)$ w.r.t. τ . Note that $\dot{\mathbf{R}}_t(\tau) = E[\dot{\mathbf{y}}(t+\tau)\mathbf{y}(t)^T]$ where the dot in $\dot{\mathbf{R}}_t(\tau)$ denotes a derivative w.r.t. τ . Consider the state-space equation in (5.45) at $t+\tau$, i.e., $\dot{\mathbf{y}}(t+\tau) = \mathbf{A}\mathbf{y}(t+\tau) + \mathbf{B}\mathbf{w}(t+\tau)$. Post-multiplying by $\mathbf{y}(t)^T$ and taking expectation,

$$\dot{\mathbf{R}}_t(\tau) = \mathbf{A}\mathbf{R}_t(\tau) + \mathbf{B}E[\mathbf{w}(t+\tau)\mathbf{y}(t)^T] \quad (5.87)$$

Following a similar procedure in obtaining (5.55), it can be shown that

$$E[\mathbf{w}(t+\tau)\mathbf{y}(t)^T] = \int_0^t \mathbf{R}_w(u+\tau)\mathbf{B}^T e^{\mathbf{A}^T u} du \quad (5.88)$$

Substituting into (5.87) yields

$$\dot{\mathbf{R}}_t(\tau) = \mathbf{A}\mathbf{R}_t(\tau) + \mathbf{P}_t(\tau) \quad t+\tau \geq 0 \quad (5.89)$$

$$\mathbf{P}_t(\tau) = \int_0^t \mathbf{B}\mathbf{R}_w(u+\tau)\mathbf{B}^T e^{\mathbf{A}^T u} du \quad (5.90)$$

Equation (5.89) is a matrix ODE in $\mathbf{R}_t(\tau)$ w.r.t. τ . It is only valid for $t + \tau \geq 0$, i.e., $\tau \geq -t$, for otherwise the time argument $t + \tau$ in $\mathbf{y}(t + \tau)$ and hence $\mathbf{R}_t(\tau) = E[\mathbf{y}(t + \tau)\mathbf{y}(t)^T]$ is before time 0, when the initial condition $\mathbf{y}(0)$ is specified. The solution is given by

$$\mathbf{R}_t(\tau) = \begin{cases} \mathbf{e}^{\mathbf{A}\tau}\mathbf{V}(t) + \int_0^\tau \mathbf{e}^{\mathbf{A}(\tau-s)}\mathbf{P}_t(s)ds & \tau \geq 0 \\ \mathbf{V}(t + \tau)\mathbf{e}^{\mathbf{A}^T|\tau|} + \int_0^{|\tau|} \mathbf{P}_{t+\tau}(s)^T \mathbf{e}^{\mathbf{A}^T(|\tau|-s)}ds & -t < \tau < 0 \end{cases} \quad (5.91)$$

where $\mathbf{V}(t) = E[\mathbf{y}(t)\mathbf{y}(t)^T]$ was determined in the last section.

In (5.91), the first expression for $\tau \geq 0$ is the Duhamel's integral solution to (5.89) starting with the initial condition $\mathbf{R}_t(0) = \mathbf{V}(t)$. To show the second expression, it can be deduced from (5.85) that $\mathbf{R}_t(\tau) = \mathbf{R}_{t+\tau}(|\tau|)^T$ for $\tau < 0$. Since the time lag $|\tau|$ in $\mathbf{R}_{t+\tau}(|\tau|)$ is positive, the first expression in (5.91) can be applied. Replacing τ by $|\tau|$ and t by $t + \tau$, the expression gives

$$\mathbf{R}_{t+\tau}(|\tau|) = \mathbf{e}^{\mathbf{A}|\tau|}\mathbf{V}(t + \tau) + \int_0^{|\tau|} \mathbf{e}^{\mathbf{A}(|\tau|-s)}\mathbf{P}_{t+\tau}(s)ds \quad (5.92)$$

Taking transpose and noting the symmetry of $\mathbf{V}(t + \tau)$ gives the second expression in (5.91).

5.4.2 Limiting Stationary Value

Taking limit $t \rightarrow \infty$ on (5.91), the correlation function at the stationary state is given by

$$\mathbf{R}_\infty(\tau) = \begin{cases} \mathbf{e}^{\mathbf{A}\tau}\mathbf{V}_\infty + \int_0^\tau \mathbf{e}^{\mathbf{A}(\tau-s)}\mathbf{P}_\infty(s)ds & \tau \geq 0 \\ \mathbf{V}_\infty\mathbf{e}^{\mathbf{A}^T|\tau|} + \int_0^{|\tau|} \mathbf{P}_\infty(s)^T \mathbf{e}^{\mathbf{A}^T(|\tau|-s)}ds & \tau < 0 \end{cases} \quad (5.93)$$

Here we show that the FT of $\mathbf{R}_\infty(\tau)$ gives the PSD matrix of $\mathbf{y}(t)$, thereby connecting the large-time limit of the transient correlation function with the stationary statistics via the Wiener-Khinchin formula.

PSD Directly From Stationary Statistics

By definition, the PSD of $\mathbf{y}(t)$ is $\mathbf{S}_y(\omega) = E[\mathcal{F}_y\mathcal{F}_y^*]$. Taking scaled FT on (5.45) and using $\mathcal{F}_{\dot{y}}(\omega) = i\omega\mathcal{F}_y(\omega)$ gives, after rearranging,

$$\mathcal{F}_y(\omega) = -(\mathbf{A} - \mathbf{i}\omega\mathbf{I})^{-1}\mathbf{B}\mathcal{F}_w(\omega) \quad (5.94)$$

where $\mathcal{F}_w(\omega)$ is the scaled FT of $\mathbf{w}(t)$ and $\mathbf{I}$ is the identity matrix. Substituting gives

$$\mathbf{S}_y(\omega) = E[\mathcal{F}_y\mathcal{F}_y^*] = (\mathbf{A} - \mathbf{i}\omega\mathbf{I})^{-1}\mathbf{B}\mathbf{S}_w(\omega)\mathbf{B}^T(\mathbf{A}^T + \mathbf{i}\omega\mathbf{I})^{-1} \quad (5.95)$$

PSD from Large-Time Limit of Transient Correlation Function

On the other hand, taking limit $t \rightarrow \infty$ of (5.89),

$$\dot{\mathbf{R}}_\infty(\tau) = \mathbf{A}\mathbf{R}_\infty(\tau) + \mathbf{P}_\infty(\tau) \quad -\infty < \tau < \infty \quad (5.96)$$

Let $\mathcal{F}_{\mathbf{R}_\infty}(\omega)$ and $\mathcal{F}_{\dot{\mathbf{R}}_\infty}(\omega)$ denote respectively the FT of $\mathbf{R}_\infty(\tau)$ and $\dot{\mathbf{R}}_\infty(\tau)$. Taking FT of (5.96) and using $\mathcal{F}_{\dot{\mathbf{R}}_\infty}(\omega) = \mathbf{i}\omega\mathcal{F}_{\mathbf{R}_\infty}(\omega)$ gives, after rearranging,

$$\mathcal{F}_{\mathbf{R}_\infty}(\omega) = -(\mathbf{A} - \mathbf{i}\omega\mathbf{I})^{-1}\tilde{\mathbf{P}}_\infty(\omega) \quad (5.97)$$

where

$$\tilde{\mathbf{P}}_\infty(\omega) = \int_{-\infty}^{\infty} \mathbf{P}_\infty(\tau)e^{-\mathbf{i}\omega\tau}d\tau \quad (5.98)$$

is the FT of $\mathbf{P}_\infty(\tau)$. Using (5.90) with $t \rightarrow \infty$ and swapping the order of integration,

$$\begin{aligned} \tilde{\mathbf{P}}_\infty(\omega) &= \int_0^\infty \mathbf{B} \underbrace{\int_{-\infty}^\infty \mathbf{R}_w(u+\tau)e^{-\mathbf{i}\omega\tau}d\tau \mathbf{B}^T e^{\mathbf{A}^T u} du}_{\mathbf{S}_w(\omega)e^{\mathbf{i}\omega u}} \\ &= \mathbf{B}\mathbf{S}_w(\omega)\mathbf{B}^T \int_0^\infty e^{(\mathbf{A}^T + \mathbf{i}\omega\mathbf{I})u} du \end{aligned} \quad (5.99)$$

where the inner integral was obtained by changing variable from τ to $s = u + \tau$; and

$$\mathbf{S}_w(\omega) = \int_{-\infty}^{\infty} \mathbf{R}_w(s)e^{-\mathbf{i}\omega s}ds \quad (5.100)$$

is the PSD of $\mathbf{w}(t)$ (Wiener-Khinchin formula). Note that

$$\int_0^{\infty} e^{(\mathbf{A}^T + \mathbf{i}\omega\mathbf{I})u} du = (\mathbf{A}^T + \mathbf{i}\omega\mathbf{I})^{-1} e^{(\mathbf{A}^T + \mathbf{i}\omega\mathbf{I})u} \Big|_0^{\infty} = -(\mathbf{A}^T + \mathbf{i}\omega\mathbf{I})^{-1} \quad (5.101)$$

where we have used $\lim_{u \rightarrow \infty} e^{(\mathbf{A}^T + \mathbf{i}\omega\mathbf{I})u} = \mathbf{0}$, which holds because the eigenvalues of $\mathbf{A}$ (and hence $\mathbf{A}^T + \mathbf{i}\omega\mathbf{I}$) have a negative real part. Substituting into (5.99),

$$\tilde{\mathbf{P}}_{\infty}(\omega) = -\mathbf{B}\mathbf{S}_{\mathbf{w}}(\omega)\mathbf{B}^T(\mathbf{A}^T + \mathbf{i}\omega\mathbf{I})^{-1} \quad (5.102)$$

Substituting into (5.97),

$$\mathcal{F}_{\mathbf{R}_{\infty}}(\omega) = (\mathbf{A} - \mathbf{i}\omega\mathbf{I})^{-1}\mathbf{B}\mathbf{S}_{\mathbf{w}}(\omega)\mathbf{B}^T(\mathbf{A}^T + \mathbf{i}\omega\mathbf{I})^{-1} \quad (5.103)$$

This is the same expression as in (5.95) and so the FT of $\mathbf{R}_{\infty}(\tau)$ is $\mathbf{S}_{\mathbf{y}}(\omega)$.

5.4.3 Response to White Noise

When $\mathbf{w}(t)$ is white noise with a constant PSD matrix $\mathbf{S}_{\mathbf{w}}$ and correlation function $\mathbf{R}_{\mathbf{w}}(\tau) = \mathbf{S}_{\mathbf{w}}\delta(\tau)$, it can be reasoned using (5.90) that $\mathbf{P}_t(s) = 0$ for $s > 0$ and $\mathbf{P}_t(0)$ has a finite value. The integrals in (5.91) are then zero, giving

$$\mathbf{R}_t(\tau) = \begin{cases} \mathbf{e}^{\mathbf{A}\tau}\mathbf{V}(t) & \tau \geq 0 \\ \mathbf{V}(t+\tau)\mathbf{e}^{\mathbf{A}^T|\tau|} & -t < \tau < 0 \end{cases} \quad (5.104)$$

where $\mathbf{V}(t)$ was obtained in Sect. 5.3.4. Taking limit $t \rightarrow \infty$ gives the correlation function at the stationary state:

$$\mathbf{R}_{\infty}(\tau) = \begin{cases} \mathbf{e}^{\mathbf{A}\tau}\mathbf{V}_{\infty} & \tau \geq 0 \\ \mathbf{V}_{\infty}\mathbf{e}^{\mathbf{A}^T|\tau|} & \tau < 0 \end{cases} \quad (5.105)$$

Example 5.5 (First order process under white noise, transient correlation)

Revisit Example 5.3 but now consider the transient correlation. Using (5.104) with $\mathbf{A} = -a$ and $V(t) = (S_w/2a)(1 - e^{-2at})$ from (5.74),

$$R_t(\tau) = E[y(t+\tau)y(t)] = \begin{cases} \frac{S_w}{2a}e^{-a\tau}(1 - e^{-2at}) & \tau \geq 0 \\ \frac{S_w}{2a}e^{a\tau}(1 - e^{-2a(t+\tau)}) & -t < \tau < 0 \end{cases} \quad (5.106)$$

Check that $R_t(0) = V(t)$ for both expressions. Also, $R_t(-t) = 0$ for the second expression, which agrees with $R_t(-t) = E[y(0)y(t)] = 0$ since the initial condition $y(0) = 0$ was assumed.

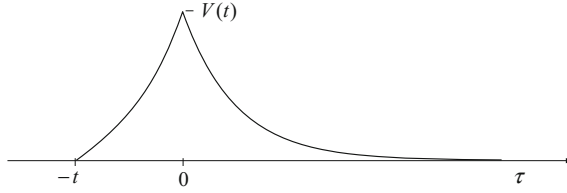


Fig. 5.4 Schematic plot of $R_t(\tau)$ for first order process under white noise

Figure 5.4 shows a schematic plot of $R_t(\tau)$. Clearly, it is not symmetric about $\tau = 0$. As t increases, the part for $\tau < 0$ extends to the left, eventually becomes symmetric to the part for $\tau > 0$ as $t \rightarrow \infty$. ■

Example 5.6 (SDOF under white noise, transient correlation) Revisit the SDOF structure in Example 5.4 with natural frequency $\omega_1 = 2\pi$ rad/s (1 Hz), damping ratio $\zeta = 1\%$ and subjected to white noise excitation with PSD $S_w = 0.04(\mu g)^2/\text{Hz}$ as before. The state vector is $\mathbf{y}(t) = [x(t), \dot{x}(t)]^T$. By definition,

$$\mathbf{R}_t(\tau) = E[\mathbf{y}(t+\tau)\mathbf{y}(t)^T] = E \begin{bmatrix} x(t+\tau)x(t) & x(t+\tau)\dot{x}(t) \\ \dot{x}(t+\tau)x(t) & \dot{x}(t+\tau)\dot{x}(t) \end{bmatrix} \quad (5.107)$$

This can be computed using (5.104). Recall from Example 3.8 in Sect. 3.6.2 that

$$\mathbf{e}^{\mathbf{A}\tau} = \begin{bmatrix} g_1(\tau) & g_2(\tau) \\ \dot{g}_1(\tau) & \dot{g}_2(\tau) \end{bmatrix} \quad (5.108)$$

where

$$g_1(\tau) = e^{-\zeta\omega_1\tau} \left(\cos \omega_d\tau + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d\tau \right) \quad g_2(\tau) = \frac{e^{-\zeta\omega_1\tau}}{\omega_d} \sin \omega_d\tau \quad (5.109)$$

and $\omega_d = \omega_1\sqrt{1-\zeta^2}$. Using (5.104), the entries of $\mathbf{R}_t(\tau)$ can be obtained by matrix-multiplying the entries of $\mathbf{e}^{\mathbf{A}\tau}$ and $\mathbf{V}(t)$. For example, the (1,1)-entry of $\mathbf{R}_t(\tau)$ is given by

$$E[x(t+\tau)x(t)] = \begin{cases} g_1(\tau)V_{11}(t) + g_2(\tau)V_{12}(t) & \tau \geq 0 \\ g_1(|\tau|)V_{11}(t+\tau) + g_2(|\tau|)V_{12}(t+\tau) & -t < \tau < 0 \end{cases} \quad (5.110)$$

where $V_{ij}(t)$ denotes the (i,j) -entry of $\mathbf{V}(t) = E[\mathbf{y}(t)\mathbf{y}(t)^T]$, which was obtained in Example 5.4. Other entries of $\mathbf{R}_t(\tau)$ can be similarly obtained. Figure 5.5 shows $E[x(t+\tau)x(t)]$ versus τ . As t increases the part for negative τ extends to the left, tending to a mirror image of that for positive τ . The value at $\tau = 0$ is equal to $E[x(t)^2]$, whose square root was plotted versus t in Fig. 5.3a. ■

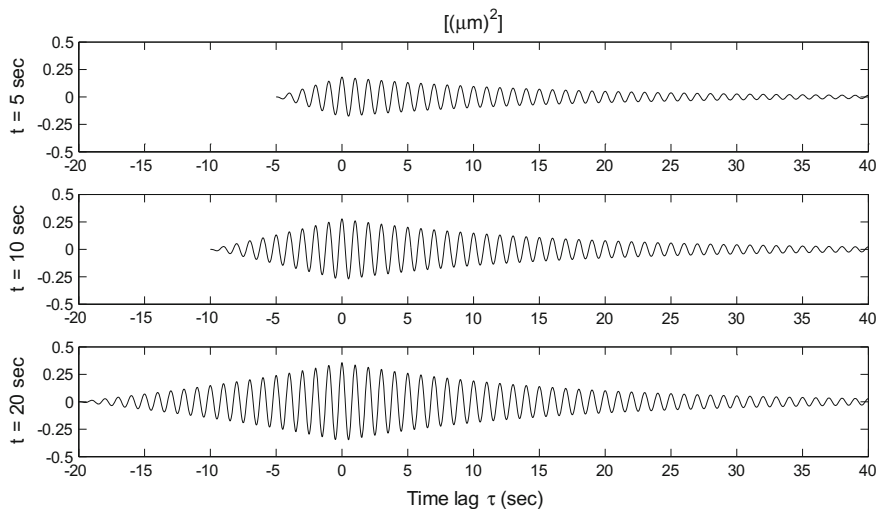


Fig. 5.5 Transient correlation $E[x(t+\tau)x(t)]$ of SDOF structure under white noise

5.5 Summary of Theories and Connections

Figure 5.6 summarizes the theories in this chapter and their connections. Relevant sections are indicated in brackets.

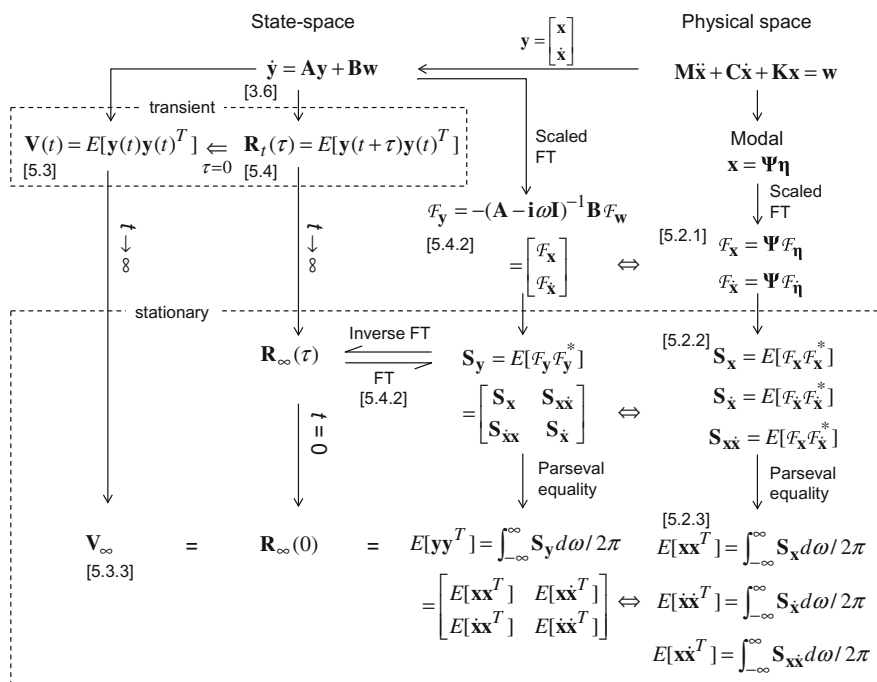


Fig. 5.6 Summary of theories in different sections and their connections

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Chapter 6

Measurement Basics

Abstract This chapter introduces the typical hardware and software process that leads to the measured ambient data in operational modal analysis. Common aspects are discussed, including sensor/hardware noise, aliasing, quantization error and synchronization. Basic concepts for calibrating measurement noise are also discussed.

Keywords Data acquisition · Noise · Aliasing · Quantization · Synchronization · Huddle test

The vibration data that we analyze on a computer is a series of digital signals supposed to carry information about the motion of the target measurement point. The data and the actual motion are never the same. Their difference is commonly referred as ‘noise’. A basic understanding of how vibration signal is picked up mechanically, transmitted electrically and eventually converted into digital form allows one to appreciate the existence and nature of noise; and to devise methods to reduce, model or account for it. This is the objective of this chapter.

We first take an overview of a typical process that produces a digital time history of ambient vibration data. Important aspects are then discussed individually. After that we discuss how to calibrate the noise of a data channel.

6.1 Data Acquisition Process

A typical data acquisition (DAQ) process that produces the data recorded on a digital computer is schematically shown in Fig. 6.1. For discussion purpose, assume that the sensor is an accelerometer producing analogue (i.e., continuous-time) voltage signal. Starting from the left, ambient excitation has a broadband power spectral density (PSD) ①. The PSD ② of the resulting structural acceleration response is equal to the product of the excitation PSD and structural dynamic amplification. It has a peak near the natural frequency due to resonance. At

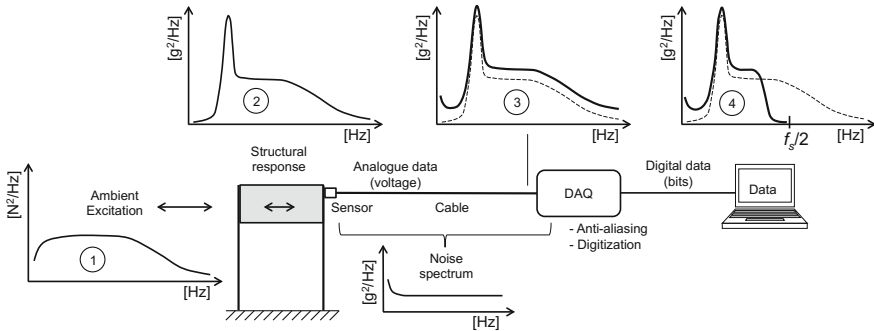


Fig. 6.1 Typical data acquisition process. PSD shown next to each quantity. For illustration, acceleration measurement is assumed. The *dashed line* in ③ and ④ is the PSD of the target in ②; f_s is the sampling rate in Hz

very high frequencies it takes the same shape as the excitation PSD because dynamic amplification approaches a constant there. The sensor measures the structural response, generating analogue data signal that is transmitted through cable to the DAQ hardware. During transmission the analogue data is contaminated with ‘noise’ arising from the sensor, cable or electronics in the DAQ hardware. The PSD in ③ is equal to the sum of the acceleration response PSD in ② and the ‘noise PSD’. To avoid aliasing, the DAQ hardware filters out frequency components in the analogue data near and beyond the Nyquist frequency (half of the sampling frequency). The resulting analogue data is digitized and subsequently recorded on a computer. This is the ‘measured data’. Its PSD in ④ is tapered off near the Nyquist frequency. To supplement the above picture, some details are next discussed individually.

6.2 Channel Noise

‘Channel noise’ refers to the difference between the measured (digital) data and the actual vibration response being measured. It may be attributed to

1. random noise in sensor/hardware (Sect. 6.3),
2. distortion from anti-aliasing filter (Sect. 6.5),
3. quantization error of digitizer (Sect. 6.6),

These are discussed in the sections indicated. For proper instrumentation, all sources should be kept small compared to the target signal. The first source is inevitable and its characteristics depend on the type of sensor/hardware. The second and third source can be managed to a negligible level by proper choice of equipment/sampling configuration. One also needs to be vigilant against other potential sources, e.g., ‘clipping’ (saturation/exceedance of measurement limit) and

electrical/magnetic interference from the test environment. Managed properly, channel noise is typically dominated by sensor/hardware noise.

6.3 Sensor/Hardware Noise

Sensor/hardware noise refers to the noise in the analogue data before it is digitized. After digitization the data can be practically considered to be transmitted without noise; or it will not be successfully transmitted at all. The noise can come from sensor, cable or DAQ hardware. Sensor/hardware noise can be considered a stationary stochastic process independent of the response being measured. Its strength is commonly characterized by the PSD, i.e., the ‘noise spectrum’. The PSD can differ from one data channel to another, although it tends to be similar for sensors or DAQ hardware of the same model. For example, the three data channels of a triaxial accelerometer usually have a similar noise spectrum.

Fig. 6.2 shows the root PSD calculated from 15 min of horizontal acceleration of a single-degree-of-freedom (SDOF) structure under laboratory environment by different sensors (with different DAQ hardware), namely, servo-accelerometer, piezoelectric accelerometer, MEMS (Micro-Electro-Mechanical Sensor) and the MEMS on a smart phone.

The square root of PSD is plotted as it scales linearly with the unit of the measured quantity ($g = 9.81 \text{ m/s}^2$ in this case). The dashed line illustrates what the ideal measured spectrum (② in Fig. 6.1) would be if there were no channel noise. It has a resonance peak at the natural frequency (about 7.5 Hz). At low frequencies the spectra of measured data are all dominated by noise. The servo-accelerometer has the smallest noise PSD, less than $1 \mu\text{g}/\sqrt{\text{Hz}}$. Its spectrum captures quite well the target near and off the resonance peak. The piezoelectric and MEMS sensor can also do so, although the band dominated by response is narrower. For the sensor on smart phone, the resonance peak appears to be ‘drowned’ by noise. In terms of PSD

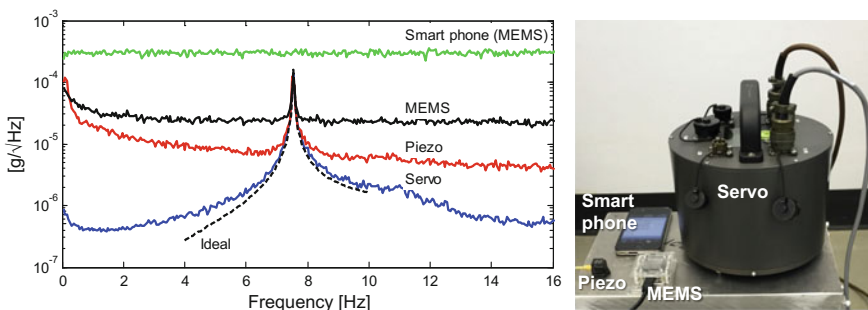


Fig. 6.2 One-sided root PSD (averaged) of acceleration measured on a SDOF structure by different sensors. *Dashed line* illustrates the ideal measured acceleration if there were no channel noise

at the natural frequency, the signal-to-noise (s/n) ratios for Servo, Piezo, MEMS and Smart phone are in the order of 10000, 1000, 100 and 1, respectively. In the general case, servo-accelerometers tend to have the smallest noise (especially for low frequencies) among the types of sensors here; but the ordering between piezoelectric and MEMS sensors do depend on the specific model.

Physical Limits

Noise is inevitable as a matter of nature. Sensing elements converting mechanical motion to electrical signal are subjected to mechanical-thermal noise associated with disturbance at molecular level (Brownian motion). The electronic noise in DAQ hardware is attributed to a variety of sources with different spectral characteristics. For example, thermal (Johnson-Nyquist) noise is associated with the random thermal motion of electrons and its PSD is approximately constant with frequency, i.e., white; ‘flicker’ noise is associated with the spatial fluctuation of electronic properties and its PSD is inversely proportional to frequency, i.e., ‘pink’. The PSD of noise discussed above increases with the absolute temperature. See Gabrielson (1993), Kogan (1996) and Levinson (2004).

Equipment Specification

Nominal information about the noise spectrum can be obtained from equipment specifications provided by manufacturer. In the absence of other information, this can be the main source for decision making at the procurement stage. Common formats are in terms of the noise PSD applicable over a certain frequency band, or RMS (root-mean-square) value. For the latter, attention should be paid to the frequency band over which the RMS is calculated; the wider the band, the higher the RMS. An average value for the noise PSD can be back-calculated as the square of RMS value divided by the bandwidth (Parseval equality).

Sensitivity and Signal-to-Noise Ratio

One common measure of data quality is the ‘signal-to-noise’ (s/n) ratio. Clearly, it can be enhanced by increasing the signal or reducing the noise. In ambient tests the level of vibration cannot be actively controlled but the measured analogue signal can be enhanced by using a sensor with a higher sensitivity. The sensitivity of accelerometers is often measured in V/g. This is more than just a factor for scaling data from V to g. A sensor with a higher sensitivity generates a higher voltage signal for the same level of vibration. For acceleration data, the s/n ratio in terms of PSD is given by

$$s/n \text{ ratio} = \frac{(\text{sensor sensitivity, V/g})^2 \times (\text{acceleration response PSD, g}^2/\text{Hz})}{\text{channel noise PSD, V}^2/\text{Hz}} \quad (6.1)$$

This can also be written as

$$s/n \text{ ratio} = \frac{\text{acceleration response PSD, g}^2/\text{Hz}}{\text{channel noise PSD, g}^2/\text{Hz}} \quad (6.2)$$

where the channel noise PSD in g^2/Hz is equal to that in V^2/Hz divided by the square of sensor sensitivity in V/g . It is possible to enhance the sensitivity by amplifying the analogue signal, although the additional hardware can also contribute to channel noise. Strictly speaking, the s/n ratio defined in terms of PSD is frequency-dependent. In practice, a representative value over a certain frequency range of interest may be reported for convenience.

As an example, in Fig. 6.2 the root PSD of structural acceleration at the natural frequency is about $200 \mu\text{g}/\sqrt{\text{Hz}}$. According to manufacturer specifications, the piezoelectric sensor has sensitivity 0.5 V/g and noise $6 \mu\text{g}/\sqrt{\text{Hz}}$ near that frequency. The DAQ hardware has noise $1 \mu\text{V}/\sqrt{\text{Hz}}$, equivalent to $1/0.5 = 2 \mu\text{g}/\sqrt{\text{Hz}}$. Neglecting the contribution from cable and assuming that the sensor and DAQ hardware noise are uncorrelated, the channel noise has a PSD of $6^2 + 2^2 = 40 (\mu\text{g})^2/\text{Hz}$. The s/n ratio in terms of PSD (near the natural frequency) is then $(200)^2/40 = 1000$. Here, the DAQ hardware contributes to 10% of noise PSD. For lower frequencies this contribution can be higher or even dominate.

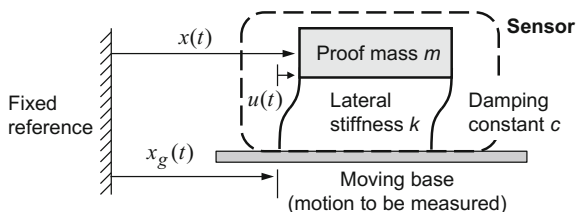
6.4 Sensor Principle

The characteristics of a sensor are attributed to its measuring principle. A basic understanding of the latter allows one to choose the appropriate type of sensor depending on application. In a basic setup, the motion to be measured causes deformation in the sensing element, which in turn induces an electrical signal carrying information of the motion. To see the relationship between the deformation and input motion, consider the idealized SDOF system of a sensor in Fig. 6.3.

For discussion purpose, suppose we want to measure the acceleration $\ddot{x}_g(t)$ of the moving base on which the sensor is placed. The relative displacement $u(t)$ (or its derivatives) of the proof mass can be detected by electro-mechanical means. For example, when subjected to dynamic stress (hence deformation), piezoelectric materials produce a charge/voltage. In capacitive accelerometers, the motion of the proof mass changes the distance between charged surfaces and hence electrical capacitance, which results in a voltage proportional to acceleration.

To see the relationship between $u(t)$ and $\ddot{x}_g(t)$, apply Newton's second law on the proof mass to give

Fig. 6.3 Schematic diagram of the mechanical part of a sensor



$$\underbrace{m\ddot{x}(t)}_{\text{inertia}} + \underbrace{c\dot{u}(t)}_{\text{damping}} + \underbrace{ku(t)}_{\text{stiffness}} = 0 \quad (6.3)$$

The inertia term is proportional to $\ddot{x}(t)$ (not $\ddot{u}(t)$) but the damping and stiffness terms are still proportional respectively to $\dot{u}(t)$ and $u(t)$ because they arise from deformation. Substituting $\ddot{x}(t) = \ddot{x}_g(t) + \ddot{u}(t)$ and rearranging gives the equation for $u(t)$:

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = -m\ddot{x}_g(t) \quad (6.4)$$

In terms of the natural frequency $\omega_1 = \sqrt{k/m}$ (rad/s) and damping ratio $\zeta = c/2\sqrt{km}$,

$$\ddot{u}(t) + 2\zeta\omega_1\dot{u}(t) + \omega_1^2u(t) = -\ddot{x}_g(t) \quad (6.5)$$

It is more convenient to study the relationship between $u(t)$ and $\ddot{x}_g(t)$ in the frequency domain. Taking Fourier Transform (FT, denoted by ‘ $\mathcal{F}$ ’) on (6.5), using $\mathcal{F}_{\ddot{u}}(\omega) = \mathbf{i}\omega\mathcal{F}_u(\omega)$ and $\mathcal{F}_{\ddot{x}_g}(\omega) = -\omega^2\mathcal{F}_u(\omega)$, and rearranging gives

$$\mathcal{F}_u(\omega) = G(\omega)\mathcal{F}_{\ddot{x}_g}(\omega) \quad (6.6)$$

where $G(\omega)$ is the transfer function between $\ddot{x}_g(t)$ and $u(t)$:

$$G(\omega) = \frac{-1}{\omega_1^2(1 - \beta^2 + 2\zeta\beta\mathbf{i})} \quad |G(\omega)| = \frac{1}{\omega_1^2\sqrt{(1 - \beta^2)^2 + (2\zeta\beta)^2}} \quad \beta = \frac{\omega}{\omega_1} \quad (6.7)$$

Figure 6.4 shows the modulus of $|G(\omega)|$ versus the frequency ratio $\beta = \omega/\omega_1$. When $\ddot{x}_g(t)$ is a simple harmonic (i.e., sine or cosine) at frequency ω , $|G(\omega)|$ tells how much the amplitude of $\ddot{x}_g(t)$ is amplified in $u(t)$. When $\ddot{x}_g(t)$ is a stationary stochastic process, $|G(\omega)|^2$ tells how much the PSD of $\ddot{x}_g(t)$ at ω is amplified in the PSD of $u(t)$. As seen in the figure, for frequencies well below the natural frequency of the sensor, i.e., $\beta \ll 1$, $|G(\omega)|$ is almost flat and this gives the ‘usable’ band for the sensor. To widen the usable band, the natural frequency ω_1 should be as high as possible and the damping ratio should be near $1/\sqrt{2} \approx 71\%$. Since $|G(\omega)| \propto 1/\omega_1^2$ and ω_1 is large, the mechanical deformation $u(t)$ is very small and so is the induced electrical signal. For effective transmission, the latter needs to be amplified. Electronic noise from the amplifier is inevitable, which can be significant or even dominate the overall noise of the sensor for low frequencies.

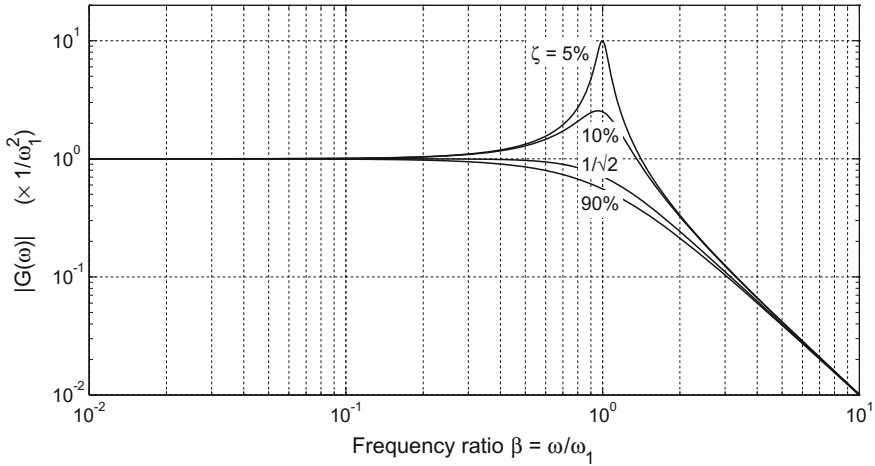


Fig. 6.4 Modulus of transfer function between input acceleration and relative displacement

Force-Balance Principle

Servo-accelerometers operate on a different mechanism, known as ‘force-balance principle’. For this reason they are also called ‘force-balance accelerometers’ (FBA). Instead of sourcing from the (small) electrical signal induced by mechanical deformation, the output signal for acceleration is associated with the required feedback force on the proof mass so that it does not move. To illustrate the idea, suppose we apply a ‘control force’ $F(t)$ on the proof mass in Fig. 6.3 so that (6.4) reads

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = -m\ddot{x}_g(t) + F(t) \quad (6.8)$$

Ideally, if we can assign $F(t)$ so that the proof mass remains stationary w.r.t. the base, i.e., $u(t) \equiv 0$, then the LHS of the equation is identically zero, giving $F(t) = m\ddot{x}_g(t)$. That is, the control force is proportional to the acceleration we want to measure. In reality this can only be achieved approximately. The control force $F(t)$ generally depends on $u(t)$ and its derivatives. The relationship is designed based on ‘feedback control theory’. In typical implementations, the detection of $u(t)$ and the generation of $F(t)$ are electro-magnetic in nature. Delay in feedback is inevitable, which can be critical for high frequency motions. This imposes an upper frequency limit on the sensor. Electronic noise is still inevitable but the electrical signal (proportional to control force) and hence s/n ratio can be much bigger now. Servo-accelerometers are especially useful for low frequencies (e.g., <1 Hz) or low level vibrations (e.g., <1 milli-g/ $\sqrt{\text{Hz}}$). See Wielandt (2012) for details on servo-accelerometers and other types of sensors.

6.5 Aliasing

A discrete-time history sampled at uniform rate f_s (Hz) can at best capture correctly the frequency components in the original analogue (continuous-time) signal up to the Nyquist frequency, equal to $f_s/2$. Components in the analogue data with frequencies higher than $f_s/2$ will be ‘aliased’ as ones with frequencies lower than $f_s/2$. In particular, a sine (or cosine) signal with frequency $f_s r \pm f$ (Hz) for any integer r and $f < f_s/2$ will appear as one with frequency f . This is because at any sampled time instant $t_j = j\Delta t$ ($\Delta t = 1/f_s$),

$$\sin[2\pi(f_s r \pm f)t_j] = \sin\left(2\pi \underbrace{f_s}_{1/\Delta t} r \underbrace{j\Delta t}_{t_j} \pm 2\pi f t_j\right) = \sin(2\pi r j \pm 2\pi f t_j) = \pm \sin(2\pi f t_j)$$

This is illustrated in Fig. 6.5. See Sects. 2.4.2 and 4.7.2 for further mathematical details on aliasing.

Paper Folding

Due to aliasing, the PSD of digital data at the Fast Fourier Transform (FFT) frequency $f_k = k/N\Delta t$ ($f_k < f_s/2$) contains the contributions in the original analogue data at frequencies $f_s r \pm f_k$ for integers $r \geq 1$. This can be visualized by folding the portion of the spectrum beyond the Nyquist frequency back and forth into the sampling band $[0, f_s/2]$; see Fig. 6.6.

Low-Pass Filter

In Fig. 6.1, the analogue data in ③ can contain harmonics beyond the Nyquist frequency. Simply taking its values at time interval $\Delta t = 1/f_s$ as data will lead to aliasing. To avoid this, ‘anti-aliasing filter’ is applied to the analogue data before it is sampled, significantly ‘attenuating’ (reducing) the components beyond the Nyquist frequency. Bounded by fundamental limits, the attenuation cannot be

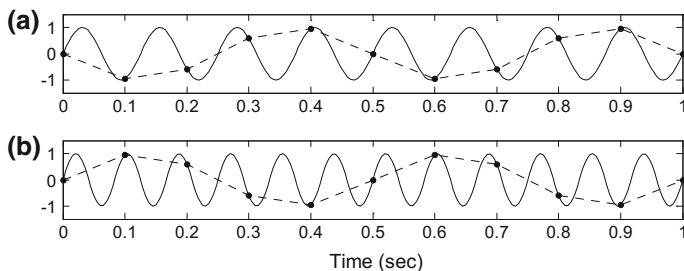


Fig. 6.5 Aliasing. *Solid line* original signal; *dashed line with dots* sampled data (dots) at 10 Hz. **a** Samples of a sine at $10 - 2 = 8$ Hz appear as a negative sine at 2 Hz. **b** Samples of a sine at $10 + 2 = 12$ Hz appear as a positive sine at 2 Hz

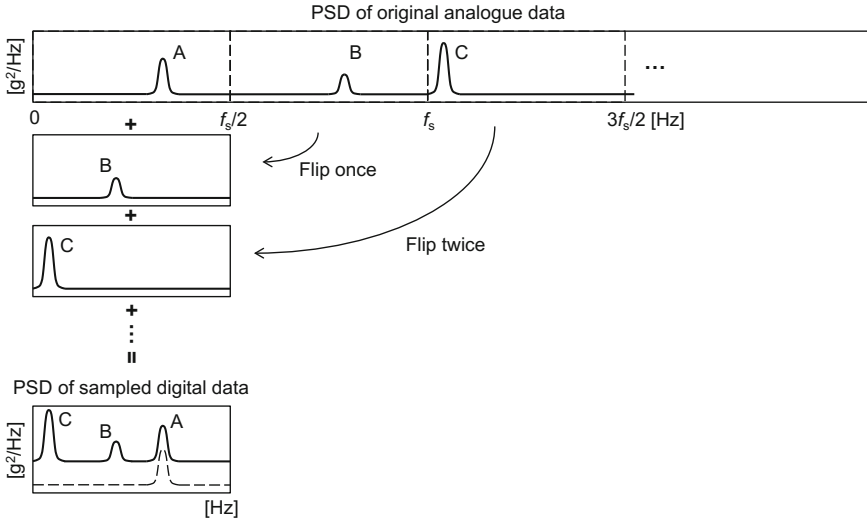


Fig. 6.6 Visualizing aliasing by paper folding. In the PSD of the sampled digital data at the bottom, the dashed line is the target PSD, solid line is the PSD of digital data distorted by aliasing

perfectly sharp at the Nyquist frequency. The components with frequencies slightly lower than the Nyquist frequency are inevitably be affected. To avoid distortion, the Nyquist frequency should be much higher than the frequencies of interest.

Decimation

Depending on DAQ hardware settings, it can happen that the raw digital data is sampled at a much higher rate than is necessary for a particular analysis. For example, to analyze a 1 Hz vibration mode, a sampling rate of 20 Hz is usually sufficient, but the raw data may be sampled at 1000 Hz. To reduce storage demand and speed up computations, one may ‘decimate’ (down-sample) the raw data to produce a set with a lower sampling rate. The basic idea is simple, e.g., taking every two sample points reduces the sampling rate (and hence data size) by half. Before this operation, however, the data needs to be low-pass filtered (digitally) to remove harmonics beyond the Nyquist frequency to avoid aliasing. Standard functions for decimation are commonly available in commercial software, e.g., `decimate` in Matlab.

6.6 Quantization Error

When analogue data is digitized by DAQ hardware, its value is converted into a binary number. The set of possible values of the digitized data is no longer a continuum but is confined to a discrete set of numbers dictated by the resolution of DAQ hardware. This is illustrated in Fig. 6.7, where the original analogue signal

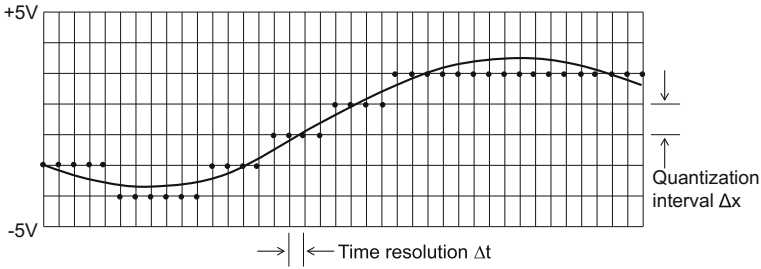


Fig. 6.7 Quantization error. *Solid line* analogue signal. *Dots* digitized data

that can range continuously over ± 5 V is sampled by a 3 bit digitizer. The digitized data can only take on $2^3 = 8$ values at the grid points along the vertical axis, whose spacing is called ‘quantization interval’. The difference between the dots and the continuous line is ‘quantization error’.

As a more realistic example, in a 24 bit digitizer, the sampled data can only have 2^{24} different values. Suppose the original analogue data ranges between ± 5 V. The range is evenly divided into $(2^{24} - 1)$ intervals, giving a quantization interval of $5 \times 2 / (2^{24} - 1) = 5.96 \times 10^{-7}$ V. If this is used with a sensor with sensitivity 0.5 V/g, the quantization interval in g is $5.96 \times 10^{-7} \text{ V} / (0.5 \text{ V/g}) \approx 1 \mu\text{g}$.

Example 6.1 (Data resolution, smart phone) The MEMS of a smart phone is set to output ± 2 g with 12 bit resolution, giving a quantization interval of $2\text{g} \times 2 / (2^{12} - 1) = 9.7680 \times 10^{-4} \text{ g} \approx 0.001 \text{ g}$. Figure 6.8 shows a set of acceleration data sampled at 50 Hz measured along the z-direction (vertical) of a smart phone placed on the floor. The value at each time instant is plotted with a dot. The horizontal clusters indicate the possible values of the sampled data. Going horizontally along each cluster, the values are not exactly the same due to numerical round off error; the data was first written to a text file with a limited number of

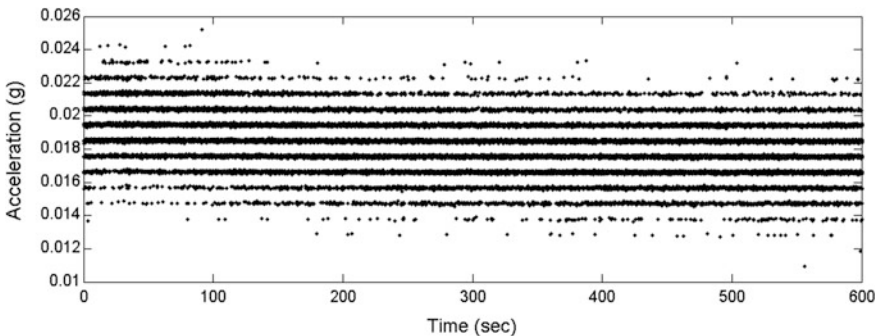


Fig. 6.8 Acceleration data along the z-direction of a smart phone put on the floor. The value at each time instant is plotted with a *dot*. The separation between horizontal clusters indicates the quantization interval

digits. The separation between neighboring clusters indicates a quantization interval of about 0.001 g, as expected. ■

6.6.1 Statistical Properties

Quantization error is a nonlinear function of the signal being digitized. If it is significant it can give distortions that render the data useless. It should be reduced to a negligible level by a suitable choice of data range and resolution of DAQ hardware. Nevertheless, it would still be desirable to assess the statistical properties of quantization error and its effect on data. This has been studied extensively in the signal processing and information theory literature. See Bennett (1948) for a seminal paper; Widrow et al. (1995) a review paper; and Widrow and Kollár (2008) a monograph. Mathematically, quantization in the input signal domain is analogous to sampling in the time domain. Fourier analysis is still a useful tool, although it is now w.r.t. ‘wave number’ (unit = 1/unit of data) instead of frequency. One classical result is that, under some ‘smoothness’ condition on the probability density function (PDF) of the input signal, for sufficiently small quantization interval Δx the quantization error

1. is uncorrelated from the input signal;
2. has a uniform distribution over $\pm\Delta x/2$; and
3. is white, i.e., independent at different sampled time instants.

This result allows the quantization error to be conveniently modeled as an additive noise and is commonly used. The smoothness condition requires that the characteristic function (i.e., Fourier Transform of PDF) of the input signal is bandlimited on $\pm 2\pi/\Delta x$. A necessary and sufficient condition has also been derived (Sripad and Snyder 1977), although it is more difficult to apply. Even when the smoothness condition is violated, these properties can still give a good approximation in many cases.

6.6.2 Power Spectral Density

Under uniform white assumption, the PSD of quantization error can be easily derived using the Parseval equality. Being uniformly distributed on $[-\Delta x/2, \Delta x/2]$, quantization error has a variance of $\Delta x^2/12$. As a white noise, it has a constant PSD S_Q (say, one-sided) up to the Nyquist frequency ($1/2\Delta t$ Hz). The area under the PSD over the sampling band is therefore $S_Q/2\Delta t$, which should also be equal to the variance $\Delta x^2/12$ (Parseval equality). This implies

$$S_Q = \frac{\Delta x^2 \Delta t}{6} \quad (\text{one-sided PSD of quantization error}) \quad (6.9)$$

Note that the noise PSD from quantization error diminishes as the sampling rate increases, essentially because the same variance is spread over a wider frequency band. This is in contrast to sensor/hardware noise, whose PSD does not depend on the sampling rate and whose variance increases with the sampling rate.

In Example 6.1, the MEMS data on smart phone was sampled at 50 Hz, i.e., $\Delta t = 1/50 = 0.02$ s, with a quantization interval of about $\Delta x = 0.001$ g. Substituting into (6.9) gives a (one-sided) PSD of $S_Q = (0.001\text{g})^2(0.02\text{s})/6 \approx 3000(\mu\text{g})^2/\text{Hz}$. This is about 3% of the overall channel noise PSD ($300^2(\mu\text{g})^2/\text{Hz}$, see Fig. 6.2), and is negligible.

6.7 Synchronization

When multiple channels of data are acquired and analyzed, they are often assumed to be ‘synchronized’, i.e., recorded simultaneously at the same time scale. This should not be taken for granted, however. Synchronization is not only to do with whether data recording starts at the same time for different channels, but also with the pace at which they are recorded afterwards. Simply recording multiple data channels on the same computer does not mean that they are synchronized. Each DAQ unit has its own clock running at its own pace, which is not perfectly constant and is temperature dependent. To be synchronized, the data at different channels need to be sampled based on the same clock. Practically, synchronization means that the time difference at which data channels are sampled is within a certain tolerance, much smaller than the periods of interested harmonics. Data channels of the same DAQ unit are often synchronized but the same need not be true for channels of different units. For example, the three data channels of triaxial acceleration measured on one smart phone are synchronized but they are not synchronized with those on another smart phone, unless special provisions are made. Instrument manufacturers often allow multiple DAQ units to be synchronized by plugging into a central module to produce a large array of synchronized data channels.

Synchronized data are much easier to process than asynchronous ones and they are conventionally assumed in mathematical theories. The cross spectral characteristics of unsynchronized data are different from synchronized ones, creating additional difficulties for modal identification. For example, asynchronous data exhibit artificial modes in the singular value spectrum and they make mode shapes more difficult to identify; see Sect. 7.5.

Common Configurations

Acquiring synchronized data comes with an overhead and has implications on test configuration. Figure 6.9 illustrates some configurations for field tests that do not

rely on communication infrastructures such as internet or Wi-Fi. Clearly they depend on instrumentation technology which is continually advancing. In the basic setting in Fig. 6.9a, analogue data from different sensors are transmitted through cables to a central DAQ console, where they are synchronously recorded. The cables required can be very long (e.g., 100 m). This can be laborious to set up or manage, especially for tests with multiple setups. It is also vulnerable to environmental noise or distortion due to signal voltage drop along the cable. Generally it is preferable to shorten the path (cable) through which the analogue data is transmitted. Once data is digitized on a DAQ unit it will no longer be contaminated by noise. In Fig. 6.9b, analogue data is recorded locally on a DAQ unit. Multiple DAQ units are synchronized digitally, e.g., by Ethernet cable. Here, the analogue cables are relatively short but the digital cables (for synchronization) are long. It is still necessary to manage long cables.

For outdoor applications, one configuration that avoids the use of long cables is to synchronize the DAQ units through GPS (Global Positioning System). In Fig. 6.9c, the DAQ units are synchronized through GPS receivers with a common GPS clock. For indoor applications where one cannot use GPS receiver, it is possible to synchronize the DAQ units with high precision clocks (Fig. 6.9d). Before field test the clocks are synchronized with a common time base (e.g., GPS).

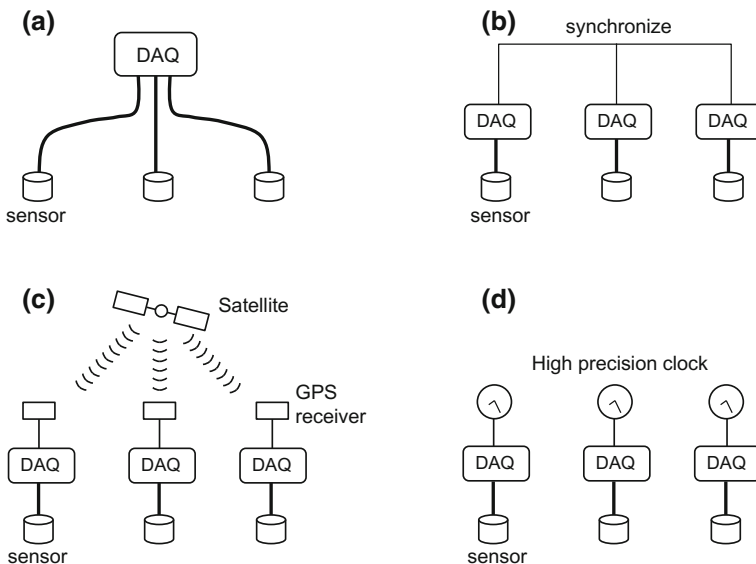


Fig. 6.9 Some test configurations for acquisition of synchronized data. *Thick and thin lines* represent analogue and digital data transmission, respectively. **d** High precision clocks are synchronized shortly before the test begins

The high precision clocks can provide effectively synchronized data for a reasonable time span (e.g., a day) after they are unplugged from the common time base.

Time Precision

To give an idea of time precision, below are some ballpark figures from published instrument specifications. A precision of 1 μs can be maintained by synchronizing with GPS. The clock on a good DAQ unit can drift by 10 min/year (20 $\mu\text{s/s}$), where a high precision clock can drift by 1 s/year (0.03 $\mu\text{s/s}$). As an example, two DAQ units synchronized in the beginning of a test but unplugged afterwards can have after 12 h a relative drift of $2 \times 20 \times 10^{-6} \times 12 \times 3600 \approx 1.6$ s, which is not acceptable unless the vibrations of interest have very long periods ($\gg 1.6$ s). Using a high precision clock, the drift is about 0.26 ms, which is acceptable unless one is interested in high frequency activities beyond (say) 100 Hz.

6.8 Channel Noise Calibration

It is a good practice to assess the noise spectrum of data to establish confidence on its use and allow proper interpretation of analysis results. This is relevant when commissioning a new piece of equipment or starting a new experiment. Specifications or calibration certificates (if available) from manufacturers provide some basic information but it is no substitute for a calibration that informs the current state of the equipment system at hand under the test environment. Calibrating the noise of seismometers (e.g., servo-accelerometers) is an important task in seismology. See Aki and Richards (2002) and Havskov and Alguacil (2016).

Assuming a linear measurement system, the input-output relationship between the scaled FTs of input motion and output data can be modeled as

$$\underset{\text{output data}}{X(\omega)} = \underset{\text{transfer function}}{G(\omega)} \underset{\text{input motion}}{Z(\omega)} + \underset{\text{channel noise}}{\varepsilon(\omega)} \quad (6.10)$$

The transfer function $G(\omega)$ accounts for both the characteristics of sensor and DAQ hardware. Assuming that the input motion and noise are independent, the PSDs are related by

$$\underset{\text{output data}}{S_x} = |G|^2 \underset{\text{input motion}}{S_z} + \underset{\text{channel noise}}{S_e} \quad (6.11)$$

where $S_x = E[XX^*]$, $S_z = E[ZZ^*]$ and $S_e = E[\varepsilon\varepsilon^*]$ are respectively the PSDs of the output data, input motion and channel noise; the dependence on ω has been omitted for simplicity. For calibration, the noise PSD S_e is the target to be determined but only S_x can be directly obtained (estimated) from data. By its very nature, channel noise is mixed with the signal being measured. It is not possible to separate noise from data unless the target signal is *negligible* or *known*. The same applies to their

PSDs. Strategies for estimating the noise PSD based on these two possibilities have been explored.

6.8.1 Base Isolation

The first idea is straightforward. In (6.11), if $|G|^2 S_z \ll S_e$ then $S_x \approx S_e$ gives the noise PSD. In Fig. 6.2, for frequencies lower than 4 Hz (say), the root PSD of MEMS data ($\approx 30 \mu\text{g}/\sqrt{\text{Hz}}$) is reflective of channel noise because the structural response there ($< 1 \mu\text{g}/\sqrt{\text{Hz}}$) is negligible. The same of course is not true near the natural frequency. Pursuing the first idea, the main challenge lies in setting up an experiment that isolates the sensor from vibration due to air (on the structure) and supports. The former can usually be well-controlled under laboratory environment, leaving the latter a dominant source. The basic strategy is then ‘base isolation’.

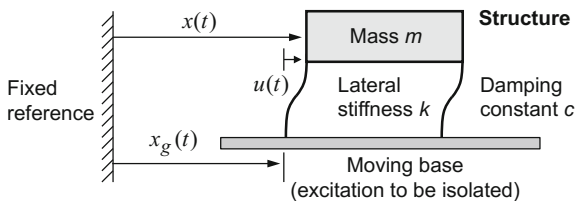
Consider the SDOF structure subjected to base motion in Fig. 6.10. This figure is the same as Fig. 6.3 for a sensor, except that the proof mass in the sensor is now the structure. Here, we want to design an ‘isolation system’ so that the acceleration of the structure picked up by accelerometer, i.e., $\ddot{x}(t)$ (not $\ddot{u}(t)$), is small regardless of the base motion $\ddot{x}_g(t)$. Proceeding in the frequency domain, $\mathcal{F}_{\ddot{x}}(\omega) = \mathcal{F}_{\ddot{x}_g}(\omega) + \mathcal{F}_{\ddot{u}}(\omega)$, where ‘ $\mathcal{F}$ ’ denotes the scaled FT of the subscripted quantity. Substituting $\mathcal{F}_{\ddot{u}}(\omega) = -\omega^2 \mathcal{F}_u(\omega)$ and using the expression of $\mathcal{F}_u(\omega)$ from (6.6) gives

$$\mathcal{F}_{\ddot{x}}(\omega) = H(\omega) \mathcal{F}_{\ddot{x}_g}(\omega) \quad (6.12)$$

where $H(\omega)$ is the (dimensionless) transfer function between $\ddot{x}_g(t)$ and $\ddot{x}(t)$:

$$H(\omega) = \frac{1 + 2\zeta\beta i}{1 - \beta^2 + 2\zeta\beta i} \quad |H(\omega)| = \sqrt{\frac{1 + 4\zeta^2\beta^2}{(1 - \beta^2)^2 + 4\zeta^2\beta^2}} \quad \beta = \frac{\omega}{\omega_1} \quad (6.13)$$

Fig. 6.10 SDOF structure subjected to base motion



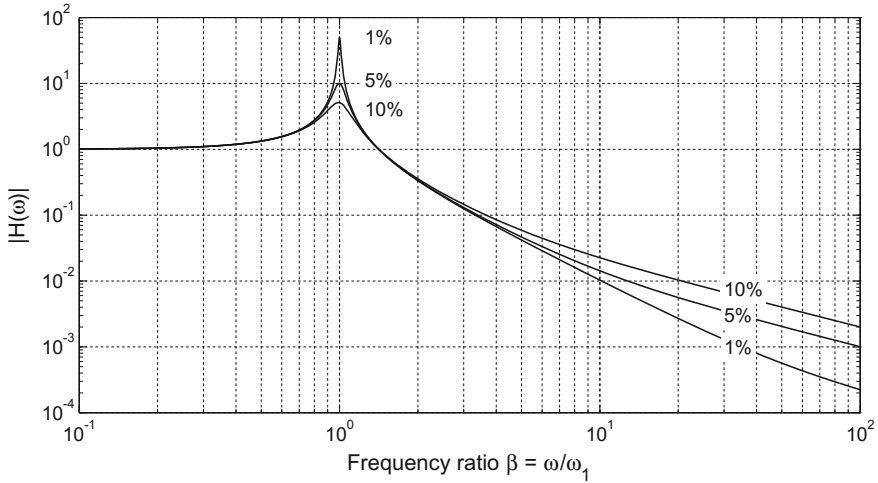


Fig. 6.11 Modulus of transfer function between base and structural acceleration

where $\omega_1 = \sqrt{k/m}$ (rad/s) and $\zeta = c/2\sqrt{mk}$ are respectively the natural frequency and damping ratio of the structure.

Figure 6.11 shows $|H(\omega)|$ versus the frequency ratio $\beta = \omega/\omega_1$. Base isolation exploits the decay of $|H(\omega)|$ for large β . By reducing ω_1 , we can shift the dominant frequencies of $\ddot{x}_g(t)$ to have high β and hence low $|H(\omega)|$. Note that $|H(\omega)| = 1$ at $\beta = \sqrt{2}$ regardless of ζ . Beyond that, $|H(\omega)| \sim 1/\beta^2$ for moderate values of $\beta \gg 1$. For very large $\beta \gg 1/4\zeta^2$, $|H(\omega)| \sim 2\zeta/\beta$ but this is of little relevance here. Based on $|H(\omega)| \sim 1/\beta^2$, to allow transmitting at most $|H(\omega)| = 1\%$ of base motion (in terms of root PSD) at 1 Hz and beyond, we need a structure with one-tenth of this frequency, i.e., 0.1 Hz. This can impose an experimental challenge. Suppose the structure is a simple pendulum with length L , whose natural frequency is $\sqrt{g/L}/2\pi$ Hz. This would require a length of $L = (9.81)/(0.1 \times 2\pi)^2 \approx 25\text{m}$, say, hanging a cable from eight stories high! More sophisticated isolation systems have been developed. See Brownjohn and Botfield (2008) that reports the performance of a laboratory-scale folded pendulum system.

6.8.2 Huddle Test

The second idea of noise calibration explores the algebraic input-output relationship of PSDs together with special experimental design. In (6.11) if $|G|^2 S_z$ is known then we can simply subtract it from S_x to obtain the noise PSD S_e . While this is usually not the case with a single data channel, possibilities exist with multiple channels.

It is possible to determine the noise PSD from multiple data channels driven by a single common input without knowing it. Intuitively, multiple channels provide

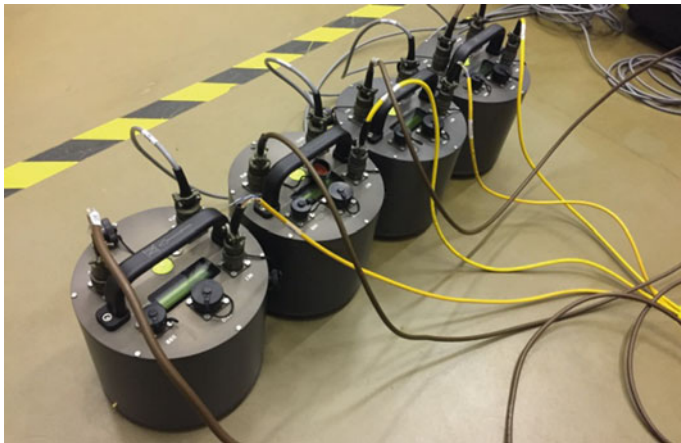


Fig. 6.12 Huddle test

redundant information about the input for distinguishing the noise. In a conventional approach called ‘huddle test’, multiple sensors with the same orientation are placed as close as possible with each other (but no contact) so that they can be assumed to measure the same motion; see Fig. 6.12. The noise PSD can be determined from the PSDs and cross PSDs of the channels measuring the same motion. The basic mathematics is as follow.

Let $\mathbf{X}(\omega)$ ($n \times 1$) comprise the scaled FT of output data which are all subjected to a common (scalar) input motion $Z(\omega)$. Then, analogous to (6.10),

$$\underset{\substack{n \times 1 \\ \text{output} \\ \text{data}}}{\mathbf{X}(\omega)} = \underset{\substack{n \times 1 \\ \text{transfer} \\ \text{function}}}{\mathbf{G}(\omega)} \underset{\substack{\text{scalar} \\ \text{input} \\ \text{motion}}}{Z(\omega)} + \underset{\substack{n \times 1 \\ \text{channel} \\ \text{noise}}}{\boldsymbol{\varepsilon}(\omega)} \quad (6.14)$$

where $\mathbf{G}(\omega)$ is a vector of transfer functions and $\boldsymbol{\varepsilon}(\omega)$ is the scaled FT of the vector of channel noise. Assume that the noise at different channels are mutually independent and they are independent of the input Z . The PSD matrices are then related by

$$\underset{n \times n}{\mathbf{S}_x} = \underbrace{\underset{n \times n}{\mathbf{G}} \underset{\text{scalar}}{\mathbf{G}^*}}_{n \times n} \underset{\text{scalar}}{S_z} + \underset{n \times n}{\mathbf{S}_e} \quad (6.15)$$

where, analogous to (6.11), $\mathbf{S}_x = E[\mathbf{X}\mathbf{X}^*]$ is the PSD matrix of the output data and $S_z = E[Z Z^*]$ is the PSD of the input motion; the dependence on ω has been omitted for simplicity;

$$\mathbf{S}_e = E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^*] = \begin{bmatrix} S_{e1} & & \\ & \ddots & \\ & & S_{en} \end{bmatrix} \quad (6.16)$$

is the PSD matrix of channel noise; S_{ei} is the noise PSD of the i th channel.

Determinacy

For given $\mathbf{S}_x$ at each frequency, (6.15) gives an algebraic system of complex-valued equations for the unknowns $\{G_i\}_{i=1}^n$, S_z and $\{S_{ei}\}_{i=1}^n$. Since G_i is complex-valued, the total number of (real-valued) unknowns is $3n + 1$. When the primary interest is on $\{S_{ei}\}_{i=1}^n$, the number of unknowns can be reduced by 1, by combining S_z and $\mathbf{G}$ to form $\mathbf{G}' = \sqrt{S_z}\mathbf{G}$ so that:

$$\mathbf{S}_x = \underbrace{\mathbf{G}'\mathbf{G}'^*}_{n \times n} + \mathbf{S}_e \quad (6.17)$$

There are now $3n$ unknowns, comprising $\{S_{ei}\}_{i=1}^n$ and the (complex-valued) entries of $\mathbf{G}'$. The number of real-valued linearly independent equation is n^2 because $\mathbf{S}_x$ is Hermitian ($\mathbf{S}_x^* = \mathbf{S}_x$). Reading the diagonal and upper off-diagonal entries of (6.17) gives

$$\begin{aligned} S_{ii} &= G'_i G'^*_i + S_{ei} & i = 1, \dots, n \\ S_{ij} &= G'_i G'^*_j & i < j \end{aligned} \quad (6.18)$$

where S_{ij} denotes the (i, j) -entry of $\mathbf{S}_x$; G'_i is the i th entry of $\mathbf{G}'$. The second equation in (6.18) is complex-valued and hence contains two real-valued equations. There are $n(n - 1)/2$ equations of this type over all possible pairs of (i, j) indicated in the equation.

Table 6.1 shows the number of (real-valued) equations and unknowns for different n . When n is 1 or 2 there are not enough equations for a unique solution. There are just enough equations when $n = 3$. When $n \geq 4$ the unknowns are over-constrained. In this case the problem is better formulated as an inference problem accounting for possible estimation and modeling errors. For example, in the actual implementation, $\mathbf{S}_x$ is substituted by its sample estimate. The latter contains estimation error and need not have the form in (6.17).

6.8.3 Three Channel Analysis

The huddle test with $n = 3$ channels is of practical significance as it requires the smallest number of channels. This possibility was discovered by Sleeman et al. (2006)

Table 6.1 Determinacy of Huddle test with a single common input

n	No. of unknowns, $3n$	No. of equations, n^2	Remark
1	3	1	Under-determined
2	6	4	Under-determined
3	9	9	Unique solution
4	12	16	Over-constrained

and has been used for calibrating a variety of seismometers (Ringler and Hutt 2010). Solving (6.18) in this case gives

$$S_{ei} = S_{ii} - \frac{S_{ji}S_{ik}}{S_{jk}} \quad i, j, k \text{ distinct} \quad (6.19)$$

In implementation, the PSDs on the RHS are replaced by their sample estimates calculated from measured data. Statistical estimation error in the sample PSDs can be controlled by using sufficient data length, trading off with frequency resolution. The huddle test relies on the single common input assumption, which should be managed by experimental setup. The accuracy of noise PSD tends to be lower at high frequencies due to lower spatial coherence (a modeling error).

Theoretically, when (6.18) holds the value of S_{ei} obtained from (6.19) is always real-valued and remains the same when j and k are swapped, because the solution is real and unique. However, this is not necessarily true in implementation because the sample PSDs need not obey (6.18). It can be easily verified that the second term in (6.19) becomes its complex conjugate when j and k are swapped; the same is also true when sample cross PSDs are substituted. This suggests that the real part of the expression should be taken in implementation.

Example 6.2 (Huddle test) The noise PSDs of the North (parallel to handles) channels of the left three triaxial servo-accelerometers in Fig. 6.12 are estimated by three channel analysis. Ambient data of 9000 s duration and sampling rate 100 Hz was collected under laboratory environment. The data is divided into 900 non-overlapping segments (each 10 s) to produce averaged sample PSD estimates at a frequency interval of $1/10 = 0.1$ Hz and a relative error of $1/\sqrt{900} \approx 3\%$. The results are shown in Fig. 6.13.

In Fig. 6.13a, the frequency axis is shown on a log-scale up to the Nyquist frequency (50 Hz). The root PSD of the three data channels almost overlap, as expected. Below 10 Hz the estimated root PSDs of noise almost overlap, suggesting similar noise characteristics among the three channels. On this log-log plot, the slope is about 1/2, demonstrating the 1/f (in PSD) nature of noise. For high frequencies, the noise PSD estimates display high fluctuations. They are not accurate because of low coherence in the input ground motion among the three sensors, violating (6.18). This is despite the fact that the sample PSDs of data at different frequencies all have the same relative error of about 3%. Figure 6.13b shows the root PSD of noise with a linear scale in the frequency on a typical range of interest. The noise PSD appears to be robust to the nature of data. For example, it is not affected by the isolated peaks in the data PSD between 12 and 16 Hz. ■

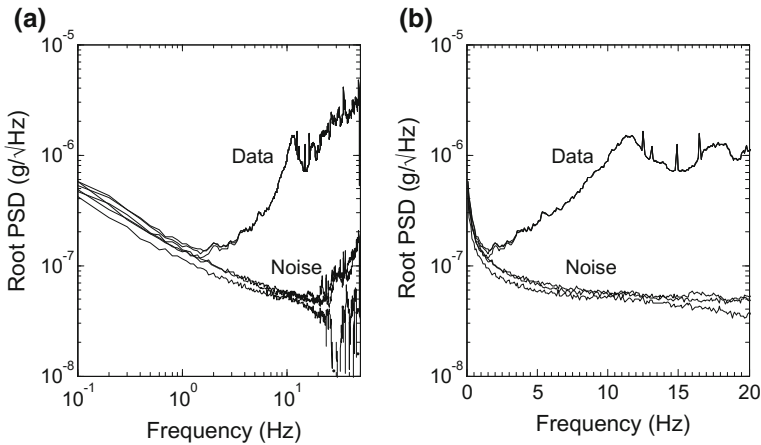


Fig. 6.13 (Two-sided) Root PSD of data and estimated root PSD of noise. **a** The slope at low frequency demonstrates $1/f$ nature (of PSD). Accuracy of noise PSD is low at high frequencies due to low spatial coherence among sensors. **b** Zooms into a typical frequency range of interest

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Chapter 7

Ambient Data Modeling and Analysis

Abstract This chapter presents some basic tools for analyzing a sample history of ambient data. Procedures for plotting the power spectral density and singular value spectra are presented. Their theoretical statistical properties near the resonance band of a structural vibration mode are discussed, allowing one to interpret and detect potential modes in applications. A detailed example with field data is presented to illustrate in a step-by-step manner the procedures and considerations in the production and analysis of these spectra. Ambient data of other nature are also discussed, including asynchronous data that exhibits artificial vibration modes in their spectra, and microtremor data that can be used to infer site characteristics. Procedures for generating synthetic ambient data are presented, which are useful for verifying algorithms or parametric studies.

Keywords Power spectral density • Singular value spectrum • Microtremor • H/V method • Site amplification

Under ambient conditions, the vibration response of a structure appears random in the time domain as it is subjected to excitations with a variety of frequency characteristics. Modeled as a stationary stochastic process, its statistical properties have been investigated in Chap. 5. Ambient vibration response can be measured, but it is inevitably contaminated with noise (Chap. 6). With these considerations in mind, in this chapter we discuss how to analyze ambient vibration data measured on a structure. For statistical analysis of time series, see, e.g., Brillinger (1981), Bendat and Piersol (1993), Kay (1993).

We first study the theoretical frequency characteristics of ambient data in the resonance band of a vibration mode. The ‘power spectral density’ (PSD) and ‘singular value’ (SV) spectrum allow one to visualize the modes potentially present in a frequency band. The picture can be different when the data is not perfectly synchronized across its channels. All structures are excited by small ground motions at their support, often known as ‘microtremor’. The nature and possible exploitation of microtremor data for identifying site characteristics are discussed.

Finally, we consider how to generate synthetic data with a given PSD. This is needed when one wants to verify an algorithm or perform parametric studies.

7.1 Resonance Band Characteristics

A proper set of ambient data measured on a structure can be assumed to comprise structural response and channel noise; and is free from distortions such as clipping, aliasing or quantization error. The structural response at a given time instant is contributed by all modes and it is generally difficult to separate them by operating in the time domain. The contributions from different modes can be naturally separated in the frequency domain because they are amplified by resonance and ‘stand out’ near their natural frequency. This is illustrated in Fig. 7.1, which shows the root PSD (no averaging) of the horizontal acceleration measured on a tall building roof. Peaks near 0.2, 0.4, 0.7, 1.2 Hz, etc., are dominated by vibration modes. The resonance band of the mode near 1.2 Hz is shaded. It can be reasoned from the dynamic amplification factor that for a mode with natural frequency f (Hz) and damping ratio ζ , the resonance band is $f(1 \pm \kappa\zeta)$ Hz for some dimensionless constant κ and hence has a width of the order of $f\zeta$ Hz.

Let $\{\hat{\mathbf{y}}_j\}_{j=0}^{N-1} (n \times 1)$ be the time-domain data with n measured channels at interval Δt (s); N is the number of sample points and so the data duration is $N\Delta t$. Let $\hat{\mathcal{F}}_k (n \times 1 \text{ complex})$ be the (two-sided) scaled Fast Fourier Transform (FFT) of $\{\hat{\mathbf{y}}_j\}_{j=0}^{N-1}$ at frequency $f_k = k/N\Delta t$ (Hz):

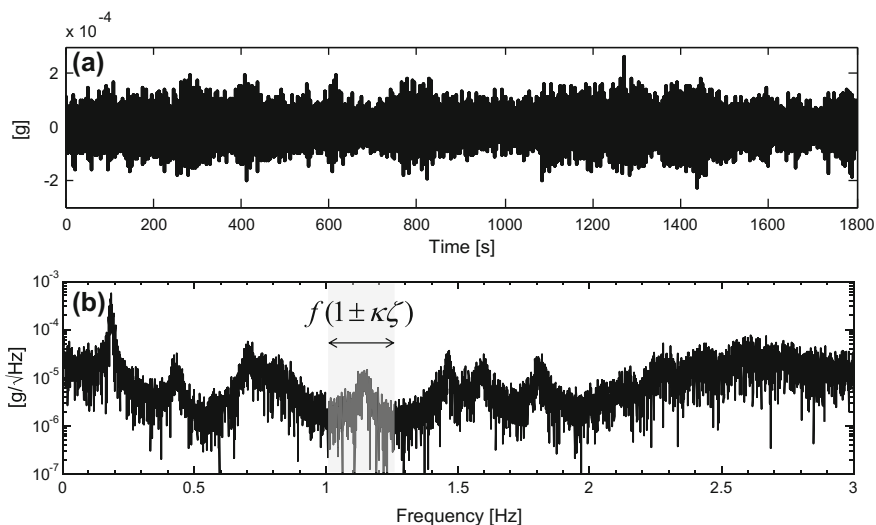


Fig. 7.1 Resonance band of a vibration mode (*shaded*); **a** Time history data; **b** Root PSD (no averaging); f = natural frequency (Hz), ζ = damping ratio, κ = dimensionless constant

$$\hat{\mathcal{F}}_k = \sqrt{\frac{\Delta t}{N}} \sum_{j=0}^{N-1} \hat{\mathbf{y}}_j e^{-2\pi i j k / N} \quad (7.1)$$

In Matlab, the sequence $\{\hat{\mathcal{F}}_k\}_{k=0}^{N-1}$ can be obtained using the function `fft` (Sect. 2.8). Only the first half of the sequence ($0 \leq k \leq \text{integer part of } N/2$) is informative. The second half is redundant as it is just the conjugate mirror image of the first half.

7.1.1 Single Mode

Consider a frequency band dominated by a single mode with natural frequency f (Hz) and damping ratio ζ . Within the band, $\hat{\mathcal{F}}_k$ can be assumed to comprise the modal response (Sect. 5.2) and channel noise:

$$\hat{\mathcal{F}}_k = \underset{n \times 1}{\boldsymbol{\Phi}} \underset{n \times 1}{h_k} \underset{n \times 1}{p_k} + \underset{n \times 1}{\boldsymbol{\epsilon}_k} \quad (7.2)$$

where $\boldsymbol{\Phi}$ ($n \times 1$) is the ‘partial mode shape’ (i.e., confined to the measured DOFs); p_k (complex) and $\boldsymbol{\epsilon}_k$ ($n \times 1$ complex) are respectively the scaled FFT of modal excitation and channel noise;

$$h_k = \frac{(2\pi i f_k)^{-q}}{1 - \beta_k^2 - 2\zeta\beta_k i} \quad \beta_k = \frac{f}{f_k} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (7.3)$$

is the modal transfer function. Assuming that p_k and $\boldsymbol{\epsilon}_k$ are independent, the PSD matrix of ambient data is given by

$$\mathbf{E}_k = E[\hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^*] = S_k D_k \underset{n \times 1}{\boldsymbol{\Phi}} \underset{1 \times n}{\boldsymbol{\Phi}^T} + \underset{n \times n}{\mathbf{S}_{ek}} \quad (\text{single mode}) \quad (7.4)$$

where $S_k = E[p_k p_k^*]$ is the modal force PSD and $\mathbf{S}_{ek} = E[\boldsymbol{\epsilon}_k \boldsymbol{\epsilon}_k^*]$ ($n \times n$ Hermitian) is the PSD matrix of channel noise;

$$D_k = \frac{(2\pi f_k)^{-2q}}{(1 - \beta_k^2)^2 + (2\zeta\beta_k)^2} \quad \beta_k = \frac{f}{f_k} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (7.5)$$

is the dynamic amplification factor.

7.1.2 Multi-mode

Consider now m contributing modes in the band, with natural frequencies $\{f_i\}_{i=1}^m$ (Hz), damping ratios $\{\zeta_i\}_{i=1}^m$ and partial mode shapes $\{\boldsymbol{\phi}_i\}_{i=1}^m$ ($n \times 1$). The scaled FFT of measured data now comprises contributions from m modes and channel noise:

$$\hat{\mathcal{F}}_k = \sum_{i=1}^m \boldsymbol{\phi}_i h_{ik} p_{ik} + \boldsymbol{\varepsilon}_k = \underbrace{\boldsymbol{\Phi} \text{diag}[\mathbf{h}_k]}_{n \times m \times m \times m} \mathbf{p}_k + \boldsymbol{\varepsilon}_k \quad (7.6)$$

where $\{p_{ik}\}_{i=1}^m$ are the scaled FFT of modal forces; $\mathbf{p}_k = [p_{1k}, \dots, p_{mk}]^T$; $\boldsymbol{\Phi} = [\boldsymbol{\phi}_1, \dots, \boldsymbol{\phi}_m]$ is the partial mode shape matrix; and $\text{diag}[\mathbf{h}_k]$ denotes a diagonal matrix with entries $\{h_{ik}\}_{i=1}^m$;

$$h_{ik} = \frac{(2\pi i f_k)^{-q}}{1 - \beta_{ik}^2 - 2\zeta_i \beta_{ik} i} \quad \beta_{ik} = \frac{f_i}{f_k} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (7.7)$$

The PSD matrix of measured data is given by

$$\begin{aligned} \mathbf{E}_k &= E[\hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^*] = \sum_{i=1}^m \sum_{j=1}^m h_{ik} h_{jk}^* S_{ijk} \underbrace{\boldsymbol{\phi}_i \boldsymbol{\phi}_j^T}_{n \times 1 \times n} + \mathbf{S}_{ek} = \underbrace{\boldsymbol{\Phi} \mathbf{H}_k \boldsymbol{\Phi}^T}_{n \times m \times m \times m \times n} + \mathbf{S}_{ek} \\ &\text{(multi-mode)} \end{aligned} \quad (7.8)$$

where $S_{ijk} = E[p_{ik} p_{jk}^*]$ is the cross PSD between the modal forces of the i th and j th mode at frequency f_k ; $\mathbf{H}_k$ is a $m \times m$ Hermitian matrix whose (i, j) -entry is $S_{ijk} h_{ik} h_{jk}^*$.

7.2 PSD Spectrum

Time histories of ambient data look erratic and it is often difficult to gain insights from them. The characteristics of data can be better analyzed in the frequency domain. The PSD of ambient data exhibits dynamic amplification near the natural frequencies of a structure, providing a useful tool for investigating the modes or generally frequency characteristics. The theoretical PSD is generally unknown but it can be estimated using the measured data. Plotting the squared modulus of the scaled FFT of data with frequency produces an erratic spectrum like Fig. 7.1 because the values at different frequencies are almost uncorrelated and their

standard deviation is as large as their mean (Sect. 4.10). The frequency characteristics are better viewed through the averaged sample PSD, which is an asymptotically unbiased and converging estimator for the theoretical PSD.

Frequency Resolution and Accuracy

To produce the averaged sample PSD using a single time history, the data is divided into M non-overlapping segments of equal length. The sample PSDs of the individual segments are calculated and averaged. The choice of M is a trade-off between frequency resolution and accuracy in the spectrum. Increasing M reduces statistical error in the averaged sample PSD at each frequency but it decreases frequency resolution and increases leakage bias. The averaged PSD over M values has a c.o.v. of $1/\sqrt{M}$ if the segments were independent; c.o.v. = coefficient of variation = ratio of standard deviation to mean. Suppose the original data has a duration of $N'\Delta t$ (s). Dividing it into M equal segments gives a segment duration of $N'\Delta t/M$ (s) and hence a FFT frequency interval of $\Delta f = M/N'\Delta t$ (Hz). To achieve a frequency interval of Δf the number of segments can be taken as

$$M = \lfloor N'\Delta t \Delta f \rfloor \quad (7.9)$$

where $\lfloor \cdot \rfloor$ denotes the integer part of the argument; e.g., $\lfloor 2.8 \rfloor = 2$. Each segment has $N_w = \lfloor N'/M \rfloor$ samples. Note that $N_w M \leq N'$. Those samples in the original data with index greater than $N_w M$ may be discarded. The wastage is insignificant for large M .

Usually a spectrum is plotted within a limited frequency band of interest, say, (f_L, f_U) Hz. For visualization it is desired to have a specified resolution, say, M_f intervals (e.g., $M_f = 300$) in the band (f_L, f_U) . The required frequency interval is then

$$\Delta f = \frac{f_U - f_L}{M_f} \text{ (Hz)} \quad (7.10)$$

7.2.1 Procedure

Given the time history data $\{\hat{y}_j\}_{j=0}^{N'-1}$ ($n \times 1$), below is a procedure that produces the (averaged) sample PSD matrix with a frequency resolution of M_f intervals displayed over the frequency band (f_L, f_U) Hz. Plotting the i th diagonal entry of the PSD matrix with frequency gives 'the sample PSD spectrum', or 'PSD spectrum' in short when understood, for the i th measured channel. For $i \neq j$, the (i, j) -entry gives the sample cross PSD between channels i and j . Statistical properties of the sample PSD spectrum are discussed in Sect. 4.10. The averaging process for a single channel ($n = 1$) is illustrated in Fig. 7.2.

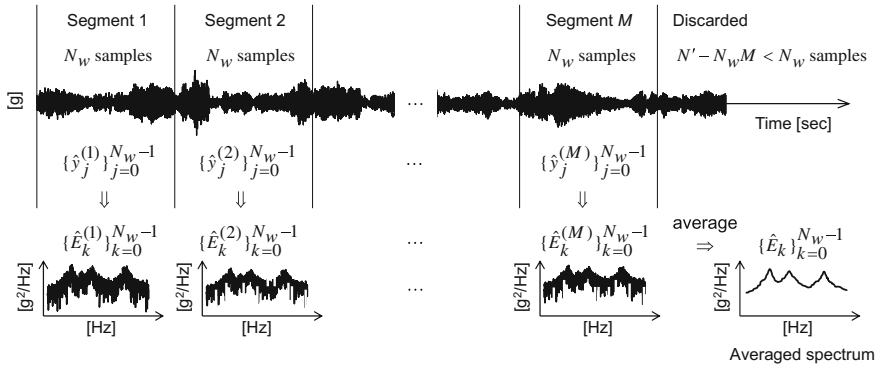


Fig. 7.2 Averaged PSD spectrum; $\{\hat{E}_k^{(r)}\}_{k=0}^{N_w-1}$ is the sample PSD of the r th segment; for illustration, data is assumed to be acceleration in g with N' samples

As a word of caution, at each frequency it is the sample PSD matrices $\{\hat{\mathcal{F}}_k^{(r)} \hat{\mathcal{F}}_k^{(r)*}\}_{r=1}^M$ that should be averaged; not the time-domain data $\{\hat{\mathbf{y}}_j^{(r)}\}_{r=1}^M$, nor the scaled FFTs $\{\hat{\mathcal{F}}_k^{(r)}\}_{r=1}^M$.

Sample PSD Matrix with M_f Intervals Over Frequency Band (f_L, f_U) Hz

0. Calculate

$$\begin{aligned} \Delta f &= (f_U - f_L)/M_f \quad (\text{frequency interval in Hz}) \\ M &= \lfloor N' \Delta t \Delta f \rfloor \quad (\text{No. of segments}) \\ N_w &= \lfloor N'/M \rfloor \quad (\text{No. of samples per segment}) \\ N &= N_w M \quad (\text{Truncated data length}) \end{aligned}$$

1. Divide the data $\{\hat{\mathbf{y}}_j\}_{j=0}^{N-1}$ into M segments, each with length N_w . Let $\{\hat{\mathbf{y}}_j^{(r)}\}_{j=0}^{N_w-1}$ be the r th segment. Then $\hat{\mathbf{y}}_j^{(r)} = \hat{\mathbf{y}}_{N_w(r-1)+j}$ ($r = 1, \dots, M; j = 0, \dots, N_w - 1$).
2. For each segment r , calculate the scaled FFT $\{\hat{\mathcal{F}}_k^{(r)}\}$ and (raw) sample PSD $\{\hat{\mathbf{E}}_k^{(r)}\}$ by

$$\hat{\mathcal{F}}_k^{(r)} = \sqrt{\frac{\Delta t}{N_w}} \sum_{j=0}^{N_w-1} \hat{\mathbf{y}}_j^{(r)} e^{-2\pi i j k / N_w} \quad \hat{\mathbf{E}}_k^{(r)} = \hat{\mathcal{F}}_k^{(r)} \hat{\mathcal{F}}_k^{(r)*} \quad (7.11)$$

3. Calculate the (averaged) sample PSD matrix $\{\hat{\mathbf{E}}_k\}$ by

$$\hat{\mathbf{E}}_k = \frac{1}{M} \sum_{r=1}^M \hat{\mathbf{E}}_k^{(r)} \quad (7.12)$$

4. Plotting $\hat{\mathbf{E}}_k(i, i)$ versus $f_k = k/N_w \Delta t$ and zooming into the band (f_L, f_U) gives the averaged sample PSD spectrum for the i th measured channel with the desired frequency resolution.

7.3 Singular Value Spectrum

A peak displaying dynamic amplification in the PSD spectrum indicates the presence of a mode but it need not indicate the number of modes. For example, three lines showing a peak at the same frequency need not indicate three modes. The number of modes can be assessed based on a plot of the *eigenvalues* of the *real part* of the PSD matrix versus frequency. The number of lines (eigenvalues) significantly above the remaining ones indicates the number of modes within the band. We call this plot the ‘singular value (SV) spectrum’. The term ‘singular value’ stems originally from ‘singular value decomposition’, which can be applied to any complex rectangular matrix. Taking the real part of the PSD matrix is a provision for detecting synchronization; see the end of Sect. 7.5.1. Singular value spectrum analysis is applicable in more general settings outside operational modal analysis (OMA). See, e.g., Elsner and Tsonis (1996), Golyandina and Zhigljavsky (2013). Figure 7.3 illustrates the difference between the PSD and SV spectrum.

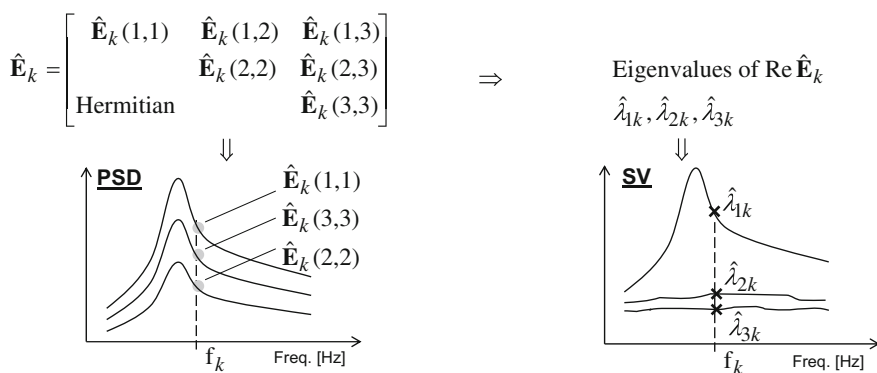


Fig. 7.3 PSD and SV spectrum for $n = 3$ channels near a well-separated mode; $\hat{\mathbf{E}}_k$ is the sample PSD matrix at frequency f_k . The ordering of $\hat{\mathbf{E}}_k(1, 1)$, $\hat{\mathbf{E}}_k(2, 2)$ and $\hat{\mathbf{E}}_k(3, 3)$ is for illustration only as it depends on data

In the resonance band, the SV spectrum has characteristic properties related to the contributing modes. When there is only one mode dominating the band, the largest eigenvalue varies as the dynamic amplification factor. Near the natural frequency, the eigenvector of the largest eigenvalue is equal to the partial mode shape. For modes with close natural frequencies, the eigenvectors need not give the partial mode shapes. Rather, they are orthogonal vectors spanning the subspace containing the partial mode shapes. Sections 7.3.1 and 7.3.2 give an account of these properties based on the theoretical PSD matrix. In the actual implementation, the SV spectrum is produced based on the (averaged) sample PSD. Conclusions will be approximately true if statistical error is small.

7.3.1 Single Mode

When there is only one mode in the frequency band, the PSD matrix of measured data, $\mathbf{E}_k$, is given by (7.4). To simplify discussion, assume i.i.d. channel noise, i.e.,

$$\mathbf{S}_{ek} = S_{ek} \mathbf{I}_n \quad (7.13)$$

where S_{ek} is the common noise PSD and $\mathbf{I}_n$ denotes the $n \times n$ identity matrix. Then

$$\mathbf{ReE}_k = S_k D_k \boldsymbol{\phi} \boldsymbol{\phi}^T + S_{ek} \mathbf{I}_n \quad (7.14)$$

This is a positive-definite and real-symmetric matrix. It can be expressed in a form that shows explicitly the eigenvalues and eigenvectors. Write $\mathbf{I}_n = \sum_{i=1}^n \mathbf{b}_i \mathbf{b}_i^T$, where $\{\mathbf{b}_i\}_{i=1}^n$ ($n \times 1$) is an orthonormal basis with $\mathbf{b}_1 = \boldsymbol{\phi}$ (assuming unit norm, i.e., $\sqrt{\boldsymbol{\phi}^T \boldsymbol{\phi}} = 1$). Substituting into (7.14) and collecting terms,

$$\mathbf{ReE}_k = \underbrace{(S_k D_k + S_{ek})}_{\text{eigenvalue}} \boldsymbol{\phi} \boldsymbol{\phi}^T + \sum_{i=2}^n \underbrace{S_{ek}}_{\text{eigenvalue}} \mathbf{b}_i \mathbf{b}_i^T \quad (7.15)$$

Since $\{\mathbf{b}_i\}_{i=1}^n$ is an orthonormal basis, the above is an eigenvector representation of $\mathbf{ReE}_k$. The largest eigenvalue is $\lambda_{1k} = S_k D_k + S_{ek}$ with eigenvector $\boldsymbol{\phi}$. The remaining $(n - 1)$ eigenvalues are all equal to S_{ek} with eigenvectors $\{\mathbf{b}_i\}_{i=2}^n$. This is illustrated in Fig. 7.4. The eigenvalues and hence SV spectrum have the same unit as the PSD.

Example 7.1 (Single mode, synthetic data) Consider synthetic data $\{\hat{\mathbf{y}}_j\}_{j=0}^{N-1}$ with three measured DOFs generated from

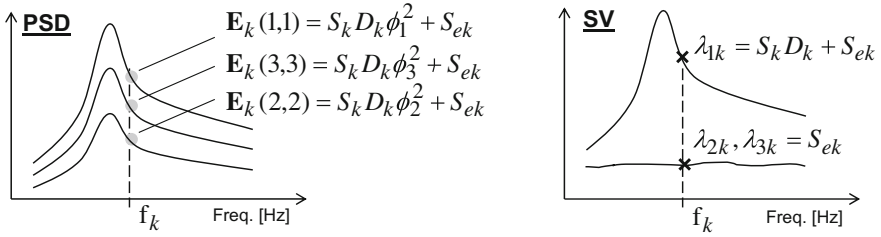


Fig. 7.4 Theoretical PSD and SV spectrum near the resonance band of a single mode; $\mathbf{E}_k = S_k D_k \boldsymbol{\phi} \boldsymbol{\phi}^T + S_{ek} \mathbf{I}_n$; $\boldsymbol{\phi} = [\phi_1, \phi_2, \phi_3]^T$ with $\phi_1^2 + \phi_2^2 + \phi_3^2 = 1$; $\lambda_{1k}, \lambda_{2k}, \lambda_{3k}$ are the eigenvalues (in descending order) of $\text{Re} \mathbf{E}_k$. Ordering of $\mathbf{E}_k(1, 1)$, $\mathbf{E}_k(2, 2)$ and $\mathbf{E}_k(3, 3)$ is for illustration only as it depends on data

$$\hat{\mathbf{y}}_j = \boldsymbol{\Phi} \ddot{\boldsymbol{\eta}}(t_j) + \boldsymbol{\varepsilon}(t_j) \quad t_j = j\Delta t, \quad j = 0, 1, 2, \dots \quad (7.16)$$

where $\Delta t = 0.01$ s; $\boldsymbol{\phi} = [1 \quad 2 \quad 2]^T/3$; and $\boldsymbol{\eta}(t)$ satisfies

$$\ddot{\boldsymbol{\eta}}(t) + 2\zeta\omega\dot{\boldsymbol{\eta}}(t) + \omega^2\boldsymbol{\eta}(t) = \mathbf{p}(t) \quad (7.17)$$

with $\omega = 2\pi$ rad/s ($f = 1$ Hz) and $\zeta = 1\%$; $\mathbf{p}(t)$ is Gaussian white noise with PSD $S = 1(\mu\text{g})^2/\text{Hz}$; and $\boldsymbol{\varepsilon}(t)$ is 3×1 Gaussian white noise vector with i.i.d. components and PSD $S_e = 1(\mu\text{g})^2/\text{Hz}$. Check that $\boldsymbol{\phi}$ has unit norm. Data is generated for 2000 s. It is divided into $M = 20$ segments of 100 s each to produce the averaged PSD matrix $\hat{\mathbf{E}}_k$. The resulting frequency interval is $\Delta f = 1/100 = 0.01$ Hz. Plotting the three diagonal entries of $\hat{\mathbf{E}}_k$ with frequency gives the PSD spectrum. Plotting the three eigenvalues of $\text{Re} \hat{\mathbf{E}}_k$ with frequency gives the SV spectrum.

PSD Spectrum

Figure 7.5a shows the PSD spectrum. The top two lines correspond to DOFs 2 and 3. They almost overlap because the mode shape values at these two DOFs are the same. Except for statistical errors, the three lines vary as $S D_k \phi_i^2 + S_e$ where D_k is the dynamic amplification factor in (7.5) (with $q = 0$) and ϕ_i is the mode shape value, equal to $1/3$, $2/3$ and $2/3$ for DOFs 1, 2 and 3, respectively. As a check, at low frequencies the PSDs are all approximately equal to $S_e = 1(\mu\text{g})^2/\text{Hz}$ because $D_k \approx 0$ there. At the natural frequency, $D_k = 1/4\zeta^2$ and the PSD of DOF 1 (lowest line) is approximately (ignoring S_e) $S D_k \phi_1^2 = (1)[1/4(0.01)^2](1/3)^2 \approx 278 (\mu\text{g})^2/\text{Hz}$.

SV Spectrum

The three lines in the PSD spectrum in Fig. 7.5a suggest a mode at around 1 Hz but they do not imply three modes. The number of modes should be examined from the SV spectrum in Fig. 7.5b. Except for statistical errors, the top line (largest eigenvalue) is equal to $S D_k + S_e$. As a check, at low frequencies it is approximately

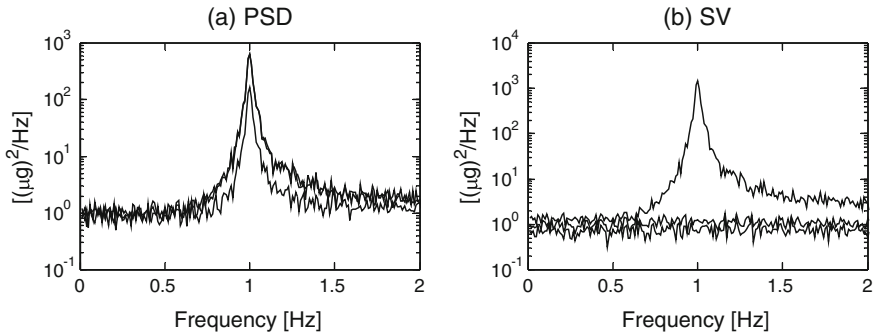


Fig. 7.5 **a** PSD spectrum and **b** SV spectrum, synthetic data with three measured DOFs and a single mode at 1 Hz

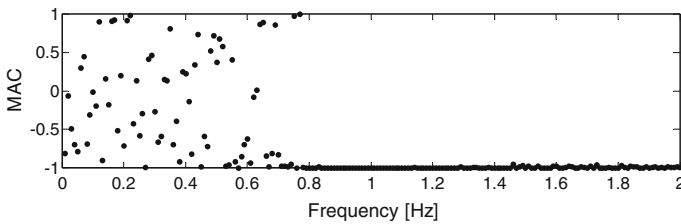


Fig. 7.6 MAC of the eigenvector (largest eigenvalue) of $\text{Re}\hat{\mathbf{E}}_k$ with the partial mode shape $\boldsymbol{\phi}$

equal to $S_e = 1 \text{ } (\mu\text{g})^2/\text{Hz}$; at resonance it is approximately equal to $SD_k = (1)[1/4(0.01)^2] = 2500 \text{ } (\mu\text{g})^2/\text{Hz}$. The remaining two lines are theoretically equal to $S_e = 1 \text{ } (\mu\text{g})^2/\text{Hz}$.

Eigenvector and Partial Mode Shape

The dot product of two vectors measures their proximity. In OMA, it is commonly called the ‘modal assurance criterion’ (MAC). It ranges between -1 and 1 . The closer the absolute value of the MAC is to 1 , the closer the direction of the two vectors (up to a sign reversal). Figure 7.6 shows the MAC between the eigenvector (largest eigenvalue) of $\text{Re}\hat{\mathbf{E}}_k$ and the partial mode shape $\boldsymbol{\phi}$. Near the resonance band the MAC is close to -1 , showing that the eigenvectors there can give a good approximation to the partial mode shape. ■

7.3.2 Multi-mode

Consider now the case when there are m contributing modes in the resonance band. As in Sect. 7.3.1, assume $\mathbf{S}_{ek} = S_{ek}\mathbf{I}_n$ (i.i.d. channel noise). Then (7.8) gives

$$\mathbf{ReE}_k = \mathbf{\Phi}(\mathbf{ReH}_k)\mathbf{\Phi}^T + S_{ek}\mathbf{I}_n \quad (7.18)$$

This is a positive-definite real-symmetric matrix. At each frequency, let $\{a_{ik}\}_{i=1}^n$ be the eigenvalues (in descending order) and $\{\mathbf{u}_{ik}\}_{i=1}^n$ ($n \times 1$) be the eigenvectors of $\mathbf{\Phi}(\mathbf{ReH}_k)\mathbf{\Phi}^T$ so that

$$\underbrace{\mathbf{\Phi}}_{n \times m} \underbrace{(\mathbf{ReH}_k)}_{m \times m} \underbrace{\mathbf{\Phi}^T}_{m \times n} = \sum_{i=1}^n a_{ik} \underbrace{\mathbf{u}_{ik}}_{n \times 1} \underbrace{\mathbf{u}_{ik}^T}_{1 \times n} \quad (7.19)$$

Let m' be the rank of $\mathbf{\Phi}(\mathbf{ReH}_k)\mathbf{\Phi}^T$. Then $a_{ik} = 0$ for $i > m'$. Substituting (7.19) and $\mathbf{I}_n = \sum_{i=1}^n \mathbf{u}_{ik}\mathbf{u}_{ik}^T$ into (7.18), gives, after rearranging,

$$\mathbf{ReE}_k = \sum_{i=1}^{m'} \underbrace{(a_{ik} + S_{ek})}_{\text{eigenvalue}} \mathbf{u}_{ik}\mathbf{u}_{ik}^T + \sum_{i=m'+1}^n \underbrace{S_{ek}}_{\text{eigenvalue}} \mathbf{u}_{ik}\mathbf{u}_{ik}^T \quad (7.20)$$

This shows that the first m' largest eigenvalues of $\mathbf{ReE}_k$ are equal to $\lambda_{ik} = a_{ik} + S_{ek}$ ($i = 1, \dots, m'$). The remaining ones are all equal to S_{ek} . For high signal-to-noise (s/n) ratio ($a_{ik} \gg S_{ek}$), $\lambda_{ik} \approx a_{ik}$ for $i = 1, \dots, m'$. Thus, in the SV spectrum the number of lines significantly above the remaining ones indicates the rank m' of $\mathbf{\Phi}(\mathbf{ReH}_k)\mathbf{\Phi}^T$. If $\mathbf{ReH}_k$ is non-singular, m' is equal to the rank of $\mathbf{\Phi}$, i.e., dimension of ‘mode shape subspace’. From elementary matrix algebra, $m' \leq \min(m, n)$, where typically $m' \leq m < n$ in OMA. Note that m' can be smaller than m because the partial mode shapes (columns of $\mathbf{\Phi}$) can be linearly dependent. In the typical case when $m < n$ and the partial modes shapes are linearly independent (but need not be orthogonal), $m' = m$ and so the number of lines significantly above the remaining ones in the SV spectrum is equal to the number of modes.

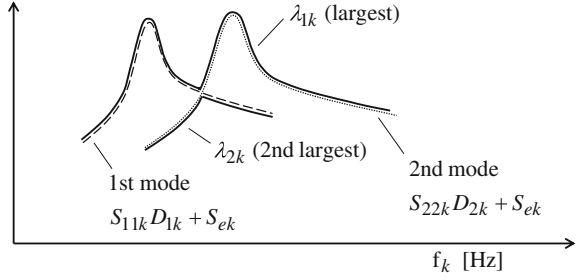
Perfectly Incoherent Modal Forces and Orthonormal Partial Mode Shapes

Consider the special case when the modal forces are perfectly incoherent and the partial mode shapes $\{\boldsymbol{\phi}_i\}_{i=1}^m$ are orthonormal. Then $\mathbf{ReH}_k = \text{diag}[\{S_{iik}D_{ik}\}_{i=1}^m]$ is a real diagonal matrix and

$$\mathbf{ReE}_k = \sum_{i=1}^m S_{iik}D_{ik}\boldsymbol{\phi}_i\boldsymbol{\phi}_i^T + S_{ek}\mathbf{I}_n \quad (7.21)$$

where $D_{ik} = |h_{ik}|^2$ is the dynamic amplification factor for mode i . Following a similar argument as before, it can be reasoned that the first m largest eigenvalues of $\mathbf{ReE}_k$ are $\{S_{iik}D_{ik} + S_{ek}\}_{i=1}^m$ with eigenvectors $\{\boldsymbol{\phi}_i\}_{i=1}^m$. The remaining eigenvalues are all equal to S_{ek} . Since D_{ik} has a peak near the natural frequency f_i , the top m lines in the SV spectrum *together* show dynamic amplification of the modes

Fig. 7.7 Eigenvalues of $\text{Re}\mathbf{E}_k$ and SV spectrum, assuming two modes, perfectly incoherent modal forces and orthonormal partial mode shapes



versus frequency. This is illustrated in Fig. 7.7. Note that $\lambda_{1k} \neq S_{11k}D_{1k} + S_{ek}$ but rather $\lambda_{1k} = \max\{S_{iik}D_{ik} + S_{ek}\}_{i=1}^m$.

In the general case when $\text{Re}\mathbf{H}_k$ is not diagonal or $\{\boldsymbol{\varphi}_i\}_{i=1}^m$ are not orthogonal, the first m largest eigenvalues of $\text{Re}\mathbf{E}_k$ need not be equal to $\{S_{iik}D_{ik} + S_{ek}\}_{i=1}^m$. The eigenvectors need not be $\{\boldsymbol{\varphi}_i\}_{i=1}^m$. Nevertheless, Fig. 7.7 still provides the basic picture for interpreting the SV spectrum.

Example 7.2 (Multi-mode, synthetic data) Consider synthetic data with two modes generated according to

$$\hat{\mathbf{y}}_j = \boldsymbol{\Phi}_1 \ddot{\eta}_1(t_j) + \boldsymbol{\Phi}_2 \ddot{\eta}_2(t_j) + \boldsymbol{\varepsilon}(t_j) \quad t_j = j\Delta t, j = 0, 1, 2, \dots \quad (7.22)$$

where $\Delta t = 0.01$ s; $\eta_i(t)$ ($i = 1, 2$) satisfies

$$\ddot{\eta}_i(t) + 2\zeta_i\omega_i\dot{\eta}_i(t) + \omega_i^2\eta_i(t) = p_i(t) \quad (7.23)$$

with $\omega_i = 2\pi f_i$ (rad/s); $f_1 = 1$ Hz, $f_2 = 1.05$ Hz, $\zeta_1 = \zeta_2 = 1\%$;

$$\boldsymbol{\Phi}_1 = [1 \quad 2 \quad 2]^T/3 \quad \boldsymbol{\Phi}_2 = [-2 \quad 1 \quad 2]^T/3 \quad (7.24)$$

$p_1(t)$ and $p_2(t)$ are modal forces, stationary Gaussian with constant PSD $S_{11} = S_{22} = 1$ (μg)²/Hz and coherence $S_{12}/\sqrt{S_{11}S_{22}} = 0.5e^{i\pi/4}$. The channel noise $\boldsymbol{\varepsilon}(t)$ has i.i.d. white components with PSD $S_e = 1$ (μg)²/Hz. Here, the modal forces have non-zero coherence and the partial mode shapes are not orthogonal.

Based on 2000 s of data, Fig. 7.8a shows the PSD spectrum with a frequency interval of 0.01 Hz (averaged over 20 segments with 100 s each). Figure 7.8b shows the SV spectrum. Theoretically the lowest line (smallest eigenvalue) is equal to $S_e = 1$ (μg)²/Hz. The top two lines (largest two eigenvalues) are contributed by the two modes. Slightly below 1 Hz, the top line is dominated by the first mode and the middle line by the second. Their roles are swapped after 1.05 Hz. Dynamic amplification of the first mode can be visualized by following the top line up to about 1 Hz, and then continuing with the middle line thereafter. Similarly, for the second mode it is the middle line up to 1 Hz and then the top line.

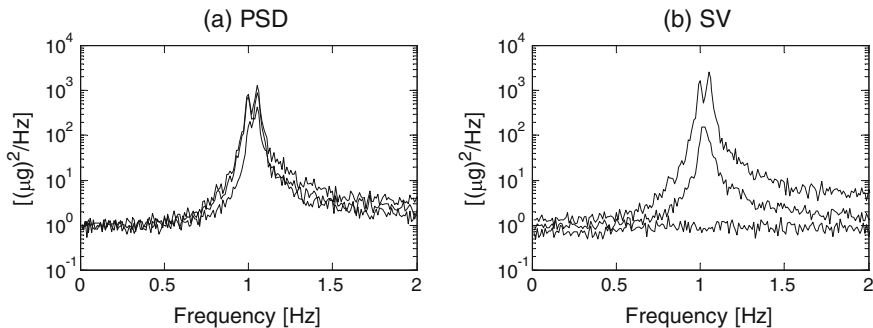


Fig. 7.8 **a** PSD spectrum and **b** SV spectrum, synthetic data with three measured DOFs and two modes at 1 and 1.05 Hz

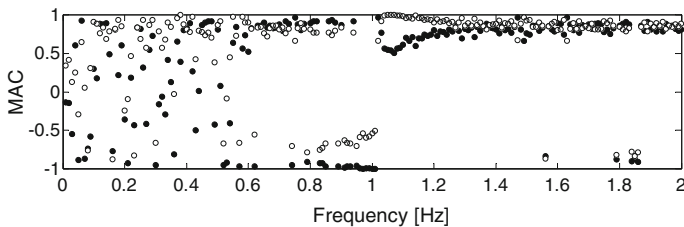


Fig. 7.9 MAC of the eigenvector (largest eigenvalue) of $\text{Re}\hat{\mathbf{E}}_k$ with the partial mode shape of mode 1 (*black circle*) and mode 2 (*white circle*)

Figure 7.9 shows the MAC of the eigenvector (largest eigenvalue) of $\text{Re}\hat{\mathbf{E}}_k$ with $\boldsymbol{\varphi}_1$ (black circle) and $\boldsymbol{\varphi}_2$ (white circle). The values have large scatter but their absolute values are close to 1 near the natural frequencies. The MAC with the first mode is close to -1 for frequencies slightly below 1 Hz. The MAC with the second mode is close to 1 for frequencies slightly above 1 Hz. In this example, the eigenvector of $\text{Re}\hat{\mathbf{E}}_k$ near the natural frequency can still give a good approximation to the partial mode shape. This should not be taken for granted in the general case, however. ■

7.4 Illustration with Field Data

In this section we illustrate the analysis of a typical set of ambient data. The data was collected from a triaxial servo-accelerometer placed near one corner on a tall building roof. It was recorded for 30 min at sampling rate 2048 Hz by a 24 bit digitizer. For analysis it is decimated to 64 Hz. This is considered adequate for the building where the modes of interest are at most a few Hz. The original

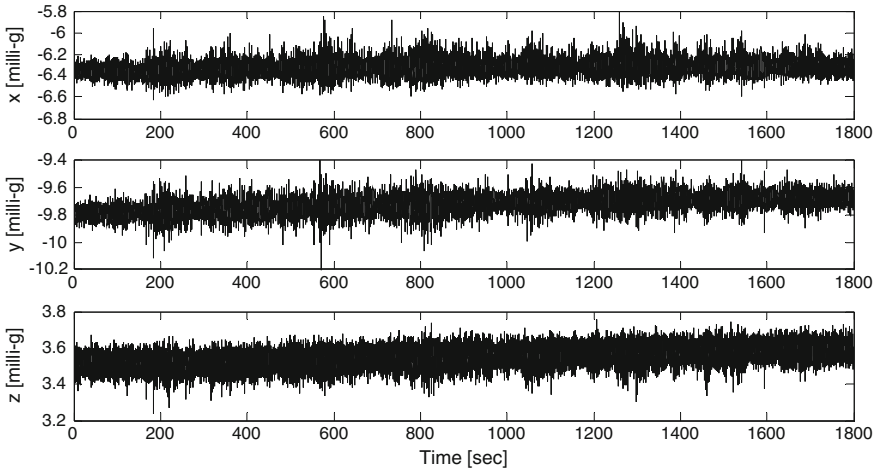


Fig. 7.10 Ambient acceleration time history at one corner on a tall building roof

analogue data is measured in volts and it is converted to be in g ($=9.81 \text{ m/s}^2$) when divided by the sensitivity of the sensor (100 V/g). All spectra shown in this section are one-sided.

7.4.1 Time Histories

Figure 7.10 shows the time histories of the raw data. The x and y direction are roughly parallel to the two faces of the building. The z direction is vertical. The data fluctuates around a non-zero value, about -6.4 , -9.8 and 3.5 milli- g for the x -, y - and z -direction, respectively. This merely results from the non-zero baseline voltage in the DAQ system. There is a slight increasing trend, or very low frequency variations, due to channel noise. This does not matter unless one is interested in very low frequency activities. The time histories could have been plotted with the very low frequency trends filtered for better viewing. It has not been done to highlight this practical aspect.

7.4.2 Sample PSD (No Averaging)

Figure 7.11 shows the root PSD spectrum (no averaging) calculated using the x -direction data. The sampling rate is 64 Hz and so the Nyquist frequency is $64/2 = 32 \text{ Hz}$. The data duration is 1800 s , giving a frequency interval of $1/1800 = 5.56 \times 10^{-4} \text{ Hz}$. The spectrum drops significantly after 27 Hz . This

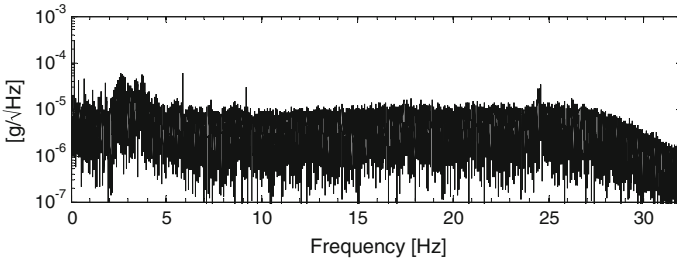


Fig. 7.11 Root PSD spectrum of x-acceleration, no averaging (frequency interval $1/1800 = 5.56 \times 10^{-4}$ Hz), shown up to the Nyquist frequency (32 Hz)

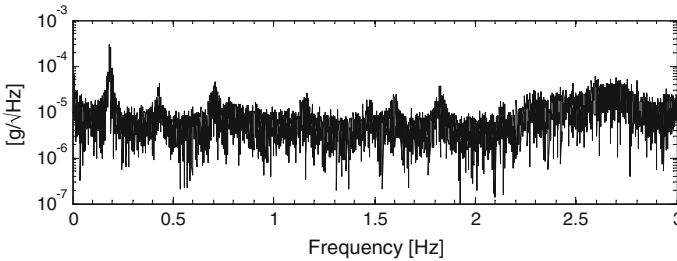


Fig. 7.12 Root PSD spectrum of x-acceleration, no averaging (frequency interval $1/1800 = 5.56 \times 10^{-4}$ Hz), shown up to 3 Hz

merely results from decimation, where the raw digital data at 2048 Hz was low-pass filtered near 32 Hz before it was down-sampled to avoid aliasing. This part of the spectrum is distorted and should not be used for analysis. The spectrum in Fig. 7.11 exhibits no continuity w.r.t. frequency because the values at different frequencies are almost uncorrelated and their standard deviation is as large as their mean (Sect. 4.10).

Viewing the spectrum over the whole sampling bandwidth, i.e., from zero to the Nyquist frequency, is not convenient as the usual interest is on vibration modes in some frequency bands. Suppose one is interested in the modes below 3 Hz. Then one would zoom into the band from 0 to 3 Hz. This is shown in Fig. 7.12. The root PSD increases as the frequency approaches zero. This is due to very low frequency noise of the data channel. The peaks at other frequencies are indicative of potential vibration modes.

7.4.3 Sample PSD (Averaged)

The resonance peaks in Fig. 7.12 are obscured by the random fluctuation of values. For better visualization, the averaged sample PSD is calculated over 18 segments

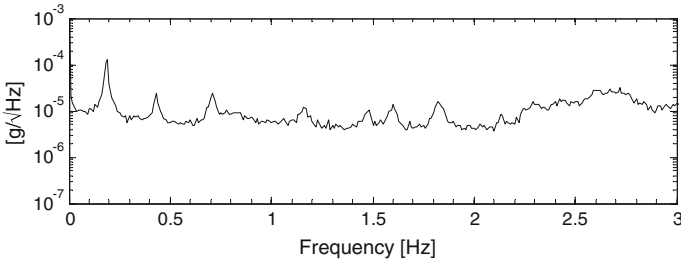


Fig. 7.13 Root PSD spectrum of x-acceleration, averaged over 18 time windows of 100 s each (frequency interval $1/100 = 0.01$ Hz)

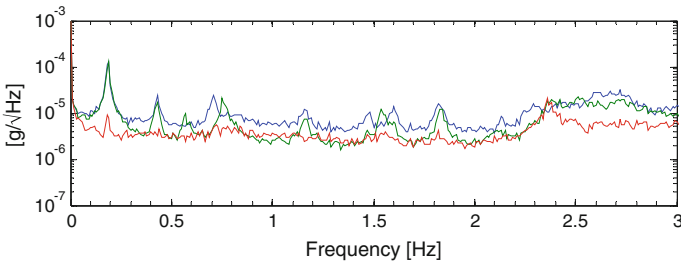


Fig. 7.14 Root PSD spectrum of acceleration data; x (blue), y (green), z (red); frequency interval 0.01 Hz

(100 s each) of the original 1800 s data. Taking square root gives the spectrum in Fig. 7.13. As a result of averaging, the c.o.v. of the sample PSD at each frequency is reduced from 100% to $100\%/\sqrt{18} = 24\%$. The c.o.v. of the root PSD is approximately half of this value, i.e., 12%. The frequency resolution of the averaged spectrum is lower now, with an interval of $1/100 = 0.01$ Hz. Resolution is sacrificed for smoothness/accuracy.

Using a similar procedure, the averaged root sample PSD of y- and z-direction data are produced and shown in Fig. 7.14. Potential modes are near 0.2, 0.4, 0.6, 0.7–0.8, 1.2, 1.4–1.7, 1.8–1.9 Hz. The part above 2.2 Hz is difficult to interpret. The resonance peaks indicate potential modes but they do not tell the number of modes. To see the latter, one needs to examine the SV spectrum.

7.4.4 Singular Value Spectrum

Figure 7.15 shows the root SV spectrum with frequency interval 0.01 Hz. In a resonance band the number of lines significantly above the remaining ones indicates the number of modes. The figure suggests two modes near 0.2 Hz but only one mode near 0.4 Hz, even though for both peaks there were two significant lines in Fig. 7.14. In this

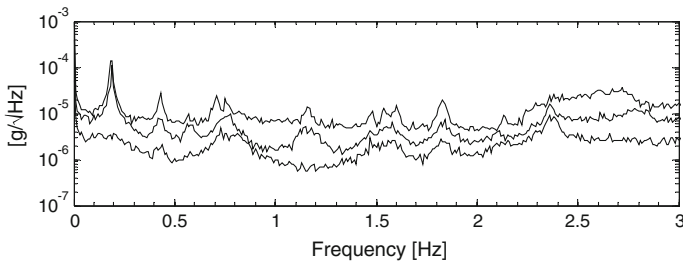


Fig. 7.15 Root SV spectrum (frequency interval 0.01 Hz)

example, it is possible to tell the nature of modes since the x- and y-directions of the sensor align with the two perpendicular translational direction of the building. The x and y spectra in Fig. 7.14 peak at almost the same frequency, suggesting that around 0.2 Hz there is one translational mode along the x-direction and another along the y-direction. The same observation at the peak near 0.4 Hz in Fig. 7.14 combined with a single mode nature suggests that it is a rotational mode. In the general case, the nature of mode may not be trivial and one will need to study the mode shape identified from data.

Other comments are as follow. There is likely to be one mode near 0.6, 1.2 and 1.8 Hz. There are likely to be two modes in the band 0.65–0.8 Hz and three modes near the band 1.4–1.7 Hz. There might be a mode near 2.15 Hz. Modes (if any) between 2.2 and 3 Hz can be difficult to identify. Note that modes appearing close in one spectrum may be distinguished as well-separated in another spectrum with a higher frequency resolution. It is prudent to examine spectra with different resolutions when in doubt.

This example reveals some practical aspects of ambient data analysis. Ascertaining the existence of modes and their number is the most important but challenging task in modal identification, requiring some human intelligence. Once the data is ascertained to fit into theory, the remaining computations are relatively well-defined.

7.5 Asynchronous Data

In Sect. 7.3 we saw that the number of potential modes in a frequency band is indicated by the number of lines displaying dynamic amplification and significantly above the remaining ones in the SV spectrum. Implicit in this result is that the data at different channels are recorded in a ‘synchronous’ manner (Sect. 6.7). Otherwise, as we will see in this section, the SV spectrum will exhibit artificial modes with the same dynamic amplification. For every mode the number of lines significantly above the remaining ones is equal to the number of measurement groups not synchronized with each other. The partial mode shapes within each group can still

be estimated from the eigenvectors but it is generally non-trivial to determine the scaling among different groups to form the overall mode shape.

Asynchronous data is generally difficult to model as it is intrinsically a non-stationary process. Here, it is modeled empirically by a stationary process with imperfect coherence in the modal response. This is found to give reasonable results while being conducive to analysis (Zhu and Au 2017).

7.5.1 Two Measurement Groups

For instructional purpose, first consider the case when there are two groups of data channels. Channels of the same group are synchronized. Channels of different groups are not. Suppose a particular frequency band is dominated by a single mode. Let $\mathbf{v}_1(n_1 \times 1)$ and $\mathbf{v}_2(n_2 \times 1)$ be the ‘local mode shapes’ of the DOFs in Groups 1 and 2, respectively. The ‘global mode shape’ comprising all measured DOFs is

$$\boldsymbol{\Phi} = \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \end{bmatrix}_{n \times 1} \quad \|\boldsymbol{\Phi}\|^2 = \|\mathbf{v}_1\|^2 + \|\mathbf{v}_2\|^2 = 1 \quad (7.25)$$

where $n = n_1 + n_2$.

The modal contribution to the response of the two groups of DOFs are $\mathbf{v}_1 \xi_1(t)$ and $\mathbf{v}_2 \xi_2(t)$, where $\xi_1(t)$ and $\xi_2(t)$ denote the modal response (displacement, velocity or acceleration) according to the individual time scale (clock) of the two groups. If the groups are synchronized, $\xi_1(t) \equiv \xi_2(t)$. Otherwise they are generally different. In the resonance band, the scaled FFT of measured data $\{\hat{\mathbf{y}}_j\}_{j=0}^{N-1}$ at frequency $f_k = k/N\Delta t$ (Hz) is modeled by

$$\hat{\mathcal{F}}_k = \begin{bmatrix} \mathbf{v}_1 \xi_{1k} \\ \mathbf{v}_2 \xi_{2k} \end{bmatrix} + \boldsymbol{\epsilon}_k \quad (7.26)$$

where ξ_{1k} and ξ_{2k} are the scaled FFT of $\{\xi_1(t_j)\}_{j=0}^{N-1}$ and $\{\xi_2(t_j)\}_{j=0}^{N-1}$, respectively; $\boldsymbol{\epsilon}_k$ ($n \times 1$ complex) is the scaled FFT of noise, with i.i.d. components and PSD S_{ek} .

Imperfect synchronization between the two groups is modeled by

$$E[\xi_{1k} \xi_{1k}^*] = E[\xi_{2k} \xi_{2k}^*] = S_k D_k \quad E[\xi_{1k} \xi_{2k}^*] = \chi_k S_k D_k \quad (7.27)$$

where S_k is the modal force PSD and D_k is the dynamic amplification factor as in (7.5); χ_k (complex, $|\chi_k| \leq 1$) is the coherence between ξ_{1k} and ξ_{2k} . That is, $\xi_1(t)$ and $\xi_2(t)$ have the same statistical properties but they are not perfectly coherent. This gives

$$\mathbf{ReE}_k = \mathbf{Re}E[\hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^*] = S_k D_k \begin{bmatrix} \mathbf{v}_1 \mathbf{v}_1^T & \chi'_k \mathbf{v}_1 \mathbf{v}_2^T \\ \chi'_k \mathbf{v}_2 \mathbf{v}_1^T & \mathbf{v}_2 \mathbf{v}_2^T \end{bmatrix} + S_{ek} \mathbf{I}_n \quad (7.28)$$

where

$$\chi'_k = \mathbf{Re} \chi_k \quad (7.29)$$

Eigenvalue Properties of $\mathbf{ReE}_k$

To see the eigenvalues of $\mathbf{ReE}_k$, write it as

$$\begin{aligned} \mathbf{ReE}_k &= S_k D_k \underbrace{\begin{bmatrix} \bar{\mathbf{v}}_1 \\ \bar{\mathbf{v}}_2 \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} c_1^2 & \chi'_k c_1 c_2 \\ \chi'_k c_2 c_1 & c_2^2 \end{bmatrix}}_{\mathbf{C}'_k} \underbrace{\begin{bmatrix} \bar{\mathbf{v}}_1 \\ \bar{\mathbf{v}}_2 \end{bmatrix}^T}_{\mathbf{U}^T} + S_{ek} \mathbf{I}_n \\ &= S_k D_k \underbrace{\mathbf{U}}_{n \times 2} \underbrace{\mathbf{C}'_k}_{2 \times 2} \underbrace{\mathbf{U}^T}_{2 \times n} + S_{ek} \mathbf{I}_n \end{aligned} \quad (7.30)$$

where

$$c_i = \|\mathbf{v}_i\| \quad \bar{\mathbf{v}}_i = \|\mathbf{v}_i\|^{-1} \mathbf{v}_i \quad \mathbf{U} = \begin{bmatrix} \bar{\mathbf{v}}_1 \\ \bar{\mathbf{v}}_2 \end{bmatrix} \quad (7.31)$$

$$\mathbf{C}'_k = \begin{bmatrix} c_1^2 & \chi'_k c_1 c_2 \\ \chi'_k c_2 c_1 & c_2^2 \end{bmatrix} = \begin{bmatrix} c_1 & \\ & c_2 \end{bmatrix} \begin{bmatrix} 1 & \chi'_k \\ \chi'_k & 1 \end{bmatrix} \begin{bmatrix} c_1 & \\ & c_2 \end{bmatrix} \quad (7.32)$$

Note that $\mathbf{C}'_k$ is real-symmetric and positive semi-definite. Let $\{\mu_{1k}, \mu_{2k} \geq 0\}$ and $\{\boldsymbol{\alpha}_{ik}\}_{i=1}^2 (2 \times 1)$ be respectively its eigenvalues (in descending order of magnitude) and eigenvectors, so that $\mathbf{C}'_k = \sum_{i=1}^2 \mu_{ik} \boldsymbol{\alpha}_{ik} \boldsymbol{\alpha}_{ik}^T$. Substituting into (7.30) and rearranging gives

$$\mathbf{ReE}_k = \sum_{i=1}^2 \underbrace{(\mu_{ik} S_k D_k + S_{ek})}_{\text{eigenvalue}} \mathbf{u}_{ik} \mathbf{u}_{ik}^T + \sum_{i=3}^n \underbrace{S_{ek}}_{\text{eigenvalue}} \mathbf{u}_{ik} \mathbf{u}_{ik}^T \quad (7.33)$$

where $\{\mathbf{u}_{ik}\}_{i=1}^n (n \times 1)$ is an orthonormal basis with $\mathbf{u}_{ik} = \mathbf{U} \boldsymbol{\alpha}_{ik}$ ($i = 1, 2$); and we have used $\mathbf{I}_n = \sum_{i=1}^n \mathbf{u}_{ik} \mathbf{u}_{ik}^T$. This shows that the largest two eigenvalues of $\mathbf{ReE}_k$ are (in descending order of magnitude)

$$\lambda_{ik} = \mu_{ik} S_k D_k + S_{ek} \quad i = 1, 2 \quad (7.34)$$

with eigenvectors $\{\mathbf{u}_{ik}\}_{i=1}^2$. The remaining eigenvalues are all equal to S_{ek} .

Perfectly Synchronized Groups

If $\chi'_k = 1$ then

$$\mathbf{C}'_k = \begin{bmatrix} c_1^2 & c_1 c_2 \\ c_2 c_1 & c_2^2 \end{bmatrix} = \underbrace{1}_{\text{eigenvalue}} \times \underbrace{\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}}_{\text{eigenvector}} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}^T \quad (7.35)$$

which has only one non-zero eigenvalue equal to 1 with eigenvector $[c_1; c_2]$. In this case, $\text{Re}\mathbf{E}_k$ has only one non-zero eigenvalue, being $\lambda_{1k} = S_k D_k + S_{ek}$; the remaining ones are all equal to S_{ek} . The eigenvector is $[c_1 \bar{v}_1; c_2 \bar{v}_2] = [v_1; v_2] = \boldsymbol{\phi}$, i.e., the global mode shape. This result agrees with that for synchronized data.

Perfectly Incoherent Groups

At the other extreme, if $\chi'_k = 0$ then

$$\mathbf{C}'_k = \begin{bmatrix} c_1^2 & 0 \\ 0 & c_2^2 \end{bmatrix} = \underbrace{c_1^2}_{\text{eigenvalue}} \times \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{\text{eigenvector}} \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T + \underbrace{c_2^2}_{\text{eigenvalue}} \times \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\text{eigenvector}} \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T \quad (7.36)$$

which has eigenvalues $\{c_i^2\}_{i=1}^2$ and eigenvectors $[1; 0]$ and $[0; 1]$. Without loss of generality, assume that $c_1 \geq c_2$. Then the largest eigenvalue of $\text{Re}\mathbf{E}_k$ is $\lambda_{1k} = c_1^2 S_k D_k + S_{ek}$ with eigenvector $[\bar{v}_1; \mathbf{0}]$. The second largest eigenvalue is $\lambda_{2k} = c_2^2 S_k D_k + S_{ek}$ with eigenvector $[\mathbf{0}; \bar{v}_2]$. For high s/n ratio ($S_k D_k / S_{ek} \gg 1$), $\lambda_{1k} / \lambda_{2k} \approx c_1^2 / c_2^2$, i.e., the ratio of eigenvalues is proportional to the scaling constants of local mode shapes.

General Case

In the general case of $0 \leq |\chi'_k| \leq 1$, it can be shown that the eigenvalues of $\mathbf{C}'_k$ are given by

$$\begin{aligned} \mu_{1k} &= \frac{1}{2}(1 + \sqrt{\Delta_k}) \\ \mu_{2k} &= \frac{1}{2}(1 - \sqrt{\Delta_k}) \end{aligned} \quad \Delta_k = 1 - 4c_1^2 c_2^2 (1 - \chi_k'^2) \quad (7.37)$$

For $\text{Re}\mathbf{E}_k$, the largest two eigenvalues λ_{1k} and λ_{2k} center at $S_k D_k / 2 + S_{ek}$ with a separation of $S_k D_k \sqrt{\Delta_k}$. Their eigenvectors do not give the global mode shape; and they depend on χ'_k , c_1 and c_2 . It can be shown that Δ_k is an increasing function of χ'_k ranging between 0 and 1. The separation between λ_{1k} and λ_{2k} therefore increases with χ'_k . Taking $\chi'_k = 0$ or $\chi'_k = 1$ gives results consistent with the previous findings for perfectly incoherent or perfectly synchronized groups, respectively.

The above results are summarized in Fig. 7.16.

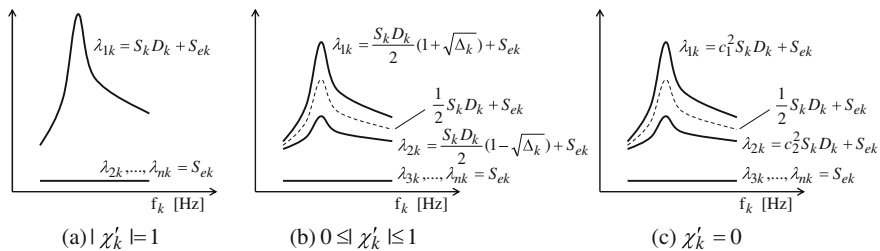


Fig. 7.16 SV spectrum in the resonance band of a single mode for two measurement groups; **a** Synchronized; **b** General case; **c** Perfectly incoherent; $\Delta_k = 1 - 4c_1^2 c_2^2 (1 - \chi_k'^2)$. In **(c)**, it is assumed that $c_1 \geq c_2$

In summary, when two measurement groups are not synchronized, the SV spectrum will show two lines significantly above the remaining ones in the resonance band of a single mode. The largest two eigenvalues of $\text{Re}\mathbf{E}_k$ depend on the coherence between the groups and the norms of local mode shapes. For these eigenvalues, the partition in the eigenvector for the DOFs of each group still gives the local mode shape but the whole eigenvector does not give the global mode shape. When s/n ratio is high and the two groups have zero coherence, the eigenvalues bear approximately the same ratio as the squares of local mode shape norms.

Why Real Part of PSD Matrix?

It is instructive to consider the eigenvalues of $\mathbf{E}_k$, which explains why in Sect. 7.3 it is the eigenvalues of $\text{Re}\mathbf{E}_k$ rather than $\mathbf{E}_k$ that are plotted in the SV spectrum. Following a similar procedure as before, it can be shown that the largest two eigenvalues of $\mathbf{E}_k$ are still given by $S_k D_k (1 \pm \sqrt{\Delta_k})/2 + S_{ek}$ but now $\Delta_k = 1 - 4c_1^2 c_2^2 (1 - |\chi_k|^2)$, i.e., with $\chi_k'^2$ replaced by $|\chi_k|^2$. Two asynchronous groups can have small $\chi_k'^2 = (\text{Re}\chi_k)^2$ but not necessarily small $|\chi_k|^2$. For example, two data channels measuring the same quantity with a constant time shift τ , say, $x(t + \tau)$ and $x(t)$, have a coherence of $\chi_k = \exp(2\pi i f_k \tau)$. This gives $|\chi_k| = 1$ regardless of τ , even though $\chi_k'^2 = \cos^2(2\pi f_k \tau)$ is a decreasing function of f_k up to $1/4\tau$ (Hz). Since $|\chi_k| = 1$, $|\Delta_k| = 1$ and so $\lambda_{1k} = S D_k + S_{ek}$ and $\lambda_{2k} = S_{ek}$, the same values for synchronized data. A plot of the eigenvalues of $\mathbf{E}_k$ then gives no sign of imperfect synchronization.

Example 7.3 (Asynchronous data) Consider three channels of data from accelerometers measuring the same horizontal acceleration of a SDOF structure with a natural frequency of about 9.85 Hz under ambient condition in the laboratory, as schematically shown in Fig. 7.17. Channels 1 and 2 were recorded on the same DAQ unit but Channel 3 was on another unit. Although the digital data of all

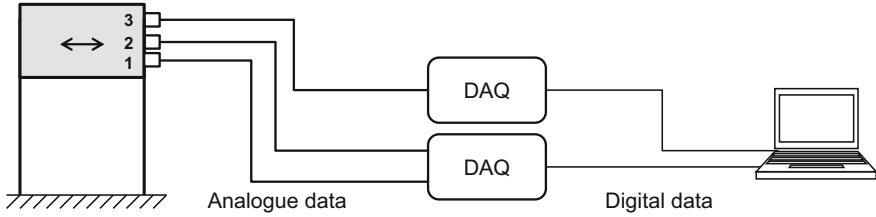


Fig. 7.17 Experimental setup in Example 7.3

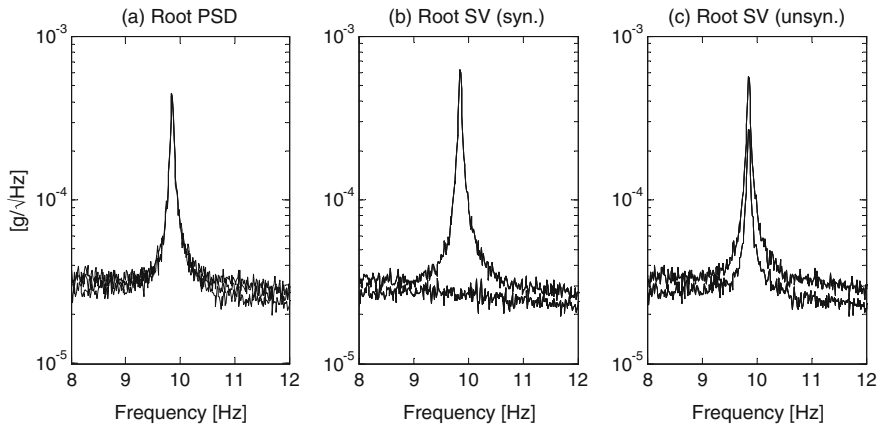


Fig. 7.18 Spectra of synchronous and asynchronous data. **a** Root PSD of Channels 1–3; **b** Root SV spectrum of Channels 1 and 2 (synchronous); **c** Root SV of Channels 1 and 3 (asynchronous)

channels were transmitted by USB cable to the same laptop and operated by the same software to start recording at the same time, only the data from Channels 1 and 2 were synchronized. The data was acquired for one hour at 2048 Hz and later decimated to 256 Hz for analysis.

Figure 7.18a shows the (averaged) PSD spectrum of all channels. The three lines almost overlap because the three channels measure the same motion. Figure 7.18b shows the SV spectrum produced from the 2×2 PSD matrix of Channels 1 and 2 only. There is one line significantly above the other, correctly suggesting a single mode. Figure 7.18c shows the SV spectrum produced from the 2×2 PSD matrix of Channels 1 and 3. There are two lines exhibiting dynamic amplification. If one did not know that the channels were unsynchronized, these two lines would have suggested erroneously two modes with almost the same frequency and damping ratio. For reference, the real part of the coherence between Channels 1 and 3 near the natural frequency is estimated to be about 0.6. Substituting this value into the theoretical eigenvalues in (7.34) produces two lines (not shown) almost overlapping with those in Fig. 7.18c. ■

7.5.2 Multiple Measurement Groups

The results for two measurement groups can be generalized to multiple, say, q groups, of data channels which are only synchronized within the group. The mathematical arguments are similar. Analogous to (7.26),

$$\hat{\mathcal{F}}_k = \begin{bmatrix} v_1 \xi_{1k} \\ \vdots \\ v_q \xi_{qk} \end{bmatrix} + \mathbf{\varepsilon}_k = \underbrace{\begin{bmatrix} \bar{v}_1 & & \\ & \ddots & \\ & & \bar{v}_q \end{bmatrix}}_{\mathbf{U}(n \times q)} \underbrace{\begin{bmatrix} c_1 & & \\ & \ddots & \\ & & c_q \end{bmatrix}}_{\text{diag}\{c_i\}_{i=1}^q (q \times q)} \underbrace{\begin{bmatrix} \xi_{1k} \\ \vdots \\ \xi_{qk} \end{bmatrix}}_{\xi_k (q \times 1)} + \mathbf{\varepsilon}_k \quad (7.38)$$

where $\bar{v}_i = v_i / \|v_i\|$ and $c_i = \|v_i\|$ ($i = 1, \dots, q$). The global mode shape is

$$\boldsymbol{\varphi} = \begin{bmatrix} v_1 \\ \vdots \\ v_q \end{bmatrix} = \begin{bmatrix} c_1 \bar{v}_1 \\ \vdots \\ c_q \bar{v}_q \end{bmatrix} \quad \|\boldsymbol{\varphi}\|^2 = \|v_1\|^2 + \dots + \|v_q\|^2 = 1 \quad (7.39)$$

Imperfect synchronization is modeled by $E[\xi_k \xi_k^*] = S_k D_k \chi_k$, where χ_k is a $q \times q$ matrix with the (i, j) -entry equal to χ_{ijk} (coherence between ξ_{ik} and ξ_{jk} ; $|\chi_{ijk}| \leq 1$). Then

$$\text{Re} \mathbf{E}_k = \text{Re} E[\hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^*] = S_k D_k \underbrace{\mathbf{U}}_{n \times q} \underbrace{\mathbf{C}'_k}_{q \times q} \mathbf{U}^T + S_{ek} \mathbf{I}_n \quad (7.40)$$

$$\mathbf{C}'_k = \text{diag}\{c_i\}_{i=1}^q \chi'_k \text{diag}\{c_i\}_{i=1}^q \quad (7.41)$$

where $\chi'_k = \text{Re} \chi_k$. It can be shown that $\text{Re} \mathbf{E}_k$ has the eigenvector representation:

$$\text{Re} \mathbf{E}_k = \sum_{i=1}^q \underbrace{(\mu_{ik} S_k D_k + S_{ek})}_{\text{eigenvalue}} \mathbf{u}_{ik} \mathbf{u}_{ik}^T + \sum_{i=q+1}^n \underbrace{S_{ek}}_{\text{eigenvalue}} \mathbf{u}_{ik} \mathbf{u}_{ik}^T \quad (7.42)$$

where $\{\mu_{ik} \geq 0\}_{i=1}^q$ are the eigenvalues (in descending order of magnitude) of $\mathbf{C}'_k$ with eigenvectors $\{\boldsymbol{\alpha}_{ik}\}_{i=1}^q$; and

$$\mathbf{u}_{ik} = \mathbf{U} \boldsymbol{\alpha}_{ik} \quad i = 1, \dots, q \quad (7.43)$$

The largest q eigenvalues (in descending order of magnitude) of $\text{Re} \mathbf{E}_k$ are therefore given by

$$\lambda_{ik} = \mu_{ik} S_k D_k + S_{ek} \quad i = 1, \dots, q \quad (7.44)$$

with eigenvectors $\{\mathbf{u}_{ik}\}_{i=1}^q$. The remaining eigenvalues are all equal to S_{ek} .

Perfectly Synchronized Groups

If $\chi_{ijk} = 1$ for all i and j , then

$$\mathbf{C}'_k = \underbrace{1}_{\text{eigenvalue}} \times \underbrace{\begin{bmatrix} c_1 \\ \vdots \\ c_q \end{bmatrix}}_{\text{eigenvector}} \begin{bmatrix} c_1 \\ \vdots \\ c_q \end{bmatrix}^T \quad (7.45)$$

which has only one non-zero eigenvalue equal to 1 with eigenvector $[c_1; \dots; c_q]$. The largest eigenvalue of ReE_k is then $\lambda_{1k} = S_k D_k + S_{ek}$ with eigenvector equal to the global mode shape $\boldsymbol{\phi} = [c_1 \bar{v}_1; \dots; c_q \bar{v}_q]$. This result agrees with that for synchronous data.

Perfectly incoherent groups

At the other extreme, if χ'_k is the identity matrix then

$$\begin{aligned} \mathbf{C}'_k &= \begin{bmatrix} c_1^2 & & \\ & \ddots & \\ & & c_q^2 \end{bmatrix} \\ &= \underbrace{c_1^2}_{\text{eigenvalue}} \times \underbrace{\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}}_{\text{eigenvector}} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}^T + \dots + \underbrace{c_q^2}_{\text{eigenvalue}} \times \underbrace{\begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}}_{\text{eigenvector}} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}^T \end{aligned} \quad (7.46)$$

which has eigenvalues $\{c_i^2\}_{i=1}^q$ and eigenvectors equal to the standard Euclidean basis vectors. Without loss of generality, assume that $c_1 \geq c_2 \geq \dots \geq c_q$. Then the q largest eigenvalues (in descending order of magnitude) and eigenvectors of ReE_k are given by

$$\lambda_{ik} = c_i^2 S_k D_k + S_{ek} \quad (i = 1, \dots, q) \quad \mathbf{u}_{1k} = \begin{bmatrix} \bar{v}_1 \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}, \mathbf{u}_{2k} = \begin{bmatrix} \mathbf{0} \\ \bar{v}_2 \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{bmatrix}, \dots, \mathbf{u}_{qk} = \begin{bmatrix} \mathbf{0} \\ \vdots \\ \mathbf{0} \\ \bar{v}_q \end{bmatrix} \quad (7.47)$$

For high s/n ratio ($S_k D_k \gg S_{ek}$), this implies $\lambda_{ik}/\lambda_{jk} \approx c_i^2/c_j^2$.

Conservation of Signal Strength

Note that $0 \leq \mu_{ik} \leq 1$ since $\mathbf{C}'_k$ is positive semi-definite and its eigenvalues are less than the maximum of diagonal entries (all ≤ 1). Correspondingly, $\lambda_{ik} \leq S_k D_k + S_{ek}$ (= largest eigenvalue for synchronous data). This implies that the top line in the SV spectrum of asynchronous data always has a smaller value than if the data were synchronous. Nevertheless, the signal strength is ‘conserved’, in the sense that the sum of the top q values in the SV spectrum remains the same, equal to $S_k D_k + q S_{ek}$. This is because the sum of eigenvalues is equal to the sum of diagonal entries, which implies

$$\sum_{i=1}^q \mu_{ik} = \sum_{i=1}^q \|v_i\|^2 = 1 \quad \sum_{i=1}^q \lambda_{ik} = S_k D_k + q S_{ek} \quad (7.48)$$

In summary, when there are q measurement groups whose data channels are only synchronized within the group, the SV spectrum will have q lines significantly above the remaining ones. The sum of the q largest eigenvalues of ReE_k is the same regardless of the coherence among groups. These eigenvalues and their eigenvectors depend in a non-trivial manner on the coherence among groups and the norms of local mode shapes. The partition in the eigenvector for the DOFs of each group still gives the local mode shape but the whole eigenvector does not give the global mode shape. When s/n ratio is high and the groups have zero coherence, the eigenvalues bear approximately the same ratio as the squares of local mode shape norms.

7.6 Microtremor Data

Structures situated on earth are subjected to support excitations due to motion of the ground. Except during earthquakes, such ground motion is typically very small, often known as ‘microtremor’. It is attributed to many sources, e.g., vibration at the bedrock, ocean waves, winds, traffic and human activities. Unlike structural vibrations, microtremors do not exhibit distinct resonance peaks in their PSD spectrum because they are affected by soil amplification and surface waves rather than structural dynamics with small damping. See Aki and Richards (2002) for physics of seismic waves and quantitative seismology.

7.6.1 Background Seismic Noise

The characteristics of microtremor are strongly influenced by local factors (e.g., soil amplification) and disturbances (e.g., culture activities). The USGS New High Noise Model (NHNM) and New Low Noise Model (NLNM) provide a reference

for the background seismic activities predominantly associated with the earth (Peterson 1993). They were produced based on the upper (NHNM) and lower (NLNM) envelopes encompassing a large collection of PSDs calculated from a global network of microtremor measurements at 75 (typically remote) locations, situated at ground levels to a couple of hundred meters below ground. NLNM is often used for benchmarking the noise of seismometers, assessing the quality of seismic stations and determining whether certain small signals can be detected.

In the original form, the PSD in dB (decibel) is given as a piecewise linear function of log-period:

$$\underbrace{10 \log_{10} S}_{\text{PSD in dB}} = a + b \underbrace{\log_{10} T}_{\text{log -period}} \tag{7.49}$$

where S is the (one-sided) PSD in $(\text{m/s}^2)^2/\text{Hz}$ and T is the period in s; a and b are coefficients taking different values on different intervals of periods. The values of a and b at the segment points are given in Table 7.1 (NHNM) and Table 7.2 (NLNM). The PSD in dB and root PSD in $\mu\text{g}/\sqrt{\text{Hz}}$ are also given. To be consistent with (7.49), linear interpolation should only be applied between the values of PSD in dB and log-period.

Figure 7.19a shows the PSD in dB versus the period. Figure 7.19b shows the root PSD in $\text{g}/\sqrt{\text{Hz}}$ versus frequency. Some features of the spectra have been explained by seismologists/geophysicists; see Bormann and Wielandt (2013). For example, the hump in Fig. 7.19a from about 3 to 30 s is associated with ocean microseisms. The right end for periods beyond 300 s is associated with free oscillations (‘normal modes’) of the earth. As mentioned before, these spectra represent an ‘envelope’ of PSD observed from a variety of sites in the database network rather than one at a particular site. That is, it is unlikely to find a site with a PSD taking the shape of the NHNM or NLNM. The microtremor level measured near urban areas can be an order of magnitude higher than the NHNM.

Table 7.1 Parameter values and PSD of NHNM at segment points

T (s)	a	b	PSD in dB	Root PSD in $\mu\text{g}/\sqrt{\text{Hz}}$
0.1	−108.73	−17.23	−91.5	2.71E+00
0.22	−150.34	−80.5	−97.4	1.37E+00
0.32	−122.31	−23.87	−110.5	3.04E−01
0.8	−116.85	32.51	−120.0	1.02E−01
3.8	−108.48	18.08	−98.0	1.28E+00
4.6	−74.66	−32.95	−96.5	1.53E+00
6.3	0.66	−127.18	−101.0	9.08E−01
7.9	−93.37	−22.42	−113.5	2.16E−01
15.4	73.54	−162.98	−120.0	1.02E−01
20	−151.52	10.01	−138.5	1.21E−02
354.8	−206.66	31.63	−126.0	5.11E−02
100,000	−206.66	31.63	−48.5	3.83E+02

Table 7.2 Parameter values and PSD of NLNM at segment points

T (s)	a	b	PSD in dB	Root PSD in $\mu\text{g}/\sqrt{\text{Hz}}$
0.1	-162.36	5.64	-168.0	4.06E-04
0.17	-166.7	0.00	-166.7	4.71E-04
0.4	-170	-8.3	-166.7	4.71E-04
0.8	-166.4	28.9	-169.2	3.53E-04
1.24	-168.6	52.48	-163.7	6.66E-04
2.4	-159.98	29.81	-148.6	3.77E-03
4.3	-141.1	0.00	-141.1	8.98E-03
5	-71.36	-99.77	-141.1	8.99E-03
6	-97.26	-66.49	-149.0	3.62E-03
10	-132.18	-31.57	-163.8	6.62E-04
12	-205.27	36.16	-166.2	4.97E-04
15.6	-37.65	-104.33	-162.1	7.98E-04
21.9	-114.37	-47.1	-177.5	1.36E-04
31.6	-160.58	-16.28	-185.0	5.74E-05
45	-187.5	0.00	-187.5	4.30E-05
70	-216.47	15.7	-187.5	4.30E-05
101	-185	0.00	-185.0	5.73E-05
154	-168.34	-7.61	-185.0	5.74E-05
328	-217.43	11.9	-187.5	4.30E-05
600	-258.28	26.6	-184.4	6.16E-05
10,000	-346.88	48.75	-151.9	2.60E-03
100,000	-346.88	48.75	-103.1	7.11E-01

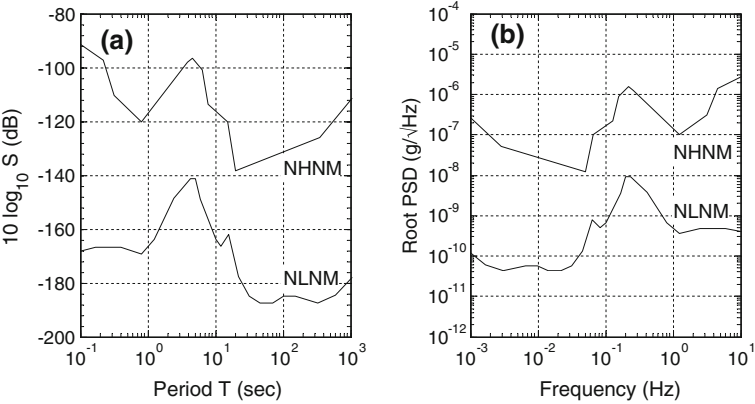


Fig. 7.19 USGS New High Noise Model (NHNM) and New Low Noise Model (NLNM). In **a**, the (one-sided) PSD S is in $(\text{m/s}^2)^2/\text{Hz}$. Taking the square root of S and dividing by 9.81 m/s^2 gives the (one-sided) root PSD in **(b)**

Example 7.4 (Ambient vibration level) How large is the ambient vibration of a structure typically generated by micro-tremors? This question is related to whether it is possible to identify vibration modes using ambient data under little or no disturbance from wind or cultural activities. Here we make a rough assessment for a shear building (see Example 3.4 in Sect. 3.2.2). For simplicity, assume uniform floor mass and linear mode shape, i.e., $\psi = [1, \dots, n_s]^T$. Then it can be reasoned that

$$\text{Root PSD at building top at natural frequency} \approx n_s q \sqrt{D_{\max} S_{\ddot{x}_g}} \quad (7.50)$$

where $D_{\max} = 1/4\zeta^2$ is the dynamic amplification at the natural frequency; $S_{\ddot{x}_g}$ is the PSD of microtremor;

$$q = \frac{\psi^T \mathbf{M} \mathbf{1}}{\psi^T \mathbf{M} \psi} = \frac{\sum_{r=1}^{n_s} r}{\sum_{r=1}^{n_s} r^2} = \frac{n_s(n_s + 1)/2}{n_s(n_s + 1)(2n_s + 1)/6} = \frac{3}{2n_s + 1} \quad (7.51)$$

is the ‘modal participation factor’; $\mathbf{M}$ denotes the mass matrix of the structure and $\mathbf{1}$ denotes a $n_s \times 1$ vector of ones. Substituting these values,

$$\text{Root PSD at building top at natural frequency} \approx \frac{3n_s}{2n_s + 1} \times \frac{\sqrt{S_{\ddot{x}_g}}}{2\zeta} \quad (7.52)$$

To assess the order of magnitude, the first factor $3n_s/(2n_s + 1)$ increases with n_s and ranges between 1 and 1.5, and so we may conservatively take it as 1. Taking $\zeta = 1\%$ and $\sqrt{S_{\ddot{x}_g}} = 1 \mu\text{g}/\sqrt{\text{Hz}}$ (typical in urban areas) then (7.52) gives a root PSD of about $50 \mu\text{g}/\sqrt{\text{Hz}}$. This is above the noise level of servo-accelerometers (typically in the order of $1 \mu\text{g}/\sqrt{\text{Hz}}$), suggesting that microtremor alone should be sufficient to produce ambient data of reasonable s/n ratio for modal identification. ■

7.6.2 Site Amplification and H/V Spectrum

When suitably measured and analyzed, microtremor data can be used as a non-destructive economic means to assess the dynamic amplification characteristics of the soil between the bedrock and the ground surface (Okada 2003; Foti et al. 2015). One empirical way is based on the ‘H/V spectrum’, which is a plot of the PSD ratios of the horizontal to the vertical vibration (Nogoshi and Igarashi 1971; Nakamura 1989, 2000). The method is often applied with either acceleration or velocity measurement obtained triaxially at the site. The premise is that the microtremor at the bedrock has similar frequency characteristics in the horizontal and vertical directions. Horizontal motions are significantly amplified near the resonance frequency of the soil layer, while vertical motion is not affected because

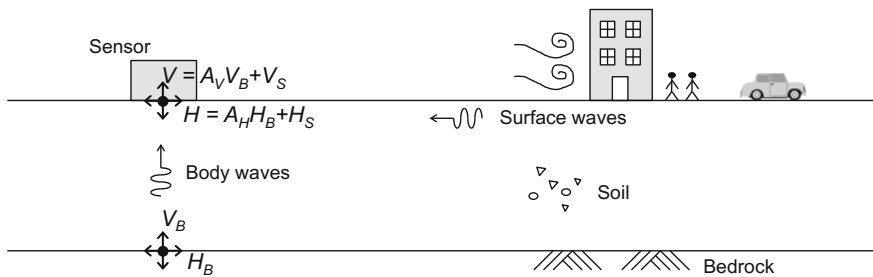


Fig. 7.20 Schematic diagram of microtremor

the soil stiffness in the vertical direction is much higher. The peak location of the H/V spectrum is insensitive to the presence of surface waves as their energy content there is small.

Figure 7.20 shows a schematic diagram where the microtremor measured on the ground surface comprises surface waves due to activities at the ground level and vibrations of the bedrock transmitted and amplified through the soil. In the frequency domain,

$$H = A_H H_B + H_S \quad V = A_V V_B + V_S \quad (7.53)$$

where H , H_B and H_S are respectively the scaled FTs of horizontal vibration (e.g., acceleration) at the ground surface, at the bed rock and due to surface waves; A_H is the site amplification (transfer function) of horizontal motion from the bed rock to the ground surface. Analogous definitions apply to V , V_B , V_S and A_V for the vertical motion. These quantities are frequency dependent, although this has been omitted in notation for simplicity.

In microtremor tests, the target is to determine $|A_H|$ but only H and V can be calculated from measurement. A plot of $|A_H|$ versus frequency is referred as the 'site spectrum'. This is an important quantity in earthquake engineering as it affects the intensity and frequency characteristics of ground motions at the site.

The 'H/V ratio' refers to $|H/V|$. The following argues that $H/V \approx A_H$ near the resonance band of the soil layer. Using (7.53),

$$\frac{H}{V} = \left(\frac{H_B}{V_B}\right) \left(\frac{A_H + H_S/H_B}{A_V + V_S/V_B}\right) \quad (7.54)$$

At the bed rock, the frequency characteristics of the horizontal and vertical motion are similar, which means $H_B/V_B \approx 1$. In typical situations where the soil is much stiffer in the vertical direction than the horizontal, $A_V \approx 1$. These imply

$$\frac{H}{V} \approx \frac{A_H + H_S/H_B}{1 + V_S/V_B} \quad (7.55)$$

If there is no surface wave then $H_S = V_S = 0$ and so $H/V \approx A_H$ as desired. In the general case, theoretical and empirical evidence reveal that the frequency content of surface waves is small near the ‘site frequency’ (where $|A_H|$ is peaked). The lowest frequency at which surface wave energy is concentrated is about twice of the site frequency, where vertical motion is dominant. As a result, the H/V ratio has a similar shape as $|A_H|$ near the site frequency and it has a trough at about twice the site frequency.

The above picture is essentially one-dimensional. It glosses over spatial variability in the subsurface soil properties, surface waves and bed rock excitation in the neighborhood of the measured location. The extent to which the H/V ratio represents site amplification depends on the degree to which the above argument applies. The method requires a proper choice of test location, quality assurance of calculated spectra and interpretation of site spectrum incorporating information from site conditions. In implementation, the H/V ratio is computed via the PSDs of microtremor:

$$\left| \frac{H}{V} \right| = \sqrt{\frac{S_x + S_y}{S_z}} \quad (7.56)$$

where S_x , S_y and S_z are the PSDs of microtremor data along the x, y and z (vertical) directions, respectively. They all depend on frequency but it has been omitted in notation.

Example 7.5 (Microtremor test) Figure 7.21 shows 900 s (sampling rate 200 Hz) of microtremor data obtained on a site in an urban area measured by a triaxial servo-accelerometer. Dividing the data into 90 segments produces an average spectrum with frequency interval 0.1 Hz and c.o.v. 10% in the PSD. The resulting root PSD spectra are shown in Fig. 7.22a. In the context of (7.56), $X = \sqrt{S_x}$, $Y = \sqrt{S_y}$ and $Z = \sqrt{S_z}$. The root PSDs along the x and y directions are very similar. Although the shape of the individual spectra is non-trivial to explain, the H/V spectrum in Fig. 7.22b exhibits dynamic amplification between 2 and 5 Hz. It suggests a site frequency of about 3 Hz. ■

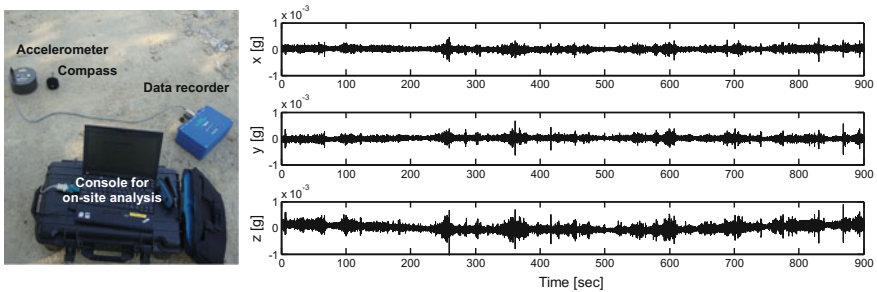


Fig. 7.21 Microtremor test and data

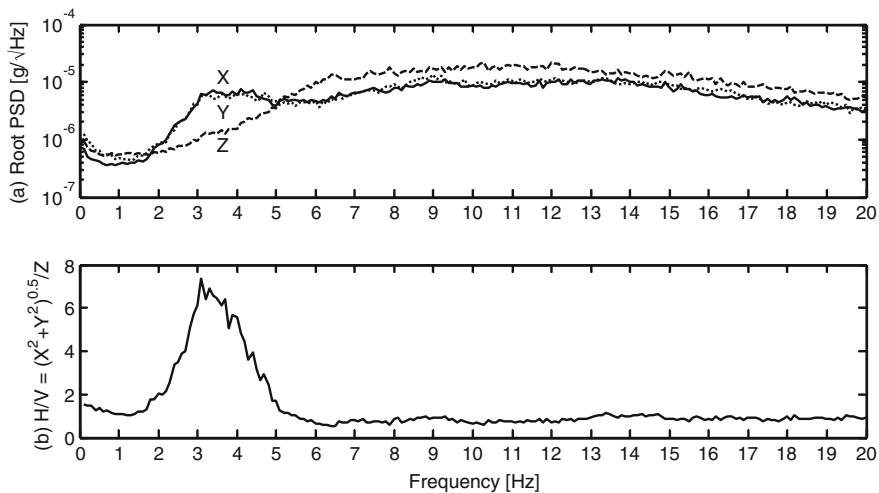


Fig. 7.22 Microtremor spectra; **a** Root PSD; **b** H/V

7.7 Simulation of Ambient Data

To verify procedures or perform parametric studies, it is necessary to generate synthetic data to provide numerical evidence of statistical significance. The time-domain data measured on a structure can be modeled as a sum of structural response and measurement noise. Given a discrete-time history of excitation, the structural response can be computed using a time integration scheme (Sects. 3.7 and 3.8). In this section we discuss how to generate a sample time history of a stationary Gaussian process with a given PSD matrix. This allows one to generate ambient excitation and measurement noise. We also discuss how to specify channel noise level to produce data of specified s/n ratio.

7.7.1 Gaussian Scalar Process

A discrete time history $\{X_j\}_{j=0}^{N-1}$ of white noise with (two-sided) PSD S and time interval Δt (s) can be generated by

$$X_j = \sqrt{\frac{S}{\Delta t}} Z_j \quad j = 0, \dots, N-1 \quad (7.57)$$

where $\{Z_j\}_{j=0}^{N-1}$ are i.i.d. standard Gaussian random variables. As a check, $\text{var}[X_j] = S/\Delta t$. This is consistent with the Parseval equality; the Nyquist frequency is $1/2\Delta t$ (Hz) and so the area under the two-sided PSD is $2S \times (1/2\Delta t) = S/\Delta t$.

In the general case, the following algorithm generates a random time history $\{X_j\}_{j=0}^{N-1}$ at time interval Δt (s) of a stochastic stationary process based on its PSD values $\{S_k \geq 0\}_{k=0}^{N_{yq}}$ at the FFT frequencies $\{f_k = k/N\Delta t\}_{k=0}^{N_{yq}}$, where N_{yq} = integer part of $N/2$, is the index at or just below the Nyquist frequency.

Gaussian Scalar Process with PSD $\{S_k\}_{k=0}^{N_{yq}}$

1. Generate an i.i.d. standard Gaussian sequence $\{Z_k\}_{k=0}^{N-1}$.
2. Generate a complex Gaussian sequence $\{W_k\}_{k=0}^{N-1}$ by

$$W_0 = \left(\frac{S_0 N}{\Delta t}\right)^{1/2} Z_0 \quad (7.58)$$

$$W_k = \left(\frac{S_k N}{2\Delta t}\right)^{1/2} (Z_k + iZ_{N-k}) \quad k = 1, \dots, N_{yq} - 1 \quad (7.59)$$

$$W_{N_{yq}} = \begin{cases} \left(\frac{S_{N_{yq}} N}{2\Delta t}\right)^{1/2} (Z_{N_{yq}} + iZ_{N-N_{yq}}) & N \text{ odd} \\ \left(\frac{S_{N_{yq}} N}{\Delta t}\right)^{1/2} Z_{N_{yq}} & N \text{ even} \end{cases} \quad (7.60)$$

$$W_k = W_{N-k}^* \quad k = N_{yq} + 1, N_{yq} + 2, \dots, N - 1 \quad (7.61)$$

3. Compute $\{X_j\}_{j=0}^{N-1}$ as the inverse FFT of $\{W_k\}_{k=0}^{N-1}$:

$$X_j = \frac{1}{N} \sum_{k=0}^{N-1} W_k e^{2\pi i j k / N} \quad j = 0, \dots, N - 1 \quad (7.62)$$

To remove residual imaginary part in $\{X_j\}_{j=0}^{N-1}$ due to round off errors, its real part should be taken as the sample history.

Proof of Algorithm

To see why the algorithm works, first note that the FFT of $\{X_j\}_{j=0}^{N-1}$ is simply $\{W_k\}_{k=0}^{N-1}$. It is real-valued because $\{W_k\}_{k=0}^{N-1}$ has conjugate mirror pattern about the index N_{yq} . By definition, the PSD of $\{X_j\}_{j=0}^{N-1}$ is

$$\tilde{S}_k = \frac{\Delta t}{N} E[W_k W_k^*] \quad (7.63)$$

This can be shown to be equal to S_k as follow. For $k = 0$,

$$\tilde{S}_0 = \frac{\Delta t}{N} E[W_0 W_0^*] = \frac{\Delta t}{N} \times \frac{S_0 N}{\Delta t} \underbrace{E[Z_0^2]}_1 = S_0 \quad (7.64)$$

For $k = 1, \dots, N_{yq} - 1$, since Z_k and Z_{N-k} are independent,

$$\tilde{S}_k = \frac{\Delta t}{N} E[W_k W_k^*] = \frac{\Delta t}{N} \times \frac{S_k N}{2\Delta t} \{ \underbrace{E[Z_k^2]}_1 + \underbrace{E[Z_{N-k}^2]}_1 \} = S_k \quad (7.65)$$

When N_{yq} is odd,

$$\tilde{S}_{N_{yq}} = \frac{\Delta t}{N} E[W_{N_{yq}} W_{N_{yq}}^*] = \frac{\Delta t}{N} \times \frac{S_{N_{yq}} N}{2\Delta t} \{ \underbrace{E[Z_{N_{yq}}^2]}_1 + \underbrace{E[Z_{N-N_{yq}}^2]}_1 \} = S_{N_{yq}} \quad (7.66)$$

When N_{yq} is even,

$$\tilde{S}_{N_{yq}} = \frac{\Delta t}{N} E[W_{N_{yq}} W_{N_{yq}}^*] = \frac{\Delta t}{N} \times \frac{S_{N_{yq}} N}{\Delta t} \underbrace{E[Z_{N_{yq}}^2]}_1 = S_{N_{yq}} \quad (7.67)$$

For $k = N_{yq} + 1, N_{yq} + 2, \dots, N - 1$, $\tilde{S}_k = S_k$ follows because $W_k = W_{N-k}^*$. ■

Example 7.6 (Conjugate mirror pattern) Here we illustrate the calculation of $\{W_k\}_{k=0}^{N-1}$ according to (7.58)–(7.61), which is slightly different depending on whether N is odd or even. Suppose $N = 7$ (odd). Then $N_{yq} = \text{integer part of } 7/2 = 3$. Equation (7.58) gives

$$W_0 = \left(\frac{S_0 N}{\Delta t}\right)^{1/2} Z_0$$

Equation (7.59) gives, for $k = 1, 2$,

$$W_1 = \left(\frac{S_1 N}{2\Delta t}\right)^{1/2} (Z_1 + \mathbf{i}Z_6) \quad W_2 = \left(\frac{S_2 N}{2\Delta t}\right)^{1/2} (Z_2 + \mathbf{i}Z_5)$$

Equation (7.60) gives (N odd)

$$W_3 = \left(\frac{S_3 N}{2\Delta t}\right)^{1/2} (Z_3 + \mathbf{i}Z_4)$$

Equation (7.61) gives the conjugate mirror image, for $k = 4, 5, 6$,

$$W_4 = W_3^* \quad W_5 = W_2^* \quad W_6 = W_1^*$$

The resulting sequence has the pattern

$$W_0, W_1, W_2, W_3, W_3^*, W_2^*, W_1^*$$

Suppose now $N = 8$ (even). Then $N_{yq} = \text{integer part of } 8/2 = 4$. Equation (7.58) gives

$$W_0 = \left(\frac{S_0 N}{2\Delta t}\right)^{1/2} Z_0$$

Equation (7.59) gives, for $k = 1, 2, 3$,

$$W_1 = \left(\frac{S_1 N}{2\Delta t}\right)^{1/2} (Z_1 + \mathbf{i}Z_7) \quad W_2 = \left(\frac{S_2 N}{2\Delta t}\right)^{1/2} (Z_2 + \mathbf{i}Z_6) \quad W_3 = \left(\frac{S_3 N}{2\Delta t}\right)^{1/2} (Z_3 + \mathbf{i}Z_5)$$

Equation (7.60) gives (N even)

$$W_4 = \left(\frac{S_4 N}{2\Delta t}\right)^{1/2} Z_4$$

Equation (7.61) gives, for $k = 5, 6, 7$,

$$W_5 = W_3^* \quad W_6 = W_2^* \quad W_7 = W_1^*$$

The resulting sequence has the pattern

$$W_0, W_1, W_2, W_3, W_4, W_3^*, W_2^*, W_1^*$$

The mirror image patterns are visualized in Fig. 7.23. ■

7.7.2 Gaussian Vector Process

The algorithm for scalar process can be extended to vector process given the PSD matrices $\{\mathbf{S}_k\}_{k=0}^{N_{yq}}$ ($n \times n$ Hermitian) at the FFT frequencies $\{f_k = k/N\Delta t\}_{k=0}^{N_{yq}}$. For each k , let $\{\lambda_{ik}\}_{i=1}^n$ be the eigenvalues of $\mathbf{S}_k$; and $\mathbf{U}_k$ be its unitary eigenmatrix, i.e., $\mathbf{U}_k^* \mathbf{U}_k = \mathbf{I}_n$. Then $\mathbf{S}_k = \mathbf{U}_k \text{diag}\{\lambda_{ik}\}_{i=1}^n \mathbf{U}_k^*$. Let $\mathbf{L}_k = \mathbf{U}_k \text{diag}\{\sqrt{\lambda_{ik}}\}_{i=1}^n$ so that $\mathbf{S}_k = \mathbf{L}_k \mathbf{L}_k^*$. The algorithm for generating a sample history $\{\mathbf{X}_j\}_{j=0}^{N-1}$ ($n \times 1$) is as follow.

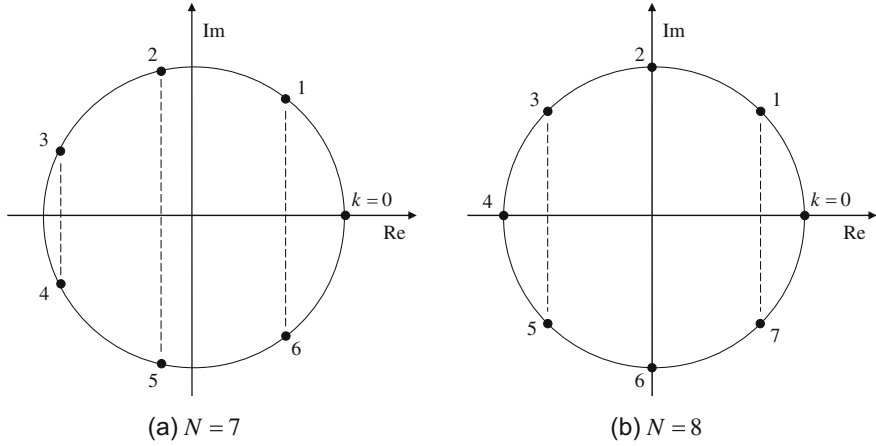


Fig. 7.23 Conjugate mirror pattern. **a** $N = 7$ (odd); **b** $N = 8$ (even)

Gaussian Vector Process with PSD Matrix $\{\mathbf{S}_k\}_{k=0}^{N_{yq}}$

1. Generate an i.i.d. standard Gaussian vector sequence $\{\mathbf{Z}_k\}_{k=0}^{N-1} (n \times 1)$.
2. Generate a complex Gaussian vector sequence $\{\mathbf{W}_k\}_{k=0}^{N-1} (n \times 1 \text{ complex})$ by

$$\mathbf{W}_0 = \left(\frac{N}{\Delta t}\right)^{1/2} \mathbf{L}_0 \mathbf{Z}_0 \quad (7.68)$$

$$\mathbf{W}_k = \left(\frac{N}{2\Delta t}\right)^{1/2} \mathbf{L}_k (\mathbf{Z}_k + i\mathbf{Z}_{N-k}) \quad k = 1, \dots, N_{yq} - 1 \quad (7.69)$$

$$\mathbf{W}_{N_{yq}} = \begin{cases} \left(\frac{N}{2\Delta t}\right)^{1/2} \mathbf{L}_{N_{yq}} (\mathbf{Z}_{N_{yq}} + i\mathbf{Z}_{N-N_{yq}}) & N \text{ odd} \\ \left(\frac{N}{\Delta t}\right)^{1/2} \mathbf{L}_{N_{yq}} \mathbf{Z}_{N_{yq}} & N \text{ even} \end{cases} \quad (7.70)$$

$$\mathbf{W}_k = \mathbf{W}_{N-k}^* \quad k = N_{yq} + 1, N_{yq} + 2, \dots, N - 1 \quad (7.71)$$

3. Compute $\{\mathbf{X}_j\}_{j=0}^{N-1} (n \times 1)$ as the inverse FFT of $\{\mathbf{W}_k\}_{k=0}^{N-1}$:

$$\mathbf{X}_j = \frac{1}{N} \sum_{k=0}^{N-1} \mathbf{W}_k e^{2\pi i j k / N} \quad j = 0, \dots, N - 1 \quad (7.72)$$

To remove residual imaginary part in $\{\mathbf{X}_j\}_{j=0}^{N-1}$ due to round off errors, its real part should be taken as the sample history.

It can be shown by similar steps in Sect. 7.7.1 that the PSD matrix of $\{\mathbf{X}_j\}_{j=0}^{N-1}$ is equal to $\mathbf{S}_k$. For example, for $k = 1, \dots, N_{yq} - 1$, the PSD matrix is given by

$$\begin{aligned}\tilde{\mathbf{S}}_k &= \frac{\Delta t}{N} E[\mathbf{W}_k \mathbf{W}_k^*] = \frac{\Delta t}{N} \times \frac{N}{2\Delta t} \mathbf{L}_k E[(\mathbf{Z}_k + i\mathbf{Z}_{N-k})(\mathbf{Z}_k^T - i\mathbf{Z}_{N-k}^T)] \mathbf{L}_k^* \\ &= \frac{1}{2} \mathbf{L}_k \underbrace{\{E[\mathbf{Z}_k \mathbf{Z}_k^T] + E[\mathbf{Z}_{N-k} \mathbf{Z}_{N-k}^T]\}}_{\mathbf{I}_n} \mathbf{L}_k^* = \mathbf{L}_k \mathbf{L}_k^* = \mathbf{S}_k\end{aligned}\quad (7.73)$$

7.7.3 Quantifying Noise Level

The performance of modal identification algorithms are often tested w.r.t. the level of noise. For OMA, a direct measure is the ratio of the PSDs of modal response to channel noise at the natural frequency. For discussion purpose, consider acceleration data in g. For a single mode with damping ratio ζ and modal force PSD S (g^2/Hz), the PSD of modal acceleration at the natural frequency is $S/4\zeta^2$. Assuming a channel noise PSD of S_e (g^2/Hz), the s/n ratio is given by

$$\gamma = \frac{S}{4S_e\zeta^2} \quad (7.74)$$

In the literature, one conventional measure for the s/n ratio is in terms of RMS (root mean square) value. For modal identification this is not a direct measure because the RMS value depends on the PSD level in the resonance band as well as other bands. It also depends on the sampling bandwidth. For example, suppose the channel noise is white with (two-sided) PSD S_e . For a sampling rate of f_s (Hz), the Nyquist frequency is $f_s/2$ and the variance of noise is $S_e f_s$, giving a RMS value of $\sqrt{S_e f_s}$. On the other hand, the RMS acceleration of the mode is about $\sqrt{\pi S f / 2\zeta}$ (Sect. 5.1.4), where f (Hz) is the natural frequency. Then

$$\frac{\text{RMS of signal}}{\text{RMS of noise}} = \sqrt{\frac{\pi S f / 2\zeta}{S_e f_s}} = \sqrt{\gamma \times \frac{2\pi f \zeta}{f_s}} \quad (7.75)$$

As this ratio depends on the sampling rate f_s , it would be difficult to directly judge the quality of data based on its order of magnitude.

To illustrate, suppose $f = 1$ Hz, $\zeta = 1\%$, $S = 1$ (μg)²/Hz and $S_e = 1$ (μg)²/Hz. Then the s/n ratio in terms of PSD at the natural frequency is $\gamma = 1/4(1)(0.01)^2 = 2500$. This gives a SV-spectrum similar to Fig. 7.5b in Example 7.1. It corresponds to very good quality data and the mode can be well-identified. Quantifying based on RMS ratio, however, may give a different impression depending on the sampling rate f_s . If $f_s = 100$ Hz, then the s/n RMS

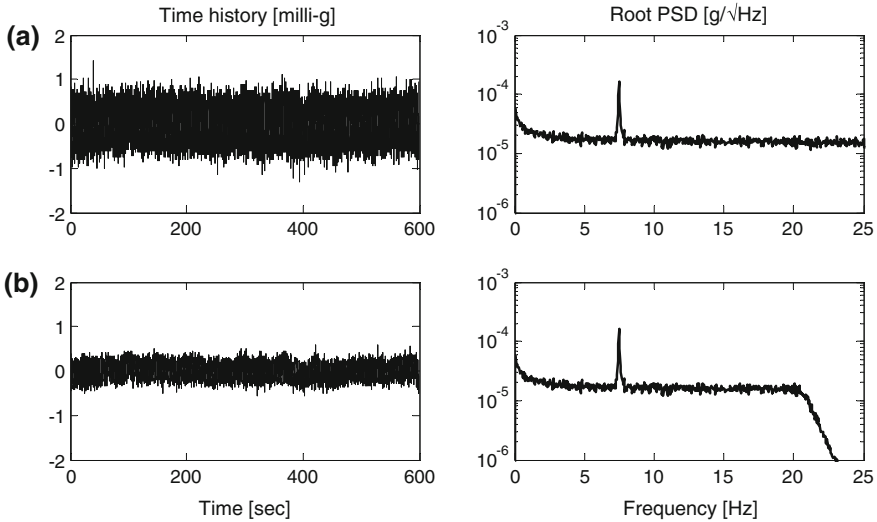


Fig. 7.24 Acceleration data sampled at **a** 250 Hz and **b** 50 Hz. Time history appears noisier in **(a)** than in **(b)** but both give essentially the same identification results for the mode near 7 Hz

ratio is about 1.25, which appears a very high noise level and suggests poor quality data. This is misleading, simply because the RMS of noise is made arbitrary large as a result of the sampling rate. If $f_s = 25$ Hz, the s/n RMS ratio is doubled to 2.5. If $f_s = 400$ Hz, the s/n RMS ratio is reduced by half to 0.63. In all these cases, the SV spectrum of data remains essentially the same as in Fig. 7.5b with a similar value of $\gamma \approx 2500$; and hence the same quality of data for modal identification.

As an example with experimental data, Fig. 7.24 shows the time history and root PSD spectrum of acceleration measured on a SDOF structure by a MEMS accelerometer, sampled at (a) 250 Hz and (b) 50 Hz. The data in (b) is obtained by decimating the data in (a). The time history in (a) appears noisier than that in (b). Nevertheless, both will give very similar identification results for the mode near 7 Hz because their spectra near the natural frequency are the essentially same. In terms of the PSD at the natural frequency, the s/n ratio of (a) and (b) are the same. However, the s/n RMS ratio in (b) is more than double of that in (a).

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Part II

Inference

Chapter 8

Bayesian Inference

Abstract This chapter introduces the Bayesian approach for system identification in a general context. The concept of ‘identifiability’ is introduced, which affects how computations should be performed for making inference. Computational strategies for ‘globally identifiable’, ‘local identifiable’ and ‘unidentifiable’ situations are further discussed in individual sections. ‘Model class selection’ is also introduced, which is an important topic for validating models beyond system identification.

Keywords Bayesian • Identifiability • Gaussian approximation • Model class selection

System identification is concerned with making inference about the parameters of interest based on the measured data and mathematical model of a system. Philosophically, it is impossible to identify the parameters exactly because of incomplete information in the data in terms of amount and precision, and the fact that the identification model can be potentially wrong. A Bayesian approach explicitly acknowledges this fact. Through Bayes’ Theorem, it provides a fundamental means for processing information in the data and making inference on the parameters in a manner consistent with modeling assumptions and probability as a logic.

Not all inference problems demand a Bayesian approach to solve. However, there are many problems where Bayesian approach offers a clear advantage over non-Bayesian ones. Examples are problems where it is non-trivial to construct an estimator for the parameters of interest; problems with significant identification uncertainty, for which it is necessary to quantify; problems where identification results depend critically on the underlying assumptions and it is necessary to quantify the effect or to determine which set of assumptions is better.

Bayesian system identification is a general topic with wide applications in science and engineering. Classical texts on Bayesian probability include Cox (1961), Jeffreys (1961), Box and Tiao (1992) and Jaynes (2003). See also book reviews with highlights of concepts, e.g., Garrett (2004), Fine (2004) and Robert et al.

(2009). A good exposition of Bayesian perspective with structural system identification background can be found in Beck (2010).

In this chapter we introduce the Bayesian approach for system identification. The problem is classified based on ‘identifiability’, which determines the computational tool that should be applied. The chapter does not cover topics such as the choice of prior distribution and Markov Chain Monte Carlo (MCMC) method. These are important general topics but are currently of little relevance to operational modal analysis (OMA). The first topic regarding priors can be found in the classical texts just mentioned. References for MCMC can be found in Sect. 8.7.

8.1 Bayes’ Theorem

Bayes’ Theorem is the fundamental equation that allows one to infer the plausibility of a statement of interest conditional on the stated information. Mathematically, it is simply a statement of conditional probability but the underlying philosophy is far reaching. In a Bayesian perspective, all probabilities are conditional on the information used to obtain them, although for notational simplicity the conditioning statement may not be fully indicated when it is understood. The probability of an event is a function of its conditioning information. There is no ‘true’ or ‘intrinsic’ probability. The probability of flipping a fair coin and getting a head is commonly perceived to have an intrinsic probability of 50%. The usual conditioning (assumption) behind this answer is that the conditions determining the outcome are unknown (all we know is the coin being flipped) and they are not expected to be biased towards head or tail. Theoretically, if one knows with perfect precision the initial conditions of flipping and all environmental conditions throughout the process of flipping and landing, it is possible to determine the outcome (imagine solving the governing equation of motion with known physical properties, initial conditions and excitation time history). Conditional on all of this information (a huge amount), the probability of obtaining a head is either 0 or 1, depending on what the information implies.

According to Bayes’ theorem, for three propositions A , B and C ,

$$P(A|B, C) = P(B|A, C) \frac{P(A|C)}{P(B|C)} \quad (8.1)$$

Here, C contains the information (e.g., a probability model) that specifies the probability of A and B in the absence of any other information. The conditional probability $P(A|B, C)$ asks question about A given the (additional) information specified by B when B is assumed to be true. Bayes’ Theorem swaps the roles of A and B , turning this question into one about B given A . This is of utility when it is non-trivial to answer the question about A using the information about B , but is feasible to do the reverse. The term $P(A|C)$ refers to the probability in the absence of the information specified by B . The term $P(B|C)$ can be interpreted similarly.

Since the probabilities in (8.1) are all conditional on the same information specified by C , such information is usually omitted for notational simplicity. Conventionally, the equation is written as

$$P(A|B) = P(B|A) \frac{P(A)}{P(B)} \quad (8.2)$$

8.2 Updating Knowledge Using Data

Let θ be a vector containing the parameters of a model to be identified from measured data $\mathbf{Z}$. Without much loss of generality, θ and $\mathbf{Z}$ are assumed to be continuous-valued. In modal identification problems where a structural dynamic model is assumed, θ may comprise the natural frequencies, damping ratios, mode shapes, etc. The data $\mathbf{Z}$ may comprise, e.g., the measured acceleration time histories. We want to ask questions about θ given the information in $\mathbf{Z}$. Using Bayes' Theorem,

$$p(\theta|\mathbf{Z}) = p(\mathbf{Z}|\theta) \frac{p(\theta)}{p(\mathbf{Z})} \quad (8.3)$$

The term $p(\theta|\mathbf{Z})$ is called the 'posterior distribution', or posterior PDF (probability density function), of θ given $\mathbf{Z}$. 'Posterior' here means that it is after using the information in $\mathbf{Z}$. The posterior distribution encapsulates all information about θ that can be inferred from $\mathbf{Z}$ in a manner consistent with the specified modeling assumptions, which have been omitted in the conditioning of probabilities in (8.3). Instead of inferring a single value of θ from the data $\mathbf{Z}$, the Bayesian approach yields the conditional distribution of θ , $p(\theta|\mathbf{Z})$, which is a function of the available data (and modeling assumptions). Based on the information in the data, the likely values that θ may take are those with a high posterior PDF value. The identification uncertainty associated with θ is related to the spread of the posterior PDF. In applications, these concepts may be quantified in terms of 'descriptive statistics', e.g., the mean and variance, associated with the posterior PDF.

Normalizing Constant, Prior and Likelihood Function

Note that $p(\theta|\mathbf{Z})$ is a probability distribution for θ . On the RHS of (8.3), the term $p(\mathbf{Z})$ does not depend on θ . Its role is a normalizing constant and it does not affect the shape of the posterior distribution. It plays no role in the identification of θ for a given set of modeling assumptions, but it is the key measure when the latter is scrutinized, as will be discussed in Sect. 8.8. The term $p(\theta)$ is called the 'prior distribution'. It reflects one's knowledge about θ in the absence of data. 'Prior' here means that it is before the information in $\mathbf{Z}$ is used. Standard distributions (e.g., Gaussian, exponential) are commonly used for assigning prior distribution. The term $p(\mathbf{Z}|\theta)$ is called the 'likelihood function'. It is derived based on modeling

assumptions that give a probabilistic description (distribution) for the possible values of $\mathbf{Z}$ for a given $\boldsymbol{\theta}$. It is the heart of the Bayesian approach and it directly affects the validity of identification results.

8.3 System Identification Framework

Solving a system identification problem using the Bayesian approach involves the following basic ingredients:

1. Formulate the likelihood function $p(\mathbf{Z}|\boldsymbol{\theta})$ based on modeling assumptions. This involves physical/logical modeling of system prediction as well as its discrepancy from measured data (prediction error).
2. Assign a prior distribution $p(\boldsymbol{\theta})$ based on available knowledge.
3. Determine the descriptive posterior statistics (e.g., mean, variance) of $\boldsymbol{\theta}$ associated with the posterior distribution $p(\boldsymbol{\theta}|\mathbf{Z})$ for convenient interpretation and decision making. This is primarily a computational problem.

The difficulty of computing the posterior statistics depends on the topology of the posterior distribution. This can be classified in terms of ‘identifiability’ of the problem, which is related to the nature of measured data in providing sufficient information for identifying the model parameters. This is discussed in the next section.

8.4 Identifiability

One way to classify a Bayesian identification problem is based on the topology of the region in the model parameter space that carries significant probability content of the posterior distribution. The problem can be classified as either ‘globally identifiable’, ‘locally identifiable’ or ‘unidentifiable’ (Beck and Katafygiotis 1998). It is ‘globally identifiable’ if the posterior PDF has a single (unique) peak in the parameter space. If there is more than one peak, the problem is said to be ‘locally identifiable’. If there is a continuum of points with the same value of the posterior PDF, the problem is ‘unidentifiable’.

Identifiability is related to whether the measured data provides sufficient information for identifying the model parameters. It is concerned with the nature of identification results rather than their precision, although these two aspects are related; unidentifiable problems often have much larger posterior uncertainty than identifiable ones. Note that if a constant prior PDF is chosen, then identifiability is determined by the likelihood function. Problems are unidentifiable because the data does not provide sufficient information about the parameters, often because of its nature rather than its amount. For example, a structural mode is unidentifiable when

it is identified based on vibration data measured near the node of the mode shape, no matter how long the data is. Identifiability is determined by the joint PDF rather than the marginal ones. It is possible for some parameters to have a unimodal marginal distribution in an unidentifiable problem.

The computational strategy for calculating posterior statistics depends on identifiability. We discuss the strategies in different cases in Sects. 8.5–8.7. Modal identification problems are globally identifiable, or else one better change the experimental setup to make it so. Readers primarily interested in OMA may just focus on Sect. 8.5. Before we leave this section we consider an example that illustrates the identifiability of the stiffness of a simple structure, depending on the amount of modal information available.

Example 8.1 (Two-DOF structure, identifiability) Consider a two-DOF structure as shown in Fig. 8.1. The masses are known and equal to 1 kg. The stiffness parameters are unknown and are denoted by θ_1 and θ_2 (N/m).

Suppose we want to identify the stiffness parameters using vibration data of the structure, denoted by $\mathbf{Z}$. A common way is to adopt a ‘two-stage approach’ where one first identifies the natural frequencies and mode shapes (Stage I), collectively denoted by ϖ , and then uses this information to identify the stiffness parameters (Stage II). Let $p_1(\varpi|\mathbf{Z})$ denote the posterior PDF of ϖ obtained in Stage I assuming a uniform prior PDF for ϖ . It can be reasoned that $p_1(\varpi|\mathbf{Z})$ is approximately a Gaussian PDF and the modal properties are independent when the signal-to-noise (s/n) ratio of data is high (Au and Zhang 2015). Let $\tilde{\varpi}(\boldsymbol{\theta})$ denote the theoretical prediction of ϖ for a given $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$. Assuming a uniform prior PDF for $\boldsymbol{\theta}$, it can be shown that

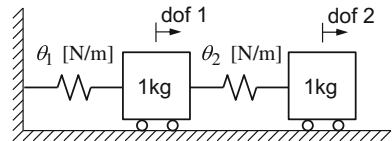
$$p(\boldsymbol{\theta}|\mathbf{Z}) \propto p_1(\tilde{\varpi}(\boldsymbol{\theta})|\mathbf{Z}) \quad (8.4)$$

Model Prediction of Modal Properties

The theoretical modal properties can be obtained by solving the eigenvalue equation $\mathbf{K}\boldsymbol{\psi} = \omega^2\mathbf{M}\boldsymbol{\psi}$, where $\mathbf{K}$ and $\mathbf{M}$ are respectively the stiffness and mass matrix; ω and $\boldsymbol{\psi}$ are respectively the natural frequency (in rad/s) and mode shape. In the present case,

$$\mathbf{K} = \begin{bmatrix} \theta_1 + \theta_2 & -\theta_2 \\ -\theta_2 & \theta_2 \end{bmatrix} \quad [\text{N/m}] \quad \mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad [\text{kg}] \quad (8.5)$$

Fig. 8.1 Two-DOF structure



Solving the eigenvalue problem yields the natural frequencies (in rad/s) as

$$\begin{aligned}\tilde{\omega}_1 &= \frac{1}{\sqrt{2}} \left[\theta_1 + 2\theta_2 - (\theta_1^2 + 4\theta_2^2)^{1/2} \right]^{1/2} \\ \tilde{\omega}_2 &= \frac{1}{\sqrt{2}} \left[\theta_1 + 2\theta_2 + (\theta_1^2 + 4\theta_2^2)^{1/2} \right]^{1/2}\end{aligned}\quad (8.6)$$

Normalizing the mode shapes to be 1 at DOF 2, they are given by

$$\tilde{\Psi}_1 = \begin{bmatrix} \tilde{u}_1 \\ 1 \end{bmatrix} \quad \tilde{\Psi}_2 = \begin{bmatrix} \tilde{u}_2 \\ 1 \end{bmatrix} \quad (8.7)$$

where

$$\tilde{u}_1 = 1 - \frac{\tilde{\omega}_1^2}{\theta_2} \quad \tilde{u}_2 = 1 - \frac{\tilde{\omega}_2^2}{\theta_2} \quad (8.8)$$

Clearly, $\tilde{\omega}_1$, $\tilde{\omega}_2$, $\tilde{u}_1$ and $\tilde{u}_2$ are determined by (θ_1, θ_2) . Figure 8.2 shows the contours of their values in the space of (θ_1, θ_2) .

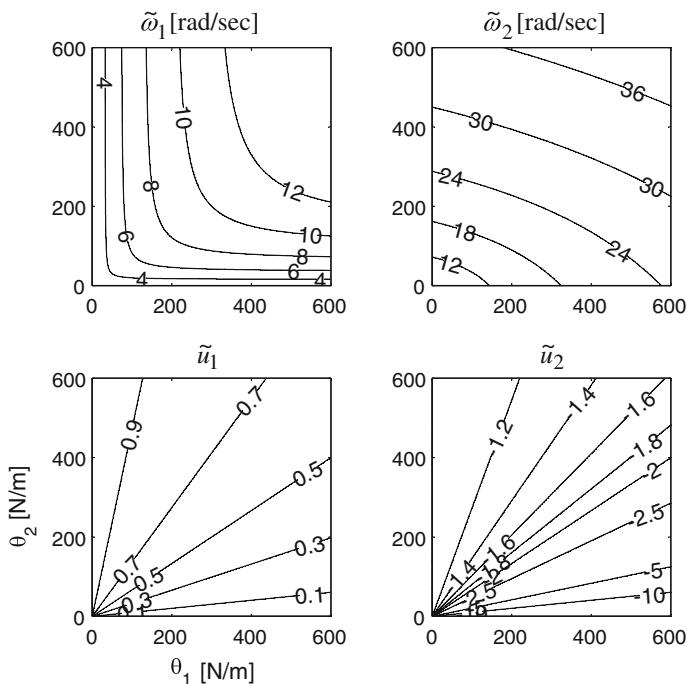


Fig. 8.2 Contour of modal properties, two-DOF structure

Posterior PDF of Stiffness Parameters

The posterior PDF of (θ_1, θ_2) depends on what modal properties are identified as well as their identification uncertainty in Stage I. Figure 8.3 shows the posterior PDF based on one hypothetical data set. Depending on the measured DOFs and identified modes, (θ_1, θ_2) can be globally identifiable ('G'), locally identifiable ('L') or unidentifiable ('U'). These are further explained in Example 8.2. ■

Example 8.2 (Two-DOF structure, identifiability explained) This example explains the identifiability in Fig. 8.3 of Example 8.1. The story is a bit long and it may be skipped on first reading. To avoid unnecessary complications, we omit numerical details of the identified modal properties during the discussion. They can be found at the end of this example.

DOF 1 Measured, Mode 1 Identified

First, consider the case when only DOF 1 is measured and only mode 1 is identified. In this case no mode shape information is available. The identified modal

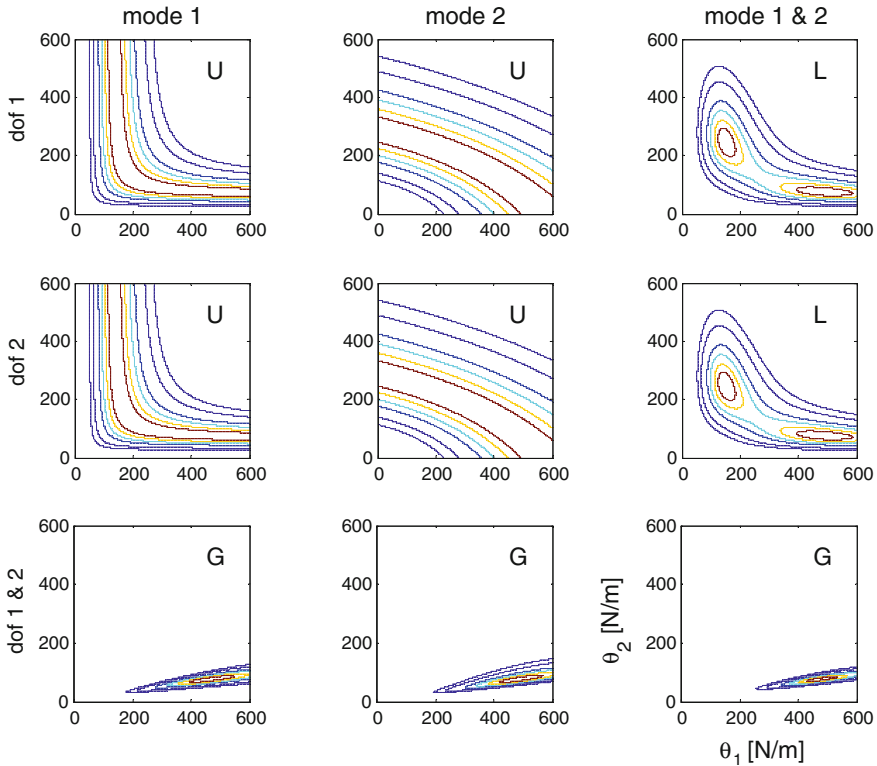


Fig. 8.3 Contour of posterior PDF of stiffness parameters (θ_1, θ_2) identified based on different modal information. Measured DOFs indicated in different rows, identified modes in different columns. 'G'—globally identifiable; 'L'—locally identifiable; 'U'—unidentifiable

properties comprise only the first mode natural frequency, i.e., $\varpi \equiv \omega_1$. Equation (8.4) reads

$$p(\boldsymbol{\theta}|\mathbf{Z}) \propto p_1(\tilde{\omega}_1(\boldsymbol{\theta})|\mathbf{Z}) = \frac{1}{\sqrt{2\pi}\hat{\sigma}_{\omega 1}} \exp\left\{-\frac{[\tilde{\omega}_1(\boldsymbol{\theta}) - \hat{\omega}_1]^2}{2\hat{\sigma}_{\omega 1}^2}\right\} \quad (8.9)$$

where $\hat{\omega}_1$ and $\hat{\sigma}_{\omega 1}^2$ are respectively the posterior MPV (most probable value) and variance of ω_1 identified in Stage I. This is an exponentially decaying function of $[\tilde{\omega}_1(\boldsymbol{\theta}) - \hat{\omega}_1]^2$ and so it peaks at those values of $\boldsymbol{\theta}$ for which $\tilde{\omega}_1(\boldsymbol{\theta}) = \hat{\omega}_1$. These values correspond to the contour line at level $\hat{\omega}_1$ (8 rad/s, see Table 8.1) in the (1,1)-subplot of Fig. 8.2. This line is reproduced in the (1,1)-subplot in Fig. 8.4. Since there is a continuum of MPVs, the stiffness parameters are unidentifiable.

Table 8.1 Numerical data of identified modal properties, two-DOF structure

Mode i	$\hat{\omega}_i$ [rad/s]	$\hat{\sigma}_{\omega i}$ [rad/s]	$\hat{u}_i$	$\hat{\sigma}_{ui}$
1	8	0.8	0.16	0.016
2	24	2.4	-6.5	0.65

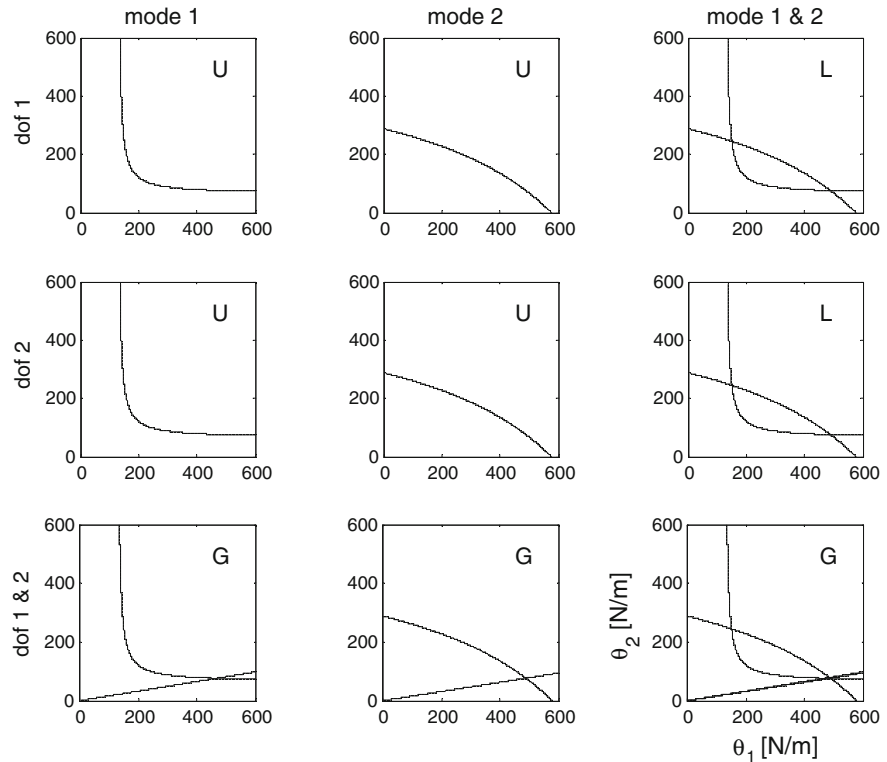


Fig. 8.4 MPV of (θ_1, θ_2) based on different modal information

DOF 1 Measured, Mode 2 Identified

The situation is similar when only DOF 1 is measured and only mode 2 is identified. In this case $\varpi \equiv \omega_2$ and

$$p(\boldsymbol{\theta}|\mathbf{Z}) = p_1(\tilde{\omega}_2(\boldsymbol{\theta})|\mathbf{Z}) = \frac{1}{\sqrt{2\pi}\hat{\sigma}_{\omega_2}} \exp\left\{-\frac{[\tilde{\omega}_2(\boldsymbol{\theta}) - \hat{\omega}_2]^2}{2\hat{\sigma}_{\omega_2}^2}\right\} \quad (8.10)$$

where $\hat{\omega}_2$ and $\hat{\sigma}_{\omega_2}^2$ are respectively the posterior MPV and variance of ω_2 identified in Stage I. The stiffness parameters $\boldsymbol{\theta}$ are unidentifiable as there is a continuum of MPVs that satisfy $\tilde{\omega}_2(\boldsymbol{\theta}) = \hat{\omega}_2$. These form the line in the (1,2)-subplot of Fig. 8.4.

DOF 1 Measured, Mode 1 and 2 Identified

When only DOF 1 is measured but both mode 1 and 2 are identified, $\varpi = \{\omega_1, \omega_2\}$ and

$$\begin{aligned} p(\boldsymbol{\theta}|\mathbf{Z}) &\propto p_1(\tilde{\omega}_1(\boldsymbol{\theta})|\mathbf{Z})p_1(\tilde{\omega}_2(\boldsymbol{\theta})|\mathbf{Z}) \\ &= \frac{1}{2\pi\hat{\sigma}_{\omega_1}\hat{\sigma}_{\omega_2}} \exp\left\{-\frac{[\tilde{\omega}_1(\boldsymbol{\theta}) - \hat{\omega}_1]^2}{2\hat{\sigma}_{\omega_1}^2} - \frac{[\tilde{\omega}_2(\boldsymbol{\theta}) - \hat{\omega}_2]^2}{2\hat{\sigma}_{\omega_2}^2}\right\} \end{aligned} \quad (8.11)$$

The MPVs of $\boldsymbol{\theta}$ are the values that satisfy both $\tilde{\omega}_1(\boldsymbol{\theta}) = \hat{\omega}_1$ and $\tilde{\omega}_2(\boldsymbol{\theta}) = \hat{\omega}_2$, i.e., the intersection of the two lines in the (1,3)-subplot of Fig. 8.4. The two intersections give two (local) MPVs, and so the problem is locally identifiable. The three cases in the second row of Fig. 8.3 when DOF 2 is measured can be explained in a similar manner.

DOFs 1 and 2 Measured, Mode 1 Identified

Consider now when DOFs 1 and 2 are measured. When only mode 1 is identified, $\varpi = \{\omega_1, u_1\}$, where u_1 is the mode shape value of mode 1 at DOF 1. Then

$$\begin{aligned} p(\boldsymbol{\theta}|\mathbf{Z}) &\propto p_1(\tilde{\omega}_1(\boldsymbol{\theta})|\mathbf{Z})p_1(\tilde{u}_1(\boldsymbol{\theta})|\mathbf{Z}) \\ &= \frac{1}{2\pi\hat{\sigma}_{\omega_1}\hat{\sigma}_{u_1}} \exp\left\{-\frac{[\tilde{\omega}_1(\boldsymbol{\theta}) - \hat{\omega}_1]^2}{2\hat{\sigma}_{\omega_1}^2} - \frac{[\tilde{u}_1(\boldsymbol{\theta}) - \hat{u}_1]^2}{2\hat{\sigma}_{u_1}^2}\right\} \end{aligned} \quad (8.12)$$

where $\hat{u}_1$ and $\hat{\sigma}_{u_1}^2$ are respectively the posterior MPV and variance of u_1 . The MPVs of $\boldsymbol{\theta}$ are the values that satisfy both $\tilde{\omega}_1(\boldsymbol{\theta}) = \hat{\omega}_1$ and $\tilde{u}_1(\boldsymbol{\theta}) = \hat{u}_1$, i.e., the intersection of the two lines in the (3,1)-subplot of Fig. 8.4. Similarly, when both DOFs 1 and 2 are measured and only mode 2 is identified, the MPVs correspond to the intersection of the two lines in the (3,2)-subplot of Fig. 8.4. In both cases there is only one intersection and so the problem is globally identifiable.

DOFs 1 and 2 Measured, Mode 1 and 2 Identified

Finally, when DOFs 1 and 2 are measured and mode 1 and 2 are identified, $\varpi = \{\omega_1, u_1, \omega_2, u_2\}$ and

$$p(\boldsymbol{\theta}|\mathbf{Z}) \propto p_1(\tilde{\omega}_1(\boldsymbol{\theta})|\mathbf{Z})p_1(\tilde{u}_1(\boldsymbol{\theta})|\mathbf{Z})p_1(\tilde{\omega}_2(\boldsymbol{\theta})|\mathbf{Z})p_1(\tilde{u}_2(\boldsymbol{\theta})|\mathbf{Z}) \quad (8.13)$$

The MPVs of $\boldsymbol{\theta}$ need not correspond to those values that satisfy $\tilde{\omega}_1(\boldsymbol{\theta}) = \hat{\omega}_1$, $\tilde{\omega}_2(\boldsymbol{\theta}) = \hat{\omega}_2$, $\tilde{u}_1(\boldsymbol{\theta}) = \hat{u}_1$ and $\tilde{u}_2(\boldsymbol{\theta}) = \hat{u}_2$, because it is generally impossible to satisfy four equations with two variables. The nature of MPVs depends on whether the identified modal properties are consistent with the theoretical models. For the (3,3)-subplot in Fig. 8.3, they are roughly consistent at the global MPV at the lower right.

Identified Modal Properties

For reference, Table 8.1 gives the numerical data of the identified modal properties used. The data is hypothetical. The posterior uncertainties of modal properties have been taken to be much larger than typical values so that the contours in Fig. 8.3 can be seen. ■

8.5 Globally Identifiable Problems

When the posterior PDF has a unique regular maximum, it may be approximated by a Gaussian PDF. The location of the maximum has the highest PDF value and is called the ‘most probable value’ (MPV), also as known as the ‘maximum a posteriori value’. The mean value of the Gaussian PDF is equal to the MPV and its covariance matrix is related to the curvature of the posterior PDF at the MPV. The main computational effort for determining the posterior statistics is spent on locating the MPV (e.g., by numerical optimization) and calculating the curvatures. For convenience, let

$$L(\mathbf{Z}, \boldsymbol{\theta}) = -\ln[p(\mathbf{Z}|\boldsymbol{\theta})p(\boldsymbol{\theta})] \quad (8.14)$$

so that the posterior PDF is expressed as

$$p(\boldsymbol{\theta}|\mathbf{Z}) \propto e^{-L(\mathbf{Z}, \boldsymbol{\theta})} \quad (8.15)$$

When $p(\boldsymbol{\theta})$ is a uniform distribution one can define $L(\mathbf{Z}, \boldsymbol{\theta}) = -\ln p(\mathbf{Z}|\boldsymbol{\theta})$. In this case, $L(\mathbf{Z}, \boldsymbol{\theta})$ is called the ‘negative log-likelihood function’ (NLLF).

Let $\hat{\boldsymbol{\theta}}(\mathbf{Z})$ be the (unique) MPV of $\boldsymbol{\theta}$, where its dependence on data $\mathbf{Z}$ has been emphasized. It maximizes the posterior PDF, or equivalently minimizes L . That is,

$$\nabla_{\boldsymbol{\theta}} L(\mathbf{Z}, \boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(\mathbf{Z})} = \mathbf{0} \quad (8.16)$$

and the Hessian $\nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(\mathbf{Z})}$ is positive definite. Here, the subscript $\boldsymbol{\theta}$ in $\nabla_{\boldsymbol{\theta}}$ indicates that the gradient is taken w.r.t. $\boldsymbol{\theta}$. Approximating L by a second order Taylor series about $\hat{\boldsymbol{\theta}}$,

$$\begin{aligned}
L(\mathbf{Z}, \boldsymbol{\theta}) &\approx L(\mathbf{Z}, \hat{\boldsymbol{\theta}}) + \underbrace{\nabla_{\boldsymbol{\theta}} L(\mathbf{Z}, \boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}}_0 (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) + \frac{1}{2} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T \nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \\
&= L(\mathbf{Z}, \hat{\boldsymbol{\theta}}) + \frac{1}{2} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T \nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})
\end{aligned} \tag{8.17}$$

Substituting into (8.15) and omitting the constant $L(\mathbf{Z}, \hat{\boldsymbol{\theta}})$ gives

$$p(\boldsymbol{\theta}|\mathbf{Z}) \propto \exp \left[-\frac{1}{2} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T \hat{\mathbf{C}}^{-1} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \right] \tag{8.18}$$

where

$$\hat{\mathbf{C}}(\mathbf{Z}) = \left[\nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \right]^{-1} \quad (\text{posterior covariance matrix}) \tag{8.19}$$

is a positive definite matrix. Equation (8.18) is just the variable part of a multi-variate Gaussian PDF with mean $\hat{\boldsymbol{\theta}}$ and covariance matrix $\hat{\mathbf{C}}$. This is sufficient to conclude that $p(\boldsymbol{\theta}|\mathbf{Z})$ can be approximated by a Gaussian PDF. The proportionality constant is $(2\pi)^{-n_{\theta}/2} |\hat{\mathbf{C}}|^{-1/2}$ (n_{θ} is the number of parameters in $\boldsymbol{\theta}$), so that $p(\boldsymbol{\theta}|\mathbf{Z})$ integrates w.r.t. $\boldsymbol{\theta}$ to 1. Thus,

$$p(\boldsymbol{\theta}|\mathbf{Z}) \approx (2\pi)^{-n_{\theta}/2} |\hat{\mathbf{C}}|^{-1/2} \exp \left[-\frac{1}{2} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T \hat{\mathbf{C}}^{-1} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}) \right] \tag{8.20}$$

As a property of Gaussian PDF, the marginal PDF of a particular parameter θ_i ($i = 1, \dots, n_{\theta}$) is also a Gaussian PDF with a mean $\hat{\theta}_i$ (the i th entry of $\hat{\boldsymbol{\theta}}$) and variance equal to the i th diagonal entry of $\hat{\mathbf{C}}$. The covariance between θ_i and θ_j is the (i, j) -entry of $\hat{\mathbf{C}}$. These are desirable properties, as they eliminate the need for integrating the joint PDF to yield the marginal PDFs.

8.5.1 Quality of Gaussian Approximation

The quality of Gaussian approximation to the posterior PDF $p(\boldsymbol{\theta}|\mathbf{Z})$ depends on how close $L(\mathbf{Z}, \boldsymbol{\theta})$ is to a quadratic function of $\boldsymbol{\theta}$. This depends on the problem, the parameter under question and data size. The posterior marginal PDF of one parameter can be approximately Gaussian while another is not. Practically, the MPV and posterior variance are of primary concern while the quality of Gaussian

approximation is of secondary importance. There are many cases where $L(\mathbf{Z}, \boldsymbol{\theta})$ systematically tends to a quadratic function of $\boldsymbol{\theta}$ as the sample size increases and hence the posterior PDF converges to a Gaussian PDF. The asymptotic behavior of functions of this type has been studied extensively, e.g., in Laplace type integrals (Bleistein and Handelsman 1986; Erdelyi 1956). For general problems, asymptotic Gaussian nature of the posterior PDF should not be taken for granted.

Example 8.3 (Mean parameter, exact posterior) Let $\{\hat{y}_k\}_{k=1}^N$, abbreviated as $\{\hat{y}_k\}$, be a set of measured data assumed to be related to an unknown parameter θ_1 by

$$\hat{y}_k = \theta_1 + \varepsilon_k \quad (8.21)$$

where $\{\varepsilon_k\}_{k=1}^N$ are prediction errors modeled as i.i.d. Gaussian variables with zero mean and unknown variance θ_2 . Although we are primarily interested in θ_1 , the likelihood function $p(\{\hat{y}_k\}|\theta_1)$ is non-trivial to formulate from first principles. We need to include θ_2 in $\boldsymbol{\theta}$ so that the likelihood function $p(\{\hat{y}_k\}|\theta_1, \theta_2)$ can be explicitly derived. According to (8.21), for given $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$, $\{\hat{y}_k\}$ are i.i.d. Gaussian variables with mean θ_1 and variance θ_2 . Thus

$$p(\{\hat{y}_k\}|\boldsymbol{\theta}) = \prod_{k=1}^N \frac{1}{\sqrt{2\pi\theta_2}} \exp\left[-\frac{(\hat{y}_k - \theta_1)^2}{2\theta_2}\right] \quad (8.22)$$

Assuming a uniform prior PDF for $\boldsymbol{\theta}$, $p(\boldsymbol{\theta}|\{\hat{y}_k\}) \propto p(\{\hat{y}_k\}|\boldsymbol{\theta})$. As a function of $\{\hat{y}_k\}$ this is a joint Gaussian PDF but as a function of $\boldsymbol{\theta}$ the dependence is less trivial. Expanding the quadratic term in the exponent, it can be written more explicitly in terms of (θ_1, θ_2) as

$$p(\boldsymbol{\theta}|\{\hat{y}_k\}) \propto p(\{\hat{y}_k\}|\boldsymbol{\theta}) = (2\pi\theta_2)^{-N/2} \exp\left\{-\frac{N}{2\theta_2} \left[(\theta_1 - \mu_1)^2 + \mu_2 - \mu_1^2\right]\right\} \quad (8.23)$$

$$\mu_1 = \frac{1}{N} \sum_{k=1}^N \hat{y}_k \quad \mu_2 = \frac{1}{N} \sum_{k=1}^N \hat{y}_k^2 \quad (8.24)$$

It can be further separated into two terms:

$$p(\boldsymbol{\theta}|\{\hat{y}_k\}) \propto \underbrace{(2\pi\theta_2)^{-(N-1)/2} \exp\left[-\frac{N(\mu_2 - \mu_1^2)}{2\theta_2}\right]}_{\text{function of } \theta_2 \text{ only}} \times \underbrace{\frac{1}{\sqrt{2\pi\theta_2/N}} \exp\left[\frac{-(\theta_1 - \mu_1)^2}{2\theta_2/N}\right]}_{\phi(\theta_1; \mu_1, \theta_2/N)} \quad (8.25)$$

where $\phi(\theta_1; \mu_1, \theta_2/N)$ denotes a Gaussian PDF with mean μ_1 , variance θ_2/N and evaluated at θ_1 . By the definition of conditional probability,

$$p(\theta_1, \theta_2 | \{\hat{y}_k\}) = p(\theta_2 | \{\hat{y}_k\}) \times p(\theta_1 | \theta_2, \{\hat{y}_k\}) \quad (8.26)$$

Comparing with (8.25), we see that

$$p(\theta_1 | \theta_2, \{\hat{y}_k\}) = \phi(\theta_1; \mu_1, \frac{\theta_2}{N}) = \frac{1}{\sqrt{2\pi\theta_2/N}} \exp\left[-\frac{(\theta_1 - \mu_1)^2}{2\theta_2/N}\right] \quad (8.27)$$

$$p(\theta_2 | \{\hat{y}_k\}) = C \theta_2^{-(N-1)/2} \exp\left[-\frac{N(\mu_2 - \mu_1^2)}{2\theta_2}\right] \quad (8.28)$$

where C is a normalizing constant so that $\int_0^\infty p(\theta_2 | \{\hat{y}_k\}) d\theta_2 = 1$. The integral on the RHS of (8.28) can be evaluated analytically, which implies

$$C = \frac{[N(\mu_2 - \mu_1^2)/2]^{(N-3)/2}}{\Gamma((N-3)/2)} \quad (8.29)$$

where $\Gamma(\cdot)$ is the Gamma function. The marginal PDF of θ_1 can be obtained by integrating out θ_2 from (8.23). The integration can be performed analytically, giving (omitting constants)

$$p(\theta_1 | \{\hat{y}_k\}) = \int_0^\infty p(\theta | \{\hat{y}_k\}) d\theta_2 \propto [(\theta_1 - \mu_1)^2 + (\mu_2 - \mu_1^2)]^{-(N-2)/2} \quad (8.30)$$

This can be written in the form (omitting constants)

$$p(\theta_1 | \{\hat{y}_k\}) \propto \left[1 + \frac{1}{v} \left(\frac{\theta_1 - \mu_1}{\sqrt{(\mu_2 - \mu_1^2)/v}}\right)^2\right]^{-(v+1)/2} \quad (8.31)$$

where $v = N - 3$. This is a (non-standard) Student's t PDF with v degrees of freedom with shift parameter μ_1 and scale parameter $\sigma = \sqrt{(\mu_2 - \mu_1^2)/v}$. That is, $t = (\theta_1 - \mu_1)/\sigma$ has a (standard) Student's t PDF with v degrees of freedom. Using the standard formula for Student's t PDF,

$$p(\theta_1 | \{\hat{y}_k\}) = \frac{\Gamma(\frac{v+1}{2})}{\Gamma(\frac{v}{2})\sqrt{\pi v}\sigma} \left[1 + \frac{1}{v} \left(\frac{\theta_1 - \mu_1}{\sigma}\right)^2\right]^{-(v+1)/2} \quad \begin{matrix} v = N - 3 \\ \sigma = \sqrt{(\mu_2 - \mu_1^2)/v} \end{matrix} \quad (8.32)$$

In this example the posterior marginal PDFs can be obtained analytically and they can be related to a standard distribution that has been studied before. For general problems this is seldom the case. It could be difficult (if possible) to

discover the joint posterior PDF as a product of marginal and conditional PDF; or to obtain an analytical expression for the integrals in the marginal PDFs or normalizing constants. The joint, conditional or the marginal PDF need not belong to any standard distribution. Computational methods are therefore usually needed for determining the descriptive statistics of the posterior PDF. ■

Example 8.4 (Mean parameter, Gaussian posterior) We revisit Example 8.3 but here we obtain the posterior MPV and covariance matrix of θ based on a Gaussian approximation of the posterior PDF. Using (8.23),

$$L(\{\hat{y}_k\}, \theta) = -\ln p(\{\hat{y}_k\}|\theta) = \frac{N}{2} \left\{ \ln(2\pi\theta_2) + \theta_2^{-1} \left[(\theta_1 - \mu_1)^2 + \mu_2 - \mu_1^2 \right] \right\} \quad (8.33)$$

Posterior MPV

Direct differentiation gives

$$\frac{\partial L}{\partial \theta_1} = N\theta_2^{-1}(\theta_1 - \mu_1) \quad \frac{\partial L}{\partial \theta_2} = \frac{N}{2} \left\{ \theta_2^{-1} - \theta_2^{-2} \left[(\theta_1 - \mu_1)^2 + \mu_2 - \mu_1^2 \right] \right\} \quad (8.34)$$

Solving $\partial L/\partial \theta_1 = 0$ and $\partial L/\partial \theta_2 = 0$ for (θ_1, θ_2) gives the posterior MPVs:

$$\hat{\theta}_1 = \mu_1 \quad \hat{\theta}_2 = \mu_2 - \mu_1^2 \quad (8.35)$$

The MPVs in (8.35) are intuitive and coincide with the estimators in classical statistics.

Posterior Covariance Matrix

Further differentiating (8.34) gives

$$\frac{\partial^2 L}{\partial \theta_1^2} = N\theta_2^{-1} \quad \frac{\partial^2 L}{\partial \theta_2^2} = \frac{N}{2} \left\{ -\theta_2^{-2} + 2\theta_2^{-3} \left[(\theta_1 - \mu_1)^2 + \mu_2 - \mu_1^2 \right] \right\} \quad (8.36)$$

$$\frac{\partial^2 L}{\partial \theta_1 \partial \theta_2} = \frac{\partial^2 L}{\partial \theta_2 \partial \theta_1} = -N\theta_2^{-2}(\theta_1 - \mu_1) \quad (8.37)$$

Substituting $\theta_1 = \hat{\theta}_1$ and $\theta_2 = \hat{\theta}_2$ from (8.35) gives, after simplifications,

$$\frac{\partial^2 \hat{L}}{\partial \theta_1^2} = N(\mu_2 - \mu_1^2)^{-1} \quad \frac{\partial^2 \hat{L}}{\partial \theta_2^2} = \frac{N}{2}(\mu_2 - \mu_1^2)^{-2} \quad \frac{\partial^2 \hat{L}}{\partial \theta_1 \partial \theta_2} = \frac{\partial^2 \hat{L}}{\partial \theta_2 \partial \theta_1} = 0 \quad (8.38)$$

where the hat ‘ $\wedge$ ’ denotes that the derivative is evaluated at the MPV. Assembling these gives the Hessian of L at the MPV:

$$\hat{\mathbf{H}}_L = \begin{bmatrix} \partial^2 \hat{L} / \partial \theta_1^2 & \partial^2 \hat{L} / \partial \theta_1 \partial \theta_2 \\ \partial^2 \hat{L} / \partial \theta_2 \partial \theta_1 & \partial^2 \hat{L} / \partial \theta_2^2 \end{bmatrix} = \begin{bmatrix} N(\mu_2 - \mu_1^2)^{-1} & 0 \\ 0 & N(\mu_2 - \mu_1^2)^{-2} / 2 \end{bmatrix} \quad (8.39)$$

Taking inverse gives the posterior covariance matrix of θ :

$$\hat{\mathbf{C}} = \hat{\mathbf{H}}_L^{-1} = \begin{bmatrix} (\mu_2 - \mu_1^2) / N & 0 \\ 0 & 2(\mu_2 - \mu_1^2)^2 / N \end{bmatrix} \quad (8.40)$$

Interpretation

In summary, given the data $\{\hat{y}_k\}_{k=1}^N$ and assuming the model in (8.21), the posterior PDF of (θ_1, θ_2) is approximately a Gaussian PDF with mean given by (8.35) and covariance matrix given by (8.40). The posterior variances of θ_1 and θ_2 are

$$\text{var}[\theta_1 | \{\hat{y}_k\}] = \hat{\mathbf{C}}(1, 1) = \frac{1}{N}(\mu_2 - \mu_1^2) \quad \text{var}[\theta_2 | \{\hat{y}_k\}] = \hat{\mathbf{C}}(2, 2) = \frac{2}{N}(\mu_2 - \mu_1^2)^2 \quad (8.41)$$

The covariance between θ_1 and θ_2 is $\hat{\mathbf{C}}(1, 2)$, which is zero. This implies that, given the data, they are uncorrelated.

As a remark, in general problems the MPV and posterior covariance matrix can seldom be obtained analytically in terms of data. The Hessian need not be a diagonal matrix and so its inverse often needs to be evaluated numerically. The covariance matrix need not be diagonal and the model parameters could be correlated according to their posterior PDF.

Quality of Gaussian Approximation

To demonstrate the quality of Gaussian approximation of the posterior PDF of (θ_1, θ_2) , we generated synthetic data according to $\hat{y}_k = 1 + \varepsilon_k$ where $N = 30$ and $\{\varepsilon_k\}_{k=1}^N$ are i.i.d. zero-mean Gaussian with variance 0.2. A particular data set gave $\mu_1 = 0.9516$ and $\mu_2 = 1.1453$. Figure 8.5 shows the contour of the posterior joint PDF using (a) the exact expression in (8.26) and (b) the Gaussian approximation. The values that generated the data ($\theta_1 = 1$, $\theta_2 = 0.2$) need not correspond to the MPV of a particular data set. Figure 8.6 shows the posterior marginal PDFs. The exact marginal PDFs can be obtained by numerically integrating the joint PDF, or in this example using (8.28) and (8.32). The Gaussian approximation is good for θ_1 but not for θ_2 . Although the exact and Gaussian PDF clearly differ, they have a similar spread. ■

Example 8.5 (linear regression, Gaussian posterior) Consider identifying $\theta = [\theta_1, \theta_2, \theta_3]^T$ from data $\{\hat{y}_k\}_{k=1}^N$, abbreviated as $\{\hat{y}_k\}$, assuming the simple linear regression model

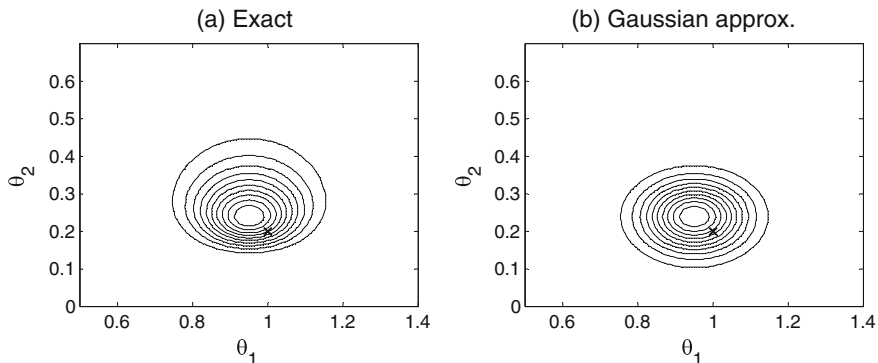


Fig. 8.5 Contour of posterior PDF in Example 8.4. **a** Exact, **b** Gaussian approximation. The value that generates the data is marked with a cross 'x'

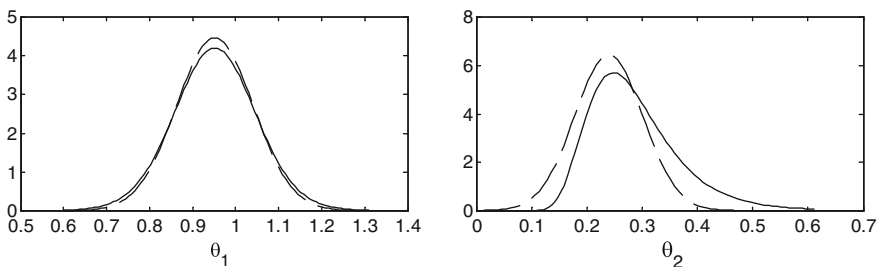


Fig. 8.6 Posterior marginal PDF in Example 8.4. *Solid line* exact; *Dashed line* Gaussian approximation

$$\hat{y}_k = \theta_1 + \theta_2 x_k + \varepsilon_k \quad (8.42)$$

where $\{x_k\}_{k=1}^N$ are given; $\{\varepsilon_k\}_{k=1}^N$ are prediction errors modeled as i.i.d. Gaussian variables with zero mean and variance θ_3 . Similar to Example 8.4, we obtain the posterior MPV and covariance matrix of $\boldsymbol{\theta}$ based on a Gaussian approximation of the posterior PDF.

Given $\boldsymbol{\theta}$, $\{\hat{y}_k\}$ are i.i.d. Gaussian, each with mean $\theta_1 + \theta_2 x_k$ and variance θ_3 . Assuming a uniform prior PDF for $\boldsymbol{\theta}$,

$$p(\boldsymbol{\theta}|\{\hat{y}_k\}) \propto p(\{\hat{y}_k\}|\boldsymbol{\theta}) = \prod_{k=1}^N \frac{1}{\sqrt{2\pi\theta_3}} \exp\left[-\frac{1}{2\theta_3} (\hat{y}_k - \theta_1 - \theta_2 x_k)^2\right] \quad (8.43)$$

$$L(\{\hat{y}_k\}, \boldsymbol{\theta}) = -\ln p(\{\hat{y}_k\}|\boldsymbol{\theta}) = \frac{N}{2} \ln(2\pi\theta_3) + \frac{1}{2\theta_3} \sum_{k=1}^N (\hat{y}_k - \theta_1 - \theta_2 x_k)^2 \quad (8.44)$$

To facilitate analysis, we expand the squares in (8.44) and rewrite it as

$$L(\{\hat{y}_k\}, \boldsymbol{\theta}) = \frac{N}{2} [\ln(2\pi\theta_3) + \theta_3^{-1}(\mu_{yy} + \theta_1^2 + \mu_{xx}\theta_2^2 - 2\mu_y\theta_1 - 2\mu_{xy}\theta_2 + 2\mu_x\theta_1\theta_2)] \quad (8.45)$$

where

$$\begin{aligned} \mu_x &= \frac{1}{N} \sum_{k=1}^N x_k & \mu_y &= \frac{1}{N} \sum_{k=1}^N \hat{y}_k \\ \mu_{xx} &= \frac{1}{N} \sum_{k=1}^N x_k^2 & \mu_{yy} &= \frac{1}{N} \sum_{k=1}^N \hat{y}_k^2 & \mu_{xy} &= \frac{1}{N} \sum_{k=1}^N x_k \hat{y}_k \end{aligned} \quad (8.46)$$

Posterior MPV

Solving $\partial L / \partial \theta_i = 0$ ($i = 1, 2, 3$) for $(\theta_1, \theta_2, \theta_3)$ gives the posterior MPV:

$$\hat{\theta}_1 = \frac{\mu_{xx}\mu_y - \mu_x\mu_{xy}}{\mu_{xx} - \mu_x^2} \quad \hat{\theta}_2 = \frac{\mu_{xy} - \mu_x\mu_y}{\mu_{xx} - \mu_x^2} \quad (8.47)$$

$$\hat{\theta}_3 = \frac{-\mu_{yy}\mu_x^2 + 2\mu_x\mu_y\mu_{xy} - \mu_{xy}^2 - \mu_{xx}\mu_y^2 + \mu_{xx}\mu_{yy}}{\mu_{xx} - \mu_x^2} \quad (8.48)$$

These MPVs are identical to the best estimates from linear regression theory.

Posterior Covariance Matrix

The Hessian of L w.r.t. $\boldsymbol{\theta}$ at the MPV can be evaluated to be

$$\hat{\mathbf{H}}_L = \frac{N}{\hat{\theta}_3} \begin{bmatrix} 1 & \mu_x & 0 \\ \mu_x & \mu_{xx} & 0 \\ 0 & 0 & 1/2\hat{\theta}_3 \end{bmatrix} \quad (8.49)$$

Unlike the last example, the Hessian is not diagonal, although the cross derivatives involving the prediction error variance θ_3 are zero as before. Taking inverse gives the posterior covariance matrix:

$$\hat{\mathbf{C}} = \hat{\mathbf{H}}_L^{-1} = \frac{\hat{\theta}_3}{N(\mu_{xx} - \mu_x^2)} \begin{bmatrix} \mu_{xx} & -\mu_x & 0 \\ -\mu_x & 1 & 0 \\ 0 & 0 & 2\hat{\theta}_3(\mu_{xx} - \mu_x^2) \end{bmatrix} \quad (8.50)$$

Interpretation

Reading off the diagonal entries of $\hat{\mathbf{C}}$ gives

$$\text{var}[\theta_1|\{\hat{y}_k\}] = \frac{\hat{\theta}_3 \mu_{xx}}{N(\mu_{xx} - \mu_x^2)} \quad \text{var}[\theta_2|\{\hat{y}_k\}] = \frac{\hat{\theta}_3}{N(\mu_{xx} - \mu_x^2)} \quad (8.51)$$

$$\text{var}[\theta_3|\{\hat{y}_k\}] = \frac{2\hat{\theta}_3^2}{N} \quad (8.52)$$

According to the posterior PDF, θ_1 and θ_2 are correlated. Their covariance is given by $\hat{\mathbf{C}}(1, 2)$:

$$\text{cov}[\theta_1, \theta_2|\{\hat{y}_k\}] = -\frac{\mu_x \hat{\theta}_3}{N(\mu_{xx} - \mu_x^2)} \quad (8.53)$$

The negative sign implies that θ_1 and θ_2 are negatively correlated when $\mu_x > 0$ (and positive correlated when $\mu_x < 0$). Intuitively, when $\mu_x > 0$ a line with a bigger slope (i.e., bigger θ_2) will need to have a lower y-intercept (i.e., smaller θ_1) in order to achieve the same ‘fit’ to data.

Illustration with Synthetic Data

Consider the synthetic data (dots) in Fig. 8.7, which were generated according to $\hat{y}_k = 1 + x_k + \varepsilon_k$, $N = 30$ and $\{\varepsilon_k\}_{k=1}^N$ being i.i.d. zero-mean Gaussian with variance 0.2. The data set gave

$$\mu_x = 0.4354 \quad \mu_y = 1.3615 \quad \mu_{xx} = 0.2801 \quad \mu_{yy} = 2.0610 \quad \mu_{xy} = 0.6604$$

Figure 8.8 shows the posterior marginal PDFs. Similar to the last example, except for the prediction error variance θ_3 , the posterior marginal PDFs are approximately Gaussian. ■

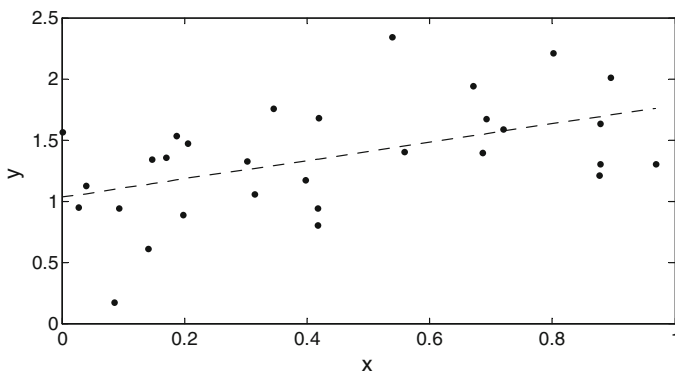


Fig. 8.7 Data in Example 8.5. Dots data. Dashed line model prediction at the MPV of parameters

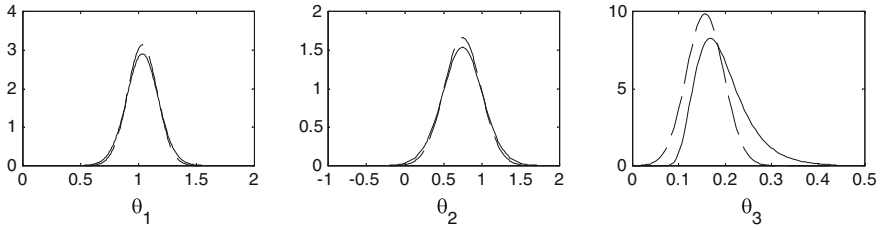


Fig. 8.8 Posterior marginal PDFs in Example 8.5. *Solid line* exact; *Dashed line* Gaussian approximation

8.6 Locally Identifiable Problems

When the posterior PDF has more than one peak, it may be approximated using a similar strategy as in the globally identifiable case. Let $\{\hat{\boldsymbol{\theta}}_i\}_{i=1}^m$ denote the local MPVs (m is their number) and $\hat{\mathbf{H}}_i$ denote the Hessian of $L(\mathbf{Z}, \boldsymbol{\theta}) = -\ln[p(\mathbf{Z}|\boldsymbol{\theta})p(\boldsymbol{\theta})]$ w.r.t. $\boldsymbol{\theta}$ and evaluated at $\hat{\boldsymbol{\theta}}_i$. One may approximate the posterior PDF as a mixture Gaussian PDFs centered on the MPVs:

$$p(\boldsymbol{\theta}|\mathbf{Z}) \approx \sum_{i=1}^m w_i \phi(\boldsymbol{\theta}; \hat{\boldsymbol{\theta}}_i, \hat{\mathbf{C}}_i) \quad (8.54)$$

where $\{w_i\}_{i=1}^m$ are weights that sum to 1; and

$$\phi(\boldsymbol{\theta}; \hat{\boldsymbol{\theta}}_i, \hat{\mathbf{C}}_i) = (2\pi)^{-n_{\boldsymbol{\theta}}/2} |\hat{\mathbf{C}}_i|^{-1/2} \exp \left[-\frac{1}{2} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_i)^T \hat{\mathbf{C}}_i^{-1} (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_i) \right] \quad (8.55)$$

is a Gaussian PDF with mean $\hat{\boldsymbol{\theta}}_i$ and covariance matrix $\hat{\mathbf{C}}_i = \hat{\mathbf{H}}_i^{-1}$. The approximation in (8.54) is unlikely to be good when the local MPVs are close because their interaction has not been accounted for. The weights $\{w_i\}_{i=1}^m$ may be determined so that the ratios of the approximated PDF at the local MPVs are the same as those of the exact posterior PDF:

$$p(\hat{\boldsymbol{\theta}}_i|\mathbf{Z}) \approx \sum_{j=1}^m w_j \phi(\hat{\boldsymbol{\theta}}_i; \hat{\boldsymbol{\theta}}_j, \hat{\mathbf{C}}_j) \approx w_i \phi(\hat{\boldsymbol{\theta}}_i; \hat{\boldsymbol{\theta}}_i, \hat{\mathbf{C}}_i) = w_i (2\pi)^{-n_{\boldsymbol{\theta}}/2} |\hat{\mathbf{C}}_i|^{-1/2} \quad (8.56)$$

where we have taken $\phi(\boldsymbol{\theta}_i; \hat{\boldsymbol{\theta}}_j, \hat{\mathbf{C}}_j) \approx 0$ for $j \neq i$. Since $p(\hat{\boldsymbol{\theta}}_i|\mathbf{Z}) \propto p(\mathbf{Z}|\hat{\boldsymbol{\theta}}_i)p(\hat{\boldsymbol{\theta}}_i)$, this implies

$$w_i \propto |\hat{\mathbf{C}}_i|^{1/2} p(\mathbf{Z}|\hat{\boldsymbol{\theta}}_i)p(\hat{\boldsymbol{\theta}}_i) \quad (8.57)$$

Although the above suggests that the strategy for globally identifiable problems can in principle be extended to the case of locally identifiable problems, in reality the computational effort can be significantly larger. This is because the number of local MPVs is unknown and how to find them is another problem. Unless a preliminary (often qualitative) analysis is performed to ascertain the number of local MPVs, practically the sum in (8.54) is only over all local MPVs that have been found. ‘Global optimization’ algorithms aim at locating multiple minima. See, e.g., Spall (2003) and Marti (2010).

8.7 Unidentifiable Problems

A system identification problem is unidentifiable when there is a manifold (continuum) of points in the model parameter space along which the posterior PDF is constant. This situation reflects that the available information is not sufficient to resolve even locally the plausibility of model parameters. Over-parameterization is one common cause. Attempts were made to devise computational algorithms to generate points on the manifold (Katafygiotis et al. 1998, 2000; Katafygiotis and Lam 2002). The amount of information required to describe the manifold grows dramatically with the number of model parameters and so the idea did not flourish.

Monte Carlo methods have been found to provide a promising computational tool for Bayesian inference problems in general and they are still under intensive research. The basic idea is to generate random samples according to the posterior PDF and estimate the posterior statistics by averaging over them. Generally, the expectation of a quantity of interest $h(\boldsymbol{\theta})$ (say) when $\boldsymbol{\theta}$ is distributed as $p(\boldsymbol{\theta}|\mathbf{Z})$ can be estimated by

$$E[h(\boldsymbol{\theta})|\mathbf{Z}] \approx \frac{1}{M} \sum_{k=1}^M h(\boldsymbol{\theta}_k) \quad (8.58)$$

where $\{\boldsymbol{\theta}_k\}_{k=1}^M$ are samples drawn from $p(\boldsymbol{\theta}|\mathbf{Z})$. This idea is simple but its efficient implementation is never straightforward. Generating samples according to an arbitrarily given distribution is a highly non-trivial task. One powerful class of methods is Markov Chain Monte Carlo simulation (MCMC), which has revolutionized the field of Bayesian inference. The random samples are generated from a specially designed Markov chain whose distribution converges to the target distribution.

MCMC was originally invented by Metropolis et al. (1953) in statistical physics. A major generalization was due to Hastings (1970). The method has found applications in a wide spectrum of disciplines involving Bayesian inference, e.g., astronautics, biostatistics, image processing, financial engineering (Malakoff 1999; Beichl and Sullivan 2000). An extensive body of research exists and is well-documented (Fishman 1996; Gilks et al. 1998; Liu 2001; Robert and Casella

2004). See Au and Wang (2014) for an introduction without pre-requisites in Markov Chain theory. See also the MCMC Preprint service available on the internet, <http://www.statslab.cam.ac.uk/~mcmc/>). The development of efficient MCMC algorithms in different disciplines is an ever growing area. Developments in structural dynamics and civil engineering may be referred to, e.g., Beck and Au (2002), Ching and Chen (2007), Cheung and Beck (2009) and Straub and Papaioannou (2015). See Green and Worden (2014) for a tutorial in the context of structural dynamics.

8.8 Model Class Selection

In the discussion so far, modeling assumptions have been omitted in conditioning statements as they are understood to be the same throughout. This conditioning matters and should be reinstalled when one compares the identification results between two Bayesian procedures formulated based on different assumptions; or when one evaluates the plausibility of a model compared to another, i.e., in a ‘model selection’ problem.

Let $\mathcal{M}$ denote a given set of modeling assumptions necessary to formulate the likelihood function $p(\mathbf{Z}|\boldsymbol{\theta})$. It is referred as a ‘model class’ (Beck 2010), i.e., the set of models induced by all possible values of $\boldsymbol{\theta}$. Indicating modeling assumptions explicitly, the posterior distribution in (8.3) reads

$$p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M}) = p(\mathbf{Z}|\boldsymbol{\theta}, \mathcal{M}) \frac{p(\boldsymbol{\theta}|\mathcal{M})}{p(\mathbf{Z}|\mathcal{M})} \quad (8.59)$$

A different model class $\mathcal{M}$ will give a different posterior distribution $p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})$ and so there is no ‘universal’ or ‘inherent’ posterior distribution. Identification results all depend on the modeling assumptions applied (and the data used). Analysis results are as good as the modeling assumptions. A natural question is what a good model class is. Based solely on probability reasoning and information from data (i.e., setting aside convenience, computational effort, etc.), this is answered by $p(\mathcal{M}|\mathbf{Z})$, which measures the plausibility of the model class for representing the system and its associated uncertainty based on the information implied by data. The higher the $p(\mathcal{M}|\mathbf{Z})$ the better.

8.8.1 Comparing Model Classes with Evidence

Using Bayes’ theorem,

$$p(\mathcal{M}|\mathbf{Z}) = p(\mathbf{Z}|\mathcal{M}) \frac{p(\mathcal{M})}{p(\mathbf{Z})} \quad (8.60)$$

The term $p(\mathcal{M})$ reflects prior information. The term $p(\mathbf{Z}|\mathcal{M})$ incorporates the information from the data and is often called the ‘evidence’ (or ‘marginal likelihood’) for the model class. It is the normalizing constant in the posterior distribution of $\boldsymbol{\theta}$ in (8.59). Interestingly, it does not play a material role in the identification of $\boldsymbol{\theta}$ for a given model class $\mathcal{M}$ but it is now the fundamental quantity that determines how plausible $\mathcal{M}$ is.

While $p(\mathcal{M})$ can be assigned based on prior information and $p(\mathbf{Z}|\mathcal{M})$ can be calculated when data is given, $p(\mathbf{Z})$ is still unknown and subjected to interpretation. Implicit in (8.60) is the set of candidate model classes such that, by the theorem of total probability,

$$p(\mathbf{Z}) = \sum_{\mathcal{M}} p(\mathbf{Z}|\mathcal{M})p(\mathcal{M}) \quad (8.61)$$

where the sum is over all candidate model class $\mathcal{M}$. Trying to determine $p(\mathcal{M}|\mathbf{Z})$ in an ‘absolute’ sense requires one to exhaust all possible candidates, which is not practical (if possible). It is possible, however, to tell in a relative sense whether one model class $\mathcal{M}_1$ is better than another model class $\mathcal{M}_2$, because $p(\mathbf{Z})$ cancels out in the ratio:

$$\frac{p(\mathcal{M}_2|\mathbf{Z})}{p(\mathcal{M}_1|\mathbf{Z})} = \frac{p(\mathbf{Z}|\mathcal{M}_2)p(\mathcal{M}_1)}{p(\mathbf{Z}|\mathcal{M}_1)p(\mathcal{M}_2)} \quad (8.62)$$

8.8.2 Model Trade-off

The computational problem in model class selection lies in the determination of $p(\mathbf{Z}|\mathcal{M})$ for a given $\mathcal{M}$. One approach is to express it via the theorem of total probability:

$$p(\mathbf{Z}|\mathcal{M}) = \int p(\mathbf{Z}|\boldsymbol{\theta}, \mathcal{M})p(\boldsymbol{\theta}|\mathcal{M})d\boldsymbol{\theta} \quad (8.63)$$

The difficulty in calculating $p(\mathbf{Z}|\mathcal{M})$ is similar to that in calculating the posterior statistics of $\boldsymbol{\theta}$ because the integrand is proportional to the posterior distribution $p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})$. Computational strategies have been investigated with insights on what constitute a good model, e.g., Beck and Yuen (2004), Muto and Beck (2008), Cheung and Beck (2010).

For globally identifiable problems, the Gaussian approximation in Sect. 8.5 gives

$$p(\mathbf{Z}|\mathcal{M}) \approx \underbrace{p(\mathbf{Z}|\hat{\boldsymbol{\theta}}, \mathcal{M})}_{\text{Prefer model fit to data}} \times \underbrace{(2\pi)^{n_\theta/2} |\hat{\mathbf{C}}|^{1/2} p(\hat{\boldsymbol{\theta}}|\mathcal{M})}_{\text{Ockham factor that penalizes parameterization}} \quad (8.64)$$

where $\hat{\boldsymbol{\theta}}$ is the MPV and $\hat{\mathbf{C}}$ is the covariance matrix associated with the posterior distribution $p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})$. The term $p(\mathbf{Z}|\hat{\boldsymbol{\theta}}, \mathcal{M})$ is larger for models that ‘fit’ the data better and hence it tends to favor those with more parameters. The term $(2\pi)^{n_\theta/2} |\hat{\mathbf{C}}|^{1/2} p(\hat{\boldsymbol{\theta}}|\mathcal{M})$ is called the ‘Ockham factor’ (Gull 1988; Jefferys and Berger 1992). It tends to decrease with the number of parameters n_θ in $\boldsymbol{\theta}$ and hence can be viewed as a penalty against parameterization. For example, when the prior distribution is Gaussian with mean $\boldsymbol{\theta}_0$ and covariance matrix $\mathbf{C}_0$,

$$\text{Ockham factor} = \frac{|\hat{\mathbf{C}}|^{1/2}}{|\mathbf{C}_0|^{1/2}} \exp \left[-\frac{1}{2} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \mathbf{C}_0^{-1} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \right] \quad (8.65)$$

When the data is informative about $\boldsymbol{\theta}$, $|\hat{\mathbf{C}}|/|\mathbf{C}_0|$ is smaller than 1 and decreases with n_θ .

Equation (8.64) spells out quantitatively the traditional wisdom that a good model trades off between the amount of parameterization and the agreement with data. In the general case, the trade-off can be stated as (see the end for proof)

$$\ln p(\mathbf{Z}|\mathcal{M}) = \underbrace{E[\ln p(\mathbf{Z}|\boldsymbol{\theta}, \mathcal{M})|\mathbf{Z}, \mathcal{M}]}_{\text{Prefer model fit to data}} - \underbrace{E \left[\ln \frac{p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})}{p(\boldsymbol{\theta}|\mathcal{M})} \middle| \mathbf{Z}, \mathcal{M} \right]}_{\text{Relative entropy} \geq 0} \quad (8.66)$$

where

$$\begin{aligned} E[\ln p(\mathbf{Z}|\boldsymbol{\theta}, \mathcal{M})|\mathbf{Z}, \mathcal{M}] &= \int [\ln p(\mathbf{Z}|\boldsymbol{\theta}, \mathcal{M})] p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M}) d\boldsymbol{\theta} \\ E \left[\ln \frac{p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})}{p(\boldsymbol{\theta}|\mathcal{M})} \middle| \mathbf{Z}, \mathcal{M} \right] &= \int \ln \left[\frac{p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})}{p(\boldsymbol{\theta}|\mathcal{M})} \right] p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M}) d\boldsymbol{\theta} \end{aligned} \quad (8.67)$$

The term $E\{\ln[p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})/p(\boldsymbol{\theta}|\mathcal{M})]|\mathbf{Z}, \mathcal{M}\}$ is the ‘relative entropy’ of the posterior distribution to the prior distribution. It reflects the amount of information gained from the data and tends to be higher for models with more parameters. Subtracting this term therefore reflects a penalty against parameterization.

Proof of (8.66) (Model Trade-Off)

Rearranging (8.59) and taking logarithm,

$$\begin{aligned}
\ln p(\mathbf{Z}|\mathcal{M}) &= \ln \left[\underbrace{p(\mathbf{Z}|\boldsymbol{\theta}, \mathcal{M}) \frac{p(\boldsymbol{\theta}|\mathcal{M})}{p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})}}_{\text{constant independent of } \boldsymbol{\theta}} \right] \\
&= \int \ln \left[p(\mathbf{Z}|\boldsymbol{\theta}, \mathcal{M}) \frac{p(\boldsymbol{\theta}|\mathcal{M})}{p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})} \right] p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M}) d\boldsymbol{\theta} \\
&= \underbrace{\int \ln [p(\mathbf{Z}|\boldsymbol{\theta}, \mathcal{M})] p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M}) d\boldsymbol{\theta}}_{E[\ln p(\mathbf{Z}|\boldsymbol{\theta}, \mathcal{M})|\mathbf{Z}, \mathcal{M}]} - \underbrace{\int \ln \left[\frac{p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})}{p(\boldsymbol{\theta}|\mathcal{M})} \right] p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M}) d\boldsymbol{\theta}}_{E \left[\ln \frac{p(\boldsymbol{\theta}|\mathbf{Z}, \mathcal{M})}{p(\boldsymbol{\theta}|\mathcal{M})} \middle| \mathbf{Z}, \mathcal{M} \right]}
\end{aligned} \tag{8.68}$$

■

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Chapter 9

Classical Statistical Inference

Abstract This chapter introduces the classical statistical approach for system identification in a general context. This is commonly referred as a ‘non-Bayesian’ approach and is currently the conventional perspective in operational modal analysis. Basic quantification of statistical estimators are presented and illustrated with examples. The Cramér-Rao bound and Fisher information matrix are presented to provide the theoretical lower bound for the variance of unbiased estimators. Maximum likelihood estimators and their asymptotic properties for large data size are discussed. The Bayesian and classical statistical approach have different philosophies but share some similarities in mathematics. These shall be discussed so that the two approaches can be applied correctly and advantage can be taken of their mathematical tools developed.

Keywords Cramér-Rao bound • Fisher information matrix • Maximum likelihood • Uncertainty law

In Chap. 8 we discussed the Bayesian approach for making inference about the parameters of interest based on available data. The ‘classical statistical approach’ is a large class of methods adopting a different perspective to the inference problem. In this approach, an unknown parameter is estimated by an ‘estimator’ constructed as a function of the available data. The quality of the estimator is assessed through its statistical properties associated with repeated experiments. This is referred as a ‘non-Bayesian’ or ‘frequentist’ inference approach, as it views probability as a relative frequency of occurrence of random events in the long run, in contrast to the Bayesian interpretation as a measure of plausibility for given information.

Within the classical statistical approach, the ‘maximum likelihood’ (ML) estimator is constructed as the model parameter value that maximizes the likelihood function. The latter refers to the probability distribution of data for a given model class and value of model parameter vector, the same definition as in the Bayesian approach. Perhaps due to the use of the likelihood function, the ML

method is sometimes perceived to be equivalent to a Bayesian method with a uniform prior distribution. This is a misconception. Their mathematical structures share some similarities for globally identifiable problems but their philosophies are fundamentally different, which affects the way identification uncertainty is measured and interpreted.

In this chapter we discuss the classical statistical approach. We clarify similarities and differences with the Bayesian approach. Discovering similarities allows mathematical tools to be shared for solving computational problems. Recognizing differences allows proper interpretation of the identification results.

As in Chap. 8, without much loss of generality, the unknown parameter θ and data $\mathbf{Z}$ are assumed to be continuous-valued. Comparing this chapter with Chap. 8, one will find that the mathematical roles of θ and $\mathbf{Z}$ are in some sense swapped. In Chap. 8, θ was a parameter value in the posterior distribution but in this chapter it is the ‘unknown true value’ that is always fixed. Data $\mathbf{Z}$ was fixed in all conditioning statements as the actual data being used but now it is treated as a random variable whose variability is averaged to obtain ensemble statistics of the estimator. Identification uncertainty is now associated with the variability of the estimator for θ , rather than θ itself. All derived results on the statistical properties of estimator do not depend on data $\mathbf{Z}$ but rather on the unknown true value θ . In the Bayesian approach, all inference results depend on $\mathbf{Z}$; and the unknown true parameter value (or such concept) is irrelevant. See Table 9.2 in Sect. 9.6.1 for a summary of comparisons between the two approaches.

A series of examples are presented to illustrate the concepts in Chaps. 8 and 9, centering around three inference problems. They are listed in Table 9.1 for easy reference.

Table 9.1 Examples in Chaps. 8 and 9 centering around three inference problems

Problem	Examples in Chap. 8	Examples in Chap. 9
Mean parameter	8.3 (Exact posterior) 8.4 (Gaussian posterior)	9.1 (Estimator) 9.4 (CRB) 9.7 (Easy CRB) 9.10 (FIM) 9.12 (Asymptotics and FIM)
Linear regression	8.5 (Gaussian posterior)	9.2 (Estimator) 9.5 (CRB) 9.8 (Easy CRB) 9.11 (FIM)
Modal identification		9.3 (MLE) 9.6 (CRB) 9.9 (Easy CRB)

CRB Cramér-Rao bound; *FIM* Fisher information matrix; *MLE* maximum likelihood estimation

9.1 Statistical Estimators

In classical statistical estimation, it is assumed that there is a ‘true’ value $\boldsymbol{\theta}$ ($n_\theta \times 1$) of the set of unknown model parameters that results in the measured data $\mathbf{Z}$. An estimator $\mathbf{G}(\mathbf{Z})$ ($n_\theta \times 1$) is designed to estimate $\boldsymbol{\theta}$. Different values of $\mathbf{Z}$ lead to different values of $\mathbf{G}(\mathbf{Z})$. Identification uncertainty is measured as the variability of $\mathbf{G}(\mathbf{Z})$ over different samples of $\mathbf{Z}$. When it is desired to assess the theoretical statistical properties of $\mathbf{G}(\mathbf{Z})$, it is assumed that $\mathbf{Z}$ is distributed as the likelihood function $p(\mathbf{Z}|\boldsymbol{\theta})$.

9.1.1 Quality Statistics

How $\mathbf{G}(\mathbf{Z})$ depends on $\mathbf{Z}$ is open to choice and so the estimator is not unique. One estimator can be better than another estimator in one sense but worse in another sense. Two common statistics for assessing the quality of an estimator are the mean $E[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}]$ and covariance matrix $\text{cov}[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}]$:

$$E[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}] = \int \mathbf{G}(\mathbf{z})p(\mathbf{z}|\boldsymbol{\theta})d\mathbf{z} \quad (9.1)$$

$$\begin{aligned} \text{cov}[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}] &= E\{[\mathbf{G}(\mathbf{Z}) - \mathbf{g}(\boldsymbol{\theta})][\mathbf{G}(\mathbf{Z}) - \mathbf{g}(\boldsymbol{\theta})]^T|\boldsymbol{\theta}\} \\ &= \int [\mathbf{G}(\mathbf{z}) - \mathbf{g}(\boldsymbol{\theta})][\mathbf{G}(\mathbf{z}) - \mathbf{g}(\boldsymbol{\theta})]^T p(\mathbf{z}|\boldsymbol{\theta})d\mathbf{z} \end{aligned} \quad (9.2)$$

where

$$\mathbf{g}(\boldsymbol{\theta}) = E[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}] \quad (9.3)$$

In these equations, the expectations are all conditional on the unknown true value $\boldsymbol{\theta}$ because the distribution of $\mathbf{Z}$ depends on $\boldsymbol{\theta}$. This dependence is usually omitted but is deliberately indicated throughout this chapter. Omitting it does not cause any mathematical problem when one is confined to the classical statistical approach, because then $\boldsymbol{\theta}$ is always considered fixed. It matters in this chapter because $\boldsymbol{\theta}$ may be viewed from a Bayesian perspective as a random vector whose probability distribution is a function of the conditioning information. According to the theorem of total probability, the expectation $E[\mathbf{G}(\mathbf{Z})]$ without the conditioning refers to

$$E[\mathbf{G}(\mathbf{Z})] = \int E[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}]p(\boldsymbol{\theta})d\boldsymbol{\theta} = \int \int \mathbf{G}(\mathbf{z})p(\mathbf{z}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\mathbf{z}d\boldsymbol{\theta} \quad (9.4)$$

This is the expectation of $\mathbf{G}(\mathbf{Z})$ under the prior distribution $p(\boldsymbol{\theta})$. It tells us what the estimator would be like in the absence of data (as in a ‘forward’ prediction

problem), instead of identifying what θ is in the presence of data (as in a ‘backward’, or ‘inverse’, inference problem).

9.1.2 Bias and Convergence

In the classical statistical approach, an estimator $\mathbf{G}(\mathbf{Z})$ is ‘unbiased’ if its expectation is equal to the true value θ . It is ‘convergent’ if its variance tends to zero as data size increases. Intuitively, if $\mathbf{G}(\mathbf{Z})$ is unbiased, the scatter of its values for different samples of $\mathbf{Z}$ should be centered around θ . If it is convergent, the extent of scatter will diminish as data size increases. Unbiasedness is a nice intuitive property but it is not a prerequisite. Many useful estimators are biased for finite sample size but asymptotically unbiased as sample size increases, which is sufficient for practical purposes. It is important for an estimator to be convergent, however.

9.1.3 Empirical Statistics

The difficulty of assessing the statistical properties of an estimator depends on its relationship with data. For complex problems, it may not be possible to obtain analytical expressions, in which case one may resort to numerical means. Empirically, the statistics can be estimated by averaging over multiple data sets, say, $\{\mathbf{Z}_k\}_{k=1}^M$:

$$E[\mathbf{G}(\mathbf{Z})|\theta] \approx \tilde{\mathbf{G}} = \frac{1}{M} \sum_{k=1}^M \mathbf{G}(\mathbf{Z}_k) \quad (9.5)$$

$$\text{cov}[\mathbf{G}|\theta] \approx \frac{1}{M} \sum_{k=1}^M [\mathbf{G}(\mathbf{Z}_k) - \tilde{\mathbf{G}}][\mathbf{G}(\mathbf{Z}_k) - \tilde{\mathbf{G}}]^T \quad (9.6)$$

Calculating these statistics requires multiple data sets, even though it is sufficient to have a single data set to estimate θ . If $\{\mathbf{Z}_k\}_{k=1}^M$ are modeled as i.i.d. random variables then by the Strong Law of Large Numbers the sample average will converge to its expectation as M increases. This is only a theoretical statement applicable for synthetic data. In reality, it need not hold because the data may not follow the assumed probability distribution, or there is no such distribution at all. Nature does not care or know about distributions. This is one fundamental difference from the Bayesian approach: probability distribution is a property (although unknown) of nature’s random events from the frequentist perspective; but it is a modeling choice describing the analyst’s uncertainty about variables from the Bayesian perspective.

9.2 Maximum Likelihood Estimator

In the maximum likelihood (ML) method, the estimator is defined as the value of θ that maximizes the likelihood function $p(\mathbf{Z}|\theta)$ for given data $\mathbf{Z}$. Here, there is an abuse of notation that θ is a (generic) parameter value of a function rather than the true value that results in the data $\mathbf{Z}$. For a globally identifiable problem, the ML estimator is mathematically the same as the posterior most probable value (MPV) in the Bayesian approach assuming a uniform prior distribution. Let $\hat{\theta}(\mathbf{Z})$ denote the ML estimator, whose dependence on $\mathbf{Z}$ has been emphasized. Then $\nabla_{\theta} p(\mathbf{Z}|\theta)|_{\theta=\hat{\theta}(\mathbf{Z})} = \mathbf{0}$ (gradient) and $\nabla_{\theta}^2 p(\mathbf{Z}|\theta)|_{\theta=\hat{\theta}(\mathbf{Z})}$ (Hessian) is a negative definite matrix. Equivalently, $\hat{\theta}(\mathbf{Z})$ minimizes the ‘negative log-likelihood function’ (NLLF) $L(\mathbf{Z}, \theta) = -\ln p(\mathbf{Z}|\theta)$ w.r.t. θ for a given $\mathbf{Z}$, i.e., $\nabla_{\theta} L(\mathbf{Z}, \theta)|_{\theta=\hat{\theta}(\mathbf{Z})} = \mathbf{0}$ and $\nabla_{\theta}^2 L(\mathbf{Z}, \theta)|_{\theta=\hat{\theta}(\mathbf{Z})}$ is a positive definite matrix. Working with the NLLF is often algebraically easier than the likelihood function.

The ML estimator has good theoretical asymptotic properties when the data is assumed to be distributed as the likelihood function. It is convergent and asymptotically unbiased as data size increases. Its variance asymptotically achieves the smallest possible value of any unbiased estimator. These are discussed in Sect. 9.5 later.

The reader may notice that the same symbol $\hat{\theta}$ is used for the ML estimator and the posterior MPV (Bayesian) in Chap. 8. This is deliberate because they are mathematically the same for globally identifiable problems with a uniform prior distribution. Nevertheless, conceptual problems arise with the ML estimator when the problem is not globally identifiable. For example, if the NLLF has two local minima, one must resort to take only one of them as the ML estimate, because it is logically impossible to have a parameter assuming two distinct (true) values.

Example 9.1 (Mean parameter, estimator) Consider estimating θ_1 from scalar data $\{\hat{y}_k\}_{k=1}^N$ modeled by

$$\hat{y}_k = \theta_1 + \varepsilon_k \quad (9.7)$$

where $\{\varepsilon_k\}_{k=1}^N$ are prediction errors. This problem was investigated in Example 8.3 using the Bayesian approach. Here we adopt a classical statistical approach.

Estimator for θ_1

Based on ‘intuition’, an estimator G_1 for θ_1 is proposed to be the average of the data:

$$G_1 = \frac{1}{N} \sum_{k=1}^N \hat{y}_k \quad (9.8)$$

To analyze the statistical properties of G_1 , we need to impose some assumptions on $\{\hat{y}_k\}_{k=1}^N$, or equivalently on $\{\varepsilon_k\}_{k=1}^N$. As in Example 8.3, assume that $\{\varepsilon_k\}_{k=1}^N$ are i.i.d. zero-mean Gaussian variables with variance θ_2 . Equation (9.7) then implies that, given $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$, $\{\hat{y}_k\}_{k=1}^N$ are i.i.d. Gaussian variables with mean θ_1 and variance θ_2 . This gives

$$E[G_1|\boldsymbol{\theta}] = \frac{1}{N} \sum_{k=1}^N \underbrace{E[\hat{y}_k|\boldsymbol{\theta}]}_{\theta_1} = \theta_1 \quad (9.9)$$

$$\text{var}[G_1|\boldsymbol{\theta}] = \frac{1}{N^2} \sum_{k=1}^N \underbrace{\text{var}[\hat{y}_k|\boldsymbol{\theta}]}_{\theta_2} = \frac{\theta_2}{N} \quad (9.10)$$

Thus, G_1 is unbiased ($E[G_1|\boldsymbol{\theta}] = \theta_1$) and convergent ($\text{var}[G_1|\boldsymbol{\theta}] \rightarrow 0$ as $N \rightarrow \infty$). Note that these expressions depend on θ_1 and θ_2 , which are unknown.

Estimator for θ_2

By intuition, an estimator G_2 for θ_2 is constructed as the average of the squared difference between $\hat{y}_k$ and G_1 :

$$G_2 = \frac{1}{N} \sum_{k=1}^N (\hat{y}_k - G_1)^2 = \left(\frac{1}{N} \sum_{k=1}^N \hat{y}_k^2 \right) - G_1^2 \quad (9.11)$$

after algebra. Taking expectation and using the identities

$$E[\hat{y}_k^2|\boldsymbol{\theta}] = \underbrace{E[\hat{y}_k|\boldsymbol{\theta}]^2}_{\theta_1^2} + \underbrace{\text{var}[\hat{y}_k|\boldsymbol{\theta}]}_{\theta_2} \quad E[G_1^2|\boldsymbol{\theta}] = \underbrace{E[G_1|\boldsymbol{\theta}]^2}_{\theta_1^2} + \underbrace{\text{var}[G_1|\boldsymbol{\theta}]}_{\theta_2/N} \quad (9.12)$$

we have

$$E[G_2|\boldsymbol{\theta}] = \left\{ \frac{1}{N} \sum_{k=1}^N \underbrace{E[\hat{y}_k^2|\boldsymbol{\theta}]}_{\theta_1^2 + \theta_2} \right\} - \underbrace{E[G_1^2|\boldsymbol{\theta}]}_{\theta_1^2 + \theta_2/N} = \left(1 - \frac{1}{N} \right) \theta_2 \quad (9.13)$$

Thus, G_2 is biased ($E[G_2|\boldsymbol{\theta}] \neq \theta_2$) for finite sample size N but asymptotically unbiased ($E[G_2|\boldsymbol{\theta}] \rightarrow \theta_2$) as $N \rightarrow \infty$. Deriving the variance of G_2 directly from (9.11) is more involved. As a short cut, one can use the fact that $NG_2/\theta_2 = \sum_{k=1}^N (\hat{y}_k - G_1)^2/\theta_2$ has a Chi-square distribution with $(N - 1)$ degrees of freedom and so its variance is equal to $2(N - 1)$. Taking variance on both sides and rearranging gives

$$\text{var}[G_2|\boldsymbol{\theta}] = \frac{2\theta_2^2}{N} \left(1 - \frac{1}{N} \right) \quad (9.14)$$

Since $\text{var}[G_2|\boldsymbol{\theta}] \rightarrow 0$ as $N \rightarrow \infty$, G_2 is convergent.

Modified Estimator

An unbiased estimator for θ_2 , denoted by G'_2 , can be constructed by correcting for the factor $(1 - 1/N)$ in the expression of $E[G_2|\boldsymbol{\theta}]$ in (9.13) that causes the bias, i.e.,

$$G'_2 = (1 - \frac{1}{N})^{-1} G_2 = \frac{1}{N-1} \sum_{k=1}^N (\hat{y}_k - G_1)^2 \quad (9.15)$$

so that $E[G'_2|\boldsymbol{\theta}] = (1 - 1/N)^{-1} E[G_2|\boldsymbol{\theta}] = \theta_2$. This is the conventional estimator used in statistics. The variance of G'_2 is

$$\text{var}[G'_2|\boldsymbol{\theta}] = (1 - \frac{1}{N})^{-2} \text{var}[G_2|\boldsymbol{\theta}] = \frac{2\theta_2^2}{N} \left(1 - \frac{1}{N}\right)^{-1} \quad (9.16)$$

Note that $\text{var}[G'_2|\boldsymbol{\theta}] > \text{var}[G_2|\boldsymbol{\theta}]$. An estimator should not be preferred simply because it has a smaller variance. Its bias needs to be taken into account.

The mathematical steps and considerations in this example are typical in classical statistics. Some observations are in order. An estimator can be intuitive but biased (G_2). The expressions of G_1 and G_2 coincide with those of the posterior MPV in Example 8.4 and so they are ML estimators. This demonstrates that an ML estimator can be unbiased (G_1) or biased (G_2). The derived statistical properties of an estimator, e.g., (9.9) and (9.10), are in terms of the unknown true values θ_1 and θ_2 . ■

Example 9.2 (Linear regression, estimator) Consider estimating θ_1 and θ_2 from scalar data $\{\hat{y}_k\}_{k=1}^N$, assuming the linear regression model

$$\hat{y}_k = \theta_1 + \theta_2 x_k + \varepsilon_k \quad (9.17)$$

where $\{x_k\}_{k=1}^N$ are given; $\{\varepsilon_k\}_{k=1}^N$ are prediction errors. This problem was investigated in Example 8.5 using the Bayesian approach. Here we adopt a ‘least squares’ approach. An estimator (G_1, G_2) for (θ_1, θ_2) is constructed as the one that minimizes the following objective function defined as the sum of squared errors:

$$J(\theta_1, \theta_2) = \sum_{k=1}^N \varepsilon_k^2 = \sum_{k=1}^N (\hat{y}_k - \theta_1 - \theta_2 x_k)^2 \quad (9.18)$$

Expanding the square gives a quadratic function of (θ_1, θ_2) . Minimizing w.r.t. (θ_1, θ_2) gives

$$G_1 = \frac{\mu_{xx}\mu_y - \mu_x\mu_{xy}}{\mu_{xx} - \mu_x^2} \quad G_2 = \frac{\mu_{xy} - \mu_x\mu_y}{\mu_{xx} - \mu_x^2} \quad (9.19)$$

where

$$\begin{aligned}\mu_x &= \frac{1}{N} \sum_{k=1}^N x_k & \mu_y &= \frac{1}{N} \sum_{k=1}^N \hat{y}_k \\ \mu_{xx} &= \frac{1}{N} \sum_{k=1}^N x_k^2 & \mu_{yy} &= \frac{1}{N} \sum_{k=1}^N \hat{y}_k^2 & \mu_{xy} &= \frac{1}{N} \sum_{k=1}^N x_k \hat{y}_k\end{aligned}\quad (9.20)$$

The least squares approach is intuitive but heuristic (e.g., why squares?). It does not even involve probability concepts. The expressions of the estimators G_1 and G_2 coincide with those of the posterior MPVs in Example 8.5 that maximize the likelihood function. They are thus ML estimators.

Statistical Properties

To analyze the statistical properties of G_1 and G_2 , it is necessary to impose probabilistic assumptions on the data. Consistent with Example 8.5, assume that $\{\varepsilon_k\}_{k=1}^N$ are i.i.d. zero-mean Gaussian variables with variance θ_3 (unknown). The derived statistical properties will then depend on the unknown true value $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^T$. Equation (9.17) implies that, given $\boldsymbol{\theta}$, $\{\hat{y}_k\}_{k=1}^N$ are i.i.d. Gaussian variables with mean $\theta_1 + \theta_2 x_k$ and variance θ_3 . Correspondingly, μ_y , μ_{yy} and μ_{xy} are random variables whose distributions depend on $\boldsymbol{\theta}$. This gives, using (9.19),

$$E[G_1|\boldsymbol{\theta}] = \frac{\mu_{xx}E[\mu_y|\boldsymbol{\theta}] - \mu_x E[\mu_{xy}|\boldsymbol{\theta}]}{\mu_{xx} - \mu_x^2} = \frac{\mu_{xx}(\theta_1 + \theta_2 \mu_x) - \mu_x(\theta_1 \mu_x + \theta_2 \mu_{xx})}{\mu_{xx} - \mu_x^2} = \theta_1 \quad (9.21)$$

$$E[G_2|\boldsymbol{\theta}] = \frac{E[\mu_{xy}|\boldsymbol{\theta}] - \mu_x E[\mu_y|\boldsymbol{\theta}]}{\mu_{xx} - \mu_x^2} = \frac{(\theta_1 \mu_x + \theta_2 \mu_{xx}) - \mu_x(\theta_1 + \theta_2 \mu_x)}{\mu_{xx} - \mu_x^2} = \theta_2 \quad (9.22)$$

where we have used

$$E[\mu_y|\boldsymbol{\theta}] = \theta_1 + \theta_2 \mu_x \quad E[\mu_{xy}|\boldsymbol{\theta}] = \theta_1 \mu_x + \theta_2 \mu_{xx} \quad (9.23)$$

Equations (9.21) and (9.22) show that G_1 and G_2 are unbiased. Their covariance matrix is more involved to derive but can be shown to be

$$\begin{bmatrix} \text{var}[G_1|\boldsymbol{\theta}] & \text{cov}[G_1, G_2|\boldsymbol{\theta}] \\ \text{cov}[G_2, G_1|\boldsymbol{\theta}] & \text{var}[G_2|\boldsymbol{\theta}] \end{bmatrix} = \frac{\theta_3}{N(\mu_{xx} - \mu_x^2)} \begin{bmatrix} \mu_{xx} & -\mu_x \\ -\mu_x & 1 \end{bmatrix} \quad (9.24)$$

It depends on θ_3 , which is unknown.

Prediction Error Variance

In the present example, there is no need to estimate θ_3 if one is only interested in θ_1 or θ_2 . When it is desired to estimate θ_3 , e.g., so that the covariance matrix in (9.24) can be estimated, one will need to construct an estimator for it. One ‘intuitive’ estimator, denoted by G_3 , is the minimized average of ε_k^2 , i.e., $J(G_1, G_2)/N$:

$$G_3 = \frac{J(G_1, G_2)}{N} = \frac{1}{N} \sum_{k=1}^N (\hat{y}_k - G_1 - G_2 x_k)^2 \quad (9.25)$$

Expanding the square and collecting terms,

$$G_3 = \mu_{yy} + G_1^2 + G_2^2 \mu_{xx} - 2G_1 \mu_y - 2G_2 \mu_{xy} + 2G_1 G_2 \mu_x \quad (9.26)$$

Substituting G_1 and G_2 from (9.19) gives, after simplifying,

$$G_3 = \frac{-\mu_{yy}\mu_x^2 + 2\mu_x\mu_y\mu_{xy} - \mu_{xy}^2 - \mu_{xx}\mu_y^2 + \mu_{xx}\mu_{yy}}{\mu_{xx} - \mu_x^2} \quad (9.27)$$

This expression coincides with the posterior MPV of θ_3 in Example 8.5. Thus, G_3 is the ML estimator for θ_3 . It can be shown that

$$E[G_3|\boldsymbol{\theta}] = \left(1 - \frac{2}{N}\right) \theta_3 \quad (9.28)$$

and so G_3 is biased for finite N . An unbiased estimator for θ_3 , denoted by G'_3 , can be constructed by correcting for the factor $(1 - 2/N)$:

$$G'_3 = \left(1 - \frac{2}{N}\right)^{-1} G_3 = \frac{1}{N-2} \sum_{k=1}^N (\hat{y}_k - G_1 - G_2 x_k)^2 \quad (9.29)$$

This is the conventional estimator used in linear regression theory. ■

Example 9.3 (Modal identification, ML estimator) Consider identifying the natural frequency and damping ratio of a vibration mode using a single channel of ambient acceleration time history $\{\hat{y}_j\}_{j=0}^{N-1}$ measured on a structure. Here we construct an ML estimator using the sample power spectral density (PSD) of data within a selected resonance band of the mode. The sample PSD at the Fast Fourier Transform (FFT) frequency $f_k = k/N\Delta t$ (Hz) is $\hat{S}_k = |\hat{\mathcal{F}}_k|^2$, where $\hat{\mathcal{F}}_k = \sqrt{\Delta t/N} \sum_{j=0}^{N-1} \hat{y}_j e^{-2\pi i j k/N}$ is the scaled FFT of data; Δt (s) is the sampling interval. The likelihood function $p(\{\hat{S}_k\}|\boldsymbol{\theta})$ is the joint PDF of $\{\hat{S}_k\}$ for a given $\boldsymbol{\theta}$ (to be defined shortly). Within the selected band, the theoretical PSD is modeled as (Sect. 7.1)

$$S_k(\boldsymbol{\theta}) = SD_k(f, \zeta) + S_e \quad (9.30)$$

where

$$D_k(f, \zeta) = [(1 - \beta_k^2)^2 + (2\zeta\beta_k)^2]^{-1} \quad \beta_k = f/f_k \quad (9.31)$$

is the dynamic amplification factor, f (Hz) is the natural frequency, ζ is the damping ratio, S is the modal force PSD and S_e is the prediction error PSD. Here, $\boldsymbol{\theta} = [f, \zeta, S, S_e]^T$. Although the primary interest is to estimate f and ζ only, S and S_e need to be included in $\boldsymbol{\theta}$ so that the likelihood function can be obtained explicitly.

To obtain the likelihood function, we assume long data and make use of the asymptotic distribution of the sample PSD. According to Sect. 4.10.1, for a given $\boldsymbol{\theta}$, $\{\hat{S}_k\}$ are asymptotically independent and each $\hat{S}_k$ is exponentially distributed with mean $S_k(\boldsymbol{\theta})$. The likelihood function is then given by

$$p(\{\hat{S}_k\}|\boldsymbol{\theta}) = \prod_k S_k(\boldsymbol{\theta})^{-1} e^{-\hat{S}_k/S_k(\boldsymbol{\theta})} \quad (9.32)$$

where the product is taken over all frequencies within the selected band. The ML estimator is the value of $\boldsymbol{\theta}$ that minimizes the NLLF:

$$L(\{\hat{S}_k\}, \boldsymbol{\theta}) = -\ln p(\{\hat{S}_k\}|\boldsymbol{\theta}) = \sum_k \left[\ln S_k(\boldsymbol{\theta}) + \frac{\hat{S}_k}{S_k(\boldsymbol{\theta})} \right] \quad (9.33)$$

This is not a quadratic function of $\boldsymbol{\theta}$ and so it does not give a least squares estimate. Its form is not suggested heuristically, but is rather derived from modeling assumptions. As this function depends in a nontrivial manner on $\boldsymbol{\theta}$, it is difficult to obtain an analytical expression for the ML estimator. It is then difficult to investigate analytically whether the estimator is biased or what its variance is. This situation is typical in applications. One strategy is to investigate the bias and variance numerically using synthetic data before the estimator can be used with confidence. The Cramér-Rao bound in the next section provides useful information on the variance of general estimators. ■

9.3 Cramér-Rao Bound

When the estimator depends on data in a complicated manner, it is difficult to derive analytical expressions for its statistical properties. The ‘Cramér-Rao bound’ (CRB) (Rao 1945; Cramér 1946) gives the smallest possible variance that can be achieved by any unbiased estimator, assuming that the data is distributed as the likelihood function. Under mild assumptions on the likelihood function, it can be shown that for *any unbiased estimator* $\mathbf{G}(\mathbf{Z})$ for $\boldsymbol{\theta}$,

$$\text{cov}[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}] \geq \mathbf{J}(\boldsymbol{\theta})^{-1} \quad (9.34)$$

where $\text{cov}[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}]$ ($n_\theta \times n_\theta$) is the covariance matrix of $\mathbf{G}(\mathbf{Z})$, as defined in (9.2);

$$\begin{aligned} \mathbf{J}(\boldsymbol{\theta}) &= E\left\{ \underbrace{[\nabla_{\boldsymbol{\theta}} L(\mathbf{Z}, \boldsymbol{\theta})]^T}_{n_\theta \times 1} \underbrace{[\nabla_{\boldsymbol{\theta}} L(\mathbf{Z}, \boldsymbol{\theta})]}_{1 \times n_\theta} \middle| \boldsymbol{\theta} \right\} \\ &= \int [\nabla_{\boldsymbol{\theta}} L(\mathbf{z}, \boldsymbol{\theta})]^T [\nabla_{\boldsymbol{\theta}} L(\mathbf{z}, \boldsymbol{\theta})] p(\mathbf{z}|\boldsymbol{\theta}) d\mathbf{z} \end{aligned} \quad (9.35)$$

is the ‘Fisher information matrix’ (FIM) for $\boldsymbol{\theta}$; $L(\mathbf{Z}, \boldsymbol{\theta}) = -\ln p(\mathbf{Z}|\boldsymbol{\theta})$ is the NLLF; and $\nabla_{\boldsymbol{\theta}} L(\mathbf{Z}, \boldsymbol{\theta})$ is the gradient of $L(\mathbf{Z}, \boldsymbol{\theta})$ w.r.t. $\boldsymbol{\theta}$. Note that the (i, j) -entry of $\mathbf{J}(\boldsymbol{\theta})$ is $E[(\partial L / \partial \theta_i)(\partial L / \partial \theta_j) | \boldsymbol{\theta}]$. The equality in (9.34) holds if and only if

$$\mathbf{G}(\mathbf{Z}) = \underbrace{\boldsymbol{\theta}}_{n_\theta \times 1} - \underbrace{\mathbf{J}(\boldsymbol{\theta})^{-1}}_{n_\theta \times n_\theta} \underbrace{\nabla_{\boldsymbol{\theta}}^T L(\mathbf{Z}, \boldsymbol{\theta})}_{n_\theta \times 1} \quad (9.36)$$

Matrix Inequality and Sylvester’s Criterion

Equation (9.34) is a ‘matrix inequality’ which should be interpreted as the LHS minus RHS being a positive semi-definite matrix. That is, all eigenvalues of the following matrix are non-negative:

$$\Delta(\boldsymbol{\theta}) = \text{cov}[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}] - \mathbf{J}(\boldsymbol{\theta})^{-1} \quad (9.37)$$

The inequalities from the CRB are seldom obtained from the eigenvalues of $\Delta(\boldsymbol{\theta})$ as the latter are difficult to determine analytically. An alternative way to use (9.34) is via the ‘Sylvester’s criterion’ (Gilbert 1991; Meyer 2000), which says that a Hermitian matrix is positive semi-definite if and only if all its principal minors are non-negative. That is, let Δ_{ij} denote the (i, j) -entry of $\Delta(\boldsymbol{\theta})$; and Δ_p denote the upper left $p \times p$ partition:

$$\Delta_p = \begin{bmatrix} \Delta_{11} & \cdots & \Delta_{1p} \\ \vdots & \ddots & \vdots \\ \Delta_{p1} & \cdots & \Delta_{pp} \end{bmatrix} \quad (9.38)$$

Then (9.34) is equivalent to

$$|\Delta_p| \geq 0 \quad \text{for all } p = 1, \dots, n_\theta \quad (9.39)$$

This provides an equivalent but easier way to derive the inequalities implied from the CRB when the determinants are easier to obtain than eigenvalues. Even easier bounds can be obtained as necessary conditions deduced from the CRB, at the expense of loosening them. This is discussed in Sect. 9.3.1 later.

Generalization

The CRB presented above is for $\mathbf{G}(\mathbf{Z})$ being an unbiased estimator for $\boldsymbol{\theta}$. It holds in a more general context where $\mathbf{G}(\mathbf{Z})$ can be any function of $\mathbf{Z}$. This is discussed in

Sect. 9.3.2 and proven in Sect. 9.3.3. For the CRB to hold, the NLLF should be twice differentiable w.r.t. $\boldsymbol{\theta}$; and the order of expectation over $\mathbf{Z}$ and differentiation w.r.t. $\boldsymbol{\theta}$ can be swapped. These conditions are often satisfied.

The CRB is only a theoretical bound, assuming that the data $\mathbf{Z}$ is distributed as the likelihood function. It need not agree with the variability of the estimator empirically estimated based on real data as in (9.6). This is simply because, besides estimation error, the data need not be distributed as the likelihood function assumed. Nevertheless, the CRB is very useful for investigating the achievable precision of estimators in classical statistics. It is also related to the leading order of the posterior covariance matrix in a Bayesian approach for asymptotically large amount of data; see Sect. 9.6.3 later.

Fisher Information Matrix via Hessian

The FIM can be written as the expectation of the Hessian of NLLF:

$$\mathbf{J}(\boldsymbol{\theta}) = E[\nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta}) | \boldsymbol{\theta}] = \int [\nabla_{\boldsymbol{\theta}}^2 L(\mathbf{z}, \boldsymbol{\theta})] p(\mathbf{z} | \boldsymbol{\theta}) d\mathbf{z} \quad (9.40)$$

This often provides an easier way for obtaining an analytical expression for $\mathbf{J}(\boldsymbol{\theta})$. The proof can be found in Sect. 9.3.3.

Example 9.4 (Mean parameter, CRB) In this example, we determine the CRB for the problem in Example 9.1. Recall that the data is modeled as $\hat{y}_k = \theta_1 + \varepsilon_k$, where $\{\varepsilon_k\}_{k=1}^N$ are i.i.d. zero-mean Gaussian variables with variance θ_2 . The likelihood function and NLLF are given by

$$p(\{\hat{y}_k\} | \boldsymbol{\theta}) = \prod_{k=1}^N \frac{1}{\sqrt{2\pi\theta_2}} \exp\left[-\frac{(\hat{y}_k - \theta_1)^2}{2\theta_2}\right] \quad (9.41)$$

$$L(\{\hat{y}_k\}, \boldsymbol{\theta}) = -\ln p(\{\hat{y}_k\} | \boldsymbol{\theta}) = \frac{N}{2} \ln(2\pi\theta_2) + \frac{1}{2\theta_2} \sum_{k=1}^N (\hat{y}_k - \theta_1)^2 \quad (9.42)$$

Fisher Information Matrix

Direct differentiation of (9.42) gives

$$\nabla_{\boldsymbol{\theta}} L = \begin{bmatrix} N\theta_2^{-1}(\theta_1 - \mu_1) & \frac{N}{2}\theta_2^{-1} - \frac{1}{2}\theta_2^{-2} \sum_{k=1}^N (\hat{y}_k - \theta_1)^2 \end{bmatrix} \quad (9.43)$$

$$\nabla_{\boldsymbol{\theta}}^2 L = \begin{bmatrix} N\theta_2^{-1} & -N\theta_2^{-2}(\theta_1 - \mu_1) \\ -N\theta_2^{-2}(\theta_1 - \mu_1) & -\frac{N}{2}\theta_2^{-2} + \theta_2^{-3} \sum_{k=1}^N (\hat{y}_k - \theta_1)^2 \end{bmatrix} \quad (9.44)$$

where $\mu_1 = \sum_{k=1}^N \hat{y}_k/N$. Using (9.40) and noting that $E[\mu_1|\boldsymbol{\theta}] = \theta_1$ and $E[(\hat{y}_k - \theta_1)^2|\boldsymbol{\theta}] = \theta_2$, the FIM is given by

$$\mathbf{J}(\boldsymbol{\theta}) = E[\nabla_{\boldsymbol{\theta}}^2 L|\boldsymbol{\theta}] = \begin{bmatrix} N\theta_2^{-1} & 0 \\ 0 & N\theta_2^{-2}/2 \end{bmatrix} \quad (9.45)$$

CRB for unbiased estimator

According to (9.34), for any unbiased estimator $\mathbf{G} = [G_1, G_2]^T$ for $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$,

$$\text{cov}[\mathbf{G}|\boldsymbol{\theta}] = \begin{bmatrix} \text{var}[G_1|\boldsymbol{\theta}] & \text{cov}[G_1, G_2|\boldsymbol{\theta}] \\ \text{sym.} & \text{var}[G_2|\boldsymbol{\theta}] \end{bmatrix} \geq \mathbf{J}(\boldsymbol{\theta})^{-1} = \begin{bmatrix} \theta_2/N & 0 \\ 0 & 2\theta_2^2/N \end{bmatrix} \quad (9.46)$$

in the sense that the LHS minus RHS is a positive semi-definite matrix. That is, the eigenvalues of the following matrix are all non-negative:

$$\Delta(\boldsymbol{\theta}) = \text{cov}[\mathbf{G}|\boldsymbol{\theta}] - \mathbf{J}(\boldsymbol{\theta})^{-1} = \begin{bmatrix} \text{var}[G_1|\boldsymbol{\theta}] - \theta_2/N & \text{cov}[G_1, G_2|\boldsymbol{\theta}] \\ \text{sym.} & \text{var}[G_2|\boldsymbol{\theta}] - 2\theta_2^2/N \end{bmatrix} \quad (9.47)$$

Let Δ_{ij} denote the (i, j) -entry of $\Delta(\boldsymbol{\theta})$. Then the eigenvalues of $\Delta(\boldsymbol{\theta})$ are given by

$$\frac{1}{2} \left[(\Delta_{11} + \Delta_{22}) \pm \sqrt{(\Delta_{11} - \Delta_{22})^2 + 4\Delta_{12}^2} \right] \quad (9.48)$$

Requiring the eigenvalues to be non-negative results in the following set of inequalities (details omitted):

$$\Delta_{11} \geq 0 \quad \Delta_{22} \geq 0 \quad \Delta_{12}^2 \leq \Delta_{11}\Delta_{22} \quad (9.49)$$

That is,

$$\begin{aligned} \text{var}[G_1|\boldsymbol{\theta}] &\geq \theta_2/N \\ \text{var}[G_2|\boldsymbol{\theta}] &\geq 2\theta_2^2/N \\ \text{cov}[G_1, G_2|\boldsymbol{\theta}]^2 &\leq \{\text{var}[G_1|\boldsymbol{\theta}] - \theta_2/N\} \{\text{var}[G_2|\boldsymbol{\theta}] - 2\theta_2^2/N\} \end{aligned} \quad (9.50)$$

Alternatively, applying the Sylvester's criterion gives the first and third equality in (9.49), from which the second inequality can be deduced.

Applying CRB to Specific Context

The bounds in (9.50) apply to any unbiased estimators G_1 and G_2 for θ_1 and θ_2 , respectively. We now demonstrate them specifically for the estimators in Example 9.1. The estimator in (9.8) for θ_1 is unbiased and so the CRB applies. From (9.10), its variance is θ_2/N and so it achieves the CRB, i.e., the minimum possible variance. For θ_2 , the CRB applies to the estimator in (9.15) but not to the one in (9.11), because the latter is biased. From (9.16), the variance of the former is $(1 - 1/N)^{-1} 2\theta_2^2/N$. For finite N , this is greater than the CRB $2\theta_2^2/N$, although it

reduces to the CRB as $N \rightarrow \infty$. The third inequality in (9.50) puts an upper bound on the covariance between any unbiased estimators G_1 and G_2 , depending on how far their variance is from the CRB. For the estimator in (9.8), $\text{var}[G_1|\boldsymbol{\theta}] = \theta_2/N$ and so $\text{cov}[G_1, G_2|\boldsymbol{\theta}] = 0$ for any unbiased G_2 , including the one in (9.15).

Fisher Information Matrix via Gradient

It is typically more tedious to evaluate the FIM using the original definition in (9.35) via the gradient. Let $(\nabla_{\boldsymbol{\theta}}^T L \nabla_{\boldsymbol{\theta}} L)_{ij}$ denote the (i, j) -entry of $\nabla_{\boldsymbol{\theta}}^T L \nabla_{\boldsymbol{\theta}} L$. Using (9.43),

$$(\nabla_{\boldsymbol{\theta}}^T L \nabla_{\boldsymbol{\theta}} L)_{11} = N^2 \theta_2^{-2} (\theta_1 - \mu_1)^2 \quad (9.51)$$

$$(\nabla_{\boldsymbol{\theta}}^T L \nabla_{\boldsymbol{\theta}} L)_{12} = (\nabla_{\boldsymbol{\theta}}^T L \nabla_{\boldsymbol{\theta}} L)_{21} = \frac{N^2}{2} \theta_2^{-2} (\theta_1 - \mu_1) - \frac{N}{2} \theta_2^{-3} (\theta_1 - \mu_1) \sum_{k=1}^N (\hat{y}_k - \theta_1)^2 \quad (9.52)$$

$$(\nabla_{\boldsymbol{\theta}}^T L \nabla_{\boldsymbol{\theta}} L)_{22} = \frac{N^2}{4} \theta_2^{-2} - \frac{N}{2} \theta_2^{-3} \sum_{k=1}^N (\hat{y}_k - \theta_1)^2 + \frac{1}{4} \theta_2^{-4} \left[\sum_{k=1}^N (\hat{y}_k - \theta_1)^2 \right]^2 \quad (9.53)$$

Taking expectation on (9.51) gives

$$E[(\nabla_{\boldsymbol{\theta}}^T L \nabla_{\boldsymbol{\theta}} L)_{11}|\boldsymbol{\theta}] = N^2 \theta_2^{-2} E[(\theta_1 - \mu_1)^2|\boldsymbol{\theta}] = N^2 \theta_2^{-2} (\theta_2/N) = N \theta_2^{-1} \quad (9.54)$$

which is equal to $E[(\nabla_{\boldsymbol{\theta}}^2 L)_{11}|\boldsymbol{\theta}]$, as expected. The expectation of $(\nabla_{\boldsymbol{\theta}}^T L \nabla_{\boldsymbol{\theta}} L)_{12}$ and $(\nabla_{\boldsymbol{\theta}}^T L \nabla_{\boldsymbol{\theta}} L)_{22}$ are more tedious to evaluate as they involve expectations of up to the fourth order of $\hat{y}_k$. Using (9.40) bypasses these difficulties. ■

Example 9.5 (Linear regression, CRB) Consider Example 9.2 again, where the data is modeled by

$$\hat{y}_k = \theta_1 + \theta_2 x_k + \varepsilon_k \quad k = 1, \dots, N \quad (9.55)$$

with $\{x_k\}_{k=1}^N$ given and $\{\varepsilon_k\}_{k=1}^N$ i.i.d. zero-mean Gaussian with variance θ_3 . Here we derive the CRB for unbiased estimator of $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^T$. Given $\boldsymbol{\theta}$, $\{\hat{y}_k\}_{k=1}^N$ are i.i.d. Gaussian with mean $\theta_1 + \theta_2 x_k$ and variance θ_3 . The likelihood function and NLLF are given by

$$p(\{\hat{y}_k\}|\boldsymbol{\theta}) = \prod_{k=1}^N \frac{1}{\sqrt{2\pi\theta_3}} \exp\left[-\frac{1}{2\theta_3} (\hat{y}_k - \theta_1 - \theta_2 x_k)^2\right] \quad (9.56)$$

$$L(\{\hat{y}_k\}, \boldsymbol{\theta}) = -\ln p(\{\hat{y}_k\}|\boldsymbol{\theta}) = \frac{N}{2} \ln(2\pi\theta_3) + \frac{1}{2\theta_3} \sum_{k=1}^N (\hat{y}_k - \theta_1 - \theta_2 x_k)^2 \quad (9.57)$$

Direct differentiation gives

$$\frac{\partial L}{\partial \theta_1} = -\theta_3^{-1} \sum_{k=1}^N (\hat{y}_k - \theta_1 - \theta_2 x_k) \quad \frac{\partial L}{\partial \theta_2} = -\theta_3^{-1} \sum_{k=1}^N (\hat{y}_k - \theta_1 - \theta_2 x_k) x_k \quad (9.58)$$

$$\frac{\partial L}{\partial \theta_3} = \frac{N}{2} \theta_3^{-1} - \frac{1}{2} \theta_3^{-2} \sum_{k=1}^N (\hat{y}_k - \theta_1 - \theta_2 x_k)^2$$

$$\frac{\partial^2 L}{\partial \theta_1^2} = N \theta_3^{-1} \quad \frac{\partial^2 L}{\partial \theta_2^2} = N \mu_{xx} \theta_3^{-1} \quad \frac{\partial^2 L}{\partial \theta_3^2} = -\frac{N}{2} \theta_3^{-2} + \theta_3^{-3} \sum_{k=1}^N (\hat{y}_k - \theta_1 - \theta_2 x_k)^2 \quad (9.59)$$

$$\frac{\partial^2 L}{\partial \theta_1 \partial \theta_2} = N \mu_x \theta_3^{-1} \quad \frac{\partial^2 L}{\partial \theta_1 \partial \theta_3} = \theta_3^{-2} \sum_{k=1}^N (\hat{y}_k - \theta_1 - \theta_2 x_k) \quad (9.60)$$

$$\frac{\partial^2 L}{\partial \theta_2 \partial \theta_3} = \theta_3^{-2} \sum_{k=1}^N (\hat{y}_k - \theta_1 - \theta_2 x_k) x_k$$

where $\mu_x = N^{-1} \sum_{k=1}^N x_k$ and $\mu_{xx} = N^{-1} \sum_{k=1}^N x_k^2$. The FIM is then given by

$$\mathbf{J}(\boldsymbol{\theta}) = E[\nabla_{\boldsymbol{\theta}}^2 L | \boldsymbol{\theta}] = \begin{bmatrix} N \theta_3^{-1} & N \mu_x \theta_3^{-1} & 0 \\ N \mu_{xx} \theta_3^{-1} & 0 & 0 \\ \text{sym.} & & N \theta_3^{-2} / 2 \end{bmatrix} \quad (9.61)$$

where we have used

$$\begin{aligned} E[\hat{y}_k - \theta_1 - \theta_2 x_k | \boldsymbol{\theta}] &= E[\varepsilon_k | \boldsymbol{\theta}] = 0 \\ E[(\hat{y}_k - \theta_1 - \theta_2 x_k)^2 | \boldsymbol{\theta}] &= E[\varepsilon_k^2 | \boldsymbol{\theta}] = \theta_3 \end{aligned} \quad (9.62)$$

The inverse of the FIM is given by

$$\mathbf{J}(\boldsymbol{\theta})^{-1} = \begin{bmatrix} \frac{\mu_{xx} \theta_3}{N(\mu_{xx} - \mu_x^2)} & \frac{-\mu_x \theta_3}{N(\mu_{xx} - \mu_x^2)} & 0 \\ & \frac{\theta_3}{N(\mu_{xx} - \mu_x^2)} & 0 \\ \text{sym.} & & \frac{2\theta_3^2}{N} \end{bmatrix} \quad (9.63)$$

Here it is difficult to obtain an analytical expression for the eigenvalue of $\Delta(\boldsymbol{\theta}) = \text{cov}[\mathbf{G}|\boldsymbol{\theta}] - \mathbf{J}(\boldsymbol{\theta})^{-1}$ in manageable form. Alternatively, using the Sylvester's criterion,

$$\begin{aligned} \Delta_{11} &\geq 0 \quad \Delta_{11}\Delta_{22} - \Delta_{12}^2 \geq 0 \\ \Delta_{11}\Delta_{22}\Delta_{33} + 2\Delta_{12}\Delta_{13}\Delta_{23} - \Delta_{11}\Delta_{23}^2 - \Delta_{22}\Delta_{13}^2 - \Delta_{33}\Delta_{12}^2 &\geq 0 \end{aligned} \quad (9.64)$$

where Δ_{ij} denote the (i, j) -entry of $\Delta(\boldsymbol{\theta})$. These inequalities can be written explicitly in terms of the variance and covariance of estimators by substituting the entries of $\text{cov}[\mathbf{G}|\boldsymbol{\theta}]$ and $\mathbf{J}(\boldsymbol{\theta})^{-1}$ into Δ_{ij} . Easier bounds can be obtained, although they are less tight. See Example 9.8 later. ■

Example 9.6 (Modal identification, CRB) In this example, we derive the CRB for unbiased estimator for the vector of modal parameters $\boldsymbol{\theta} = [f, \zeta, S, S_e]^T$ in Example 9.3. Recall the NLLF in (9.33):

$$L(\{\hat{S}_k\}, \boldsymbol{\theta}) = \sum_k \left[\ln S_k(\boldsymbol{\theta}) + \frac{\hat{S}_k}{S_k(\boldsymbol{\theta})} \right] \quad (9.65)$$

where $S_k(\boldsymbol{\theta}) = SD_k + S_e$ and D_k is the dynamic amplification factor in (9.31) that depends on f and ζ . The gradient and Hessian of L w.r.t. $\boldsymbol{\theta}$ are respectively given by

$$\nabla_{\boldsymbol{\theta}} L = \sum_k (S_k^{-1} - S_k^{-2} \hat{S}_k) \nabla S_k \quad (9.66)$$

$$\nabla_{\boldsymbol{\theta}}^2 L = \sum_k (2S_k^{-3} \hat{S}_k - S_k^{-2}) \nabla^T S_k \nabla S_k + \sum_k (S_k^{-1} - S_k^{-2} \hat{S}_k) \nabla^2 S_k \quad (9.67)$$

where the dependence of S_k on $\boldsymbol{\theta}$ has been omitted for notational simplicity; ∇S_k and $\nabla^2 S_k$ are respectively the gradient and Hessian of S_k (w.r.t. $\boldsymbol{\theta}$). The FIM is given by

$$\begin{aligned} \mathbf{J}(\boldsymbol{\theta}) &= E[\nabla_{\boldsymbol{\theta}}^2 L|\boldsymbol{\theta}] \\ &= \sum_k \{2S_k^{-3} \underbrace{E[\hat{S}_k|\boldsymbol{\theta}]}_{S_k} - S_k^{-2}\} \nabla^T S_k \nabla S_k + \sum_k \{S_k^{-1} - S_k^{-2} \underbrace{E[\hat{S}_k|\boldsymbol{\theta}]}_{S_k}\} \nabla^2 S_k \\ &= \sum_k S_k^{-2} \nabla^T S_k \nabla S_k \end{aligned} \quad (9.68)$$

Here, it is difficult to obtain an analytical expression for $\mathbf{J}(\boldsymbol{\theta})^{-1}$ in manageable form. Easier but looser bounds are obtained in Example 9.9 later. ■

9.3.1 Easier but Looser Bounds

Determining the FIM and taking inverse are two major difficulties in obtaining the CRB. As illustrated in Examples 9.5 and 9.6, it is not always feasible to obtain analytical expressions in manageable form. It is possible to obtain ‘easier’ bounds at the expense of loosening the inequality, i.e., a bound that is a necessary consequence of the original CRB and is therefore less tight.

The key is that inequalities similar to the original CRB also apply to a subset of the parameters. Let α denote a subset of the index set $\{1, \dots, n_\theta\}$; $\mathbf{G}_\alpha$ and $\mathbf{J}_\alpha$ denote the partition of $\mathbf{G}$ (estimator) and $\mathbf{J}$ (FIM) corresponding to the indices in α . Then the CRB in (9.34) implies that

$$\text{cov}[\mathbf{G}_\alpha|\boldsymbol{\theta}] \geq \underbrace{[\mathbf{J}^{-1}]_\alpha}_{\text{partition of inverse}} \geq \underbrace{\mathbf{J}_\alpha^{-1}}_{\text{inverse of partition}} \quad (9.69)$$

where $[\cdot]_\alpha$ denotes the partition of the argument matrix corresponding to the index set α .

The first inequality in (9.69) follows from the fact that any partition of a positive semi-definite matrix is also positive semi-definite, and is consistent with the Sylvester’s criterion. It is just a partitioned version of (9.34). The second inequality is a general result that holds for any positive definite matrix (see Sect. C.5.7).

Equation (9.69) provides an easier way for deriving lower bounds because the FIM involved ($\mathbf{J}_\alpha$) is now of a smaller dimension. As α expands to include more indices, the dimension of the FIM increases and the resulting bound tightens. The tightest (but most difficult) bound is the one involving all parameters in the original CRB in (9.34), i.e., $\alpha = \{1, \dots, n_\theta\}$. The loosest (but easiest) bound is obtained by considering a particular parameter. Taking $\alpha = i$ in (9.69) gives

$$\text{var}[G_i|\boldsymbol{\theta}] \geq (\mathbf{J}^{-1})_{ii} \geq J_{ii}^{-1} \quad i = 1, \dots, n_\theta \quad (9.70)$$

where J_{ii} and $(\mathbf{J}^{-1})_{ii}$ denote respectively the (i, i) -entry of $\mathbf{J}$ and $\mathbf{J}^{-1}$. Note that this bound holds only for diagonal entries. For $i \neq j$, it is not necessary that $\text{cov}[G_i, G_j|\boldsymbol{\theta}] \geq (\mathbf{J}^{-1})_{ij}$ or $\geq J_{ij}^{-1}$.

Example 9.7 (Mean parameter, easy CRB) In Example 9.4, the bounds in (9.50) were obtained by requiring the eigenvalues of $\boldsymbol{\Lambda}(\boldsymbol{\theta}) = \text{cov}[\mathbf{G}|\boldsymbol{\theta}] - \mathbf{J}(\boldsymbol{\theta})^{-1}$ to be non-negative. Easier bounds can be obtained by using (9.70). From (9.45), $\mathbf{J}$ is a diagonal matrix and so $(\mathbf{J}^{-1})_{ii} = J_{ii}^{-1}$. The middle and rightmost term in (9.70) are then the same. Applying (9.70) with $\alpha = 1, 2$, for any unbiased estimator $\mathbf{G} = [G_1, G_2]^T$,

$$\text{var}[G_1|\boldsymbol{\theta}] \geq \underbrace{\theta_2/N}_{1/J_{11}} \quad \text{var}[G_2|\boldsymbol{\theta}] \geq \underbrace{2\theta_2^2/N}_{1/J_{22}} \quad (9.71)$$

which coincide with the first two inequalities in (9.50). Here, easier bounds are obtained at the expense of not discovering the third inequality in (9.50). ■

Example 9.8 (Linear regression, easy CRB) In Example 9.5, the CRB was obtained in terms of the entries of $\Delta(\boldsymbol{\theta}) = \text{cov}[\mathbf{G}|\boldsymbol{\theta}] - \mathbf{J}(\boldsymbol{\theta})^{-1}$ in (9.64), which is recalled here:

$$\begin{aligned} \Delta_{11} &\geq 0 & \Delta_{11}\Delta_{22} - \Delta_{12}^2 &\geq 0 \\ \Delta_{11}\Delta_{22}\Delta_{33} + 2\Delta_{12}\Delta_{13}\Delta_{23} - \Delta_{11}\Delta_{23}^2 - \Delta_{22}\Delta_{13}^2 - \Delta_{33}\Delta_{12}^2 &\geq 0 \end{aligned} \quad (9.72)$$

Easier bounds can be obtained by applying the first inequality in (9.70) with $\boldsymbol{\alpha} = 1, 2, 3$:

$$\Delta_{11} \geq 0 \quad \Delta_{22} \geq 0 \quad \Delta_{33} \geq 0 \quad (9.73)$$

These bounds are at most as tight as those in (9.72) because they can be deduced from it. The first inequality in (9.73) is the same as the first inequality in (9.72). The second inequality in (9.73) can be deduced from the first and second inequality in (9.72). The third inequality in (9.73) can be deduced from the second and third inequality in (9.72). Substituting the entries of $\mathbf{J}(\boldsymbol{\theta})^{-1}$ in (9.63), the inequalities in (9.73) read:

$$\text{var}[G_1|\boldsymbol{\theta}] \geq \frac{\mu_{xx}\theta_3}{N(\mu_{xx} - \mu_x^2)} \quad \text{var}[G_2|\boldsymbol{\theta}] \geq \frac{\theta_3}{N(\mu_{xx} - \mu_x^2)} \quad \text{var}[G_3|\boldsymbol{\theta}] \geq \frac{2\theta_3^2}{N} \quad (9.74)$$

The easiest bounds that do not require inverting $\mathbf{J}(\boldsymbol{\theta})$ are obtained by the rightmost term in (9.70) with $\boldsymbol{\alpha} = 1, 2, 3$:

$$\text{var}[G_1|\boldsymbol{\theta}] \geq \underbrace{\frac{\theta_3}{N}}_{1/J_{11}} \quad \text{var}[G_2|\boldsymbol{\theta}] \geq \underbrace{\frac{\theta_3}{N\mu_{xx}}}_{1/J_{22}} \quad \text{var}[G_3|\boldsymbol{\theta}] \geq \underbrace{\frac{2\theta_3^2}{N}}_{1/J_{33}} \quad (9.75)$$

where J_{ii} denotes the (i, i) -entry of $\mathbf{J}(\boldsymbol{\theta})$. The values on the RHS of the first two inequalities in (9.75) are smaller than their counterparts in (9.74). This implies that the first two bounds in (9.75) are looser than (i.e., necessary consequence of) those in (9.74), as expected. The third inequality in (9.75) is the same as that in (9.74). This stems from the fact that in this example $(\mathbf{J}^{-1})_{33} = J_{33}^{-1}$. ■

Example 9.9 (Modal identification, easy CRB) Recall the FIM in (9.68) of Example 9.6:

$$\mathbf{J}(\boldsymbol{\theta}) = \sum_k S_k^{-2} \nabla^T S_k \nabla S_k \quad (9.76)$$

where $\boldsymbol{\theta} = [f, \zeta, S, S_e]^T$, $S_k(\boldsymbol{\theta}) = SD_k + S_e$ and D_k is the dynamic amplification factor in (9.31) that depends on f and ζ . As it is difficult to invert $\mathbf{J}(\boldsymbol{\theta})$ analytically, we shall obtain the easiest bound by the rightmost term of (9.70). Note that the i th diagonal entry of $\mathbf{J}(\boldsymbol{\theta})$ is $J_{ii} = \sum_k S_k^{-2} (\partial S_k / \partial \theta_i)^2$. For any unbiased estimator $\mathbf{G} = [G_1, G_2, G_3, G_4]^T$ for $\boldsymbol{\theta}$, the easiest CRB gives

$$\text{var}[G_i | \boldsymbol{\theta}] \geq \left[\sum_k S_k^{-2} \left(\frac{\partial S_k}{\partial \theta_i} \right)^2 \right]^{-1} \quad i = 1, \dots, 4 \quad (9.77)$$

Note that

$$\frac{\partial S_k}{\partial f} = S \frac{\partial D_k}{\partial f} \quad \frac{\partial S_k}{\partial \zeta} = S \frac{\partial D_k}{\partial \zeta} \quad \frac{\partial S_k}{\partial S} = D_k \quad \frac{\partial S_k}{\partial S_e} = 1 \quad (9.78)$$

Substituting into (9.77) gives

$$\text{var}[G_1 | \boldsymbol{\theta}] \geq \left[\sum_k (SD_k + S_e)^{-2} S^2 \left(\frac{\partial D_k}{\partial f} \right)^2 \right]^{-1} \quad (\text{frequency}) \quad (9.79)$$

$$\text{var}[G_2 | \boldsymbol{\theta}] \geq \left[\sum_k (SD_k + S_e)^{-2} S^2 \left(\frac{\partial D_k}{\partial \zeta} \right)^2 \right]^{-1} \quad (\text{damping}) \quad (9.80)$$

$$\text{var}[G_3 | \boldsymbol{\theta}] \geq \left[\sum_k (SD_k + S_e)^{-2} D_k^2 \right]^{-1} \quad (\text{modal force PSD}) \quad (9.81)$$

$$\text{var}[G_4 | \boldsymbol{\theta}] \geq \left[\sum_k (SD_k + S_e)^{-2} \right]^{-1} \quad (\text{prediction error PSD}) \quad (9.82)$$

The expressions on the RHS can be further simplified by considering long data and small damping. The techniques are discussed in Sect. 16.2. ■

9.3.2 General Form

The CRB in (9.34) is a special form of a more general result where $\mathbf{G}(\mathbf{Z})$ need not even be an estimator for $\boldsymbol{\theta}$. Let $\mathbf{G}(\mathbf{Z})$ ($m \times 1$) now be a general function of $\boldsymbol{\theta}$ where m can be different from n_θ . As in (9.3), let $\mathbf{g}(\boldsymbol{\theta}) = E[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}]$ with $\mathbf{Z}$ distributed as $p(\mathbf{Z}|\boldsymbol{\theta})$. Then it can be shown that

$$\underbrace{\text{cov}[\mathbf{G}(\mathbf{Z})|\boldsymbol{\theta}]}_{m \times m} \geq \underbrace{[\nabla \mathbf{g}(\boldsymbol{\theta})]}_{m \times n_\theta} \underbrace{\mathbf{J}(\boldsymbol{\theta})^{-1}}_{n_\theta \times n_\theta} \underbrace{[\nabla \mathbf{g}(\boldsymbol{\theta})]^T}_{n_\theta \times m} \quad (9.83)$$

where

$$\underbrace{\nabla \mathbf{g}}_{m \times n_\theta} = \left[\frac{\partial \mathbf{g}}{\partial \theta_1}, \dots, \frac{\partial \mathbf{g}}{\partial \theta_{n_\theta}} \right] \quad (9.84)$$

is the gradient of $\mathbf{g}(\boldsymbol{\theta})$; and $\mathbf{J}(\boldsymbol{\theta})$ is the FIM in (9.35) as before. The equality in (9.83) holds if and only if

$$\underbrace{\mathbf{G}(\mathbf{Z})}_{m \times 1} = \underbrace{\mathbf{g}(\boldsymbol{\theta})}_{m \times 1} - \underbrace{[\nabla \mathbf{g}(\boldsymbol{\theta})]}_{m \times n_\theta} \underbrace{\mathbf{J}(\boldsymbol{\theta})^{-1}}_{n_\theta \times n_\theta} \underbrace{\nabla_{\boldsymbol{\theta}}^T L(\mathbf{Z}, \boldsymbol{\theta})}_{n_\theta \times 1} \quad (9.85)$$

where $L(\mathbf{Z}, \boldsymbol{\theta}) = -\ln p(\mathbf{Z}|\boldsymbol{\theta})$ is the NLLF. Equations (9.83) and (9.85) reduce respectively to (9.34) and (9.36) when $\mathbf{G}(\mathbf{Z})$ is an unbiased estimator for $\boldsymbol{\theta}$, in which case $\mathbf{g}(\boldsymbol{\theta}) = \boldsymbol{\theta}$ and $\nabla \mathbf{g}$ is the $n_\theta \times n_\theta$ identity matrix.

9.3.3 Derivation

Here we derive the general form of CRB in (9.83) and the alternative formula for the FIM in (9.40). The derivation is based on the fundamental property that the likelihood function is the (assumed) PDF of data. Recall $L(\mathbf{z}, \boldsymbol{\theta}) = -\ln p(\mathbf{z}|\boldsymbol{\theta})$. Since $p(\mathbf{z}|\boldsymbol{\theta}) = \exp[-L(\mathbf{z}, \boldsymbol{\theta})]$ is the PDF of data $\mathbf{Z}$ for a given $\boldsymbol{\theta}$, its integral over $\mathbf{z}$ is always equal to 1:

$$\int e^{-L(\mathbf{z}, \boldsymbol{\theta})} d\mathbf{z} = 1 \quad \text{for any } \boldsymbol{\theta} \quad (9.86)$$

Taking gradient w.r.t. $\boldsymbol{\theta}$ and swapping the order of gradient and integration,

$$\int [\nabla_{\boldsymbol{\theta}} L(\mathbf{z}, \boldsymbol{\theta})] e^{-L(\mathbf{z}, \boldsymbol{\theta})} d\mathbf{z} = \mathbf{0} \quad (9.87)$$

This identity can be written in terms of the ‘score function’:

$$\mathbf{S}(\mathbf{Z}, \boldsymbol{\theta}) = -\nabla_{\boldsymbol{\theta}}^T L(\mathbf{Z}, \boldsymbol{\theta}) \quad (9.88)$$

$n_{\theta} \times 1$

When $\mathbf{Z}$ is distributed as $p(\mathbf{z}|\boldsymbol{\theta})$, (9.87) says that

$$E[\mathbf{S}(\mathbf{Z}, \boldsymbol{\theta})|\boldsymbol{\theta}] = \mathbf{0} \quad (9.89)$$

The FIM in (9.35) is just the covariance of $\mathbf{S}(\mathbf{Z}, \boldsymbol{\theta})$:

$$\mathbf{J}(\boldsymbol{\theta}) = E[\nabla_{\boldsymbol{\theta}}^T L(\mathbf{Z}, \boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}} L(\mathbf{Z}, \boldsymbol{\theta})|\boldsymbol{\theta}] = E[\mathbf{S}(\mathbf{Z}, \boldsymbol{\theta}) \mathbf{S}(\mathbf{Z}, \boldsymbol{\theta})^T|\boldsymbol{\theta}] \quad (9.90)$$

Consider the cross covariance between $\mathbf{G}(\mathbf{Z})$ and $\mathbf{S}(\mathbf{Z}, \boldsymbol{\theta})$:

$$\underbrace{\text{cov}[\mathbf{G}, \mathbf{S}|\boldsymbol{\theta}]}_{m \times n_{\theta}} = E[\mathbf{G} \mathbf{S}^T|\boldsymbol{\theta}] - E[\mathbf{G}|\boldsymbol{\theta}] \underbrace{E[\mathbf{S}^T|\boldsymbol{\theta}]}_{\mathbf{0}} = \int \mathbf{G}(\mathbf{z}) \mathbf{S}(\mathbf{z}, \boldsymbol{\theta})^T p(\mathbf{z}|\boldsymbol{\theta}) d\mathbf{z} \quad (9.91)$$

Substituting (9.88),

$$\text{cov}[\mathbf{G}, \mathbf{S}|\boldsymbol{\theta}] = - \int \mathbf{G}(\mathbf{z}) \underbrace{\nabla_{\boldsymbol{\theta}} L(\mathbf{z}, \boldsymbol{\theta}) p(\mathbf{z}|\boldsymbol{\theta})}_{-\nabla_{\boldsymbol{\theta}} p(\mathbf{z}|\boldsymbol{\theta})} d\mathbf{z} = \nabla_{\boldsymbol{\theta}} \underbrace{\int \mathbf{G}(\mathbf{z}) p(\mathbf{z}|\boldsymbol{\theta}) d\mathbf{z}}_{\mathbf{g}(\boldsymbol{\theta})} = \nabla \mathbf{g} \quad (9.92)$$

where we have swapped the order of gradient and integration.

On the other hand, $\text{cov}[\mathbf{G}, \mathbf{S}|\boldsymbol{\theta}]$ can be related to $\text{cov}[\mathbf{G}|\boldsymbol{\theta}]$ and $\text{cov}[\mathbf{S}|\boldsymbol{\theta}]$ through a matrix inequality. For any two random column vectors X and Y with positive definite covariance matrices, it can be shown that (Sect. C.5.8)

$$E[YY^T] \geq E[YX^T] E[XX^T]^{-1} E[XY^T] \quad (9.93)$$

where equality holds if and only if $Y = E[YX^T] E[XX^T]^{-1} X$. This is a matrix inequality that should be interpreted as the LHS minus RHS being a positive semi-definite matrix. Taking $X = \mathbf{S}$ and $Y = \mathbf{G} - \mathbf{g}$, we have $E[YY^T] = \text{cov}[\mathbf{G}|\boldsymbol{\theta}]$, $E[YX^T] = \text{cov}[\mathbf{G}, \mathbf{S}|\boldsymbol{\theta}] = \nabla \mathbf{g}$ [from (9.92)] and $E[XX^T] = \mathbf{J}$ [from (9.90)]. Substituting these into (9.93) gives (9.83), where equality holds if and only if $\mathbf{G} - \mathbf{g} = (\nabla \mathbf{g}) \mathbf{J}^{-1} \mathbf{S}$, i.e., (9.85).

Fisher Information Matrix via Hessian

The expression of $\mathbf{J}(\boldsymbol{\theta})$ in (9.40) can be shown by taking gradient w.r.t. $\boldsymbol{\theta}$ of (9.87) and swapping the order of gradient and integration:

$$\underbrace{\int [\nabla_{\boldsymbol{\theta}}^2 L(\mathbf{z}, \boldsymbol{\theta})] e^{-L(\mathbf{z}, \boldsymbol{\theta})} d\mathbf{z}}_{E[\nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta}) | \boldsymbol{\theta}]} - \underbrace{\int [\nabla_{\boldsymbol{\theta}}^T L(\mathbf{z}, \boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}} L(\mathbf{z}, \boldsymbol{\theta})] e^{-L(\mathbf{z}, \boldsymbol{\theta})} d\mathbf{z}}_{\mathbf{J}(\boldsymbol{\theta})} = \mathbf{0} \quad (9.94)$$

9.4 Fisher Information Matrix for Gaussian Data

A general expression can be derived for the FIM for Gaussian data with mean and covariance matrix that depend on model parameters, which is frequently encountered in applications.

9.4.1 Real Gaussian

Let data $\mathbf{Z}$ ($N \times 1$) be modeled as a real-valued Gaussian vector with mean $\boldsymbol{\mu}(\boldsymbol{\theta}) = E[\mathbf{Z} | \boldsymbol{\theta}]$ ($N \times 1$) and covariance matrix $\mathbf{C}(\boldsymbol{\theta}) = E[(\mathbf{Z} - \boldsymbol{\mu})(\mathbf{Z} - \boldsymbol{\mu})^T | \boldsymbol{\theta}]$ ($N \times N$), both possibly depending on $\boldsymbol{\theta}$ ($n_{\theta} \times 1$). Then the likelihood function and NLLF are respectively given by

$$p(\mathbf{Z} | \boldsymbol{\theta}) = (2\pi)^{-N/2} |\mathbf{C}(\boldsymbol{\theta})|^{-1/2} \exp \left\{ -\frac{1}{2} [\mathbf{Z} - \boldsymbol{\mu}(\boldsymbol{\theta})]^T \mathbf{C}(\boldsymbol{\theta})^{-1} [\mathbf{Z} - \boldsymbol{\mu}(\boldsymbol{\theta})] \right\} \quad (9.95)$$

$$L(\mathbf{Z}, \boldsymbol{\theta}) = -\ln p(\mathbf{Z} | \boldsymbol{\theta}) = \frac{N}{2} \ln 2\pi + \frac{1}{2} \ln |\mathbf{C}(\boldsymbol{\theta})| + \frac{1}{2} [\mathbf{Z} - \boldsymbol{\mu}(\boldsymbol{\theta})]^T \mathbf{C}(\boldsymbol{\theta})^{-1} [\mathbf{Z} - \boldsymbol{\mu}(\boldsymbol{\theta})] \quad (9.96)$$

Let J_{ij} denote the (i, j) -entry of the FIM $\mathbf{J}(\boldsymbol{\theta})$. Then it can be shown that

$$J_{ij} = \underbrace{\frac{\partial \boldsymbol{\mu}^T}{\partial \theta_i}}_{1 \times N} \underbrace{\mathbf{C}^{-1}}_{N \times N} \underbrace{\frac{\partial \boldsymbol{\mu}}{\partial \theta_j}}_{N \times 1} + \frac{1}{2} \underbrace{tr \left(\mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_i} \mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_j} \right)}_{N \times N} \quad (9.97)$$

where $tr(\cdot)$ denotes the trace (i.e., sum of diagonal entries) of the argument matrix. Using the symmetry of $\mathbf{C}^{-1}$ and the cyclic property of trace ($tr(XYZ) = tr(ZXY)$), it can be verified that $J_{ij} = J_{ji}$, as expected.

If $\mathbf{C}$ does not depend on $\boldsymbol{\theta}$, the second term in (9.97) is zero. The first term can then be assembled into a matrix to give the FIM in a compact form:

$$\mathbf{J}_{n_{\boldsymbol{\theta}} \times n_{\boldsymbol{\theta}}} = \underbrace{(\nabla \boldsymbol{\mu})^T}_{n_{\boldsymbol{\theta}} \times N} \underbrace{\mathbf{C}^{-1}}_{N \times N} \underbrace{(\nabla \boldsymbol{\mu})}_{N \times n_{\boldsymbol{\theta}}} \quad \nabla \boldsymbol{\mu}_{N \times n_{\boldsymbol{\theta}}} = \left[\frac{\partial \boldsymbol{\mu}}{\partial \theta_1}, \dots, \frac{\partial \boldsymbol{\mu}}{\partial \theta_{n_{\boldsymbol{\theta}}}} \right] \quad (9.98)$$

Proof of (9.97) (FIM for real Gaussian data)

To show (9.97), we evaluate J_{ij} as the expectation of the second derivative of NLLF in (9.96). The second derivative of the log-determinant term is given by (Sect. C.5.3)

$$\frac{\partial^2 \ln |\mathbf{C}|}{\partial \theta_i \partial \theta_j} = \text{tr} \left(\mathbf{C}^{-1} \frac{\partial^2 \mathbf{C}}{\partial \theta_i \partial \theta_j} \right) - \text{tr} \left(\mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_i} \mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_j} \right) \quad (9.99)$$

For the quadratic term, applying chain-rule on the product of the three terms leads to nine terms:

$$\begin{aligned} \frac{\partial^2 [(\mathbf{Z} - \boldsymbol{\mu})^T \mathbf{C}^{-1} (\mathbf{Z} - \boldsymbol{\mu})]}{\partial \theta_i \partial \theta_j} = & \\ & - \frac{\partial \boldsymbol{\mu}^T}{\partial \theta_i} \frac{\partial \mathbf{C}^{-1}}{\partial \theta_j} (\mathbf{Z} - \boldsymbol{\mu}) \quad - (\mathbf{Z} - \boldsymbol{\mu})^T \frac{\partial \mathbf{C}^{-1}}{\partial \theta_i} \frac{\partial \boldsymbol{\mu}}{\partial \theta_j} \quad + \frac{\partial \boldsymbol{\mu}^T}{\partial \theta_i} \mathbf{C}^{-1} \frac{\partial \boldsymbol{\mu}}{\partial \theta_j} \\ & - \frac{\partial \boldsymbol{\mu}^T}{\partial \theta_j} \frac{\partial \mathbf{C}^{-1}}{\partial \theta_i} (\mathbf{Z} - \boldsymbol{\mu}) \quad - (\mathbf{Z} - \boldsymbol{\mu})^T \frac{\partial \mathbf{C}^{-1}}{\partial \theta_j} \frac{\partial \boldsymbol{\mu}}{\partial \theta_i} \quad + \frac{\partial \boldsymbol{\mu}^T}{\partial \theta_j} \mathbf{C}^{-1} \frac{\partial \boldsymbol{\mu}}{\partial \theta_i} \\ & - \frac{\partial^2 \boldsymbol{\mu}^T}{\partial \theta_i \partial \theta_j} \mathbf{C}^{-1} (\mathbf{Z} - \boldsymbol{\mu}) \quad + (\mathbf{Z} - \boldsymbol{\mu})^T \frac{\partial^2 \mathbf{C}^{-1}}{\partial \theta_i \partial \theta_j} (\mathbf{Z} - \boldsymbol{\mu}) \quad - (\mathbf{Z} - \boldsymbol{\mu})^T \mathbf{C}^{-1} \frac{\partial^2 \boldsymbol{\mu}}{\partial \theta_i \partial \theta_j} \end{aligned}$$

The randomness of these terms comes from the data $\mathbf{Z}$. There are six terms that are linear in $(\mathbf{Z} - \boldsymbol{\mu})$. Their expectations are zero because $E[\mathbf{Z}|\boldsymbol{\theta}] = \boldsymbol{\mu}$. Thus, only the expectations of the third, sixth and eighth term are non-zero. Since $\mathbf{C}^{-1}$ is symmetric, the third and sixth terms are the same. The above considerations give

$$E \left[\frac{\partial^2 (\mathbf{Z} - \boldsymbol{\mu})^T \mathbf{C}^{-1} (\mathbf{Z} - \boldsymbol{\mu})}{\partial \theta_i \partial \theta_j} \middle| \boldsymbol{\theta} \right] = 2 \frac{\partial \boldsymbol{\mu}^T}{\partial \theta_i} \mathbf{C}^{-1} \frac{\partial \boldsymbol{\mu}}{\partial \theta_j} + E \left[(\mathbf{Z} - \boldsymbol{\mu})^T \frac{\partial^2 \mathbf{C}^{-1}}{\partial \theta_i \partial \theta_j} (\mathbf{Z} - \boldsymbol{\mu}) \middle| \boldsymbol{\theta} \right] \quad (9.100)$$

It remains to evaluate the expectation on the RHS. The term inside the bracket is a scalar and so is equal to its trace. Using the cyclic property ($\text{tr}(XYZ) = \text{tr}(ZXY)$) and linearity ($E[\text{tr}(\cdot)] = \text{tr}\{E[\cdot]\}$) of trace,

$$\begin{aligned}
E \left[(\mathbf{Z} - \boldsymbol{\mu})^T \frac{\partial^2 \mathbf{C}^{-1}}{\partial \theta_i \partial \theta_j} (\mathbf{Z} - \boldsymbol{\mu}) \middle| \boldsymbol{\theta} \right] &= E \left\{ \text{tr} \left[(\mathbf{Z} - \boldsymbol{\mu})^T \frac{\partial^2 \mathbf{C}^{-1}}{\partial \theta_i \partial \theta_j} (\mathbf{Z} - \boldsymbol{\mu}) \right] \middle| \boldsymbol{\theta} \right\} \\
&= E \left\{ \text{tr} \left[(\mathbf{Z} - \boldsymbol{\mu})(\mathbf{Z} - \boldsymbol{\mu})^T \frac{\partial^2 \mathbf{C}^{-1}}{\partial \theta_i \partial \theta_j} \right] \middle| \boldsymbol{\theta} \right\} \\
&= \text{tr} \left\{ E[(\mathbf{Z} - \boldsymbol{\mu})(\mathbf{Z} - \boldsymbol{\mu})^T | \boldsymbol{\theta}] \frac{\partial^2 \mathbf{C}^{-1}}{\partial \theta_i \partial \theta_j} \right\} = \text{tr} \left(\mathbf{C} \frac{\partial^2 \mathbf{C}^{-1}}{\partial \theta_i \partial \theta_j} \right)
\end{aligned} \tag{9.101}$$

Note that (Sect. C.5.3)

$$\frac{\partial^2 \mathbf{C}^{-1}}{\partial \theta_i \partial \theta_j} = \mathbf{C}^{-1} \left(\frac{\partial \mathbf{C}}{\partial \theta_i} \mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_j} + \frac{\partial \mathbf{C}}{\partial \theta_j} \mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_i} - \frac{\partial^2 \mathbf{C}}{\partial \theta_i \partial \theta_j} \right) \mathbf{C}^{-1} \tag{9.102}$$

Substituting into (9.101) and using the cyclic property of trace and symmetry of $\mathbf{C}$ gives

$$E \left[(\mathbf{Z} - \boldsymbol{\mu})^T \frac{\partial^2 \mathbf{C}^{-1}}{\partial \theta_i \partial \theta_j} (\mathbf{Z} - \boldsymbol{\mu}) \middle| \boldsymbol{\theta} \right] = 2 \text{tr} \left(\mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_i} \mathbf{C}^{-1} \frac{\partial \mathbf{C}}{\partial \theta_j} \right) - \text{tr} \left(\mathbf{C}^{-1} \frac{\partial^2 \mathbf{C}}{\partial \theta_i \partial \theta_j} \right) \tag{9.103}$$

Substituting into (9.100) and then combining with (9.99) gives (9.97). ■

Example 9.10 (Mean parameter, FIM) Consider Example 9.4 again but now we use (9.97) to obtain the FIM. Here, $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$, $\mathbf{Z} = [\hat{y}_1, \dots, \hat{y}_N]^T$, $\boldsymbol{\mu}(\boldsymbol{\theta}) = \theta_1 \mathbf{1}_N$ and $\mathbf{C}(\boldsymbol{\theta}) = \theta_2 \mathbf{I}_N$, where $\mathbf{1}_N = [1, \dots, 1]^T$ ($N \times 1$) and $\mathbf{I}_N$ is the $N \times N$ identity matrix. Direct differentiation gives

$$\frac{\partial \boldsymbol{\mu}}{\partial \theta_1} = \mathbf{1}_N \quad \frac{\partial \boldsymbol{\mu}}{\partial \theta_2} = \mathbf{0} \quad \frac{\partial \mathbf{C}}{\partial \theta_1} = \mathbf{0} \quad \frac{\partial \mathbf{C}}{\partial \theta_2} = \mathbf{I}_N \tag{9.104}$$

Using (9.97),

$$J_{11} = (\mathbf{1}_N)^T (\theta_2 \mathbf{I}_N)^{-1} (\mathbf{1}_N) + \frac{1}{2} \text{tr} \left[(\theta_2 \mathbf{I}_N)^{-1} (\mathbf{0}) (\theta_2 \mathbf{I}_N)^{-1} (\mathbf{0}) \right] = \frac{N}{\theta_2} \tag{9.105}$$

$$J_{22} = (\mathbf{0})^T (\theta_2 \mathbf{I}_N)^{-1} (\mathbf{0}) + \frac{1}{2} \text{tr} \left[(\theta_2 \mathbf{I}_N)^{-1} (\mathbf{I}_N) (\theta_2 \mathbf{I}_N)^{-1} (\mathbf{I}_N) \right] = \frac{N}{2\theta_2^2} \tag{9.106}$$

$$J_{12} = (\mathbf{1}_N)^T (\theta_2 \mathbf{I}_N)^{-1} (\mathbf{0}) + \frac{1}{2} \text{tr} \left[(\theta_2 \mathbf{I}_N)^{-1} (\mathbf{0}) (\theta_2 \mathbf{I}_N)^{-1} (\mathbf{I}_N) \right] = 0 \tag{9.107}$$

and $J_{21} = J_{12}$ by symmetry. Assembling these into a matrix gives the same result as in (9.45). ■

Example 9.11 (Linear regression, FIM) Consider Example 9.5 again but now we use (9.97) to obtain the FIM. Here, $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^T$, $\mathbf{Z} = [\hat{y}_1, \dots, \hat{y}_N]^T$, $\boldsymbol{\mu}(\boldsymbol{\theta}) = \theta_1 \mathbf{1}_N + \theta_2 \mathbf{x}$, $\mathbf{x} = [x_1, \dots, x_N]^T$ and $\mathbf{C}(\boldsymbol{\theta}) = \theta_3 \mathbf{I}_N$, where $\mathbf{1}_N = [1, \dots, 1]^T$ ($N \times 1$) and $\mathbf{I}_N$ is the $N \times N$ identity matrix. Direct differentiation gives

$$\frac{\partial \boldsymbol{\mu}}{\partial \theta_1} = \mathbf{1}_N \quad \frac{\partial \boldsymbol{\mu}}{\partial \theta_2} = \mathbf{x} \quad \frac{\partial \boldsymbol{\mu}}{\partial \theta_3} = \mathbf{0} \quad \frac{\partial \mathbf{C}}{\partial \theta_1} = \frac{\partial \mathbf{C}}{\partial \theta_2} = \mathbf{0} \quad \frac{\partial \mathbf{C}}{\partial \theta_3} = \mathbf{I}_N \quad (9.108)$$

Using (9.97) and recalling $\mu_x = N^{-1} \sum_{k=1}^N x_k$ and $\mu_{xx} = N^{-1} \sum_{k=1}^N x_k^2$,

$$J_{11} = (\mathbf{1}_N)^T (\theta_3 \mathbf{I}_N)^{-1} (\mathbf{1}_N) + \frac{1}{2} \text{tr} \left[(\theta_3 \mathbf{I}_N)^{-1} (\mathbf{0}) (\theta_3 \mathbf{I}_N)^{-1} (\mathbf{0}) \right] = N \theta_3^{-1} \quad (9.109)$$

$$J_{22} = \mathbf{x}^T (\theta_3 \mathbf{I}_N)^{-1} \mathbf{x} + \frac{1}{2} \text{tr} \left[(\theta_3 \mathbf{I}_N)^{-1} (\mathbf{0}) (\theta_3 \mathbf{I}_N)^{-1} (\mathbf{0}) \right] = \theta_3^{-1} \mathbf{x}^T \mathbf{x} = N \mu_{xx} \theta_3^{-1} \quad (9.110)$$

$$J_{33} = (\mathbf{0})^T (\theta_3 \mathbf{I}_N)^{-1} (\mathbf{0}) + \frac{1}{2} \text{tr} \left[(\theta_3 \mathbf{I}_N)^{-1} (\mathbf{I}_N) (\theta_3 \mathbf{I}_N)^{-1} (\mathbf{I}_N) \right] = \frac{1}{2} N \theta_3^{-2} \quad (9.111)$$

$$J_{12} = (\mathbf{1}_N)^T (\theta_3 \mathbf{I}_N)^{-1} (\mathbf{x}) + \frac{1}{2} \text{tr} \left[(\theta_3 \mathbf{I}_N)^{-1} (\mathbf{0}) (\theta_3 \mathbf{I}_N)^{-1} (\mathbf{0}) \right] = N \mu_x \theta_3^{-1} \quad (9.112)$$

$$J_{13} = (\mathbf{1}_N)^T (\theta_3 \mathbf{I}_N)^{-1} (\mathbf{0}) + \frac{1}{2} \text{tr} \left[(\theta_3 \mathbf{I}_N)^{-1} (\mathbf{0}) (\theta_3 \mathbf{I}_N)^{-1} (\mathbf{I}_N) \right] = 0 \quad (9.113)$$

and similarly $J_{23} = 0$. Other entries can be determined by symmetry. Assembling these into a matrix gives the same result in (9.61). ■

9.4.2 Complex Gaussian

The result in (9.97) can be extended to $\mathbf{Z} = \boldsymbol{\mu}(\boldsymbol{\theta}) + \mathbf{Z}'$ where $\boldsymbol{\mu}(\boldsymbol{\theta}) = E[\mathbf{Z}|\boldsymbol{\theta}]$ ($N \times 1$ complex) and $\mathbf{Z}'$ ($N \times 1$ complex) is a (zero-mean) circularly-symmetric complex Gaussian vector with (complex) covariance matrix $\mathbf{E}(\boldsymbol{\theta}) = E[\mathbf{Z}'\mathbf{Z}'^*|\boldsymbol{\theta}]$. See Appendix A for details on complex Gaussian vectors. The likelihood function and NLLF are given respectively by

$$p(\mathbf{Z}|\boldsymbol{\theta}) = \pi^{-N} |\mathbf{E}(\boldsymbol{\theta})|^{-1} \exp \left\{ -[\mathbf{Z} - \boldsymbol{\mu}(\boldsymbol{\theta})]^* \mathbf{E}(\boldsymbol{\theta})^{-1} [\mathbf{Z} - \boldsymbol{\mu}(\boldsymbol{\theta})] \right\} \quad (9.114)$$

$$L(\mathbf{Z}, \boldsymbol{\theta}) = -\ln p(\mathbf{Z}|\boldsymbol{\theta}) = N \ln \pi + \ln |\mathbf{E}(\boldsymbol{\theta})| + [\mathbf{Z} - \boldsymbol{\mu}(\boldsymbol{\theta})]^* \mathbf{E}(\boldsymbol{\theta})^{-1} [\mathbf{Z} - \boldsymbol{\mu}(\boldsymbol{\theta})] \quad (9.115)$$

Let J_{ij} denote the (i, j) -entry of the FIM $\mathbf{J}(\boldsymbol{\theta})$. Then it can be shown that

$$J_{ij} = 2\text{Re}\left(\frac{\partial \boldsymbol{\mu}^*}{\partial \theta_i} \mathbf{E}^{-1} \frac{\partial \boldsymbol{\mu}}{\partial \theta_j}\right) + \text{tr}\left(\mathbf{E}^{-1} \frac{\partial \mathbf{E}}{\partial \theta_i} \mathbf{E}^{-1} \frac{\partial \mathbf{E}}{\partial \theta_j}\right) \quad (9.116)$$

where $\boldsymbol{\mu}^*$ denotes the conjugate transpose of $\boldsymbol{\mu}$. Using the Hermitian nature of $\mathbf{E}$ and the cyclic property of trace, it can be verified that $J_{ij} = J_{ji}$, as expected. Despite its appearance, it can be shown that the second term on the RHS is always real-valued. Equation (9.116) is used in Chap. 15 for deriving the long-data asymptotic expressions for the posterior covariance matrix of modal properties in the Bayesian approach.

9.5 Asymptotic Properties of ML Estimator

The ML estimator has good theoretical asymptotic properties for large sample size. Under mild assumptions, it is asymptotically unbiased and Gaussian; and its covariance matrix achieves the CRB. Specifically, assuming that $\mathbf{Z}$ is distributed as the likelihood function $p(\mathbf{Z}|\boldsymbol{\theta})$, the ML estimator $\hat{\boldsymbol{\theta}}(\mathbf{Z})$ is asymptotically Gaussian with mean $\boldsymbol{\theta}$ and covariance matrix $\mathbf{J}(\boldsymbol{\theta})^{-1}$ as the sample size increases. Roughly speaking, for the asymptotic properties to hold, the data should have limited correlation so that their amount of information increases indefinitely as sample size increases.

The asymptotic properties can be reasoned from the definition of ML estimator and using the Strong Law of Large Numbers and Central Limit Theorem. Here we illustrate the argument assuming $\mathbf{Z} = [\mathbf{Z}_1; \dots; \mathbf{Z}_N]$, where $\{\mathbf{Z}_k\}$ are i.i.d., each with identical NLLF $L_0(\mathbf{Z}_k, \boldsymbol{\theta}) = -\ln p(\mathbf{Z}_k|\boldsymbol{\theta})$. In this case, the NLLF based on all data $\mathbf{Z}$ is the sum of the individual ones:

$$L(\mathbf{Z}, \boldsymbol{\theta}) = \sum_{k=1}^N L_0(\mathbf{Z}_k, \boldsymbol{\theta}) \quad (9.117)$$

The FIM based on $\mathbf{Z}$ is then given by

$$\mathbf{J}(\boldsymbol{\theta}) = E[\nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta})|\boldsymbol{\theta}] = \sum_{k=1}^N \underbrace{E[\nabla_{\boldsymbol{\theta}}^2 L_0(\mathbf{Z}_k, \boldsymbol{\theta})|\boldsymbol{\theta}]}_{\mathbf{J}_0(\boldsymbol{\theta})} = N\mathbf{J}_0(\boldsymbol{\theta}) \quad (9.118)$$

where $\mathbf{J}_0(\boldsymbol{\theta})$ is the FIM based on each $\mathbf{Z}_k$. For a given $\mathbf{Z}$, since $L(\mathbf{Z}, \boldsymbol{\theta})$ is minimized at $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}(\mathbf{Z})$

$$\nabla_{\boldsymbol{\theta}} L(\mathbf{Z}, \boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(\mathbf{Z})} = \mathbf{0} \quad (9.119)$$

On the other hand, using a first order Taylor expansion,

$$\nabla_{\boldsymbol{\theta}} L(\mathbf{Z}, \boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(\mathbf{Z})} = \nabla_{\boldsymbol{\theta}} L(\mathbf{Z}, \boldsymbol{\theta}) + [\hat{\boldsymbol{\theta}}(\mathbf{Z}) - \boldsymbol{\theta}]^T \nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta}) + \text{Remainder} \quad (9.120)$$

where the remainder on the RHS is second order in $[\hat{\boldsymbol{\theta}}(\mathbf{Z}) - \boldsymbol{\theta}]$. Equation (9.119) and (9.120) imply

$$\sqrt{N}[\hat{\boldsymbol{\theta}}(\mathbf{Z}) - \boldsymbol{\theta}] \sim -\mathbf{Y}^{-1}\mathbf{X} \quad (9.121)$$

where ‘ $\sim$ ’ denotes that the LHS and RHS are asymptotically the same when $N \rightarrow \infty$;

$$\mathbf{X} = \frac{1}{\sqrt{N}} \nabla_{\boldsymbol{\theta}}^T L(\mathbf{Z}, \boldsymbol{\theta}) = \frac{1}{\sqrt{N}} \sum_{k=1}^N \nabla_{\boldsymbol{\theta}}^T L_0(\mathbf{Z}_k, \boldsymbol{\theta}) \quad (9.122)$$

$$\mathbf{Y} = \frac{1}{N} \nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta}) = \frac{1}{N} \sum_{k=1}^N \nabla_{\boldsymbol{\theta}}^2 L_0(\mathbf{Z}_k, \boldsymbol{\theta}) \quad (9.123)$$

Consider the distributions of $\mathbf{X}$ and $\mathbf{Y}$ when data $\mathbf{Z}$ is distributed as $p(\mathbf{Z}|\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ is the true parameter value. From (9.87), $E[\mathbf{X}|\boldsymbol{\theta}] = \mathbf{0}$. The covariance matrix of $\mathbf{X}$ is

$$E[\mathbf{X}\mathbf{X}^T|\boldsymbol{\theta}] = \frac{1}{N} E[\nabla_{\boldsymbol{\theta}}^T L(\mathbf{Z}, \boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}} L(\mathbf{Z}, \boldsymbol{\theta})|\boldsymbol{\theta}] = \frac{1}{N} \mathbf{J}(\boldsymbol{\theta}) = \mathbf{J}_0(\boldsymbol{\theta}) \quad (9.124)$$

Since $\{\mathbf{Z}_k\}_{k=1}^N$ and hence $\{\nabla_{\boldsymbol{\theta}} L_0(\mathbf{Z}_k, \boldsymbol{\theta})\}_{k=1}^N$ are i.i.d., by the Central Limit Theorem, $\mathbf{X}$ is asymptotically Gaussian with mean zero and covariance matrix $\mathbf{J}_0(\boldsymbol{\theta})$ as $N \rightarrow \infty$. On the other hand, $E[\mathbf{Y}|\boldsymbol{\theta}] = N^{-1} \mathbf{J}(\boldsymbol{\theta}) = \mathbf{J}_0(\boldsymbol{\theta})$. Since $\mathbf{Y}$ is the average of the i.i.d. matrices $\{\nabla_{\boldsymbol{\theta}}^2 L_0(\mathbf{Z}_k, \boldsymbol{\theta})\}_{k=1}^N$, by the Strong Law of Large Numbers it tends to its expectation $\mathbf{J}_0(\boldsymbol{\theta})$ as $N \rightarrow \infty$. The asymptotic properties of $\mathbf{X}$ and $\mathbf{Y}$ imply that $\sqrt{N}[\hat{\boldsymbol{\theta}}(\mathbf{Z}) - \boldsymbol{\theta}] \sim -\mathbf{Y}^{-1}\mathbf{X}$ is asymptotically Gaussian with zero mean and covariance matrix equal to $\mathbf{J}_0(\boldsymbol{\theta})^{-1}$. Equation (9.121) then implies that $\hat{\boldsymbol{\theta}}(\mathbf{Z})$ is asymptotically Gaussian with mean $\boldsymbol{\theta}$ (i.e., asymptotically unbiased) and covariance matrix $[N\mathbf{J}_0(\boldsymbol{\theta})]^{-1}$, i.e., $\mathbf{J}(\boldsymbol{\theta})^{-1}$, reaching the CRB.

The asymptotic properties of the ML estimator hold in more general situations where the data are neither independent nor identically distributed. As seen in the above reasoning, the conditions are related to those that ensure the Central Limit Theorem to hold for $\mathbf{X} = \nabla_{\boldsymbol{\theta}}^T L(\mathbf{Z}, \boldsymbol{\theta})/\sqrt{N}$ and the Strong Law of Large Numbers to hold for $\mathbf{Y} = \nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta})/N$. Roughly speaking, the conditions ensure that data are not too strongly correlated so that the effective number of independent observations

can increase indefinitely as $N \rightarrow \infty$; and that probabilistically the sum is not dominated by a small number of terms whose magnitudes are significantly larger than the remaining ones.

Example 9.12 (Mean parameter, asymptotics of MLE) In Example 9.1, the estimator G_2 for the prediction error variance was found to be the ML estimator but it was biased for finite sample size N . Nevertheless, since $E[G_2|\boldsymbol{\theta}] = (1 - 1/N)^{-1}\theta_2 \rightarrow \theta_2$ as $N \rightarrow \infty$, G_2 is asymptotically unbiased. These results were concluded based on the expression for the variance of estimator, which may not be available in general problems. Here we illustrate the asymptotic behavior of the NLLF that accounts for the asymptotic properties of the ML estimator G_2 .

Recall from (9.42), (9.43) and (9.44) that

$$L(\{\hat{y}_k\}, \boldsymbol{\theta}) = -\ln p(\{\hat{y}_k\}|\boldsymbol{\theta}) = \frac{N}{2} \ln(2\pi\theta_2) + \frac{1}{2\theta_2} \sum_{k=1}^N (\hat{y}_k - \theta_1)^2 \quad (9.125)$$

$$\nabla_{\boldsymbol{\theta}} L = \begin{bmatrix} N\theta_2^{-1}(\theta_1 - \mu_1) & \frac{N}{2}\theta_2^{-1} - \frac{1}{2}\theta_2^{-2} \sum_{k=1}^N (\hat{y}_k - \theta_1)^2 \end{bmatrix} \quad (9.126)$$

$$\nabla_{\boldsymbol{\theta}}^2 L = \begin{bmatrix} N\theta_2^{-1} & -N\theta_2^{-2}(\theta_1 - \mu_1) \\ -N\theta_2^{-2}(\theta_1 - \mu_1) & -\frac{N}{2}\theta_2^{-2} + \theta_2^{-3} \sum_{k=1}^N (\hat{y}_k - \theta_1)^2 \end{bmatrix} \quad (9.127)$$

where $\mu_1 = \sum_{k=1}^N \hat{y}_k/N$. Given $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$, $\{\hat{y}_k\}_{k=1}^N$ are i.i.d. Gaussian with mean θ_1 and variance θ_2 . By the Strong Law of Large Numbers, as $N \rightarrow \infty$,

$$\mathbf{Y} = \frac{1}{N} \nabla_{\boldsymbol{\theta}}^2 L \rightarrow \begin{bmatrix} \theta_2^{-1} & 0 \\ 0 & \theta_2^{-2}/2 \end{bmatrix} = \frac{1}{N} \mathbf{J}(\boldsymbol{\theta}) \quad (9.128)$$

since $\mu_1 \rightarrow \theta_1$ and $N^{-1} \sum_{k=1}^N (\hat{y}_k - \theta_1)^2 \rightarrow \theta_2$; $\mathbf{J}(\boldsymbol{\theta})$ is the FIM in (9.45).

On the other hand, it can be reasoned from (9.126) that $\nabla_{\boldsymbol{\theta}} L$ has zero expectation. Its covariance matrix is by definition equal to $\mathbf{J}(\boldsymbol{\theta})$. The first entry of $\nabla_{\boldsymbol{\theta}} L$ is Gaussian because μ_1 is. By the Central Limit Theorem, the sum $\sum_{k=1}^N (\hat{y}_k - \theta_1)^2$ and hence the second entry of $\nabla_{\boldsymbol{\theta}} L$ is asymptotically Gaussian. Consequently, $\mathbf{X} = \nabla_{\boldsymbol{\theta}} L / \sqrt{N}$ is asymptotically a Gaussian vector with zero mean and covariance matrix $\mathbf{J}(\boldsymbol{\theta})/N$. The asymptotic properties of $\mathbf{X}$ and $\mathbf{Y}$ in this example agree with those discussed in the general case. ■

9.6 Comparison with Bayesian Inference

The foregoing discussions of the classical statistical approach reveals many similarities in mathematics with the Bayesian approach in Chap. 8, e.g., the likelihood function, its gradient and Hessian. Nevertheless, there are important differences that distinguish the two approaches fundamentally and that matter in the interpretation of results and application to general inference problems. Here we compare the two approaches within the scope of system identification. General discussions can be found in the references mentioned in the introduction of Chap. 8.

To allow a proper comparison, assume that when applying the Bayesian or classical statistical approaches the same (modeled) likelihood function $p(\mathbf{Z}|\boldsymbol{\theta})$ is used. In the discussion of the CRB and the asymptotic behavior of ML estimators, we also assume that data $\mathbf{Z}$ is indeed distributed as $p(\mathbf{Z}|\boldsymbol{\theta})$ for some unknown true parameter value $\boldsymbol{\theta}$.

9.6.1 Philosophical Perspectives

Given the measured data $\mathbf{Z}$, the Bayesian approach asks the question:

‘What do we know probabilistically about $\boldsymbol{\theta}$?’

The answer to this question is encapsulated in the posterior distribution $p(\boldsymbol{\theta}|\mathbf{Z})$, which is directly related to the likelihood function. In a Bayesian context, $\boldsymbol{\theta}$ is always uncertain and modeled as a random variable. The presence of data does not make it precisely known. Its distribution depends on available information, which is indicated in the conditioning statement. In the absence of data, or without using the information in data, the distribution is $p(\boldsymbol{\theta})$ (prior PDF). When information in the data is used, the distribution is $p(\boldsymbol{\theta}|\mathbf{Z})$ (posterior PDF). When interpreting or using this function, $\mathbf{Z}$ is fixed at the available data and $\boldsymbol{\theta}$ is a variable. Under the same modeling assumptions, data and prior information, the posterior distribution is unique. This is all one needs to address the system identification question for a given model and data. The remaining issues are only on calculating the descriptive statistics (e.g., mean and variance) of the posterior distribution for use in decision making. These issues are important but nevertheless of computational nature only.

The classical statistical approach constructs a statistical estimator $\mathbf{G}(\mathbf{Z})$ whose value can be calculated for given measured data $\mathbf{Z}$ and is intended to be used as a proxy for $\boldsymbol{\theta}$. This estimator is not unique. It is subjected to the developer’s mathematical facility, imagination, familiarity with the problem, etc. Assessing the quality of this estimator (e.g., bias, convergence) then becomes relevant.

Identification uncertainty in the classical statistical approach refers to the ensemble variability of estimator $\mathbf{G}(\mathbf{Z})$ (not the unknown parameter $\boldsymbol{\theta}$) over hypothetical repeated experiments. This uncertainty does not depend on the particular data set and is therefore conceptually ‘inherent’ in nature. Analytical expression for the variance of estimator, when it is feasible to derive, assumes that

$\mathbf{Z}$ is distributed as the likelihood function conditional on the ‘true’ parameter value, i.e., $p(\mathbf{Z}|\boldsymbol{\theta})$. Here $\boldsymbol{\theta}$ is not known but mathematically it is always fixed in all statistical statements about $\mathbf{G}(\mathbf{Z})$. For example, the expression for the variance of $\mathbf{G}(\mathbf{Z})$ depends on $\boldsymbol{\theta}$. Since it is unknown in applications, it is replaced by the value of the estimate (calculated from available data) to give a value that approximates the ‘true’ variance. When it is not possible to derive an analytical expression, the variance of estimator can be estimated by the sample average of estimator values over repeated experiments. In this case, it need not be the same as the theoretical one because the data in reality need not be distributed as the likelihood function. The resulting sample variance consists of modeling error as well, although it is generally not trivial to distinguish between identification error (for a given model class) and modeling error.

The posterior distribution in the Bayesian approach is conditional on a given model but it does not involve the concept of a ‘true parameter value’. Neither are ‘true variance’ or ‘inherent uncertainty’ relevant. The posterior distribution can always be calculated for a given set of data (and model class). It is a function of data $\mathbf{Z}$ and so there is no concept of intrinsic distribution or uncertainty. Different $\mathbf{Z}$ s give different posterior distributions. Except for computational reasons, it is irrelevant to talk about ‘estimating’ the posterior distribution or posterior variance. The ensemble average concept using hypothetical repeated experiments is irrelevant. Averaging the posterior distribution for different $\mathbf{Z}$ s has no direct meaning.

Table 9.2 gives a summary of comparisons of the Bayesian and classical statistical approaches for system identification.

9.6.2 *Maximum Likelihood Estimator*

Within the classical statistical perspective, consider now ML estimators. To avoid conceptual problems arising from non-uniqueness of the ML estimator value, we consider globally identifiable situations. Being the value that maximizes the likelihood function for given data, the ML estimator is mathematically the same as the posterior MPV in a Bayesian context assuming a uniform prior distribution. As discussed in Sect. 9.5, the ML estimator is asymptotically unbiased and Gaussian as the data size increases. Its covariance matrix asymptotically approaches the inverse of the FIM. The latter is equal to the expectation of Hessian of NLLF. These are theoretical statements that assume the data is indeed distributed as the likelihood function conditional on the true parameter value. In a Bayesian context, within a Gaussian approximation of the posterior distribution w.r.t. the model parameters, the posterior covariance matrix is equal to the inverse of the Hessian of NLLF at the posterior MPV.

The distribution of the ML estimator should not be confused with the posterior distribution of model parameters. Neither should the asymptotic Gaussian nature of the ML estimator be confused with the Gaussian approximation of the posterior distribution. The latter is merely an approximation arising from the second order

Table 9.2 Comparison of the Bayesian and classical statistical approaches

Bayesian	Classical statistics
θ is unknown and modeled as a random variable whose distribution depends on available information	θ is the ‘true’ parameter value, unknown but fixed
Identification result is the posterior PDF $p(\theta \mathbf{Z})$	Identification result is estimator $\mathbf{G}(\mathbf{Z})$, suggested by a developer
Different data $\mathbf{Z}$ gives different $p(\theta \mathbf{Z})$	Different data $\mathbf{Z}$ gives different values of $\mathbf{G}(\mathbf{Z})$
Identification uncertainty is related to the spread of $p(\theta \mathbf{Z})$ w.r.t. θ for given $\mathbf{Z}$	Identification uncertainty is the variability of $\mathbf{G}(\mathbf{Z})$ over hypothetical repeated trials of $\mathbf{Z}$
Identification uncertainty (for a give model class) does not tell modeling error	Variability of $\mathbf{G}(\mathbf{Z})$ estimated from real data is an aggregate of identification uncertainty and other effects (e.g., changing conditions, modeling error)

Taylor approximation of the NLLF w.r.t. the model parameters about the MPV; and it is computational in nature. No asymptotic concept is involved. The approximation can be exact even for finite data size if the NLLF is indeed a quadratic function of the model parameters. On the other hand, the posterior distribution can be non-Gaussian even if the data size increases to infinity.

Being the inverse of the FIM, the asymptotic covariance matrix of the ML estimator shares some similarity with the posterior covariance matrix in the Gaussian approximation, as they are both related to the Hessian of the NLLF. There are some important differences, however. For the posterior covariance matrix, the Hessian is evaluated at the posterior MPV and so it is always positive definite. The Hessian is calculated for a given set of data and no expectation (ensemble average) is involved. For the FIM, the Hessian is evaluated at the unknown true parameter value, and so it is not necessarily positive definite; only its expectation is. It cannot be calculated using a single set of data, as it involves expectation over data.

Figure 9.1 compares Bayesian and ML inference. To avoid confusion in notation the posterior distribution is denoted by $p(\xi|\mathbf{Z})$ (instead of $p(\theta|\mathbf{Z})$), where ξ denotes a dummy value of θ . The asymptotic property of the posterior covariance matrix in relation to the FIM is discussed in the next section.

9.6.3 Cramér-Rao Bound and Uncertainty Law

Recall from Sect. 8.5 that, for globally identifiable problems, the posterior distribution for given data $\mathbf{Z}$ can be approximated by a Gaussian PDF with covariance matrix $\hat{\mathbf{C}}(\mathbf{Z})$ given by:

$$\hat{\mathbf{C}}(\mathbf{Z}) = \left[\nabla_{\theta}^2 L(\mathbf{Z}, \theta) \big|_{\theta=\hat{\theta}(\mathbf{Z})} \right]^{-1} \quad (9.129)$$

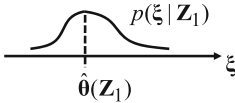
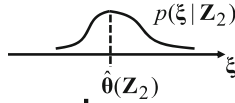
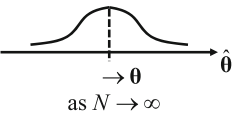
Experiment	Data	Identification result	Properties
Trial 1	$\mathbf{Z}_1$		Posterior PDF of $\boldsymbol{\theta}$ given $\mathbf{Z}$ $p(\xi \mathbf{Z}) \approx$ Gaussian PDF with mean $\hat{\boldsymbol{\theta}}(\mathbf{Z})$ and covariance matrix $\hat{\mathbf{H}}(\mathbf{Z})^{-1}$ (Hessian of NLLF at $\hat{\boldsymbol{\theta}}(\mathbf{Z})$)
Trial 2	$\mathbf{Z}_2$		When $\mathbf{Z}$ is distributed as $p(\mathbf{Z} \boldsymbol{\theta})$ $\hat{\mathbf{H}}(\mathbf{Z}) \sim \mathbf{J}(\boldsymbol{\theta})[\mathbf{I} + O_{\mathbf{Z}}(N^{-1/2})]$ as $N \rightarrow \infty$
$\vdots$	$\vdots$	$\vdots$	
Ensemble statistics of $\hat{\boldsymbol{\theta}}(\mathbf{Z})$			PDF of $\hat{\boldsymbol{\theta}}(\mathbf{Z})$ Asymp. Gaussian with mean $\boldsymbol{\theta}$ and covariance matrix $\mathbf{J}(\boldsymbol{\theta})^{-1}$ as $N \rightarrow \infty$
Assume $\mathbf{Z}$ distributed as $p(\mathbf{Z} \boldsymbol{\theta})$		$\rightarrow \boldsymbol{\theta}$ as $N \rightarrow \infty$	

Fig. 9.1 Comparison of Bayesian and ML inference, assuming data $\mathbf{Z}$ is distributed as the likelihood function $p(\mathbf{Z}|\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ is the true value of the unknown parameter. The posterior PDF is denoted by $p(\xi|\mathbf{Z})$ where ξ denotes a dummy parameter value of $\boldsymbol{\theta}$. NLLF = Negative log-likelihood function; $\mathbf{J}(\boldsymbol{\theta}) = E[\nabla_{\boldsymbol{\theta}}^2 \ln p(\mathbf{Z}|\boldsymbol{\theta})] =$ Fisher information matrix

where the dependence on $\mathbf{Z}$ has been emphasized; $\boldsymbol{\theta}$ here is a dummy variable for the model parameters. On the other hand, in classical statistical estimation, the CRB is given by the inverse of the FIM:

$$\mathbf{J}(\boldsymbol{\theta})^{-1} = E[\nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta})|\boldsymbol{\theta}]^{-1} \quad (9.130)$$

where $\boldsymbol{\theta}$ here is the true parameter value. Under similar assumptions for the asymptotic properties of the ML estimator, it is possible to connect mathematically the posterior covariance matrix and the (tightest) CRB. For a large sample size, denoted by N , the ‘leading order’ of the posterior covariance matrix is asymptotically equal to the CRB, i.e.,

$$\hat{\mathbf{C}}(\mathbf{Z}) = \underbrace{\mathbf{J}(\boldsymbol{\theta})^{-1}}_{\text{leading order}} [\mathbf{I} + O_{\mathbf{Z}}(N^{-1/2})] \quad N \rightarrow \infty \quad (9.131)$$

where $\mathbf{I}$ denotes the identity matrix; $O_{\mathbf{Z}}(N^{-1/2})$ denotes that the remainder depends on data $\mathbf{Z}$ and is of order $N^{-1/2}$.

Equation (9.131) can be deduced as follow. First, note that $\hat{\boldsymbol{\theta}}(\mathbf{Z})$ is asymptotically unbiased and its covariance matrix is $O(N^{-1})$. We can then write $\hat{\boldsymbol{\theta}}(\mathbf{Z}) = [\mathbf{I} + O_{\mathbf{Z}}(N^{-1/2})]\boldsymbol{\theta}$. Assuming that the third order derivatives of NLLF exist,

$$\frac{1}{N} \nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(\mathbf{Z})} = \frac{1}{N} \nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta}) [\mathbf{I} + O_{\mathbf{Z}}(N^{-1/2})] \rightarrow \frac{1}{N} \mathbf{J}(\boldsymbol{\theta}) [\mathbf{I} + O_{\mathbf{Z}}(N^{-1/2})] \quad (9.132)$$

since $N^{-1} \nabla_{\boldsymbol{\theta}}^2 L(\mathbf{Z}, \boldsymbol{\theta}) \rightarrow N^{-1} \mathbf{J}(\boldsymbol{\theta})$ (Sect. 9.5). Equation (9.131) then follows by noting that matrix inverse is a continuous mapping. The leading order term of the posterior covariance matrix is called ‘uncertainty law’ and will be discussed in Chaps. 15 and 16 for operational modal analysis.

Example 9.13 (Mean parameter, asymptotics of posterior covariance matrix)

In this example, we demonstrate the mathematical connection between the posterior covariance matrix (under Gaussian approximation of the posterior PDF) and the inverse of the FIM for the problem in Example 9.4. Recall that $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$ is to be identified from data modeled as $\hat{y}_k = \theta_1 + \varepsilon_k$, where $\{\varepsilon_k\}_{k=1}^N$ are assumed to be i.i.d. zero-mean Gaussian with variance θ_2 . The FIM was obtained in (9.45):

$$\mathbf{J}(\boldsymbol{\theta}) = \begin{bmatrix} N\theta_2^{-1} & 0 \\ 0 & N\theta_2^{-2}/2 \end{bmatrix} \quad (9.133)$$

The Hessian of NLLF at the posterior MPV was obtained in Example 8.4:

$$\hat{\mathbf{H}}_L = \begin{bmatrix} N(\mu_2 - \mu_1^2)^{-1} & 0 \\ 0 & N(\mu_2 - \mu_1^2)^{-2}/2 \end{bmatrix} \quad (9.134)$$

where $\mu_1 = \sum_{k=1}^N \hat{y}_k/N$ and $\mu_2 = \sum_{k=1}^N \hat{y}_k^2/N$. This can be written as

$$\hat{\mathbf{H}}_L = \mathbf{J}(\boldsymbol{\theta}) \begin{bmatrix} X_1 & 0 \\ 0 & X_2 \end{bmatrix} \quad X_1 = \theta_2(\mu_2 - \mu_1^2)^{-1} \quad X_2 = \theta_2^2(\mu_2 - \mu_1^2)^{-2} \quad (9.135)$$

Assume the data is indeed distributed according to the likelihood function for a true parameter $\boldsymbol{\theta}$. It can then be reasoned that $\mu_2 - \mu_1^2$ is asymptotically Gaussian with mean θ_2 and variance $2\theta_2/N$ as $N \rightarrow \infty$. This implies that asymptotically X_1 and X_2 have a mean of 1 and a variance of $O(N^{-1})$, i.e., a standard deviation of $O(N^{-1/2})$. Then we can write

$$\begin{bmatrix} X_1 & 0 \\ 0 & X_2 \end{bmatrix} = \mathbf{I}_2 + O_{\mathbf{Z}}(N^{-1/2}) \quad (9.136)$$

where $\mathbf{I}_2$ denotes the 2×2 identity matrix and $O_{\mathbf{Z}}(N^{-1/2})$ denotes the remainder term that depends on data $\mathbf{Z}$ and is $O(N^{-1/2})$. Substituting into (9.135),

$$\hat{\mathbf{H}}_L = \mathbf{J}(\boldsymbol{\theta})[\mathbf{I}_2 + O_{\mathbf{Z}}(N^{-1/2})] \quad (9.137)$$

A similar relationship holds for its inverse, as in (9.131). ■

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Chapter 10

Bayesian OMA Formulation

Abstract This chapter develops Bayesian formulations in operational modal analysis for data of different natures, including synchronous data in a single setup (the conventional setting), data from multiple setups and asynchronous data in a single setup. Modeling assumptions and their justifications are discussed.

Keywords OMA formulation • Single mode • Multi-mode • Multi-setup • Asynchronous

In this chapter we formulate Bayesian inference problems for operational modal analysis (OMA) using the scaled Fast Fourier Transform (FFT) of ambient vibration data. This involves stating the modeling assumptions, the parameters to be identified and deriving the likelihood function, which is practically the posterior probability density function (PDF) of parameters. Typical scenarios are considered, including data from a single setup, data from multiple setups at different time periods, and asynchronous data in a single setup. This chapter covers formulations but not computational issues in the determination of posterior statistics. The latter are discussed in a general context in Chap. 11. Efficient algorithms in specific contexts are considered in Chaps. 12–14. See Sect. 1.5.4 for a literature note.

10.1 Single Setup Data

Let $\{\hat{\mathbf{y}}_j\}_{j=0}^{N-1}$ ($n \times 1$) be the ambient vibration time history data measured at n DOFs of a structure, where N is the number of samples per data channel. It is modeled as a stationary stochastic process. Its scaled FFT at frequency $f_k = k/N\Delta t$ (Hz) is defined as

$$\hat{\mathcal{F}}_k = \sqrt{\frac{\Delta t}{N}} \sum_{j=0}^{N-1} \hat{\mathbf{y}}_j e^{-2\pi i j k / N} \quad (10.1)$$

where Δt (s) is the sampling time interval. The FFTs on a selected frequency band, denoted by $\{\hat{\mathcal{F}}_k\}$, is used for identifying the modal properties. Typically, the selected band covers the resonance band of the mode(s) of interest.

Let $\boldsymbol{\theta}$ denote the set of modal parameters to be identified. It primarily comprises the natural frequencies, damping ratios, mode shapes and other parameters (see later) necessary to determine explicitly the distribution of $\{\hat{\mathcal{F}}_k\}$. Using Bayes' Theorem, the posterior PDF of $\boldsymbol{\theta}$ given $\{\hat{\mathcal{F}}_k\}$ is

$$p(\boldsymbol{\theta}|\{\hat{\mathcal{F}}_k\}) = p(\{\hat{\mathcal{F}}_k\})^{-1} p(\{\hat{\mathcal{F}}_k|\boldsymbol{\theta}\}) p(\boldsymbol{\theta}) \quad (10.2)$$

The term $p(\{\hat{\mathcal{F}}_k\})^{-1}$ is a normalizing constant and is immaterial to the posterior PDF. For typical data size, the likelihood function $p(\{\hat{\mathcal{F}}_k|\boldsymbol{\theta}\})$ is fast varying compared to the prior PDF $p(\boldsymbol{\theta})$. Without loss of generality, $p(\boldsymbol{\theta})$ is assumed to be a uniform distribution. The posterior PDF is then directly proportional to the likelihood function:

$$p(\boldsymbol{\theta}|\{\hat{\mathcal{F}}_k\}) \propto p(\{\hat{\mathcal{F}}_k|\boldsymbol{\theta}\}) \quad (10.3)$$

10.1.1 Likelihood Function

The likelihood function is the joint PDF of $\{\hat{\mathcal{F}}_k\}$ for given $\boldsymbol{\theta}$. Assuming long data (Sect. 4.8.3), $\{\hat{\mathcal{F}}_k\}$ are (circularly symmetric) complex Gaussian and independent at different frequencies. This gives

$$p(\{\hat{\mathcal{F}}_k|\boldsymbol{\theta}\}) = \prod_k p(\hat{\mathcal{F}}_k|\boldsymbol{\theta}) \quad (10.4)$$

where the product is taken over all frequencies in the selected band with N_f FFT points, assumed to be large compared to 1 (long data); and

$$p(\hat{\mathcal{F}}_k|\boldsymbol{\theta}) = \frac{\pi^{-n}}{|\mathbf{E}_k(\boldsymbol{\theta})|} \exp\left[-\hat{\mathcal{F}}_k^* \mathbf{E}_k(\boldsymbol{\theta})^{-1} \hat{\mathcal{F}}_k\right] \quad (10.5)$$

$\mathbf{E}_k(\boldsymbol{\theta}) = E[\hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^* | \boldsymbol{\theta}]$ ($n \times n$ Hermitian) is the theoretical power spectral density (PSD) matrix of data for given $\boldsymbol{\theta}$. This complex Gaussian PDF is central to Bayesian OMA as it holds for general stationary ambient data regardless of the

frequency content of contributing activities. The latter affects only $\mathbf{E}_k(\boldsymbol{\theta})$ but not the form of the PDF.

Resonance band modeling

Deriving $\mathbf{E}_k(\boldsymbol{\theta})$ involves modeling the ambient data. Within the selected frequency band, it is assumed that

$$\hat{\mathcal{F}}_k = \mathcal{F}_k + \boldsymbol{\varepsilon}_k \quad (10.6)$$

where $\mathcal{F}_k$ and $\boldsymbol{\varepsilon}_k$ denote respectively the scaled FFT of the theoretical structural dynamic response and prediction error (e.g., data noise). Their distributions depend on $\boldsymbol{\theta}$. Suppose the selected band is dominated by m vibration modes, referred as mode 1, 2, ..., m . Then $\mathcal{F}_k = \sum_{i=1}^m \boldsymbol{\varphi}_i \zeta_{ik}$ where $\boldsymbol{\varphi}_i$ ($n \times 1$) is the partial mode shape (i.e., confined to measured DOFs) of mode i , and ζ_{ik} is the scaled FFT of modal response at frequency f_k . Let p_{ik} be the scaled FFT of the i th modal force at frequency f_k . Then $\zeta_{ik} = h_{ik} p_{ik}$ (Sect. 5.2.1) and

$$\hat{\mathcal{F}}_k = \sum_{i=1}^m \boldsymbol{\varphi}_i h_{ik} p_{ik} + \boldsymbol{\varepsilon}_k \quad (10.7)$$

where, for mode i ,

$$h_{ik} = \frac{(2\pi i f_k)^{-q}}{1 - \beta_{ik}^2 - 2\zeta_i \beta_{ik} i} \quad \beta_{ik} = \frac{f_i}{f_k} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (10.8)$$

is the transfer function, f_i (Hz) and ζ_i are respectively the natural frequency and damping ratio. The modal forces are assumed to have a constant PSD matrix $\mathbf{S}$ ($m \times m$ Hermitian) in the selected band, whose (i, j) -entry is $S_{ij} = E[p_{ik} p_{jk}^* | \boldsymbol{\theta}]$. The prediction errors at different measured DOFs are assumed to be i.i.d. with a constant PSD S_e in the band:

$$E[\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^* | \boldsymbol{\theta}] = S_e \mathbf{I}_n \quad (10.9)$$

where $\mathbf{I}_n$ denotes the $n \times n$ identity matrix. They are also assumed to be independent of the modal forces. Consequently,

$$\mathbf{E}_k(\boldsymbol{\theta}) = E[\mathcal{F}_k \mathcal{F}_k^* | \boldsymbol{\theta}] + E[\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^* | \boldsymbol{\theta}] = \sum_{i=1}^m \sum_{j=1}^m h_{ik} h_{jk}^* S_{ij} \boldsymbol{\varphi}_i \boldsymbol{\varphi}_j^T + S_e \mathbf{I}_n \quad (10.10)$$

This can be written in matrix form as

$$\mathbf{E}_k = \underbrace{\mathbf{\Phi} \mathbf{H}_k \mathbf{\Phi}^T}_{\text{modal contribution}} + \underbrace{S_e \mathbf{I}_n}_{\text{prediction error}} \quad (10.11)$$

where

$$\mathbf{\Phi} = [\boldsymbol{\varphi}_1, \dots, \boldsymbol{\varphi}_m] \quad (10.12)$$

is the partial mode shape matrix;

$$\mathbf{H}_k = \text{diag}(\mathbf{h}_k) \mathbf{S} \text{diag}(\mathbf{h}_k)^* \quad \mathbf{h}_k = [h_{1k}, \dots, h_{mk}] \quad (10.13)$$

and $\text{diag}(\mathbf{h}_k)$ denotes a diagonal matrix of the entries in $\mathbf{h}_k$. The (i, j) -entry of $\mathbf{H}_k$ is $S_{ij} h_{ik} h_{jk}^*$.

Posterior PDF and negative log-likelihood function

Combining (10.3)–(10.5), the posterior PDF of $\boldsymbol{\theta}$ is given by

$$p(\boldsymbol{\theta} | \{\hat{\mathcal{F}}_k\}) \propto p(\{\hat{\mathcal{F}}_k\} | \boldsymbol{\theta}) = \frac{\pi^{-nN_f}}{\prod_k |\mathbf{E}_k(\boldsymbol{\theta})|} \exp \left[- \sum_k \hat{\mathcal{F}}_k^* \mathbf{E}_k(\boldsymbol{\theta})^{-1} \hat{\mathcal{F}}_k \right] \quad (10.14)$$

where $\mathbf{E}_k(\boldsymbol{\theta})$ is given by (10.11). The set of modal parameters $\boldsymbol{\theta}$ comprises $\{f_i\}_{i=1}^m$, $\{\zeta_i\}_{i=1}^m$, $\mathbf{S}$, S_e and $\mathbf{\Phi}$. For convenience in analysis and computation, the posterior PDF is expressed as

$$p(\boldsymbol{\theta} | \{\hat{\mathcal{F}}_k\}) \propto p(\{\hat{\mathcal{F}}_k\} | \boldsymbol{\theta}) = e^{-L(\boldsymbol{\theta})} \quad (10.15)$$

where

$$L(\boldsymbol{\theta}) = nN_f \ln \pi + \sum_k \ln |\mathbf{E}_k(\boldsymbol{\theta})| + \sum_k \hat{\mathcal{F}}_k^* \mathbf{E}_k(\boldsymbol{\theta})^{-1} \hat{\mathcal{F}}_k \quad (10.16)$$

is the ‘negative log-likelihood function’ (NLLF).

For typical data size, modal identification problem is globally identifiable (Sect. 8.5). The posterior PDF has a unique maximum at the most probable value (MPV). Equivalently, the NLLF has a unique minimum at the MPV. A second

order Taylor approximation of the NLLF at the MPV leads to a Gaussian approximation of the posterior PDF:

$$p(\boldsymbol{\theta}|\{\hat{\mathcal{F}}_k\}) \approx (2\pi)^{-n_\theta/2} |\hat{\mathbf{C}}|^{-1/2} \exp\left[-\frac{1}{2}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T \hat{\mathbf{C}}^{-1}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})\right] \quad (10.17)$$

where $\hat{\boldsymbol{\theta}}$ is the MPV; $\hat{\mathbf{C}}$ is the covariance matrix, equal to the inverse of the Hessian of NLLF at the MPV; n_θ is the number of parameters in $\boldsymbol{\theta}$. Both $\hat{\boldsymbol{\theta}}$ and $\hat{\mathbf{C}}$ depend on the FFT data $\{\hat{\mathcal{F}}_k\}$ and can be calculated when it is given.

10.1.2 Single Mode

The expression of $\mathbf{E}_k$ in (10.11) simplifies when there is only one mode in the selected band ($m = 1$):

$$\mathbf{E}_k = S D_k \underset{n \times 1}{\boldsymbol{\varphi}} \underset{1 \times n}{\boldsymbol{\varphi}^T} + S_e \mathbf{I}_n \quad (10.18)$$

where, omitting mode index i , S is the modal force PSD, $\boldsymbol{\varphi}$ ($n \times 1$) is the partial mode shape;

$$D_k = \frac{(2\pi f_k)^{-2q}}{(1 - \beta_k^2)^2 + (2\zeta\beta_k)^2} \quad \beta_k = \frac{f}{f_k} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (10.19)$$

is the dynamic amplification factor; f (Hz) is the natural frequency and ζ is the damping ratio. The set of parameters $\boldsymbol{\theta}$ reduces to comprise f , ζ , S , S_e and $\boldsymbol{\varphi}$.

10.2 Remarks to Formulation

A Bayesian formulation and its identification results are as good as the underlying assumptions. Understanding the latter helps one interpret and judge the validity of results in applications. Several aspects are discussed in the following.

10.2.1 *Complex Gaussian FFT*

The formulation has made use of the long-data asymptotic result that the scaled FFT of a stationary stochastic process has a (circularly symmetric) complex Gaussian distribution and they are independent at different frequencies. Data is considered ‘long’ if the number of FFT points in the resonance band is large compared to 1. See Sect. 4.8.4 for details, and Example 4.7 that demonstrates the correlation of FFTs under non-asymptotic situations.

The Gaussian nature of FFT is a robust result supported by several layers of arguments by virtue of the Central Limit Theorem. In the first place, FFT is Gaussian whenever the time-domain modal response, modal excitation or physical excitation is Gaussian; but it can be so even if none of them is Gaussian, provided that the data is long. Modal excitation is a linear combination of the physical excitations at various DOFs of the structure and so it can be Gaussian even if the physical excitations are not. Even if the modal excitation is not Gaussian, the resulting time-domain modal response can still be Gaussian because the response at a given time instant is a linear combination of the past excitations. Even if the time-domain modal response is not Gaussian, its FFT can still be Gaussian because FFT is a linear combination of values in the time domain.

10.2.2 *Selected Frequency Band*

The selected frequency band determines the FFTs that are included in the likelihood function for modal identification. It can be determined with the help of a singular value (SV) spectrum (Sect. 7.3), from which the number of modes m dominating the band can also be decided. The choice of the selected band is a trade-off between the information included for modal identification and modeling error risk. Widening the band includes more FFTs in the likelihood function and hence potentially more information for identification. As long as modeling assumptions are valid in the selected band, this can reduce identification uncertainty. Since the FFT of modal response diminishes as the frequency moves away from the resonance peak, including the FFTs at the tail has only a diminishing marginal effect on reducing identification uncertainty. Widening the selected band always increases modeling error risk, e.g., with regard to a constant PSD in the modal force or prediction error. The computed posterior statistics are conditional on modeling assumptions and they do not reflect modeling errors. They are likely to be biased when the selected band is excessively wide. As a trade-off, for a band dominated by a single mode, it may be chosen such that in the SV spectrum the top line (which reflects modal response) almost meets the second top line (which reflects prediction error). Further widening it increases modeling error risk with little or no gain in information for modal identification. A similar rule can be applied for the general case of multiple (possibly close) modes.

Ambient data contaminated with isolated harmonic signals can still be used for modal identification. If the frequency lies outside the selected band then the signal does not affect the FFT data used in the likelihood function. Otherwise the FFT at or near the isolated frequency can be simply excluded from the selected band. The selected band need not be continuous.

10.2.3 Prediction Error Model

The prediction error ϵ_k has been modeled to account primarily for measurement noise. Instead of being white over the whole sampling band, it is assumed to have a constant PSD S_e within the selected band only. This is a robust assumption as the selected band typically covers the resonance band of the modes of interest. The bandwidth (in Hz) is $O(f_\zeta)$, much smaller than the sampling bandwidth ($1/2\Delta t$). This is an advantage of operating in the frequency domain rather than the time domain.

In reality, different data channels can have different noise levels and so it is reasonable to assume that their prediction errors have different PSDs. A more realistic assumption than (10.9) is

$$E[\epsilon_k \epsilon_k^* | \theta] = \begin{bmatrix} S_{e1} & & \\ & \ddots & \\ & & S_{en} \end{bmatrix} \quad (10.20)$$

where S_{ei} is the prediction error PSD of the i th measured DOF. Despite the obvious violation of reality, the one parameter prediction error model in (10.9) is adopted because it leads to a simpler algebraic structure in the likelihood function, simplifying analysis and computation. Numerical experiments showed that identification results are insensitive to this assumption for data with reasonable signal-to-noise (s/n) ratios.

It is tempting to think that the prediction error can account for any discrepancy between the modeled response and the measured data. *It cannot.* It can only account for discrepancies whose statistical properties agree with the probability model assumed. For example it does not account for discrepancy due to unmodeled modes. The presence of unmodeled modes generally increases the MPV of S_e but its effect on identification uncertainty need not be fully reflected in the posterior covariance matrix.

10.2.4 Measurement Type

The type of measurement affects only the transfer function h_{ik} in (10.8) through the parameter q . In structural vibration problems, velocity or acceleration data are common, although reliable displacement measurements are becoming viable using laser or high precision camera. From first glance, it may seem that the type of measurement is immaterial because data of one type can be converted to another, by numerical differentiation/integration in the time domain, or equivalently by multiplying/dividing by $2\pi f_k i$ in the frequency domain. For example, multiplying the FFT of displacement data by $(2\pi f_k i)^2$ gives the FFT of acceleration data. For modal identification, however, the type of measurement matters to the formulation. This is because the associated prediction error model is different. If the prediction error in displacement data has a constant PSD, then the derived acceleration data will have a prediction error whose PSD is proportional to f_k^4 . This is illustrated in Fig. 10.1, where the displacement data PSD in (a) has a flat noise floor but the derived acceleration data PSD in (b) has a noise floor increasing with frequency. Using the derived acceleration data and assuming a constant prediction error PSD will lead to modeling error. The effect of this modeling error depends on the width of the selected band (the wider the worse) and s/n ratio (the lower the worse).

The type of data that should be assumed in the formulation should be determined based on the validity of prediction error model rather than the actual type of measured raw data. For example, suppose the raw data is velocity but upon investigation of PSD one has reason to believe that its data noise PSD is indeed inversely proportional to f_k^2 . Then it should be converted to acceleration data for modal identification and take $q = 0$ in (10.8).

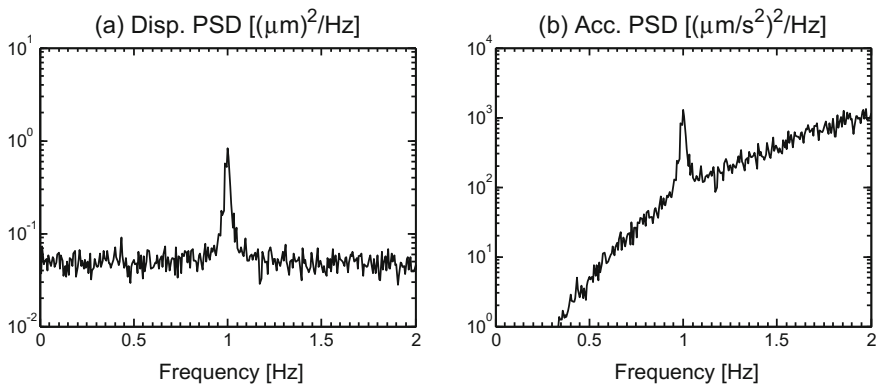


Fig. 10.1 **a** Displacement PSD calculated from synthetic data (3600 s at 100 Hz) of a SDOF structure (1 Hz natural frequency, 1% damping) with constant data noise PSD; **b** acceleration PSD produced by multiplying the displacement PSD by $(2\pi f_k)^4$

10.2.5 Mode Shape Scaling

In (10.7), the scaling of the partial mode shape $\boldsymbol{\phi}_i$ affects the scaling of the modal response ξ_{ik} and modal force p_{ik} . As discussed in Sect. 5.2.4, when $\boldsymbol{\phi}_i$ is multiplied by a factor, ξ_{ik} and p_{ik} will be divided by the same factor; and the modal force PSD S_{ii} will be divided by the square of the factor. A robust way is to scale $\boldsymbol{\phi}_i$ to have unit norm, i.e., $\boldsymbol{\phi}_i^T \boldsymbol{\phi}_i = 1$. In this case, the mode shape value of a particular DOF will decrease when a new measured DOF is added. The modal response and modal force PSD will then increase. This issue matters when one wants to compare the intensity of the modal response or modal force identified from data of different measurement arrays. For this purpose, the two sets of measurement arrays must have some DOF(s) in common. Only modal quantities with the same mode shape scaling can be compared. This can be done by scaling them so that the norm of the common DOFs is equal to 1.

Example 10.1 (Scaling of modal PSD) Consider the horizontal vibration of a ten-storied shear building with uniform mass 1000 tons per floor, interstory stiffness 1767 kN/mm and damping ratio 1% in all modes. The first three modes have natural frequencies 1, 2.98 and 4.89 Hz. The structure is subjected to i.i.d. white noise excitation at all floors, each with PSD $S_w = 96.2 \text{ N}^2/\text{Hz}$. The equation of motion is $\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) = \mathbf{F}(t)$ where $\mathbf{x}(t)$ is the displacement vector; $\mathbf{M}$, $\mathbf{C}$ and $\mathbf{K}$ are respectively the mass, damping and stiffness matrix; $\mathbf{F}(t)$ is the force vector. Consider the first mode with mode shape $\boldsymbol{\psi}$ (10×1). The modal force PSD is given by (Sect. 5.2.2)

$$S_p = \frac{\boldsymbol{\psi}^T \mathbf{S}_F \boldsymbol{\psi}}{(\boldsymbol{\psi}^T \mathbf{M} \boldsymbol{\psi})^2} \quad (10.21)$$

where $\mathbf{S}_F$ is the PSD matrix of $\mathbf{F}(t)$. Here, $\mathbf{S}_F = S_w \mathbf{I}_{10} = 96.2 \mathbf{I}_{10}$ (N^2/Hz) where $\mathbf{I}_{10}$ denotes the 10×10 identity matrix. The mass matrix is $\mathbf{M} = 1000 \times 10^3 \mathbf{I}_{10}$ (kg).

Let $\boldsymbol{\psi} = [\psi_1, \dots, \psi_{10}]^T$ and suppose it is scaled to have unit norm. Solving the eigenvalue problem $\mathbf{K}\boldsymbol{\psi} = \omega^2 \mathbf{M}\boldsymbol{\psi}$ numerically gives the mode shape in Table 10.1. Check that $\sum_{i=1}^{10} \psi_i^2 = 1$. Substituting $\mathbf{M}$, $\mathbf{S}_F$ and $\boldsymbol{\psi}$ into (10.21) gives

$$\begin{aligned} S_p &= \frac{\sum_{i=1}^{10} \psi_i^2 (96.2)}{[\sum_{i=1}^{10} \psi_i^2 (1000 \times 10^3)]^2} = \frac{96.2}{10^{12}} \left[\frac{\text{N}^2/\text{Hz}}{\text{kg}^2} \right] = \frac{96.2}{10^{12} (9.81)^2} [\text{g}^2/\text{Hz}] \\ &= 1.00 (\mu\text{g})^2/\text{Hz} \end{aligned}$$

Table 10.1 Mode shape

i	1	2	3	4	5	6	7	8	9	10
ψ_i	0.0650	0.1286	0.1894	0.2459	0.2969	0.3412	0.3780	0.4063	0.4255	0.4352

This value of modal force PSD is seldom identified from data because the number of measured DOFs is limited, i.e., the full mode shape vector ψ is not known.

Suppose the acceleration on 5/F and the roof is measured. The partial mode shape ϕ (2×1) at the measured DOFs is scaled to have unit norm. Under this scaling,

$$\phi = \frac{1}{\sqrt{\psi_5^2 + \psi_{10}^2}} \begin{bmatrix} \psi_5 \\ \psi_{10} \end{bmatrix} \quad \psi = \frac{1}{\sqrt{\psi_5^2 + \psi_{10}^2}} \begin{bmatrix} \psi_1 \\ \vdots \\ \psi_{10} \end{bmatrix} \quad (10.22)$$

In reality, ψ is only referred conceptually because it cannot be identified. Since ψ is now divided by $\sqrt{\psi_5^2 + \psi_{10}^2}$, the modal force PSD is multiplied by the square of this factor:

$$S = S_p(\psi_5^2 + \psi_{10}^2) = (1.00)(0.2969^2 + 0.4352^2) = 0.278(\mu\text{g})^2/\text{Hz} \quad (10.23)$$

This value can be identified using the measured data, in contrast to S_p . It will increase if more DOFs are measured. ■

Example 10.2 (Comparing modal force PSDs) Consider the five-storied building in Fig. 10.2. In Test 1, DOFs 2, 3, 5 were measured. In Test 2, DOFs 1, 2, 3 were measured. The identified mode shapes and modal force PSDs are shown in Table 10.2. Which test has a higher modal force intensity in terms of PSD?

The modal force PSDs in Table 10.2 cannot be compared directly because their mode shapes are scaled differently. To compare, the mode shapes are scaled so that the sum of values at the common DOFs (2 and 3) is equal to 1. The mode shape in Test 1 is divided by $\sqrt{0.3982^2 + 0.5567^2} = 0.6844$ and that in Test 2 by

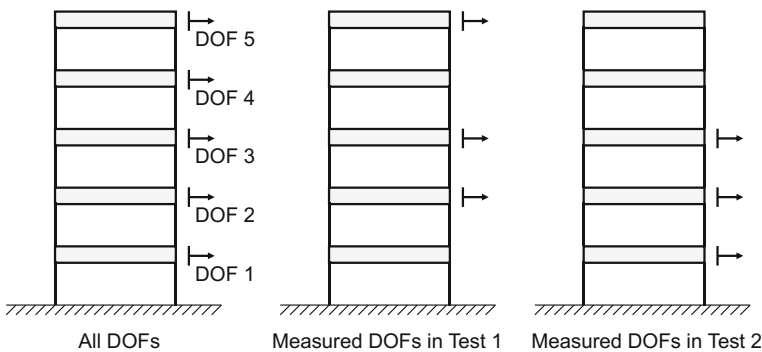


Fig. 10.2 Five-storied building

Table 10.2 Identified mode shape (normalized to unit norm) and modal force PSD

		Test 1	Test 2
Modal force PSD ($(\mu\text{g})^2/\text{Hz}$)		1.0	0.6
Identified mode shape	DOF 1	–	0.2901
	DOF 2	0.3982	0.5568
	DOF 3	0.5567	0.7783
	DOF 4	–	–
	DOF 5	0.7291	–
	Norm	1	1

Table 10.3 Identified mode shape and modal force PSD

		Test 1	Test 2
Modal force PSD ($(\mu\text{g})^2/\text{Hz}$)		0.4685	0.5495
Identified mode shape	DOF 1	–	0.3032
	DOF 2	0.5818	0.5818
	DOF 3	0.8133	0.8133
	DOF 4	–	–
	DOF 5	1.0652	–
	Norm	1.461	1.045

Mode shape normalized so that the sum of squares of DOF 2 and 3 is equal to 1

$\sqrt{0.5568^2 + 0.7783^2} = 0.9570$. Correspondingly, the modal force PSD in Test 1 is multiplied by 0.6844^2 and that in Test 2 by 0.9570^2 . The results under the new scaling are shown in Table 10.3. The identified mode shapes no longer have a unit norm but this is immaterial for the purpose here. They now have the same scaling and their modal force PSDs can be compared. In Table 10.2, although Test 1 appears to have a higher modal force PSD than Test 2, its intensity of modal force is actually smaller, as evidenced by the lower modal force PSD value in Table 10.3.

As a remark, in this example the mode shape values identified in Setups 1 and 2 at the common DOFs (2 and 3) are exactly the same (see Table 10.3). In reality, this is generally not true due to identification error but the principle of comparison remains the same. ■

10.2.6 Leakage

FFT has leakage due to finite data duration (Sect. 4.7.2). The formulation in Sect. 10.1 uses the long-data asymptotic form of the PSD matrix $\mathbf{E}_k$, and so it does not account for leakage in FFT. In principle, this modeling error can be eliminated by simply replacing $\mathbf{E}_k$ with its finite-duration counterpart. The latter is equal to the long-data asymptotic PSD in (10.11) convoluted with a sampling kernel. This gives the same form as before:

$$\mathbf{E}_k = \Phi \mathbf{H}'_k \Phi^T + S_e \mathbf{I}_n \quad (10.24)$$

except now

$$\mathbf{H}'_k = \int_{-\infty}^{\infty} \mathbf{H}(\mathbf{f}) F_N(\mathbf{f}_k - \mathbf{f}) d\mathbf{f} \quad (10.25)$$

where $\mathbf{H}(\mathbf{f})$ is a $m \times m$ matrix whose (i, j) -entry is equal to $S_{ij} h_i(\mathbf{f}) h_j(\mathbf{f})^*$;

$$h_i(\mathbf{f}) = \frac{(2\pi i \mathbf{f})^{-q}}{1 - \beta_i^2 - 2\zeta_i \beta_i i} \quad \beta_i = \frac{f_i}{f} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (10.26)$$

and

$$F_N(\mathbf{f}) = \frac{\Delta t \sin^2(N\pi \mathbf{f} \Delta t)}{N \sin^2(\pi \mathbf{f} \Delta t)} \quad (10.27)$$

is the sampling kernel. Note that $\mathbf{H}(\mathbf{f})$ and $h_i(\mathbf{f})$ are continuous-frequency counterparts of $\mathbf{H}_k$ and h_{ik} , respectively. The convolution in (10.25) can be computed numerically on a grid of frequencies for $\mathbf{f}$ with a finer resolution than $\mathbf{f}_k$.

Although the above is computationally feasible, it is not necessary in practice. This is because the long-data criterion for suppressing leakage imposes a similar requirement on data length for reasonable accuracy in the damping ratio, which is the property most affected by leakage. If data is not long then the identification uncertainty in damping will be unreasonably large, in which case leakage bias will be relatively insignificant. In short, use a data length so that the identification uncertainty in damping is small, in which case leakage is automatically suppressed. See Chap. 15 for guidance.

10.3 Multi-setup Data

The formulation in Sect. 10.1 assumes that the data at different measured DOFs are collected in a synchronous manner during the same time span. This requires an array of sensors and data acquisition channels large enough to cover all the DOFs of interest in a single setup. To identify a mode shape with more DOFs than the available number of synchronous data channels, one common strategy is to perform multiple setups with each one covering a different part of the structure while sharing some reference DOFs in common.

10.3.1 Global and Local Mode Shape

Let $\boldsymbol{\phi}_i$ ($n \times 1$) be the ‘global mode shape’ of the i th mode ($i = 1, \dots, m$) covering all DOFs of interest. Assume that there are q setups, each covering a possibly different set of DOFs. For Setup r ($r = 1, \dots, q$), let $\mathbf{L}_r$ ($n_r \times n$) be the ‘selection matrix’ that relate $\boldsymbol{\phi}_i$ to the ‘local mode shape’ $\mathbf{v}_{ri}$ ($n_r \times 1$) covering only the n_r measured DOFs in the setup:

$$\mathbf{v}_{ri} = \mathbf{L}_r \boldsymbol{\phi}_i \quad (10.28)$$

$\begin{matrix} n_r \times 1 & & n_r \times n & n \times 1 \end{matrix}$

The (j, k) -entry of $\mathbf{L}_r$ is equal to 1 if DOF k in $\boldsymbol{\phi}_i$ is measured by data channel j in Setup r ; and zero otherwise. Since some DOFs are measured in more than one setup, $n < \sum_{r=1}^q n_r$. Each DOF in $\boldsymbol{\phi}_i$ can be measured by more than one data channel in a given setup, but each channel can measure only one DOF in $\boldsymbol{\phi}_i$. That is, for each $\mathbf{L}_r$, there can be more than one ‘1’ in each column, but only one ‘1’ in each row.

10.3.2 Reference DOFs

A measured DOF that appears in more than one setup is called a ‘reference DOF’. Its presence allows the global mode shape to be determined from multiple setup data. Having the same set of reference DOFs in all setups is easy to implement but is not a necessary requirement. Nor is it necessary to have some DOFs measured in all setups. A necessary and sufficient condition is that the setups form a ‘connected graph’. Two setups are ‘connected’ if they share at least one reference DOF with non-zero mode shape value. Let C be a $q \times q$ matrix whose (r, s) -entry is 1 if Setups r and s are connected. Then C must not be a block diagonal matrix.

Example 10.3 (Six-storied building, multi-setup plan) Consider the six-storied building in Fig. 10.3. A global mode shape with $n = 6$ DOFs is identified using three uniaxial sensors (A, B, C). The setup scheme appears somewhat unsystematic but it can result from equipment or logistic constraints. Here is a hypothetical story. Originally, A and B were planned to act as reference DOFs in all setups. Placing A at the roof is intuitive as the response there is expected to be high. Placing B on 3/F allows more modes to be captured in all setups. By roving C to other floors, a global mode shape with six stories can be identified in four setups. C started on 4/F in Setup 1. It was roved to 2/F in Setup 2. In preparing for Setup 3, A was found to be defective. Also, the roof was no longer accessible (this was not known at planning stage). Setups 3 and 4 were then re-planned on site. In Setup 3, B was placed on 5/F (where response is expected to be high) while C remained on 2/F. In Setup 4, C was roved to 1/F.

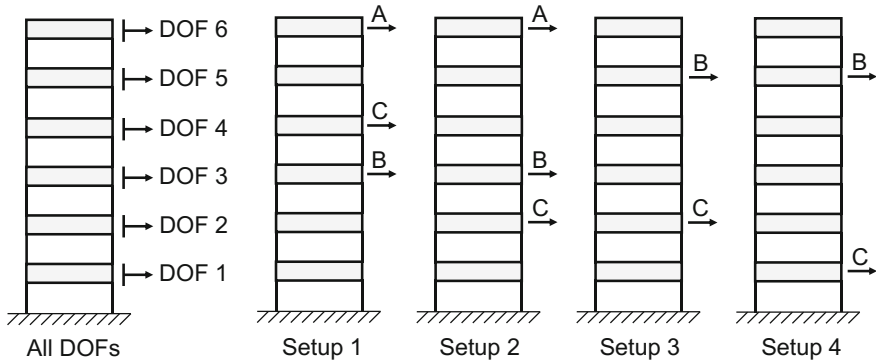


Fig. 10.3 Setup plan, six-storied building example

Assume that in all setups the data from A, B and C are recorded in channels 1, 2 and 3, respectively. The global mode shape is $\boldsymbol{\phi} = [\phi_1, \dots, \phi_6]^T$. The local mode shapes (arranged according to data channels) are

$$\mathbf{v}_1 = \begin{bmatrix} \phi_6 \\ \phi_3 \\ \phi_4 \end{bmatrix} \begin{matrix} \text{A} \\ \text{B} \\ \text{C} \end{matrix} \quad \mathbf{v}_2 = \begin{bmatrix} \phi_6 \\ \phi_3 \\ \phi_2 \end{bmatrix} \begin{matrix} \text{A} \\ \text{B} \\ \text{C} \end{matrix} \quad \mathbf{v}_3 = \begin{bmatrix} \phi_5 \\ \phi_2 \end{bmatrix} \begin{matrix} \text{B} \\ \text{C} \end{matrix} \quad \mathbf{v}_4 = \begin{bmatrix} \phi_5 \\ \phi_1 \end{bmatrix} \begin{matrix} \text{B} \\ \text{C} \end{matrix} \quad (10.29)$$

The selection matrices are

$$\mathbf{L}_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \quad \mathbf{L}_2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (10.30)$$

$$\mathbf{L}_3 = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{L}_4 = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Check that $\mathbf{v}_r = \mathbf{L}_r \boldsymbol{\phi}$, $r = 1, \dots, 4$. Note the following: the setups do not have the same number of measured DOFs; two setups can share more than one reference DOF; no DOFs appear in all setups. These are all admissible. ■

10.3.3 Parameters in Different Setups

In multi-setup problems, different setups are assumed to be performed in different time periods. This reflects the typical situation and has implications on modeling assumptions. The data measured in the same setup are assumed to be synchronous, while those in different setups are not. The ambient excitations and sensor noise in different setups are assumed to be statistically independent; and so are their

measured data. Except for the mode shapes, the natural frequencies, damping ratios, modal excitations and sensor noise characteristics in different setups need not be the same. The set of parameters $\boldsymbol{\theta}$ therefore comprises:

$$\{\{f_i^{(r)}\}_{i=1}^m, \{s_i^{(r)}\}_{i=1}^m, \mathbf{S}^{(r)}, S_e^{(r)}\}_{r=1}^q \quad \{\boldsymbol{\Phi}_i\}_{i=1}^m$$

where $f_i^{(r)}$ denotes the natural frequency of mode i in Setup r ; the same notation applies to other quantities. There are m norm constraints on the global mode shapes, $\boldsymbol{\Phi}_i^T \boldsymbol{\Phi}_i = 1$, $i = 1, \dots, m$.

10.3.4 Likelihood Function

Let $\{\hat{\mathbf{y}}_j^{(r)}\}_{j=0}^{N_r-1}$ ($n_r \times 1$) be the ambient vibration data in Setup r and $\{\hat{\mathcal{F}}_k^{(r)}\}$ ($n_r \times 1$ complex) be the collection of scaled FFTs in the selected frequency band:

$$\hat{\mathcal{F}}_k^{(r)} = \sqrt{\frac{\Delta t_r}{N_r}} \sum_{j=0}^{N_r-1} \hat{\mathbf{y}}_j^{(r)} e^{-2\pi i j k / N_r} \quad \text{frequency } f_k^{(r)} = \frac{k}{N_r \Delta t_r} \text{ Hz} \quad (10.31)$$

where Δt_r is the sampling interval of Setup r . Different setups can have different measured DOFs, sampling interval or duration. Their selected frequency band can also differ. Let the full data set be $\mathcal{D} = \{\{\hat{\mathcal{F}}_k^{(1)}\}, \dots, \{\hat{\mathcal{F}}_k^{(q)}\}\}$. Assuming a uniform prior PDF for $\boldsymbol{\theta}$, $p(\boldsymbol{\theta}|\mathcal{D}) \propto p(\mathcal{D}|\boldsymbol{\theta})$. Assume that the prediction errors and modal forces in different setups are independent. This implies that the scaled FFT in different setups are independent, giving

$$p(\boldsymbol{\theta}|\mathcal{D}) \propto \prod_{r=1}^q p(\{\hat{\mathcal{F}}_k^{(r)}\}|\boldsymbol{\theta}) \quad (10.32)$$

For long data in each setup r , $\{\hat{\mathcal{F}}_k^{(r)}\}$ are i.i.d. complex Gaussian with a joint PDF given by

$$p(\{\hat{\mathcal{F}}_k^{(r)}\}|\boldsymbol{\theta}) = \frac{\pi^{-n_r N_f^{(r)}}}{\prod_k |\mathbf{E}_k^{(r)}|} \exp \left[- \sum_k \hat{\mathcal{F}}_k^{(r)*} \mathbf{E}_k^{(r)-1} \hat{\mathcal{F}}_k^{(r)} \right] \quad (10.33)$$

$$\mathbf{E}_k^{(r)} = \underset{n_r \times m}{\boldsymbol{\Phi}_r} \underset{m \times m}{\mathbf{H}_k^{(r)}} \underset{m \times n_r}{\boldsymbol{\Phi}_r^T} + \underset{n_r \times n_r}{S_e^{(r)} \mathbf{I}_{n_r}} \quad (10.34)$$

$$\boldsymbol{\Phi}_r = [\mathbf{v}_{r1}, \dots, \mathbf{v}_{rm}] \quad \mathbf{v}_{ri} = \mathbf{L}_r \boldsymbol{\Phi}_i \quad (10.35)$$

$$\mathbf{H}_k^{(r)} = \text{diag}(\mathbf{h}_k^{(r)}) \mathbf{S}^{(r)} \text{diag}(\mathbf{h}_k^{(r)})^* \quad \mathbf{h}_k^{(r)} = [h_{1k}^{(r)}, \dots, h_{mk}^{(r)}] \quad (10.36)$$

$$h_{ik}^{(r)} = \frac{(2\pi \mathbf{f}_k^{(r)})^{-q}}{1 - \beta_{ik}^{(r)2} - 2\zeta_i^{(r)} \beta_{ik}^{(r)} \mathbf{i}} \quad \beta_{ik}^{(r)} = \frac{f_i^{(r)}}{\mathbf{f}_k^{(r)}} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (10.37)$$

The NLLF is given by

$$L(\boldsymbol{\theta}) = -\ln p(\mathcal{D}|\boldsymbol{\theta}) = \sum_{r=1}^q L_r(\boldsymbol{\theta}_r) \quad (10.38)$$

$$L_r(\boldsymbol{\theta}_r) = n_r N_f^{(r)} \ln \pi + \sum_k \ln |\mathbf{E}_k^{(r)}| + \sum_k \hat{\mathcal{F}}_k^{(r)*} \mathbf{E}_k^{(r)-1} \hat{\mathcal{F}}_k^{(r)} \quad (10.39)$$

where $\boldsymbol{\theta}_r = \{\{f_i^{(r)}\}_{i=1}^m, \{\zeta_i^{(r)}\}_{i=1}^m, \mathbf{S}^{(r)}, S_e^{(r)}, \boldsymbol{\Phi}_r\}$ comprises the parameters relevant to Setup r . For $r \neq s$, $\boldsymbol{\theta}_r$ and $\boldsymbol{\theta}_s$ are not necessarily distinct. They may share in common some mode shape parameters at the reference DOFs.

10.3.5 Single Mode

The expression of $\mathbf{E}_k^{(r)}$ in (10.34) simplifies when there is only one mode in the selected band ($m = 1$):

$$\mathbf{E}_k^{(r)} = S^{(r)} D_k^{(r)} \mathbf{v}_r \mathbf{v}_r^T + S_e^{(r)} \mathbf{I}_{n_r} \quad (10.40)$$

$n_r \times 1 \quad 1 \times n_r$

where (omitting mode index i) for Setup r , $S^{(r)}$ is the modal force PSD, $\mathbf{v}_r$ ($n_r \times 1$) is the mode shape;

$$D_k^{(r)} = \frac{(2\pi \mathbf{f}_k^{(r)})^{-2q}}{(1 - \beta_k^{(r)2})^2 + (2\zeta^{(r)} \beta_k^{(r)})^2} \quad \beta_k^{(r)} = \frac{f^{(r)}}{\mathbf{f}_k^{(r)}} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (10.41)$$

is the dynamic amplification factor, $f^{(r)}$ (Hz) is the natural frequency and $\zeta^{(r)}$ is the damping ratio. The set of parameters $\boldsymbol{\theta}$ reduces to comprise $\{f^{(r)}, \zeta^{(r)}, S^{(r)}, S_e^{(r)}\}_{r=1}^q$ and $\boldsymbol{\Phi}$, subjected to constraint $\boldsymbol{\Phi}^T \boldsymbol{\Phi} = 1$.

10.4 Asynchronous Data

The formulation in Sect. 10.1 assumes that the data at different DOFs are measured in a synchronous manner. In this section we consider the case when they are not perfectly synchronized. The analysis of asynchronous data is discussed in Sect. 7.5. In a given setup suppose there are q groups of measured DOFs. Data channels of the same group are synchronized but those of different groups are not. The total number of measured DOFs is $n = \sum_{r=1}^q n_r$ where n_r is the number in Group r . As before let $\{\hat{\mathcal{F}}_k\}$ ($n \times 1$ complex) denote the scaled FFT of data in the selected frequency band and N_f be their number. Drifting of digital clock is a statistically nonstationary process and so is asynchronous data. Despite this, the data is assumed to be stationary. This is approximate but is found to give reasonable results with real data when the relative drift between data channels is small compared to the natural period of interest. Under this assumption, for long data, $\{\hat{\mathcal{F}}_k\}$ is still complex Gaussian and independent among different frequencies. Assuming a uniform prior PDF, the posterior PDF of the set of modal parameters $\boldsymbol{\theta}$ is still given by (10.14):

$$p(\boldsymbol{\theta}|\{\hat{\mathcal{F}}_k\}) \propto p(\{\hat{\mathcal{F}}_k\}|\boldsymbol{\theta}) = \frac{\pi^{-nN_f}}{\prod_k |\mathbf{E}_k(\boldsymbol{\theta})|} \exp \left[- \sum_k \hat{\mathcal{F}}_k^* \mathbf{E}_k(\boldsymbol{\theta})^{-1} \hat{\mathcal{F}}_k \right] \quad (10.42)$$

where $\mathbf{E}_k(\boldsymbol{\theta}) = E[\hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^* | \boldsymbol{\theta}]$ is the PSD matrix of data. The departure point lies in the expression of $\mathbf{E}_k(\boldsymbol{\theta})$.

10.4.1 PSD Matrix

Deriving $\mathbf{E}_k(\boldsymbol{\theta})$ requires modeling the spectral properties of asynchronous modal response. This is done by associating different modal responses with different groups. Let $\boldsymbol{\varphi}_i$ be partitioned as

$$\boldsymbol{\varphi}_i = \begin{bmatrix} \mathbf{v}_{1i} \\ \vdots \\ \mathbf{v}_{qi} \end{bmatrix} \begin{matrix} \} n_1 \times 1 \\ \vdots \\ \} n_q \times 1 \end{matrix} \quad (10.43)$$

That is, $\mathbf{v}_{ri}$ ($n_r \times 1$) is the local mode shape of the measured DOFs in Group r . The scaled FFT is modeled as

$$\hat{\mathcal{F}}_k = \sum_{i=1}^m \begin{bmatrix} \mathbf{v}_{1i} \zeta_{ik}^{(1)} \\ \vdots \\ \mathbf{v}_{qi} \zeta_{ik}^{(q)} \end{bmatrix} + \boldsymbol{\varepsilon}_k \quad (10.44)$$

where $\xi_{ik}^{(r)}$ ($r = 1, \dots, q$) is the scaled FFT of the i th modal response following the clock of Group r . Imperfect coherence between groups is modeled by

$$E[\xi_{ik}^{(r)} \xi_{jk}^{(s)} | \boldsymbol{\theta}] = h_{ik} h_{jk}^* S_{ij} \chi_{ijrs} \quad (10.45)$$

where χ_{ijrs} (complex) is a coherence factor; S_{ij} (modal force cross PSD) and h_{ik} (transfer function) are defined as before in Sect. 10.1. By definition, $|\chi_{ijrs}| \leq 1$ and $\chi_{ijrr} = 1$. It has been assumed that χ_{ijrs} and S_{ij} are constant in the selected frequency band, i.e., they do not depend on the frequency index k .

Equation (10.44) can be written in a more compact form as

$$\hat{\mathcal{F}}_k = \sum_{i=1}^m \mathbf{u}_i \xi_{ik} + \boldsymbol{\varepsilon}_k \quad \mathbf{u}_i = \begin{bmatrix} \mathbf{v}_{1i} \\ n_1 \times 1 \\ \vdots \\ \mathbf{v}_{qi} \\ n_q \times 1 \end{bmatrix} \quad \xi_{ik} = \begin{bmatrix} \xi_{ik}^{(1)} \\ \vdots \\ \xi_{ik}^{(q)} \end{bmatrix} \quad (10.46)$$

Correspondingly,

$$\mathbf{E}_k = E[\hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^T | \boldsymbol{\theta}] = \sum_{i=1}^m \sum_{j=1}^m h_{ik} h_{jk}^* S_{ij} \mathbf{u}_i \boldsymbol{\chi}_{ij} \mathbf{u}_j^T + \text{diag}(\{S_{er} \mathbf{I}_{n_r}\}_{r=1}^q) \quad (10.47)$$

where the (r, s) -entry of $\boldsymbol{\chi}_{ij}$ is χ_{ijrs} ; $\text{diag}(\{S_{er} \mathbf{I}_{n_r}\}_{r=1}^q)$ denotes a block-diagonal matrix formed by $\{S_{er} \mathbf{I}_{n_r}\}_{r=1}^q$. The double sum over modes can be further assembled to give

$$\mathbf{E}_k = \mathbf{U} \underbrace{[\text{diag}(\mathbf{h}_k) \otimes \mathbf{I}_q]}_{mq \times mq} \mathbf{S}_{\boldsymbol{\chi}} \underbrace{[\text{diag}(\mathbf{h}_k)^* \otimes \mathbf{I}_q]}_{mq \times mq} \mathbf{U}^T + \underbrace{\text{diag}(\{S_{er} \mathbf{I}_{n_r}\}_{r=1}^q)}_{n \times n} \quad (10.48)$$

where ' $\otimes$ ' denotes the Kronecker product, $\mathbf{h}_k = [h_{1k}, \dots, h_{mk}]$ and $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_m]$; $\mathbf{S}_{\boldsymbol{\chi}}$ is a matrix with $m \times m$ partitions and its (i, j) -partition is $S_{ij} \boldsymbol{\chi}_{ij}$ ($q \times q$):

$$\mathbf{S}_{\boldsymbol{\chi}} = \begin{bmatrix} S_{11} \boldsymbol{\chi}_{11} & S_{12} \boldsymbol{\chi}_{12} & \cdots & S_{1m} \boldsymbol{\chi}_{1m} \\ & S_{22} \boldsymbol{\chi}_{22} & & S_{2m} \boldsymbol{\chi}_{2m} \\ & & \ddots & \vdots \\ \text{block sym.} & & & S_{mm} \boldsymbol{\chi}_{mm} \end{bmatrix} \quad (10.49)$$

The set of modal parameters θ comprises

$$\begin{array}{ccccc}
 \{f_i\}_{i=1}^m & \{\zeta_i\}_{i=1}^m & \mathbf{S} & \{S_{er}\}_{r=1}^q & \{\boldsymbol{\phi}_i\}_{i=1}^m \\
 \text{Natural frequencies} & \text{Damping ratios} & \begin{array}{c} m \times m \\ \text{Hermitian} \\ \text{modal force} \\ \text{PSD matrix} \end{array} & \text{Prediction error PSD} & n \times 1 \text{ Mode shapes} \\
 \\
 \{\chi_{ij} : j \geq i\}_{i=1}^m & & & & \\
 \begin{array}{c} q \times q \\ \text{Hermitian} \\ \text{coherence} \\ \text{matrix} \end{array} & & & &
 \end{array}$$

The diagonal entries of χ_{ij} are equal to 1. The mode shapes are subjected to norm constraints, $\boldsymbol{\phi}_i^T \boldsymbol{\phi}_i = 1$, $i = 1, \dots, m$.

10.4.2 Single Mode

The expression of $\mathbf{E}_k$ in (10.47) simplifies when there is only one mode in the selected band ($m = 1$):

$$\mathbf{E}_k = S D_k \mathbf{u} \boldsymbol{\chi} \mathbf{u}^T + \text{diag}(\{S_{er} \mathbf{I}_{n_r}\}_{r=1}^q) \quad (10.50)$$

where (omitting mode index i)

$$\begin{array}{l}
 \boldsymbol{\phi}_{n \times 1} = \begin{bmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_q \end{bmatrix} \begin{array}{l} \} n_1 \times 1 \\ \vdots \\ \} n_q \times 1 \end{array} \quad \mathbf{u}_{n \times q} = \begin{bmatrix} \mathbf{v}_1 & & & \\ & \ddots & & \\ & & \ddots & \\ & & & \mathbf{v}_q \end{bmatrix} \begin{array}{l} n_1 \times 1 \\ \\ \\ n_q \times 1 \end{array} \\
 \boldsymbol{\chi}_{q \times q} = \begin{bmatrix} 1 & \chi_{12} & \cdots & \chi_{1q} \\ & 1 & & \chi_{2q} \\ & & \ddots & \vdots \\ \text{Herm.} & & & 1 \end{bmatrix}
 \end{array} \quad (10.51)$$

The dynamic amplification factor D_k and modal force PSD S have the same definition in Sect. 10.1. The prediction error PSD S_{er} is analogous to S_e . The set of modal parameters θ reduces to comprise

$$\begin{array}{ccccc}
 f & \zeta & S & \{S_{er}\}_{r=1}^q & \boldsymbol{\phi} & \boldsymbol{\chi} \\
 \text{Natural frequency} & \text{Damping ratio} & \text{Modal force PSD} & \text{Prediction error PSD} & \begin{array}{c} n \times 1 \\ \text{Mode shape} \end{array} & \begin{array}{c} q \times q \\ \text{Hermitian} \\ \text{coherence} \\ \text{matrix} \end{array}
 \end{array}$$

Chapter 11

Bayesian OMA Computation

Abstract This chapter discusses the computational issues in Bayesian operational modal analysis and presents their solutions in a general context. The main theoretical issue lies in the computation of the posterior covariance matrix of modal parameters under constraints, which can be addressed systematically using Lagrange multiplier concepts. The uncertainty quantification of mode shape, which is of vectorial nature under scaling constraint, is also discussed. The computational framework in this chapter will be followed in Chaps. 12–14 where specific algorithms are developed.

Keywords OMA computation • Posterior covariance matrix • Constrained Hessian

In this chapter we discuss the computational issues in determining the posterior statistics of modal parameters in Bayesian operational modal analysis (OMA). Modal identification problem is globally identifiable and so Gaussian approximation can be applied to the posterior probability density function (PDF) of modal parameters. The posterior statistics can then be conveniently described in terms of the most probable value (MPV) and covariance matrix. General issues are discussed in this chapter; efficient algorithms for specific contexts (single mode, multi-mode, etc.) are postponed to Chap. 12–14. Mode shapes, among other parameters, have a special role because of their vectorial nature. Issues on their computation and identification uncertainty quantification are discussed. Although this chapter is presented in the context of OMA, its theory is generally applicable for globally identifiable problems of Bayesian inference.

Let $\mathcal{D}$ denote the data for Bayesian inference. Assuming a uniform prior PDF, the posterior PDF of the set of parameters $\boldsymbol{\theta}$ to be identified can be written as

$$p(\boldsymbol{\theta}|\mathcal{D}) \propto e^{-L(\boldsymbol{\theta})} \quad (11.1)$$

where

$$L(\boldsymbol{\theta}) = -\ln p(\mathcal{D}|\boldsymbol{\theta}) \quad (11.2)$$

is the negative log-likelihood function (NLLF), whose expression has been derived in Chap. 10 for OMA under different situations. The posterior MPV of $\boldsymbol{\theta}$ minimizes the NLLF. A second order approximation of NLLF at the MPV leads to a Gaussian approximation of the posterior PDF centered at the MPV and with a covariance matrix equal to the inverse of the Hessian of NLLF at the MPV (Sect. 8.5). Computational issues on the MPV are concerned with how it can be determined efficiently from the NLLF without brute-force numerical optimization. Issues with the posterior covariance matrix are concerned with how it can be computed correctly when the parameters are under constraints (e.g., from mode shape scaling).

11.1 Posterior Most Probable Value

Determining the MPV by brute-force numerical minimization of the NLLF is far from being efficient, or defective in many cases. For example, consider the case of single setup data in Sect. 10.1. The NLLF is given by

$$L(\boldsymbol{\theta}) = nN_f \ln \pi + \sum_k \ln |\mathbf{E}_k(\boldsymbol{\theta})| + \sum_k \hat{\mathcal{F}}_k^* \mathbf{E}_k(\boldsymbol{\theta})^{-1} \hat{\mathcal{F}}_k \quad (11.3)$$

where $\{\hat{\mathcal{F}}_k\}$ ($n \times 1$ complex) is the scaled Fast Fourier Transform (FFT) of data in the selected frequency band with N_f points and n measured DOFs; and $\mathbf{E}_k$ ($n \times n$ Hermitian) is the theoretical power spectral density (PSD) matrix of data at FFT frequency f_k (Hz) for given $\boldsymbol{\theta}$. Typically, $\mathbf{E}_k$ is almost singular because it is dominated by the contribution from modal response. The latter has a rank of at most m (the number of modes), often smaller than the dimension n of $\mathbf{E}_k$. The number of modal parameters, n_θ , is typically large. It increases with m and n . The increase with m is quadratic, although m is typically small. For well-separated modes, one can identify each mode separately and so $m = 1$. In general, m only need to be the number of close modes, which often does not exceed three. Arising from mode shapes, n_θ increases linearly with n , which can be large in applications (e.g., >10). An additional complication is that the parameters in $\boldsymbol{\theta}$ are subjected to constraints, e.g., arising from the scaling of mode shapes.

The NLLF can be rewritten in a form that facilitates analysis and computation. The basic strategy is to write it as a quadratic function (or alike) of the mode shapes using linear algebra techniques, from which the most probable mode shapes can be determined analytically or semi-analytically in terms of the remaining parameters. In doing so, the singularity of $\mathbf{E}_k$ can be resolved analytically so that the matrices in the resulting expression are all well-conditioned. Structured matrices such as the

modal force PSD matrix (for multiple modes) and coherence matrix (for asynchronous data) can be parameterized by a necessary set of free parameters so that they can be optimized directly. ‘Partial optimal solutions’, i.e., in terms of the remaining parameters, allow the development of iterative algorithms where the MPV of different groups of parameters are updated in turn until convergence.

Example 11.1 (Singularity of PSD matrix) Consider single setup acceleration data with only one mode in the selected band, i.e., $m = 1$. From Sect. 10.1, the theoretical PSD matrix of data is

$$\mathbf{E}_k = SD_k \boldsymbol{\phi} \boldsymbol{\phi}^T + S_e \mathbf{I}_n = SD_k (\boldsymbol{\phi} \boldsymbol{\phi}^T + \frac{S_e}{SD_k} \mathbf{I}_n) \quad (11.4)$$

where S is the modal force PSD, $D_k = [(1 - \beta_k^2)^2 + (2\zeta\beta_k)^2]^{-1}$, $\beta_k = f/f_k$, f is the natural frequency (Hz), ζ is the damping ratio and $\boldsymbol{\phi}$ ($n \times 1$) is the mode shape. The term S_e/SD_k represents a noise-to-signal ratio in terms of PSD at frequency f_k . At $f_k = f$, $D_k = 1/4\zeta^2$ and so $S_e/SD_k = 4S_e\zeta^2/S$. For good quality data this ratio is small compared to 1. Consequently $\mathbf{E}_k \approx SD_k \boldsymbol{\phi} \boldsymbol{\phi}^T$ has rank 1. Suppose $\boldsymbol{\phi} = [1, 2, 3]^T / \sqrt{14}$ and assume the following typical values: $\zeta = 1\%$, $S_e = 10^{-12} \text{ g}^2/\text{Hz}$ and $S = 10^{-12} \text{ g}^2/\text{Hz}$. Then $S_e/SD_k = 4 \times 10^{-4} \ll 1$ at $f_k = f$. The eigenvalues of $\mathbf{E}_k$ are 10^{-12} , 10^{-12} and $2.50 \times 10^{-9} \text{ g}^2/\text{Hz}$. The first two eigenvalues are significantly smaller than the last one. In this sense, $\mathbf{E}_k$ is almost singular with rank 1. ■

Example 11.2 (Counting modal parameters) In a single setup test suppose we use six triaxial sensors. Then there are $n = 6 \times 3 = 18$ measured DOFs. Suppose there are $m = 2$ modes in the selected frequency band. The number of parameters are counted as follow: frequencies $\{f_1, f_2\}$, damping ratios $\{\zeta_1, \zeta_2\}$, prediction error PSD S_e , mode shapes

$$\boldsymbol{\Phi}_1 = \begin{bmatrix} \phi_{11} \\ \phi_{21} \\ \vdots \\ \phi_{18,1} \end{bmatrix} \quad \boldsymbol{\Phi}_2 = \begin{bmatrix} \phi_{12} \\ \phi_{22} \\ \vdots \\ \phi_{18,2} \end{bmatrix} \quad (11.5)$$

The modal force PSD matrix is Hermitian and so it can be represented as

$$\mathbf{S} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} = \begin{bmatrix} S_{11} & u + \mathbf{v}\mathbf{i} \\ u - \mathbf{v}\mathbf{i} & S_{22} \end{bmatrix} \quad (11.6)$$

where $u = \text{Re}S_{12}$ and $v = \text{Im}S_{12}$. It has four (real-valued) parameters (S_{11}, S_{22}, u, v). The total number of parameters is

$$n_\theta = \underbrace{2}_{\text{frequencies}} + \underbrace{2}_{\text{damping ratios}} + \underbrace{1}_{\text{prediction error PSD}} + \underbrace{36}_{\text{mode shapes}} + \underbrace{4}_{\text{modal force PSD}} = 45$$

There are two norm constraints on mode shapes:

$$\boldsymbol{\phi}_1^T \boldsymbol{\phi}_1 = \phi_{11}^2 + \phi_{21}^2 + \dots + \phi_{18,1}^2 = 1 \quad \boldsymbol{\phi}_2^T \boldsymbol{\phi}_2 = \phi_{12}^2 + \phi_{22}^2 + \dots + \phi_{18,2}^2 = 1 \quad (11.7)$$

■

11.2 Posterior Covariance Matrix

Under Gaussian approximation of the posterior PDF, its covariance matrix is equal to the inverse of the Hessian of NLLF at the MPV. The Hessian can be calculated using finite difference approximation or analytical expressions for the derivatives of NLLF. The former is approximate but requires only the values of the NLLF. The latter is exact but has overheads in derivation and programing.

Modal parameters can be subjected to equality constraints such as those arising from norm constraints on mode shapes. In some formulation (e.g., multi-setup) the set of modal parameters is deliberately expanded so that the partial derivatives of the NLLF can be obtained more easily. In this case there are additional constraints arising from the relationships among the expanded set of parameters. The constraints need to be accounted for when calculating the Hessian of the NLLF.

11.2.1 Mapping with Free Parameters

The posterior covariance matrix of $\boldsymbol{\theta}$ under constraints can be calculated systematically via the covariance matrix of a set of ‘free’ (i.e., unconstrained) parameters. Suppose there are n_c independent constraints given by

$$G_i(\boldsymbol{\theta}) = 0 \quad i = 1, \dots, n_c \quad (11.8)$$

Let $\mathbf{v}_c(\mathbf{u})$ ($n_\theta \times 1$) be a function that maps $\mathbf{u}$ ($p \times 1$ free parameters) to $\boldsymbol{\theta}$ so that it always satisfies the constraints, i.e.,

$$G_i(\mathbf{v}_c(\mathbf{u})) = 0 \quad i = 1, \dots, n_c \text{ for any } \mathbf{u} \quad (11.9)$$

With n_θ parameters in $\boldsymbol{\theta}$ subjected to n_c constraints, the dimension of the admissible space is $n_\theta - n_c$. The number of parameters in $\mathbf{u}$ must then be

$$p \geq n_\theta - n_c \quad (11.10)$$

Suppose the Bayesian inference problem is formulated in terms of $\mathbf{u}$ instead of $\boldsymbol{\theta}$. The NLLF for $\mathbf{u}$, denoted by $L_c(\mathbf{u})$, is then given by

$$L_c(\mathbf{u}) = L(\mathbf{v}_c(\mathbf{u})) \quad (11.11)$$

11.2.2 Transformation of Covariance Matrix

The MPV of $\mathbf{u}$ is related to the MPV of $\boldsymbol{\theta}$ by $\hat{\boldsymbol{\theta}} = \mathbf{v}_c(\hat{\mathbf{u}})$, where a hat ‘ $\wedge$ ’ denotes the MPV. Under Gaussian approximation of the posterior PDF of $\mathbf{u}$, its posterior covariance matrix is the inverse of the Hessian of $L_c(\mathbf{u})$ at $\hat{\mathbf{u}}$. From this the posterior covariance matrix of $\boldsymbol{\theta} = \mathbf{v}_c(\mathbf{u})$ can be obtained by transformation of variables. Let $\Delta\mathbf{u}$ and $\Delta\boldsymbol{\theta}$ be uncertain variations in $\mathbf{u}$ and $\boldsymbol{\theta}$ from their MPV, respectively. To the first order,

$$\Delta\boldsymbol{\theta}_{n_\theta \times 1} = \sum_{i=1}^p \frac{\partial \hat{\mathbf{v}}_c}{\partial u_i} \Delta u_i = \underbrace{\begin{bmatrix} \frac{\partial \hat{\mathbf{v}}_c}{\partial u_1} & \cdots & \frac{\partial \hat{\mathbf{v}}_c}{\partial u_p} \end{bmatrix}}_{\nabla \hat{\mathbf{v}}_c} \underbrace{\begin{bmatrix} \Delta u_1 \\ \vdots \\ \Delta u_p \end{bmatrix}}_{\Delta \mathbf{u}} = \nabla \hat{\mathbf{v}}_c \Delta \mathbf{u}_{n_\theta \times p \quad p \times 1} \quad (11.12)$$

where the hat ‘ $\wedge$ ’ in $\partial \hat{\mathbf{v}}_c / \partial u_i$ denotes that it is evaluated at the MPV. The posterior covariance matrix of $\boldsymbol{\theta}$ is then

$$\hat{\mathbf{C}} = E[\Delta\boldsymbol{\theta}\Delta\boldsymbol{\theta}^T | \mathcal{D}] = (\nabla \hat{\mathbf{v}}_c) \underbrace{E[\Delta \mathbf{u} \Delta \mathbf{u}^T | \mathcal{D}]}_{\hat{\mathbf{C}}_u} (\nabla \hat{\mathbf{v}}_c)^T = (\nabla \hat{\mathbf{v}}_c) \hat{\mathbf{C}}_u (\nabla \hat{\mathbf{v}}_c)^T \quad (11.13)$$

where $\hat{\mathbf{C}}_u$ ($p \times p$) is the posterior covariance matrix of $\mathbf{u}$, equal to the inverse of $\nabla^2 \hat{L}_c$ (Hessian of L_c at the MPV). Due to the mapping $\mathbf{v}_c(\mathbf{u})$, $\nabla^2 \hat{L}_c$ ($p \times p$) only has rank $n_\theta - n_c$ and so is singular when $p > n_\theta - n_c$. Nevertheless this singularity is immaterial to the posterior uncertainty of $\boldsymbol{\theta}$. Section 11.2.6 shows that

$$\hat{\mathbf{C}}_{n_\theta \times n_\theta} = (\nabla \hat{\mathbf{v}}_c)_{n_\theta \times p} \underbrace{(\nabla^2 \hat{L}_c)^+}_{p \times p} (\nabla \hat{\mathbf{v}}_c)^T_{p \times n_\theta} \quad (11.14)$$

where ‘ $+$ ’ denotes the ‘pseudo-inverse’, i.e., evaluated via eigenvector representation ignoring the $p - n_\theta + n_c$ zero eigenvalue components arising from constraints.

11.2.3 Hessian of Composite Function

Since $L_c(\mathbf{u}) = L(\mathbf{v}_c(\mathbf{u}))$ is a composite function of $\mathbf{u}$, its Hessian $\nabla^2 L_c$ (w.r.t. $\mathbf{u}$) is analytically more complicated to derive than the Hessian $\nabla^2 L$ (w.r.t. $\boldsymbol{\theta}$). It is

possible to obtain $\nabla^2 L_c$ in terms of the derivatives of $\mathbf{v}_c(\mathbf{u})$ and $L(\boldsymbol{\theta})$ so that their information can be combined systematically. It can be shown that (Sect. B.1)

$$\nabla^2 L_c = (\nabla \mathbf{v}_c)^T \nabla^2 L (\nabla \mathbf{v}_c) + \underbrace{(\mathbf{I}_p \otimes \nabla L)}_{p \times n_0 p} \nabla^2 \mathbf{v}_c \quad (11.15)$$

where ‘ $\otimes$ ’ denotes the Kronecker product; $\mathbf{I}_p$ denotes the $p \times p$ identity matrix;

$$\nabla \mathbf{v}_c = \begin{bmatrix} \frac{\partial \mathbf{v}_c}{\partial u_1} & \dots & \frac{\partial \mathbf{v}_c}{\partial u_p} \end{bmatrix}_{n_0 \times p} \quad \nabla^2 \mathbf{v}_c = \begin{bmatrix} \frac{\partial^2 \mathbf{v}_c}{\partial u_1^2} & \frac{\partial^2 \mathbf{v}_c}{\partial u_1 \partial u_2} & \dots & \frac{\partial^2 \mathbf{v}_c}{\partial u_1 \partial u_p} \\ & \ddots & & \frac{\partial^2 \mathbf{v}_c}{\partial u_2 \partial u_p} \\ & & \ddots & \vdots \\ \text{block sym.} & & & \frac{\partial^2 \mathbf{v}_c}{\partial u_p^2} \end{bmatrix} \quad (11.16)$$

$$\nabla L = \nabla L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\mathbf{v}_c(\mathbf{u})} \quad \nabla^2 L = \nabla^2 L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\mathbf{v}_c(\mathbf{u})} \quad (11.17)$$

Equation (11.15) is applicable for any $\mathbf{u}$, not just at the MPV. It involves the first two derivatives of $L(\boldsymbol{\theta})$ and $\mathbf{v}_c(\mathbf{u})$. The second term involves the product of two large matrices, although the first matrix is sparse.

Generally, ∇L does not vanish at the MPV due to the presence of constraints. The Hessian $\nabla^2 L_c$ at the MPV is not equal to the first term in (11.15). Rather, it can be shown using Lagrange multiplier concept that at the MPV (Sect. B.2)

$$\nabla^2 \hat{L}_c = (\nabla \hat{\mathbf{v}}_c)^T (\nabla^2 \hat{J}) \nabla \hat{\mathbf{v}}_c \quad (11.18)$$

where

$$J(\boldsymbol{\theta}) = L(\boldsymbol{\theta}) + \sum_{i=1}^{n_c} \lambda_i G_i(\boldsymbol{\theta}) \quad (11.19)$$

is the ‘Lagrangian’; $\{\lambda_i\}_{i=1}^{n_c}$ are Lagrange multipliers;

$$\nabla^2 \hat{J} = \nabla^2 \hat{L} + \sum_{i=1}^{n_c} \hat{\lambda}_i \nabla^2 \hat{G}_i \quad (11.20)$$

is the Hessian of J at the MPV;

$$\nabla \hat{\mathbf{v}}_c = \nabla \mathbf{v}_c(\mathbf{u})|_{\mathbf{u}=\hat{\mathbf{u}}} \quad \nabla \hat{G}_i = \nabla G_i(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \quad \nabla^2 \hat{G}_i = \nabla^2 G_i(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \quad (11.21)$$

$$\hat{\boldsymbol{\lambda}} = \begin{bmatrix} \hat{\lambda}_1 \\ \vdots \\ \hat{\lambda}_{n_c} \end{bmatrix} = - \underbrace{[(\nabla \hat{\mathbf{G}}) (\nabla \hat{\mathbf{G}})^T]^{-1}}_{n_c \times n_c} \underbrace{(\nabla \hat{\mathbf{G}})}_{n_c \times n_\theta} \underbrace{(\nabla \hat{L})^T}_{n_\theta \times 1} \quad (11.22)$$

$$\nabla \hat{L} = \nabla L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \quad \nabla^2 \hat{L} = \nabla^2 L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \quad \nabla \hat{\mathbf{G}} = \begin{bmatrix} \nabla \hat{G}_1 \\ \vdots \\ \nabla \hat{G}_{n_c} \end{bmatrix} \quad (11.23)$$

Equation (11.18) involves the first two derivatives of $L(\boldsymbol{\theta})$ but only the first derivatives of $\mathbf{v}_c(\mathbf{u})$. It also involves the first two derivatives of $\{G_i\}_{i=1}^{n_c}$. The definition of $\mathbf{v}_c$ and $\{G_i\}_{i=1}^{n_c}$ are not unique but they do not affect $\nabla^2 \hat{L}_c$; see next section.

11.2.4 Transformation Invariance

The posterior covariance matrix of $\boldsymbol{\theta}$ is invariant to the choice of the constraint functions $\{G_i\}_{i=1}^{n_c}$ and mapping $\mathbf{v}_c(\mathbf{u})$. For the former, it can be shown that (Sect. B.2.2) $\nabla^2 \hat{L}_c$ remains the same when G_i is replaced by $H_i(G_i(\boldsymbol{\theta}))$ for any scalar function H_i with a non-zero derivative at 0. On the other hand, if one works with a new set of free parameters $\mathbf{a}$ instead of $\mathbf{u}$, the resulting posterior covariance matrix of $\boldsymbol{\theta}$ remains the same.

To illustrate, let $\mathbf{u} = \mathbf{T}(\mathbf{a})$ where $\mathbf{T}$ is a transformation; and $\mathbf{w}(\mathbf{a}) = \mathbf{v}_c(\mathbf{T}(\mathbf{a}))$. To be admissible, the transformation should be such that $\boldsymbol{\theta} = \mathbf{w}(\mathbf{a})$ always satisfies constraints. The NLLF for $\mathbf{a}$ is $K(\mathbf{a}) = L(\mathbf{w}(\mathbf{a}))$. Applying (11.14) but now taking $\mathbf{a} = [a_1, \dots, a_q]^T$ as the set of free parameters, the posterior covariance matrix of $\boldsymbol{\theta}$ is given by

$$\hat{\mathbf{C}}' = (\nabla \hat{\mathbf{w}})(\nabla^2 \hat{K})^+ (\nabla \hat{\mathbf{w}})^T \quad (11.24)$$

where

$$\nabla^2 \hat{K} = (\nabla \hat{\mathbf{w}})^T (\nabla^2 J) (\nabla \hat{\mathbf{w}}) \quad (11.25)$$

Note that

$$\frac{\partial \mathbf{w}}{\partial a_i} = \sum_{j=1}^p \frac{\partial \mathbf{v}_c}{\partial u_j} \frac{\partial u_j}{\partial a_i} = \underbrace{\begin{bmatrix} \frac{\partial \mathbf{v}_c}{\partial u_1} & \cdots & \frac{\partial \mathbf{v}_c}{\partial u_p} \end{bmatrix}}_{\nabla_{\mathbf{v}_c}} \underbrace{\begin{bmatrix} \frac{\partial u_1}{\partial a_i} \\ \vdots \\ \frac{\partial u_p}{\partial a_i} \end{bmatrix}}_{\partial \mathbf{T} / \partial a_i} = \nabla_{\mathbf{v}_c} \frac{\partial \mathbf{T}}{\partial a_i} \quad (11.26)$$

$$\nabla \mathbf{w} = \begin{bmatrix} \frac{\partial \mathbf{w}}{\partial a_1} & \cdots & \frac{\partial \mathbf{w}}{\partial a_q} \end{bmatrix} = \nabla_{\mathbf{v}_c} \begin{bmatrix} \frac{\partial \mathbf{T}}{\partial a_1} & \cdots & \frac{\partial \mathbf{T}}{\partial a_q} \end{bmatrix} = \nabla_{\mathbf{v}_c} \nabla \mathbf{T} \quad (11.27)$$

Thus, $\nabla \hat{\mathbf{w}} = \nabla \hat{\mathbf{v}}_c \nabla \hat{\mathbf{T}}$, where the hat ‘^’ denotes that it is evaluated at the MPV. Substituting into (11.25) and then into (11.24) gives

$$\begin{aligned} \hat{\mathbf{C}}' &= (\nabla \hat{\mathbf{w}}) [(\nabla \hat{\mathbf{w}})^T (\nabla^2 \hat{J}) (\nabla \hat{\mathbf{w}})]^+ (\nabla \hat{\mathbf{w}})^T \\ &= \nabla \hat{\mathbf{v}}_c \nabla \hat{\mathbf{T}} \left[(\nabla \hat{\mathbf{T}})^T \underbrace{(\nabla \hat{\mathbf{v}}_c)^T (\nabla^2 \hat{J}) (\nabla \hat{\mathbf{v}}_c)}_{\nabla^2 \hat{L}_c} (\nabla \hat{\mathbf{T}}) \right]^+ (\nabla \hat{\mathbf{T}})^T (\nabla \hat{\mathbf{v}}_c)^T \quad (11.28) \end{aligned}$$

Using singular value decomposition, it can be shown that this gives the same expression as $\hat{\mathbf{C}}$ in (11.14) and so the posterior covariance matrix of $\boldsymbol{\theta}$ remains the same (details omitted). Essentially, the effect of the $\nabla \hat{\mathbf{T}}$ s inside the pseudo-inverse ‘cancels’ out the effect of those outside.

11.2.5 Constraint Singularity

As a result of $\mathbf{v}_c(\mathbf{u})$ mapping $\mathbf{u}$ (dimension $p \geq n_\theta - n_c$) to the admissible space of lower dimension $s = n_\theta - n_c$, $\nabla^2 \hat{L}_c$ has $p - n_\theta + n_c$ zero eigenvalues. To see this, take gradient w.r.t. $\mathbf{u}$ on $G_i(\mathbf{v}_c(\mathbf{u})) = 0$, transpose and evaluate at the MPV to obtain

$$\underbrace{(\nabla \hat{\mathbf{v}}_c)^T}_{p \times n_\theta} \underbrace{(\nabla \hat{G}_i)^T}_{n_\theta \times 1} = \mathbf{0} \quad i = 1, \dots, n_c \quad (11.29)$$

Assuming that $\{\nabla \hat{G}_i\}_{i=1}^{n_c}$ are linearly independent, this implies that $(\nabla \hat{\mathbf{v}}_c)^T$ has a null space of dimension (i.e., nullity) n_c . The rank-nullity theorem (rank + nullity = number of columns) implies that $(\nabla \hat{\mathbf{v}}_c)^T$ (with n_θ columns) has a rank of $s = n_\theta - n_c$. The same is also true for $\nabla \hat{\mathbf{v}}_c$ (column rank = row rank). Using the rank-nullity theorem again, $\nabla \hat{\mathbf{v}}_c$ (with p columns) then has a nullity of $p - s$. Right-multiplying $\nabla^2 \hat{L}_c = (\nabla \hat{\mathbf{v}}_c)^T (\nabla^2 \hat{J}) (\nabla \hat{\mathbf{v}}_c)$ in (11.18) with $p - s$ linearly

independent null vectors of $\nabla \hat{\mathbf{v}}_c$ gives a zero vector, implying that they are also null vectors of $\nabla^2 \hat{L}_c$. This is referred as ‘constraint singularity’. Note that $\nabla^2 L_c$ need not be singular when it is not evaluated at the MPV.

11.2.6 Pseudo-inverse

Apparently, constraint singularity causes legitimacy problem when taking the inverse of $\nabla^2 \hat{L}_c$. As far as posterior uncertainty is concerned, however, this singularity can be ignored, in the sense that the posterior covariance matrix can be evaluated via the eigenvector representation of $\nabla^2 \hat{L}_c$ where the zero eigenvalue components associated with constraint singularity are ignored. This is legitimate because to the first order $\boldsymbol{\theta} = \mathbf{v}_c(\mathbf{u})$ is invariant to variations of $\mathbf{u}$ in the null space of $\nabla^2 \hat{L}_c$.

Specifically, let $\{\mathbf{b}_i\}_{i=1}^p$ be the (orthonormal) eigenvectors of $\nabla^2 \hat{L}_c$ with eigenvalues $\{\kappa_i\}_{i=1}^p$ (in descending order of magnitude). Then $\nabla^2 \hat{L}_c = \sum_{i=1}^p \kappa_i \mathbf{b}_i \mathbf{b}_i^T$. Replacing the eigenvalues by their reciprocals gives (symbolically) the inverse, $(\nabla^2 \hat{L}_c)^{-1} = \sum_{i=1}^p \kappa_i^{-1} \mathbf{b}_i \mathbf{b}_i^T$. The posterior covariance matrix of $\boldsymbol{\theta}$ is then (symbolically)

$$\hat{\mathbf{C}} = (\nabla \hat{\mathbf{v}}_c) \left(\sum_{i=1}^p \kappa_i^{-1} \mathbf{b}_i \mathbf{b}_i^T \right) (\nabla \hat{\mathbf{v}}_c)^T = \sum_{i=1}^p \kappa_i^{-1} (\nabla \hat{\mathbf{v}}_c \mathbf{b}_i) (\nabla \hat{\mathbf{v}}_c \mathbf{b}_i)^T \quad (11.30)$$

Due to constraint singularity, for $i = s+1, \dots, p$, $\kappa_i = 0$ and $\mathbf{b}_i$ is a null vector of $\nabla \hat{\mathbf{v}}_c$, i.e., $\nabla \hat{\mathbf{v}}_c \mathbf{b}_i = \mathbf{0}$. According to (11.12), any uncertain variations in $\mathbf{u}$ from the MPV along the directions $\{\mathbf{b}_i\}_{i=s+1}^p$ will not lead to any variation in $\boldsymbol{\theta}$. Thus, despite the reciprocal of zero eigenvalues, the components along the directions $\{\mathbf{b}_i\}_{i=s+1}^p$ are mapped to zero in $\hat{\mathbf{C}}$. The summands in (11.30) for $i = s+1, \dots, p$ are then zero, giving

$$\hat{\mathbf{C}} = (\nabla \hat{\mathbf{v}}_c) \left(\sum_{i=1}^s \kappa_i^{-1} \mathbf{b}_i \mathbf{b}_i^T \right) (\nabla \hat{\mathbf{v}}_c)^T \quad (11.31)$$

Symbolically, this is written as

$$\hat{\mathbf{C}} = (\nabla \hat{\mathbf{v}}_c) (\nabla^2 \hat{L}_c)^+ (\nabla \hat{\mathbf{v}}_c)^T \quad (11.32)$$

where

$$(\nabla^2 \hat{L}_c)^+ = \sum_{i=1}^s \kappa_i^{-1} \mathbf{b}_i \mathbf{b}_i^T \quad (11.33)$$

denotes the ‘pseudo-inverse’ of $\nabla^2 \hat{L}_c$ obtained by ignoring the zero eigenvalue components along $\{\mathbf{b}_i\}_{i=s+1}^p$.

Example 11.3 (Pseudo-inverse) Let $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^T$ and $\mathbf{A}$ be a 2×2 real symmetric and positive definite matrix with eigenvalues $\{a_1, a_2\}$ (in ascending order of magnitude) and eigenvectors $\{\mathbf{b}_1, \mathbf{b}_2\}$ (with unit norm). Consider minimizing

$$F(\boldsymbol{\theta}) = \theta_1^2 + \mathbf{x}^T \mathbf{A} \mathbf{x} \quad \mathbf{x} = \begin{bmatrix} \theta_2 \\ \theta_3 \end{bmatrix} \quad (11.34)$$

subjected to constraint $\theta_2^2 + \theta_3^2 = 1$. It can be reasoned that the optimal point is $\hat{\boldsymbol{\theta}} = [0; \mathbf{b}_1]$. The constraint can be expressed as $G(\boldsymbol{\theta}) = 0$ where $G(\boldsymbol{\theta}) = 1 - \theta_2^2 - \theta_3^2$. For $\mathbf{u} = [u_1, u_2, u_3]^T$, the mapping

$$\mathbf{v}_c(\mathbf{u}) = \begin{bmatrix} u_1 \\ u_2 / \sqrt{u_2^2 + u_3^2} \\ u_3 / \sqrt{u_2^2 + u_3^2} \end{bmatrix} = \begin{bmatrix} u_1 \\ \|\mathbf{x}\|^{-1} \mathbf{x} \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} \quad \|\mathbf{x}\| = \sqrt{u_2^2 + u_3^2} \quad (11.35)$$

always satisfies the constraint, i.e., $G(\mathbf{v}_c(\mathbf{u})) = 0$ for any $\mathbf{u}$. Define the composite function

$$F_c(\mathbf{u}) = F(\mathbf{v}_c(\mathbf{u})) = u_1^2 + \frac{\mathbf{x}^T}{\|\mathbf{x}\|} \mathbf{A} \frac{\mathbf{x}}{\|\mathbf{x}\|} = u_1^2 + \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{x}^T \mathbf{x}} \quad \mathbf{x} = \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} \quad (11.36)$$

By direct differentiation and using Sect. C.5.4 for the derivative of $(\mathbf{x}^T \mathbf{A} \mathbf{x})/(\mathbf{x}^T \mathbf{x})$, it can be shown that the Hessian of $F_c(\mathbf{u})$ at the optimum is

$$\nabla^2 \hat{F}_c = \begin{bmatrix} 2 & \\ & 2(\mathbf{A} - a_1 \mathbf{I}_2) \end{bmatrix} \quad (11.37)$$

where $\mathbf{I}_2$ denotes the 2×2 identity matrix. Substituting $\mathbf{I}_2 = \mathbf{b}_1 \mathbf{b}_1^T + \mathbf{b}_2 \mathbf{b}_2^T$ and $\mathbf{A} = a_1 \mathbf{b}_1 \mathbf{b}_1^T + a_2 \mathbf{b}_2 \mathbf{b}_2^T$, one can write

$$\nabla^2 \hat{F}_c = 0 \times \begin{bmatrix} 0 \\ \mathbf{b}_1 \end{bmatrix} \begin{bmatrix} 0 \\ \mathbf{b}_1 \end{bmatrix}^T + 2(a_2 - a_1) \times \begin{bmatrix} 0 \\ \mathbf{b}_2 \end{bmatrix} \begin{bmatrix} 0 \\ \mathbf{b}_2 \end{bmatrix}^T + 2 \times \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix}^T \quad (11.38)$$

This shows that $\nabla^2 \hat{F}_c$ has eigenvalues $\{0, 2(a_2 - a_1), 2\}$ and eigenvectors $[0; \mathbf{b}_1]$, $[0; \mathbf{b}_2]$, $[1, 0, 0]^T$. Its pseudo-inverse is

$$(\nabla^2 \hat{F}_c)^+ = \frac{1}{2(a_2 - a_1)} \begin{bmatrix} 0 \\ \mathbf{b}_2 \end{bmatrix} \begin{bmatrix} 0 \\ \mathbf{b}_2 \end{bmatrix}^T + \frac{1}{2} \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix}^T = \frac{1}{2} \begin{bmatrix} 1 & \\ & (a_2 - a_1)^{-1} \mathbf{b}_2 \mathbf{b}_2^T \end{bmatrix} \quad (11.39)$$

■

11.2.7 Singular Vector Formula

The posterior covariance matrix of $\boldsymbol{\theta}$ can be expressed in terms of the singular vectors of $\nabla \hat{\mathbf{v}}_c$. Consider the singular value decomposition of $\nabla \hat{\mathbf{v}}_c$:

$$\nabla \hat{\mathbf{v}}_c = \underset{n_\theta \times p}{\mathbf{L}} \underset{n_\theta \times s}{\boldsymbol{\Lambda}} \underset{s \times p}{\mathbf{R}^T} \quad (11.40)$$

where $\boldsymbol{\Lambda}$ is a diagonal matrix of the $s = n_\theta - n_c$ singular values of $\nabla \hat{\mathbf{v}}_c$; $\mathbf{L}$ and $\mathbf{R}$ are orthonormal matrices, i.e., $\mathbf{L}^T \mathbf{L} = \mathbf{I}_s$ and $\mathbf{R}^T \mathbf{R} = \mathbf{I}_s$; and their columns are singular vectors of $\nabla \hat{\mathbf{v}}_c$. Substituting into (11.18) and evaluating the pseudo-inverse in (11.14), it can be shown that

$$\hat{\mathbf{C}} = \underset{n_\theta \times n_\theta}{\mathbf{L}} \underset{n_\theta \times s}{\left[\underset{s \times n_\theta}{\mathbf{L}^T} \underset{n_\theta \times n_\theta}{\nabla^2 \hat{\mathbf{J}}} \underset{n_\theta \times s}{\mathbf{L}} \right]^{-1}} \underset{s \times n_\theta}{\mathbf{L}^T} \quad (11.41)$$

The bracketed matrix has full rank and hence is invertible.

Proof of (11.41) (Singular Vector Formula)

Substituting (11.40) into (11.18) and noting that $\boldsymbol{\Lambda} = \boldsymbol{\Lambda}^T$,

$$\nabla^2 \hat{\mathbf{L}}_c = \underset{p \times s}{\mathbf{R}} \underbrace{[\boldsymbol{\Lambda} \mathbf{L}^T (\nabla^2 \hat{\mathbf{J}}) \boldsymbol{\Lambda}]_{s \times s}}_{s \times s} \underset{s \times p}{\mathbf{R}^T} \quad (11.42)$$

The bracketed matrix is real symmetric. Let $\{\sigma_i\}_{i=1}^s$ and $\{\mathbf{a}_i\}_{i=1}^s$ ($s \times 1$) be its eigenvalues (real) and eigenvectors (orthonormal). Replacing the bracketed matrix by $\sum_{i=1}^s \sigma_i \mathbf{a}_i \mathbf{a}_i^T$ gives $\nabla^2 \hat{\mathbf{L}}_c = \sum_{i=1}^s \sigma_i (\mathbf{R} \mathbf{a}_i) (\mathbf{R} \mathbf{a}_i)^T$. Since $\mathbf{R}^T \mathbf{R} = \mathbf{I}_s$ and $\{\mathbf{a}_i\}_{i=1}^s$ are orthonormal, $(\mathbf{R} \mathbf{a}_i)^T (\mathbf{R} \mathbf{a}_j) = \mathbf{a}_i^T \mathbf{R}^T \mathbf{R} \mathbf{a}_j = \mathbf{a}_i^T \mathbf{a}_j$ is equal to 1 if $i = j$ and zero otherwise. This implies that the eigenvalues and eigenvectors of $\nabla^2 \hat{\mathbf{L}}_c$ are respectively σ_i and $\mathbf{R} \mathbf{a}_i$, $i = 1, \dots, s$; the remaining eigenvalues are zero. The pseudo-inverse of $\nabla^2 \hat{\mathbf{L}}_c$ is then

$$\begin{aligned} (\nabla^2 \hat{\mathbf{L}}_c)^+ &= \sum_{i=1}^s \sigma_i^{-1} (\mathbf{R} \mathbf{a}_i) (\mathbf{R} \mathbf{a}_i)^T \\ &= \mathbf{R} \left(\sum_{i=1}^s \sigma_i^{-1} \mathbf{a}_i \mathbf{a}_i^T \right) \mathbf{R}^T \\ &= \mathbf{R} \underbrace{[\boldsymbol{\Lambda} \mathbf{L}^T (\nabla^2 \hat{\mathbf{J}}) \boldsymbol{\Lambda}]_{s \times s}}_{s \times s}^{-1} \mathbf{R}^T \\ &= \mathbf{R} \boldsymbol{\Lambda}^{-1} [\mathbf{L}^T (\nabla^2 \hat{\mathbf{J}}) \mathbf{L}]^{-1} \boldsymbol{\Lambda}^{-1} \mathbf{R}^T \end{aligned} \quad (11.43)$$

Substituting this and (11.40) into (11.14) gives (11.41). ■

11.2.8 Dimensionless Hessian

Modal parameters can have different units and hence potentially different orders of magnitude and sensitivity in the likelihood function. Large disparity in magnitude among the entries in $\nabla^2 \hat{L}_c$ leads to ill-conditioning in computation. One way to circumvent this issue is to normalize the parameters in $\mathbf{u}$ by their MPV (except for mode shapes) so that $\nabla^2 \hat{L}_c$ is dimensionless. Consider $\boldsymbol{\theta} = \mathbf{v}_c(\mathbf{u})$ as before but now $\mathbf{u} = \mathbf{T}\mathbf{a}$ where $\mathbf{T}$ ($p \times p$, invertible) transforms the set of dimensionless free parameters $\mathbf{a}$ ($p \times 1$) to give $\mathbf{u}$. When the problem is formulated in terms of $\mathbf{a}$ the NLLF is $L_c(\mathbf{a}) = L(\mathbf{v}_c(\mathbf{T}(\mathbf{a})))$. To the first order, uncertain variations of $\boldsymbol{\theta}$ and $\mathbf{a}$ from their MPVs are related by $\Delta\boldsymbol{\theta} = (\nabla \hat{\mathbf{v}}_c \mathbf{T}) \Delta\mathbf{a}$. Following the same argument in Sect. 11.2.3, $\nabla \hat{L}_c$ and $\hat{\mathbf{C}}$ are now given by (11.18) and (11.14) with $\nabla \hat{\mathbf{v}}_c$ replaced by $\nabla \hat{\mathbf{v}}_c \mathbf{T}$. Thus

$$\hat{\mathbf{C}}_{n_\theta \times n_\theta} = (\nabla \hat{\mathbf{v}}_c \mathbf{T})_{n_\theta \times p} [(\nabla \hat{\mathbf{v}}_c \mathbf{T})_{p \times n_\theta}^T (\nabla^2 \hat{J})_{n_\theta \times n_\theta} (\nabla \hat{\mathbf{v}}_c \mathbf{T})_{n_\theta \times p} + (\nabla \hat{\mathbf{v}}_c \mathbf{T})_{p \times n_\theta}^T] \quad (11.44)$$

The bracketed matrix is now dimensionless. Note that $\mathbf{a}$ is only conceptually involved in arriving at the result. All quantities in the formula have the same definition as before. The singular vector formula (11.41) still applies, but (11.40) now reads $\nabla \hat{\mathbf{v}}_c \mathbf{T} = \mathbf{L}\mathbf{A}\mathbf{R}^T$, from which $\mathbf{A}$, $\mathbf{L}$ and $\mathbf{R}$ should be obtained accordingly.

Example 11.4 (Quadratic likelihood, different methods) Here we use a simple example to illustrate different ways of calculating the Hessian and posterior covariance matrix under constraints. Let $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$ ($\theta_1, \theta_2 \geq 0$) be under constraint $\theta_1 \theta_2 = 1$. Suppose the NLLF is $L(\boldsymbol{\theta}) = \theta_1^2 + 2\theta_2^2$. What is the posterior covariance matrix of $\boldsymbol{\theta}$?

By Elimination

Without using the theory in this section, it is not difficult in this example to determine the posterior covariance matrix by eliminating θ_2 . Since $\theta_2 = 1/\theta_1$, the problem can be formulated in terms of only θ_1 with the NLLF $L_1(\theta_1) = \theta_1^2 + 2\theta_1^{-2}$. Minimizing L_1 gives the MPV $\hat{\theta}_1 = 2^{1/4}$. Direct differentiation gives $\partial^2 L_1 / \partial^2 \theta_1 = 2 + 12\theta_1^{-4}$, which is equal to 8 at the MPV. Under Gaussian approximation, the posterior variance of θ_1 is therefore 1/8. Its c.o.v. (coefficient of variation) is $\sqrt{1/8}/2^{1/4} = 2^{-7/4} \approx 30\%$. Using $\theta_2 = 1/\theta_1$, the MPV of θ_2 is $\hat{\theta}_2 = 2^{-1/4}$. Taking log and differential of $\theta_2 = 1/\theta_1$ gives $\Delta\theta_2/\theta_2 = -\Delta\theta_1/\theta_1$ and so to the first order the c.o.v. of θ_2 is equal to that of θ_1 . The variance of θ_2 is then equal to $(\text{c.o.v.} \times \text{MPV})^2 = (2^{-7/4} \times 2^{-1/4})^2 = 1/16$. Clearly, θ_1 and θ_2 are fully negatively correlated and so their covariance is equal to the negative of the product

of their standard deviations, i.e., $-\sqrt{1/8} \times \sqrt{1/16} = -\sqrt{2}/16$. In summary, the posterior covariance matrix of $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$ is

$$\hat{\mathbf{C}} = \begin{bmatrix} 1/8 & -\sqrt{2}/16 \\ -\sqrt{2}/16 & 1/16 \end{bmatrix} \quad (11.45)$$

General Formula

We now determine the posterior covariance matrix of $\boldsymbol{\theta}$ using (11.14) where $\nabla^2 \hat{L}_c$ is calculated using (11.15) at the MPV. The constraint is expressed as $G(\boldsymbol{\theta}) = 0$ where $G(\boldsymbol{\theta}) = 1 - \theta_1 \theta_2$. There are $n_\theta = 2$ parameters and $n_c = 1$ constraint and so $\mathbf{u}$ must have $p \geq n_\theta - n_c = 1$ parameters. Suppose we define $\mathbf{u} = [u_1 \ u_2]^T$ and $\mathbf{v}_c(\mathbf{u}) = [u_1 \ u_1^{-1}]^T$ so that $G(\mathbf{v}_c(\mathbf{u})) = 1 - u_1 u_1^{-1} = 0$ for any $\mathbf{u}$. Direct differentiation gives

$$\nabla L = 2[\theta_1 \ 2\theta_2] \quad \nabla G = -[\theta_2 \ \theta_1] \quad \nabla \mathbf{v}_c = \begin{bmatrix} 1 & 0 \\ -u_1^{-2} & 0 \end{bmatrix} \quad (11.46)$$

$$\nabla^2 L = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \quad \nabla^2 G = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \quad \nabla^2 \mathbf{v}_c = \begin{bmatrix} 0 & 0 \\ 2u_1^{-3} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (11.47)$$

Substituting into (11.15) gives $\nabla^2 L_c$ for any $\mathbf{u}$. Evaluating at the MPV $\hat{\mathbf{u}} = [2^{1/4} \ 2^{-1/4}]^T$ gives $\nabla^2 \hat{L}_c$. The results are

$$\nabla^2 L_c = \begin{bmatrix} 12u_1^{-4} + 2 & 0 \\ 0 & 0 \end{bmatrix} \quad \nabla^2 \hat{L}_c = \begin{bmatrix} 8 & 0 \\ 0 & 0 \end{bmatrix} \quad (11.48)$$

These expressions check with the Hessian obtained by direct differentiation of $L_c(\mathbf{u}) = L(\mathbf{v}_c(\mathbf{u})) = u_1^2 + 2u_1^{-2}$. The eigenvector representation and pseudo-inverse of $\nabla^2 \hat{L}_c$ are respectively given by

$$\nabla^2 \hat{L}_c = 0 \times \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T + 8 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T \quad (\nabla^2 \hat{L}_c)^+ = \frac{1}{8} \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T \quad (11.49)$$

Substituting into (11.14) gives the same result in (11.45).

Lagrange Multiplier Formula

Consider using (11.14) but now $\nabla^2 \hat{L}_c$ is calculated using (11.18). Evaluating (11.46) and (11.47) at the MPV and substituting into (11.22) gives

$$\hat{\lambda} = -[(\nabla \hat{G})(\nabla \hat{G})^T]^{-1}(\nabla \hat{G})(\nabla \hat{L})^T = \frac{6\hat{\theta}_1 \hat{\theta}_2}{\hat{\theta}_1^2 + \hat{\theta}_2^2} = 2^{3/2} \quad (11.50)$$

$$\nabla^2 \hat{J} = \nabla^2 \hat{L} + \hat{\lambda} \nabla^2 \hat{G} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} + 2^{3/2} \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -2^{3/2} \\ -2^{3/2} & 4 \end{bmatrix} \quad (11.51)$$

Equation (11.18) then gives

$$\begin{aligned} \nabla^2 \hat{L}_c &= (\nabla \hat{\mathbf{v}}_c)^T (\nabla^2 \hat{J}) \nabla \hat{\mathbf{v}}_c \\ &= \begin{bmatrix} 1 & 0 \\ -2^{-1/2} & 0 \end{bmatrix}^T \begin{bmatrix} 2 & -2^{3/2} \\ -2^{3/2} & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2^{-1/2} & 0 \end{bmatrix} = \begin{bmatrix} 8 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned} \quad (11.52)$$

as before. Substituting into (11.14) gives the same result in (11.45).

Singular Vector Formula

Consider using (11.41). Here, $\mathbf{L}$ in (11.40) can be obtained by observing that

$$\begin{aligned} \nabla \hat{\mathbf{v}}_c &= \begin{bmatrix} 1 & 0 \\ -2^{-1/2} & 0 \end{bmatrix} = \underbrace{\sqrt{\frac{2}{3}} \begin{bmatrix} 1 \\ -2^{-1/2} \end{bmatrix}}_{\text{unit norm}} \times \underbrace{\sqrt{\frac{3}{2}} \times \begin{bmatrix} 1 & 0 \end{bmatrix}}_{\text{unit norm}} \\ &= \underbrace{\begin{bmatrix} \sqrt{2/3} \\ -1/\sqrt{3} \end{bmatrix}}_{\mathbf{L}} \times \underbrace{\sqrt{\frac{3}{2}}}_{\Lambda} \times \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\mathbf{R}^T} \end{aligned} \quad (11.53)$$

Check that $\mathbf{L}^T \mathbf{L}$ and $\mathbf{R}^T \mathbf{R}$ are both equal to 1. Substituting into (11.41) gives

$$\begin{aligned} \hat{\mathbf{C}} &= \mathbf{L} [\mathbf{L}^T (\nabla^2 \hat{J}) \mathbf{L}]^{-1} \mathbf{L}^T \\ &= \begin{bmatrix} \sqrt{2/3} \\ -1/\sqrt{3} \end{bmatrix} \left\{ \begin{bmatrix} \sqrt{2/3} \\ -1/\sqrt{3} \end{bmatrix}^T \begin{bmatrix} 2 & -2^{3/2} \\ -2^{3/2} & 4 \end{bmatrix} \begin{bmatrix} \sqrt{2/3} \\ -1/\sqrt{3} \end{bmatrix} \right\}^{-1} \begin{bmatrix} \sqrt{2/3} \\ -1/\sqrt{3} \end{bmatrix}^T \end{aligned} \quad (11.54)$$

which returns the same result in (11.45). Note that the term in the brace is just a scalar.

Dimensionless Hessian

Consider using (11.44) with dimensionless $\mathbf{a} = [\theta_1/\hat{\theta}_1 \quad \theta_2/\hat{\theta}_2]^T$. Here, $\mathbf{u} = \mathbf{T}\mathbf{a}$ where

$$\mathbf{T} = \begin{bmatrix} \hat{\theta}_1 & \\ & \hat{\theta}_2 \end{bmatrix} = \begin{bmatrix} 2^{1/4} & \\ & 2^{-1/4} \end{bmatrix} \quad \nabla \hat{\mathbf{v}}_c \mathbf{T} = \begin{bmatrix} 2^{1/4} & 0 \\ -2^{-1/4} & 0 \end{bmatrix} \quad (11.55)$$

Substituting into (11.44) gives

$$\begin{aligned}\hat{\mathbf{C}} &= (\nabla \hat{\mathbf{v}}_c \mathbf{T}) [(\nabla \hat{\mathbf{v}}_c \mathbf{T})^T (\nabla^2 \hat{J}) (\nabla \hat{\mathbf{v}}_c \mathbf{T})] + (\nabla \hat{\mathbf{v}}_c \mathbf{T})^T \\ &= \begin{bmatrix} 2^{1/4} & 0 \\ -2^{-1/4} & 0 \end{bmatrix} \left\{ \begin{bmatrix} 2^{1/4} & 0 \\ -2^{-1/4} & 0 \end{bmatrix}^T \begin{bmatrix} 2 & -2^{3/2} \\ -2^{3/2} & 4 \end{bmatrix} \begin{bmatrix} 2^{1/4} & 0 \\ -2^{-1/4} & 0 \end{bmatrix} \right\}^+ \begin{bmatrix} 2^{1/4} & 0 \\ -2^{-1/4} & 0 \end{bmatrix}^T\end{aligned}\quad (11.56)$$

which returns the same result in (11.45). Note that the matrix in the brace is dimensionless.

Remark In this problem the minimum number of parameters required in $\mathbf{u}$ is $p = 1$. The problem could have been solved by defining $\mathbf{u} = u_1$ while $\mathbf{v}_c(\mathbf{u})$ and $G(\boldsymbol{\theta})$ are the same as before. In the calculation, only the following quantities are affected:

$$\begin{aligned}\nabla \hat{\mathbf{v}}_c &= \begin{bmatrix} 1 \\ -2^{-1/2} \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{2/3} \\ -1/\sqrt{3} \end{bmatrix}}_{\mathbf{L}} \times \underbrace{\sqrt{3/2}}_{\Lambda} \times \underbrace{\begin{bmatrix} 1 \end{bmatrix}}_{\mathbf{R}^T} & \nabla^2 \hat{\mathbf{v}}_c &= \begin{bmatrix} 0 \\ 2^{1/4} \end{bmatrix} & \nabla^2 \hat{L}_c &= 8 \\ \mathbf{T} &= 2^{1/4}\end{aligned}$$

In this case, $\nabla^2 \hat{L}_c$ is no longer singular. Substituting these into (11.14), (11.41) or (11.44) gives the same result as before. ■

11.3 Mode Shape Uncertainty

Mode shape is a vector quantity and it is subjected to norm constraint. Unlike other scalar parameters such as natural frequency or damping ratio, the standard deviation or c.o.v. of the mode shape value at a particular DOF is not a direct indication of uncertainty. For example, if there is only one DOF then the mode shape value is always 1 and so its uncertainty is irrelevant. If there are two DOFs, the mode shape values are correlated because their squares must sum to 1 to satisfy norm constraint.

Consider a particular mode shape $\boldsymbol{\varphi}$ ($n \times 1$) and let the set of modal parameters be partitioned as $\boldsymbol{\theta} = [\boldsymbol{\varpi}; \boldsymbol{\varphi}]$ where $\boldsymbol{\varpi}$ (a $(n_\theta - n) \times 1$ vector) contains the remaining parameters (it can contain other mode shapes). Let $G(\boldsymbol{\theta})$ be the constraint function enforcing unit norm in $\boldsymbol{\varphi}$; and $\mathbf{v}_c(\mathbf{u})$ be the mapping such that $G(\mathbf{v}_c(\mathbf{u})) = 0$ for any $\mathbf{u}$. One simple choice is

$$G(\boldsymbol{\theta}) = 1 - \boldsymbol{\varphi}^T \boldsymbol{\varphi} \quad \mathbf{v}_c(\mathbf{u}) = \begin{bmatrix} \boldsymbol{\varpi} \\ ||\boldsymbol{\varphi}||^{-1} \boldsymbol{\varphi} \end{bmatrix} \quad (11.57)$$

The discussion in this subsection is based on this choice but the resulting conclusions are applicable in general because the posterior covariance matrix is invariant to it (Sect. 11.2.4).

11.3.1 Norm Constraint Singularity

As mode shapes are subjected to norm constraints, the posterior covariance matrix of $\boldsymbol{\theta}$ and its partition corresponding to each mode shape is singular along the most probable mode shape directions. Specifically, let $\hat{\boldsymbol{\theta}} = [\hat{\boldsymbol{w}}; \hat{\boldsymbol{\phi}}]$ and $\hat{\mathbf{C}}$ ($n_\theta \times n_\theta$) be respectively the posterior MPV and covariance matrix of $\boldsymbol{\theta}$; and $\hat{\mathbf{C}}_\phi$ ($n \times n$) be the covariance matrix of $\boldsymbol{\phi}$, equal to the $n \times n$ partition of $\hat{\mathbf{C}}$ corresponding to $\boldsymbol{\phi}$. Then

$$\hat{\mathbf{C}} \begin{bmatrix} \mathbf{0} \\ \hat{\boldsymbol{\phi}} \end{bmatrix} = \mathbf{0} \quad \hat{\mathbf{C}}_\phi \hat{\boldsymbol{\phi}} = \mathbf{0} \quad (11.58)$$

The second equation can be obtained by multiplying out the partitions in the first equation. To show the first equation, since $G(\mathbf{v}_c(\mathbf{u})) = \mathbf{0}$ for any $\mathbf{u}$, taking gradient and evaluating at the MPV gives $\nabla \hat{G} \nabla \hat{\mathbf{v}}_c = \mathbf{0}$. Using (11.14),

$$\hat{\mathbf{C}}(\nabla \hat{G})^T = (\nabla \hat{\mathbf{v}}_c)(\nabla^2 \hat{L}_c) + \underbrace{(\nabla \hat{G} \nabla \hat{\mathbf{v}}_c)^T}_{\mathbf{0}} = \mathbf{0} \quad (11.59)$$

and so $\hat{\mathbf{C}}$ is singular along the direction $(\nabla \hat{G})^T$. The argument follows by noting that $(\nabla \hat{G})^T = [\mathbf{0}_{n_\theta-n}; 2\hat{\boldsymbol{\phi}}]$. Applying the same argument to different modes, it follows that $\hat{\mathbf{C}}$ has as many singularities as the number of modes along their most probable mode shape directions.

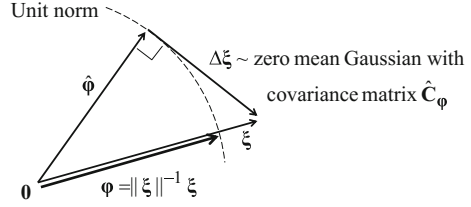
11.3.2 Stochastic Representation

According to the mapping $\mathbf{v}_c(\mathbf{u})$ in (11.57), an uncertain mode shape distributed as the posterior PDF can be represented as

$$\boldsymbol{\phi} = \|\boldsymbol{\xi}\|^{-1} \boldsymbol{\xi} \quad (11.60)$$

where $\boldsymbol{\xi}$ is a Gaussian vector with mean $\hat{\boldsymbol{\phi}}$ (unit norm) and covariance matrix $\hat{\mathbf{C}}_\phi$. The relationship between $\boldsymbol{\phi}$ and $\boldsymbol{\xi}$ is illustrated in Fig. 11.1. To the first order, any uncertain deviation $\Delta \boldsymbol{\xi}$ is perpendicular to $\hat{\boldsymbol{\phi}}$.

Fig. 11.1 Uncertain mode shape $\boldsymbol{\phi}$



Since $\hat{\mathbf{C}}_{\boldsymbol{\phi}}$ is positive semi-definite and has a zero eigenvalue with eigenvector $\hat{\boldsymbol{\phi}}$, it has the eigenvector representation

$$\hat{\mathbf{C}}_{\boldsymbol{\phi}} = \sum_{i=2}^n \delta_i^2 \mathbf{a}_i \mathbf{a}_i^T \quad (11.61)$$

where $\{\delta_i^2\}_{i=1}^n$ and $\{\mathbf{a}_i\}_{i=1}^n$ denote respectively the eigenvalues (in ascending order of magnitude) and orthonormal eigenvectors of $\hat{\mathbf{C}}_{\boldsymbol{\phi}}$; $\delta_1 = 0$ and $\mathbf{a}_1 = \hat{\boldsymbol{\phi}}$. Then $\boldsymbol{\xi}$ can be represented as

$$\boldsymbol{\xi} = \hat{\boldsymbol{\phi}} + \sum_{i=2}^n Z_i \delta_i \mathbf{a}_i \quad (11.62)$$

where Z_i s are i.i.d. standard Gaussian random variables. Using orthogonality and unit norm property of $\{\mathbf{a}_i\}_{i=1}^n$,

$$\|\boldsymbol{\xi}\|^2 = 1 + \sum_{i=2}^n Z_i^2 \delta_i^2 \quad (11.63)$$

Substituting into (11.60),

$$\boldsymbol{\phi} = (1 + \sum_{i=2}^n Z_i^2 \delta_i^2)^{-1/2} (\hat{\boldsymbol{\phi}} + \sum_{i=2}^n Z_i \delta_i \mathbf{a}_i) \quad (11.64)$$

This can be used for generating random samples of uncertain mode shape distributed as the posterior PDF.

11.3.3 Expected MAC and Mode Shape c.o.v

The ‘modal assurance criterion’ (MAC) is commonly used for quantifying the discrepancy between two mode shapes. It is defined as the dot product between two mode shape vectors, or equivalently, the cosine of the hyper-angle between them. The higher the MAC value the closer the two mode shapes. This idea can be

extended to quantifying the posterior uncertainty of mode shape in a Bayesian context. Consider the MAC between the uncertain mode shape $\boldsymbol{\varphi}$ and its MPV $\hat{\boldsymbol{\varphi}}$. If the uncertainty in $\boldsymbol{\varphi}$ is small, it will be close to $\hat{\boldsymbol{\varphi}}$ and their MAC will be close to 1 (in a statistical sense). Using (11.64) and noting $\hat{\boldsymbol{\varphi}}^T \hat{\boldsymbol{\varphi}} = 1$ and $\hat{\boldsymbol{\varphi}}^T \mathbf{a}_i = 0$ ($i = 2, \dots, n$),

$$\text{MAC between } \boldsymbol{\varphi} \text{ and } \hat{\boldsymbol{\varphi}} = \frac{\hat{\boldsymbol{\varphi}}^T \boldsymbol{\varphi}}{\|\hat{\boldsymbol{\varphi}}\| \|\boldsymbol{\varphi}\|} = \frac{1}{\sqrt{1 + \sum_{i=2}^n Z_i^2 \delta_i^2}} \quad (11.65)$$

This is a random variable whose expectation, denoted by ρ , is bounded above by 1. It can be reasoned that $\rho = 1$ if and only if all δ_i s are zero. Analytical expression for ρ is not available but it can be shown that asymptotically (Shi et al. 2010)

$$\rho \sim \frac{1}{\sqrt{1 + \delta_\varphi^2}} \quad n \rightarrow \infty \text{ or all } \delta_i \rightarrow 0 \quad (11.66)$$

where

$$\delta_\varphi^2 = \sum_{i=2}^n \delta_i^2 \quad (11.67)$$

is the sum of the eigenvalues of $\hat{\mathbf{C}}_\varphi$ (note that $\delta_1 = 0$). Typically, ρ is close to 1. One can interpret $\rho = \cos \alpha$ where α is a hyper-angle reflecting the identification uncertainty of $\boldsymbol{\varphi}$. Note that $\cos \alpha \sim 1 - \alpha^2/2$ for small α . From (11.66), $\rho \sim 1 - \delta_\varphi^2/2$ for small δ_φ and so $\alpha \sim \delta_\varphi$. Thus, δ_φ can be used as a measure of mode shape uncertainty, the lower the better. Being the square root sum of eigenvalues of $\hat{\mathbf{C}}_\varphi$, it can be interpreted as the ‘mode shape c.o.v.’, analogous to the c.o.v. of a scalar parameter:

$$\text{Mode shape c.o.v.} = \delta_\varphi = \sqrt{\frac{\text{Sum of eigenvalues of}}{\text{mode shape covariance matrix}}} \quad (11.68)$$

Reference

Shi X, Wu Y, Liu Y (2010) A note on asymptotic approximations of inverse moments of nonnegative random variables. *Stat Probab Lett* 80:1260–1264

Part III

Algorithms

Chapter 12

Single Mode Problem

Abstract This chapter develops the computational algorithm for determining the most probable value (MPV) and covariance matrix of modal parameters in Bayesian operational modal analysis with data in a single setup and for well-separated modes. The MPV of the mode shape is analytically derived in terms of other parameters, which effectively reduces the numerical optimization problem for MPV to involve only four parameters independent of the number of measured degrees of freedom. The asymptotic behavior of MPV for modes with high signal-to-noise ratio is analyzed. Analytical formulas are derived for systematically computing the covariance matrix of parameters given the measured data. Examples with synthetic data, laboratory data and field data are presented to illustrate the algorithms and their applications.

Keywords Single mode • Tall building • Signal-to-noise ratio

In this chapter we discuss efficient methods for calculating the posterior most probable values (MPV) and covariance matrix of modal parameters in Bayesian operational modal analysis (OMA) with data collected in a single setup. The mode of interest is assumed to be well-separated from others so that it dominates the selected frequency band. The general case of multiple modes is considered in Chap. 13. Similar techniques in this chapter are used for developing algorithms for well-separated modes from multi-setup data in Chap. 14.

Mathematically, the posterior MPV minimizes the negative log-likelihood function (NLLF) and so its determination involves an optimization problem. The algorithm for MPV makes use of the fact that the NLLF can be written as a quadratic function of mode shape. This allows the most probable mode shape to be determined analytically in terms of the remaining parameters. The dimension of the optimization problem is then reduced to four only, regardless of the number of measured DOFs. A good initial guess can be obtained by considering the asymptotic behavior of the MPV when the signal-to-noise ratio of data is high.

The algorithm presented in this chapter is based on Au (2011); see also Zhang and Au (2013). The presentation has been simplified and reorganized. Computer

programming of the posterior covariance matrix using the analytical formulas in Sect. 12.4 is simpler as it has taken advantage of a more systematic procedure based on Lagrange multiplier concepts (Sect. 11.2.3).

Recall the problem context from Sect. 10.1 where the measured ambient vibration history is denoted by $\{\hat{\mathbf{y}}_j\}_{j=0}^{N-1}$ ($n \times 1$); n is the number of measured DOFs and N is the number of samples per data channel. The scaled Fast Fourier Transform (FFT) of $\{\hat{\mathbf{y}}_j\}_{j=0}^{N-1}$ at frequency $f_k = k/N\Delta t$ (Hz) is

$$\hat{\mathcal{F}}_k = \sqrt{\frac{\Delta t}{N}} \sum_{j=0}^{N-1} \hat{\mathbf{y}}_j e^{-2\pi i j k / N} \quad (12.1)$$

where Δt (s) is the sampling interval. The set of modal parameters $\boldsymbol{\theta}$ comprises

f	ζ	S	S_e	$\boldsymbol{\phi}$
Natural frequency (Hz)	Damping ratio	Modal force PSD	Prediction error PSD	$n \times 1$ Mode shape

The mode shape is subjected to unit norm constraint, $\boldsymbol{\phi}^T \boldsymbol{\phi} = 1$. The modal force PSD has the same unit as the PSD of acceleration. The prediction error PSD has the same unit as the PSD of data.

12.1 Alternative Form of NLLF

The NLLF can be rewritten in a form that facilitates the determination of the most probable mode shape. Recall the NLLF from Sect. 10.1:

$$L(\boldsymbol{\theta}) = nN_f \ln \pi + \sum_k \ln |\mathbf{E}_k(\boldsymbol{\theta})| + \sum_k \hat{\mathcal{F}}_k^* \mathbf{E}_k(\boldsymbol{\theta})^{-1} \hat{\mathcal{F}}_k \quad (12.2)$$

where $\mathbf{E}_k(\boldsymbol{\theta})$ is the theoretical PSD matrix of data for given $\boldsymbol{\theta}$. Assuming a single mode in the selected frequency band,

$$\mathbf{E}_k = S D_k \boldsymbol{\phi} \boldsymbol{\phi}^T + S_e \mathbf{I}_n \quad (12.3)$$

$$D_k = \frac{(2\pi f_k)^{-2q}}{(1 - \beta_k^2)^2 + (2\zeta\beta_k)^2} \quad \beta_k = \frac{f}{f_k} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (12.4)$$

where $\mathbf{I}_n$ denotes the $n \times n$ identity matrix. The key is to express $|\mathbf{E}_k|$ and $\mathbf{E}_k^{-1}$ in explicit form using the eigenvector representation of $\mathbf{E}_k$. Let $\{\mathbf{b}_i\}_{i=1}^n$ ($n \times 1$) be an orthonormal basis with $\mathbf{b}_1 = \boldsymbol{\phi}$. Substituting $\mathbf{I}_n = \sum_{i=1}^n \mathbf{b}_i \mathbf{b}_i^T$ into (12.3) and collecting terms,

$$\mathbf{E}_k = \underbrace{(SD_k + S_e)}_{\text{eigenvalue}} \boldsymbol{\phi} \boldsymbol{\phi}^T + \sum_{i=2}^n \underbrace{S_e}_{\text{eigenvalue}} \mathbf{b}_i \mathbf{b}_i^T \quad (12.5)$$

This shows that $\mathbf{E}_k$ has eigenvalues $\{SD_k + S_e, S_e, \dots, S_e\}$ and eigenvectors $\{\boldsymbol{\phi}, \mathbf{b}_2, \dots, \mathbf{b}_n\}$. The determinant of $\mathbf{E}_k$ is equal to the product of its eigenvalues:

$$|\mathbf{E}_k| = (SD_k + S_e) S_e^{n-1} \quad (12.6)$$

Replacing the eigenvalues in (12.5) by their reciprocals gives $\mathbf{E}_k^{-1}$:

$$\mathbf{E}_k^{-1} = (SD_k + S_e)^{-1} \boldsymbol{\phi} \boldsymbol{\phi}^T + \sum_{i=2}^n S_e^{-1} \mathbf{b}_i \mathbf{b}_i^T \quad (12.7)$$

Substituting $\sum_{i=2}^n \mathbf{b}_i \mathbf{b}_i^T = \mathbf{I}_n - \boldsymbol{\phi} \boldsymbol{\phi}^T$ and rearranging gives

$$\mathbf{E}_k^{-1} = S_e^{-1} \left[\mathbf{I}_n - \left(1 + \frac{S_e}{SD_k} \right)^{-1} \boldsymbol{\phi} \boldsymbol{\phi}^T \right] \quad (12.8)$$

Substituting (12.6) and (12.8) into (12.2) and noting

$$\underbrace{\hat{\mathcal{F}}_k^*}_{1 \times n} \underbrace{\boldsymbol{\phi}}_{n \times 1} \underbrace{\boldsymbol{\phi}^T}_{1 \times n} \underbrace{\hat{\mathcal{F}}_k}_{n \times 1} = \underbrace{(\boldsymbol{\phi}^T \hat{\mathcal{F}}_k)}_{\text{scalar}} \underbrace{(\hat{\mathcal{F}}_k^* \boldsymbol{\phi})}_{\text{scalar}} = \underbrace{\boldsymbol{\phi}^T \text{Re}(\hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^*)}_{\substack{1 \times n \\ n \times n \text{ real sym.}}} \underbrace{\boldsymbol{\phi}}_{n \times 1} \quad (12.9)$$

gives

$$L = nN_f \ln \pi + \sum_k \ln(SD_k + S_e) + (n-1)N_f \ln S_e + S_e^{-1} d - S_e^{-1} \boldsymbol{\phi}^T \mathbf{A} \boldsymbol{\phi} \quad (12.10)$$

where

$$\mathbf{A} = \sum_k \left(1 + \frac{S_e}{SD_k} \right)^{-1} \hat{\mathbf{D}}_k \quad (12.11)$$

$$\hat{\mathbf{D}}_k = \text{Re}(\hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^*) \quad d = \sum_k \hat{\mathcal{F}}_k^* \hat{\mathcal{F}}_k \quad (12.12)$$

In deriving (12.10), we have assumed $\boldsymbol{\phi}^T \boldsymbol{\phi} = 1$. When using this form of the NLLF for determining the MPV by optimization, the mode shape substituted in the expression should always have unit norm.

12.2 Algorithm for MPV

In (12.10), only the term $\boldsymbol{\varphi}^T \mathbf{A} \boldsymbol{\varphi}$ depends on $\boldsymbol{\varphi}$. The most probable mode shape, denoted by $\hat{\boldsymbol{\varphi}}$, should therefore maximize the term subjected to $\boldsymbol{\varphi}^T \boldsymbol{\varphi} = 1$. As $\mathbf{A}$ is positive definite, it follows that $\hat{\boldsymbol{\varphi}}$ is the eigenvector of $\mathbf{A}$ with the largest eigenvalue (Section C.4). Evaluating L at $\boldsymbol{\varphi} = \hat{\boldsymbol{\varphi}}$, the MPV of $\{f, \zeta, S, S_e\}$ can be obtained by minimizing

$$\tilde{L}(f, \zeta, S, S_e) = nN_f \ln \pi + \sum_k \ln(SD_k + S_e) + (n-1)N_f \ln S_e + S_e^{-1}(d - \hat{\lambda}) \quad (12.13)$$

where $\hat{\lambda} = \hat{\boldsymbol{\varphi}}^T \mathbf{A} \hat{\boldsymbol{\varphi}}$ is the largest eigenvalue of $\mathbf{A}$. Note that $\hat{\lambda}$ depends on $\{f, \zeta, S, S_e\}$ but the dependence has been omitted for notational simplicity.

Below are two algorithms for calculating the MPV of $\boldsymbol{\theta} = \{f, \zeta, S, S_e, \boldsymbol{\varphi}\}$. The first one follows directly from the considerations above. The second one is iterative. It is a special case of the multi-mode algorithm in Chap. 13.

Algorithm 1

0. Set initial guess for $\{f, \zeta, S, S_e\}$. See Sect. 12.3.1.
1. Determine the MPV of $\{f, \zeta, S, S_e\}$ by minimizing $\tilde{L}$ in (12.13).
2. Determine the MPV of $\boldsymbol{\varphi}$ as the eigenvector (largest eigenvalue) of $\mathbf{A}$ in (12.11) at the MPV of $\{f, \zeta, S, S_e\}$.

Algorithm 2

The following is an iterative algorithm. At any one step, only the mentioned parameters are updated; the remaining ones are kept at their current value.

0. Set initial guess for $\{f, \zeta, S, S_e, \boldsymbol{\varphi}\}$. See Sect. 12.3.1.
1. Update $\{f, \zeta, S, S_e\}$ by minimizing L in (12.10) w.r.t. $\{f, \zeta, S, S_e\}$.
2. Update $\boldsymbol{\varphi}$ as the eigenvector (largest eigenvalue) of $\mathbf{A}$ in (12.11).

Repeat Steps 1 and 2 until convergence.

12.3 High s/n Asymptotics of MPV

The MPV of modal parameters exhibits characteristic behavior when the signal-to-noise (s/n) ratio of data is high, in the sense that

$$\gamma_k = \frac{SD_k}{S_e} \gg 1 \quad (12.14)$$

In this case $(1 + S_e/SD_k) \sim 1$. Substituting into (12.11),

$$\mathbf{A} \sim \mathbf{A}_0 = \sum_k \hat{\mathbf{D}}_k = \text{Re} \sum_k \hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^* \quad (12.15)$$

The most probable mode shape is therefore asymptotically given by the eigenvector (largest eigenvalue) of $\mathbf{A}_0$, which can be directly computed from data without iterations.

Replacing $\hat{\lambda}$ in the expression of $\tilde{L}$ in (12.13) by the largest eigenvalue of $\mathbf{A}_0$ results in an expression that does not yield a reasonable value for the MPV of S or S_e , suggesting that the leading order w.r.t. these parameters has been lost in the approximation. To obtain the correct asymptotic form, the first order of $\mathbf{A}$ should be retained. Substituting $(1 + S_e/SD_k)^{-1} \sim 1 - S_e/SD_k$ into (12.11) gives

$$\mathbf{A} \sim \sum_k \hat{\mathbf{D}}_k - \frac{S_e}{S} \sum_k D_k^{-1} \hat{\mathbf{D}}_k \quad (12.16)$$

For the summand in (12.13),

$$\begin{aligned} \ln(SD_k + S_e) &= \ln \left[SD_k \left(1 + \frac{S_e}{SD_k} \right) \right] \\ &= \ln SD_k + \ln \left(1 + \frac{S_e}{SD_k} \right) \sim \ln S + \ln D_k + \frac{S_e}{SD_k} \end{aligned} \quad (12.17)$$

Substituting (12.16) and (12.17) into (12.13) gives

$$\begin{aligned} \tilde{L}(f, \zeta, S, S_e) &\sim nN_f \ln \pi + \sum_k \ln D_k + \frac{S_e}{S} \sum_k D_k^{-1} \\ &+ \left[(n-1)N_f \ln S_e + S_e^{-1}(d - \hat{\lambda}_0) \right] + \left[N_f \ln S + S^{-1} \sum_k \frac{\hat{d}_k}{D_k} \right] \end{aligned} \quad (12.18)$$

$$\hat{\lambda}_0 = \hat{\boldsymbol{\Phi}}^T \mathbf{A}_0 \hat{\boldsymbol{\Phi}} \quad \hat{d}_k = \hat{\boldsymbol{\Phi}}^T \hat{\mathbf{D}}_k \hat{\boldsymbol{\Phi}} \quad (12.19)$$

The variation of $S_e S^{-1} \sum_k D_k^{-1}$ w.r.t. S_e or S is asymptotically small compared to the bracketed terms and so $\tilde{L}$ depends on S_e dominantly through the first bracket and on S through the second bracket. Minimizing the first and second bracket gives the MPV of S_e and S , respectively:

$$\hat{S}_e \sim \frac{d - \hat{\lambda}_0}{(n-1)N_f} \quad \hat{S} \sim \frac{1}{N_f} \sum_k \frac{\hat{d}_k}{D_k} \quad (12.20)$$

12.3.1 Initial Guess of MPV

The initial guess of MPV for numerical optimization can be set as follow. The initial guess for f can be picked from the singular value (SV) spectrum of data (Sect. 7.3). The initial guess for ζ can be set as 1%. The initial guess for S and S_e can be calculated using (12.20). The initial guess for $\boldsymbol{\phi}$ can be taken as the eigenvector (largest eigenvalue) of $\mathbf{A}_0$ in (12.15).

12.4 Posterior Covariance Matrix

Under a Gaussian approximation of the posterior PDF, the posterior covariance matrix of the set of modal parameters $\boldsymbol{\theta}$ ($n_\theta \times 1$) satisfying constraint (on mode shape norm) is given by (Sect. 11.2)

$$\hat{\mathbf{C}}_{n_\theta \times n_\theta} = (\nabla \hat{\mathbf{v}}_c)_{n_\theta \times p} \underbrace{(\nabla^2 \hat{L}_c)^+}_{p \times p} (\nabla \hat{\mathbf{v}}_c)^T_{p \times n_\theta} \quad (12.21)$$

where ‘+’ denotes the pseudo-inverse, i.e., ignoring the zero eigenvalues from constraint singularity; $\nabla^2 \hat{L}_c$ is the Hessian of $L_c(\mathbf{u}) = L(\mathbf{v}_c(\mathbf{u}))$ w.r.t. a set of free parameters $\mathbf{u}$ ($p \times 1$) at the MPV; $\mathbf{v}_c(\mathbf{u})$ maps $\mathbf{u}$ to $\boldsymbol{\theta}$ that always satisfies the constraint $G(\mathbf{v}_c(\mathbf{u})) = 0$; $G(\boldsymbol{\theta})$ is the norm constraint equation on mode shape; $\nabla \hat{\mathbf{v}}_c$ is the gradient of $\mathbf{v}_c(\mathbf{u})$ at the MPV. A hat ‘^’ on quantity denotes that it is evaluated at the MPV. Using the formula based on Lagrange multiplier (Sect. 11.2.3),

$$\nabla^2 \hat{L}_c = \underbrace{(\nabla \hat{\mathbf{v}}_c)^T}_{p \times n_\theta} \underbrace{(\nabla^2 \hat{L} + \hat{\lambda}_1 \nabla^2 \hat{G})}_{n_\theta \times n_\theta} \underbrace{(\nabla \hat{\mathbf{v}}_c)}_{n_\theta \times p} \quad (12.22)$$

where $\nabla^2 \hat{L}$ is the Hessian of the NLLF in (12.10) (ignoring mode shape norm constraint) and

$$\hat{\lambda}_1 = - \frac{(\nabla \hat{G})(\nabla \hat{L})^T}{(\nabla \hat{G})(\nabla \hat{G})^T} \quad (12.23)$$

is the value of the Lagrange multiplier at the MPV.

The terms in (12.22) can be determined as follow. Let $\varpi = [f, \zeta, S, S_e]^T$ comprise the modal parameters other than mode shape. One can take

$$\mathbf{u} = \begin{bmatrix} \varpi \\ \boldsymbol{\phi} \end{bmatrix} \quad \boldsymbol{\theta} = \begin{bmatrix} \varpi \\ \boldsymbol{\phi} \end{bmatrix} \quad \mathbf{v}_c(\mathbf{u}) = \begin{bmatrix} \varpi \\ \boldsymbol{\phi}/\|\boldsymbol{\phi}\| \end{bmatrix} \quad G(\boldsymbol{\theta}) = 1 - \boldsymbol{\phi}^T \boldsymbol{\phi} \quad (12.24)$$

so that $G(\mathbf{v}_c(\mathbf{u})) \equiv 0$ for any $\mathbf{u}$. Direct differentiation and evaluating at the MPV gives

$$\nabla \hat{G} = [\mathbf{0}_{1 \times 4} \quad -2\hat{\boldsymbol{\phi}}^T] \quad \nabla^2 \hat{G} = \begin{bmatrix} \mathbf{0}_{4 \times 4} & \\ & -2\mathbf{I}_n \end{bmatrix} \quad (12.25)$$

$$\nabla \hat{L} = [\nabla_{\varpi} \hat{L} \quad -2S_e^{-1} \hat{\boldsymbol{\phi}}^T \hat{\mathbf{A}}] \quad \nabla \hat{\mathbf{v}}_c = \begin{bmatrix} \mathbf{I}_{4 \times 4} & \\ & \mathbf{I}_n - \hat{\boldsymbol{\phi}} \hat{\boldsymbol{\phi}}^T \end{bmatrix} \quad (12.26)$$

where ∇_{ϖ} denotes the gradient w.r.t. ϖ ; $\hat{\mathbf{A}}$ denotes the value of $\mathbf{A}$ at the MPV. Substituting into (12.23) gives

$$\hat{\lambda}_1 = -S_e^{-1} \hat{\lambda} \quad (12.27)$$

where $\hat{\lambda}$ is the largest eigenvalue of $\hat{\mathbf{A}}$. Due to mode shape norm constraint, $\nabla \hat{L}_c$ has one zero eigenvalue with eigenvector $[\mathbf{0}_{4 \times 1}; \hat{\boldsymbol{\phi}}]$.

It remains to determine $\nabla^2 \hat{L}$, which can be obtained via finite difference approximation or analytical expression. The latter is discussed in the next two sections. Section 12.4.1 expresses the derivatives of L in terms of those of $\mathbf{E}_k$. It has the advantage of unifying derivation and computer coding effort with other formulations. Section 12.4.2 presents alternative expressions based on a condensed form of the NLLF. It can speed up computation especially when n is large, but requires additional coding effort specific to the present context.

12.4.1 General Expressions

Let a superscripted variable in parenthesis denote a derivative w.r.t. the variable. For any parameters x and y ,

$$\begin{aligned} L^{(x)} &= \sum_k (\ln|\mathbf{E}_k|)^{(x)} + \sum_k \hat{\mathcal{F}}_k^*(\mathbf{E}_k^{-1})^{(x)} \hat{\mathcal{F}}_k \\ L^{(xy)} &= \sum_k (\ln|\mathbf{E}_k|)^{(xy)} + \sum_k \hat{\mathcal{F}}_k^*(\mathbf{E}_k^{-1})^{(xy)} \hat{\mathcal{F}}_k \end{aligned} \quad (12.28)$$

The derivatives of $\ln|\mathbf{E}_k|$ and $\mathbf{E}_k^{-1}$ can be expressed in terms of the derivatives of $\mathbf{E}_k$. For any x and y ,

$$\begin{aligned}
(\ln |\mathbf{E}_k|)^{(x)} &= tr[\mathbf{E}_k^{-1} \mathbf{E}_k^{(x)}] & (\mathbf{E}_k^{-1})^{(x)} &= -\mathbf{E}_k^{-1} \mathbf{E}_k^{(x)} \mathbf{E}_k^{-1} \\
(\ln |\mathbf{E}_k|)^{(xy)} &= tr[\mathbf{E}_k^{-1} \mathbf{E}_k^{(xy)} - \mathbf{E}_k^{-1} \mathbf{E}_k^{(x)} \mathbf{E}_k^{-1} \mathbf{E}_k^{(y)}] \\
(\mathbf{E}_k^{-1})^{(xy)} &= \mathbf{E}_k^{-1} [\mathbf{E}_k^{(x)} \mathbf{E}_k^{-1} \mathbf{E}_k^{(y)} + \mathbf{E}_k^{(y)} \mathbf{E}_k^{-1} \mathbf{E}_k^{(x)} - \mathbf{E}_k^{(xy)}] \mathbf{E}_k^{-1}
\end{aligned} \tag{12.29}$$

The derivatives of $\mathbf{E}_k$ are given in Table 12.1. The expressions involve the derivatives of D_k , which are given in Table 12.2. For better conditioning, $\mathbf{E}_k^{-1}$ should be calculated using the formula below (recalled from (12.8)), where the ill-conditioning due to small S_e has been segregated out in the factor S_e^{-1} :

$$\mathbf{E}_k^{-1} = S_e^{-1} \left[\mathbf{I}_n - \left(1 + \frac{S_e}{SD_k} \right)^{-1} \boldsymbol{\Phi} \boldsymbol{\Phi}^T \right] \tag{12.30}$$

12.4.2 Condensed Expressions

Computing the derivatives of the NLLF via those of $\mathbf{E}_k$ involves matrix computations of size n (the number of DOFs) which can be large in applications. It is

Table 12.1 Derivatives of $\mathbf{E}_k = SD_k \boldsymbol{\Phi} \boldsymbol{\Phi}^T + S_e \mathbf{I}_n$

$\mathbf{E}_k^{(xy)}$		$y =$			
		f, ζ	S	S_e	ϕ_r
$x =$	f, ζ	$SD_k^{(xy)} \boldsymbol{\Phi} \boldsymbol{\Phi}^T$	$D_k^{(x)} \boldsymbol{\Phi} \boldsymbol{\Phi}^T$	$\mathbf{0}_n$	$SD_k^{(x)} [\mathbf{e}_r \boldsymbol{\Phi}^T + \boldsymbol{\Phi} \mathbf{e}_r^T]$
	S		$\mathbf{0}_n$	$\mathbf{0}_n$	$D_k [\mathbf{e}_r \boldsymbol{\Phi}^T + \boldsymbol{\Phi} \mathbf{e}_r^T]$
	S_e			$\mathbf{0}_n$	$\mathbf{0}_n$
	ϕ_i	Sym.			$SD_k [\mathbf{e}_{ir} + \mathbf{e}_{ir}^T]$
$\mathbf{E}_k^{(y)}$		$SD_k^{(y)} \boldsymbol{\Phi} \boldsymbol{\Phi}^T$	$D_k \boldsymbol{\Phi} \boldsymbol{\Phi}^T$	$\mathbf{I}_n$	$SD_k [\mathbf{e}_r \boldsymbol{\Phi}^T + \boldsymbol{\Phi} \mathbf{e}_r^T]$

See Table 12.2. for the derivatives of D_k ; $\boldsymbol{\Phi} = [\phi_1, \dots, \phi_n]^T$; $\mathbf{e}_r$ is a $n \times 1$ vector with the r th entry equal to 1 (zero elsewhere); $\mathbf{e}_{ir}$ is a $n \times n$ matrix with the (i, r) -entry equal to 1 (zero elsewhere)

Table 12.2 Derivatives of $D_k^{-1} = [(1 - \beta_k^2)^2 + (2\zeta\beta_k)^2](2\pi f_k)^{2q}$ and D_k

$D_k^{(x)}$		$-D_k^2 (D_k^{-1})^{(x)}$ for any x	
$D_k^{(xy)}$		$2D_k^3 (D_k^{-1})^{(x)} (D_k^{-1})^{(y)} - D_k^2 (D_k^{-1})^{(xy)}$ for any x, y	
$(D_k^{-1})^{(xy)}$		$y =$	
		f	ζ
$x =$	f	$4f_k^{-2} (3\beta_k^2 - 1 + 2\zeta^2) (2\pi f_k)^{2q}$	$16f_k^{-1} \zeta \beta_k (2\pi f_k)^{2q}$
	ζ	Sym.	$8\beta_k^2 (2\pi f_k)^{2q}$
$(D_k^{-1})^{(y)}$		$4f_k^{-1} [\beta_k^3 - \beta_k (1 - 2\zeta^2)] (2\pi f_k)^{2q}$	$8\zeta \beta_k^2 (2\pi f_k)^{2q}$

$\beta_k = f/f_k$; $q = 0, 1, 2$ for acceleration, velocity and displacement data

possible to obtain the derivatives in a condensed form using (12.10) to speed up computations. Let $\varpi = [f, \zeta, S, S_e]^T$ and $\nabla^2 L$ be partitioned as

$$\nabla^2 L = \begin{bmatrix} L_{4 \times 4}^{(\varpi\varpi)} & L_{4 \times n}^{(\varpi\Phi)} \\ \text{sym.} & L_{n \times n}^{(\Phi\Phi)} \end{bmatrix} \quad (12.31)$$

The partition $L^{(\Phi\Phi)}$ involves $L^{(x\Phi)}$ for $x = f, \zeta, S, S_e$. These are given in Table 12.3. The partition $L^{(\varpi\varpi)}$ involves $L^{(xy)}$ for $x, y = f, \zeta, S, S_e$. To facilitate presentation, the NLLF in (12.10) is separated into two parts:

$$L = L_D + L_Q \quad (12.32)$$

$$L_D = nN_f \ln \pi + \sum_k \ln(SD_k + S_e) + (n-1)N_f \ln S_e \quad (12.33)$$

$$L_Q = S_e^{-1}(d - \Phi^T \mathbf{A} \Phi) = S_e^{-1}(d - \sum_k B_k q_k) \quad (12.34)$$

$$B_k = (1 + \frac{S_e}{SD_k})^{-1} \quad q_k = \Phi^T \hat{\mathbf{D}}_k \Phi \quad (12.35)$$

Note that B_k only depends on $\{f, \zeta, S, S_e\}$ and q_k only on Φ . The second derivatives of L_D and L_Q are given in Tables 12.4 and 12.5, respectively. They involve the derivatives of D_k and B_k , which are given in Tables 12.2 and 12.6, respectively. Figure 12.1 shows the information flow. The quantities D_k , B_k , $\gamma_k = SD_k/S_e$ and $q_k = \Phi^T \hat{\mathbf{D}}_k \Phi$ appear repeatedly and so they may be computed and stored upfront. The expressions in Tables 12.3, 12.4, 12.5 and 12.6 are applicable for general parameter values, not necessarily at the MPV.

Table 12.3 Derivatives $L^{(x\Phi)}$ for $x = f, \zeta, S, S_e$

	$x =$	
	f, ζ, S	S_e
$L^{(x\Phi)}$	$-2\Phi^T S_e^{-1} \sum_k B_k^{(x)} \hat{\mathbf{D}}_k$	$-2\Phi^T S_e^{-1} \sum_k B_k^{(S_e)} \hat{\mathbf{D}}_k + 2S_e^{-2} \Phi^T \mathbf{A}$

See Table 12.6 for derivatives of B_k

Table 12.4 Derivatives of $L_D = nN_f \ln \pi + \sum_k \ln(SD_k + S_e) + (n-1)N_f \ln S_e$

$L_D^{(xy)}$		$y =$		
		f, ζ	S	S_e
$x =$	f, ζ	$\sum_k [B_k D_k^{-1} D_k^{(xy)} - B_k^2 D_k^{-2} D_k^{(x)} D_k^{(y)}]$	$S^{-1} \sum_k \gamma_k^{-1} B_k^2 D_k^{-1} D_k^{(x)}$	$-S_e^{-1} \sum_k \gamma_k^{-1} B_k^2 D_k^{-1} D_k^{(x)}$
	S		$-S^{-2} \sum_k B_k^2$	$-S^{-1} S_e^{-1} \sum_k \gamma_k^{-1} B_k^2$
	S_e	Sym.		$-S_e^{-2} N_f (n-1)$ $-S_e^{-2} \sum_k \gamma_k^{-2} B_k^2$

See Table 12.2 for derivatives of D_k

Table 12.5 Derivatives of $L_Q = S_e^{-1}(d - \Phi^T \mathbf{A} \Phi)$

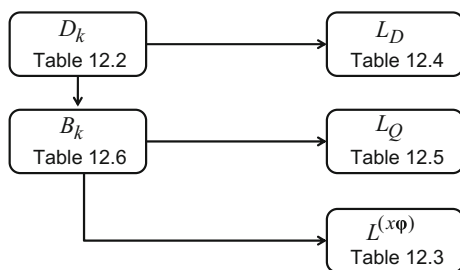
$L_Q^{(xy)}$		$y =$	
		f, ζ, S	S_e
$x =$	f, ζ, S	$-S_e^{-1} \sum_k B_k^{(xy)} q_k$	$S_e^{-2} \sum_k B_k^{(x)} q_k - S_e^{-1} \sum_k B_k^{(xS_e)} q_k$
	S_e	Sym.	$2S_e^{-3}(d - \sum_k B_k q_k) + 2S_e^{-2} \sum_k B_k^{(S_e)} q_k - S_e^{-1} \sum_k B_k^{(S_e S_e)} q_k$

See Table 12.6 for derivatives of B_k

Table 12.6 Derivatives of $B_k^{-1} = (1 + S_e/SD_k)$ and B_k

$B_k^{(x)}$		$-B_k^2(B_k^{-1})^{(x)}$ for any x		
$B_k^{(xy)}$		$2B_k^3(B_k^{-1})^{(x)}(B_k^{-1})^{(y)} - B_k^2(B_k^{-1})^{(xy)}$ for any x, y		
$(B_k^{-1})^{(xy)}$		$y =$		
		f, ζ	S	S_e
$x =$	f, ζ	$\gamma_k^{-1} D_k (D_k^{-1})^{(xy)}$	$S^{-1} \gamma_k^{-1} D_k^{-1} D_k^{(x)}$	$-\gamma_k^{-1} S_e^{-1} D_k^{-1} D_k^{(x)}$
	S		$2\gamma_k^{-1} S^{-2}$	$-\gamma_k^{-1} S_e^{-1} S^{-1}$
	S_e	Sym.		0
$(B_k^{-1})^{(y)}$		$-\gamma_k^{-1} D_k^{-1} D_k^{(y)}$	$-\gamma_k^{-1} S^{-1}$	$\gamma_k^{-1} S_e^{-1}$

See Table 12.2 for derivatives of D_k

Fig. 12.1 Information flow of derivative calculations

12.5 Synthetic Data Examples

In this section, we present examples on Bayesian OMA with well-separated modes based on synthetic data. Example 12.1 provides a basic illustration of Bayesian modal identification results and their interpretation. Examples 12.2 and 12.3 illustrate the effect of modeling error and the choice of bandwidth, respectively.

Example 12.1 (Ten-storied building) Consider the horizontal vibration of a ten-storied shear building with uniform mass 1000 tons per floor, interstory stiffness 1767 kN/mm and damping ratio 1% in all modes. The first three modes have natural frequencies 1, 2.98 and 4.89 Hz. The structure is subjected to i.i.d. white noise excitation at all floors, each with PSD $S_w = 96.2 \text{ N}^2/\text{Hz}$. This structure was

considered in Example 10.1. The acceleration on 5/F and the roof are measured for 600 s at 100 Hz. They are contaminated by i.i.d. data noise of PSD $S_e = 1(\mu\text{g})^2/\text{Hz}$. Figure 12.2 shows the root PSD and SV spectrum of data. The frequency band used for modal identification is indicated by the horizontal bar below each peak.

Posterior Statistics

Table 12.7 summarizes the modal identification results. The exact value of modal force PSD of the first mode was calculated in Example 10.1; the values for other

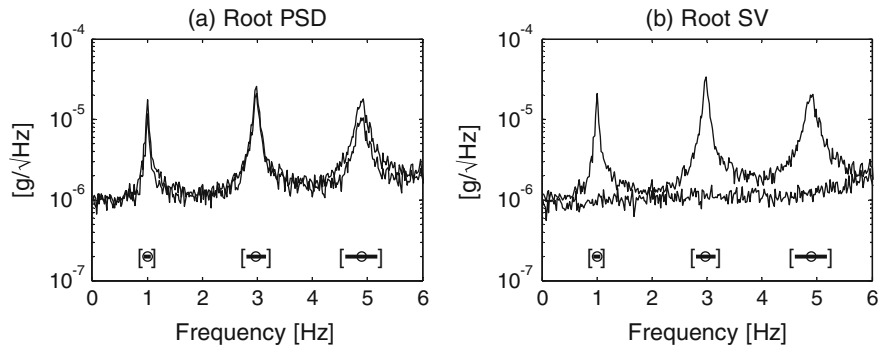


Fig. 12.2 **a** Root PSD and **b** root SV spectrum using data on 5/F and the roof, ten-storied building (synthetic data). *Horizontal bar* indicates selected frequency band

Table 12.7 Modal identification results, ten-storied building (synthetic data)

		Mode 1	Mode 2	Mode 3
Modal s/n ratio γ		643	1013	297
Frequency f	Exact (Hz)	1.000	2.978	4.889
	MPV (Hz)	1.003	2.977	4.891
	c.o.v. (%)	0.190	0.092	0.091
Damping ζ	Exact (%)	1	1	1
	MPV (%)	1.04	0.78	1.13
	c.o.v. (%)	21	13	10
Modal force PSD S	Exact $[(\mu\text{g})^2/\text{Hz}]$	0.28	0.30	0.23
	MPV $[(\mu\text{g})^2/\text{Hz}]$	0.30	0.33	0.25
	c.o.v. (%)	16	9	8
Prediction error PSD S_e	Exact $[(\mu\text{g})^2/\text{Hz}]$	1	1	1
	MPV $[(\mu\text{g})^2/\text{Hz}]$	1.1	1.3	1.6
	c.o.v. (%)	12	7	5
Mode shape ϕ	c.o.v. (%)	0.67	0.35	0.43
Angle between exact and MPV mode shape	(%)	0.55	0.31	0.25

modes can be calculated similarly. The MPVs are close to the exact values in a manner consistent with the posterior c.o.v.s (coefficient of variation = standard deviation/MPV). The exact values are generally within two standard deviations from the MPVs. The MPV of the prediction error PSD appears to be biased high as the mode number increases. This is because the frequency band of the higher modes contains unmodeled pseudo-static contribution from the lower modes.

The most probable mode shapes (omitted here) are very close to the exact mode shapes. In Table 12.7, the mode shape c.o.v. is equal to the square root sum of eigenvalues of the mode shape covariance matrix (Sect. 11.3.3). It is of the same order of magnitude as the hyper-angle between the exact and most probable mode shape.

Table 12.8 shows the posterior correlation coefficients among f , ζ , S and S_e . Correlation is generally small, with the only exception between ζ and S , of about 66%. This is no coincidence and can be explained based on uncertainty laws (Chap. 15).

Posterior marginal PDF

Figure 12.3 shows the posterior marginal PDFs of modal parameters. The exact marginal PDF (solid line) was obtained by numerically integrating the likelihood

Table 12.8 Posterior correlation coefficient (%), ten-storied building (synthetic data)

	f	ζ	S	S_e
f	100	2.1	4.8	0.29
ζ		100	66	−1.65
S			100	−2.94
S_e	Sym.			100

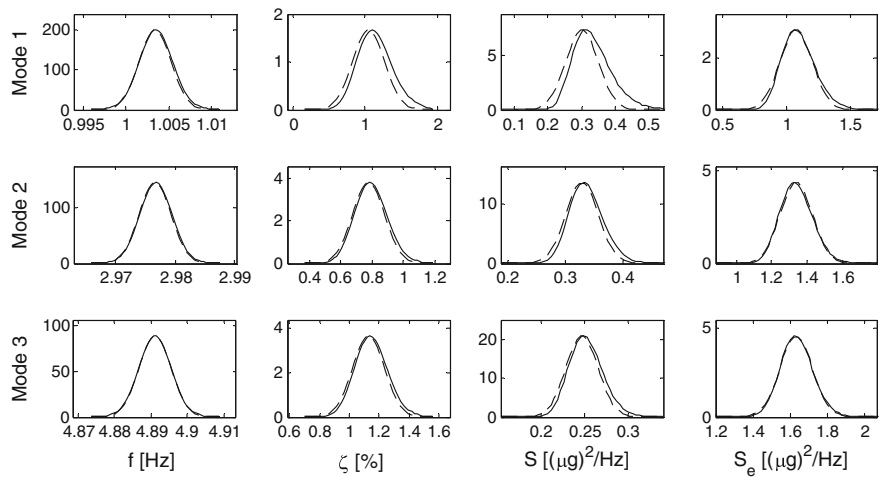


Fig. 12.3 Posterior marginal PDF of modal parameters. *Solid line* exact by numerical integration; *dashed line* Gaussian approximation

function w.r.t. the remaining parameters and normalizing so that its area is 1. The approximate PDF (dashed line) is a Gaussian PDF centered at the MPV and with a standard deviation obtained from the posterior covariance matrix. Obtaining the exact marginal PDF is computationally expensive. It is only done here for comparison. The figure shows that the Gaussian approximation is generally good.

Effect of Data Duration

Modal identification results depend on data, and clearly its duration. Figure 12.4 shows the results for data durations of 600, 1200, 2400 and 4800 s. The MPV (dot) need not coincide with the exact value but the latter is generally within the $\pm$ two standard deviation error bar. As data duration increases, the MPV converges (in a random manner) to the exact value and the error bar shortens. In real applications the exact value (dashed line) is unknown (or does not even exist) and so identification precision is assessed primarily based on the error bars.

Effect of Signal-to-Noise Ratio

Identification uncertainty depends on data noise intensity. Figure 12.5 shows the posterior c.o.v. of modal parameters of the first mode when the data is contaminated with measurement noise of PSD S_e , ranging from $100 (\mu\text{g})^2/\text{Hz}$ to $0.01 (\mu\text{g})^2/\text{Hz}$. The results are plotted w.r.t. the ‘modal s/n ratio’ (Sect. 7.7.3)

$$\gamma = \frac{S}{4S_e\zeta^2} \quad (12.36)$$

calculated using the MPVs determined in each case. For reference, in the nominal case, $S_e = 1 (\mu\text{g})^2/\text{Hz}$ and $\gamma \approx 700$. The figure shows that the posterior c.o.v.

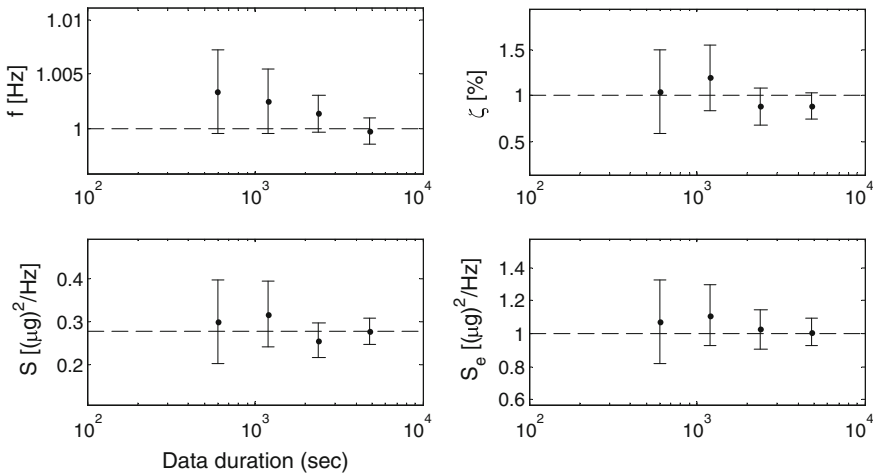


Fig. 12.4 Modal identification results for different data durations. *Dot* MPV; *error bar* $\pm$ two standard deviations. *Dashed line* exact value

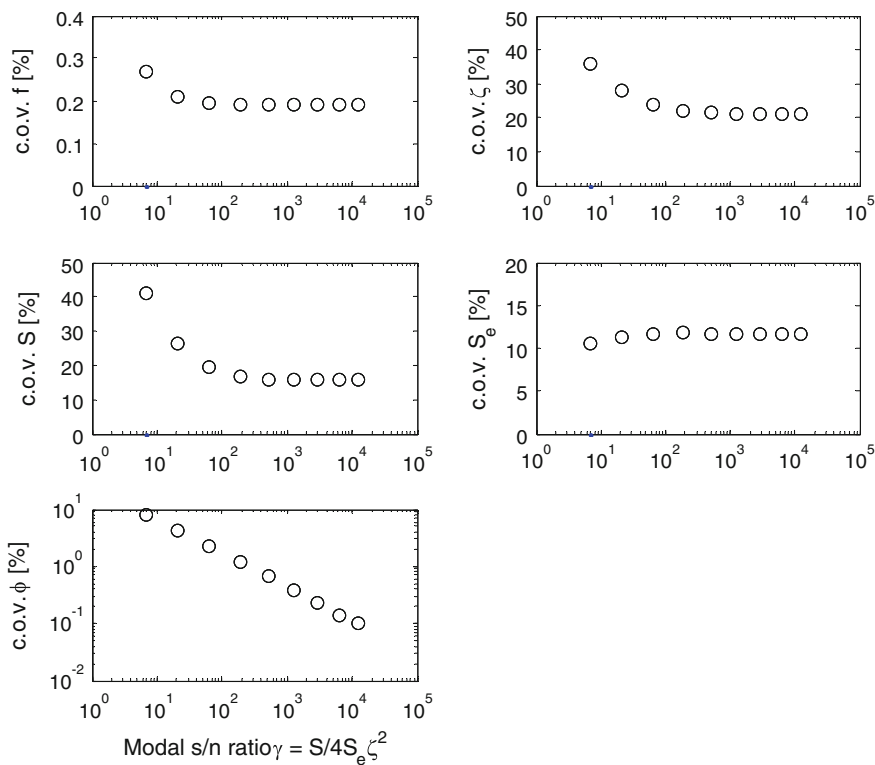


Fig. 12.5 Posterior c.o.v. of modal parameters (first mode) for different S_e

decreases significantly with γ when γ is small (say, <100). When γ is large the posterior c.o.v. settles at a non-zero value. Intuitively, even when there is no measurement noise, there is still uncertainty in the modal parameters because the input excitation is unknown. ■

Example 12.2 (Modeling error from unmodeled modes) Consider synthetic data generated at 100 Hz for 1200 s according to

$$\hat{\mathbf{y}}_j = \boldsymbol{\Phi}_1 \ddot{\eta}_1(t_j) + \boldsymbol{\Phi}_2 \ddot{\eta}_2(t_j) + \boldsymbol{\varepsilon}(t_j) \quad (12.37)$$

where $\ddot{\eta}_i(t)$ ($i = 1, 2$) is the modal acceleration satisfying

$$\ddot{\eta}_i(t) + 2\zeta_i \omega_i \dot{\eta}_i(t) + \omega_i^2 \eta_i(t) = p_i(t) \quad (12.38)$$

with $\omega_i = 2\pi f_i$ (rad/s); $f_1 = 1$ Hz, $f_2 = 1.5$ Hz, $\zeta_1 = \zeta_2 = 1\%$;

$$\boldsymbol{\varphi}_1 = [1 \quad 2 \quad 2]^T/3 \quad \boldsymbol{\varphi}_2 = [-2 \quad 1 \quad 2]^T/3 \quad (12.39)$$

and $p_i(t)$ is the modal excitation, stationary Gaussian with PSD S_i ; $S_1 = 1(\mu\text{g})^2/\text{Hz}$; S_2 will be specified later. The prediction error $\boldsymbol{\varepsilon}(t)$ is i.i.d. Gaussian white noise with PSD $S_e = 1(\mu\text{g})^2/\text{Hz}$. The mode at 1 Hz is identified using FFT in the band $[0.95, 1.00]$ Hz and assuming single mode. Here we investigate the effect of modeling error due to the unmodeled mode at 1.5 Hz.

When $S_2 = 0$ the second mode is absent and there is no modeling error. The PSD and SV spectrum of data in this case are shown in Fig. 12.6. The lowest two lines in (b) reflect data noise PSD. When $S_2 > 0$, the contribution of the second mode in the selected band induces modeling error. Figure 12.7 shows the root PSD and SV spectrum when $S_2 = 300(\mu\text{g})^2/\text{Hz}$. As seen in (b), the presence of the second mode increases the second largest singular value. This appears like an increase in the noise PSD in the selected band.

Figure 12.8 shows the modal identification results for different values of S_2 up to $300(\mu\text{g})^2/\text{Hz}$. The results are plotted w.r.t. the ‘modeling error ratio’, defined as the PSD ratio of the second to the first mode at 1 Hz. When the ratio is 10^{-5} the results essentially correspond to $S_2 = 0$. Even in this case the MPV need not be equal to the exact value because the data duration is finite. Departure of identification results from this nominal (‘model error free’) case reflects modeling error. For $\{f_1, \zeta_1, S_1\}$, there is a slight departure in MPV from the nominal value but it is still small compared to identification uncertainty (error bar). For S_e , when the modeling error ratio is small, (say $< 10^{-3}$), the identification result reflects the PSD of data noise ($1(\mu\text{g})^2/\text{Hz}$). For larger ratios it reflects the PSD contribution of the second mode. In all cases the identified mode shapes (not shown) are close to the exact ones, in the worst case with a c.o.v. of about 10%. ■

Example 12.3 (Bandwidth trade-off) Consider Example 12.2 again with the largest modeling error ratio, i.e., $S_2 = 300(\mu\text{g})^2/\text{Hz}$. It was demonstrated that the

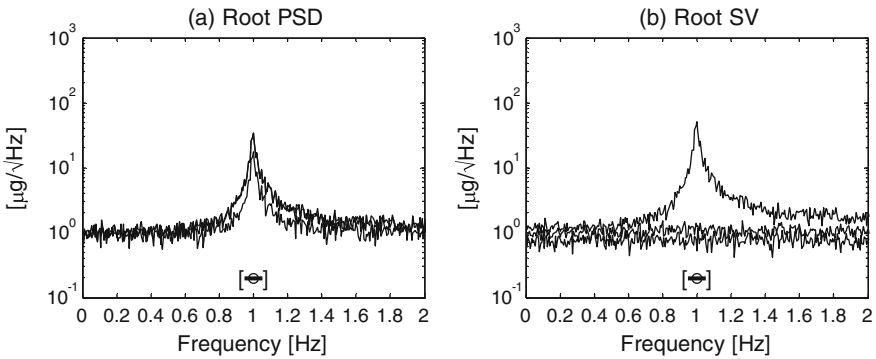


Fig. 12.6 **a** Root PSD and **b** root SV spectrum when $S_2 = 0$

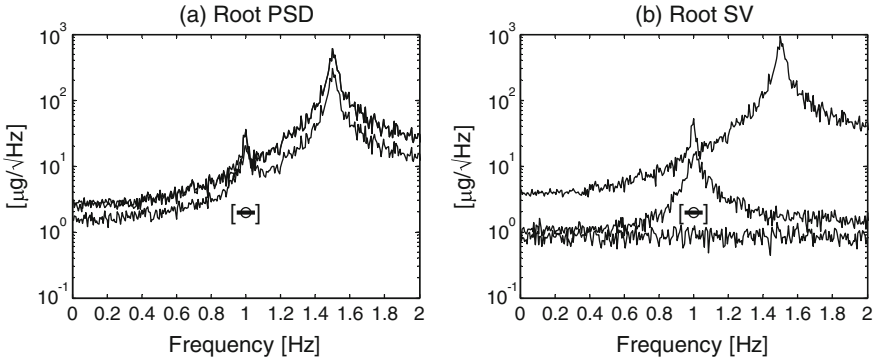


Fig. 12.7 **a** Root PSD and **b** root SV spectrum when $S_2 = 300(\mu\text{g})^2/\text{Hz}$

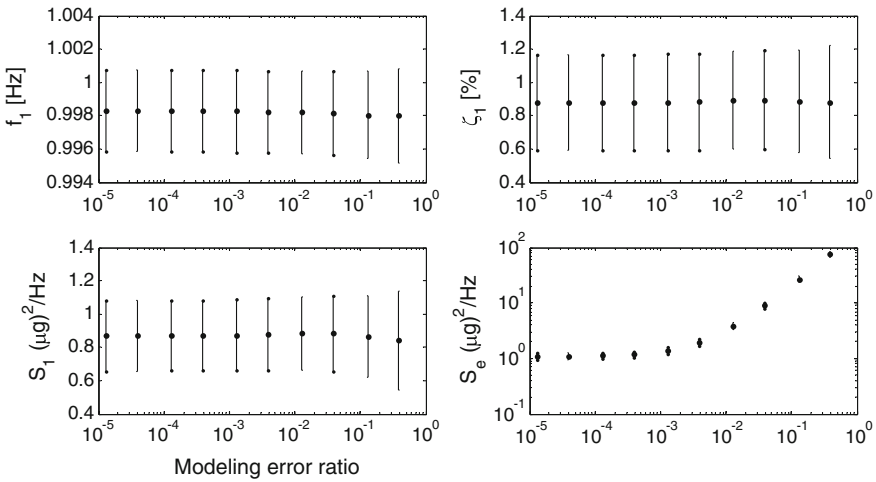


Fig. 12.8 Modal identification results for different values of S_2 , plotted versus modeling error ratio, defined as the PSD ratio of the second to the first mode at 1 Hz. *Dot* MPV; *error bar* $\pm$ two standard deviations

effect of the unmodeled second mode on the identification results of $\{f_1, \zeta_1, S_1\}$ was insignificant. This example further illustrates that this in fact depends on the selected bandwidth, which in general should play a balance between identification precision (the wider the better) and modeling error (the narrower the better).

Figure 12.9 shows the modal identification results for different selected bands, parameterized as $(1 \pm 0.01\kappa)$ where κ is a ‘bandwidth factor’. For reference, the selected band in the last example was $[0.95, 1.05]$, i.e., $\kappa = 5$. As seen in the figure, as κ increases from small values, identification uncertainty (error bar) decreases, reflecting the positive effect of increasing FFT data size. Identification uncertainty is

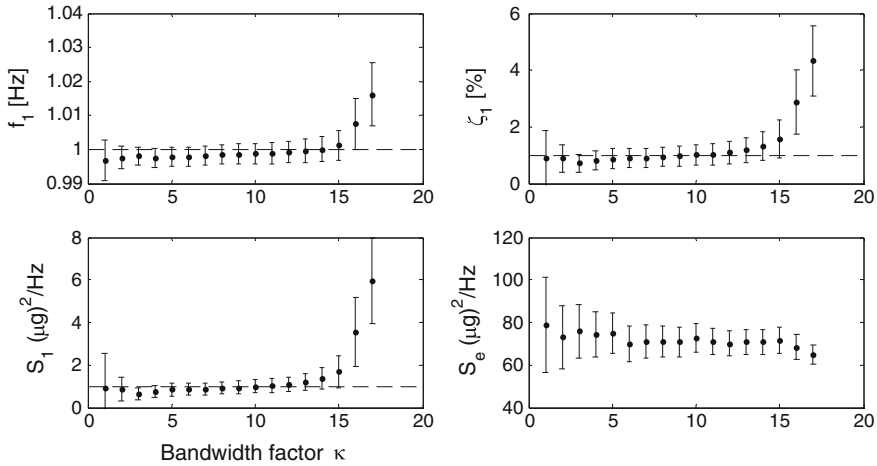


Fig. 12.9 Modal identification results with different selected bands ($1 \pm 0.01\kappa$)

insensitive to κ when $5 < \kappa < 10$, where it is trading off with modeling error. Bias is not significant until $\kappa > 15$, as reflected by the error bar not covering the exact value (dashed line). Thus, when the selected band is narrow, modeling error is small but identification uncertainty could be unacceptably large. When the band is too wide, results can be significantly biased due to modeling error. The band should trade off these two factors. Here a value between 5 and 10 is appropriate. ■

12.6 Laboratory/Field Data Examples

In this section we present examples on Bayesian OMA with well-separated modes based on experimental data measured under laboratory or field environment. They illustrate the typical results one may obtain in reality and how to interpret them, taking into account potential modeling errors.

Example 12.4 (Laboratory shear frame) Consider the three-storied laboratory shear frame in Fig. 12.10. The four corners of each floor are instrumented with two piezoelectric accelerometers along the horizontal x- and y-directions. This gives a total of $3 \times 4 \times 2 = 24$ measured DOFs. Ambient data was recorded for 600 s at 2048 Hz. It was later decimated to 512 Hz for analysis.

Data and spectra

Figure 12.11 shows the measured time histories at DOFs 1 and 2. Figure 12.12 shows the root PSD and SV spectrum. The spectra indicate a noise PSD in the order

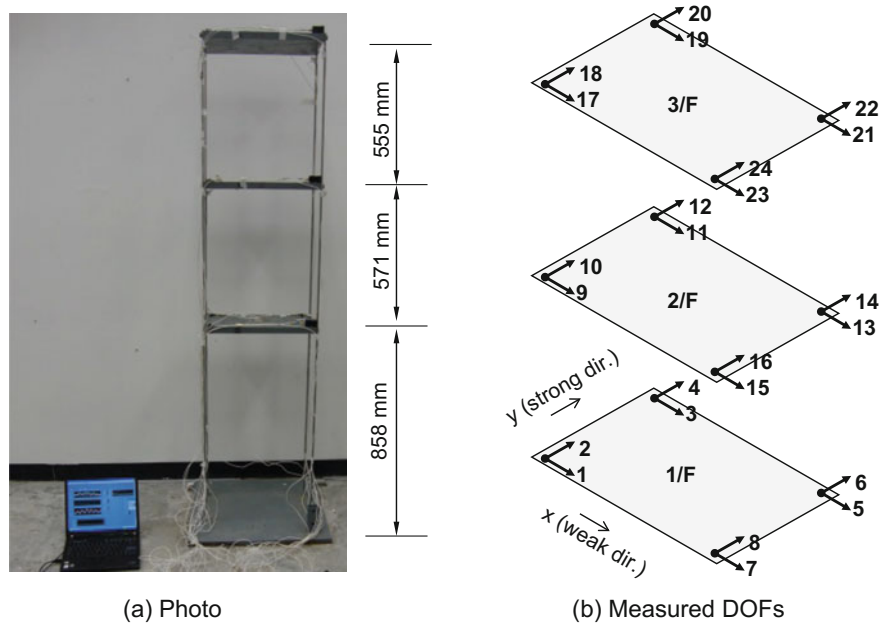


Fig. 12.10 Laboratory shear frame; **a** photo; **b** measured DOFs

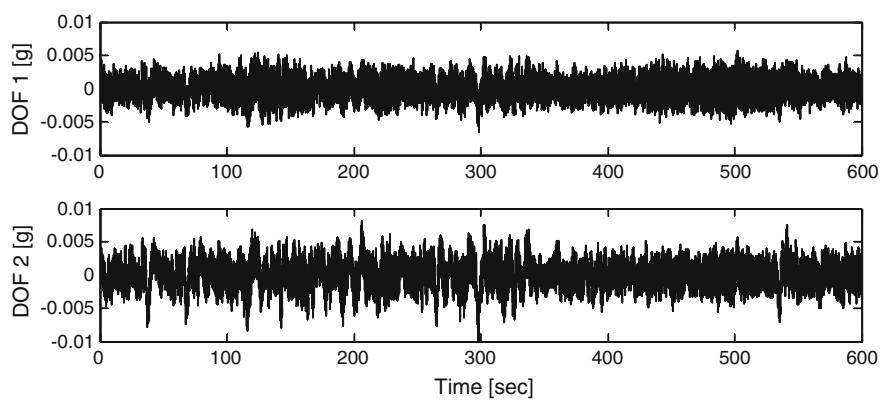


Fig. 12.11 Measured time histories at DOFs 1 and 2, laboratory frame (detrended)

of $100 \mu\text{g}/\sqrt{\text{Hz}}$ to $20 \mu\text{g}/\sqrt{\text{Hz}}$ from low to high frequencies. The peaks of the top line in (b) indicate potential modes. Ten well-separated modes can be easily recognized.

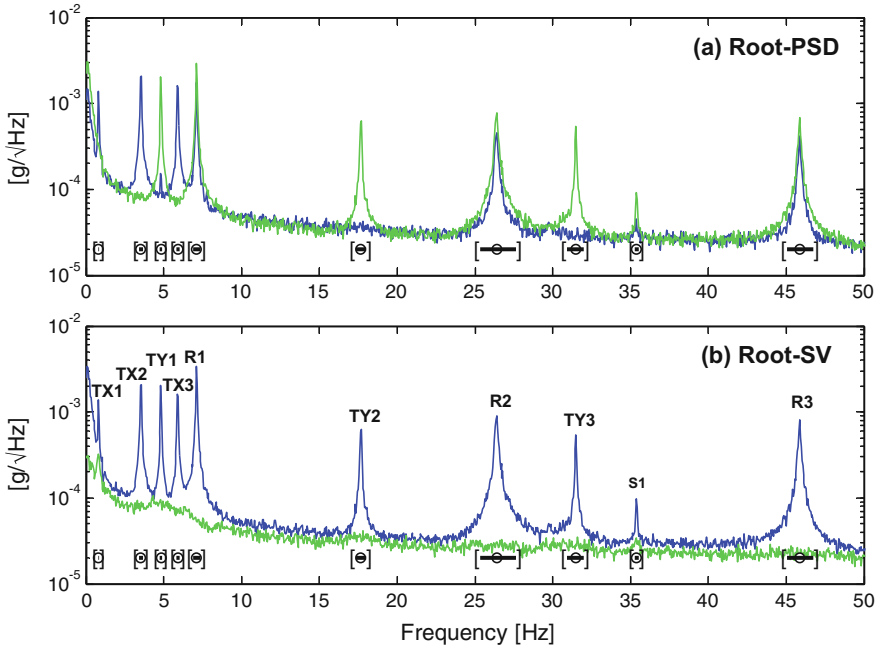


Fig. 12.12 One-sided Root PSD **a** and root SV **b** spectrum of data at DOFs 1 and 2 of laboratory frame. *Horizontal bar* indicates selected frequency band

Modal Identification Using DOFs 1 and 2

The measured data at DOFs 1 and 2 are already sufficient for reliable modal identification. The horizontal bars below the peaks in Fig. 12.12b indicate the bands selected for modal identification. They are hand-picked by zooming in the individual bands. The identification results are summarized in Table 12.9. The c.o.v. of frequencies are all below 0.1%. The c.o.v. of damping ratio ranges from a few to fifty percent. The c.o.v. of modal force PSD is of a similar order as the c.o.v. of damping ratio. The MPV of prediction error PSD decreases with the mode number, which is consistent with the spectra in Fig. 12.12. The modal s/n ratio $\gamma = S/4S_e\zeta^2$ is defined as before in (12.36), calculated using the MPVs in each case. It is generally quite high, ranging from a few hundreds to a few thousands.

Modal Identification Using All Measured DOFs

Identifying modes using more DOFs allows one to examine their nature through the mode shapes. The results using all the 24 measured DOFs are summarized in Table 12.10. The increase in the number of measured DOFs directly increases the modal force PSD S and modal s/n ratio $\gamma = S/4S_e\zeta^2$. Despite the substantial increase in γ , there is little difference in the MPV and c.o.v. of the natural frequency and damping ratio.

Table 12.9 Modal identification results using DOFs 1 and 2, laboratory frame

Mode	s/n ratio	Frequency		Damping		Root modal force PSD		Root prediction error PSD		Mode shape
	γ	f		ζ		$\sqrt{S}$		$\sqrt{S_e}$		Φ
		MPV	c.o.v.	MPV	c.o.v.	MPV	c.o.v.	MPV	c.o.v.	c.o.v.
		(Hz)	(%)	(%)	(%)	$\mu\text{g}/\sqrt{\text{Hz}}$	(%)	$\mu\text{g}/\sqrt{\text{Hz}}$	(%)	(%)
1	335	0.81	0.10	0.26	47	24	17	250	12	2.07
2	7647	3.52	0.03	0.13	26	15	4	69	4	0.28
3	6173	4.82	0.02	0.06	31	7.3	5	72	5	0.38
4	3293	5.90	0.02	0.12	22	8.5	5	62	4	0.35
5	9496	7.11	0.03	0.16	17	17	3	52	3	0.16
6	650	17.7	0.01	0.12	13	2.1	3	35	2	0.45
7	1100	26.4	0.02	0.25	7	4.7	2	28	1	0.20
8	519	31.5	0.01	0.08	12	1.0	3	28	2	0.46
9	14	35.4	0.01	0.06	19	0.1	10	32	4	3.49
10	1156	45.9	0.01	0.12	8	1.9	2	23	2	0.21

Table 12.10 Modal identification results using all 24 measured DOFs, laboratory frame

Mode	s/n ratio	Frequency		Damping		Root modal force PSD		Root prediction error PSD		Mode shape
	γ	f		ζ		$\sqrt{S}$		$\sqrt{S_e}$		Φ
		MPV	c.o.v.	MPV	c.o.v.	MPV	c.o.v.	MPV	c.o.v.	c.o.v.
		(Hz)	(%)	(%)	(%)	$\mu\text{g}/\sqrt{\text{Hz}}$	(%)	$\mu\text{g}/\sqrt{\text{Hz}}$	(%)	(%)
1	631	0.81	0.10	0.25	47	91	16	731	3	7.33
2	34,960	3.52	0.03	0.12	26	40	4	86	1	0.64
3	68,359	4.82	0.02	0.07	30	32	5	90	1	0.53
4	47,781	5.90	0.02	0.12	22	43	5	84	1	0.44
5	89,022	7.11	0.03	0.16	17	68	3	70	1	0.25
6	8200	17.7	0.01	0.13	12	5.8	3	26	1	0.59
7	13,376	26.4	0.02	0.25	7	12	1	21	0.3	0.27
8	19,308	31.5	0.01	0.08	11	4.8	2	22	0.4	0.36
9	502	35.4	0.01	0.07	14	0.6	5	20	1	2.34
10	48,835	45.9	0.01	0.12	8	8.5	2	16	0.3	0.15

Figure 12.13 shows the identified mode shapes (MPV) with their nature indicated, e.g., TX2 for the second translational mode along the x direction; R1 for the first torsional mode. Based on lumped mass structural dynamics, the frame theoretically has nine modes, comprising three translational modes in the x and the y direction, and three rotational modes. Figure 12.13 contains an additional mode S1 whose frequency and mode shape lie between R2 and R3. ■

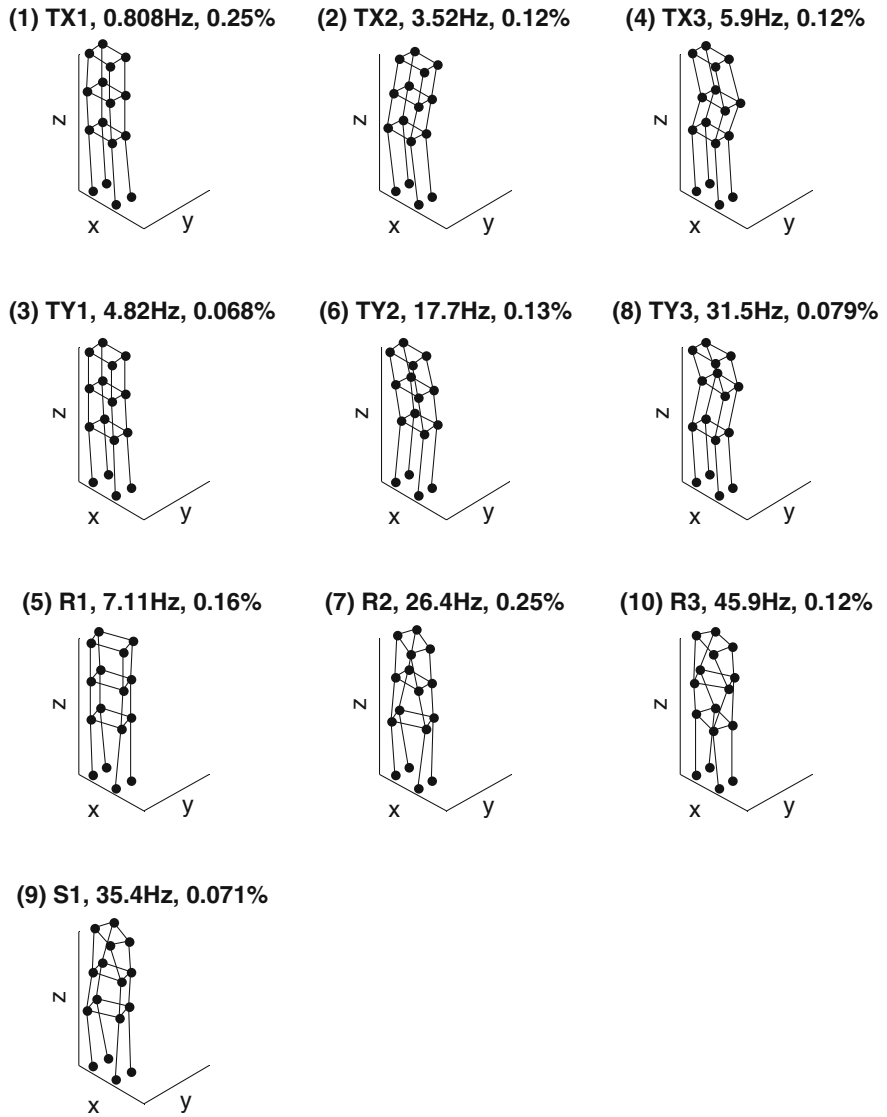


Fig. 12.13 Mode shapes (MPV) identified using all 24 measured DOFs. Title above each plot shows the mode number, nature, frequency and damping ratio (MPV)

Example 12.5 (Tall building) Consider a tall building measuring 50 m × 50 m in plan and over 300 m in height. Accelerations along two horizontal (x and y) directions at four corners on the roof were measured for 1800 s under normal wind situation in the late afternoon. Figure 12.14 shows the measured time histories at one corner. The root PSD and SV spectrum are shown in Fig. 12.15. The SV

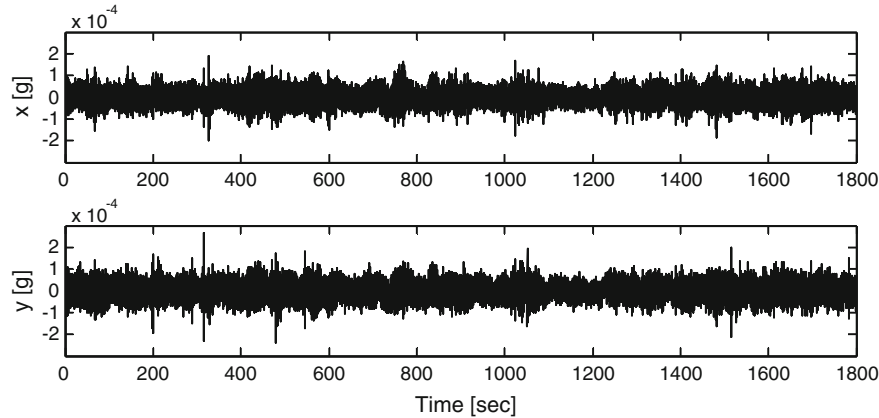


Fig. 12.14 Biaxial (x and y) measured acceleration time histories (detrended) at one corner on the roof of a tall building

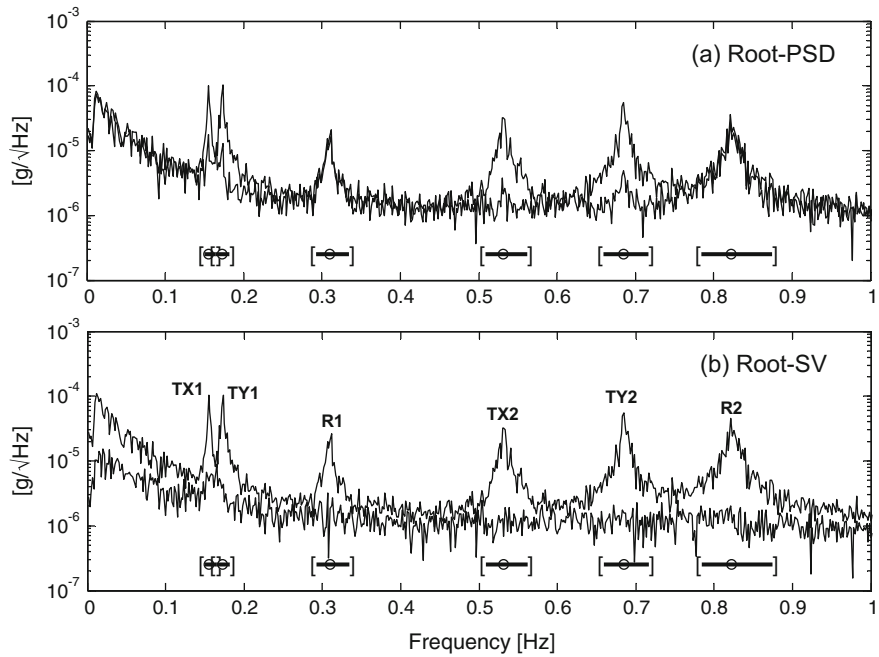


Fig. 12.15 One-sided root PSD (a) and root SV spectrum (b) of biaxial (x and y) data measured at one corner on the roof of a tall building. *Horizontal bar* indicates selected band

spectrum suggests six potential modes in the bands shown. The nature of each mode is determined from the mode shape found later.

Modal Identification

The modal identification results using the biaxial data at one corner on the roof are shown in Table 12.11. The MPV of $\sqrt{S_e}$ is a few $\mu\text{g}/\sqrt{\text{Hz}}$ and tends to decrease with the mode number. This is consistent with the spectra in Fig. 12.15. The modal s/n ratio ranges from a few hundreds to a thousand. The modal force has a root PSD of about $1\ \mu\text{g}/\sqrt{\text{Hz}}$. The c.o.v. of damping ratio tends to decrease with the mode number. This is partly because the effective data length as a multiple of natural period is longer for higher modes with higher frequencies.

Table 12.12 shows the modal identification results using data at all four corners (8 DOFs). Comparing with Table 12.11, comments similar to Example 12.4 can be made. Using all measured DOFs directly increases the modal force PSD and modal

Table 12.11 Modal identification results using biaxial data at one corner on the roof of tall building

Mode	s/n ratio	Frequency		Damping		Root modal force PSD		Root prediction error PSD		Mode shape
	γ	f		ζ		$\sqrt{S}$		$\sqrt{S_e}$		Φ
		MPV	c.o.v.	MPV	c.o.v.	MPV	c.o.v.	MPV	c.o.v.	c.o.v.
		(Hz)	(%)	(%)	(%)	$\mu\text{g}/\sqrt{\text{Hz}}$	(%)	$\mu\text{g}/\sqrt{\text{Hz}}$	(%)	(%)
1	650	0.155	0.11	0.21	66	0.72	13	6.6	10	2.13
2	689	0.173	0.16	0.43	40	1.06	11	4.7	9	1.35
3	233	0.309	0.16	0.74	24	0.37	8	1.6	6	1.35
4	556	0.532	0.10	0.54	21	0.34	6	1.3	5	0.78
5	1226	0.685	0.07	0.36	22	0.36	6	1.4	5	0.56
6	636	0.823	0.08	0.56	17	0.39	5	1.4	4	0.57

Table 12.12 Modal identification results using data at four corners on the roof of tall building

Mode	s/n ratio	Frequency		Damping		Root modal force PSD		Root prediction error PSD		Mode shape
	γ	f		ζ		$\sqrt{S}$		$\sqrt{S_e}$		Φ
		MPV	c.o.v.	MPV	c.o.v.	MPV	c.o.v.	MPV	c.o.v.	c.o.v.
		(Hz)	(%)	(%)	(%)	$\mu\text{g}/\sqrt{\text{Hz}}$	(%)	$\mu\text{g}/\sqrt{\text{Hz}}$	(%)	(%)
1	1400	0.155	0.11	0.22	61	1.49	12	8.9	4	3.7
2	1214	0.173	0.16	0.46	39	2.17	10	6.8	3	2.6
3	235	0.309	0.16	0.68	25	0.66	8	3.2	2	3.7
4	706	0.532	0.10	0.52	21	0.68	6	2.5	2	1.9
5	1592	0.685	0.07	0.36	22	0.70	6	2.4	2	1.3
6	714	0.823	0.09	0.56	17	0.75	5	2.5	1	1.4

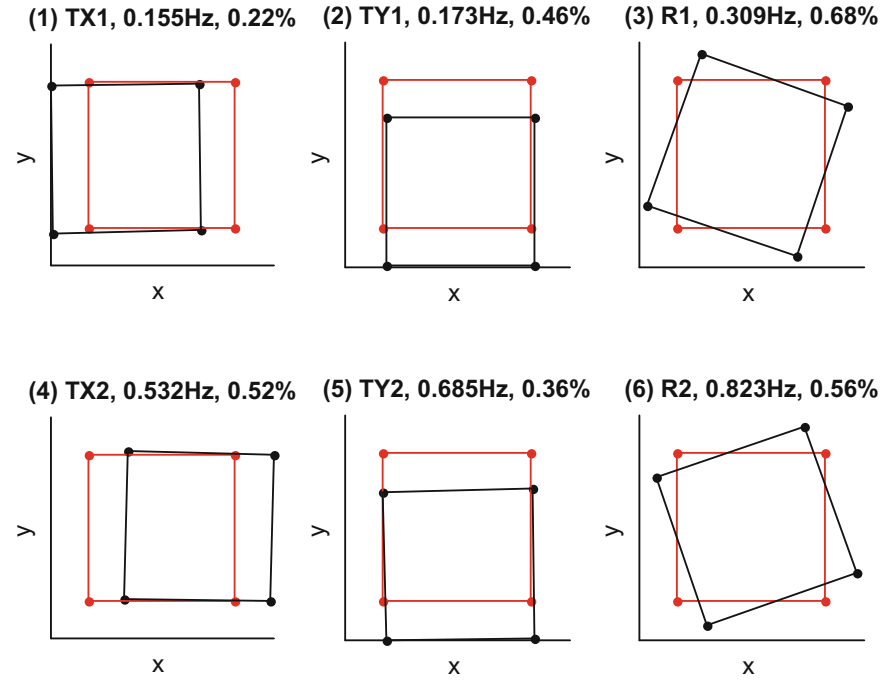


Fig. 12.16 Identified mode shapes (MPV) using data at four corners on the roof of tall building. Title above each plot shows the mode number, nature, frequency and damping ratio (MPV)

s/n ratio but it makes little difference in the MPV and posterior c.o.v.s. The root modal force PSD roughly doubles because the number of DOFs with similar mode shape values is quadrupled. Figure 12.16 shows the identified mode shapes (MPV). ■

Example 12.6 (Smart phone data, footbridge) This example demonstrates ambient modal identification using triaxial acceleration data recorded on a smart phone. As shown in Fig. 12.17, the phone was placed on one side of the bridge deck at 1/4 span of the Queen’s Park Bridge in Chester, UK. Figure 12.18 shows the measured time history collected for 300 s at 50 Hz. Figure 12.19 shows the root PSD and SV spectrum of data. The spectra suggest a noise PSD of a few hundreds of $\mu\text{g}/\sqrt{\text{Hz}}$, consistent with Fig. 6.2 in Sect. 6.3. The peaks suggest potential modes between 0.5 and 5 Hz. The band around 0.8 Hz is likely to contain a single mode. The situation is less clear with other bands, which can possibly contain close modes. This can usually be resolved using sensors from multiple locations.

Modal Identification

Table 12.13 shows the modal identification results. The most probable mode shapes are shown in Table 12.14, where their likely nature are also indicated. These results should be interpreted with a level of confidence consistent with the quality of data,

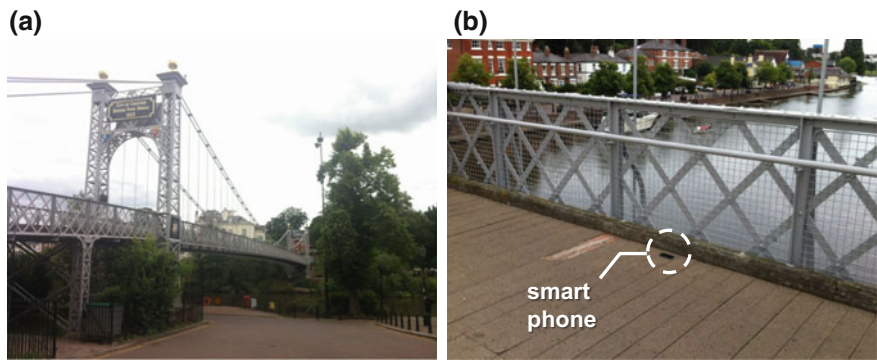


Fig. 12.17 **a** Queen's Park Bridge and **b** location of sensor (smart phone)

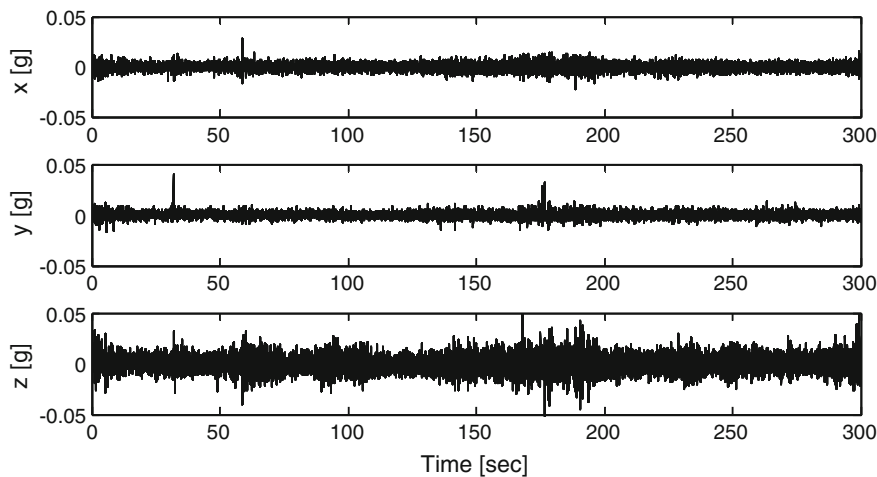


Fig. 12.18 Triaxial data (detrended) at 1/4 span of Queen's Park Bridge. The x, y and z direction correspond to the transverse, longitudinal and vertical direction of the bridge

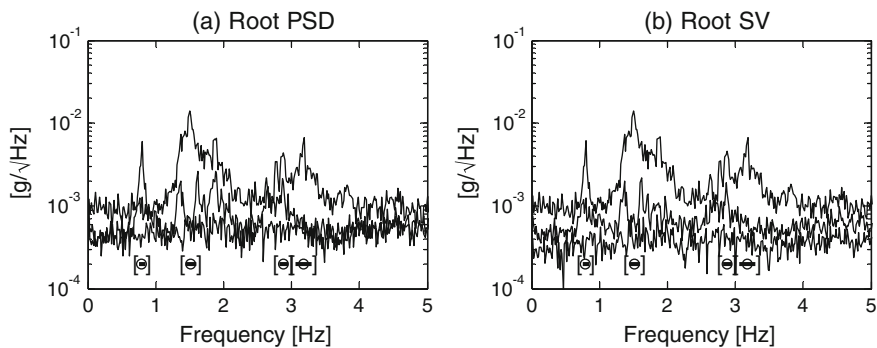


Fig. 12.19 One-sided root PSD **(a)** and SV **(b)** spectrum of triaxial data, Queen's Park Bridge. Horizontal bar indicates selected band

Table 12.13 Modal identification results, triaxial smart phone data, Queen’s Park Bridge

Mode	s/n ratio	Frequency		Damping		Root modal force PSD		Root prediction error PSD		Mode shape
	γ	f		ζ		$\sqrt{S}$		$\sqrt{S_e}$		Φ
		MPV	c.o.v.	MPV	c.o.v.	MPV	c.o.v.	MPV	c.o.v.	c.o.v.
		(Hz)	(%)	(%)	(%)	$\mu\text{g}/\sqrt{\text{Hz}}$	(%)	$\mu\text{g}/\sqrt{\text{Hz}}$	(%)	(%)
1	77	0.794	0.44	1.7	34	193	16	659	6.7	3.6
2	415	1.501	0.43	2.4	26	637	15	655	5.1	1.0
3	25	2.872	0.19	0.9	30	77	17	909	5.6	5.0
4	92	3.163	0.24	1.6	22	168	13	545	4.2	1.8

Table 12.14 Most probable mode shapes, triaxial smart phone data, Queen’s Park Bridge

Mode	Nature	X	Y	Z
1	Transverse	−0.9972	−0.0340	0.0671
2	Vertical	−0.0117	0.0243	0.9996
3	Torsional	−0.2482	0.0266	0.9684
4	Vertical	−0.0315	−0.0335	0.9989

which is much lower than the previous two examples. For reference, the quantization resolution is about 0.001 g (Example 6.1) and its contribution to the root PSD of noise is about 60 $\mu\text{g}/\sqrt{\text{Hz}}$ (Sect. 6.6.2). The overall noise PSD is a few hundreds of $\mu\text{g}/\sqrt{\text{Hz}}$; see Fig. 12.19 or Table 12.13 ($\sqrt{S_e}$). The identification results assume a single mode dominating the selected band, which is debatable in the current case. The Bayesian algorithm always returns results as long as a converged solution in the MPV can be obtained. The posterior c.o.v. need not account for modeling error. ■

References

Au SK (2011) Fast Bayesian FFT method for ambient modal identification with separated modes. J Eng Mech ASCE 137(3):214–226
Zhang FL, Au SK (2013) Erratum for fast Bayesian FFT method for ambient modal identification with separated modes. J Eng Mech ASCE 139(4):545–545

Chapter 13

Multi-mode Problem

Abstract This chapter develops the computational algorithm for determining the most probable value (MPV) and covariance matrix of modal parameters in Bayesian operational modal analysis with data in a single setup and for the general case of multiple (possibly close) modes. The partial mode shapes, i.e., confined to the measured degrees of freedom, are not necessarily orthogonal and this presents difficulty in their efficient identification. Representing them via an orthogonal basis of a ‘mode shape subspace’ allows efficient determination in terms of such basis and the associated coordinates. An iterative algorithm is presented for determining the MPV, where the MPVs of different groups of parameters are updated until convergence. The asymptotic behavior of the MPV for modes with high signal-to-noise ratio is analyzed. Analytical formulas are derived for systematically computing the covariance matrix of parameters given the measured data. Examples with synthetic data and field data are presented to illustrate the algorithms and their applications.

Keywords Multi-mode • Mode shape subspace • Tall building • Signal-to-noise ratio

In this chapter we discuss efficient methods for calculating the posterior most probable value (MPV) and covariance matrix of modal parameters in Bayesian operational modal analysis (OMA) using data collected in a single setup and considering the general case of multiple (possibly close) modes in the selected frequency band. Mode shapes present the major hurdle as they contain a large number of parameters. Confined to the measured DOFs, they are not necessarily orthogonal. One strategy is to represent them via a lower dimensional orthonormal basis and then develop techniques for determining the MPV of the basis and associated coordinates. This leads to an iterative algorithm where different groups of parameters are updated sequentially until convergence.

Due to tangling of modes, the considerations and algorithms in this chapter are much more complicated than those for well-separated modes in Chap. 12. It is based on Au (2012a, b). The presentation has been simplified and reorganized.

Computer programming of the posterior covariance matrix using the analytical formulas in Sect. 13.7 is simpler as it has taken advantage of a more systematic procedure based on Lagrange multiplier concepts (Sect. 11.2.3).

Recall the problem context from Sect. 10.1 where the measured ambient vibration history is denoted by $\{\hat{\mathbf{y}}_j\}_{j=0}^{N-1}$ ($n \times 1$); n is the number of measured DOFs and N is the number of samples per data channel. The scaled Fast Fourier Transform (FFT) of $\{\hat{\mathbf{y}}_j\}_{j=0}^{N-1}$ at frequency $f_k = k/N\Delta t$ (Hz) is

$$\hat{\mathcal{F}}_k = \sqrt{\frac{\Delta t}{N}} \sum_{j=0}^{N-1} \hat{\mathbf{y}}_j e^{-2\pi i j k / N} \quad (13.1)$$

where Δt (s) is the sampling interval. The set of modal parameters $\boldsymbol{\theta}$ comprises

$$\begin{array}{llll} \{f_i\}_{i=1}^m & \{\zeta_i\}_{i=1}^m & \mathbf{S} & S_e \\ \text{Natural frequencies (Hz)} & \text{Damping ratios} & m \times m \text{ Hermitian modal force PSD} & \text{Prediction error PSD} \\ \{\boldsymbol{\phi}_i\}_{i=1}^m & & & \\ n \times 1 \text{ Mode shapes} & & & \end{array}$$

The mode shapes are subjected to unit norm constraint, $\boldsymbol{\phi}_i^T \boldsymbol{\phi}_i = 1$ ($i = 1, \dots, m$). The modal force PSD has the same unit as the PSD of acceleration. The prediction error PSD has the same unit as the PSD of data.

Using the scaled FFT $\{\hat{\mathcal{F}}_k\}$ in a selected frequency band, the negative log-likelihood function (NLLF) is given by:

$$L(\boldsymbol{\theta}) = nN_f \ln \pi + \sum_k \ln |\mathbf{E}_k(\boldsymbol{\theta})| + \sum_k \hat{\mathcal{F}}_k^* \mathbf{E}_k(\boldsymbol{\theta})^{-1} \hat{\mathcal{F}}_k \quad (13.2)$$

where N_f is the number of FFT points in the selected band;

$$\mathbf{E}_k = \underset{n \times n}{\boldsymbol{\Phi}} \underset{n \times m}{\mathbf{H}_k} \underset{m \times m}{\boldsymbol{\Phi}^T} + \underset{n \times n}{S_e} \mathbf{I}_n \quad (13.3)$$

is the theoretical PSD matrix of measured data for given $\boldsymbol{\theta}$; $\mathbf{I}_n$ denotes the $n \times n$ identity matrix; $\boldsymbol{\Phi} = [\boldsymbol{\phi}_1, \dots, \boldsymbol{\phi}_m]$ ($n \times m$) is the (partial) mode shape matrix;

$$\underset{m \times m}{\mathbf{H}_k} = \underset{m \times m}{\text{diag}(\mathbf{h}_k)} \underset{m \times m}{\mathbf{S}} \underset{m \times m}{\text{diag}(\mathbf{h}_k^*)} \quad \mathbf{h}_k = [h_{1k}, \dots, h_{mk}] \quad (13.4)$$

$$h_{ik} = \frac{(2\pi i f_k)^{-q}}{1 - \beta_{ik}^2 - 2\zeta_i \beta_{ik} i} \quad \beta_{ik} = \frac{f_i}{f_k} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (13.5)$$

13.1 Mode Shape Subspace

The most probable mode shapes can be determined efficiently by separately identifying the subspace they lie in and the associated coordinates. The ‘mode shape subspace’ is the subspace in R^n (the n -dimensional Euclidean space) spanned by $\{\boldsymbol{\phi}_i\}_{i=1}^m$. Its dimension, denoted by m' , is equal to the rank of $\boldsymbol{\Phi} = [\boldsymbol{\phi}_1, \dots, \boldsymbol{\phi}_m]$. Elementary linear algebra shows that $m' \leq \min(n, m)$. Table 13.1 lists the possible ordering of m' , m and n . The most typical case is $m' = m < n$. It is possible for $m' < m$ when some mode shapes are linearly dependent. For example, when all DOFs measure the same quantity then trivially $\boldsymbol{\phi}_i = [1, \dots, 1]^T / \sqrt{n}$ for all modes and so $m' = 1$ regardless of n and m .

The mode shape dimension m' is part of the modeling assumptions. Assuming m' larger than the actual dimension increases computational effort. Assuming m' less than the actual dimension over-constrains the inference problem and leads to bias in results. A robust choice is $m' = m$. When it is justified to do so, assuming $m' < m$ may speed up convergence of the MPV. The number of lines (eigenvalues) significantly above the remaining ones in the singular value (SV) spectrum gives an indication of m' (Sect. 7.3).

Example 13.1 (Dimension of mode shape subspace) Consider identifying the horizontal modes of a building with sensors placed on the same floor. Figure 13.1 shows the dimension of the mode shape subspace (m') depending on the measured DOFs and selected frequency band. For reference, the mode shapes confined to two locations are shown at the bottom of the table. ■

13.1.1 Orthonormal Basis Representation

Let $\mathbf{B}' = [\mathbf{b}_1, \dots, \mathbf{b}_{m'}]$ ($n \times m'$) contain in its columns an orthonormal basis $\{\mathbf{b}_i\}_{i=1}^{m'}$ ($n \times 1$) in the mode shape subspace. That is, $\mathbf{b}_i^T \mathbf{b}_j = 1$ if $i = j$ and zero otherwise. The mode shape matrix $\boldsymbol{\Phi} = [\boldsymbol{\phi}_1, \dots, \boldsymbol{\phi}_m]$ ($n \times m$) can then be represented as

$$\boldsymbol{\Phi} = \mathbf{B}' \boldsymbol{\alpha} \quad (13.6)$$

where

$$\boldsymbol{\alpha} = \begin{bmatrix} \boldsymbol{\alpha}_1 & \dots & \boldsymbol{\alpha}_m \\ m' \times m & & m' \times 1 \end{bmatrix} \quad (13.7)$$

Table 13.1 Possible ordering of m' , m and n

Case	I	II ^a	III	IV	V	VI
Condition	$m' < m < n$	$m' = m < n$	$m' = m = n$	$m' < m = n$	$m' < n < m$	$m' = n < m$

^aMost typical

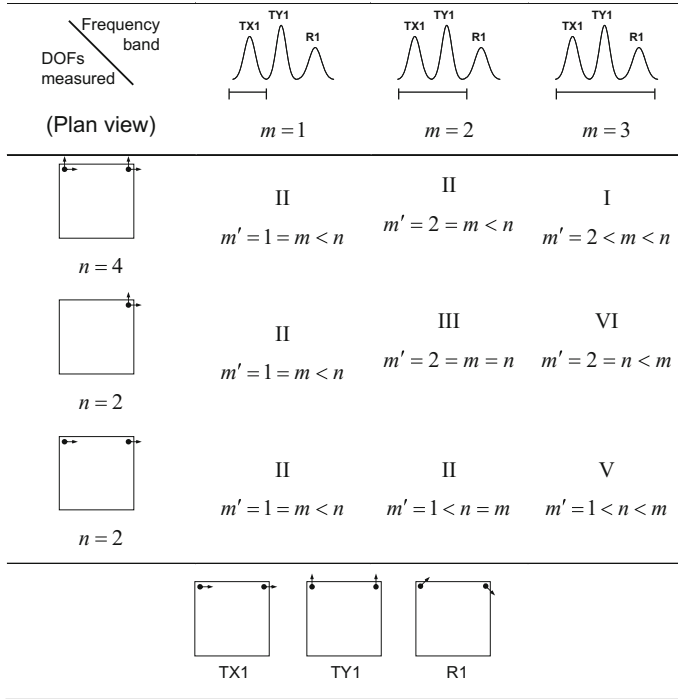


Fig. 13.1 Dimension of mode shape subspace in different situations

and α_i contains the coordinates of ϕ_i w.r.t. the basis $\mathbf{B}'$. Let $\mathbf{B}'_{\perp} = [\mathbf{b}_{m'+1}, \dots, \mathbf{b}_n]$ ($n \times (n - m')$) where $\{\mathbf{b}_i\}_{i=m'+1}^n$ ($n \times 1$) are orthonormal and are orthogonal to $\{\mathbf{b}_i\}_{i=1}^{m'}$. Combining $\mathbf{B}'$ and $\mathbf{B}'_{\perp}$ gives a matrix containing in its columns an orthonormal basis in R^n , $\mathbf{B} = [\mathbf{B}', \mathbf{B}'_{\perp}]$ ($n \times n$). Comparing (13.6) with the singular value decomposition $\Phi = \mathbf{L}\mathbf{\Lambda}\mathbf{R}^T$ shows that $\mathbf{B}' = \mathbf{L}$ and $\alpha = \mathbf{\Lambda}\mathbf{R}^T$.

13.2 Alternative Form of NLLF

Analogous to the case of single mode, the NLLF can be written in a form to facilitate determination of the most probable basis in terms of the remaining parameters. The key is to express $|\mathbf{E}_k|$ and $\mathbf{E}_k^{-1}$ via the eigenvector representation of $\mathbf{E}_k$. First, we write

$$\Phi = [\mathbf{B}', \mathbf{B}'_{\perp}] \begin{bmatrix} \boldsymbol{\alpha} \\ \mathbf{0} \end{bmatrix} = \mathbf{B} \begin{bmatrix} \boldsymbol{\alpha} \\ \mathbf{0} \end{bmatrix} \quad (13.8)$$

Substituting this and $\mathbf{I}_n = \mathbf{B}\mathbf{I}_n\mathbf{B}^T$ into (13.3),

$$\mathbf{E}_k = \mathbf{B} \begin{bmatrix} \boldsymbol{\alpha} \\ \mathbf{0} \end{bmatrix} \mathbf{H}_k \begin{bmatrix} \boldsymbol{\alpha} \\ \mathbf{0} \end{bmatrix}^T \mathbf{B}^T + S_e \mathbf{B} \mathbf{I}_n \mathbf{B}^T = \mathbf{B} \begin{bmatrix} \mathbf{E}'_k & \mathbf{0} \\ \mathbf{0} & S_e \mathbf{I}_{n-m'} \end{bmatrix} \mathbf{B}^T \quad (13.9)$$

where

$$\mathbf{E}'_k = \underset{m' \times m'}{\boldsymbol{\alpha}} \underset{m' \times m}{\mathbf{H}_k} \underset{m \times m'}{\boldsymbol{\alpha}^T} + S_e \underset{m' \times m'}{\mathbf{I}_{m'}} \quad (13.10)$$

As the determinant of a block diagonal matrix is equal to the product of determinants of the individual blocks,

$$|\mathbf{E}_k| = |\mathbf{B}| |\mathbf{E}'_k| |S_e \mathbf{I}_{n-m'}| |\mathbf{B}^T| = S_e^{n-m'} |\mathbf{E}'_k| \quad (13.11)$$

since $|\mathbf{B}| = |\mathbf{B}^T| = 1$ and $|S_e \mathbf{I}_{n-m'}| = S_e^{n-m'}$. On the other hand, using $\mathbf{B}^{-1} = \mathbf{B}^T$ and $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$ for invertible matrices A , B and C ,

$$\mathbf{E}_k^{-1} = \mathbf{B} \begin{bmatrix} \mathbf{E}'_k{}^{-1} & \mathbf{0} \\ \mathbf{0} & S_e^{-1} \mathbf{I}_{n-m'} \end{bmatrix} \mathbf{B}^T \quad (13.12)$$

Substituting $\mathbf{B} = [\mathbf{B}', \mathbf{B}'_{\perp}]$ and expanding,

$$\mathbf{E}_k^{-1} = [\mathbf{B}', \mathbf{B}'_{\perp}] \begin{bmatrix} \mathbf{E}'_k{}^{-1} & \mathbf{0} \\ \mathbf{0} & S_e^{-1} \mathbf{I}_{n-m'} \end{bmatrix} [\mathbf{B}', \mathbf{B}'_{\perp}]^T = \mathbf{B}' \mathbf{E}'_k{}^{-1} \mathbf{B}'^T + S_e^{-1} \mathbf{B}'_{\perp} \mathbf{B}'_{\perp}{}^T \quad (13.13)$$

Substituting $\mathbf{B}'_{\perp} \mathbf{B}'_{\perp}{}^T = \mathbf{I}_n - \mathbf{B}' \mathbf{B}'^T$ and rearranging,

$$\mathbf{E}_k^{-1} = S_e^{-1} \mathbf{I}_n - S_e^{-1} \mathbf{B}' (\mathbf{I}_{m'} - S_e \mathbf{E}'_k{}^{-1}) \mathbf{B}'^T \quad (13.14)$$

Substituting (13.11) and (13.14) into (13.2),

$$\begin{aligned} L = & nN_f \ln \pi + (n - m')N_f \ln S_e + S_e^{-1}d \\ & + \sum_k \ln |\mathbf{E}'_k| - S_e^{-1} \underbrace{\sum_k \hat{\mathcal{F}}_k^* \mathbf{B}' (\mathbf{I}_{m'} - S_e \mathbf{E}'_k{}^{-1}) \mathbf{B}'^T \hat{\mathcal{F}}_k}_{\text{the only term that depends on } \mathbf{B}'} \end{aligned} \quad (13.15)$$

where

$$d = \sum_k \hat{\mathcal{F}}_k^* \hat{\mathcal{F}}_k \quad (13.16)$$

The significance of (13.15) in comparison with (13.2) is that it involves $\mathbf{E}'_k$ whose dimension m' is typically much smaller than n . The mode shape basis $\mathbf{B}$ has been segregated out and its spatial effect has been condensed through the projection of the scaled FFT onto the mode shape subspace, i.e., $\mathbf{B}^T \hat{\mathcal{F}}_k$. In (13.2), $\mathbf{E}_k$ is almost singular for small S_e but now $\mathbf{E}'_k$ has full rank. The ill-conditioning due to small S_e has been segregated out in the factor S_e^{-1} .

13.3 Most Probable Mode Shape Basis

When $m = 1$ (single mode), $m' = 1$ and, trivially, $\alpha = 1$ and $\Phi = \mathbf{B}'$. It can then be shown that the MPV of $\mathbf{B}'$ is simply equal to the eigenvector of $\mathbf{A}$ in Sect. 12.1 with the largest eigenvalue. If $m' = n$ then the mode shape subspace is trivially R^n and $\mathbf{B}' = \mathbf{I}_n$. In the general case where $1 < m' < n$, the MPV of $\mathbf{B}' = [\mathbf{b}_1, \dots, \mathbf{b}_{m'}]$ minimizes the quadratic form in (13.15) subjected to orthonormal constraints. The latter can be handled by representing $\mathbf{B}'$ by a necessary and sufficient set of (free) hyper angles that can be determined by an iterative algorithm. Details are explained next in Sects. 13.3.1–13.3.3.

13.3.1 Hyper Angle Representation

Apparently, the number of free parameters in $\mathbf{B}'$ is $m'n - m'(m' + 1)/2$ because it contains $m'n$ parameters under $m'(m' + 1)/2$ distinct constraints, i.e., $\mathbf{b}_i^T \mathbf{b}_j = 1$ if $i = j$ and zero otherwise ($i = 1, \dots, m', j \leq i$). However, the MPV of $\mathbf{B}'$ is determined only up to arbitrary rotations within the planes formed by any two of its columns because the mode shape subspace is invariant under such rotations. There are $m'(m' - 1)/2$ such rotations (m' choose 2). The same number of additional constraints can be applied to remove this arbitrariness. Subtracting $m'(m' - 1)/2$ from $m'n - m'(m' + 1)/2$, the number of free parameters is $m'(n - m')$.

We next characterize $\mathbf{B}'$ with $m'(n - m')$ hyper angles from a reference basis $\mathbf{B}_0$, which can play the role of the current value of $\mathbf{B} = [\mathbf{B}', \mathbf{B}'_\perp]$ in an iterative algorithm for determining the MPV of $\mathbf{B}'$. Let $\mathbf{B}_0 = [\mathbf{B}'_0, \mathbf{B}'_{0\perp}]$ ($n \times n$) contain in its columns an orthonormal basis in R^n , where $\mathbf{B}'_0$ ($n \times m'$) contains the first m' vectors and $\mathbf{B}'_{0\perp}$ ($n \times (n - m')$) has the remaining $n - m'$ vectors. Then the $m'(n - m')$ angles can be defined as the rotations within the planes formed by all possible column pairs,

one from $\mathbf{B}'_0$ and the other from $\mathbf{B}'_{0\perp}$. To see this, $\mathbf{B}$ can be obtained by transforming $\mathbf{B}_0$ through $n(n-1)/2$ rotations (n choose 2) within all possible planes spanned by any two columns in $\mathbf{B}_0$. These rotations comprise three types:

- (1) $m'(m'-1)/2$ rotations within the planes formed by any two columns in $\mathbf{B}'_0$;
- (2) $(n-m')(n-m'-1)/2$ rotations within the planes formed by any two columns in $\mathbf{B}'_{0\perp}$;
- (3) $m'(n-m')$ rotations within the planes formed by one column in $\mathbf{B}'_0$ and the other in $\mathbf{B}'_{0\perp}$.

Check that the total number of rotations is $n(n-1)/2$. Only rotations of the third type are necessary because rotations of the first two types do not change the mode shape subspace.

13.3.2 Rotation Matrix

Let a_{ij} ($i = 1, \dots, m'; j = 1, \dots, n-m'$) be the rotation (in radian) from the i th column vector in $\mathbf{B}'_0$ towards the j th column vector in $\mathbf{B}'_{0\perp}$. The $m'(n-m')$ angles characterizing $\mathbf{B}'$ are collected in the vector

$$\mathbf{a} = [a_{11}, a_{12}, \dots, a_{1(n-m')}, a_{21}, a_{22}, \dots, a_{2(n-m')}, \dots, a_{m'1}, a_{m'2}, \dots, a_{m'(n-m')}]^T \quad (13.17)$$

The rotations a_{ij} are applied in the same sequence as they appear in $\mathbf{a}$, i.e., a_{11} first, a_{12} second, and so on.

The basis $\mathbf{B}$ can be written in terms of $\mathbf{a}$ and $\mathbf{B}_0$ by

$$\mathbf{B}(\mathbf{a}) = \mathbf{P}(\mathbf{a})\mathbf{B}_0 \quad (13.18)$$

where

$$\begin{aligned} \mathbf{P}(\mathbf{a}) &= \mathbf{P}_{m',(n-m')} \mathbf{P}_{m',(n-m')-1} \dots \mathbf{P}_{m'1} \\ &\times \mathbf{P}_{m'-1,(n-m')} \mathbf{P}_{m'-1,(n-m')-1} \dots \mathbf{P}_{m'-1,1} \\ &\times \dots \mathbf{P}_{1,(n-m')} \mathbf{P}_{1,(n-m')-1} \dots \mathbf{P}_{11} \end{aligned} \quad (13.19)$$

is the rotation matrix that transforms $\mathbf{B}_0$ through the $m'(n-m')$ angles in $\mathbf{a}$;

$$\mathbf{P}_{ij} = \mathbf{B}_0 \mathbf{R}_{ij} \mathbf{B}_0^T \quad (13.20)$$

is the rotation matrix associated with a_{ij} ;

$$\mathbf{R}_{ij} = (\mathbf{e}_i \mathbf{e}_i^T + \mathbf{e}_j \mathbf{e}_j^T) \cos a_{ij} + (\mathbf{e}_j \mathbf{e}_i^T - \mathbf{e}_i \mathbf{e}_j^T) \sin a_{ij} + \sum_{k \neq i, j}^n \mathbf{e}_k \mathbf{e}_k^T \quad j' = m' + j \quad (13.21)$$

is the matrix that rotates an angle a_{ij} from the i th axis towards the j' th axis in R^n . Here, $\mathbf{e}_i$ denotes a $n \times 1$ vector with the i th entry being the only nonzero entry equal to 1. The sequence in which the a_{ij} s appear in (13.17) is exactly the opposite of the $\mathbf{P}_{ij}$ s in (13.19), because the rotation matrices on the right take action first.

Substituting (13.20) into (13.19) and then into (13.18), and noting $\mathbf{B}_0^T \mathbf{B}_0 = \mathbf{I}_n$, we have

$$\mathbf{B}(\mathbf{a}) = \mathbf{B}_0 \mathbf{R}(\mathbf{a}) \quad (13.22)$$

where

$$\begin{aligned} \mathbf{R}(\mathbf{a}) &= \mathbf{R}_{m', (n-m')} \mathbf{R}_{m', (n-m')-1} \cdots \mathbf{R}_{m', 1} \\ &\quad \times \mathbf{R}_{m'-1, (n-m')} \mathbf{R}_{m'-1, (n-m')-1} \cdots \mathbf{R}_{m'-1, 1} \\ &\quad \times \cdots \mathbf{R}_{1, (n-m')} \mathbf{R}_{1, (n-m')-1} \cdots \mathbf{R}_{1, 1} \end{aligned} \quad (13.23)$$

Note that $\mathbf{B}(\mathbf{0}) = \mathbf{B}_0$ because $\mathbf{R}(\mathbf{0}) = \mathbf{I}_n$.

13.3.3 Newton Iteration

Given the remaining parameters, consider determining the MPV of $\mathbf{B}'$ by representing it via the hyper angles in $\mathbf{a}$ and minimizing the NLLF in (13.15) w.r.t. $\mathbf{a}$. This can be done by Newton iteration. Suppose the reference basis $\mathbf{B}_0$ is sufficiently close to the MPV so that the optimal value of $\mathbf{a}$ is expected to be small. This is justified in an iterative algorithm where $\mathbf{B}_0$ is continually being updated. Thus, consider the Taylor approximation of the gradient of NLLF about $\mathbf{a} = \mathbf{0}$ (omitting dependence on other parameters):

$$\nabla_{\mathbf{a}}^T L(\hat{\mathbf{a}}) \approx \nabla_{\mathbf{a}}^T L(\mathbf{0}) + [\nabla_{\mathbf{a}}^2 L(\mathbf{0})](\hat{\mathbf{a}} - \mathbf{0}) \quad (13.24)$$

where $\hat{\mathbf{a}}$ is the MPV of $\mathbf{a}$ given the remaining parameters. Solving $\nabla_{\mathbf{a}} L(\hat{\mathbf{a}}) = \mathbf{0}$ for $\hat{\mathbf{a}}$ gives the approximation

$$\hat{\mathbf{a}} = -[\nabla_{\mathbf{a}}^2 L(\mathbf{0})]^{-1} \nabla_{\mathbf{a}}^T L(\mathbf{0}) \quad (13.25)$$

The gradient and Hessian of the NLLF w.r.t. $\mathbf{a}$ at $\mathbf{0}$ can be derived analytically:

$$\frac{\partial L(\mathbf{0})}{\partial a_{ij}} = -2S_e^{-1} \text{Re} \left[\sum_k \mathbf{A}_k F_k^{(1)} F_k^{(2)*} \right]_{(ij)} \quad (13.26)$$

$$\frac{\partial^2 L(\mathbf{0})}{\partial a_{ij} \partial a_{rs}} = 2S_e^{-1} \delta_{js} \text{Re} \left[\sum_k \mathbf{A}_k F_k^{(1)} F_k^{(1)*} \right]_{(i,r)} - 2S_e^{-1} \text{Re} \left\{ \sum_k \mathbf{A}_k(r, i) \left[F_k^{(2)} F_k^{(2)*} \right]_{(j,s)} \right\} \quad (13.27)$$

where δ_{js} denotes the Kronecker Delta (equal to 1 if $j = s$ and zero otherwise); $[\cdot]_{(ij)}$ denotes the (i, j) -entry of the argument matrix;

$$\mathbf{A}_k = \mathbf{I}_{m'} - S_e \mathbf{E}_k'^{-1} \quad F_k^{(1)} = \mathbf{B}_0'^T \hat{\mathcal{F}}_k \quad F_k^{(2)} = \mathbf{B}_{0\perp}'^T \hat{\mathcal{F}}_k \quad (13.28)$$

$m' \times m' \quad m' \times 1 \quad m' \times n \quad n \times 1 \quad (n-m') \times 1 \quad (n-m') \times n \quad n \times 1$

Equation (13.27) assumes that a_{rs} is either a_{ij} or appears after (i.e., to the right of) a_{ij} in (13.17). It allows the diagonal and upper triangular entries of the Hessian $\nabla_{\mathbf{a}}^2 L(\mathbf{0})$ to be determined. The lower triangular entries can be determined by symmetry.

Proof (Derivatives of NLLF w.r.t. $\mathbf{a}$ at $\mathbf{0}$)

Let

$$q_k(\mathbf{a}) = \hat{\mathcal{F}}_k^* \mathbf{B}' \mathbf{A}_k \mathbf{B}'^T \hat{\mathcal{F}}_k \quad (13.29)$$

where $\mathbf{A}_k$ is defined in (13.28), so that the NLLF in (13.15) and its derivatives can be written as (omitting dependence on other parameters)

$$L(\mathbf{a}) = -S_e^{-1} \sum_k q_k(\mathbf{a}) + \text{terms independent of } \mathbf{a} \quad (13.30)$$

$$\frac{\partial L(\mathbf{0})}{\partial a_{ij}} = -S_e^{-1} \sum_k \frac{\partial q_k(\mathbf{0})}{\partial a_{ij}} \quad \frac{\partial^2 L(\mathbf{0})}{\partial a_{ij} \partial a_{rs}} = -S_e^{-1} \sum_k \frac{\partial^2 q_k(\mathbf{0})}{\partial a_{ij} \partial a_{rs}} \quad (13.31)$$

We shall show that the first two derivatives of $q_k(\mathbf{a})$ at $\mathbf{a} = \mathbf{0}$ are given by

$$\frac{\partial q_k(\mathbf{0})}{\partial a_{ij}} = 2\text{Re} \left[\mathbf{A}_k F_k^{(1)} F_k^{(2)*} \right]_{(ij)} \quad (13.32)$$

$$\frac{\partial^2 q_k(\mathbf{0})}{\partial a_{ij} \partial a_{rs}} = -2\delta_{js} \text{Re} \left[\mathbf{A}_k F_k^{(1)} F_k^{(1)*} \right]_{(i,r)} + 2\text{Re} \left\{ \mathbf{A}_k(r, i) \left[F_k^{(2)} F_k^{(2)*} \right]_{(j,s)} \right\} \quad (13.33)$$

Substituting these into (13.31) gives (13.26) and (13.27).

We first write q_k explicitly in terms of the rotation matrix $\mathbf{R}(\mathbf{a})$. Using (13.22),

$$\overbrace{\begin{bmatrix} \mathbf{B}' \\ \mathbf{B}'_{\perp} \end{bmatrix}}^{\mathbf{B}} = \mathbf{B}_0 \mathbf{R} = \mathbf{B}_0 \mathbf{R} \underbrace{\mathbf{B}_0^T \mathbf{B}_0}_{\mathbf{I}_n} = \mathbf{B}_0 \mathbf{R} \mathbf{B}_0^T \overbrace{\begin{bmatrix} \mathbf{B}'_0 \\ \mathbf{B}'_{0\perp} \end{bmatrix}}^{\mathbf{B}_0} \quad (13.34)$$

$n \times m'$ $n \times (n-m')$ $n \times m'$ $n \times (n-m')$

Reading the first m' columns gives

$$\mathbf{B}' = \mathbf{B}_0 \mathbf{R} \mathbf{B}_0^T \mathbf{B}'_0 \quad (13.35)$$

On the other hand, let $\mathbf{R}$ be partitioned as

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_1 & \mathbf{R}_2 \end{bmatrix} \quad (13.36)$$

$n \times n$ $n \times m'$ $n \times (n-m')$

Substituting (13.35) and (13.36) into (13.29),

$$q_k = \underbrace{\hat{\mathcal{F}}_k^* \mathbf{B}_0}_{F_k^*} \underbrace{\mathbf{R}}_{[\mathbf{R}_1, \mathbf{R}_2]} \underbrace{\mathbf{B}_0^T \mathbf{B}'_0}_{\begin{bmatrix} \mathbf{I}_{m'} \\ \mathbf{0} \end{bmatrix}} \mathbf{A}_k \underbrace{\mathbf{B}_0^T \mathbf{B}_0}_{\begin{bmatrix} \mathbf{I}_{m'} & \mathbf{0} \end{bmatrix}} \underbrace{\mathbf{R}^T}_{\begin{bmatrix} \mathbf{R}_1^T \\ \mathbf{R}_2^T \end{bmatrix}} \underbrace{\mathbf{B}_0^T \hat{\mathcal{F}}_k}_{F_k} = F_k^* \mathbf{R}_1 \mathbf{A}_k \mathbf{R}_1^T F_k \quad (13.37)$$

where $F_k = \mathbf{B}_0^T \hat{\mathcal{F}}_k$. Differentiating and noting that $\mathbf{A}_k$ is Hermitian,

$$\frac{\partial q_k}{\partial a_{ij}} = F_k^* \frac{\partial \mathbf{R}_1}{\partial a_{ij}} \mathbf{A}_k \mathbf{R}_1^T F_k + F_k^* \mathbf{R}_1 \mathbf{A}_k \frac{\partial \mathbf{R}_1^T}{\partial a_{ij}} F_k = 2\text{Re} \left[F_k^* \frac{\partial \mathbf{R}_1}{\partial a_{ij}} \mathbf{A}_k \mathbf{R}_1^T F_k \right] \quad (13.38)$$

$$\frac{\partial^2 q_k}{\partial a_{ij} \partial a_{rs}} = 2\text{Re} \left[F_k^* \frac{\partial^2 \mathbf{R}_1}{\partial a_{ij} \partial a_{rs}} \mathbf{A}_k \mathbf{R}_1^T F_k \right] + 2\text{Re} \left[F_k^* \frac{\partial \mathbf{R}_1}{\partial a_{ij}} \mathbf{A}_k \frac{\partial \mathbf{R}_1^T}{\partial a_{rs}} F_k \right] \quad (13.39)$$

Using (13.21) and (13.23), and noting that $\mathbf{R}_1$ contains the first m' columns of $\mathbf{R}$,

$$\frac{\partial \mathbf{R}(\mathbf{0})}{\partial a_{ij}} = \mathbf{e}_{j'} \mathbf{e}_i^T - \mathbf{e}_i \mathbf{e}_{j'}^T \quad \frac{\partial \mathbf{R}_1(\mathbf{0})}{\partial a_{ij}} = \mathbf{e}_{j'} \mathbf{e}_i'^T \quad j' = m' + j \quad (13.40)$$

$n \times 1$ $1 \times n$ $n \times 1$ $1 \times n$ $n \times 1$ $1 \times m'$

where $\mathbf{e}'_i$ contains the first m' entries of $\mathbf{e}_i$. The second derivative $\partial^2 \mathbf{R}(\mathbf{0}) / \partial a_{ij} \partial a_{rs}$ depends on the ordering of a_{ij} and a_{rs} . Assume that a_{rs} appears after (i.e., to the right of) a_{ij} in (13.17). Then

$$\mathbf{R} = \dots \mathbf{R}_{rs} \dots \mathbf{R}_{ij} \dots \quad (13.41)$$

This gives

$$\begin{aligned}\frac{\partial^2 \mathbf{R}(\mathbf{0})}{\partial a_{ij} \partial a_{rs}} &= (\mathbf{e}_s \mathbf{e}_r^T - \mathbf{e}_r \mathbf{e}_s^T)(\mathbf{e}_j \mathbf{e}_i^T - \mathbf{e}_i \mathbf{e}_j^T) = -\delta_{ir} \mathbf{e}_s \mathbf{e}_j^T - \delta_{js} \mathbf{e}_r \mathbf{e}_i^T \quad j' = m' + j \\ \frac{\partial^2 \mathbf{R}_1(\mathbf{0})}{\partial a_{ij} \partial a_{rs}} &= -\delta_{js} \mathbf{e}_r \mathbf{e}_i^T \quad s' = m' + s\end{aligned}\quad (13.42)$$

where (13.40) and the following have been used:

$$\mathbf{e}_r^T \mathbf{e}_{j'} = \mathbf{e}_s^T \mathbf{e}_i = 0 \quad \mathbf{e}_r^T \mathbf{e}_i = \delta_{ir} \quad \mathbf{e}_s^T \mathbf{e}_{j'} = \delta_{js} \quad (13.43)$$

It can be reasoned that this result is also applicable when $(i, j) = (r, s)$. Substituting $\partial \mathbf{R}_1(\mathbf{0}) / \partial a_{ij}$ from (13.40) into (13.38) at $\mathbf{a} = \mathbf{0}$ and noting $\mathbf{R}(\mathbf{0}) = \mathbf{I}_n$,

$$\begin{aligned}\frac{\partial q_k(\mathbf{0})}{\partial a_{ij}} &= 2\text{Re} \left[\underbrace{F_k^* \mathbf{e}_{j'}}_{\text{scalar}} \underbrace{\mathbf{e}_i^T \mathbf{A}_k \mathbf{R}_1(\mathbf{0})^T F_k}_{\text{scalar}} \right] = 2\text{Re} \left[\mathbf{e}_i^T \mathbf{A}_k \underbrace{\mathbf{R}_1(\mathbf{0})^T F_k}_{F_k^{(1)}} \underbrace{F_k^*}_{[F_k^{(1)*}, F_k^{(2)*}]} \mathbf{e}_{j'} \right] \\ &= 2\text{Re} \left\{ \mathbf{e}_i^T \left[\mathbf{A}_k F_k^{(1)} F_k^{(1)*} \quad \mathbf{A}_k F_k^{(1)} F_k^{(2)*} \right] \mathbf{e}_{j'} \right\} = 2\text{Re} \left[\mathbf{A}_k F_k^{(1)} F_k^{(2)*} \right]_{(i,j)}\end{aligned}\quad (13.44)$$

which is (13.32). Similarly, substituting $\partial \mathbf{R}_1(\mathbf{0}) / \partial a_{ij}$ from (13.40) and $\partial^2 \mathbf{R}_1(\mathbf{0}) / \partial a_{ij} \partial a_{rs}$ from (13.42) into (13.39) at $\mathbf{a} = \mathbf{0}$ gives two terms, where

$$\begin{aligned}\text{1st term} &= -2\delta_{js} \text{Re} \left[\underbrace{F_k^* \mathbf{e}_r}_{\text{scalar}} \underbrace{\mathbf{e}_i^T \mathbf{A}_k \mathbf{R}_1(\mathbf{0})^T F_k}_{\text{scalar}} \right] \\ &= -2\delta_{js} \text{Re} \left[\mathbf{e}_i^T \mathbf{A}_k \underbrace{\mathbf{R}_1(\mathbf{0})^T F_k}_{F_k^{(1)}} \underbrace{F_k^*}_{[F_k^{(1)*}, F_k^{(2)*}]} \mathbf{e}_r \right] \\ &= -2\delta_{js} \text{Re} \left\{ \mathbf{e}_i^T \left[\mathbf{A}_k F_k^{(1)} F_k^{(1)*} \quad \mathbf{A}_k F_k^{(1)} F_k^{(2)*} \right] \mathbf{e}_r \right\} \\ &= -2\delta_{js} \text{Re} \left[\mathbf{A}_k F_k^{(1)} F_k^{(1)*} \right]_{(i,r)}\end{aligned}\quad (13.45)$$

$$\begin{aligned}\text{2nd term} &= 2\text{Re} \left[\underbrace{F_k^* \mathbf{e}_{j'}}_{\text{scalar}} \underbrace{\mathbf{e}_i^T \mathbf{A}_k \mathbf{e}_r \mathbf{e}_s^T F_k}_{\text{scalar}} \right] = 2\text{Re} \left[\mathbf{e}_i^T \mathbf{A}_k \mathbf{e}_r \underbrace{\mathbf{e}_s^T F_k F_k^*}_{[F_k^{(2)} F_k^{(2)*}]_{(s,j)}} \mathbf{e}_{j'} \right] \\ &= 2\text{Re} \left\{ \mathbf{A}_k(r, i) \left[F_k^{(2)} F_k^{(2)*} \right]_{(j,s)} \right\}\end{aligned}\quad (13.46)$$

since $\text{Re} z = \text{Re} \bar{z}$, $\overline{\mathbf{A}_k(i, r)} = \mathbf{A}_k(r, i)$ and $\overline{[F_k^{(2)} F_k^{(2)*}]_{(s,j)}} = [F_k^{(2)} F_k^{(2)*}]_{(j,s)}$. Combining the first and second term gives (13.33). ■

13.4 Most Probable Spectral Parameters

Consider now determining the MPV of the remaining parameters for given $\mathbf{B}'$:

$$\{f_i\}_{i=1}^m \quad \{\zeta_i\}_{i=1}^m \quad \boldsymbol{\alpha} \quad \mathbf{S} \quad S_e$$

The quadratic term of the NLLF in (13.15) is re-grouped to read

$$L = nN_f \ln \pi + \underbrace{(n - m')N_f \ln S_e + S_e^{-1}(d - d')}_{\text{sensitive to } S_e} + \underbrace{\sum_k \ln |\mathbf{E}'_k| + \sum_k \hat{\mathcal{F}}_k^{J*} \mathbf{E}'_k{}^{-1} \hat{\mathcal{F}}_k'}_{\text{sensitive to } \{f_i, \zeta_i\}, \boldsymbol{\alpha}, \mathbf{S}} \quad (13.47)$$

where

$$\hat{\mathcal{F}}_k' = \mathbf{B}'^T \hat{\mathcal{F}}_k \quad d' = \sum_k \hat{\mathcal{F}}_k^{J*} \hat{\mathcal{F}}_k' \quad (13.48)$$

For high s/n ratio, $\mathbf{E}'_k$ is not sensitive to S_e . Equation (13.47) then roughly separates the effect of S_e and that of the remaining parameters. The MPV of S_e effectively minimizes the second and third term; see Sect. 13.6 later. These two terms measure the spatial discrepancy arising from incomplete dimension ($m' \leq n$) of the mode shape subspace. They are identically zero when $m' = n$ because then $\mathbf{B}'$ has full rank, $\hat{\mathcal{F}}_k^{J*} \hat{\mathcal{F}}_k' = \hat{\mathcal{F}}_k^* \mathbf{B}' \mathbf{B}'^T \hat{\mathcal{F}}_k = \hat{\mathcal{F}}_k^* \hat{\mathcal{F}}_k$ and hence $d' = d$. In the general case, $(d - d')$ reflects the spectral contribution in the measured response that cannot be accounted by the mode shape subspace.

13.4.1 Parameterizing Structured Matrices

While $\{f_i\}_{i=1}^m$, $\{\zeta_i\}_{i=1}^m$ and S_e are free parameters, $\boldsymbol{\alpha}$ and $\mathbf{S}$ are subjected to constraints. The entries of $\mathbf{S}$ are subjected to $m(m+1)/2$ constraints so that $\mathbf{S}$ is Hermitian. Each column of $\boldsymbol{\alpha}$, say, $\boldsymbol{\alpha}_i$ ($m' \times 1$), is subjected to unit norm constraint because $\boldsymbol{\varphi}_i = \mathbf{B}' \boldsymbol{\alpha}_i$ and

$$1 = \boldsymbol{\varphi}_i^T \boldsymbol{\varphi}_i = \boldsymbol{\alpha}_i^T \mathbf{B}'^T \mathbf{B}' \boldsymbol{\alpha}_i = \boldsymbol{\alpha}_i^T \boldsymbol{\alpha}_i \quad i = 1, \dots, m \quad (13.49)$$

The following represents $\boldsymbol{\alpha}$ and $\mathbf{S}$ by free parameters. Write

$$\mathbf{S} = \text{diag}(\mathbf{S})^{1/2} \mathbf{S}' \text{diag}(\mathbf{S})^{1/2} \quad (13.50)$$

where $\mathbf{S}'$ ($m \times m$ Hermitian) is a dimensionless form of $\mathbf{S}$ and its (i,j) -entry is $S'_{ij} = S_{ij} / \sqrt{S_{ii}S_{jj}}$, i.e., the coherence between the i th and j th modal force. Clearly, $S'_{ii} = 1$, $S'_{ij} = S'_{ji}^*$ and $|S'_{ij}| \leq 1$. One can represent

$$S'_{ij} = e^{iv_{ij}} \sin u_{ij} \quad i = 1, \dots, m; j < i \quad (13.51)$$

where u_{ij} and v_{ij} are free parameters. Note that

$$u_{ij} = \sin^{-1} |S'_{ij}| \quad v_{ij} = \angle S'_{ij} \quad (13.52)$$

Substituting (13.50) into (13.10), $\mathbf{E}'_k$ can be written as

$$\mathbf{E}'_k = \boldsymbol{\alpha}_S \mathbf{H}'_k \boldsymbol{\alpha}_S^T + S_e \mathbf{I}_{m'} \quad (13.53)$$

where

$$\mathbf{H}'_k = \text{diag}(\mathbf{h}_k) \mathbf{S}' \text{diag}(\mathbf{h}_k^*) \quad (13.54)$$

$$\boldsymbol{\alpha}_S = \boldsymbol{\alpha} \text{diag}(\mathbf{S})^{1/2} = [S_{11}^{1/2} \boldsymbol{\alpha}_1, S_{22}^{1/2} \boldsymbol{\alpha}_2, \dots, S_{mm}^{1/2} \boldsymbol{\alpha}_m] \quad (13.55)$$

The m diagonal entries of $\mathbf{S}$ and $m'm$ parameters in $\boldsymbol{\alpha}$ under m norm constraints are now combined into $m'm$ free parameters in $\boldsymbol{\alpha}_S$.

Based on the above, the MPV of $\boldsymbol{\alpha}_S$ ($m' \times m$) and $\{u_{ij}, v_{ij}; i = 1, \dots, m; j < i\}$ are first determined by minimizing the NLLF where $\mathbf{E}'_k$ is given by (13.53). The MPV of $\boldsymbol{\alpha}$ and $\mathbf{S}$ can then be recovered. Let $\boldsymbol{\alpha}_S(i)$ denote the i th column of $\boldsymbol{\alpha}_S$. Then

$$\boldsymbol{\alpha}(i) = \frac{\boldsymbol{\alpha}_S(i)}{\|\boldsymbol{\alpha}_S(i)\|} \quad S_{ii} = \|\boldsymbol{\alpha}_S(i)\|^2 \quad i = 1, \dots, m \quad (13.56)$$

$$S_{ij} = e^{iv_{ij}} \sqrt{S_{ii}S_{jj}} \sin u_{ij} \quad S_{ji} = \overline{S_{ij}} \quad i = 1, \dots, m; j < i \quad (13.57)$$

13.5 Algorithm for MPV

Below is an iterative algorithm for determining the MPV of modal parameters. The following groups of parameters are optimized sequentially given the remaining ones and iterated until convergence:

$$\{f_i\}_{i=1}^m \quad \{\zeta_i\}_{i=1}^m \quad \boldsymbol{\alpha}_S \quad \{u_{ij}; i = 1, \dots, m; j < i\} \quad \{v_{ij}; i = 1, \dots, m; j < i\} \quad S_e \quad \mathbf{a}$$

Iterative algorithm for MPV

0. Set initial guess for $\{f_i\}$, $\{\zeta_i\}$, α_S , $\{u_{ij}\}$, $\{v_{ij}\}$, S_e and $\mathbf{B}$. See Sect. 13.6.1.
1. Update α_S and $\{u_{ij}\}$ by minimizing (13.47).
2. Update $\{v_{ij}\}$ by minimizing (13.47).
3. Update $\{f_i\}$ and $\{\zeta_i\}$ by minimizing (13.47).
4. Update S_e by minimizing (13.47).
5. Update $\mathbf{a} = \{a_{ij}\}$ in (13.17) using (13.25). Calculate $\mathbf{R}$ in (13.23) and update $\mathbf{B}$ to be $\mathbf{B}\mathbf{R}$.

Repeat Steps 1–5 until convergence.

Steps 1–4 can be performed using conventional optimization algorithm, e.g., `fminsearch` in Matlab. They can be arranged in any order. Optimization is better performed w.r.t. non-dimensional parameters, e.g., as a multiplier of the last iterated value. Updating the mode shape basis last (via $\mathbf{a}$) preserves norm constraint singularity in the posterior covariance matrix (Sect. 11.3.1).

13.6 High s/n Asymptotics of MPV

When the modal s/n ratio is high, the MPV of $\mathbf{B}'$ and S_e can be obtained analytically. For multiple modes, high modal s/n asymptotics refers to $\mathbf{I}_{m'} - S_e \mathbf{E}_k'^{-1} \sim \mathbf{I}_{m'}$. Let $\mathbf{b}_i$ be the i th column of $\mathbf{B}'$ so that $\mathbf{B}' = [\mathbf{b}_1, \dots, \mathbf{b}_{m'}]$. Then in (13.15)

$$\begin{aligned}
 \sum_k \hat{\mathcal{F}}_k^* \underbrace{\mathbf{B}' (\mathbf{I}_{m'} - S_e \mathbf{E}_k'^{-1}) \mathbf{B}'^T}_{\sim \sum_{i=1}^{m'} \mathbf{b}_i \mathbf{b}_i^T} \hat{\mathcal{F}}_k &\sim \sum_k \sum_{i=1}^{m'} \underbrace{\hat{\mathcal{F}}_k^* \mathbf{b}_i}_{\text{scalar}} \underbrace{\mathbf{b}_i^T \hat{\mathcal{F}}_k}_{\text{scalar}} = \underbrace{\sum_k \sum_{i=1}^{m'} \mathbf{b}_i^T \hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^* \mathbf{b}_i}_{\text{real scalar}} \\
 &= \sum_{i=1}^{m'} \mathbf{b}_i^T \mathbf{D} \mathbf{b}_i
 \end{aligned} \tag{13.58}$$

where

$$\mathbf{D} = \text{Re} \sum_k \hat{\mathcal{F}}_k \hat{\mathcal{F}}_k^* \tag{13.59}$$

In (13.58), there are no cross terms between $\mathbf{b}_i$ and $\mathbf{b}_j$ for $i \neq j$. Under orthonormal constraints on $\{\mathbf{b}_i\}_{i=1}^{m'}$, it is maximized (and hence the NLLF is minimized) when $\{\mathbf{b}_i\}_{i=1}^{m'}$ are the eigenvectors (with unit norm) corresponding to the m' largest eigenvalues of $\mathbf{D}$. Thus, the asymptotic MPV of $\mathbf{B}' = [\mathbf{b}_1, \dots, \mathbf{b}_{m'}]$ can be

determined directly from data without iterations. The dimension of the mode shape subspace (m') can be empirically determined based on the eigenvalues of $\mathbf{D}$.

On the other hand, for high s/n ratio, $\mathbf{E}'_k$ in (13.10) is dominated by the first term, which is insensitive to S_e . The MPV of S_e then minimizes the second and third term in (13.47), giving

$$\hat{S}_e \sim \frac{d - d'}{(n - m')N_f} \quad (m' < n) \quad (13.60)$$

Using the norm-preserving property of $\mathbf{B}$, it can be shown that $d' \leq d$ and so $\hat{S}_e$ is non-negative. When $n = m'$, the second and third term in (13.47) are identically zero. The remaining terms then become the leading order, rendering the asymptotic MPV of S_e non-trivial.

13.6.1 Initial Guess of MPV

Initial guess for the iterative algorithm in Sect. 13.5 may be set as follow. The initial guess for the natural frequencies $\{f_j\}$ can be picked from the singular value (SV) spectrum (Sect. 7.3). The initial guess for damping ratios $\{\zeta_i\}$ may be assigned as 1%. By noting that at the natural frequency the top line in the SV spectrum is approximately $S_{ii}/4\zeta_i^2(2\pi f_i)^{2q}$, multiplying it by $4\zeta_i^2(2\pi f_i)^{2q}$ gives the initial guess for S_{ii} ; $q = 0, 1, 2$ for acceleration, velocity and displacement data, respectively. The asymptotic MPV in (13.60) can be used as the initial guess for S_e . The initial guess for $\mathbf{B} = [\mathbf{B}', \mathbf{B}'_\perp]$ can be taken as the eigenmatrix of $\mathbf{D}$ in (13.59) with eigenvalues arranged in descending order of magnitude. The initial guess for $\boldsymbol{\varphi}_i$ can be set as the eigenvector (largest eigenvalue) of the real part of sample PSD matrix of data at the initial guess of natural frequency. With the initial guess for $\mathbf{B}'$ and $\boldsymbol{\Phi} = [\boldsymbol{\varphi}_1, \dots, \boldsymbol{\varphi}_m]$, the initial guess for $\boldsymbol{\alpha}$ can be determined as the least squares solution to $\boldsymbol{\Phi} = \mathbf{B}'\boldsymbol{\alpha}$, i.e., $\boldsymbol{\alpha} = (\mathbf{B}'^T \mathbf{B}')^{-1} \mathbf{B}'^T \boldsymbol{\Phi}$. Combine $\{S_{ii}\}$ and $\boldsymbol{\alpha}$ to form the initial guess for $\boldsymbol{\alpha}_S$ according to (13.55). The initial guess for $\{u_{ij}\}$ and $\{v_{ij}\}$ may be assigned as 0.1.

13.7 Posterior Covariance Matrix

Under a Gaussian approximation of the posterior PDF, the posterior covariance matrix of modal parameters satisfying constraints (on mode shape norms) is given by (Sect. 11.2)

$$\hat{\mathbf{C}} = (\nabla \hat{\mathbf{v}}_c)_{n_\theta \times p} \underbrace{(\nabla^2 \hat{L}_c)^+}_{p \times p} (\nabla \hat{\mathbf{v}}_c)^T_{p \times n_\theta} \quad (13.61)$$

where ‘+’ denotes the pseudo-inverse, i.e., ignoring zero eigenvalues from constraint singularity; $\nabla^2 \hat{L}_c$ is the Hessian of $L_c(\mathbf{u}) = L(\mathbf{v}_c(\mathbf{u}))$ w.r.t. a set of free parameters $\mathbf{u}$ ($p \times 1$); $\mathbf{v}_c(\mathbf{u})$ maps $\mathbf{u}$ to $\boldsymbol{\theta}$ that always satisfies the constraints $G_i(\mathbf{v}_c(\mathbf{u})) = 0$, $i = 1, \dots, m$; $\{G_i(\boldsymbol{\theta})\}_{i=1}^m$ are the norm constraint equations on mode shapes. A hat ‘^’ on quantity denotes that it is evaluated at the MPV. Using the formula based on Lagrange multiplier (Sect. 11.2.3),

$$\nabla^2 \hat{L}_c = (\nabla \hat{\mathbf{v}}_c)^T \left(\nabla^2 \hat{L} + \sum_{i=1}^m \hat{\lambda}_i \nabla^2 \hat{G}_i \right) (\nabla \hat{\mathbf{v}}_c) \quad (13.62)$$

$p \times p \quad p \times n_\theta \quad n_\theta \times n_\theta \quad m \quad n_\theta \times n_\theta \quad n_\theta \times p$

where $\nabla^2 \hat{L}$ is the Hessian of the NLLF in (13.2) (ignoring mode shape norm constraints) and

$$\hat{\boldsymbol{\lambda}} = \begin{bmatrix} \hat{\lambda}_1 \\ \vdots \\ \hat{\lambda}_m \end{bmatrix} = - \underbrace{[\nabla \hat{\mathbf{G}} \quad (\nabla \hat{\mathbf{G}})^T]^{-1}}_{m \times m} \begin{pmatrix} \nabla \hat{\mathbf{G}} \\ (\nabla \hat{L})^T \end{pmatrix} \quad (13.63)$$

$m \times n_\theta \quad n_\theta \times m \quad m \times n_\theta \quad n_\theta \times 1$

$$\nabla \hat{L} = \nabla L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \quad \nabla^2 \hat{L} = \nabla^2 L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \quad \nabla \hat{\mathbf{G}} = \begin{bmatrix} \nabla \hat{G}_1 \\ \vdots \\ \nabla \hat{G}_m \end{bmatrix} \quad (13.64)$$

$m \times n_\theta$

In the present case, one can take $\mathbf{u}$ and $\boldsymbol{\theta}$ to comprise the same set of parameters:

$$\{f_i\}_{i=1}^m \quad \{\zeta_i\}_{i=1}^m \quad \{S_{ij}\}_{i=1}^m \quad \{U_{ij}\}_{j<i;i=1}^m \quad \{V_{ij}\}_{j<i;i=1}^m \quad S_e \quad \boldsymbol{\Phi} = [\boldsymbol{\phi}_1, \dots, \boldsymbol{\phi}_m]$$

where $U_{ij} = \text{Re } S_{ij}$ and $V_{ij} = \text{Im } S_{ij}$ so that $S_{ij} = U_{ij} + \mathbf{i}V_{ij}$. Let ϖ comprise the modal parameters other than mode shapes. One can take

$$G_i(\boldsymbol{\theta}) = 1 - \boldsymbol{\phi}_i^T \boldsymbol{\phi}_i \quad (i = 1, \dots, m) \quad \mathbf{v}_c(\boldsymbol{\theta}) = \begin{bmatrix} \varpi \\ ||\boldsymbol{\phi}_1||^{-1} \boldsymbol{\phi}_1 \\ \vdots \\ ||\boldsymbol{\phi}_m||^{-1} \boldsymbol{\phi}_m \end{bmatrix} \quad (13.65)$$

so that that $G_i(\mathbf{v}_c(\mathbf{u})) \equiv 0$ ($i = 1, \dots, m$) for any $\mathbf{u}$. Direct differentiation and evaluating at the MPV gives

$$\nabla_{\varpi} \hat{G}_i = \mathbf{0} \quad \nabla_{\boldsymbol{\phi}_i} \hat{G}_i = \mathbf{0} \quad \nabla_{\boldsymbol{\phi}_j} \hat{G}_i = -2\hat{\boldsymbol{\phi}}_i^T \delta_{ij} \quad \nabla_{\boldsymbol{\phi}_j}^2 \hat{G}_i = -2\mathbf{I}_n \delta_{ij} \quad (13.66)$$

where δ_{ij} is the Kronecker Delta (1 if $i = j$, zero otherwise). Using these results and $\nabla_{\boldsymbol{\phi}_i} (||\boldsymbol{\phi}_i||^{-1} \boldsymbol{\phi}_i) = \mathbf{I}_n - \boldsymbol{\phi}_i \boldsymbol{\phi}_i^T$ for $||\boldsymbol{\phi}_i|| = 1$,

$$\hat{\lambda}_i = \frac{1}{2} (\nabla_{\boldsymbol{\phi}_i} \hat{L}) \hat{\boldsymbol{\phi}}_i \quad (13.67)$$

$$\sum_{i=1}^m \hat{\lambda}_i \nabla^2 \hat{G}_i = \begin{bmatrix} \mathbf{0}_{n_\varpi \times n_\varpi} & & & \\ & -2\hat{\lambda}_1 \mathbf{I}_n & & \\ & & \ddots & \\ & & & -2\hat{\lambda}_m \mathbf{I}_n \end{bmatrix} \quad (13.68)$$

$$\nabla \hat{\mathbf{v}}_c = \begin{bmatrix} \mathbf{I}_{n_\varpi \times n_\varpi} & & & \\ & \mathbf{I}_n - \hat{\boldsymbol{\phi}}_1 \hat{\boldsymbol{\phi}}_1^T & & \\ & & \ddots & \\ & & & \mathbf{I}_n - \hat{\boldsymbol{\phi}}_m \hat{\boldsymbol{\phi}}_m^T \end{bmatrix} \quad (13.69)$$

Due to norm constraints, $\nabla^2 \hat{L}_c$ has m zero eigenvalues with eigenvectors equal to the null vectors of $\nabla \hat{\mathbf{v}}_c$, which are given by

$$\begin{bmatrix} \mathbf{0}_{n_\varpi \times 1} \\ \hat{\boldsymbol{\phi}}_1 \end{bmatrix} \quad \begin{bmatrix} \mathbf{0}_{n_\varpi \times 1} \\ \hat{\boldsymbol{\phi}}_2 \end{bmatrix} \quad \cdots \quad \begin{bmatrix} \mathbf{0}_{n_\varpi \times 1} \\ \hat{\boldsymbol{\phi}}_m \end{bmatrix} \quad (13.70)$$

It remains to determine $\nabla \hat{L}$ and $\nabla^2 \hat{L}$, which can be via finite difference approximation or analytical expressions. The latter is discussed in Sects. 13.7.1 and 13.7.2. Section 13.7.1 discusses a method that expresses the results in terms of the derivatives of $\mathbf{E}_k$. It is generic but can be slow for large n . Section 13.7.2 discusses an alternative method which is more efficient for $n > m$.

13.7.1 General Expressions

For any parameters x and y ,

$$\begin{aligned} L^{(x)} &= \sum_k (\ln |\mathbf{E}_k|)^{(x)} + \sum_k \hat{\mathcal{F}}_k^* (\mathbf{E}_k^{-1})^{(x)} \hat{\mathcal{F}}_k \\ L^{(xy)} &= \sum_k (\ln |\mathbf{E}_k|)^{(xy)} + \sum_k \hat{\mathcal{F}}_k^* (\mathbf{E}_k^{-1})^{(xy)} \hat{\mathcal{F}}_k \end{aligned} \quad (13.71)$$

The derivatives of $\ln |\mathbf{E}_k|$ and $\mathbf{E}_k^{-1}$ can be expressed in terms of the derivatives of $\mathbf{E}_k$. For any parameters x and y ,

Table 13.2 Derivatives of $\mathbf{E}_k = \Phi \mathbf{H}_k \Phi^T + S_e \mathbf{I}_n$

$\mathbf{E}_k^{(xy)}$		$y =$		
		$f_j, \zeta_j, S_{jj}, U_{rs}, V_{rs}$	S_e	Φ_{rs}
$x =$	$f_i, \zeta_i, S_{ii}, U_{ij}, V_{ij}$	$\Phi \mathbf{H}_k^{(xy)} \Phi^T$	$\mathbf{0}_n$	$\Phi \mathbf{H}_k^{(x)} \mathbf{e}_{rs}^T + (\cdot)^*$
	S_e		$\mathbf{0}_n$	$\mathbf{0}_n$
	Φ_{ij}	Sym.		$\mathbf{e}_{ij} \mathbf{H}_k \mathbf{e}_{rs}^T + (\cdot)^*$
	$\mathbf{E}_k^{(y)}$	$\Phi \mathbf{H}_k^{(y)} \Phi^T$	$\mathbf{I}_n$	$\Phi \mathbf{H}_k \mathbf{e}_{rs}^T + (\cdot)^*$

See Table 13.3 for derivatives of $\mathbf{H}_k$; $(\cdot)^*$ denotes a term equal to the complex conjugate of the previous one.

$$\begin{aligned}
 (\ln |\mathbf{E}_k|)^{(x)} &= \text{tr}[\mathbf{E}_k^{-1} \mathbf{E}_k^{(x)}] & (\mathbf{E}_k^{-1})^{(x)} &= -\mathbf{E}_k^{-1} \mathbf{E}_k^{(x)} \mathbf{E}_k^{-1} \\
 (\ln |\mathbf{E}_k|)^{(xy)} &= \text{tr}[\mathbf{E}_k^{-1} \mathbf{E}_k^{(xy)} - \mathbf{E}_k^{-1} \mathbf{E}_k^{(x)} \mathbf{E}_k^{-1} \mathbf{E}_k^{(y)}] \\
 (\mathbf{E}_k^{-1})^{(xy)} &= \mathbf{E}_k^{-1} [\mathbf{E}_k^{(x)} \mathbf{E}_k^{-1} \mathbf{E}_k^{(y)} + \mathbf{E}_k^{(y)} \mathbf{E}_k^{-1} \mathbf{E}_k^{(x)} - \mathbf{E}_k^{(xy)}] \mathbf{E}_k^{-1}
 \end{aligned} \tag{13.72}$$

The derivatives of $\mathbf{E}_k$ are given in Table 13.2. The expressions involve the derivatives of $\mathbf{H}_k$ in Table 13.3, which in turn depends on the derivatives of the transfer function h in Table 13.4. Figure 13.2 shows the information flow. Since $\mathbf{E}_k$ is almost singular, its inverse should be calculated using the eigenvector representation in (13.14), where the ill-conditioning due to small S_e has been segregated out in the factor S_e^{-1} :

$$\mathbf{E}_k^{-1} = \mathbf{B}' \mathbf{E}_k'^{-1} \mathbf{B}'^T + S_e^{-1} (\mathbf{I}_m - \mathbf{B}' \mathbf{B}'^T) \tag{13.73}$$

13.7.2 Condensed Expressions

Computing the derivatives of the NLLF via those of $\mathbf{E}_k$ involves matrix computations of size n (the number of DOFs) which can be large in applications. It is possible to obtain the derivatives in a condensed form that involves matrix computations of size m (the number of modes) only. This can be done using the Matrix Inverse Lemma and Matrix Determinant Theorem (Sect. C.5.1), where for any matrices A , C , U and V of appropriate size with A and C invertible:

Table 13.3 Derivatives of $\mathbf{H}_k = \text{diag}(\mathbf{h}_k)\mathbf{S}\text{diag}(\mathbf{h}_k^*)$

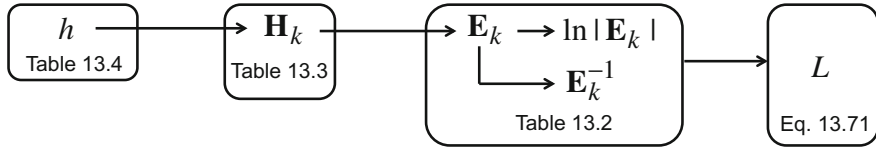
$\mathbf{H}_k^{(xy)}$	$y =$			
	f_i, ζ_j		S_{ij}	U_{rs}, V_{rs}
$x =$	f_i, ζ_i	$[S_{ij}h_{ik}^{(x)}h_{jk}^{(y)*}\mathbf{e}_{ij} + \delta_{ij}h_{ik}^{(xy)*}\text{diag}(\mathbf{h}_k)\mathbf{S}\mathbf{e}_{ii}] + (\cdot)^*$	$\delta_{ij}h_{ik}^{(x)}h_{jk}^{(y)*}\mathbf{e}_{ii} + (\cdot)^*$	$\begin{cases} c h_{ik}^{(x)}h_{sk}^*\mathbf{e}_{rs} + (\cdot)^* & i = r \\ c h_{ik}^{*(x)}h_{rk}\mathbf{e}_{rs} + (\cdot)^* & i = s \\ c = 1 \text{ for } U_{rs}; c = \mathbf{i} \text{ for } V_{rs} \end{cases}$
	S_{ii}		$\mathbf{0}$	$\mathbf{0}$
	U_{ij}, V_{ij}	Sym.		$\mathbf{0}$
$\mathbf{H}_k^{(y)}$		$h_{jk}^{(y)*}\text{diag}(\mathbf{h}_k)\mathbf{S}\mathbf{e}_{jj} + (\cdot)^*$	$h_{jk}h_{jk}^*\mathbf{e}_{jj}$	$\begin{cases} c h_{rk}h_{sk}^*\mathbf{e}_{rs} + (\cdot)^* \\ c = 1 \text{ for } U_{rs}; c = \mathbf{i} \text{ for } V_{rs} \end{cases}$

See Table 13.4 for derivatives of h ; $(\cdot)^*$ denotes a term equal to the complex conjugate of the previous one.

Table 13.4 Derivatives of $h(\beta, \zeta) = (1 - \beta^2 - 2\zeta\beta\mathbf{i})^{-1}(2\pi f_k \mathbf{i})^{-q}$

$h^{(xy)}$		$y =$	
		β	ζ
$x =$	β	$2(3\beta^2 + 1 - 4\zeta^2 + 6\mathbf{i}\zeta\beta)h^3(2\pi f_k \mathbf{i})^{-q}$	$2\mathbf{i}(3\beta^2 + 1 + 2\mathbf{i}\zeta\beta)h^3(2\pi f_k \mathbf{i})^{-q}$
	ζ	Sym.	$-8\beta^2 h^3(2\pi f_k \mathbf{i})^{-q}$
$h^{(y)}$		$2(\beta + \mathbf{i}\zeta)h^2(2\pi f_k \mathbf{i})^{-q}$	$2\mathbf{i}\beta h^2(2\pi f_k \mathbf{i})^{-q}$

$q = 0, 1, 2$ for acceleration data, velocity data and displacement data, respectively

**Fig. 13.2** Information flow of derivative calculations

$$(UCV + A)^{-1} = A^{-1} - A^{-1}U(VA^{-1}U + C^{-1})^{-1}VA^{-1} \quad (13.74)$$

$$|UCV + A| = |A||C||VA^{-1}U + C^{-1}| \quad (13.75)$$

Using these with $A = S_e \mathbf{I}_n$, $C = \mathbf{H}_k$, $U = \Phi$ and $V = \Phi^T$,

$$|\mathbf{E}_k| = S_e^{-m} |\mathbf{H}_k| |\mathbf{P}_k| \quad \mathbf{E}_k^{-1} = S_e^{-1} \mathbf{I}_n - S_e^{-1} \Phi \mathbf{P}_k^{-1} \Phi^T \quad (13.76)$$

$$\mathbf{P}_k = \underbrace{\Phi^T}_{m \times m} \underbrace{\Phi}_{m \times n} \underbrace{\mathbf{H}_k^{-1}}_{n \times m} + S_e \underbrace{\mathbf{H}_k^{-1}}_{m \times m} \quad (13.77)$$

Substituting into the NLLF in (13.2),

$$L = nN_f \ln \pi + (n - m)N_f \ln S_e + \sum_k \ln |\mathbf{H}_k| + \sum_k \ln |\mathbf{P}_k| + S_e^{-1} \left(d - \sum_k \mathbf{r}_k^* \mathbf{P}_k^{-1} \mathbf{r}_k \right) \quad (13.78)$$

where

$$\mathbf{r}_k = \underbrace{\Phi^T}_{m \times n} \underbrace{\hat{\mathcal{F}}_k}_{n \times 1} \quad (13.79)$$

and d is given by (13.16). Comparing (13.78) with (13.2), it only involves $\mathbf{H}_k$ and $\mathbf{P}_k$ of dimension m rather than n . The matrices $\mathbf{P}_k$ and $\mathbf{H}_k$ depend on all parameters, while $\mathbf{r}_k$ only depends on mode shapes.

To facilitate presentation of derivatives, the NLLF in (13.78) is written as

$$L = L_e + L_H + L_P + L_q \quad (13.80)$$

where

$$L_e = nN_f \ln \pi + (n - m)N_f \ln S_e \quad L_H = \sum_k \ln |\mathbf{H}_k| \quad L_P = \sum_k \ln |\mathbf{P}_k| \quad (13.81)$$

$$L_q = S_e^{-1}(d - q) \quad q = \sum_k \mathbf{r}_k^* \mathbf{P}_k^{-1} \mathbf{r}_k \quad (13.82)$$

Clearly, L_e depends only on S_e . Its derivatives are non-zero only when taken w.r.t. S_e :

$$L_e^{(S_e)} = (n - m)N_f S_e^{-1} \quad L_e^{(S_e S_e)} = -(n - m)N_f S_e^{-2} \quad (13.83)$$

The derivatives of L_H are simply equal to the sum of the derivatives of $\ln |\mathbf{H}_k|$. The latter can be obtained by exactly the same formulas in (13.72) (replace $\mathbf{E}_k$ by $\mathbf{H}_k$) in terms of the derivatives of $\mathbf{H}_k$, which are given in Table 13.3. The derivatives of L_P can be obtained in the same way in terms of the derivatives of $\mathbf{P}_k$, which are given in Table 13.5. The derivatives of L_q are given in Table 13.6. The expressions depend in a hierarchical manner on the derivatives of q (Table 13.7), $\mathbf{P}_k$, $\mathbf{H}_k$ and h . In the development of software it is recommended that these derivatives be coded and checked separately as they are used repeatedly. Figure 13.3 illustrates the information flow.

Table 13.5 Derivatives of $\mathbf{P}_k = \Phi^T \Phi + S_e \mathbf{H}_k^{-1}$

$\mathbf{P}_k^{(xy)}$		$y =$		
		$f_j, \zeta_j,$ S_{jj}, U_{rs}, V_{rs}	S_e	Φ_{rs}
$x =$	$f_i, \zeta_i,$ S_{ii}, U_{ij}, V_{ij}	$S_e (\mathbf{H}_k^{-1})^{(xy)}$	$(\mathbf{H}_k^{-1})^{(x)}$	$\mathbf{0}$
	S_e		$\mathbf{0}$	$\mathbf{0}$
	Φ_{ij}	Sym.		$\delta_{ir} \mathbf{e}_{js} + (\cdot)^T$
$\mathbf{P}_k^{(y)}$		$S_e (\mathbf{H}_k^{-1})^{(y)}$	$\mathbf{H}_k^{-1}$	$\Phi^T \mathbf{e}_{rs} + (\cdot)^T$

See Table 13.3 for derivatives of $\mathbf{H}_k^{-1}$; $(\cdot)^T$ denotes a term equal to the transpose of the previous one.

Table 13.6 Derivatives of $L_q = S_e^{-1}(d - q)$

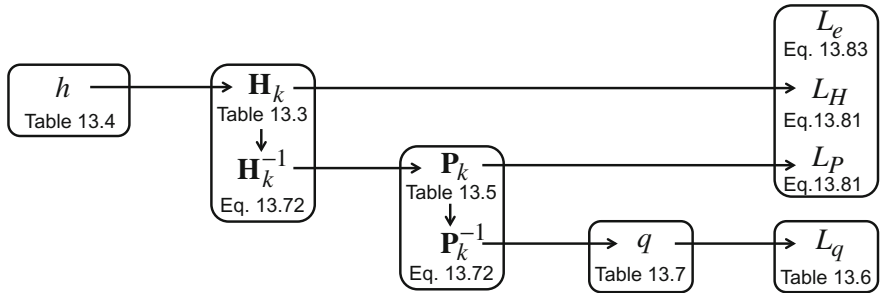
$L_q^{(xy)}$		$y =$	
		$f_j, \zeta_j, S_{jj},$ $U_{rs}, V_{rs}, \Phi_{rs}$	S_e
$x =$	$f_i, \zeta_i, S_{ii},$ $U_{ij}, V_{ij}, \Phi_{ij}$	$-S_e^{-1}q^{(xy)}$	$S_e^{-2}q^{(x)} - S_e^{-1}q^{(xS_e)}$
	S_e	Sym.	$2S_e^{-3}(d - q)$ $+ 2S_e^{-2}q^{(S_e)} - S_e^{-1}q^{(S_e S_e)}$
$L_q^{(y)}$		$-S_e^{-1}q^{(y)}$	$-S_e^{-2}(d - q) - S_e^{-1}q^{(S_e)}$

See Table 13.7 for derivatives of q .

Table 13.7 Derivatives of $q = \sum_k \mathbf{r}_k^* \mathbf{P}_k^{-1} \mathbf{r}_k$

$q^{(xy)}$		$y =$	
		$f_j, \zeta_j, S_{jj},$ U_{rs}, V_{rs}, S_e	Φ_{rs}
$x =$	$f_i, \zeta_i, S_{ii},$ U_{ij}, V_{ij}, S_e	$\sum_k \mathbf{r}_k^* (\mathbf{P}_k^{-1})^{(xy)} \mathbf{r}_k$	$\sum_k \mathbf{r}_k^* (\mathbf{P}_k^{-1})^{(x\Phi_{rs})} \mathbf{r}_k$ $+ 2\text{Re} \sum_k \hat{\mathcal{F}}_k(r) \mathbf{r}_k^* (\mathbf{P}_k^{-1})^{(x)}(s)$
	Φ_{ij}	Sym.	$\sum_k \mathbf{r}_k^* (\mathbf{P}_k^{-1})^{(\Phi_{ij}\Phi_{rs})} \mathbf{r}_k$ $+ 2\text{Re} \sum_k \hat{\mathcal{F}}_k(r) \mathbf{r}_k^* (\mathbf{P}_k^{-1})^{(\Phi_{ij})}(s)$ $+ 2\text{Re} \sum_k \hat{\mathcal{F}}_k(i) \mathbf{r}_k^* (\mathbf{P}_k^{-1})^{(\Phi_{rs})}(j)$ $+ 2\text{Re} \sum_k \hat{\mathcal{F}}_k(r)^* \hat{\mathcal{F}}_k(i) \mathbf{P}_k^{-1}(s, j)$
$q^{(y)}$		$\sum_k \mathbf{r}_k^* (\mathbf{P}_k^{-1})^{(y)} \mathbf{r}_k$	$\sum_k \mathbf{r}_k^* (\mathbf{P}_k^{-1})^{(\Phi_{rs})} \mathbf{r}_k$ $+ 2\text{Re} \sum_k \hat{\mathcal{F}}_k(r) \mathbf{r}_k^* \mathbf{P}_k^{-1}(s)$

See (13.72) for derivatives of $\mathbf{P}_k^{-1}$ (replace $\mathbf{E}_k$ by $\mathbf{P}_k$) and Table 13.5 for derivatives of $\mathbf{P}_k$; $\hat{\mathcal{F}}_k(r)$ denotes the r th entry of $\hat{\mathcal{F}}_k$; $\mathbf{P}_k^{-1}(s)$ denotes the s th column of $\mathbf{P}_k^{-1}$; $\mathbf{P}_k^{-1}(s, j)$ denotes the (s, j) -entry of $\mathbf{P}_k^{-1}$.

**Fig. 13.3** Information flow of derivative calculations in condensed form

13.8 Illustrative Examples

In this section we present examples on Bayesian OMA with close modes. The first example is based on synthetic data. The second example is based on field data from a tall building.

Example 13.2 (Synthetic data) Consider synthetic data with two modes generated at 100 Hz by

$$\hat{\mathbf{y}}_j = \boldsymbol{\Phi}_1 \ddot{\eta}_1(t_j) + \boldsymbol{\Phi}_2 \ddot{\eta}_2(t_j) + \boldsymbol{\varepsilon}(t_j) \quad (13.84)$$

where $\ddot{\eta}_i(t)$ ($i = 1, 2$) is the modal acceleration satisfying

$$\ddot{\eta}_i(t) + 2\zeta_i \omega_i \dot{\eta}_i(t) + \omega_i^2 \eta_i(t) = p_i(t) \quad (13.85)$$

with natural frequencies $\omega_i = 2\pi f_i$ (rad/s), $f_1 = 1$ Hz, $f_2 = 1.02$ Hz; and damping ratios $\zeta_1 = \zeta_2 = 1\%$; $p_1(t)$ and $p_2(t)$ are modal excitations, stationary Gaussian with PSD $S_{11} = S_{22} = 1(\mu\text{g})^2/\text{Hz}$ and cross PSD $S_{12} = 0.5e^{i\pi/4}(\mu\text{g})^2/\text{Hz}$;

$$\boldsymbol{\Phi}_1 = [1 \quad 2 \quad 2]^T/3 \quad \boldsymbol{\Phi}_2 = [-2 \quad 1 \quad 2]^T/3 \quad (13.86)$$

Figure 13.4 shows the root PSD and SV spectrum of 600 s data. The top two lines in (b) indicate two modes with very similar spectrum frequencies. Table 13.8 summarizes the modal identification results. The posterior c.o.v. is about 0.2% for the natural frequencies and 20% for the damping ratios. The modal force PSDs have a c.o.v. of about 15%, which is similar to the c.o.v. of the damping ratios. The c.o.v. of S_{12} is quite high, about 100% in the real part and 30% in the imaginary part. Table 13.9 shows that the exact and most probable mode shapes are close, with a hyper angle of about 10%. The mode shape c.o.v. is of the same order of magnitude.

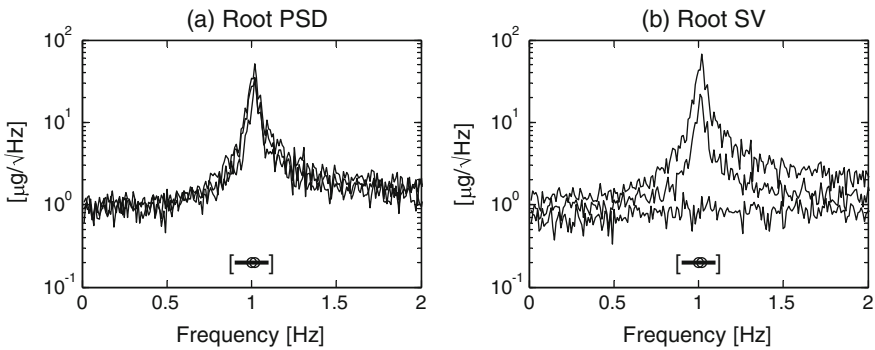


Fig. 13.4 Root PSD (a) and root SV spectrum (b), synthetic data. Horizontal bar indicates selected band

Table 13.8 Modal identification results, synthetic data, close modes

Parameter		Mode 1	Mode 2
Frequency f	Exact (Hz)	1.000	1.020
	MPV (Hz)	1.002	1.021
	c.o.v. (%)	0.19	0.16
Damping ζ	Exact (%)	1.000	1.000
	MPV (%)	1.12	0.86
	c.o.v. (%)	18	21
Modal force PSD S_{ii}	Exact ($(\mu\text{g})^2/\text{Hz}$)	1.000	1.000
	MPV ($(\mu\text{g})^2/\text{Hz}$)	1.086	1.035
	c.o.v. (%)	16	15
Cross PSD, real part $\text{Re}S_{12}$	Exact ($(\mu\text{g})^2/\text{Hz}$)	0.35	
	MPV ($(\mu\text{g})^2/\text{Hz}$)	0.17	
	c.o.v. (%)	97	
Cross PSD, imaginary part $\text{Im}S_{12}$	Exact ($(\mu\text{g})^2/\text{Hz}$)	0.35	
	MPV ($(\mu\text{g})^2/\text{Hz}$)	0.25	
	c.o.v. (%)	31	
Prediction error PSD S_e	Exact ($(\mu\text{g})^2/\text{Hz}$)	1.000	
	MPV ($(\mu\text{g})^2/\text{Hz}$)	1.05	
	c.o.v. (%)	9	

Table 13.9 Identified mode shape (MPV), synthetic data, close modes

	Mode 1			Mode 2		
Exact	0.333	0.667	0.667	-0.667	0.333	0.667
MPV	0.235	0.667	0.707	-0.611	0.373	0.698
Angle between exact and MPV mode shape (%)	10.6			7.6		
Mode shape c.o.v. (%)	8.1			7.7		

Figure 13.5 shows the results for different data durations. The results generally converge to the exact value that generated the data. The error bars generally cover the exact values, reflecting consistency of the posterior c.o.v. with the MPV. ■

Example 13.3 (Field data, tall building with close modes) Consider a tall building measuring 50 m $\times$ 50 m in plan and over 300 m in height. This building has similar dimensions as the one in Example 12.5 but its first two modes are much closer. Triaxial acceleration (x, y and z) at four corners on the roof were measured for 1800 s under normal wind situation in the late afternoon. The x and y directions are parallel to the two perpendicular sides of the building. Figure 13.6 shows the measured time histories at one corner. The root PSD and SV spectrum are shown in Fig. 13.7. The spectra suggest six potential modes. Their nature is determined based on mode shape information found later. The first two modes are very close and so they are identified

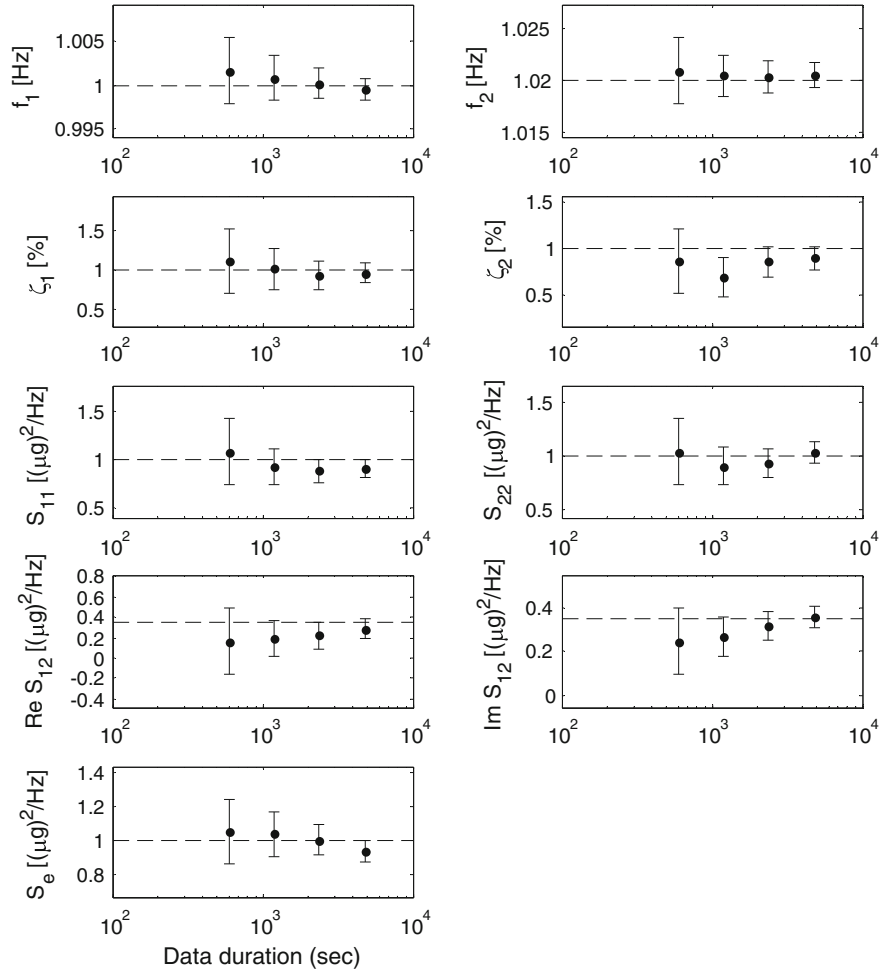


Fig. 13.5 Modal identification results for different data durations. *Dot* MPV; *error bar* $\pm$ two standard deviations. *Dashed line* exact value

together assuming $m = 2$ modes in the selected band. The third and fourth mode are well-separated from others and can be identified assuming a single mode ($m = 1$). The fifth and sixth modes are not very close but their resonance bands do overlap. They are identified on the same band assuming $m = 2$ modes.

Results Using Triaxial Data at One Corner

Table 13.10 shows the modal identification results using triaxial data at one corner only. The root modal force PSD is in the order of $1 \mu\text{g}/\sqrt{\text{Hz}}$. The root prediction error $\sqrt{S_e}$ is a few $\mu\text{g}/\sqrt{\text{Hz}}$, which is consistent with the noise PSD reflected in Fig. 13.7. The values for modes 1 and 2 are by definition the same because they are

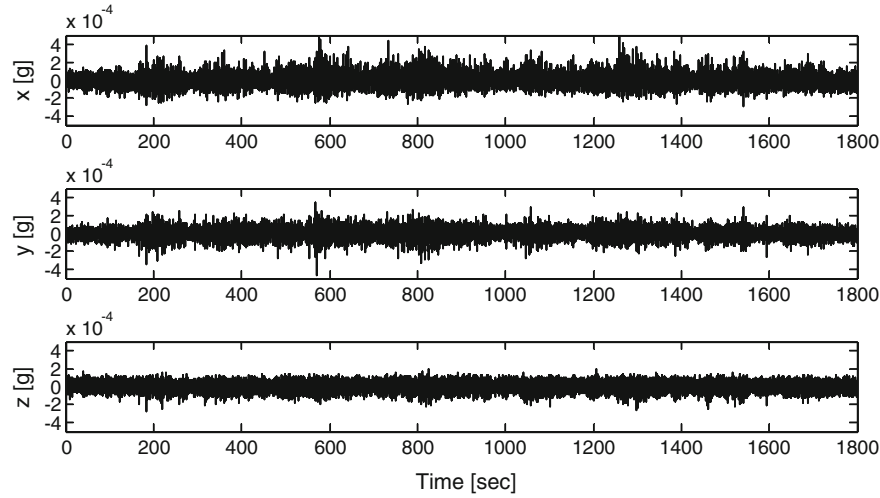


Fig. 13.6 Measured data (detrended) at one corner on the roof of a tall building

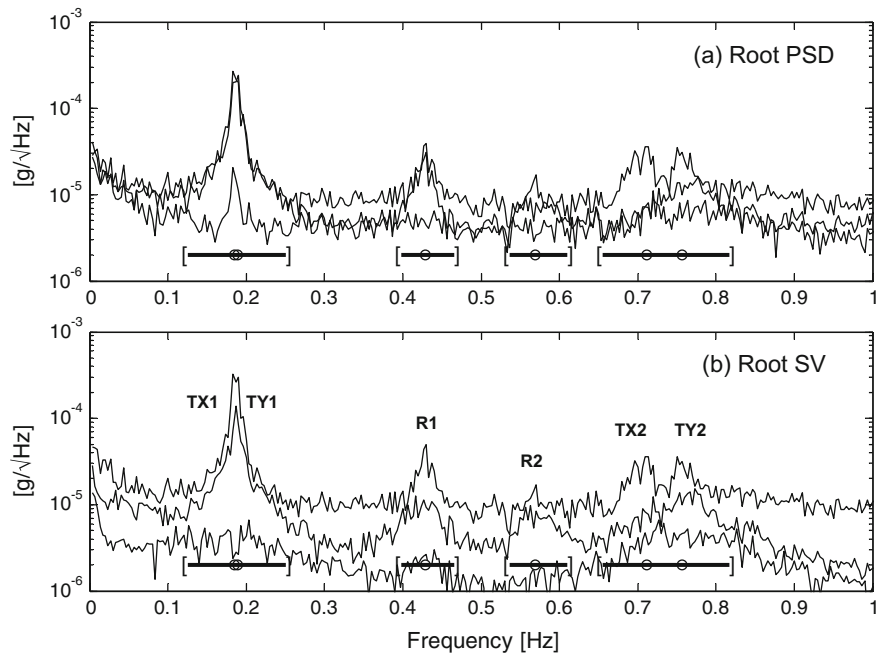


Fig. 13.7 One-sided root PSD spectrum (a) and root SV spectrum (b) of triaxial data at one corner on the roof of a tall building. *Horizontal bar indicates selected band*

Table 13.10 Modal identification results using triaxial data at one corner of tall building

Mode	s/n ratio γ	Frequency f		Damping ζ		Root modal force PSD $\sqrt{S_{ii}}$		Root prediction error PSD $\sqrt{S_e}$		Mode shape ϕ
		MPV (Hz)	c.o.v. (%)	MPV (%)	c.o.v. (%)	MPV $\mu\text{g}/\sqrt{\text{Hz}}$	c.o.v. (%)	MPV $\mu\text{g}/\sqrt{\text{Hz}}$	c.o.v. (%)	
1	10745	0.184	0.17	0.5	30	5.7	6	5.1	3.6	7
2	3968	0.189	0.23	0.9	23	6.0	6			8
3	44	0.428	0.20	1.2	19	1.1	8	6.7	3.1	3
4	6	0.567	0.24	1.1	31	0.3	19	5.9	2.9	11
5	21	0.712	0.33	3.6	12	1.9	7	5.5	2.6	3
6	50	0.758	0.13	0.8	17	0.6	6			7

identified on the same band. To supplement, the MAC (modal assurance criterion) of the most probable mode shapes of modes 1 and 2 is 0.127; they are not orthogonal. The MPV of modal force coherence is $-0.435 - 0.178i$ for the first two modes; and $-0.275 - 0.116i$ for the last two modes.

Results Using Triaxial Data at Four Corners

Table 13.11 shows the modal identification results when triaxial data at four corners (12 measured DOFs) are used. The identified mode shapes are shown in Fig. 13.8. Comments similar to Example 12.5 can be made. The root modal force PSD is roughly doubled because there are now four sensors and their mode shape values are similar in magnitude. The modal s/n ratio is roughly quadrupled. Despite this, there is no significant reduction in the posterior c.o.v.s. One exception is mode 4 where the damping c.o.v. reduces from 31 to 16%. This is partly attributed to the increase in the s/n ratio from a low value before. ■

Table 13.11 Modal identification results using triaxial data at four corners on the roof of tall building

Mode	s/n ratio γ	Frequency f		Damping ζ		Root modal force PSD $\sqrt{S_{ii}}$		Root prediction error PSD $\sqrt{S_e}$		Mode shape ϕ
		MPV (Hz)	c.o.v. (%)	MPV (%)	c.o.v. (%)	MPV $\mu\text{g}/\sqrt{\text{Hz}}$	c.o.v. (%)	MPV $\mu\text{g}/\sqrt{\text{Hz}}$	c.o.v. (%)	
1	45885	0.184	0.18	0.6	28	12.6	6	4.5	1.1	7
2	18210	0.189	0.24	1.1	20	13.4	5			8
3	245	0.427	0.19	1.3	17	2.1	6	4.9	1.4	3
4	48	0.568	0.22	1.8	16	0.9	8	3.5	1.3	5
5	217	0.711	0.24	2.8	10	2.9	5	3.6	0.9	3
6	532	0.758	0.12	1.0	13	1.6	4			4

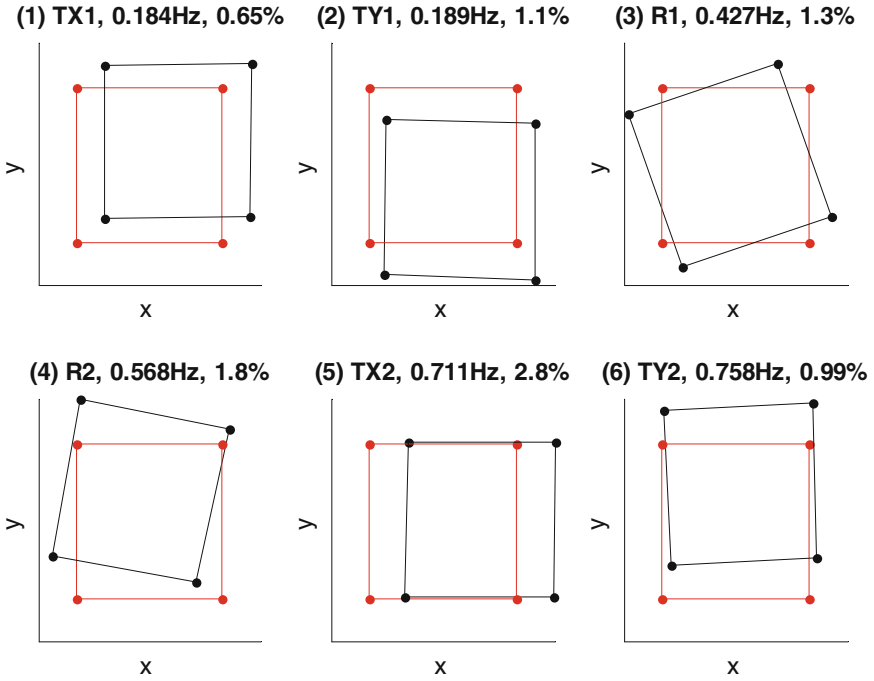


Fig. 13.8 Identified mode shapes (MPV) using triaxial data at four corners on the roof of tall building. Title above each plot shows the mode number, nature, frequency and damping ratio (MPV)

References

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Chapter 14

Multi-setup Problem

Abstract This chapter discusses operational modal analysis using data from multiple setups, each covering a possibly different set of measured degrees of freedom. Motivated from practical applications, the focus is on obtaining a ‘global mode shape’ covering a potentially large number of degrees of freedom by incorporating the information from different setups that are measured with a small number of sensors. Least squares methods based only on the ‘local mode shapes’ in different setups are first discussed, which offer a mathematically simple yet reliable solution for modes with high signal-to-noise ratios. Bayesian algorithm is then developed to offer a more robust solution, incorporating all information in different setups and accounting for their data quality. The asymptotic behavior of the MPV for modes with high signal-to-noise ratio is analyzed, which can be mathematically connected with the least squares solution. Analytical formulas are derived for systematically computing the covariance matrix of parameters given the measured data. Field data examples are presented to illustrate the practical aspects of ambient vibration tests with multiple setups and their modal identification.

Keywords Local least squares • Global least squares • Footbridge • Coupled slab • Velodrome

In this chapter we discuss efficient methods for Bayesian operational modal analysis (OMA) using data from multiple setups performed in different time periods and covering possibly different sets of DOFs. This context is demanded by practical situations where it is desired to obtain the ‘global mode shape’ of a structure with more measured DOFs than the number of synchronous data channels. The global mode shape is the main target, as other modal properties (e.g., natural frequency, damping ratio) can otherwise be identified separately using the data in individual setups.

This chapter is structured somewhat differently from the previous two. It first introduces (non-Bayesian) least squares methods for assembling the global mode shape directly from local ones. These heuristic methods are easier to understand and allow one to appreciate the basic nature of the mode shape assembly problem. They yield a global mode shape asymptotically the same as the most probable mode

shape in a Bayesian approach when the quality of local mode shapes is high. As least squares methods only make use of local mode shapes, they ignore the information in other modal properties, which matters when the quality of data in some setup is poor. The Bayesian algorithm developed in this chapter is limited to well-separated modes. The techniques for determining the posterior statistics leverage on those for single-setup problem (Chap. 12). Efficient algorithm for multiple modes with multiple setup data has not been developed.

We first recall the problem context from Sect. 10.3, simplifying for the case of a single mode dominating the selected frequency band. Ambient vibration data is collected from q setups, each performed in different time periods and measuring a possibly different set of DOFs. Modal properties other than mode shape can possibly vary in different setups. The local mode shape in Setup r , i.e., confined to the measured DOFs in the setup, is denoted by $\mathbf{v}_r$ ($n_r \times 1$). The global mode shape $\boldsymbol{\phi}$ ($n \times 1$) covers the measured DOFs in all setups. It is scaled to have unit norm, i.e., $\boldsymbol{\phi}^T \boldsymbol{\phi} = 1$. Each $\mathbf{v}_r$ is related to $\boldsymbol{\phi}$ by $\mathbf{v}_r = \mathbf{L}_r \boldsymbol{\phi}$, where $\mathbf{L}_r$ ($n_r \times n$) is a selection matrix. The (i, j) -entry of $\mathbf{L}_r$ is 1 if the i th data channel in Setup r measures DOF j of $\boldsymbol{\phi}$. Different setups are connected through some ‘reference DOFs’ they have measured in common, although the same set of reference DOFs need not appear in all setups. The basic mechanism of mode shape assembly is to enforce a common scaling among the local mode shapes via the reference DOFs.

14.1 Local Least Squares

Least squares approach provides a heuristic means for determining the global mode shape from local ones. Let $\tilde{\mathbf{v}}_r$ ($n_r \times 1$) be the local mode shape identified using the data in Setup r , with unit norm, i.e., $\tilde{\mathbf{v}}_r^T \tilde{\mathbf{v}}_r = 1$. It can be the most probable mode shape obtained from a single-setup algorithm in Chaps. 12 or 13, or more generally the mode shape estimate from a non-Bayesian method. Given $\{\tilde{\mathbf{v}}_r\}_{r=1}^q$, the ‘best’ global mode shape $\tilde{\boldsymbol{\phi}}$ is determined so that its local counterparts $\{\mathbf{L}_r \tilde{\boldsymbol{\phi}}\}_{r=1}^q$ are ‘closest’ to $\{\tilde{\mathbf{v}}_r\}_{r=1}^q$ in a predefined least squares sense. This only makes use of the identified local mode shapes, regardless of whether the mode is well-separated from others or not. Simplicity is a clear advantage, at the expense of being heuristic (e.g., why ‘square’) and ignoring the information of other modal properties that are potentially useful.

In the ‘local least squares method’, the identified mode shape of one setup is scaled relative to another so that their common DOFs agree in a least squares sense. Applying this to different pairs of setups and combining the results from all setups gives the global mode shape. Without loss of generality, let the setups be 1 and 2, and let $\tilde{\mathbf{u}}$ and $\tilde{\mathbf{v}}$ be respectively their mode shapes confined to the reference DOFs they share in common. If one takes Setup 1 as reference, then $\tilde{\mathbf{v}}$ is scaled by c so that $c\tilde{\mathbf{v}}$ is closest to $\tilde{\mathbf{u}}$ in a least squares sense. That is, c minimizes the following objective function:

$$J(c) = (c\tilde{\mathbf{v}} - \tilde{\mathbf{u}})^T(c\tilde{\mathbf{v}} - \tilde{\mathbf{u}}) = (\tilde{\mathbf{v}}^T\tilde{\mathbf{v}})c^2 - 2(\tilde{\mathbf{v}}^T\tilde{\mathbf{u}})c + (\tilde{\mathbf{u}}^T\tilde{\mathbf{u}}) \tag{14.1}$$

Solving for $dJ/dc = 0$ gives the least squares estimate of c :

$$\tilde{c} = \frac{\tilde{\mathbf{v}}^T\tilde{\mathbf{u}}}{\tilde{\mathbf{v}}^T\tilde{\mathbf{v}}} \quad (\tilde{\mathbf{u}} = \text{reference}) \tag{14.2}$$

Check that $\partial^2 J/\partial c^2 = 2(\tilde{\mathbf{v}}^T\tilde{\mathbf{v}}) > 0$ and so $\tilde{c}$ minimizes J . To obtain the global mode shape comprising the DOFs of Setups 1 and 2, the mode shape value of a reference DOF is determined as the average of the two setups. The mode shape value at a non-reference DOF is simply equal to the scaled value. The global mode shape is normalized to have unit norm.

Example 14.1 (Six-storied building, local least squares) Revisit the multi-setup scenario in Example 10.3, now shown in Fig. 14.1. Suppose the data in each setup was analyzed individually, yielding the local mode shapes in Table 14.1 that have been scaled to have unit norm. To obtain the global mode shape, we take Setup 2 (say) as reference and scale the mode shape of Setup 1 so that the common DOFs 3 and 6 best fit those of Setup 2. Using (14.2), $\tilde{\mathbf{u}} = [0.420, 0.804]^T$ and $\tilde{\mathbf{v}} = [-0.454, -0.642]^T$, which gives $\tilde{c} = -1.143$. The negative sign reflects that the local mode shapes of Setups 1 and 2 are given in opposite sense. The scaled mode shape of Setup 1 is shown in Table 14.2, which is now in the same sense of Setup 2. Its norm shown at the bottom is equal to the absolute value of the

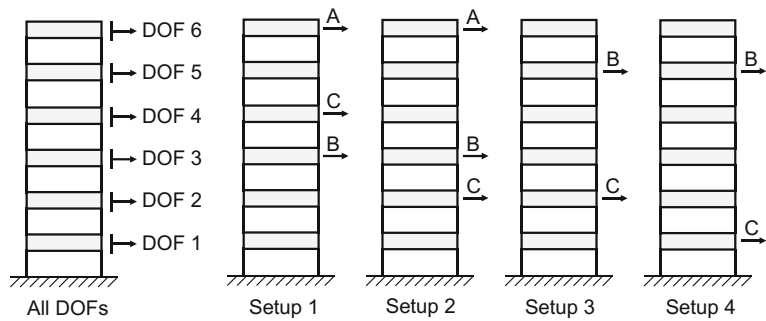


Fig. 14.1 Setup plan of six-storied building example

Table 14.1 Local mode shape (with unit norm) in each setup

DOF/Setup	1	2	3	4
1				0.284
2		0.421	0.470	
3	-0.454	0.420		
4	-0.618			
5			0.883	0.959
6	-0.642	0.804		
Norm	1	1	1	1

Table 14.2 Scaled local mode shapes and global mode shape (*last column*)

DOF\Setup	1	2	3	4	Averaged	Normalized
1				0.234	0.234	0.159
2		0.421	0.421		0.421	0.286
3	0.519	0.420			0.469	0.319
4	0.707				0.707	0.480
5			0.791	0.791	0.791	0.537
6	0.734	0.804			0.769	0.522
Norm	1.143	1.000	0.896	0.825	1.472	1.000

multiplier because the original local mode shape in Table 14.1 has unit norm. No scaling is applied to Setup 2 (reference) and so its columns in Tables 14.1 and 14.2 are the same. Setup 3 is also scaled using Setup 2 as reference, where the common DOF is 2. Using Eq. (14.2), $\tilde{\mathbf{u}} = 0.421$ and $\tilde{\mathbf{v}} = 0.470$, which gives $\tilde{c} = 0.896$. Note that this is simply the ratio $0.421/0.470$ because DOF 2 is the only common DOF. For Setup 4, it does not have any DOF in common with Setup 2 and so the latter cannot be taken as reference. Instead, Setup 3 is used as reference, with common DOF 5. Since Setup 3 is already scaled, matching at DOF 5 using (14.2) corresponds to $\tilde{\mathbf{u}} = 0.791$ (not 0.883) and $\tilde{\mathbf{v}} = 0.959$, which gives $\tilde{c} = 0.825$. The local mode shapes in Table 14.2 are said to have been scaled to best fit in the local least squares sense described above. The global mode shape is obtained by averaging the mode shape values across the setups. The result is shown under the column ‘Averaged’. Dividing this mode shape by its norm (1.472, shown at the bottom) gives the final assembled global mode shape with unit norm, shown under the column ‘Normalized’.

14.2 Global Least Squares

Local least squares method requires the choice of a reference setup. When there is no setup that shares some DOFs with all other setups, it must be applied repeatedly to different setup pairs, which is not convenient for implementation. The choice of reference setup(s) affects the quality of results. Its non-uniqueness is an indication of a possible suboptimal solution where the information in the local mode shapes has not been fully utilized. These issues motivated the ‘global least squares method’ that aims at fitting the local mode shapes $\{\tilde{\mathbf{v}}_r\}_{r=1}^q$ in an overall sense. Specifically, the global mode shape is determined as the $\boldsymbol{\phi}$ that minimizes the following objective function subjected to unit norm constraint $\boldsymbol{\phi}^T \boldsymbol{\phi} = 1$:

$$J(\boldsymbol{\phi}) = \sum_{r=1}^q w_r (\mathbf{v}_r - \|\mathbf{v}_r\| \tilde{\mathbf{v}}_r)^T (\mathbf{v}_r - \|\mathbf{v}_r\| \tilde{\mathbf{v}}_r) \quad \mathbf{v}_r = \mathbf{L}_r \boldsymbol{\phi} \quad (14.3)$$

where $\{w_r \geq 0\}_{r=1}^q$ are chosen weights with $\sum_{r=1}^q w_r = 1$. In the summand in (14.3), $\mathbf{v}_r = \mathbf{L}_r \boldsymbol{\phi}$ and $\|\mathbf{v}_r\| = \boldsymbol{\phi}^T \mathbf{L}_r^T \mathbf{L}_r \boldsymbol{\phi}$ are functions of $\boldsymbol{\phi}$ but these are not written explicitly for simplicity. The difference $\mathbf{v}_r - \|\mathbf{v}_r\| \tilde{\mathbf{v}}_r$ measures the discrepancy between the local mode shape $\mathbf{v}_r$ and its identified counterpart $\|\mathbf{v}_r\| \tilde{\mathbf{v}}_r$. The latter has been scaled to have the same norm as $\mathbf{v}_r$ (recall that $\|\tilde{\mathbf{v}}_r\| = 1$), ensuring a proper measure of vector proximity.

The norm $\|\mathbf{v}_r\|$ in (14.3) makes the optimal solution of $\boldsymbol{\phi}$ non-trivial to find. This can be resolved by introducing the auxiliary variables $c_r = \|\mathbf{v}_r\|^2$ ($r = 1, \dots, q$). Using Lagrange multiplier method (Sect. C.3), the global mode shape is determined from the stationary solution of the following objective function (Lagrangian):

$$J(\boldsymbol{\phi}, \mathbf{c}, \boldsymbol{\lambda}) = \sum_{r=1}^q w_r \underbrace{(\mathbf{v}_r - \sqrt{c_r} \tilde{\mathbf{v}}_r)^T (\mathbf{v}_r - \sqrt{c_r} \tilde{\mathbf{v}}_r)}_{\text{discrepancy in local mode shape}} + \sum_{r=1}^q \underbrace{\lambda_r (\mathbf{v}_r^T \mathbf{v}_r - c_r)}_{\text{enforce } c_r = \|\mathbf{v}_r\|^2} + \underbrace{\lambda_0 (1 - \boldsymbol{\phi}^T \boldsymbol{\phi})}_{\text{enforce } \|\boldsymbol{\phi}\|^2 = 1} \quad (14.4)$$

$(\mathbf{v}_r = \mathbf{L}_r \boldsymbol{\phi})$

where $\mathbf{c} = [c_1, \dots, c_q]$ and $\boldsymbol{\lambda} = [\lambda_0, \lambda_1, \dots, \lambda_q]$ comprises the Lagrange multipliers.

It has not been possible to determine simultaneously the optimal value of $\{\boldsymbol{\phi}, \mathbf{c}, \boldsymbol{\lambda}\}$ in closed form. The next section shows that the optimal value of each parameter can be obtained in terms of the remaining ones. These partial solutions can be used to determine the optimal solution through an iterative algorithm.

14.2.1 Partial Solutions

The optimal solution of $\{c_r\}_{r=1}^q$ and $\{\lambda_r\}_{r=1}^q$ can be obtained in terms of $\boldsymbol{\phi}$ as follow. Solving for $\partial J / \partial c_r = 0$ from (14.4) gives

$$c_r = \left(\frac{\tilde{\mathbf{v}}_r^T \mathbf{v}_r}{1 - \lambda_r / w_r} \right)^2 \quad (14.5)$$

Combining with the constraint $c_r = \|\mathbf{v}_r\|^2$ gives $\lambda_r = w_r (1 \pm |\tilde{\mathbf{v}}_r^T \mathbf{v}_r| / \|\mathbf{v}_r\|)$. Evaluating $\partial^2 J / \partial c_r^2$ at (14.5) gives $\partial^2 J / \partial c_r^2 = w_r (1 - \lambda_r / w_r)^3 / 2 (\tilde{\mathbf{v}}_r^T \mathbf{v}_r)^2$. Requiring this to be positive (looking for minimum) eliminates the larger root of λ_r . In terms of $\boldsymbol{\phi}$ (through $\mathbf{v}_r = \mathbf{L}_r \boldsymbol{\phi}$), the optimal values of c_r and λ_r are then given by

$$c_r = \|\mathbf{v}_r\|^2 \quad \lambda_r = w_r \left(1 - \frac{|\tilde{\mathbf{v}}_r^T \mathbf{v}_r|}{\|\mathbf{v}_r\|} \right) \quad r = 1, \dots, q \quad (14.6)$$

Since $|\tilde{\mathbf{v}}_r^T \mathbf{v}_r| / \|\mathbf{v}_r\| \leq \|\tilde{\mathbf{v}}_r\| \|\mathbf{v}_r\| / \|\mathbf{v}_r\| = 1$,

$$0 \leq \tilde{\lambda}_r \leq w_r \quad (14.7)$$

To derive the optimal solution of $\boldsymbol{\phi}$ in terms of $\mathbf{c}$ and $\boldsymbol{\lambda}$, rewrite (14.4) explicitly as a quadratic function of $\boldsymbol{\phi}$:

$$J = \boldsymbol{\phi}^T \mathbf{A} \boldsymbol{\phi} + 2\boldsymbol{\phi}^T \mathbf{b} + \lambda_0(1 - \boldsymbol{\phi}^T \boldsymbol{\phi}) + C \quad (14.8)$$

where

$$\mathbf{A}_{n \times n} = \sum_{r=1}^q (w_r + \lambda_r) \mathbf{L}_r^T \mathbf{L}_r \quad \mathbf{b}_{n \times 1} = - \sum_{r=1}^q w_r \sqrt{c_r} \mathbf{L}_r^T \tilde{\mathbf{v}}_r \quad C = \sum_{r=1}^q (w_r - \lambda_r) c_r \quad (14.9)$$

Note that $\mathbf{A}$ is positive definite since $w_r + \lambda_r \geq 0$ and for any $\boldsymbol{\phi}$ there is at least one setup with local mode shape norm $\|\mathbf{v}_r\| = \|\mathbf{L}_r \boldsymbol{\phi}\| > 0$ so that

$$\boldsymbol{\phi}^T \mathbf{A} \boldsymbol{\phi} = \sum_{r=1}^q (w_r + \lambda_r) \boldsymbol{\phi}^T \mathbf{L}_r^T \mathbf{L}_r \boldsymbol{\phi} = \sum_{r=1}^q (w_r + \lambda_r) \|\mathbf{L}_r \boldsymbol{\phi}\|^2 > 0 \quad (14.10)$$

Setting the gradient of J w.r.t. $\boldsymbol{\phi}$ to zero gives a constrained eigenvalue problem:

$$\mathbf{A} \boldsymbol{\phi} + \mathbf{b} = \lambda_0 \boldsymbol{\phi} \quad (14.11)$$

This can be solved by constructing an augmented vector that satisfies the standard eigenvalue problem (Sect. C.4). As a result, for given $\mathbf{c}$ and $\boldsymbol{\lambda}$, the optimal global mode shape can be obtained as the first half of the eigenvector with the smallest real eigenvalue of

$$\mathbf{D}(\mathbf{c}, \boldsymbol{\lambda}) = \begin{bmatrix} \mathbf{A}(\boldsymbol{\lambda}) & \mathbf{b}(\mathbf{c})\mathbf{b}(\mathbf{c})^T \\ \mathbf{I}_n & \mathbf{A}(\boldsymbol{\lambda}) \end{bmatrix}_{2n \times 2n} \quad (14.12)$$

where the dependence of $\mathbf{A}$ on $\boldsymbol{\lambda}$ and $\mathbf{b}$ on $\mathbf{c}$ has been indicated.

14.2.2 Limiting Behavior of Solution

When the identified local mode shapes $\{\tilde{\mathbf{v}}_r\}_{r=1}^q$ match well at the reference DOFs, $\mathbf{v}_r / \|\mathbf{v}_r\| \sim \tilde{\mathbf{v}}_r$ at the optimum. Equation (14.6) then implies $\lambda_r \sim 0$ and (14.5) implies $c_r \sim (\tilde{\mathbf{v}}_r^T \mathbf{v}_r)^2 = (\tilde{\mathbf{v}}_r^T \mathbf{L}_r \boldsymbol{\phi})^2$. Substituting these into (14.9) gives, at the optimum,

$$\mathbf{A} \sim \sum_{r=1}^q w_r \mathbf{L}_r^T \mathbf{L}_r \quad \mathbf{b} \sim - \sum_{r=1}^q w_r \mathbf{L}_r^T \tilde{\mathbf{v}}_r \tilde{\mathbf{v}}_r^T \mathbf{L}_r \boldsymbol{\phi} \quad (14.13)$$

Substituting into (14.8) and optimizing w.r.t. $\boldsymbol{\phi}$ shows that the optimal global mode shape is the eigenvector of the following matrix with the smallest eigenvalue:

$$\mathbf{A}_0 = \sum_{r=1}^q w_r \mathbf{L}_r^T \left(\mathbf{I}_{n_r} - \tilde{\mathbf{v}}_r \tilde{\mathbf{v}}_r^T \right) \mathbf{L}_r \quad (14.14)$$

$n \times n$ $n \times n_r$ $n_r \times n_r$ $n_r \times 1$ $1 \times n_r$ $n_r \times n$

This matrix can be computed directly from the identified local mode shapes $\{\tilde{\mathbf{v}}_r\}_{r=1}^q$. The optimal global mode shape in the limiting case can then be obtained without iterations.

14.2.3 Iterative Algorithm

The global mode shape $\boldsymbol{\phi}$ that best fits the identified local mode shapes $\{\tilde{\mathbf{v}}_r\}_{r=1}^q$ ($\|\tilde{\mathbf{v}}_r\| = 1$) in a global least squares sense can be determined by the following iterative algorithm. In a given setup, parameters not updated are kept at their latest value.

Global Least Squares Algorithm

- Step 0. Set initial guess of $\boldsymbol{\phi}$ as the eigenvector (smallest eigenvalue) of $\mathbf{A}_0$ in (14.14)
 - Step 1. Update $\{c_r\}_{r=1}^q$ and $\{\lambda_r\}_{r=1}^q$ using (14.6)
 - Step 2. Update $\boldsymbol{\phi}$ as the first half of the eigenvector (smallest real eigenvalue) of $\mathbf{D}$ in (14.12).
- Repeat Steps 1 and 2 until convergence.

When the quality of local mode shapes is good (they agree at the common DOFs) the results are insensitive to the choice of weights $\{w_r\}_{r=1}^q$. In the general case, Sect. 14.3.6 suggests that the weights may be chosen proportional to the modal signal-to-noise (s/n) ratio of the setup so that the global least squares solution coincides asymptotically with the most probable global mode shape in the Bayesian approach.

Example 14.2 (Six-storied building, global least squares) Consider applying the global least squares method to Example 14.1. The selection matrices are

$$\begin{aligned} \mathbf{L}_1 &= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} & \mathbf{L}_2 &= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \\ \mathbf{L}_3 &= \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} & \mathbf{L}_4 &= \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned} \quad (14.15)$$

From Table 14.1, the identified local mode shapes (with unit norm) are

$$\tilde{\mathbf{v}}_1 = \begin{bmatrix} -0.642 \\ -0.454 \\ -0.618 \end{bmatrix} \quad \tilde{\mathbf{v}}_2 = \begin{bmatrix} 0.804 \\ 0.420 \\ 0.421 \end{bmatrix} \quad \tilde{\mathbf{v}}_3 = \begin{bmatrix} 0.883 \\ 0.470 \end{bmatrix} \quad \tilde{\mathbf{v}}_4 = \begin{bmatrix} 0.959 \\ 0.284 \end{bmatrix} \quad (14.16)$$

where the local DOFs follow the alphabetical order (A, B, C). Using uniform weights, i.e., $w_r = 1/4$ ($r = 1, \dots, 4$), the iterative algorithm yields the best global mode shape (with unit norm) as

$$\boldsymbol{\phi} = [0.162 \quad 0.286 \quad 0.315 \quad 0.477 \quad 0.546 \quad 0.516]^T.$$

■

14.2.4 Reference Condensation

The global least squares algorithm involves repeated solution (during iterations) of the eigenvalue problem for $\mathbf{D}$ in (14.12) of dimension n (the total number of DOFs), which can be large in applications. It is possible to reduce the problem size by observing the theoretical fact that non-reference DOFs retain their ‘shape’ in the assembled global mode shape (Au 2011). Specifically, let $\boldsymbol{\phi}_R$ ($n'' \times 1$) denote the mode shape containing the reference DOFs (no repetitions) from all setups; $\boldsymbol{\phi}'_r$ denote the local mode shape in Setup r confined to non-reference DOFs and $\tilde{\boldsymbol{\phi}}'_r$ denote its identified counterpart. Note that a reference DOF is one that is measured by more than one data channel, regardless of whether it is in the same or different setups. Suppose the global mode shape is arranged as $\boldsymbol{\phi} = [\boldsymbol{\phi}_R; \boldsymbol{\phi}'_1; \dots; \boldsymbol{\phi}'_q]$. Then it can be reasoned that, for the optimal global mode shape, $\boldsymbol{\phi}'_r = a_r \tilde{\boldsymbol{\phi}}'_r / \|\tilde{\boldsymbol{\phi}}'_r\|$ where a_r is a multiplier to be determined. The optimal global mode shape is then of the form

$$\boldsymbol{\varphi} = \begin{bmatrix} \boldsymbol{\varphi}_R \\ a_1 \tilde{\boldsymbol{\varphi}}'_1 / \|\tilde{\boldsymbol{\varphi}}'_1\| \\ \vdots \\ a_q \tilde{\boldsymbol{\varphi}}'_q / \|\tilde{\boldsymbol{\varphi}}'_q\| \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{I}_{n''} & & \\ & \tilde{\boldsymbol{\varphi}}'_1 / \|\tilde{\boldsymbol{\varphi}}'_1\| & \\ & & \ddots \\ & & & \tilde{\boldsymbol{\varphi}}'_q / \|\tilde{\boldsymbol{\varphi}}'_q\| \end{bmatrix}}_{\mathbf{L}_c} \underbrace{\begin{bmatrix} \boldsymbol{\varphi}_R \\ a_1 \\ \vdots \\ a_q \end{bmatrix}}_{\boldsymbol{\varphi}_c} = \mathbf{L}_c \boldsymbol{\varphi}_c \quad (14.17)$$

$n \times n' \quad n' \times 1$

where n' ($n'' \leq n' \leq n$) is the number of condensed parameters. Determining $\boldsymbol{\varphi}$ then reduces to determining $\boldsymbol{\varphi}_c$. Note that $\boldsymbol{\varphi}'_r$ and a_r can be null (empty) if all DOFs in Setup r are reference DOFs. As $\mathbf{L}_c^T \mathbf{L}_c = \mathbf{I}_{n'}$ (identity matrix), $\boldsymbol{\varphi}_c$ has unit norm because $\boldsymbol{\varphi}_c^T \boldsymbol{\varphi}_c = \boldsymbol{\varphi}_c^T \mathbf{L}_c^T \mathbf{L}_c \boldsymbol{\varphi}_c = \boldsymbol{\varphi}^T \boldsymbol{\varphi} = 1$. Substituting $\boldsymbol{\varphi} = \mathbf{L}_c \boldsymbol{\varphi}_c$ into (14.11) and pre-multiplying by $\mathbf{L}_c^T$ gives a constrained eigenvalue problem of dimension n' :

$$\mathbf{A}_c \boldsymbol{\varphi}_c + \mathbf{b}_c = \lambda_0 \boldsymbol{\varphi}_c \quad (14.18)$$

$$\mathbf{A}_c = \mathbf{L}_c^T \mathbf{A} \mathbf{L}_c \quad \mathbf{b}_c = \mathbf{L}_c^T \mathbf{b} \quad (14.19)$$

$n' \times n' \quad n' \times n \quad n \times n' \quad n' \times 1 \quad n' \times n \quad n \times 1$

where $\mathbf{A}$ and $\mathbf{b}$ are given by (14.9). The solution of $\boldsymbol{\varphi}_c$ is given by the first half of the real eigenvector (smallest real eigenvalue, unit norm) of

$$\mathbf{D}_c = \begin{bmatrix} \mathbf{L}_c^T \mathbf{A} \mathbf{L}_c & \mathbf{L}_c^T \mathbf{b} \mathbf{b}^T \mathbf{L}_c \\ \mathbf{I}_{n'} & \mathbf{L}_c^T \mathbf{A} \mathbf{L}_c \end{bmatrix} \quad (14.20)$$

$2n' \times 2n'$

Step 2 of the algorithm in Sect. 14.2.3 can be modified to first determining $\boldsymbol{\varphi}_c$ and then recovering $\boldsymbol{\varphi} = \mathbf{L}_c \boldsymbol{\varphi}_c$. It is more efficient especially when $n \gg n'$. This condensed procedure assumes that the DOFs are arranged as $\boldsymbol{\varphi} = [\boldsymbol{\varphi}_R; \boldsymbol{\varphi}'_1; \dots; \boldsymbol{\varphi}'_q]$, i.e., reference DOFs first, followed by non-reference DOFs of each setup. In the actual implementation, a re-mapping of DOFs is necessary to fit the analyst's preference of DOF ordering.

In Example 14.1, DOFs 2, 3, 5 and 6 appear in more than one setup and so $\boldsymbol{\varphi}_R = [\phi_2, \phi_3, \phi_5, \phi_6]^T$. For the non-reference DOFs, $\boldsymbol{\varphi}'_1 = [\phi_4]$, $\boldsymbol{\varphi}'_2$ and $\boldsymbol{\varphi}'_3$ are null, and $\boldsymbol{\varphi}'_4 = [\phi_1]$. This gives, in (14.17), $n'' = 4$, $\mathbf{L}_c = \mathbf{I}_6$ and $\boldsymbol{\varphi}_c = [\phi_2, \phi_3, \phi_5, \phi_6, a_1, a_4]^T$. In this example the condensed procedure has no computational advantage because the number of condensed parameters ($n' = 6$) is the same as the total number of DOFs ($n = 6$).

14.3 Bayesian Method

We now consider an algorithm for identifying the global mode shape using data from multiple setups, following a Bayesian approach and assuming a single mode in the selected frequency band. Recall the context from Sect. 10.3. Let

$\{\hat{\mathbf{y}}_j^{(r)}\}_{j=0}^{N_r-1}$ ($n_r \times 1$) be the ambient vibration data in Setup r and $\{\hat{\mathcal{F}}_k^{(r)}\}$ ($n_r \times 1$ complex) be the collection of scaled Fast Fourier Transform (FFT) in the selected band:

$$\hat{\mathcal{F}}_k^{(r)} = \sqrt{\frac{\Delta t_r}{N_r}} \sum_{j=0}^{N_r-1} \hat{\mathbf{y}}_j^{(r)} e^{-2\pi i j k / N_r} \quad (14.21)$$

where Δt_r is the sampling interval. Using the FFTs from all setups, the negative log-likelihood function (NLLF) for identifying the set of modal parameters $\boldsymbol{\theta}$ is

$$L(\boldsymbol{\theta}) = -\ln p(\{\hat{\mathcal{F}}_k^{(1)}\}, \dots, \{\hat{\mathcal{F}}_k^{(q)}\} | \boldsymbol{\theta}) = \sum_{r=1}^q L_r \quad (14.22)$$

where, for Setup r ,

$$L_r = n_r N_f^{(r)} \ln \pi + \sum_k \ln |\mathbf{E}_k^{(r)}| + \sum_k \hat{\mathcal{F}}_k^{(r)*} \mathbf{E}_k^{(r)-1} \hat{\mathcal{F}}_k^{(r)} \quad (14.23)$$

is the NLLF; $N_f^{(r)}$ is the number of FFT points in the selected band;

$$\mathbf{E}_k^{(r)} = S^{(r)} D_k^{(r)} \mathbf{v}_r \mathbf{v}_r^T + S_e^{(r)} \mathbf{I}_{n_r} \quad \mathbf{v}_r = \mathbf{L}_r \boldsymbol{\phi} \quad (14.24)$$

is the theoretical power spectral density (PSD) matrix of data;

$$D_k^{(r)} = \frac{(2\pi f_k^{(r)})^{-2q}}{(1 - \beta_k^{(r)2})^2 + (2\zeta^{(r)} \beta_k^{(r)})^2} \quad \beta_k^{(r)} = \frac{f^{(r)}}{f_k^{(r)}} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (14.25)$$

$f^{(r)}$ is the natural frequency in Hz, $\zeta^{(r)}$ is the damping ratio, $S^{(r)}$ is the modal force PSD and $S_e^{(r)}$ is the prediction error PSD. The set of modal parameters $\boldsymbol{\theta}$ comprises $\{f^{(r)}, \zeta^{(r)}, S^{(r)}, S_e^{(r)}\}_{r=1}^q$ and $\boldsymbol{\phi}$, subjected to unit norm constraint $\boldsymbol{\phi}^T \boldsymbol{\phi} = 1$.

14.3.1 Alternative Form of NLLF

To facilitate determining the posterior MPV of $\boldsymbol{\theta}$, the NLLF is rewritten using similar techniques in Chap. 12. Let

$$\bar{\mathbf{v}}_r = \|\mathbf{v}_r\|^{-1} \mathbf{v}_r \quad (14.26)$$

be the normalized counterpart of $\mathbf{v}_r$; and

$$S'^{(r)} = S^{(r)} \|\mathbf{v}_r\|^2 \quad (14.27)$$

be the modal force PSD of Setup r when the local mode shape is scaled to have unit norm. Note that $S^{(r)}$ is the modal force PSD of Setup r when the global mode shape is scaled to have unit norm. Working with $\{S'^{(r)}\}_{r=1}^q$ instead of $\{S^{(r)}\}_{r=1}^q$ leads to the same form of the NLLF as in a single-setup problem, allowing a similar mathematical treatment.

Using (14.26) and (14.27), (14.24) is rewritten as

$$\mathbf{E}_k^{(r)} = S'^{(r)} D_k^{(r)} \bar{\mathbf{v}}_r \bar{\mathbf{v}}_r^T + S_e^{(r)} \mathbf{I}_{n_r} \quad (14.28)$$

Since $\|\bar{\mathbf{v}}_r\| = 1$, this is of the same form as the theoretical PSD in a single-setup problem and so the results in Sect. 12.1 can be applied to give

$$|\mathbf{E}_k^{(r)}| = \left[S'^{(r)} D_k^{(r)} + S_e^{(r)} \right] S_e^{(r) n_r - 1} \quad (14.29)$$

$$\mathbf{E}_k^{(r)-1} = S_e^{(r)-1} \left\{ \mathbf{I}_{n_r} - \left[1 + \frac{S_e^{(r)}}{S'^{(r)} D_k^{(r)}} \right]^{-1} \bar{\mathbf{v}}_r \bar{\mathbf{v}}_r^T \right\} \quad (14.30)$$

Substituting into (14.23) gives the same alternative form in a single-setup problem:

$$\begin{aligned} L_r = & n_r N_f^{(r)} \ln \pi + \sum_k \ln \left[S'^{(r)} D_k^{(r)} + S_e^{(r)} \right] \\ & + (n_r - 1) N_f^{(r)} \ln S_e^{(r)} + \frac{1}{S_e^{(r)}} \left[\hat{d}^{(r)} - \bar{\mathbf{v}}_r^T \mathbf{A}^{(r)} \bar{\mathbf{v}}_r \right] \end{aligned} \quad (14.31)$$

$$\mathbf{A}_{n_r \times n_r}^{(r)} = \sum_k \left[1 + \frac{S_e^{(r)}}{S'^{(r)} D_k^{(r)}} \right]^{-1} \hat{\mathbf{D}}_{n_r \times n_r}^{(r)} \quad (14.32)$$

$$\hat{\mathbf{D}}_{n_r \times n_r}^{(r)} = \text{Re} \left[\hat{\mathcal{F}}_{n_r \times 1}^{(r)} \hat{\mathcal{F}}_{1 \times n_r}^{(r)*} \right] \quad \hat{d}_{\text{scalar}}^{(r)} = \sum_k \hat{\mathcal{F}}_{1 \times n_r}^{(r)*} \hat{\mathcal{F}}_{n_r \times 1}^{(r)} \quad (14.33)$$

Using $L = \sum_{r=1}^q L_r$, substituting $\bar{\mathbf{v}}_r = \mathbf{L}_r \boldsymbol{\phi} / \|\mathbf{v}_r\|$ and collecting terms leads to a form for determining the most probable global mode shape:

$$\begin{aligned}
L = & (\ln \pi) \sum_{r=1}^q n_r N_f^{(r)} + \sum_{r=1}^q (n_r - 1) N_f^{(r)} \ln S_e^{(r)} + \sum_{r=1}^q \frac{\hat{d}^{(r)}}{S_e^{(r)}} \\
& + \sum_{r=1}^q \sum_k \ln \left[S^{(r)} D_k^{(r)} + S_e^{(r)} \right] - \boldsymbol{\phi}^T \mathbf{A} \boldsymbol{\phi}
\end{aligned} \tag{14.34}$$

$$\mathbf{A}_{n \times n} = \sum_{r=1}^q \|\mathbf{v}_r\|^{-2} S_e^{(r)-1} \mathbf{L}_r^T \mathbf{A}^{(r)} \mathbf{L}_r \tag{14.35}$$

$n \times n_r \quad n_r \times n_r \quad n_r \times n$

14.3.2 Partial MPV of Global Mode Shape

The NLLF in (14.34) depends on $\boldsymbol{\phi}$ through $\boldsymbol{\phi}^T \mathbf{A} \boldsymbol{\phi}$. This term is not quadratic in $\boldsymbol{\phi}$ because $\mathbf{A}$ in (14.35) depends on $\boldsymbol{\phi}$ through $\|\mathbf{v}_r\|^2 = \boldsymbol{\phi}^T \mathbf{L}_r^T \mathbf{L}_r \boldsymbol{\phi}$. To obtain the MPV of $\boldsymbol{\phi}$ in terms of the remaining parameters, introduce the auxiliary parameters $\{c_r\}_{r=1}^q$ to replace $\{\|\mathbf{v}_r\|^2\}_{r=1}^q$ so that $\mathbf{A}$ reads

$$\mathbf{A} = \sum_{r=1}^q c_r^{-1} S_e^{(r)-1} \mathbf{L}_r^T \mathbf{A}^{(r)} \mathbf{L}_r \tag{14.36}$$

The NLLF should now be minimized w.r.t. $\{f^{(r)}, \zeta^{(r)}, S^{(r)}, S_e^{(r)}\}_{r=1}^q$, $\boldsymbol{\phi}$ and $\{c_r\}_{r=1}^q$ subjected to constraints $\boldsymbol{\phi}^T \boldsymbol{\phi} = 1$ and

$$c_r = \boldsymbol{\phi}^T \mathbf{L}_r^T \mathbf{L}_r \boldsymbol{\phi} \quad r = 1, \dots, q \tag{14.37}$$

The constraints can be incorporated by considering the Lagrangian:

$$J = L + \lambda_0 (1 - \boldsymbol{\phi}^T \boldsymbol{\phi}) + \sum_{r=1}^q \lambda_r (\boldsymbol{\phi}^T \mathbf{L}_r^T \mathbf{L}_r \boldsymbol{\phi} - c_r) \tag{14.38}$$

where $\{\lambda_r\}_{r=0}^q$ are Lagrange multipliers.

The MPV of $\boldsymbol{\phi}$ and $\{\lambda_r\}_{r=1}^q$ can be determined analytically in terms of the remaining parameters. Setting the gradient of J w.r.t. $\boldsymbol{\phi}$ to zero gives the standard eigenvalue problem

$$\mathbf{B} \boldsymbol{\phi} = \lambda_0 \boldsymbol{\phi} \tag{14.39}$$

$$\mathbf{B} = \left(\sum_{r=1}^q \lambda_r \mathbf{L}_r^T \mathbf{L}_r \right) - \mathbf{A} \tag{14.40}$$

where $\mathbf{A}$ is given by (14.36). Note that $\mathbf{B}$ depends on $\{c_r\}_{r=1}^q$ (through $\mathbf{A}$) and $\{\lambda_r\}_{r=1}^q$. Evaluating L at an eigenvector of $\mathbf{B}$ gives a value proportional to λ_0 . This implies that the MPV of $\boldsymbol{\phi}$ is the eigenvector of $\mathbf{B}$ with the smallest eigenvalue.

Setting the derivative of J w.r.t. c_r and solving for λ_r gives $\lambda_r = \boldsymbol{\phi}^T \mathbf{L}_r^T \mathbf{A}^{(r)} \mathbf{L}_r \boldsymbol{\phi} / c_r^2 S_e^{(r)}$. Substituting $c_r = \boldsymbol{\phi}^T \mathbf{L}_r^T \mathbf{L}_r \boldsymbol{\phi}$ gives the MPV in terms of other parameters

$$\lambda_r = \frac{\mathbf{v}_r^T \mathbf{A}^{(r)} \mathbf{v}_r}{S_e^{(r)} (\mathbf{v}_r^T \mathbf{v}_r)^2} \quad \mathbf{v}_r = \mathbf{L}_r \boldsymbol{\phi} \quad r = 1, \dots, q \quad (14.41)$$

14.3.3 Algorithm for MPV

The MPV of $\boldsymbol{\theta} = \{\{f^{(r)}, \zeta^{(r)}, S'^{(r)}, S_e^{(r)}\}_{r=1}^q, \boldsymbol{\phi}\}$ can be determined by the iterative algorithm below. After that, $\{S^{(r)}\}_{r=1}^q$ can be recovered from $S^{(r)} = S'^{(r)} / \|\mathbf{L}_r \boldsymbol{\phi}\|^2$. In a given step, parameters not updated are kept at their latest value.

Iterative Algorithm for MPV

Step 0. Set initial guess of $\{f^{(r)}, \zeta^{(r)}, S'^{(r)}, S_e^{(r)}\}_{r=1}^q, \boldsymbol{\phi}, \{\lambda_r\}_{r=1}^q$ (see Sect. 14.3.5).

Step 1. For $r = 1, \dots, q$, update $\{f^{(r)}, \zeta^{(r)}, S'^{(r)}, S_e^{(r)}\}$ by minimizing L_r in (14.31).

Step 2. Update $\boldsymbol{\phi}$ as the eigenvector (smallest eigenvalue, unit norm) of $\mathbf{B}$ in (14.40).

Step 3. Update $\{\lambda_r\}_{r=1}^q$ using (14.41) and $\{c_r\}_{r=1}^q$ using (14.37).

Repeat Steps 1 to 3 until convergence.

This algorithm is a modified version of that in Au and Zhang (2012). By working with $S'^{(r)}$ instead of $S^{(r)}$, it involves in Step 1 optimizing the same NLLF as in a single-setup problem (Chap. 12).

14.3.4 High s/n Asymptotic MPV

As in a single-setup problem, the MPV of modal parameters for multi-setup problem exhibits characteristic behavior when the modal signal-to-noise (s/n) ratio is high, in the sense that in the selected frequency band,

$$\gamma_k^{(r)} = \frac{S'^{(r)} D_k^{(r)}}{S_e^{(r)}} \gg 1 \quad r = 1, \dots, q \quad (14.42)$$

Since $L = \sum_{r=1}^q L_r$ and each NLLF L_r depends on $S'^{(r)}$ and $S_e^{(r)}$ in the same manner as in the case of single-setup problem, one can apply the results in Sect. 12.3 to obtain the asymptotic MPV as

$$\hat{S}_e^{(r)} \sim \frac{d_r - \hat{d}^{(r)}}{(n_r - 1)N_f^{(r)}} \quad \hat{S}'^{(r)} \sim \frac{1}{N_f^{(r)}} \sum_k \frac{\hat{d}_k^{(r)}}{D_k^{(r)}} \quad (14.43)$$

where $\hat{d}_k^{(r)} = \hat{\mathbf{v}}_r^T \mathbf{D}_k^{(r)} \hat{\mathbf{v}}_r / \hat{\mathbf{v}}_r^T \hat{\mathbf{v}}_r$ and $\hat{\mathbf{v}}_r = \mathbf{L}_r \hat{\boldsymbol{\phi}}$; a hat (' $\hat{}$ ') denotes that the quantity is at the MPV. These are the same as those in a single-setup problem.

Determining the asymptotic MPV of the global mode shape requires a somewhat different argument from that for single-setup problem because of the Lagrange multiplier terms. Applying (14.42) on (14.32),

$$\mathbf{A}^{(r)} \sim \mathbf{A}_0^{(r)} = \sum_k \hat{\mathbf{D}}_k^{(r)} \quad (14.44)$$

Let $\{\alpha_i^{(r)}\}_{i=1}^{n_r}$ be the eigenvalues (in ascending order of magnitude) of $\mathbf{A}_0^{(r)}$. As in a single-setup problem, it can be reasoned that asymptotically $\hat{\mathbf{v}}_r$ is the eigenvector with the largest eigenvalue of $\mathbf{A}_0^{(r)}$. Using (14.41) at the MPV,

$$\hat{\lambda}_r \sim \frac{\alpha_{n_r}^{(r)}}{\hat{S}_e^{(r)} \|\hat{\mathbf{v}}_r\|^2} \quad r = 1, \dots, q \quad (14.45)$$

since $\hat{\mathbf{v}}_r^T \mathbf{A}_0^{(r)} \hat{\mathbf{v}}_r / \hat{\mathbf{v}}_r^T \hat{\mathbf{v}}_r = \alpha_{n_r}^{(r)}$. Substituting into (14.40) and using

$$\mathbf{A} \sim \sum_{r=1}^q \|\hat{\mathbf{v}}_r\|^{-2} \hat{S}_e^{(r)-1} \mathbf{L}_r^T \mathbf{A}_0^{(r)} \mathbf{L}_r \quad (14.46)$$

implied from (14.35), we have

$$\mathbf{B} \sim \sum_{r=1}^q \hat{S}_e^{(r)-1} \|\hat{\mathbf{v}}_r\|^{-2} \mathbf{L}_r^T \left[\alpha_{n_r}^{(r)} \mathbf{I}_{n_r} - \mathbf{A}_0^{(r)} \right] \mathbf{L}_r \quad (14.47)$$

The bracketed term in the summand is positive semi-definite because all eigenvalues of $\mathbf{A}_0^{(r)}$ are no greater than $\alpha_{n_r}^{(r)}$. Its smallest eigenvalue is zero with eigenvector $\hat{\mathbf{v}}_r = \mathbf{L}_r \hat{\boldsymbol{\phi}}$. It follows that $\mathbf{B} \hat{\boldsymbol{\phi}} \sim \mathbf{0}$ and so asymptotically the most probable global mode shape $\hat{\boldsymbol{\phi}}$ is a null vector of $\mathbf{B}$.

14.3.5 Initial Guess

Initial guess for the iterative algorithm in Sect. 14.3.3 can be set as follow. The initial guess for the natural frequency $f^{(r)}$ can be picked from the singular value (SV) spectrum of data in Setup r . The initial guess for the damping ratio can be set as 1%. The initial guess for the global mode shape can be taken as the eigenvector (unit norm) of $\mathbf{B}_0$ with the smallest eigenvalue:

$$\mathbf{B}_0 = \sum_{r=1}^q \mathbf{L}_r^T \left[\alpha_{n_r}^{(r)} \mathbf{I}_{n_r} - \mathbf{A}_0^{(r)} \right] \mathbf{L}_r \quad (14.48)$$

where $\alpha_{n_r}^{(r)}$ is the largest eigenvalue of $\mathbf{A}_0^{(r)} = \sum_k \hat{\mathbf{D}}_k^{(r)}$ in (14.44). The matrix $\mathbf{B}_0$ is adapted from $\mathbf{B}$ in (14.47) by removing the unknown factor $\hat{S}_e^{(r)} \|\hat{\mathbf{v}}_r\|^2$, which asymptotically does not affect the null vector. The initial guess for the local mode shapes can be extracted from that for the global mode shape. Substituting the initial guess for the local mode shapes into (14.43) gives the initial guess for $S^{(r)}$ and $S_e^{(r)}$. The initial guess for λ_r can be determined from (14.45).

14.3.6 Asymptotic Weight for Global Least Squares

For high modal s/n ratio, the most probable global mode shape is closely related to the global least squares estimate in Sect. 14.2. The relationship leads to a simple choice for the weights in the global least squares method. Let $\{\mathbf{b}_i^{(r)}\}_{i=1}^{n_r}$ be the eigenvectors (unit norm, in ascending order of eigenvalues) of $\mathbf{A}_0^{(r)} = \sum_k \hat{\mathbf{D}}_k^{(r)}$ in (14.44). Then $\mathbf{b}_{n_r}^{(r)} = \hat{\mathbf{v}}_r / \|\hat{\mathbf{v}}_r\| = \hat{\mathbf{v}}_r$ (say). Substituting $\mathbf{A}_0^{(r)} = \alpha_{n_r}^{(r)} \hat{\mathbf{v}}_r \hat{\mathbf{v}}_r^T + \sum_{i=1}^{n_r-1} \alpha_i^{(r)} \mathbf{b}_i^{(r)} \mathbf{b}_i^{(r)T}$ into (14.47),

$$\mathbf{B} \sim \sum_{r=1}^q \frac{\alpha_{n_r}^{(r)}}{S_e^{(r)} \|\hat{\mathbf{v}}_r\|^2} \mathbf{L}_r^T (\mathbf{I}_{n_r} - \hat{\mathbf{v}}_r \hat{\mathbf{v}}_r^T) \mathbf{L}_r - \sum_{r=1}^q \sum_{i=1}^{n_r-1} \frac{\alpha_i^{(r)}}{S_e^{(r)} \|\hat{\mathbf{v}}_r\|^2} \mathbf{L}_r^T \mathbf{b}_i^{(r)} \mathbf{b}_i^{(r)T} \mathbf{L}_r \quad (14.49)$$

Post-multiplying the second sum by $\hat{\boldsymbol{\phi}}$ gives a zero vector because $\mathbf{L}_r \hat{\boldsymbol{\phi}} = \hat{\mathbf{v}}_r = \|\hat{\mathbf{v}}_r\| \mathbf{b}_{n_r}^{(r)}$ is orthogonal to $\{\mathbf{b}_i^{(r)}\}_{i=1}^{n_r-1}$. This implies that asymptotically the most probable global mode shape is a null vector of the first sum. Comparing the first sum with (14.14) and taking $\tilde{\mathbf{v}}_r = \hat{\mathbf{v}}_r$ suggests that the weights $\{w_r\}_{r=1}^q$ in the global least squares method may be chosen as

$$w_r \propto \frac{\alpha_{n_r}^{(r)}}{S_e^{(r)} \|\hat{\mathbf{v}}_r\|^2} \quad (14.50)$$

so that its estimate is asymptotically the same as the most probable global mode shape in the Bayesian approach. The quantities on the RHS can all be determined by analyzing individually the data in Setup r .

14.3.7 Posterior Covariance Matrix

Under a Gaussian approximation of the posterior PDF, the posterior covariance matrix of the set of modal parameters $\boldsymbol{\theta}$ ($n_\theta \times 1$) satisfying constraints is given by (Sect. 11.2)

$$\hat{\mathbf{C}} = (\nabla \hat{\mathbf{v}}_c)_{n_\theta \times n_\theta} (\nabla^2 \hat{L}_c)_{n_\theta \times p}^+ (\nabla \hat{\mathbf{v}}_c)_{p \times n_\theta}^T \quad (14.51)$$

where ‘+’ denotes the pseudo-inverse, i.e., ignoring zero eigenvalues from constraint singularity; $\nabla^2 \hat{L}_c$ is the Hessian of $L_c(\mathbf{u}) = L(\mathbf{v}_c(\mathbf{u}))$ w.r.t. a set of free parameters $\mathbf{u}$ ($p \times 1$); $\mathbf{v}_c(\mathbf{u})$ maps $\mathbf{u}$ to $\boldsymbol{\theta}$ that always satisfies the constraints $G_i(\mathbf{v}_c(\mathbf{u})) = 0$ ($i = 1, \dots, n_c$); $\mathbf{G}(\boldsymbol{\theta}) = [G_1(\boldsymbol{\theta}), \dots, G_{n_c}(\boldsymbol{\theta})]^T$. A hat ‘^’ on quantity denotes that it is evaluated at the MPV. Using the formula based on Lagrange multiplier (Sect. 11.2.3),

$$\nabla^2 \hat{L}_c = (\nabla \hat{\mathbf{v}}_c)_{p \times n_\theta}^T \left(\nabla^2 \hat{L} + \sum_{i=1}^{n_c} \hat{\lambda}_i \nabla^2 \hat{G}_i \right)_{n_\theta \times p} \nabla \hat{\mathbf{v}}_c \quad (14.52)$$

where $\nabla^2 \hat{L}$ is the Hessian of the NLLF in (14.22) (ignoring constraints); and

$$\hat{\boldsymbol{\lambda}} = \begin{bmatrix} \hat{\lambda}_1 \\ \vdots \\ \hat{\lambda}_{n_c} \end{bmatrix} = - \underbrace{[\nabla \hat{\mathbf{G}}]_{n_c \times n_\theta} (\nabla \hat{\mathbf{G}})^T_{n_\theta \times n_c}}_{n_c \times n_c}^{-1} (\nabla \hat{\mathbf{G}})_{n_c \times n_\theta} (\nabla \hat{L})_{n_\theta \times 1}^T \quad (14.53)$$

is a vector of Lagrange multipliers at the MPV. Note that $\{\hat{\lambda}_i\}_{i=1}^{n_c}$ here are not those in Sect. 14.3.2 when obtaining the MPV.

The main derivation and programing effort for calculating $\hat{\mathbf{C}}$ stems from $\nabla^2 \hat{L}$. This depends on the definition of $\boldsymbol{\theta}$, which affects how it appears in $L(\boldsymbol{\theta})$ and the definitions of $\mathbf{v}_c(\mathbf{u})$ and $\mathbf{G}(\boldsymbol{\theta})$. It is possible to leverage on the computer code for single-setup problem (Chap. 12), by making use of the additive relationship:

$$L(\boldsymbol{\theta}) = \sum_{r=1}^q L_r(\boldsymbol{\theta}_r) \quad (14.54)$$

where L_r is the NLLF in (14.31) that depends only on the parameters in Setup r , i.e., $\boldsymbol{\theta}_r = \{f^{(r)}, \zeta^{(r)}, S^{(r)}, S_e^{(r)}, \mathbf{v}_r\}$. Define

$$\boldsymbol{\varpi}_r = [f^{(r)}, \zeta^{(r)}, S^{(r)}, S_e^{(r)}]^T \quad (14.55)$$

$$\begin{aligned}
\mathbf{u} &= \begin{bmatrix} \varpi_1 \\ \vdots \\ \varpi_q \\ \boldsymbol{\phi} \end{bmatrix} & \boldsymbol{\theta} &= \begin{bmatrix} \varpi_1 \\ \vdots \\ \varpi_q \\ \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_q \\ \boldsymbol{\phi} \end{bmatrix} & \mathbf{v}_c(\mathbf{u}) &= \begin{bmatrix} \varpi_1 \\ \vdots \\ \varpi_q \\ \mathbf{L}_1 \boldsymbol{\phi} / \|\boldsymbol{\phi}\| \\ \vdots \\ \mathbf{L}_q \boldsymbol{\phi} / \|\boldsymbol{\phi}\| \\ \boldsymbol{\phi} / \|\boldsymbol{\phi}\| \end{bmatrix} & \mathbf{G}(\boldsymbol{\theta}) &= \begin{bmatrix} \mathbf{v}_1 - \mathbf{L}_1 \boldsymbol{\phi} \\ \vdots \\ \mathbf{v}_q - \mathbf{L}_q \boldsymbol{\phi} \\ 1 - \boldsymbol{\phi}^T \boldsymbol{\phi} \end{bmatrix}
\end{aligned} \tag{14.56}$$

Here, $\mathbf{u}$ has $p = 4q + n$ parameters; $\boldsymbol{\theta}$ has $n_\theta = 4q + \Sigma n_r + n$ parameters ($\Sigma n_r = n_1 + \dots + n_r$); and $\mathbf{G}$ contains $n_c = \Sigma n_r + 1$ constraints. Check that $\mathbf{G}(\mathbf{v}_c(\mathbf{u})) = \mathbf{0}$ for any $\mathbf{u}$. Since $\{\mathbf{v}_r\}_{r=1}^q$ are explicitly contained in $\boldsymbol{\theta}$, they can be treated as independent variables when taking partial derivatives of L . The partial derivative of $L = \sum_{r=1}^q L_r$ w.r.t. $\boldsymbol{\phi}$ is zero because L_r depends on $\boldsymbol{\phi}$ only through $\mathbf{v}_r$, but $\mathbf{v}_r$ has already been included in $\boldsymbol{\theta}$. The global mode shape $\boldsymbol{\phi}$ is deliberately included in $\boldsymbol{\theta}$, for otherwise it is difficult to express systematically in $\mathbf{G}$ the constraints among the local mode shapes $\{\mathbf{v}_r\}_{r=1}^q$. Since $\bar{\mathbf{v}}_r = \mathbf{v}_r / \|\mathbf{v}_r\|$ instead of $\mathbf{v}_r$ appears in the expression of L_r in (14.31), the gradient and Hessian of L_r w.r.t. $\boldsymbol{\theta}_r = \{f^{(r)}, \zeta^{(r)}, S^{(r)}, S_e^{(r)}, \mathbf{v}_r\}$ is equal to those of the NLLF for single-setup problem *with unit norm constraint* in the local mode shape. This can be computed by exactly the same computer code developed for single-setup problem.

To clarify, apply the procedure in Sect. 12.4 for Setup r to obtain the Hessian of L_r w.r.t. $\boldsymbol{\theta}_r$ with constraint $\|\mathbf{v}_r\| = 1$. Expand it (filling zeros) to a $n_\theta \times n_\theta$ matrix so that it corresponds to the Hessian of L_r w.r.t. $\boldsymbol{\theta}$. The sum of the resulting matrices ($r = 1, \dots, q$) is the Hessian of L w.r.t. $\boldsymbol{\theta}$, whose pseudo-inverse gives the posterior covariance matrix of $\boldsymbol{\theta}$.

The terms in (14.52) are determined as follow. Direct differentiation and evaluating at the MPV gives

$$\nabla \hat{\mathbf{v}}_c = \begin{bmatrix} \mathbf{I}_{4q \times 4q} \\ \underbrace{\mathbf{L}_1(\mathbf{I}_n - \hat{\boldsymbol{\phi}}\hat{\boldsymbol{\phi}}^T)}_{n_1 \times n} \\ \vdots \\ \underbrace{\mathbf{L}_q(\mathbf{I}_n - \hat{\boldsymbol{\phi}}\hat{\boldsymbol{\phi}}^T)}_{n_q \times n} \\ \underbrace{\mathbf{I}_n - \hat{\boldsymbol{\phi}}\hat{\boldsymbol{\phi}}^T}_{n \times n} \end{bmatrix} \tag{14.57}$$

$$\nabla \hat{\mathbf{G}} = \begin{bmatrix} \mathbf{0}_{n_1 \times 4q} & \mathbf{I}_{n_1 \times n_1} & & -\mathbf{L}_1_{n_1 \times n} \\ \vdots & & \ddots & \vdots \\ \mathbf{0}_{n_q \times 4q} & & & \mathbf{I}_{n_q \times n_q} - \mathbf{L}_q_{n_q \times n} \\ \mathbf{0}_{1 \times 4q} & \mathbf{0}_{1 \times n_1} & \cdots & \mathbf{0}_{1 \times n_q} - 2\hat{\boldsymbol{\Phi}}^T_{1 \times n} \end{bmatrix} \quad (14.58)$$

$$\begin{aligned} \nabla^2 \hat{G}_i &= \mathbf{0}_{n_\theta \times n_\theta} \quad i = 1, \dots, \Sigma n_r \\ \nabla^2 \hat{G}_{1+\Sigma n_r} &= \begin{bmatrix} \mathbf{0}_{(4q+\Sigma n_r) \times (4q+\Sigma n_r)} \\ -2\mathbf{I}_{n \times n} \end{bmatrix} \end{aligned} \quad (14.59)$$

Note that $\boldsymbol{\theta}_r = [\boldsymbol{\varpi}_r; \mathbf{v}_r]$ and so the gradient and Hessian of $L_r(\boldsymbol{\theta}_r)$ w.r.t. $\boldsymbol{\theta}_r$ at the MPV can be partitioned as

$$\nabla \hat{L}_r = \begin{bmatrix} \hat{L}_r^{(\boldsymbol{\varpi}_r)}_{1 \times 4} & \hat{L}_r^{(\mathbf{v}_r)}_{1 \times n_r} \end{bmatrix} \quad \nabla^2 \hat{L}_r = \begin{bmatrix} \hat{L}_r^{(\boldsymbol{\varpi}_r \boldsymbol{\varpi}_r)}_{4 \times 4} & \hat{L}_r^{(\boldsymbol{\varpi}_r \mathbf{v}_r)}_{4 \times n_r} \\ \text{sym.} & \hat{L}_r^{(\mathbf{v}_r \mathbf{v}_r)}_{n_r \times n_r} \end{bmatrix} \quad (14.60)$$

These can be obtained from the computer code developed for single-setup problem. The gradient and Hessian of $L(\boldsymbol{\theta}) = \sum_{r=1}^q L_r(\boldsymbol{\theta}_r)$ w.r.t. $\boldsymbol{\theta}$ at the MPV are related to those of $\{L_r\}_{r=1}^q$ by

$$\nabla \hat{L} = \begin{bmatrix} \hat{L}_1^{(\boldsymbol{\varpi}_1)}_{1 \times 4} & \cdots & \hat{L}_q^{(\boldsymbol{\varpi}_q)}_{1 \times 4} & \hat{L}_1^{(\mathbf{v}_1)}_{1 \times n_1} & \cdots & \hat{L}_q^{(\mathbf{v}_q)}_{1 \times n_q} & \mathbf{0}_{1 \times n} \end{bmatrix} \quad (14.61)$$

$$\nabla^2 \hat{L} = \begin{bmatrix} \hat{L}_1^{(\boldsymbol{\varpi}_1 \boldsymbol{\varpi}_1)}_{4 \times 4} & & & \hat{L}_1^{(\boldsymbol{\varpi}_1 \mathbf{v}_1)}_{4 \times n_1} & & & \\ & \ddots & & & \ddots & & \\ & & \hat{L}_q^{(\boldsymbol{\varpi}_q \boldsymbol{\varpi}_q)}_{4 \times 4} & & & \hat{L}_q^{(\boldsymbol{\varpi}_q \mathbf{v}_q)}_{4 \times n_q} & \\ & & & \hat{L}_1^{(\mathbf{v}_1 \mathbf{v}_1)}_{n_1 \times n_1} & & & \\ & & & & \ddots & & \\ & & & & & \hat{L}_q^{(\mathbf{v}_q \mathbf{v}_q)}_{n_q \times n_q} & \\ \text{sym.} & & & & & & \mathbf{0}_{n \times n} \end{bmatrix} \quad (14.62)$$

Coding the posterior covariance matrix using the procedure here is simpler than using the formulas in Zhang et al. (2015) due to the use of Lagrange multiplier formula and leverage on computer code developed for single-setup problem.

14.4 Representative Statistics

In multi-setup problems, it is assumed that all modal properties except the mode shape can possibly change over different setups. A direct way to report identification results is in terms of the posterior MPV and c.o.v. (standard deviation/MPV) in each setup. While this gives a detailed picture of each setup, in applications it is also useful to have some representative statistics for the modal properties without distinguishing between setups. One empirical way is to think of the modal property as having equal chance of taking the value in one of the setups (Zhang and Au 2016). Specifically, let θ be the subject modal parameter. It is modeled as

$$\theta = \theta^{(I)} \quad (14.63)$$

where $\theta^{(I)}$ denotes the modal parameter in Setup I ; and I is a discrete-random variable uniformly distributed on $\{1, \dots, q\}$; q is the number of setups. Conditional on $I = r$, θ has the same distribution as the posterior distribution of $\theta^{(r)}$. For a given Setup r , let the posterior MPV and variance of $\theta^{(r)}$ be $\hat{\theta}^{(r)}$ and $\hat{v}^{(r)}$, respectively. Using conditional expectation, it can be shown that the expectation μ and variance v of θ are given by

$$\mu = \frac{1}{q} \sum_{r=1}^q \hat{\theta}^{(r)} \quad (14.64)$$

$$v = \underbrace{\frac{1}{q} \sum_{r=1}^q [\hat{\theta}^{(r)} - \mu]^2}_{\text{setup-setup variability}} + \underbrace{\frac{1}{q} \sum_{r=1}^q \hat{v}^{(r)}}_{\text{identification uncertainty}} \quad (14.65)$$

The expectation μ is the sample mean of the MPV $\hat{\theta}^{(r)}$ among the setups. The variance v is the sum of the sample variance of $\hat{\theta}^{(r)}$ among the setups (first term) and the sample mean of the posterior variances $\{\hat{v}^{(r)}\}_{r=1}^q$ (second term). Clearly, the second term reflects identification uncertainty. Although the first term is also affected by identification uncertainty, it is dominated by the variability of the modal parameter among different setups when it is significantly larger than the identification uncertainty. For this reason, the first term may be empirically referred as a ‘setup-setup variability’.

More generally, if the modal parameter is modeled to have value in one of the setups with probability $\{w_r\}_{r=1}^q$ ($\sum_{r=1}^q w_r = 1$), its mean and variance are given by

$$\mu = \sum_{r=1}^q w_r \hat{\theta}^{(r)} \quad v = \sum_{r=1}^q w_r [\hat{\theta}^{(r)} - \mu]^2 + \sum_{r=1}^q w_r \hat{v}^{(r)} \quad (14.66)$$

This is relevant, e.g., when the setups have significantly different durations.

14.5 Field Applications

In this section, we consider several field applications where global mode shapes with a large number of DOFs are identified using a small number of sensors. These examples illustrate some practical aspects and the quality of mode shapes that can be identified with limited resources under ambient conditions. The identified global mode shapes are interesting in themselves and give a good idea of dynamics. When the modal s/n ratio and hence the quality of local mode shape is high in all setups, the global mode shape is of good quality regardless of the method used. Otherwise, there can be significant difference between the results from different methods. The first example illustrates that Bayesian method produces the most reasonable result even when the modal s/n ratio in some setups is low.

Practical Considerations

Field vibration tests with multiple setups require substantially more planning in equipment and logistics. In a typical test, a visit was made to the site to observe accessible locations and environment. A setup plan was made that took into account equipment, human resources and time constraints. Locations to be measured were numbered and details such as sensor locations and orientations were clearly indicated. Team members were briefed of the schedule and their responsibility before the test. For calibration, data with sensors placed together and oriented in the same direction were obtained and their PSDs examined to spot any anomalies, e.g., whether measurement noise was reasonable, whether data channels measuring the same direction had similar PSDs. This was done in the laboratory before equipment was transported to the site, and in the site before regular test setups. The latter also gave an idea of potential modes and ambient environment. The data of each setup was analyzed shortly after it was acquired. The first few modes were identified to produce global mode shapes covering the DOFs measured so far. Contingent planning of setups were made when necessary, e.g., due to failure of data channels (Example 14.4). Equipment was powered by battery and did not rely on power supply from the site. In all tests, triaxial servo-accelerometers (noise about $1 \mu\text{g}/\sqrt{\text{Hz}}$) were used and DAQ (data acquisition) hardware had a 24 bit resolution. Example 14.4 had six sensors with analogue data transmitted through cables (about 30 m) to a central console and recorded synchronously. In Example 14.3 and Example 14.5, four sensors were deployed and their data were synchronized by GPS. In Example 14.6, five sensors with local DAQ and high precision clocks were deployed. Their clocks were synchronized shortly before the test but afterwards they were disconnected and left to run independently during the test.

Example 14.3 (Footbridge) Consider a footbridge 216 m long and continuously supported by six intermediate piers over the Tolo Harbor in Hong Kong, as shown in Fig. 14.2a. A three-dimensional global mode shape with 37 locations on each side of the bridge was identified with four triaxial servo-accelerometers. The total number of measured DOFs is $n = 37 \times 2 \times 3 = 222$. As shown in Fig. 14.2b, two sensors were placed near the mid- and quarter span in all setups as reference. The

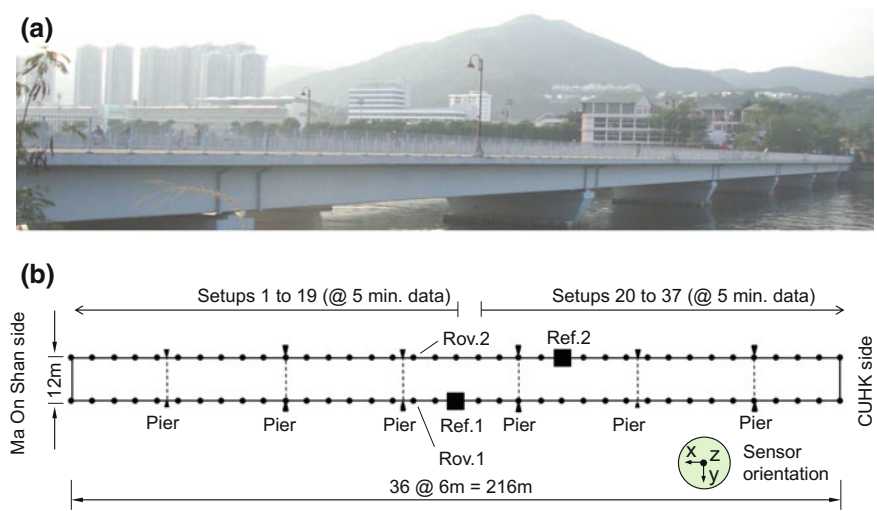


Fig. 14.2 **a** Overview of footbridge. **b** Setup plan. *Square* reference sensor location; *dot* rover sensor location. Rover positions shown for Setup 3

other two sensors were placed on two sides and roved through the length of the bridge, towards the left side in Setups 1–19 and then the right side in Setups 20–37. When the location of rover sensor coincided with that of the reference, the rover sensor was placed at the next (for Setup 1) or the previous location (for Setup 24). At each location, acceleration data was acquired and stored locally by a data recorder, as shown in Fig. 14.3. Timing at different recorders was synchronized



Fig. 14.3 Equipment per location

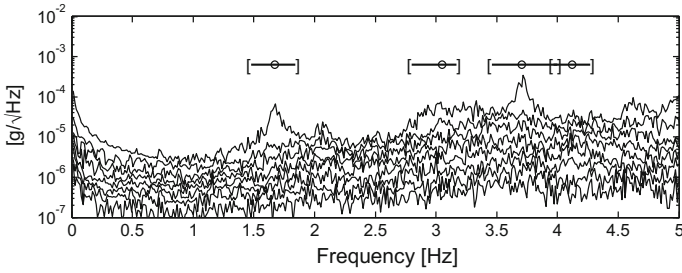


Fig. 14.4 One-sided SV spectrum of footbridge data in Setup 1. *Horizontal bar* indicates the selected frequency band for modal identification. The *bands* for other setups are similar

through GPS receivers. The sensors were placed with the same orientation in all setups.

Five minutes of data at 200 Hz were acquired in each setup. This was expected to provide at least a thousand natural periods of data, as the fundamental natural frequency of the bridge was expected to be a few Hz. The transition between setups, including transporting equipment, sensor orientation and levelling typically took 5 min. Each setup thus required 10 min. Including initial setup and calibration, the whole series of tests took 7 h (0930–1630 h).

Modal Identification

Figure 14.4 shows the SV spectrum of the data in Setup 1. Four well-separated modes in the spectrum are indicated. The fourth mode indicated in the figure is not apparent in this setup but it is more so in other setups. The modes are identified separately using the FFT in their bands indicated for this setup. The bands for other setups are similar.

Figure 14.5 shows the identified global mode shapes using Bayesian method incorporating the data in all setups (Sect. 14.3). The first mode shows that the DOFs at the piers do have transverse movements, probably due to expansion joints. Although there is no exact answer for the mode shapes, they generally look reasonable and make physical sense. The representative statistics of natural frequency and damping ratio are also indicated for each mode in the figure. The mean value μ is calculated based on (14.64), i.e., the sample mean of the MPVs among the setups. The c.o.v. is equal to $\sqrt{v}/\mu$, where v is the ‘total variance’ calculated according to (14.65). Recall that v is attributed to setup-setup variability and identification uncertainty in different setups. The percentage contribution of these two sources are shown in Table 14.3. In the present case, the total variance is dominated by setup-setup variability, especially for the natural frequency.

Comparison with Least Squares Solution

The global mode shapes are also estimated using least squares methods based on the local mode shapes (MPV) identified in individual setups. The quality depends on the quality of local mode shapes, which varies over modes and setups. For modes 1 to 3, the modal s/n ratio is consistently high in all setups and the local

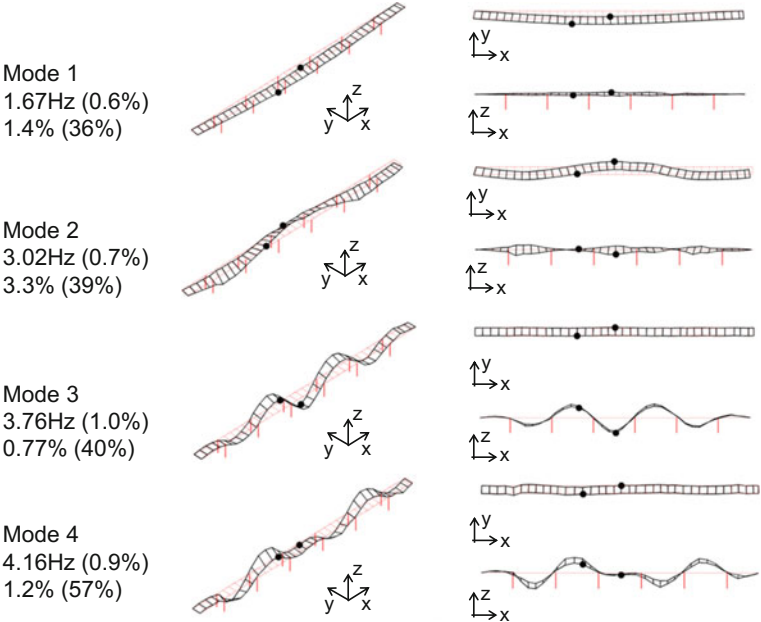


Fig. 14.5 Identified mode shapes (MPV) of footbridge. Representative statistics of natural frequency and damping ratio are indicated in the form ‘mean (c.o.v.)’

Table 14.3 Contribution of setup-setup variability and identification uncertainty to total variance

Contribution\mode	Natural frequency				Damping ratio			
	1	2	3	4	1	2	3	4
Setup-setup var. (%)	86	56	98	93	67	65	78	80
ID uncertainty (%)	14	44	2	7	33	35	22	20

mode shapes can be well identified. Least squares methods (local or global) then give similar results as in Fig. 14.5. The same is not true for mode 4, where some setups have low modal s/n ratio, essentially when the roving sensors were near the left or right end of the bridge. For local least squares method, the quality also depends on the choice of the reference setup, as demonstrated in Fig. 14.6a–c. The mode shape in Fig. 14.6d by global least squares method (with uniform weights) has a kink near the mid-spans, which appears unreasonable. Without additional information, it is difficult to judge and may be misinterpreted as an indication of structural damage.

Posterior Uncertainty

Figure 14.7 shows the posterior statistics of the natural frequency and damping ratio of mode 1 in different setups. Results obtained by analyzing the setups together (the method in Sect. 14.3) and individually (the method in Chap. 12) are shown for comparison. The MPVs vary in different setups, revealing the necessity

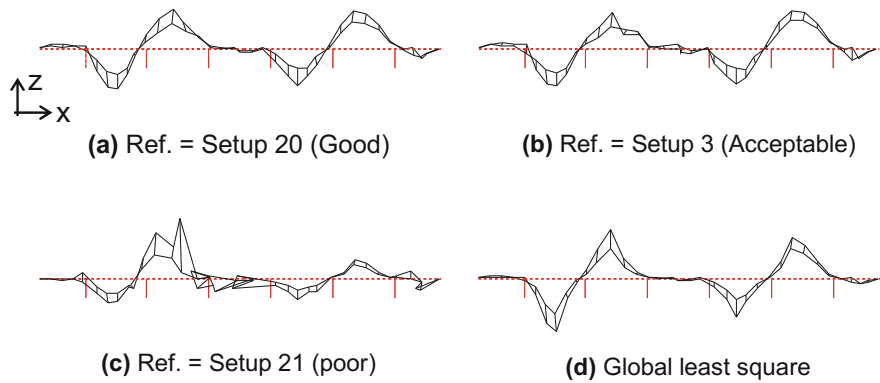


Fig. 14.6 Global mode shapes by local least squares method (a–c) with different setups as reference; and by global least squares method (d)

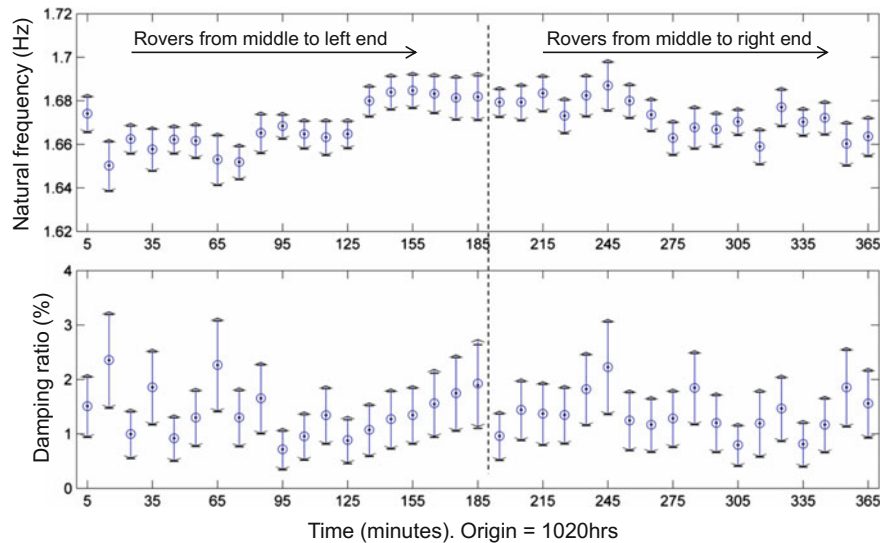


Fig. 14.7 Posterior statistics of mode 1 frequency and damping in different setups; (—o—): setups analyzed separately; |—•—|: setups analyzed together

for separate parameterization of modal properties. The figure suggests that it makes no difference to the posterior statistics whether the setups are analyzed together or separately. This is only true when the modal s/n ratio is high, which is the case for mode 1. Figure 14.8 shows the results for mode 4 where significant discrepancy appears in setups with low modal s/n ratio. When the setups are analyzed separately, the ‘problematic’ setups (shaded) have larger (unreasonable) fluctuation in the MPV with time and larger posterior uncertainty. This suggests that, not only

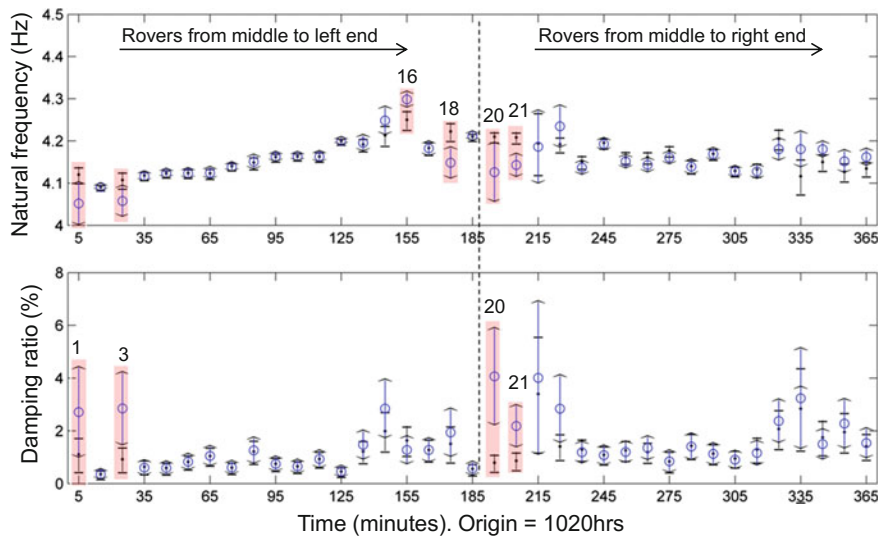


Fig. 14.8 Posterior statistics of mode 4 frequency and damping in different setups; (—o—): setups analyzed separately; | — • — |: setups analyzed together; shaded: setups with large discrepancy

does analyzing setups together produce a better global mode shape, it also improves the identification results in the natural frequency and damping ratios. This cannot be achieved by the least squares methods in Sects. 14.1 or 14.2 because they only assemble the global mode shape from local ones. Further details on this example are referred to Au and Zhang (2012) and Zhang et al. (2015). ■

Example 14.4 (Coupled floor slab) Consider a coupled floor slab shown schematically in Fig. 14.9. It is formed by two mega-truss systems on 2/F and 3/F of a community recreation center. Different from typical situations, the vertical vibrations of the two floors are coupled due to the interior steel columns. A series of tests (ambient, shaker, occupant jumping) were performed during and after construction to update serviceability vibration performance of the floors. Figure 14.10a, b show the conditions on 2/F and 3/F during the ambient test. Six

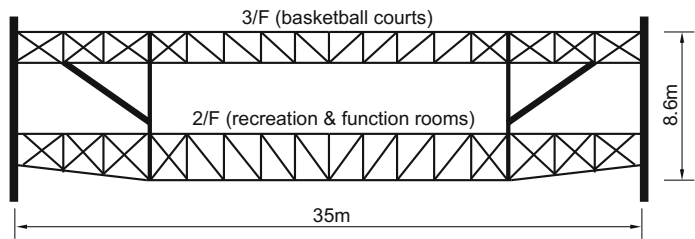


Fig. 14.9 Coupled floor slab (elevation)

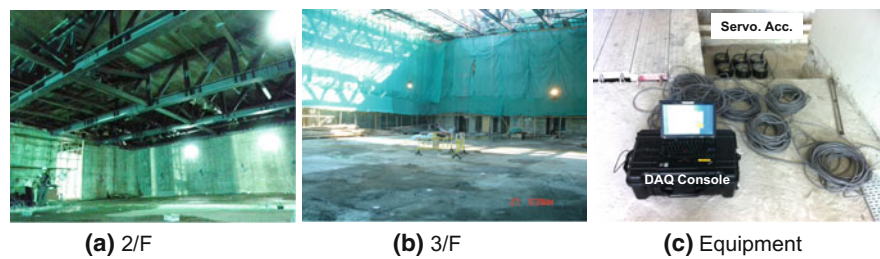


Fig. 14.10 Condition on 2/F (a) and 3/F (b), and equipment (c) for ambient vibration test

triaxial servo-accelerometers were deployed with their analogue data transmitted through cables to a DAQ console. Data was acquired at 2048 Hz and later decimated to 256 Hz for analysis.

Setup Plan

Figure 14.11 shows the setup plan. A three-dimensional global mode shape covering $63 \times 2 = 126$ locations were to be obtained, with $n = 126 \times 3 = 378$ DOFs. One reference sensor was placed on 2/F and 3/F at the reference locations (square) to connect their mode shapes. On 2/F, each roving sensor (A, B, C, D) basically covers two column lines of locations. The console was placed on the floor instrumented. To avoid using cable of excessive length (which increases data noise/distortion and safety risk), a hole was drilled on 3/F (feasible during construction) to allow passage of sensor cable to 2/F. Each setup typically took 20 min (15 min for data and 5 min for transition). Including initial calibration and setup, the sixteen setups on 2/F required one day (0800–1800 h) to finish.

Originally, the setups on 3/F followed the same plan on 2/F. However, Sensor A (or its data channel) was found to be defective after Setup 8. In the interest of time,

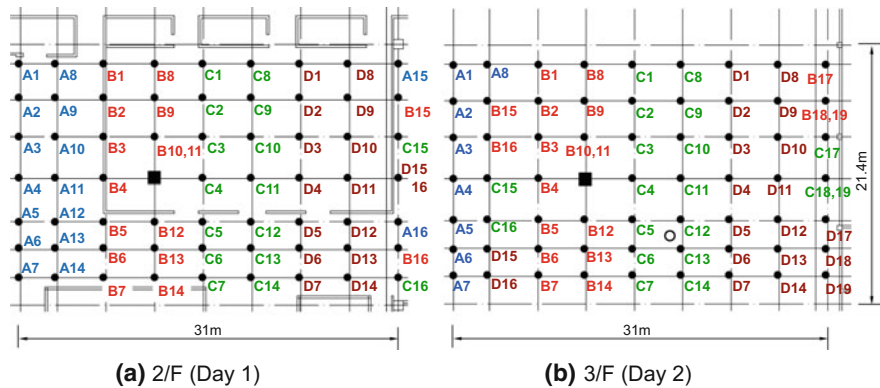


Fig. 14.11 Setup plan for coupled floor slab. *Square* reference sensor locations; *dot* roving sensor locations (A/B/C/D, followed by setup number). In (b), *hollow circle* indicates a hole drilled on 3/F to pass cable to 2/F

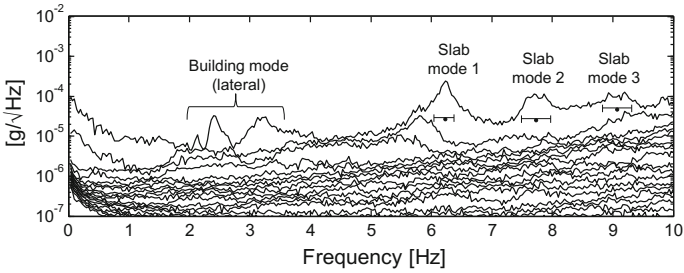


Fig. 14.12 One-sided SV spectrum of Setup 1 on 2/F, coupled floor slab

the setups were continued as planned with the remaining sensors, during which a contingent plan was made. Changes were made from Setup 15 onwards to make up for the locations missed by Sensor A with the remaining three roving sensors. This required three additional setups. The test finished by 1800 h.

Modal Identification

Figure 14.12 shows the SV spectrum of data in Setup 1 on 2/F. The modes below 4 Hz and the one just below 6 Hz were attributed to lateral vibration of the building (not of concern), as confirmed from the PSD of data. The first three modes of slab vertical vibration have frequencies around 6, 8 and 9 Hz. Their global mode shapes identified using Bayesian method with all setups incorporated are shown in Fig. 14.13. The quality is generally good. The results by global least squares or local least squares method (with reasonable choice of reference setup) are similar (not shown here). Similar to the last example, setup-setup variability dominates the

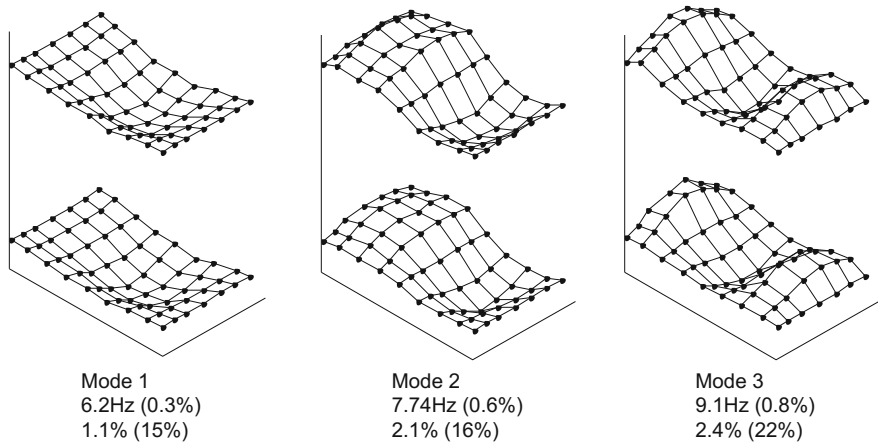


Fig. 14.13 Identified global mode shapes of coupled floor slab (vertical not to scale with horizontal). Representative statistics of natural frequency and damping ratio are indicated in the form ‘mean (c.o.v.)’



Fig. 14.14 Velodrome during test

representative variance in the modal properties. For the natural frequencies, it accounts for over 95% of the total variance. For the damping ratios, it accounts for over 80%.

Further details on ambient vibration test and shaker test of this structure can be found in Au et al. (2012). The identified modal properties were used for structural model updating (Lam et al. 2014, 2016). Design considerations of the slabs for floor vibration against rhythmic activities are discussed in Li et al. (2011). ■

Example 14.5 (Velodrome roof) Consider a velodrome roof with principal diameter $120\text{ m} \times 80\text{ m}$, as shown in Fig. 14.14a. It is formed by mega trusses spanning over the minor diameter. Ambient vibration tests were performed on the roof to identify the modal properties. Scaffold mounted on the top side of the mega truss provided a platform feasible for instrumentation during construction. See Fig. 14.14b viewing from below and Fig. 14.15a on the platform.

Setup Plan

As shown in Fig. 14.15b, the same set of equipment with four triaxial sensors in Example 14.3 was deployed, where the data of different sensors were synchronized by GPS. Sensors were leveled on mounts tied securely on truss members (Fig. 14.15c). Horizontally they were aligned with their y-axis (local North) along the local truss member axis. Half of the roof was measured at 48 locations as shown

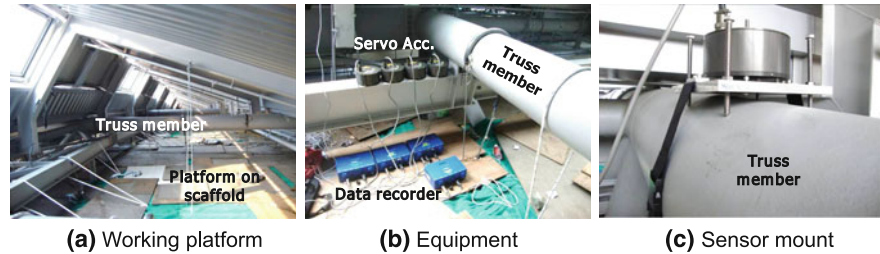


Fig. 14.15 Instrumentation details of velodrome roof

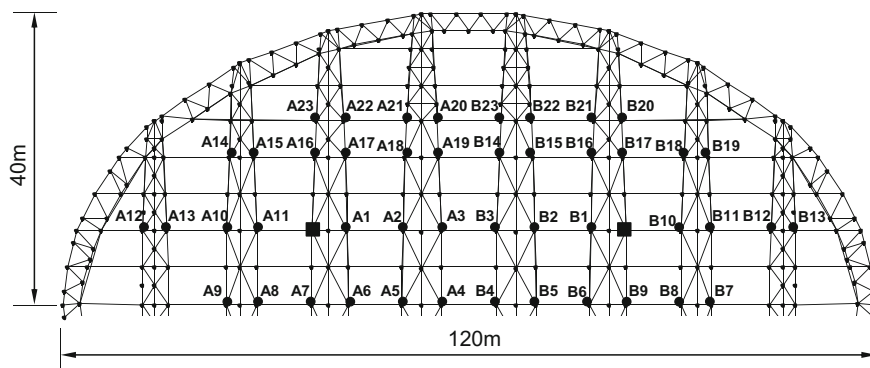


Fig. 14.16 Setup plan. *Square* reference sensor locations; *dot* roving sensor locations (A/B, followed by setup number); Setups 1–6 on Day 1, 7–13 on Day 2; 14–23 on Day 3

in Fig. 14.16, producing a three-dimensional mode shape with $n = 48 \times 3 = 144$ DOFs. There were two reference sensors, indicated by the squares in the figure. The other two roving sensors covered the left and right part of the roof. The locations near the left and right end of the major axis were not measured as they could not be accessed during the test.

Setup planning was much constrained by environmental and construction activities. The data was to be acquired with minimal logistics assistance from contractor and no intervention of construction activities. Except for the reference sensor locations, the roving sensor locations reflect more of what was feasible on the test day rather than what was planned. Mobility was limited on the platform. Most setup transitions took place along the major axis direction (left–right in Fig. 14.16), as the minor axis direction was blocked by sky-window panels (see left side of Fig. 14.15a). To go to another row one had to go to the perimeter first. The data of different sensors were synchronized by GPS receivers, which required good exposure to the sky. During the test, the (automatic) sky-windows were not in full commission yet. The locations of windows that could be opened were only known on the test day or the day before. Each setup took 30 min (15 min for data and 15 min for transition). The whole test with twenty three setups spanned over three consecutive days, with six setups on Day 1, seven on Day 2 and ten on Day 3.

Modal Identification

Figure 14.17 shows the SV spectrum of data in Setup 1. Four potential modes are indicated. Their global mode shapes identified with all data incorporated are shown in Fig. 14.18. The quality is generally good. Local least squares (with reasonable reference) and global least squares method gave similar results, except for occasional slope discontinuity in some higher modes. Setup-setup variability accounts for 85–95% of the total variance for the natural frequencies, and 50–70% for the damping ratios. ■

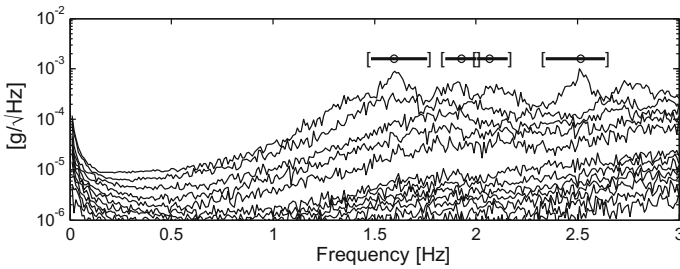


Fig. 14.17 One-sided SV spectrum of data in Setup 1, velodrome roof

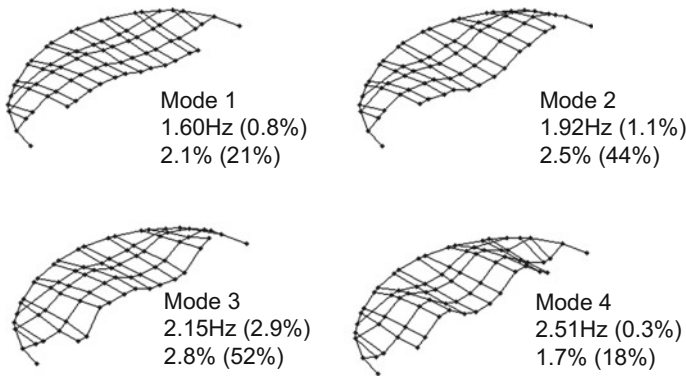


Fig. 14.18 Identified mode shapes of velodrome roof. Representative statistics of natural frequency and damping ratio are indicated in the form ‘mean (c.o.v.)’

Example 14.6 (Brodie Tower) Consider the Brodie Tower at the University of Liverpool (UK) in Fig. 14.19. It is a seven-storied reinforced concrete building with a basement. A three-dimensional global mode shape with 28 locations was identified with five triaxial servo-accelerometers. The total number of measured DOFs is $n = 28 \times 3 = 84$.

Setup Plan

A reference sensor was placed on 7/F. The remaining four sensors roved from 7/F down to 1/F, covering all measured DOFs in seven setups; see Fig. 14.20a. The acceleration data at each location was acquired and stored locally on the sensor with integrated DAQ. The sensors were equipped with high precision clocks and so they do not need to be synchronized by wired or wireless means; see Fig. 14.20b. Synchronized shortly before the test by GPS, their data were practically synchronized within the test duration. Twenty minutes of data at 50 Hz were acquired in each setup. Including initial setup and calibration, the whole series of tests took 4 h in the afternoon.

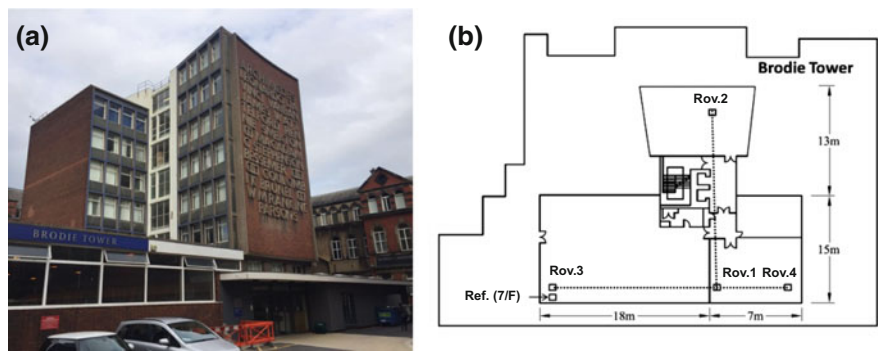


Fig. 14.19 Brodie Tower. **a** View from NE of the plan in (b). **b** Typical floor plan

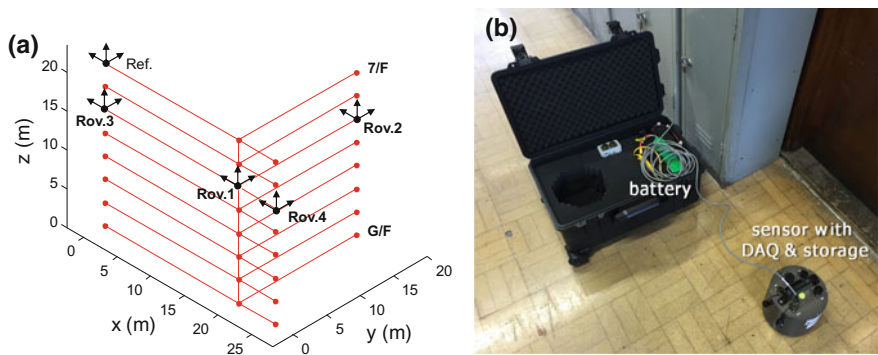


Fig. 14.20 Setup plan. **a** Setup 3. **b** Equipment for measurement per location

Modal Identification

Figure 14.21 shows the SV spectrum of the data in Setup 1. Three potential modes are indicated. Their global mode shapes identified with all data incorporated are shown in Fig. 14.22. The first two modes correspond to translations primarily along the x and y direction, respectively. The third mode is rotational. The quality of these modes is generally good. Local least squares and global least squares method gave similar results. Setup-setup variability accounts for over 75% of the total variance for the natural frequencies, and about 50% for the damping ratios. Higher modes can also be identified (omitted here), although the results from different methods differ for modes with low modal s/n ratio, similar to Example 14.3. ■

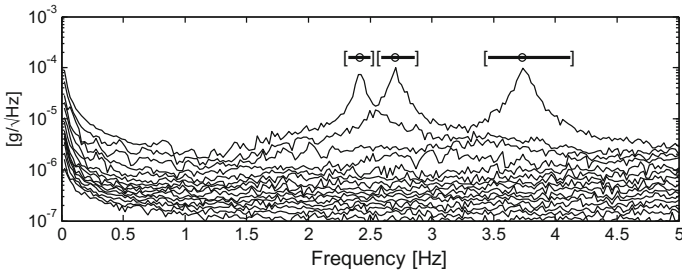


Fig. 14.21 One-sided SV spectrum of data in Setup 1

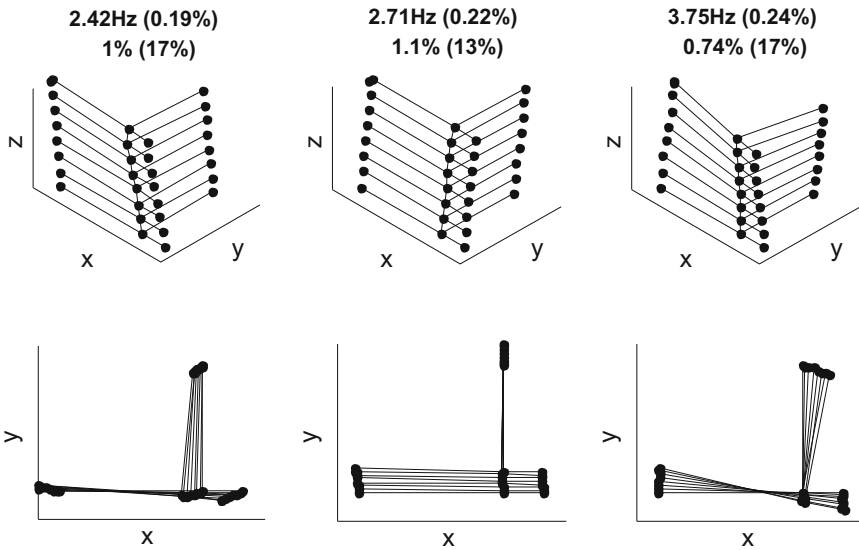


Fig. 14.22 Identified mode shapes (MPV) of Brodie Tower. Representative statistics of natural frequency and damping ratio are indicated in the form ‘mean (c.o.v.)’

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Part IV

Uncertainty Laws

Chapter 15

Managing Identification Uncertainties

Abstract This chapter presents closed form analytical formulas that govern the identification uncertainty of modal parameters in operational modal analysis for well-separated modes. These are collectively referred as ‘uncertainty laws’, which is one of the latest advances in operational modal analysis. Underlying assumptions and contexts are discussed to allow proper interpretation and application of the formulas. The uncertainty laws govern the achievable precision of operational modal analysis and they can be used for planning ambient vibration tests. The chapter is written for general readers with minimal preparation in mathematics.

Keywords Uncertainty laws • Asymptotic • Bandwidth • Uncertainty management • Test planning

Suppose we are going to perform an ambient vibration test on a structure to identify its modal properties. One typical question is,

How much data do we need?

‘How much’ here practically refers to the sampling rate, data duration and the number of sensors. A more general question is,

What is the required test configuration?

In addition to data size, this question now involves other factors such as the quality of sensors, their locations, the quality of data acquisition hardware, or even the number of setups if available sensors are not enough to cover all required DOFs in a single setup.

These questions are not trivial to answer. They depend on the required precision in the identification results. Answering them requires insights into how identification uncertainty depends on test configuration. Experience certainly helps but a quantitative account will be a valuable tool.

The Bayesian algorithms for operational modal analysis (OMA) discussed in Chaps. 12–14 allow us to *calculate* the most probable value and identification uncertainty of modal parameters when *data is given*. The results are numerical and *point-wise* in nature. Suppose the posterior c.o.v. of damping ratio is calculated to be 30% for a given data set. This helps us judge how accurate the damping ratio has

been estimated from the given data. This 30%, however, is only specific to the data obtained, for the particular mode, of the particular structure, and under the particular ambient environment. It gives little or no insight about *why* it is 30%, or more generally, how identification uncertainty depends on test configuration. Knowing the *why* will allow us to master the identification uncertainty in OMA, which has both scientific and engineering significance. For example, what is the achievable precision of modal parameters identified from ambient data? Would adding a sensor significantly reduce the identification uncertainty of the damping ratio? If so, by how much? How about using a sensor with much lower noise?

One way to master the relationship between test configuration and identification uncertainty is to have simple expressions or graphs relating them to comprehend and build insights. Glancing over the analytical expressions for programming the posterior covariance matrix in Chaps. 12–14, the reader would be convinced that this is by no means a simple task, if at all possible. The posterior covariance matrix is the inverse of the Hessian of NLLF (negative log-likelihood function). The Hessian contains second derivatives which are no simple expressions and involve stochastic data. Also, how to quantify test configuration?

The general situation may not admit simple expressions, but it turns out that under some ‘asymptotic’ conditions, namely, *long data* and *small damping*, the posterior covariance matrix does reduce to mathematically tractable form that allows us to build insights. Asymptotic expressions related to the posterior uncertainty are referred as ‘uncertainty laws’. This chapter discusses the uncertainty laws so that they can be properly interpreted and applied for planning field tests. The scope is limited to well-separated modes with single setup data. Deriving uncertainty laws is technical and tedious. It is postponed until Chap. 16, which may be skipped for readers primarily interested in applications.

Figure 15.1 explains how the objective of uncertainty laws differs from that of conventional modal identification. We first present the key results of uncertainty laws and discuss their meaning. Examples are given to demonstrate the quality of approximation in general non-asymptotic situations. Their use for planning ambient vibration tests are discussed and illustrated through examples.

15.1 Context and Key Formulas

Consider a classically-damped mode with natural frequency f (Hz) and damping ratio ζ , and is well-separated from other modes. It is identified using the scaled Fast Fourier Transform (FFT) of ambient data on a selected frequency band. Within the band, the scaled FFT of data is assumed to be the sum of modal response and prediction error. The modal contribution of displacement response (in the time domain) is $\boldsymbol{\phi}\eta(t)$, where $\boldsymbol{\phi}$ ($n \times 1$, with unit norm) is the partial mode shape vector confined to the measured DOFs, and $\eta(t)$ (scalar) is the modal response. The latter satisfies the modal equation of motion $\ddot{\eta}(t) + 2\zeta\omega\dot{\eta}(t) + \omega^2\eta(t) = p(t)$, where $p(t)$

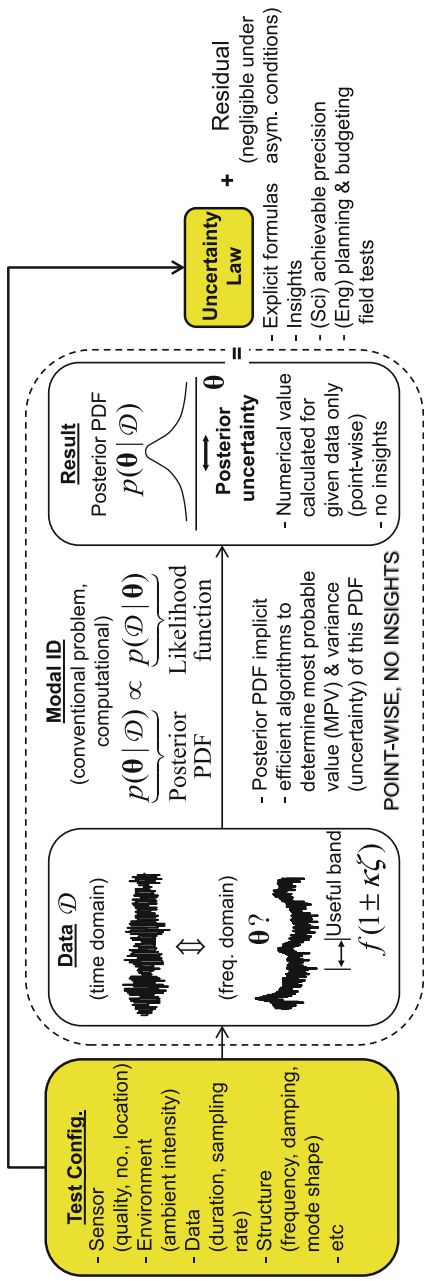


Fig. 15.1 Objective of uncertainty laws

is the modal force and $\omega = 2\pi f$. The modal force and prediction error are assumed to be uncorrelated and have respectively a constant power spectral density (PSD) of S and S_e within the selected band. The above context corresponds to that of Chap. 12. In this chapter, the prediction error is assumed to be solely attributed to channel (measurement) noise and so it is simply referred as channel noise. The modal properties to identify are

f	ζ	S	S_e	Φ
Natural frequency (Hz)	Damping ratio (dimensionless)	Modal force PSD (same unit as acceleration)	Channel noise PSD (same unit as data)	$n \times 1$ Mode shape (sum of squares equal to 1)

Let x denote a modal parameter, i.e., f , ζ , S , S_e or Φ . The c.o.v. of x is denoted by δ_x ; c.o.v. = coefficient of variation = standard deviation/mean. Assume that the selected frequency band is $f(1 \pm \kappa\zeta)$ Hz, where κ is a ‘bandwidth factor’ (e.g., $\kappa = 1$ corresponds to the half-power band). Let T_d (s) be the data duration and N_f be the number of FFT points in the selected band. Since the FFT frequency interval is $\Delta f = 1/T_d$ and the selected bandwidth is $2\kappa\zeta f$,

$$N_f = 2\kappa\zeta f T_d \quad (15.1)$$

It can be shown that for long data ($N_f \rightarrow \infty$) and small damping ($\zeta \rightarrow 0$),

$$\delta_x^2 \sim \delta_{x1}^2 = \delta_{x0}^2 \left(1 + \frac{a_x}{\gamma}\right) \quad N_f \rightarrow \infty, \zeta \rightarrow 0 \quad (15.2)$$

where δ_{x0}^2 and δ_{x1}^2 are referred respectively as the ‘zeroth order law’ and ‘first order law’; ‘ $\sim$ ’ (read as ‘asymptotic to’) denotes that the ratio of the two sides tends to 1;

$$\gamma = \frac{1}{4v\zeta^2} \quad (15.3)$$

is the (dimensionless) ‘modal signal-to-noise (s/n) ratio’;

$$v = \frac{S_e}{S} (2\pi f)^{2q} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (15.4)$$

is the ‘noise-to-environment’ (n/e) ratio; a_x is the ‘first order coefficient’ associated with the first order effect of γ when it is finite. The expressions for δ_{x0}^2 and a_x for different parameters are summarized in Table 15.1.

Figure 15.2 illustrates the above context for acceleration data ($q = 0$) in an idealized singular value spectrum, i.e., a plot of eigenvalues of the PSD matrix of data (Sect. 7.3). In this case the modal s/n ratio is $\gamma = S/4S_e\zeta^2$. At the natural

Table 15.1 Key results of uncertainty laws

Parameter x	Zeroth order law δ_{x0}^2	Data length factor $B_x(\kappa)$	First order law coefficient $a_x(\kappa)$
Frequency f	$\frac{\zeta}{2\pi N_c B_x}$	$\frac{2}{\pi} \left(\tan^{-1} \kappa - \frac{\kappa}{\kappa^2 + 1} \right)$	$\frac{4(\kappa - \tan^{-1} \kappa)}{\tan^{-1} \kappa - \frac{\kappa}{\kappa^2 + 1}}$
Damping ζ	$\frac{1}{2\pi \zeta N_c B_x}$	$\frac{2}{\pi} \left[\tan^{-1} \kappa + \frac{\kappa}{\kappa^2 + 1} - \frac{2(\tan^{-1} \kappa)^2}{\kappa} \right]$	$\frac{4(\kappa^2 + 1)(3 \tan^{-1} \kappa - 3\kappa + \kappa^2 \tan^{-1} \kappa) \tan^{-1} \kappa}{3[(\kappa^2 + 1)(\kappa - 2 \tan^{-1} \kappa) \tan^{-1} \kappa + \kappa^2]}$
Modal force PSD S	$\frac{1}{N_f B_x}$	$1 - \frac{2}{\kappa} (\tan^{-1} \kappa)^2 \left(\tan^{-1} \kappa + \frac{\kappa}{\kappa^2 + 1} \right)^{-1}$	$2 + \frac{2}{3} \kappa^2 - \frac{(\tan^{-1} \kappa)^2}{2(\tan^{-1} \kappa)^2 - b\kappa} \left[\frac{8 \tan^{-1} \kappa}{b} - \frac{8\kappa}{\tan^{-1} \kappa} + \frac{4}{3} \kappa^2 + 4 \right]$ $b = \tan^{-1} \kappa + \frac{\kappa}{\kappa^2 + 1}$
Channel noise PSD S_e	$\frac{1}{(n-1)N_f B_x}$	1	0
Mode shape ϕ	Covariance matrix $\hat{\mathbf{C}}_\phi = \frac{2v_\zeta}{\pi N_c B_x} (\mathbf{I}_n - \phi \phi^T)$ Square of mode shape c.o.v. $\delta_\phi^2 = \frac{2(n-1)v_\zeta}{\pi N_c B_x}$	$\frac{2}{\pi} \tan^{-1} \kappa$	–

Generally, $\delta_x^2 \sim \delta_{x0}^2 (1 + a_x/\gamma)$ where $\gamma = 1/4v_\zeta^2$ is the modal signal/noise ratio, $v = (2\pi f)^{2q} S_e/S$ is the noise/environment ratio, $q = 0, 1, 2$ for acceleration, velocity and displacement data, respectively
The selected band for modal identification is $f(1 \pm \kappa \zeta)$ where κ is the bandwidth factor, $N_c = N_f/2\zeta\kappa = \text{Data length/natural period}$
Symbols f , ζ , S , S_e and ϕ here denote the *actual* value of modal properties rather than the variable in the likelihood function

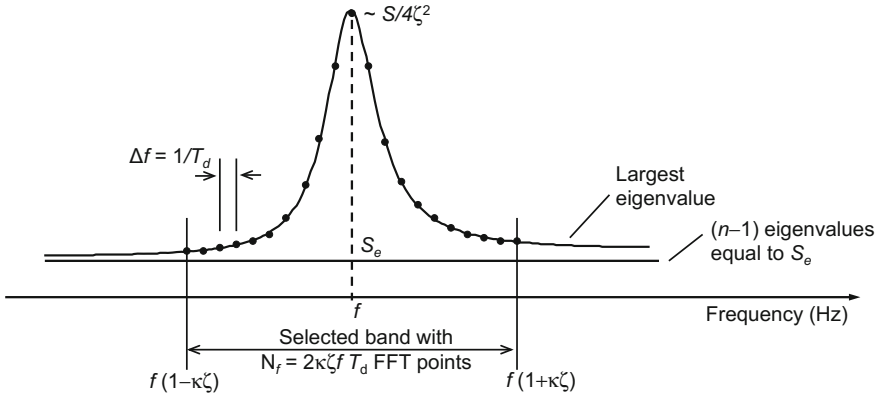


Fig. 15.2 Idealized singular value spectrum of acceleration data ($q = 0$), illustrating the context of uncertainty laws. At the natural frequency, the ratio of the *top* to *second top* line is approximately equal to the modal s/n ratio $\gamma = S/4S_e\zeta^2$

frequency, the largest eigenvalue is $S/4\zeta^2 + S_e$ and the remaining eigenvalues are all equal to S_e . The ratio of the top to the second top line in the singular value spectrum is then approximately equal to the modal s/n ratio when it is large.

In the presentation of uncertainty laws, for notational convenience, f , ζ , S , S_e and $\boldsymbol{\varphi}$ denote the *actual* values of modal parameters that result in the data, rather than the variables in the likelihood function. In Table 15.1, $\hat{\mathbf{C}}_{\boldsymbol{\varphi}}$ is the uncertainty law for the posterior covariance matrix of mode shape $\boldsymbol{\varphi}$ (with unit norm). The mode shape c.o.v., denoted by $\delta_{\boldsymbol{\varphi}}$, is defined in Sect. 11.3.3 as the square root sum of eigenvalues of $\hat{\mathbf{C}}_{\boldsymbol{\varphi}}$. According to the uncertainty laws, given the data, significant correlation exists only between ζ and S , which is $O(\kappa^{-1/2})$. The correlation between any other parameter pair is asymptotically small, at most $O(\zeta)$.

15.2 Understanding Uncertainty Laws

Uncertainty laws are asymptotic expressions in the sense that the ratio of their value to the exact value of posterior uncertainty tends to 1 under the stated asymptotic conditions. The symbol ‘ $\sim$ ’ in (15.2) denotes that the ratio of the LHS to the RHS, i.e., δ_x^2/δ_{x1}^2 , tends to 1 as $N_f \rightarrow \infty$ and $\zeta \rightarrow 0$. Here, δ_x is the posterior c.o.v. that we want to understand and δ_{x1} is the asymptotic analytical expression that we have derived, from which we can build insights. Equation (15.2) implies

$$\frac{\delta_x^2}{\delta_{x0}^2} = 1 + \frac{a_x}{\gamma} + \text{Remainder} \quad N_f \rightarrow \infty, \zeta \rightarrow 0 \quad (15.5)$$

The remainder depends on data, as δ_x^2 does. However, it vanishes asymptotically, leaving only the ‘information content’ of data captured in the uncertainty law $\delta_{x1}^2 = \delta_{x0}^2(1 + a_x/\gamma)$. Note that $a_x/\gamma \rightarrow 0$ because $\gamma = 1/4v\zeta^2 \rightarrow \infty$, and so $\delta_x^2/\delta_{x0}^2 \rightarrow 1$ as well. Nevertheless δ_{x1} captures the effect of γ when it is finite and is more useful than δ_{x0} .

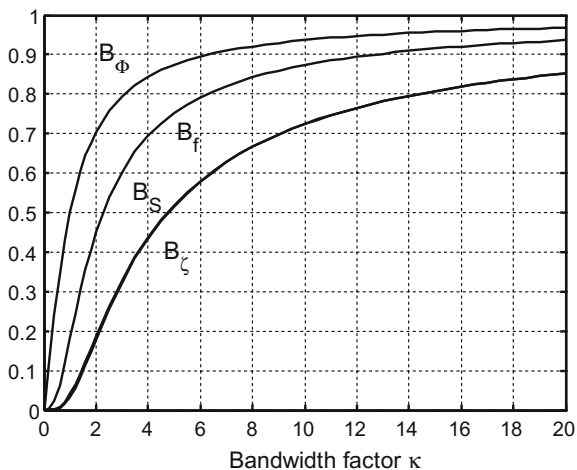
Data is not available before it is measured. In deriving the uncertainty laws, the ambient data is modeled by a stochastic process that obeys the assumptions in Bayesian OMA. In other words the uncertainty laws do not account for modeling errors. They reflect identification uncertainty if the modeling assumptions are correct.

15.2.1 Data Length and Usable Bandwidth

The uncertainty laws depend on the following (dimensionless) scales: ζ , v , κ , N_c or N_f . Damping ratio ζ is a property of the subject structure. The n/e ratio v represents a modal noise-to-signal ratio excluding structural dynamic amplification. The posterior c.o.v.s are inversely proportional to the square root of data length in terms of either N_f or $N_c = N_f/2\zeta\kappa$. This is common in statistical estimation. The ‘normalized’ data length N_c is related to the maximum amount of information available in data for inferring the mode of interest. On the other hand, κ and N_f are related to the amount of information that can be actually utilized. The bandwidth factor κ must trade off modeling error and information included for modal identification. One can have a long time history of data so that N_c is large. However, in the neighborhood of the natural frequency, other unknown colored noise exists. Then only a small bandwidth can be used to avoid modeling error. This limits κ and N_f .

The uncertainty laws in Table 15.1 have been written in a form that isolates the effect of κ into the ‘data length factor’ $B_x(\kappa)$. The latter is an increasing function of κ from 0 to 1. The remaining part in the formula represents the lower limit of the posterior uncertainty when the full bandwidth can be utilized. For example, $\delta_f^2 \geq \zeta/2\pi N_c$ for the natural frequency. In reality, one is not able to make use of the full bandwidth for modal identification, limiting κ to be finite. The term $N_c B_f(\kappa)$ represents an effective (dimensionless) data length accounting for limited bandwidth.

Figure 15.3 shows the variation of the data length factors with κ . Note that $B_\phi > B_f > B_S > B_\zeta$; B_ζ and B_S almost overlap. For $\kappa = 6$ (typical) the value is about 80% for the natural frequency and 90% for the mode shape. The values for ζ and S are smaller, about 60%. The figure suggests that the accuracy in the mode shape or natural frequency is typically less sensitive to the bandwidth than ζ or S , which can be explained intuitively.

Fig. 15.3 Data length factor

15.2.2 Signal-to-Noise Ratio

The term ‘signal-to-noise ratio’ (s/n ratio) is often used to indicate the contribution of the target signal compared to ‘noise’. The latter arises from sensors, cables transmitting the signal (especially for analogue signals), data acquisition (DAQ) hardware, etc. Different definitions have been used depending on application. The definition in (15.3), combined with (15.4), stems from the theory in Chap. 12:

$$\gamma = \frac{S}{4S_e\zeta^2(2\pi f)^{2q}} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (15.6)$$

This is the PSD ratio of the response at resonance ($S/4\zeta^2(2\pi f)^{2q}$) to measurement noise (S_e). The modal force PSD S depends on environmental excitation intensity and (less trivially) on the measured DOFs. Due to mode shape scaling, it is proportional to the sum of squares of mode shape values at the measured DOFs.

The posterior c.o.v. δ_x does not vanish even when the modal s/n ratio γ is infinite. Rather, it stays at a non-zero level given by the zeroth order law δ_{x0} . High s/n ratio only removes uncertainty due to measurement noise, but not uncertainty due to modal force, which is not measured and hence cannot be eliminated. The zeroth order laws therefore reflect the achievable precision of modal parameters in OMA.

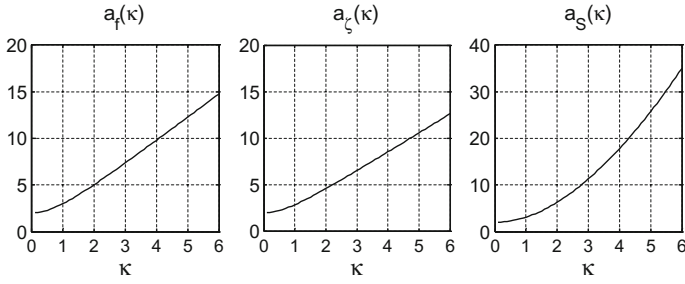


Fig. 15.4 First order coefficients a_f , a_ζ and a_S

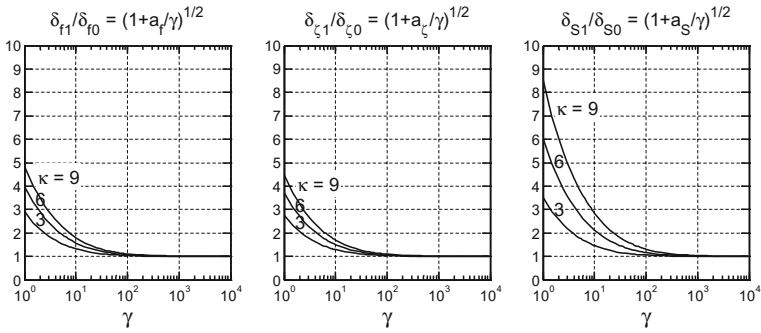


Fig. 15.5 Ratio of first order law to zeroth order law

15.2.3 First Order Effect of Modal s/n Ratio

The first order law $\delta_{x1}^2 = \delta_{x0}^2(1 + a_x/\gamma)$ captures the first order effect of the modal s/n ratio γ through the term a_x/γ . The first order law coefficients for f , ζ and S are plotted in Fig. 15.4. Figure 15.5 shows the ratio of the first order law to the zeroth order law. Regardless of κ , the ratio converges to 1 as γ increases because the first order correction a_x/γ vanishes in the limit.

15.2.4 Governing Uncertainty

The dependence of the posterior c.o.v. of a modal parameter on ζ has important implications on how difficult it can be estimated from ambient data. From Table 15.1, δ_{f0}^2 and δ_ϕ^2 are proportional to ζ , while on the contrary $\delta_{\zeta0}^2$ is inversely proportional to ζ . For small ζ encountered in applications, say, 0.5–5%, this means that the identification uncertainty in the damping ratio is much larger than that in the

natural frequency or mode shape, and so its accuracy requirement is likely to govern planning decisions.

The dependence on ζ can be explained intuitively. Smaller ζ implies that resonance oscillations decay slower and stay longer in the data. In the frequency domain, the resonance peak is more pronounced and sharper, giving better accuracy in the natural frequency. The quality of mode shape also improves because resonance oscillations dominate the measured vibration signal. Vanishingly small damping makes it impossible to identify to the same relative accuracy, in the presence of uncertainty arising from unknown modal excitation. The posterior standard deviation of the damping ratio does reduce in a square root manner as the damping ratio decreases. Nevertheless the reduction rate is slower than linear, and so on a relative basis the identification uncertainty increases.

15.3 Demonstrative Examples

Generally, uncertainty laws capture well identification uncertainties even under non-asymptotic situations. This section provides some demonstrative examples based on synthetic, laboratory and field data. The field data example also gives an idea of the order of magnitude of the governing scales encountered in practice.

Example 15.1 (Synthetic data, ten-storied shear building) Revisit the ten-storied building in Example 12.1, where synthetic acceleration data ($q = 0$) was generated at 100 Hz for 600 s. The selected band for identification was [0.94, 1.06] Hz ($\kappa = 6$). Figure 15.6 compares the uncertainty laws (cross) with the exact values of the posterior c.o.v. (for the given data) recalled from Fig. 12.5. Data was generated with different values of S_e , giving different values of modal s/n ratio $\gamma = S/4S_e\zeta^2$. The values of uncertainty laws are obtained by substituting modal parameter values with the posterior MPV determined in each case. The figure shows that the uncertainty laws agree with the exact values of the posterior c.o.v., especially for large modal s/n ratios. The posterior c.o.v. generally decreases to a constant level (equal to the zeroth order law) as the modal s/n ratio increases. ■

Example 15.2 (Laboratory data, three-storied shear frame) Revisit the three-storied laboratory shear frame in Example 12.4, where each floor was measured biaxially at four corners, giving a total of $3 \times 4 \times 2 = 24$ measured DOFs. Ambient acceleration data was recorded for ten minutes. Consider performing modal identification with an increasing set of measured DOFs, starting from the smallest set with DOFs 1 and 2. Adding one DOF at a time, this gives for each mode 23 cases, corresponding to DOFs {1, 2}, {1, 2, 3} and so on, until {1, 2, ..., 24}. These cases have different modal s/n ratios because their modal force PSDs are different.

The plots on the left column of Fig. 15.7 show the posterior c.o.v. of modal parameters of Mode 1 (TX1) calculated from data (exact, circles) and those from

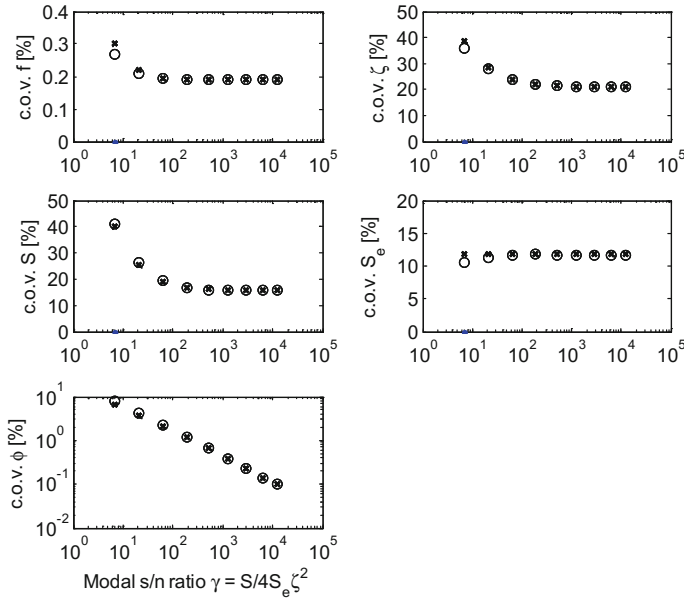


Fig. 15.6 Posterior c.o.v. of modal parameters of ten-storied building (synthetic data). Circle: exact value (δ_x); cross: uncertainty law (δ_{x1}); $x = f, \zeta, S, S_e, \phi$

uncertainty laws (crosses). The uncertainty laws give a good approximation of the exact values. The apparent random fluctuation in the trend between the posterior c.o.v. and modal s/n ratio stems from the fluctuation in the MPV of modal parameters determined from data with different measured DOFs. If the MPVs were the same in all cases, the points would have formed a smooth decaying trend to the zeroth order law. The plots on the right column of Fig. 15.7 illustrate this point, where the ratio of the posterior c.o.v. to the zeroth order law is plotted. For the first order law (cross), this ratio is simply $1 + a_x/\gamma$, which only depends on the bandwidth factor κ (through a_x) and modal s/n ratio γ . As the selected band has been fixed, κ is relatively constant and so the ratio is essentially a function of γ only. Again, the circles (exact/zeroth order) and the crosses (first order/zeroth order) agree with each other, both converging to 1 as γ increases. Figure 15.8 shows the results of mode 9 (S1), analogous to Fig. 15.7. This mode has the lowest γ among all modes and so it illustrates more clearly the decaying trend of the posterior c.o.v. with γ . ■

Example 15.3 (Field data, tall building) Revisit the tall building in Example 12.5. Acceleration data is now collected from a triaxial servo-accelerometer ($n = 3$ DOFs) placed in a room on the roof near the core, acquired for 18 h at 50 Hz under normal wind condition. The data is divided into 36 non-overlapping segments, each 30 min long. Modal identification is performed for the six modes below 1 Hz for each segment, each giving 36 sets of MPVs and posterior c.o.v.s. Figure 15.9

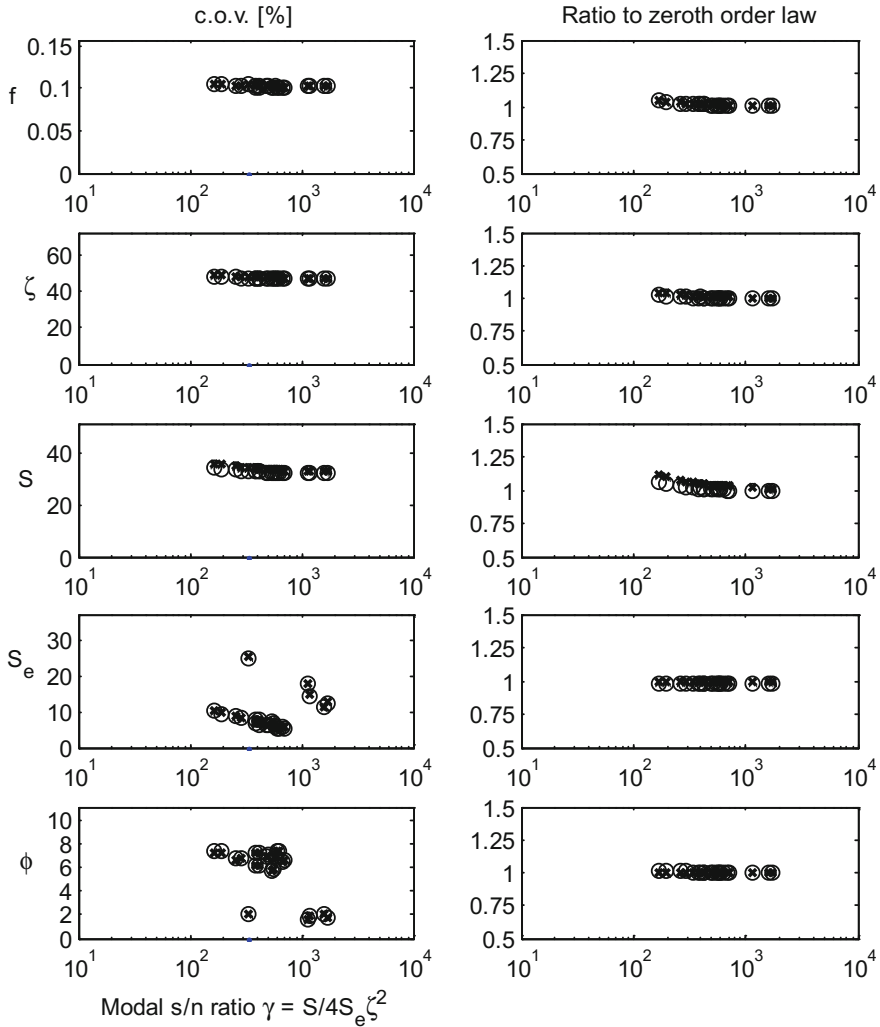


Fig. 15.7 Posterior c.o.v. of modal parameters versus modal s/n ratio γ , Mode 1 (TX1), laboratory frame data. *Left column* circle = exact value (δ_x), cross = uncertainty law (δ_{x1}), $x = f, \zeta, S, S_e, \phi$; *right column* ratio to zeroth order law, i.e., δ_x/δ_{x0} (circle) and δ_{x1}/δ_{x0} (cross)

compares the exact values of the posterior c.o.v. with the values from the uncertainty laws, calculated using the MPVs in each data set. The agreement is generally good. Table 15.2 shows the statistics of the 36 identified values (MPV) of S_e , S and n/e ratio $v = S_e/S$, giving an idea of their order of magnitude in the field test. ■

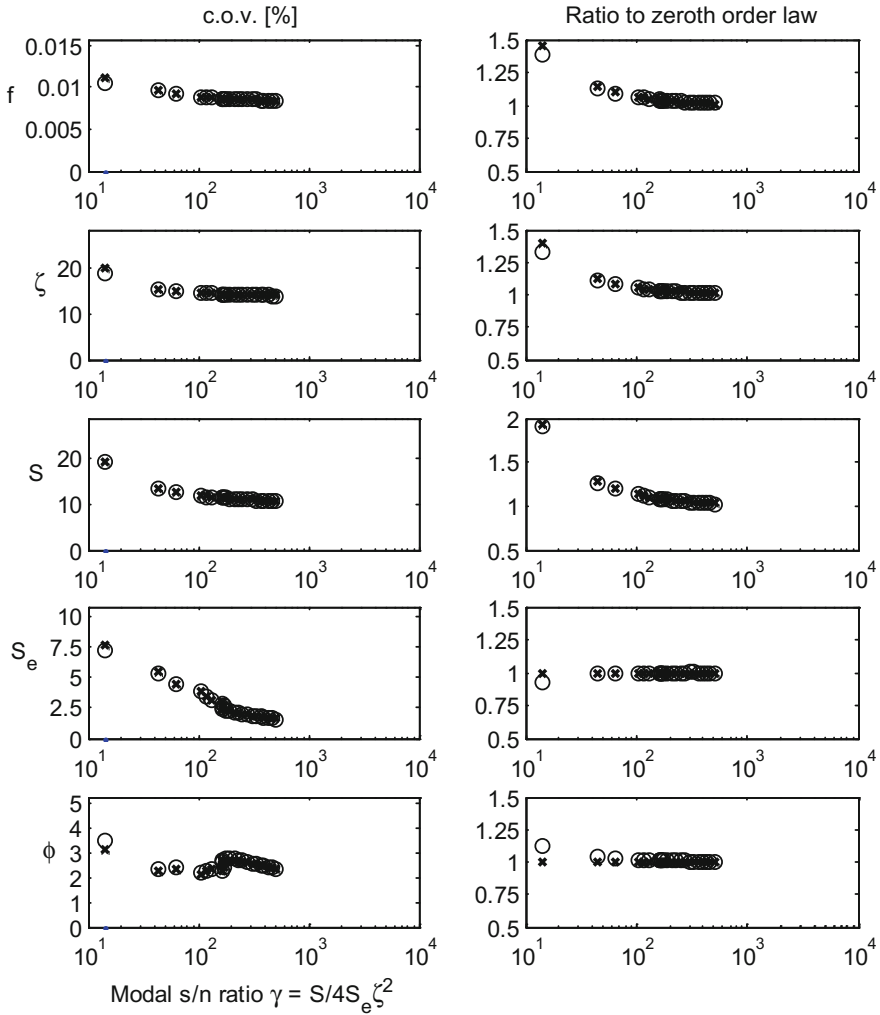


Fig. 15.8 Posterior c.o.v. of modal parameters versus modal s/n ratio γ , Mode 9 (S1), laboratory frame data. *Left column* circle = exact value (δ_x), cross = uncertainty law (δ_{x1}), $x = f, \zeta, S, S_e, \phi$; *right column* ratio to zeroth order law, i.e., δ_x/δ_{x0} (circle) and δ_{x1}/δ_{x0} (cross)

15.4 Planning Ambient Vibration Tests

In this section, we discuss some implications of the uncertainty laws with regard to planning for ambient vibration tests. According to the uncertainty laws, the posterior c.o.v.s of modal parameters depend only on the following scales:

1. damping ratio ζ
2. bandwidth factor κ

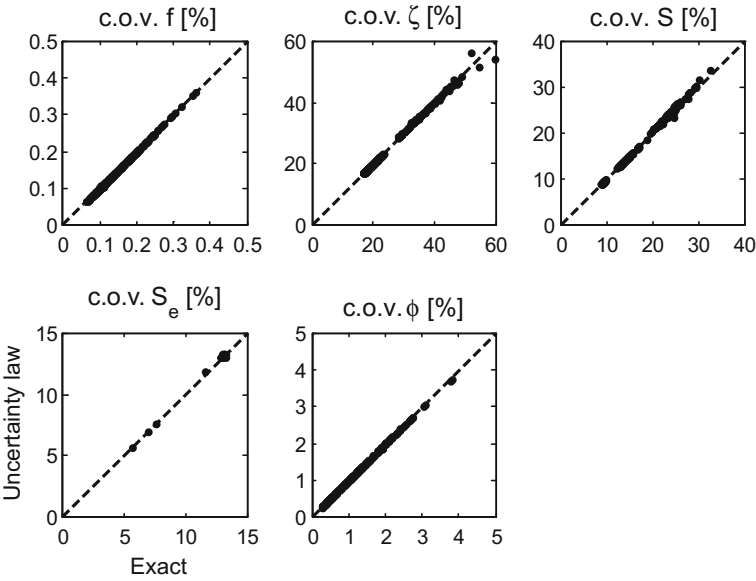


Fig. 15.9 Comparison of exact posterior c.o.v. with uncertainty law

Table 15.2 Statistics of S_e , S (MPV, one-sided) and $v = S_e/S$, tall building example

Mode	S_e [$(\mu\text{g})^2/\text{Hz}$]			S [$(\mu\text{g})^2/\text{Hz}$]			$v = S_e/S$		
	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max
1	68	301	977	4.0	19	61	5.2	17	41
2	44	208	501	4.6	19	61	6.2	13	36
3	0.8	2.2	7.9	0.1	0.4	1.2	3.0	5.5	11
4	0.4	0.8	2.1	0.2	0.6	1.5	0.8	1.5	2.5
5	0.6	2.3	6.4	0.1	0.4	1.0	4.2	6.7	11
6	0.3	1.6	6.1	0.02	0.1	0.2	10	19	36

- 3. normalized data length $N_c = T_d f$
- 4. noise-to-environment (n/e) ratio $v = (2\pi f)^{2q} S_e/S$, $q = 0$ (acceleration noise), 1 (velocity noise) or 2 (displacement noise)

The damping ratio ζ is a property of the subject structure. Assuming that one always uses the widest possible bandwidth for modal identification, the bandwidth factor κ is a result of test configuration rather than a parameter than can be actively changed. Thus, only the data length N_c and n/e ratio v are related to test configuration, through the following:

1. data duration T_d
2. channel noise level S_e
3. measured DOFs (number and location)

The sophistication of decision-making process depends on the available information or assumption one is willing to make. The damping ratio has the largest posterior uncertainty and so it is the governing factor for planning decisions.

15.4.1 Simple Rule of Thumb

The zeroth order law of damping uncertainty provides a simple decision rule for the data duration independent of equipment configuration, as it is the limit for infinite modal s/n ratio. The required data length as a multiple of natural period to achieve a posterior c.o.v. of δ in the damping ratio is given by

$$N_c = \frac{1}{2\pi\zeta\delta^2 B_\zeta(\kappa)} \quad (15.7)$$

where $B_\zeta(\kappa)$ can be found in Table 15.1. For $\zeta = 1\%$, $\delta = 30\%$ and $\kappa = 6$ (for which $B_\zeta \approx 0.6$), (15.7) gives a rule of thumb of 300 natural periods. To allow for deviations from these assumptions, one may use some convenient multiple of this rule of thumb as long as the measured data can still be considered stationary, e.g., 500 natural periods or 1000 natural periods.

15.4.2 Accounting for Channel Noise and Measured DOFs

To account for channel noise and measured DOFs, one needs to go beyond the rule of thumb that is based on the zeroth order law. One can apply the first order law of damping uncertainty to obtain the required data length:

$$N_c = \frac{1 + a_\zeta(\kappa)\gamma^{-1}}{2\pi\zeta\delta^2 B_\zeta(\kappa)} \quad (15.8)$$

where $a_\zeta(\kappa)$ can be found in Table 15.1 and $\gamma = S/4S_e\zeta^2(2\pi f)^{2q}$ is the modal s/n ratio. For $\kappa = 6$, $a_\zeta \approx 12.7$. To assess γ , assume (say) acceleration data ($q = 0$) and a nominal value of 1% for the damping ratio. Here, S_e is the channel noise PSD in the vicinity of the natural frequency. This depends on the quality of sensor and data acquisition system (cables, DAQ hardware, etc.). For example, well-managed systems with servo-accelerometers have a channel noise PSD S_e in the order of 1 $(\mu\text{g})^2/\text{Hz}$. The accelerometer data acquired by smart phone in Sect. 6.3 has a S_e in

the order of $10^5(\mu\text{g})^2/\text{Hz}$. The modal force PSD S depends on environmental excitation intensity and the measured DOFs. On a normal day, S may range (say) between 0.01 to $10(\mu\text{g})^2/\text{Hz}$ for a single DOF suitably placed for the mode. If the additional DOFs have a mode shape value less than the existing measured DOF, adding such DOFs can at most increase the value of S linearly.

The sensitivity of $\delta_{\zeta 1}$ to γ determines whether the posterior c.o.v. can be further reduced effectively by improving the quality of equipment or increasing the number of measured DOFs. For this purpose, plotting $\delta_{\zeta 1}/\delta_{\zeta 0} = \sqrt{1 + a_{\zeta}/\gamma}$ and locating where the existing configuration stands on the curve can give an idea of sensitivity. Increasing data duration can always reduce identification uncertainty in an inverse square root manner, although this needs to be traded-off with model error risk in terms of, e.g., stationarity of measured response (the longer the higher).

Example 15.4 (Planning for tall building) Here we consider a hypothetical ambient test on a tall building to illustrate the use of uncertainty laws in determining whether a mode can be identified reliably under different situations. Consider the fundamental translational mode with natural frequency $f = 0.2$ Hz and damping ratio $\zeta = 1\%$. In terms of data length requirement, the fundamental mode is the most demanding because it has the longest period.

Low channel noise

Suppose we use a triaxial servo-accelerometer (i.e., 3 DOFs) with a noise PSD of $S_e = 1(\mu\text{g})^2/\text{Hz}$, typical of good quality data channel for this purpose. On a normal day with a sensor placed at one corner on the roof, assume a modal force PSD of $S = 0.1(\mu\text{g})^2/\text{Hz}$ (microtremor level). Then $v = 1/0.1 = 10$ and $\gamma = 1/4(10)(0.01)^2 = 250$. From Fig. 15.5 the ratio $\delta_{\zeta 1}/\delta_{\zeta 0}$ is flat for $\gamma > 100$ (regardless of κ) and so the posterior c.o.v. is insensitive to the measured DOFs and quality of data channel. They are already good enough to give a high modal s/n ratio. The zeroth order and first order law give practically the same posterior c.o.v. value. Assume $\kappa = 6$ and so $B_{\zeta} \approx 0.6$. With 300 natural periods of data, i.e., $(5) \times (300) = 1500$ s,

$$\delta_{\zeta} \approx \delta_{\zeta 1} \approx \delta_{\zeta 0} = 1/\sqrt{2\pi(0.01)(300)(0.6)} \approx 30\%$$

Increasing the data duration (while trading off with modeling error risk) is the only effective means that can further reduce identification uncertainty. Adding sensors or further reducing channel noise (e.g., better sensor) will have little or no effect.

High channel noise

Suppose now the sensor or data acquisition system has lower quality, in the sense that $S_e = 100(\mu\text{g})^2/\text{Hz}$ (100 times higher than before). This gives $v = 100/0.1 = 1000$ and $\gamma = 1/(4)(1000)(0.01)^2 = 2.5$. The zeroth order law is the same as before (about 30%) because it does not depend on the modal s/n ratio. For $\kappa = 6$, $a_{\zeta} \approx 12.7$ and so the first order law gives

$$\delta_{\zeta 1} = \delta_{\zeta 0} \sqrt{1 + a_{\zeta}/\gamma} = (0.3) \sqrt{1 + 12.7/2.5} = 74\%$$

This is quite large. From Fig. 15.5, the ratio $\delta_{\zeta 1}/\delta_{\zeta 0}$ is sensitive to γ when $\gamma < 10$ and so the posterior c.o.v. may be effectively reduced by improving the quality of sensor or DAQ system, or increasing the number of measured DOFs. Some options are considered below:

Reducing channel noise. Reducing the channel noise PSD S_e from 100 to 10 $(\mu\text{g})^2/\text{Hz}$ reduces ν to 100 and increases γ to 25. The zeroth order law $\delta_{\zeta 0}$ is unaffected but the ratio $\delta_{\zeta 1}/\delta_{\zeta 0}$ is reduced from $\sqrt{1 + 12.7/2.5} = 2.47$ to $\sqrt{1 + 12.7/25} = 1.23$, i.e., by half. As a result, the posterior c.o.v. of damping ratio is reduced from 74% to 37%. Further reducing the channel noise PSD to 1 $(\mu\text{g})^2/\text{Hz}$ or lower can at best bring $\delta_{\zeta 1}$ down to 30% and so has diminishing return.

Additional sensors. Suppose we keep using the same sensor and DAQ system, i.e., S_e remains at 100 $(\mu\text{g})^2/\text{Hz}$, but now we install additional triaxial sensors at other corners on the roof. Since the mode shape values at other corners is similar to the existing ones, modal force PSD S grows roughly linearly with the number of sensors. Suppose we put another three sensors at the remaining three corners on the roof. This will increase S by four times from 0.1 to 0.4 $(\mu\text{g})^2/\text{Hz}$ and hence γ from 2.5 to 10. The resulting ratio of $\delta_{\zeta 1}/\delta_{\zeta 0}$ reduces from 2.47 to $\sqrt{1 + 12.7/10} = 1.5$ and the posterior c.o.v. $\delta_{\zeta 1}$ reduces from 74% to (30%) (1.5) = 45%.

Longer data length. Without additional sensors, the posterior c.o.v. can be reduced effectively by increasing data duration. Doubling data duration from 300 to 600 natural periods reduces the posterior c.o.v to $1/\sqrt{2}$ of its original value (i.e., by about 30%) to 32%, which may be acceptable. The resulting data duration is (5) (600) = 3000 s (about an hour). One needs to examine the data to see if a stochastic stationary assumption is justified. This is especially relevant for damping ratios and modal force PSDs. ■

15.5 Common Sense

Quick comments are in order, some we knew and now confirmed by uncertainty laws; and some we were puzzled and now obtain surprising answer.

Identification uncertainty in terms of c.o.v. reduces in an inverse square root manner with data length. To reduce c.o.v. by half, one needs to quadruple data length. The natural frequency and mode shape are much easier to identify than the damping ratio, because the posterior c.o.v. of the former is $O(\sqrt{\zeta})$ but that of the latter is $O(1/\sqrt{\zeta})$. These agree with our intuitions.

If we want to obtain mode shape at more locations, then obviously we need more sensors. Discovering more modes or separating close modes are also common reasons for deploying more sensors. Otherwise, improving the precision of modal

properties is not a good justification for deploying more sensors. One either has good enough accuracy or it would not help with just a few more sensors. This is because if the quality of sensor is good (say servo-accelerometer with $\text{micro-g}/\sqrt{\text{Hz}}$ noise), then chances are the modal s/n ratio is already good enough even with one or two sensors. Otherwise, a material reduction in the posterior c.o.v. of natural frequency or damping ratio would require an order of magnitude increase in the modal s/n ratio and hence the number of sensors. Thus, a structural health monitoring (SHM) system with only one or two good accelerometers can give practically the same accuracy for the natural frequency or damping ratio as another one with hundreds. If the sensors have high noise, they are simply not fit for the purpose and should be substituted by better ones. Trying to make this up by using more sensors is not cost-effective.

Chapter 16

Theory of Uncertainty Laws

Abstract This chapter discusses the theory behind the uncertainty laws in Chap. 15. It is intended for researchers who are interested in understanding the mathematics of the theory or learning the techniques for further development.

Keywords Uncertainty laws • Asymptotics • Fisher information matrix

In this chapter we discuss the reasoning and derivation behind the uncertainty laws for operational modal analysis (OMA) in Chap. 15. Recall the context of identifying well-separated modes with single setup data in Chap. 12. The set of modal parameters is $\boldsymbol{\theta} = \{f, \zeta, S, S_e, \boldsymbol{\phi}\}$, where f is the natural frequency (Hz), ζ is the damping ratio, S is the modal force PSD, S_e is the prediction error PSD and $\boldsymbol{\phi}$ ($n \times 1$) is the mode shape (confined to the measured DOFs); n is the number of measured DOFs. Assuming a uniform prior probability density function (PDF), the posterior PDF of $\boldsymbol{\theta}$ given the scaled Fast Fourier Transform (FFT) $\{\hat{\mathcal{F}}_k\}$ ($n \times 1$ complex) of data in a selected frequency band containing the mode is $p(\boldsymbol{\theta}|\{\hat{\mathcal{F}}_k\}) \propto \exp[-L(\{\hat{\mathcal{F}}_k\}, \boldsymbol{\theta})]$, where

$$L(\{\hat{\mathcal{F}}_k\}, \boldsymbol{\theta}) = nN_f \ln \pi + \sum_k \ln |\mathbf{E}_k| + \sum_k \hat{\mathcal{F}}_k^* \mathbf{E}_k^{-1} \hat{\mathcal{F}}_k \quad (16.1)$$

is the negative log-likelihood function (NLLF), whose dependence on $\{\hat{\mathcal{F}}_k\}$ and $\boldsymbol{\theta}$ has been explicitly indicated; N_f is the number of FFT points in the selected band;

$$\mathbf{E}_k = SD_k \bar{\boldsymbol{\phi}} \bar{\boldsymbol{\phi}}^T + S_e \mathbf{I}_n \quad (16.2)$$

is the theoretical PSD matrix of data for given $\boldsymbol{\theta}$;

$$D_k = \frac{(2\pi f_k)^{-2q}}{(1 - \beta_k^2)^2 + (2\zeta\beta_k)^2} \quad \beta_k = \frac{f}{f_k} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (16.3)$$

is the dynamic amplification factor at FFT frequency $f_k = k/T_d$; T_d is the data duration (s); and

$$\bar{\boldsymbol{\phi}} = ||\boldsymbol{\phi}||^{-1} \boldsymbol{\phi} \quad (16.4)$$

is the normalized mode shape. Here $\mathbf{E}_k$ is written in terms of $\bar{\boldsymbol{\phi}}$ so that the Hessian of L w.r.t. $\boldsymbol{\phi}$ automatically accounts for mode shape norm constraint.

The posterior covariance matrix of $\boldsymbol{\theta}$ is equal to the pseudo-inverse (i.e., ignoring norm constraint singularity) of the Hessian of L w.r.t. $\boldsymbol{\theta}$ at the posterior most probable value (MPV). Uncertainty laws are asymptotic expressions for the leading order of the posterior covariance matrix for long data ($N_f \rightarrow \infty$) and small damping ($\zeta \rightarrow 0$). The FFT data $\{\hat{\mathcal{F}}_k\}$ is unknown before test but it is assumed to be distributed as the likelihood function.

Long data asymptotics, small damping asymptotics, and asymptotic decoupling are three important concepts in the derivation. Long data asymptotics retains in the leading order of the Hessian of NLLF the ‘information content’ of data (rather than its specific details), equal to the Fisher information matrix. The latter involves sums of dynamic amplification factor and derivatives over the frequency band. Consideration of small damping and hence high modal signal-to-noise (s/n) ratio leads to closed form expressions. Taking inverse of the resulting Hessian is still algebraically intractable. For long data, the mode shape is decoupled from the remaining parameters, in the sense that the cross derivatives vanish asymptotically. For long data and small damping, the prediction error PSD and natural frequency are also decoupled from the remaining parameters. These findings reduce the dimension of inverse, making it algebraically manageable and eventually leading to closed form expressions.

The uncertainty laws are generally applicable for displacement, velocity or acceleration data. Acceleration data is assumed in the main derivation. Necessary modifications to the derivation for velocity or displacement data are discussed in Sect. 16.6.

16.1 Long Data Asymptotics

The leading order of the posterior covariance matrix for long data can be obtained from first principle or by using its connection with the Fisher information matrix (FIM). From first principle, substituting the stochastic representation of scaled FFT into the second derivatives of the NLLF leads to an expression comprising sums of random terms, whose leading order is its expectation for large N_f . This approach was adopted when the uncertainty laws were first derived (Au 2014). A more systematic approach is to make use of the connection with the FIM (Sect. 9.6.3). Let $\hat{L}^{(xy)}$ denote the derivative of the NLLF w.r.t. variables x and y , and evaluated at the MPV for given $\{\hat{\mathcal{F}}_k\}$ distributed as the likelihood function $p(\{\hat{\mathcal{F}}_k\}|\boldsymbol{\theta})$. Then

$$\hat{L}^{(xy)} = J_{xy}(\boldsymbol{\theta})[1 + O(N_f^{-1/2})] \quad N_f \rightarrow \infty \quad (16.5)$$

where

$$J_{xy}(\boldsymbol{\theta}) = E[L^{(xy)}|\boldsymbol{\theta}] \quad (16.6)$$

is the entry corresponding to parameter pair (x, y) in the FIM. This implies that $\hat{L}^{(xy)} \sim J_{xy}$ if $J_{xy} \neq 0$. Otherwise, the leading order of $\hat{L}^{(xy)}$ will need to be determined by other means. Since the likelihood function is a complex Gaussian PDF, one can apply the standard result in Sect. 9.4.2 to give

$$J_{xy} = \sum_k \text{tr}[\mathbf{E}_k^{-1} \mathbf{E}_k^{(x)} \mathbf{E}_k^{-1} \mathbf{E}_k^{(y)}] \quad (16.7)$$

where a superscripted variable denotes a partial derivative w.r.t. it; $\text{tr}(\cdot)$ denotes the trace (i.e., sum of diagonal entries) of the argument matrix.

Abuse in Notation

Some abuse in notation has been used to simplify presentation but it should be clarified to avoid conceptual confusion. In (16.5) and (16.6), $\boldsymbol{\theta}$ denotes the *actual* parameter value that gives rise to the FFT data $\{\hat{\mathcal{F}}_k\}$. In (16.7), x and y (which come from $\boldsymbol{\theta}$) are variables, for otherwise it does not make sense to take derivative of $\mathbf{E}_k$ w.r.t. them. The derived expression of J_{xy} will involve x and y , and hence $\boldsymbol{\theta}$, which should then be interpreted as the actual parameter value again. This type of confusion or interpretation arises from the classical statistical context of FIM.

16.1.1 Fisher Information Matrix

The trace in (16.7) can be evaluated effectively using the eigenvector representation of $\mathbf{E}_k$. Let $\{\mathbf{b}_i\}_{i=1}^n$ be an orthonormal basis in R^n with $\mathbf{b}_1 = \bar{\boldsymbol{\varphi}}$. Clearly, it depends on $\bar{\boldsymbol{\varphi}}$ but not on other parameters (f, ζ, S, S_e). Substituting $\mathbf{I}_n = \sum_{i=1}^n \mathbf{b}_i \mathbf{b}_i^T$ into $\mathbf{E}_k$ in (16.2) and rearranging gives the eigenvector representation of $\mathbf{E}_k$, from which $\mathbf{E}_k^{-1}$ can also be obtained:

$$\begin{aligned} \mathbf{E}_k &= \underbrace{(SD_k + S_e)}_{\text{eigenvalue}} \mathbf{b}_1 \mathbf{b}_1^T + \underbrace{S_e}_{\text{eigenvalue}} \sum_{i=2}^n \mathbf{b}_i \mathbf{b}_i^T \\ \mathbf{E}_k^{-1} &= \underbrace{(SD_k + S_e)^{-1}}_{\text{eigenvalue}} \mathbf{b}_1 \mathbf{b}_1^T + \underbrace{S_e^{-1}}_{\text{eigenvalue}} \sum_{i=2}^n \mathbf{b}_i \mathbf{b}_i^T \end{aligned} \quad (16.8)$$

Terms Not Involving Mode Shape

For $x = f, \zeta, S, S_e$, differentiation simply operates on the eigenvalues:

$$\mathbf{E}_k^{(x)} = (SD_k + S_e)^{(x)} \mathbf{b}_1 \mathbf{b}_1^T + S_e^{(x)} \sum_{i=2}^n \mathbf{b}_i \mathbf{b}_i^T \quad (16.9)$$

The same also applies to the product. For $x, y = f, \zeta, S, S_e$,

$$\mathbf{E}_k^{-1} \mathbf{E}_k^{(x)} \mathbf{E}_k^{-1} \mathbf{E}_k^{(y)} = \frac{(SD_k + S_e)^{(x)} (SD_k + S_e)^{(y)}}{(SD_k + S_e)^2} \mathbf{b}_1 \mathbf{b}_1^T + \frac{S_e^{(x)} S_e^{(y)}}{S_e^2} \sum_{i=2}^n \mathbf{b}_i \mathbf{b}_i^T \quad (16.10)$$

Taking trace, noting $\text{tr}(\mathbf{b}_i \mathbf{b}_i^T) = 1$ ($i = 1, \dots, n$) and summing over k ,

$$J_{xy} = \sum_k \frac{(SD_k + S_e)^{(x)} (SD_k + S_e)^{(y)}}{(SD_k + S_e)^2} + (n-1) N_f \frac{S_e^{(x)} S_e^{(y)}}{S_e^2} \quad x, y = f, \zeta, S, S_e \quad (16.11)$$

Terms Involving Mode Shape

The terms $J_{x\phi}$ ($x = f, \zeta, S, S_e$) and $J_{\phi\phi}$ involve differentiation w.r.t. ϕ . Analysis can be simplified by making use of orthogonality between ϕ and the gradient of $\bar{\phi} = \|\phi\|^{-1} \phi$. Let $\phi = [\phi_1, \dots, \phi_n]^T$. Then it can be shown that (Section C.5.5)

$$\nabla \bar{\phi} = \begin{bmatrix} \frac{\partial \bar{\phi}}{\partial \phi_1} & \dots & \frac{\partial \bar{\phi}}{\partial \phi_n} \end{bmatrix} = \mathbf{I}_n - \bar{\phi} \bar{\phi}^T \quad (\|\phi\| = 1) \quad (16.12)$$

It follows that

$$(\nabla \bar{\phi})^T = \nabla \bar{\phi} \quad (\nabla \bar{\phi})(\nabla \bar{\phi}) = \nabla \bar{\phi} \quad (\nabla \bar{\phi})\phi = \mathbf{0} \quad \phi^T(\nabla \bar{\phi}) = \mathbf{0} \quad (16.13)$$

$$\mathbf{E}_k^{(\phi_i)} = SD_k[\bar{\phi}^{(\phi_i)} \bar{\phi}^T + \bar{\phi} \bar{\phi}^{(\phi_i)T}] = SD_k[(\nabla \bar{\phi}) \mathbf{e}_i \bar{\phi}^T + \bar{\phi} \mathbf{e}_i^T (\nabla \bar{\phi})] \quad (16.14)$$

where $\mathbf{e}_i$ is a $n \times 1$ vector whose i th entry is the only non-zero entry equal to 1. Using these, it can be shown that for $\|\phi\| = 1$ and $x = f, \zeta, S, S_e$,

$$\mathbf{E}_k^{-1} \mathbf{E}_k^{(x)} = \frac{(SD_k + S_e)^{(x)}}{SD_k(1 + e_k)} \bar{\phi} \bar{\phi}^T + \frac{S_e^{(x)}}{S_e} \nabla \bar{\phi} \quad (16.15)$$

$$\mathbf{E}_k^{-1} \mathbf{E}_k^{(\phi_i)} = (1 + e_k)^{-1} \bar{\phi} \mathbf{e}_i^T (\nabla \bar{\phi}) + e_k^{-1} (\nabla \bar{\phi}) \mathbf{e}_i \bar{\phi}^T$$

$$\mathbf{E}_k^{-1} \mathbf{E}_k^{(x)} \mathbf{E}_k^{-1} \mathbf{E}_k^{(\phi_i)} = \frac{(SD_k + S_e)^{(x)}}{SD_k(1 + e_k)^2} \bar{\phi} \mathbf{e}_i^T (\nabla \bar{\phi}) + \frac{S_e^{(x)}}{S_e e_k} (\nabla \bar{\phi}) \mathbf{e}_i \bar{\phi}^T \quad (16.16)$$

$$\mathbf{E}_k^{-1} \mathbf{E}_k^{(\phi_i)} \mathbf{E}_k^{-1} \mathbf{E}_k^{(\phi_j)} = (1 + e_k)^{-1} e_k^{-1} \left[\bar{\boldsymbol{\phi}} \mathbf{e}_i^T (\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_j \bar{\boldsymbol{\phi}}^T + (\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_i \mathbf{e}_j^T (\nabla \bar{\boldsymbol{\phi}}) \right] \quad (16.17)$$

where

$$e_k = \frac{S_e}{SD_k} \quad (16.18)$$

Taking trace on (16.16) and summing over k gives

$$J_{x\phi_i} = 0 \quad x = f, \zeta, S, S_e \quad (16.19)$$

since $\text{tr}[\bar{\boldsymbol{\phi}} \mathbf{e}_i^T (\nabla \bar{\boldsymbol{\phi}})] = \text{tr}[(\nabla \bar{\boldsymbol{\phi}}) \bar{\boldsymbol{\phi}} \mathbf{e}_i^T] = 0$ and $\text{tr}[(\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_i \bar{\boldsymbol{\phi}}^T] = \text{tr}[\bar{\boldsymbol{\phi}}^T (\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_i] = 0$ (cyclic property of trace). This indicates that the leading order of $\hat{L}^{(x\phi_i)}$ is not given by the FIM. Direct investigation of $\hat{L}^{(x\phi_i)}$, which is a sum of N_f random terms, shows that it is $O(N_f^{1/2})$.

On the other hand, taking trace on (16.17) and summing over k gives

$$J_{\phi_i \phi_j} = 2 \sum_k (1 + e_k)^{-1} e_k^{-1} [\mathbf{e}_i^T (\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_j] \quad (16.20)$$

since, using the cyclic property of trace,

$$\begin{aligned} \text{tr}[\bar{\boldsymbol{\phi}} \mathbf{e}_i^T (\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_j \bar{\boldsymbol{\phi}}^T] &= \text{tr}[\underbrace{\bar{\boldsymbol{\phi}}^T \bar{\boldsymbol{\phi}}}_1 \underbrace{\mathbf{e}_i^T (\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_j}_{\text{scalar}}] = \mathbf{e}_i^T (\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_j \\ \text{tr}[(\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_i \mathbf{e}_j^T (\nabla \bar{\boldsymbol{\phi}})] &= \text{tr}[\underbrace{\mathbf{e}_j^T (\nabla \bar{\boldsymbol{\phi}}) (\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_i}_{\text{scalar}}] = \mathbf{e}_j^T \underbrace{(\nabla \bar{\boldsymbol{\phi}})}_{\text{sym.}} \mathbf{e}_i = \mathbf{e}_i^T (\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_j \end{aligned} \quad (16.21)$$

Since $\mathbf{e}_i^T (\nabla \bar{\boldsymbol{\phi}}) \mathbf{e}_j$ is simply the (i, j) -entry of $\nabla \bar{\boldsymbol{\phi}}$, (16.20) implies

$$J_{\boldsymbol{\phi}\boldsymbol{\phi}} = 2 \sum_k (1 + e_k)^{-1} e_k^{-1} (\nabla \bar{\boldsymbol{\phi}}) = 2 \sum_k (1 + e_k)^{-1} e_k^{-1} [\mathbf{I}_n - \bar{\boldsymbol{\phi}} \bar{\boldsymbol{\phi}}^T] \quad (16.22)$$

Table 16.1 summarizes the long data asymptotic expressions for the second derivatives of the NLLF at the MPV. The derivatives involving D_k have been written in terms of those of D_k^{-1} , as the latter are easier to analyze. For acceleration data ($q = 0$),

Table 16.1 Leading order of $\hat{L}^{(xy)}$ for $N_f \rightarrow \infty$; $e_k = S_e/SD_k$; $\|\Phi\| = 1$

$y =$		S	S_e	Φ
$x =$	f, ζ	$\sum_k \frac{D_k^2 (D_k^{-1})^{(x)} (D_k^{-1})^{(y)}}{(1+e_k)^2}$		Sym.
	S	$-S^{-1} \sum_k \frac{D_k (D_k^{-1})^{(y)}}{(1+e_k)^2}$		
	S_e	$-S^{-1} \sum_k \frac{(D_k^{-1})^{(y)}}{(1+e_k)^2}$	$S_e^{-2} \left[(n-1)N_f + \sum_k \frac{e_k^2}{(1+e_k)^2} \right]$	
	Φ	$O(N_f^{1/2})$	$O(N_f^{1/2})$	$\left[2 \sum_k (1+e_k)^{-1} e_k^{-1} \right] (\mathbf{1}_n - \Phi \Phi^T)$

$$(D_k^{-1})^{(f)} = 4f^{-1}\beta_k^2(\beta_k^2 - 1 + 2\zeta^2) \quad (D_k^{-1})^{(\zeta)} = 8\zeta\beta_k^2 \quad (16.23)$$

For $\hat{L}^{(S_e S_e)}$, the second term in the bracket can be omitted as it is dominated by the first term. It is retained to be consistent with the expression of J_{xy} in (16.11).

16.2 Small Damping Asymptotics

The long data asymptotic expressions in Table 16.1 involve discrete sums over frequencies. More insightful expressions can be obtained for small damping. Using the Taylor approximation $(1 + e_k)^{-1} \sim 1 - e_k$, the leading order of the sums can be written in the standard form $\sum_k D_k^a (\beta_k - 1)^b \beta_k^c$, where a , b and c are integers. Viewed as a Riemann sum, this can be shown to be asymptotic to an integral:

$$\sum_k D_k^a (\beta_k - 1)^b \beta_k^c \sim (-1)^b 2^{-2a} N_c \zeta^{b-2a+1} \int_{-\kappa}^{\kappa} \frac{(1 + \zeta u)^{4a-b-c} u^b}{[u^2(1 + \zeta u/2)^2 + (1 + \zeta u)^2]^a} du$$

$N_f \rightarrow \infty$

(16.24)

Analytical expression for the integral can be obtained for $\zeta \rightarrow 0$. Details can be found in Appendix A of this chapter. Using these results, closed-form asymptotic expressions for the second derivatives of the NLLF at the MPV can be obtained. Details can be found in Appendix B of this chapter. The results are summarized in Table 16.2.

16.3 Asymptotic Decoupling

Inverting the Hessian of the NLLF is analytically intractable even when using the leading order expressions in Table 16.2. Further simplification results by discovering that the mode shape, prediction error PSD and natural frequency are asymptotically ‘decoupled’ from the remaining parameters, reducing the dimension of inverse.

16.3.1 Scalar Parameter

A parameter θ_i is ‘perfectly decoupled’ from the remaining ones if $\hat{L}^{(\theta_i \theta_j)} = 0$ for all $j \neq i$. In general, coupling can be considered small if $|q_{\theta_i \theta_j}| \ll 1$ for all $j \neq i$, where

Table 16.2 Leading (zeroth) order of $\hat{L}^{(xy)}$ for $N_f \rightarrow \infty$ and $\zeta \rightarrow 0$; $\|\boldsymbol{\phi}\| = 1$

$y =$		ζ	S	S_e	$\boldsymbol{\phi}$
$x =$	f	$\frac{4N_c}{f^2\zeta} \left(\tan^{-1} \kappa - \frac{\kappa}{\kappa^2 + 1} \right)$			Sym.
	ζ	$\frac{4N_c}{f} \left[\tan^{-1} \kappa + \frac{2\kappa^3 + \kappa}{(\kappa^2 + 1)^2} \right]$	$\frac{4N_c}{\zeta} \left(\tan^{-1} \kappa + \frac{\kappa}{\kappa^2 + 1} \right)$		
	S	$-\frac{2N_c\zeta}{fS} \left(2 \tan^{-1} \kappa + \frac{\kappa^3}{\kappa^2 + 1} \right)$	$\frac{N_f}{S^2}$		
	S_e	$-\frac{8N_c\zeta^3}{3fS} (7\kappa^3 + 6\kappa)$	$\frac{8N_c\zeta^3}{3S^2} (\kappa^3 + 3\kappa)$	$\frac{(n-1)N_f}{S_e^2}$	
	$\boldsymbol{\phi}$	$O(N_f^{1/2})$	$O(N_f^{1/2})$	$O(N_f^{1/2})$	$\frac{N_c \tan^{-1} \kappa}{v_\zeta} (\mathbf{I}_n - \bar{\boldsymbol{\phi}} \bar{\boldsymbol{\phi}}^T)$

$$q_{\theta_i\theta_j} = \frac{\hat{L}^{(\theta_i\theta_j)}}{[\hat{L}^{(\theta_i\theta_i)}]^{1/2}[\hat{L}^{(\theta_j\theta_j)}]^{1/2}} \quad (16.25)$$

is the (dimensionless) ‘cross sensitivity coefficient’ between θ_i and θ_j .

When θ_i is asymptotically decoupled from all the remaining parameters, its leading order posterior variance is simply given by $1/\hat{L}^{(\theta_i\theta_i)}$. This can be reasoned as follow. The full Hessian of L at the MPV, denoted by $\hat{H}$ ($n_\theta \times n_\theta$), can be factored as

$$\hat{H} = \text{diag}[\{\sqrt{\hat{L}^{(\theta_j\theta_j)}}\}_{j=1}^{n_\theta}] \mathbf{Q} \text{diag}[\{\sqrt{\hat{L}^{(\theta_j\theta_j)}}\}_{j=1}^{n_\theta}] \quad (16.26)$$

where $\text{diag}[\{\cdot\}_{j=1}^{n_\theta}]$ denotes a diagonal matrix formed by the entries in the argument; $\mathbf{Q}$ is the sensitivity matrix whose (i,j) -entry is $q_{\theta_i\theta_j}$. The diagonal entries of $\mathbf{Q}$ are all equal to 1. Taking inverse,

$$\hat{H}^{-1} = \text{diag}[\{1/\sqrt{\hat{L}^{(\theta_j\theta_j)}}\}_{j=1}^{n_\theta}] \mathbf{Q}^{-1} \text{diag}[\{1/\sqrt{\hat{L}^{(\theta_j\theta_j)}}\}_{j=1}^{n_\theta}] \quad (16.27)$$

Without loss of generality, suppose the cross sensitivity coefficient of θ_1 with all other parameters is small, i.e., $q_{\theta_1\theta_j} \ll 1$ for all $j \neq 1$. Let $\mathbf{Q}$ and $\mathbf{Q}^{-1}$ be partitioned as

$$\mathbf{Q} = \begin{bmatrix} 1 & \mathbf{q}^T \\ \mathbf{q} & \mathbf{Q}_{-1} \end{bmatrix} \quad \mathbf{Q}^{-1} = \begin{bmatrix} a & \mathbf{b}^T \\ \mathbf{b} & \mathbf{B} \end{bmatrix} \quad (16.28)$$

$\begin{matrix} & & 1 \times (n_\theta - 1) \\ & & \mathbf{Q}_{-1} \\ (n_\theta - 1) \times 1 & & (n_\theta - 1) \times (n_\theta - 1) \end{matrix}$

where $\mathbf{q}^T = [q_{\theta_1\theta_2}, q_{\theta_1\theta_3}, \dots, q_{\theta_1\theta_{n_\theta}}]$ and a is a scalar. Substituting (16.28) into $\mathbf{Q}\mathbf{Q}^{-1} = \mathbf{I}$ (identity matrix) and solving gives $a = 1 + \mathbf{q}^T \mathbf{B} \mathbf{q}$. The posterior variance of θ_1 is then given by

$$\sigma_{\theta_1}^2 = \hat{H}^{-1}(1, 1) = \frac{a}{\hat{L}^{(\theta_1\theta_1)}} = \frac{1 + \mathbf{q}^T \mathbf{B} \mathbf{q}}{\hat{L}^{(\theta_1\theta_1)}} \quad (16.29)$$

Thus, if $q_{\theta_1\theta_j} \ll 1$ for all $j \neq 1$, then $\mathbf{q}^T \mathbf{B} \mathbf{q} \ll 1$ and so $\sigma_{\theta_1}^2 \sim 1/\hat{L}^{(\theta_1\theta_1)}$.

16.3.2 Vector-Valued Parameter

For a vector-valued parameter such as mode shape $\boldsymbol{\phi}$ ($n \times 1$), it is asymptotically decoupled from the remaining parameters if the cross sensitivity

$$q_{\theta_j \boldsymbol{\Phi}} = \max_{\substack{\mathbf{u} \in R^n \\ \|\mathbf{u}\| = 1}} \frac{\hat{L}^{(\theta_j \boldsymbol{\Phi})} \mathbf{u}}{[\hat{L}^{(\theta_j \theta_j)}]^{1/2} [\mathbf{u}^T \hat{L}^{(\boldsymbol{\Phi} \boldsymbol{\Phi})} \mathbf{u}]^{1/2}} \quad (16.30)$$

is small with all other parameters θ_j . This can be reasoned as follow. Consider the second order Taylor series of L for small perturbations $\Delta\theta_j$ and $\Delta\boldsymbol{\Phi}$ about the MPV:

$$\begin{aligned} \Delta L &= L(\hat{\theta}_j + \Delta\theta_j, \hat{\boldsymbol{\Phi}} + \Delta\boldsymbol{\Phi}) - L(\hat{\theta}_j, \hat{\boldsymbol{\Phi}}) \\ &\approx \frac{1}{2} \underbrace{[\hat{L}^{(\theta_j \theta_j)} \Delta\theta_j^2]}_{Q_{\theta_j \theta_j}} + 2 \underbrace{\Delta\theta_j \hat{L}^{(\theta_j \boldsymbol{\Phi})} \Delta\boldsymbol{\Phi}}_{Q_{\theta_j \boldsymbol{\Phi}}} + \underbrace{\Delta\boldsymbol{\Phi}^T \hat{L}^{(\boldsymbol{\Phi} \boldsymbol{\Phi})} \Delta\boldsymbol{\Phi}}_{Q_{\boldsymbol{\Phi} \boldsymbol{\Phi}}} \end{aligned} \quad (16.31)$$

where the first order terms vanish because L is minimized at the MPV. Writing in complete-square form,

$$\Delta L \approx \frac{1}{2} (Q_{\theta_j \theta_j}^{1/2} + Q_{\boldsymbol{\Phi} \boldsymbol{\Phi}}^{1/2})^2 - Q_{\theta_j \theta_j}^{1/2} Q_{\boldsymbol{\Phi} \boldsymbol{\Phi}}^{1/2} (1 - q) \quad (16.32)$$

where $q = Q_{\theta_j \boldsymbol{\Phi}} / Q_{\theta_j \theta_j}^{1/2} Q_{\boldsymbol{\Phi} \boldsymbol{\Phi}}^{1/2}$. Writing $\Delta\boldsymbol{\Phi} = \|\Delta\boldsymbol{\Phi}\| \mathbf{u}$, where $\mathbf{u}$ is a unit vector, we have $q = \hat{L}^{(\theta_j \boldsymbol{\Phi})} \mathbf{u} / [\hat{L}^{(\theta_j \theta_j)}]^{1/2} [\mathbf{u}^T \hat{L}^{(\boldsymbol{\Phi} \boldsymbol{\Phi})} \mathbf{u}]^{1/2}$. Thus, if $|q| \ll 1$ for any unit vector $\mathbf{u}$ then the second term in (16.32) is approximately $Q_{\theta_j \theta_j}^{1/2} Q_{\boldsymbol{\Phi} \boldsymbol{\Phi}}^{1/2}$ and hence $\Delta L \approx (Q_{\theta_j \theta_j}^2 + Q_{\boldsymbol{\Phi} \boldsymbol{\Phi}}^2)/2$, which does not involve any cross derivative term between θ_j and $\boldsymbol{\Phi}$. This condition is equivalent to $|q_{\theta_j \boldsymbol{\Phi}}| \ll 1$ defined in (16.30).

16.4 Leading Order Uncertainty

Using Table 16.2, the order of magnitude of cross sensitivity coefficients can be assessed. The results are summarized in Table 16.3. For cross sensitivities involving $\boldsymbol{\Phi}$, only the scaling order w.r.t. N_f is indicated. Terms with ζ or N_f^{-1} to some power are asymptotically zero.

Posterior Variance of f , S_e and $\boldsymbol{\Phi}$

Among all parameter pairs, the only non-vanishing sensitivity coefficient is $q_{\zeta S}$. Thus, f , S_e and $\boldsymbol{\Phi}$ are asymptotically decoupled from the remaining parameters. This implies that their posterior variances are asymptotically given by

$$\sigma_f^2 \sim \frac{1}{\hat{L}(f)} \quad \sigma_{S_e}^2 \sim \frac{1}{\hat{L}(S_e S_e)} \quad \hat{\mathbf{C}}_{\boldsymbol{\Phi}} \sim [\hat{L}^{(\boldsymbol{\Phi} \boldsymbol{\Phi})}]^{-1} \quad (16.33)$$

Substituting $\hat{L}^{(ff)}$, $\hat{L}^{(S_e S_e)}$ and $\hat{L}^{(\Phi\Phi)}$ from Table 16.2 gives their leading (zeroth) order posterior covariance in Table 16.4. The inverse of $\hat{L}^{(\Phi\Phi)}$ is obtained via its eigenvector representation.

Posterior Variance of ζ and S

The parameters ζ and S are coupled and need to be considered together. Their posterior covariance matrix is asymptotically the inverse of

$$\begin{bmatrix} \hat{L}^{(\zeta\zeta)} & \hat{L}^{(\zeta S)} \\ \hat{L}^{(\zeta S)} & \hat{L}^{(SS)} \end{bmatrix} = \begin{bmatrix} \sqrt{\hat{L}^{(\zeta\zeta)}} & \\ & \sqrt{\hat{L}^{(SS)}} \end{bmatrix} \begin{bmatrix} 1 & q_{\zeta S} \\ q_{\zeta S} & 1 \end{bmatrix} \begin{bmatrix} \sqrt{\hat{L}^{(\zeta\zeta)}} & \\ & \sqrt{\hat{L}^{(SS)}} \end{bmatrix} \quad (16.34)$$

where $q_{\zeta S} = \hat{L}^{(\zeta S)} / \sqrt{\hat{L}^{(\zeta\zeta)} \hat{L}^{(SS)}}$. Substituting $\hat{L}^{(\zeta\zeta)}$, $\hat{L}^{(SS)}$ and $\hat{L}^{(\zeta S)}$ in Table 16.2 gives

$$q_{\zeta S} \sim -\sqrt{2}(\tan^{-1} \kappa) \kappa^{-1/2} (\tan^{-1} \kappa + \frac{\kappa}{\kappa^2 + 1})^{-1/2} \quad (16.35)$$

Taking inverse on (16.34),

$$\begin{bmatrix} \hat{L}^{(\zeta\zeta)} & \hat{L}^{(\zeta S)} \\ \hat{L}^{(\zeta S)} & \hat{L}^{(SS)} \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{\sqrt{\hat{L}^{(\zeta\zeta)}}} & \\ & \frac{1}{\sqrt{\hat{L}^{(SS)}}} \end{bmatrix} \frac{1}{1 - q_{\zeta S}^2} \begin{bmatrix} 1 & -q_{\zeta S} \\ -q_{\zeta S} & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{\hat{L}^{(\zeta\zeta)}}} & \\ & \frac{1}{\sqrt{\hat{L}^{(SS)}}} \end{bmatrix} \quad (16.36)$$

Reading off the (1,1) and (2,2) entries gives the posterior variances of ζ and S :

$$\sigma_{\zeta}^2 \sim \frac{1}{\hat{L}^{(\zeta\zeta)}(1 - q_{\zeta S}^2)} \quad \sigma_S^2 \sim \frac{1}{\hat{L}^{(SS)}(1 - q_{\zeta S}^2)} \quad (16.37)$$

Substituting $\hat{L}^{(\zeta\zeta)}$ and $\hat{L}^{(SS)}$ from Table 16.2 and $q_{\zeta S}$ from (16.35) gives the corresponding posterior variances in Table 16.4.

Table 16.3 Leading order of sensitivity coefficients q_{xy}

		y =				
		f	ζ	S	S_e	Φ
x =	f	1				
	ζ	$O(\zeta)$	1		Sym.	
	S	$O(\kappa^{1/2}\zeta)$	$O(\kappa^{-1/2})$	1		
	S_e	$O(\nu\kappa^{5/2}\zeta^3 n^{-1/2})$	$O(\nu\kappa^{1/2}\zeta^2 n^{-1/2})$	$O(\nu\kappa^2\zeta^2 n^{-1/2})$	1	
	Φ	$O(N_f^{-1/2})$	$O(N_f^{-1/2})$	$O(N_f^{-1/2})$	$O(N_f^{-1/2})$	1

Table 16.4 Leading (zeroth) order posterior statistics for $N_f \rightarrow \infty$ and $\zeta \rightarrow 0$. Variance on diagonal; correlation coefficient on off-diagonal

	ζ	S	S_e	Φ
f	$\frac{f^2 \zeta}{4N_c} \left(\tan^{-1} \kappa - \frac{\kappa}{\kappa^2 + 1} \right)^{-1}$			
ζ	$O(\zeta)$	$\frac{\zeta}{4N_c} \left[\tan^{-1} \kappa + \frac{\kappa}{\kappa^2 + 1} - \frac{2(\tan^{-1} \kappa)^2}{\kappa} \right]^{-1}$	Sym.	
S	$O(\kappa^{1/2} \zeta)$	$O(\kappa^{-1/2})$		
S_e	$O(v\kappa^{5/2} \zeta^3 n^{-1/2})$	$O(v\kappa^{3/2} \zeta^2 n^{-1/2})$	$\frac{S_e^2}{(n-1)N_f}$	
Φ	$O(N_f^{-1/2})$	$O(N_f^{-1/2})$	$O(N_f^{-1/2})$	$\frac{v \zeta}{N_c \tan^{-1} \kappa} (\mathbf{I}_n - \bar{\Phi} \bar{\Phi}^T)$

For correlations involving Φ only the scaling order w.r.t. N_f is indicated

Posterior Correlations

Among all parameter pairs, only ζ and S are asymptotically correlated. From (16.36), their posterior correlation coefficient is $\rho_{\zeta S} \sim -q_{\zeta S}$. The correlation coefficient between any other parameter pair is asymptotically zero. The order of magnitude can be assessed as follow. The correlation of $\mathbf{\Phi}$ with any other parameter is clearly $O(N_f^{-1/2})$. The correlation among other parameters (f, ζ, S, S_e) does not depend on N_f . It can be investigated through the sensitivity matrix. Let

$$Q = \begin{bmatrix} 1 & q_{f\zeta} & q_{fS} & q_{fS_e} \\ & 1 & q_{\zeta S} & q_{\zeta S_e} \\ & & 1 & q_{SS_e} \\ \text{sym.} & & & 1 \end{bmatrix} \quad (16.38)$$

Under long data asymptotics where $\mathbf{\Phi}$ is decoupled, the correlation between f and ζ is given by $\rho_{f\zeta} = Q^{-1}(1, 2) / \sqrt{Q^{-1}(1, 1)Q^{-1}(2, 2)}$ where $Q^{-1}(i, j)$ denotes the (i, j) -entry of Q^{-1} . It can be reasoned that $Q^{-1}(1, 1)$ and $Q^{-1}(2, 2)$ are $O(1)$. On the other hand, $Q^{-1}(1, 2) = -c_{12}/|Q|$, where

$$\begin{aligned} c_{12} &= \left| \begin{bmatrix} q_{f\zeta} & q_{\zeta S} & q_{\zeta S_e} \\ q_{fS} & 1 & q_{SS_e} \\ q_{fS_e} & q_{SS_e} & 1 \end{bmatrix} \right| \\ &= q_{f\zeta} - q_{fS}q_{\zeta S} - q_{fS_e}q_{\zeta S_e} - q_{f\zeta}^2 q_{SS_e}^2 + q_{fS}q_{\zeta S_e}q_{SS_e} + q_{fS_e}q_{\zeta S}q_{SS_e} \end{aligned} \quad (16.39)$$

is the cofactor. It can be reasoned that $|Q| = O(1)$ and so $\rho_{f\zeta} = O(c_{12})$. The order of magnitude of c_{12} can be determined using those of the sensitivity coefficients in Table 16.3. The correlation of other pairs can be determined using the same reasoning above. The results are summarized in the off-diagonal entries of Table 16.4. In most cases the correlation is of the same order as the sensitivity coefficient. The only exception is $\rho_{\zeta S_e}$, which is $O(\kappa^{3/2})$ but $q_{\zeta S_e} = O(\kappa^{1/2})$.

16.5 First Order Effect of Signal-to-Noise Ratio

The asymptotic expressions of the NLLF derivatives and posterior covariance matrix derived in Sect. 16.4 are of the ‘zeroth order’, in the sense that they (except for the mode shape) do not depend on the n/e ratio $v = S_e/S$ and hence modal s/n ratio $\gamma = 1/4v\zeta^2$. The leading order posterior variance in Table 16.4 reflects the achievable identification precision limit of modal parameters when the modal excitation is not known but assumed to be broadband random. To investigate the effect of test configuration, it is necessary to capture the first order behavior w.r.t. v . The basic procedure is to retain the first order term e_k in the long data asymptotic expressions in Table 16.1 and apply small damping asymptotics to obtain the first order terms of the NLLF derivatives. The first order terms of the posterior variances

are then obtained by perturbation w.r.t. the NLLF derivatives. For $\hat{L}^{(S_e S_e)}$, it can be reasoned from the expression in Table 16.1 that the effect of v is second order. On the other hand, v already appears in the leading order of $\hat{L}^{(\Phi\Phi)}$. It therefore remains to consider f , ζ and S .

As a refined statement of (16.33) and (16.37), the posterior variances of f , ζ , S are given by

$$\sigma_f^2 \sim \frac{1 + O(q^2)}{\hat{L}^{(ff)}} \quad \sigma_\zeta^2 \sim \frac{1 + O(q^2)}{\hat{L}^{(\zeta\zeta)}(1 - q_{\zeta S}^2)} \quad \sigma_S^2 \sim \frac{1 + O(q^2)}{\hat{L}^{(SS)}(1 - q_{\zeta S}^2)} \quad (16.40)$$

where $O(q^2)$ denotes a remainder term that is generally a quadratic function of the sensitivity coefficients. For example, for σ_f^2 , $O(q^2)$ is the leading order among any product of $q_{f\zeta}$, q_{fS} and q_{fS_e} . Table 16.5 shows the order of these products based on Table 16.3. The leading order among these products is $O(\zeta^2)$ and so for σ_f^2 the remainder term $O(q^2)$ is $O(\zeta^2)$. Using a similar reasoning, it can be deduced that the remainder terms for σ_ζ^2 and σ_S^2 are also $O(\zeta^2)$. These remainder terms do not contain v . The first order effect of v on the posterior variances therefore stems purely from the first order terms in $\hat{L}^{(ff)}$, $\hat{L}^{(\zeta\zeta)}$, $\hat{L}^{(SS)}$ and $q_{\zeta S}$.

First Order Perturbation

Taking differential on the natural log of (16.40), the first order term (denoted with a prefix Δ) of σ_x^2 ($x = f, \zeta, S$) about the zeroth order is related to those of $\hat{L}^{(xx)}$ and $q_{\zeta S}^2$ by

$$\begin{aligned} \frac{\Delta(\sigma_f^2)}{\sigma_{f0}^2} &= -\frac{\Delta\hat{L}^{(ff)}}{\hat{L}_0^{(ff)}} \\ \frac{\Delta(\sigma_\zeta^2)}{\sigma_{\zeta0}^2} &= -\frac{\Delta\hat{L}^{(\zeta\zeta)}}{\hat{L}_0^{(\zeta\zeta)}} + \frac{\Delta(q_{\zeta S}^2)}{1 - q_{\zeta S0}^2} \\ \frac{\Delta(\sigma_S^2)}{\sigma_{S0}^2} &= -\frac{\Delta\hat{L}^{(SS)}}{\hat{L}_0^{(SS)}} + \frac{\Delta(q_{\zeta S}^2)}{1 - q_{\zeta S0}^2} \end{aligned} \quad (16.41)$$

Table 16.5 Scaling order of $q_1 q_2$. Only the scaling w.r.t. ζ and v is indicated

		q_2		
		$q_{f\zeta} = O(\zeta)$	$q_{fS} = O(\zeta)$	$q_{fS_e} = O(v\zeta^3)$
q_1	$q_{f\zeta} = O(\zeta)$	$O(\zeta^2)$	$O(\zeta^2)$	$O(v\zeta^4)$
	$q_{fS} = O(\zeta)$		$O(\zeta^2)$	$O(v\zeta^4)$
	$q_{fS_e} = O(v\zeta^3)$	Sym.		$O(v^2\zeta^6)$

where a subscript ‘0’ denotes the zeroth order. See Table 16.2 for $\hat{L}_0^{(xy)}$ ($x, y = f, \zeta, S$) and Table 16.4 for σ_{x0}^2 ($x = f, \zeta, S$). Also, using (16.35),

$$q_{\zeta S0}^2 = \frac{(\hat{L}_0^{(\zeta S)})^2}{\hat{L}_0^{(\zeta \zeta)} \hat{L}_0^{(SS)}} = 2(\tan^{-1} \kappa)^2 \kappa^{-1} (\tan^{-1} \kappa + \frac{\kappa}{\kappa^2 + 1})^{-1} \quad (16.42)$$

Using $(1 + e_k)^{-1} \sim 1 - e_k$, where $e_k = S_e / SD_k = vD_k^{-1}$, it can be shown that generally (see Appendix C of this chapter)

$$\hat{L}^{(xy)} \sim \hat{L}_0^{(xy)} (1 - c_{xy} \gamma^{-1}) \quad x, y = f, \zeta, S \quad (16.43)$$

where $\gamma = S/4S_e \zeta^2$ is the modal s/n ratio; and, for the relevant terms here,

$$c_{ff} = \frac{4(\kappa - \tan^{-1} \kappa)}{\tan^{-1} \kappa - \frac{\kappa}{\kappa^2 + 1}} \quad c_{\zeta \zeta} = \frac{4 \tan^{-1} \kappa}{\tan^{-1} \kappa + \frac{\kappa}{\kappa^2 + 1}} \quad c_{SS} = 2 \left(1 + \frac{\kappa^2}{3} \right) \quad c_{\zeta S} = \frac{2\kappa}{\tan^{-1} \kappa} \quad (16.44)$$

Equation (16.43) indicates that

$$\frac{\Delta \hat{L}^{(xy)}}{\hat{L}_0^{(xy)}} = -c_{xy} \gamma^{-1} \quad (16.45)$$

On the other hand, taking differential of the natural log of $q_{\zeta S}^2 = (\hat{L}^{(\zeta S)})^2 / \hat{L}^{(\zeta \zeta)} \hat{L}^{(SS)}$ about the zeroth order,

$$\frac{\Delta(q_{\zeta S}^2)}{q_{\zeta S0}^2} = 2 \frac{\Delta \hat{L}^{(\zeta S)}}{\hat{L}_0^{(\zeta S)}} - \frac{\Delta \hat{L}^{(\zeta \zeta)}}{\hat{L}_0^{(\zeta \zeta)}} - \frac{\Delta \hat{L}^{(SS)}}{\hat{L}_0^{(SS)}} = (c_{\zeta \zeta} + c_{SS} - 2c_{\zeta S}) \gamma^{-1} \quad (16.46)$$

after using (16.45). Substituting (16.45) and (16.46) into (16.41) and using $\sigma_x^2 \sim \sigma_{x0}^2 [1 + \Delta(\sigma_x^2) / \sigma_{x0}^2]$ gives the asymptotic expression of the posterior variance with first order term in the form

$$\sigma_x^2 \sim \sigma_{x0}^2 (1 + a_x \gamma^{-1}) \quad x = f, \zeta, S \quad (16.47)$$

where

$$\begin{aligned} a_f &= c_{ff} \quad a_\zeta = c_{\zeta \zeta} + \frac{q_{\zeta S0}^2}{1 - q_{\zeta S0}^2} (c_{\zeta \zeta} + c_{SS} - 2c_{\zeta S}) \\ a_S &= c_{SS} + \frac{q_{\zeta S0}^2}{1 - q_{\zeta S0}^2} (c_{\zeta \zeta} + c_{SS} - 2c_{\zeta S}) \end{aligned} \quad (16.48)$$

Substituting c_{ff} , $c_{\zeta\zeta}$, c_{SS} and $c_{\zeta S}$ in (16.44) and $q_{\zeta S0}^2$ in (16.42) gives, after algebra:

$$a_f = \frac{4(\kappa - \tan^{-1} \kappa)}{\tan^{-1} \kappa - \frac{\kappa}{\kappa^2 + 1}} \quad (16.49)$$

$$a_\zeta = \frac{4(\kappa^2 + 1)(3 \tan^{-1} \kappa - 3\kappa + \kappa^2 \tan^{-1} \kappa) \tan^{-1} \kappa}{3[(\kappa^2 + 1)(\kappa - 2 \tan^{-1} \kappa) \tan^{-1} \kappa + \kappa^2]} \quad (16.50)$$

$$a_S = 2 + \frac{2}{3}\kappa^2 - \frac{(\tan^{-1} \kappa)^2}{2(\tan^{-1} \kappa)^2 - b\kappa} \left[\frac{8 \tan^{-1} \kappa}{b} - \frac{8\kappa}{\tan^{-1} \kappa} + \frac{4}{3}\kappa^2 + 4 \right] \quad (16.51)$$

where $b = \tan^{-1} \kappa + \kappa/(\kappa^2 + 1)$.

16.6 Other Data Types

The foregoing derivations assume acceleration data. In the general case, the same derivation and result still apply, except that the dynamic amplification factor is now given by the general expression

$$D_k = \frac{(2\pi f_k)^{-2q}}{(1 - \beta_k^2)^2 + (2\zeta\beta_k)^2} \quad q = \begin{cases} 0 & \text{acceleration data} \\ 1 & \text{velocity data} \\ 2 & \text{displacement data} \end{cases} \quad (16.52)$$

and the noise-to-environment ratio v should be defined as

$$v = \frac{S_e}{S} (2\pi f)^{2q} \quad (16.53)$$

Check that v is still dimensionless because the unit of the channel noise PSD S_e changes with data type. For example, for velocity data ($q = 1$) the unit is m/s and the unit of S_e is $(\text{m/s})^2/\text{Hz}$. Multiplying by $(2\pi f)^2$ (with unit s^{-2}) gives a unit of $(\text{m/s}^2)^2/\text{Hz}$, the same unit as acceleration PSD and hence modal force PSD.

The derivation in the general case can be reasoned as follow. Any possible difference must result from the factor $(2\pi f_k)^{-2q}$ in D_k . The long data asymptotic results in Table 16.1 were derived for generic D_k and so are still applicable. Except for $\hat{L}^{(\Phi\Phi)}$ and cross derivatives involving S_e , i.e., $\hat{L}^{(fS_e)}$, $\hat{L}^{(\zeta S_e)}$ and $\hat{L}^{(SS_e)}$, the expressions for $\hat{L}^{(xy)}$ are invariant to the scaling of D_k because their leading order does not depend on $e_k = S_e/SD_k$, or the scaling cancels out due to the co-appearance of D_k and D_k^{-1} . They therefore have the same small damping asymptotic expression in Table 16.2. For $\hat{L}^{(\Phi\Phi)}$, substituting the general expression of D_k in (16.52) and following the same steps in small damping asymptotics gives

the same asymptotic expression in Table 16.2 with v defined as in (16.53). For $\hat{L}^{(S_e)}$, $\hat{L}^{(\zeta S_e)}$ and $\hat{L}^{(SS_e)}$, their small damping asymptotic expressions for $q \neq 0$ are different from the ones in Table 16.2 but they still result in the same (vanishing) order of magnitude in the sensitivity coefficients in Table 16.3. That is, asymptotic decoupling still applies when $q \neq 0$. As a result the leading order expressions in Table 16.4 still apply in the general case for $q \neq 0$, provided that v is defined as in (16.53). Similarly, it can be shown that $\hat{L}_1^{(ff)}$, $\hat{L}_1^{(\zeta\zeta)}$, $\hat{L}_1^{(SS)}$ and $\hat{L}_1^{(\zeta S)}$ remain the same, provided that v is given by (16.53) and correspondingly $\gamma^{-1} = 4v\zeta^2 = 4S_e\zeta^2(2\pi f)^{2q}/S$.

Appendix A Asymptotics of $\sum_k D_k^a(\beta_k - 1)^b \beta_k^c$

This appendix derives an asymptotic expression for $\sum_k D_k^a(\beta_k - 1)^b \beta_k^c$, where a, b and c are integers and $b \geq 0$; $D_k = [(1 - \beta_k^2)^2 + (2\zeta\beta_k)^2]^{-1}$ is the dynamic amplification factor (acceleration data); $\beta_k = f/f_k$; $f_k = k/T_d$ is the FFT frequency; T_d is the data duration (s); and the sum is over f_k between $f(1 \pm \zeta\kappa)$. The strategy is to express it as a Riemann sum and then approximate by an integral. This is asymptotically correct when $N_f \rightarrow \infty$. A closed form expression can be obtained by further considering $\zeta \rightarrow 0$.

Long Data

The frequency ratios $\{\beta_k\}$ are not evenly spaced, i.e., $\beta_{k+1} - \beta_k$ is not a constant of k . In order to write as a Riemann sum we define and work with an evenly spaced (dimensionless) coordinate. Let $u_k = (f_k/f) - 1 = (1/\beta_k) - 1$ so that $\{u_k\}$ are evenly spaced at $\Delta u = u_{k+1} - u_k = \Delta f/f = 1/N_c$; $\Delta f = T_d^{-1}$ is the FFT frequency interval and $N_c = \text{data duration/natural period}$. Since f_k ranges over $f(1 \pm \kappa\zeta)$, u_k ranges over $\pm\kappa\zeta$. Substituting $\beta_k = 1/(1 + u_k)$ and rearranging gives

$$D_k = \frac{(1 + u_k)^4}{u_k^2(2 + u_k)^2 + 4\zeta^2(1 + u_k)^2} \quad (16.54)$$

$$\sum_k D_k^a(\beta_k - 1)^b \beta_k^c = (-1)^b 2^{-2a} N_c \sum_k \frac{(1 + u_k)^{4a-b-c} u_k^b}{[u_k^2(1 + u_k/2)^2 + \zeta^2(1 + u_k)^2]^a} \Delta u \quad (16.55)$$

The Riemann sum on the RHS tends to an integral, which is asymptotically correct when $\Delta u = N_c^{-1} \ll 2\kappa\zeta$, or equivalently, $N_f \gg 1$. Thus

$$\sum_k D_k^a (\beta_k - 1)^b \beta_k^c \sim (-1)^b 2^{-2a} N_c \int_{-\kappa\zeta}^{\kappa\zeta} \frac{(1+u)^{4a-b-c} u^b}{[u^2(1+u/2)^2 + \zeta^2(1+u)^2]^a} du \quad N_f \rightarrow \infty \quad (16.56)$$

Changing variable from u to u/ζ removes ζ from the integration limit:

$$\sum_k D_k^a (\beta_k - 1)^b \beta_k^c \sim (-1)^b 2^{-2a} N_c \zeta^{b-2a+1} \int_{-\kappa}^{\kappa} \frac{(1+\zeta u)^{4a-b-c} u^b}{[u^2(1+\zeta u/2)^2 + (1+\zeta u)^2]^a} du \quad N_f \rightarrow \infty \quad (16.57)$$

Small Damping

For small ζ the absolute value of the integrand is almost symmetric about the origin. When b is even, the integrals on the positive and negative side are both positive. When b is odd they tend to cancel out. Their difference can be captured by a Taylor approximation w.r.t. ζ . It can be neglected when b is even but it becomes the leading order when b is odd.

Separate the integral in (16.57) on two intervals, $[0, \kappa]$ and $[-\kappa, 0]$. For the latter, change integration variable from u to $-u$. This gives

$$\sum_k D_k^a (\beta_k - 1)^b \beta_k^c = (-1)^b 2^{-2a} N_c \zeta^{b-2a+1} \times \left[\int_0^{\kappa} \frac{(1+\zeta u)^{4a-b-c} u^b}{[u^2(1+\zeta u/2)^2 + (1+\zeta u)^2]^a} + \frac{(1-\zeta u)^{4a-b-c} u^b (-1)^b}{[u^2(1-\zeta u/2)^2 + (1-\zeta u)^2]^a} du \right] \quad (16.58)$$

The two integrands mainly differ by terms involving $\pm\zeta$. For small ζ ,

$$\frac{(1 \pm \zeta u)^{4a-b-c} u^b}{[u^2(1 \pm \zeta u/2)^2 + (1 \pm \zeta u)^2]^a} = \frac{u^b}{(1+u^2)^a} \left\{ 1 \pm \frac{\zeta u}{1+u^2} [(2a-b-c) + (3a-b-c)u^2] \right\} + O(\zeta^2) \quad (16.59)$$

When b is even the $O(\zeta)$ terms in (16.58) cancel out, giving

$$\sum_k D_k^a (\beta_k - 1)^b \beta_k^c \sim 2^{-2a+1} N_c \zeta^{b-2a+1} \int_0^\kappa \frac{u^b}{(1+u^2)^a} du \quad (16.60)$$

$b \text{ even}, N_f \rightarrow \infty, \zeta \rightarrow 0$

When b is odd, the zeroth order terms cancel out, leaving the $O(\zeta)$ terms as the leading order:

$$\sim -2^{-2a+1} N_c \zeta^{b-2a+2} \int_0^\kappa \frac{u^{b+1}}{(1+u^2)^{a+1}} [(2a-b-c) + (3a-b-c)u^2] du$$

$b \text{ odd}, N_f \rightarrow \infty, \zeta \rightarrow 0$

(16.61)

The integrals in (16.60) and (16.61) can be evaluated analytically. Table 16.6 shows some results for $c = 0$ that are frequently used.

Caution with Shortcuts

When analyzing the asymptotic behavior of $\sum_k D_k^a (\beta_k - 1)^b \beta_k^c$ it is tempting to take $\beta_k \sim 1$ or $\beta_k - 1 \sim 0$ but this need not be correct. When b is even it is correct to take $\beta_k^c \sim 1$, i.e.,

$$\sum_k D_k^a (\beta_k - 1)^b \beta_k^c \sim \sum_k D_k^a (\beta_k - 1)^b \quad b \text{ even} \quad (16.62)$$

because (16.60) does not depend on c . This rule can be generalized to

$$\sum_k D_k^a (\beta_k - 1)^b P(\beta_k) \sim \sum_k D_k^a (\beta_k - 1)^b P(1) \quad b \text{ even} \quad (16.63)$$

for any polynomial $P(\beta_k)$ with $P(1) \neq 0$. When b is odd, this rule does not apply and (16.61) should be used. On the other hand, terms of $(\beta_k - 1)$ should not be

Table 16.6 Asymptotic expression for $\sum_k D_k^a(\beta_k - 1)^b$ for $N_f \rightarrow \infty$ and $\zeta \rightarrow 0$

$a \backslash b$	0	1	2
1	$\frac{N_c}{2\zeta} \tan^{-1} \kappa$	$-\frac{N_c \zeta}{4} \left(4\kappa - 5 \tan^{-1} \kappa + \frac{\kappa}{\kappa^2 + 1} \right)$	$\frac{N_c \zeta}{2} (\kappa - \tan^{-1} \kappa)$
2	$\frac{N_c}{16\zeta^3} \left(\tan^{-1} \kappa + \frac{\kappa}{\kappa^2 + 1} \right)$	$-\frac{N_c}{32\zeta} \left[9 \tan^{-1} \kappa - \frac{\kappa(11\kappa^2 + 9)}{(\kappa^2 + 1)^2} \right]$	$\frac{N_c}{16\zeta} \left(\tan^{-1} \kappa - \frac{\kappa}{\kappa^2 + 1} \right)$
3	$\frac{N_c}{256\zeta^5} \left[3 \tan^{-1} \kappa + \frac{\kappa(3\kappa^2 + 5)}{(\kappa^2 + 1)^2} \right]$	$-\frac{N_c}{512\zeta^3} \left[13 \tan^{-1} \kappa + \frac{\kappa(13\kappa^4 - 8\kappa^2 - 13)}{(\kappa^2 + 1)^3} \right]$	$\frac{N_c}{256\zeta^3} \left[\tan^{-1} \kappa + \frac{\kappa(\kappa^2 - 1)}{(\kappa^2 + 1)^2} \right]$

approximated by zero, since then their asymptotic order is grossed out. The sum $\sum_k D_k^a (\beta_k - 1)^b$ should be evaluated using (16.60) or (16.61).

Appendix B Derivation of Small Damping Asymptotics (Zeroth Order)

This appendix derives the small damping (in addition to long data) asymptotic expressions in Table 16.2.

Auto-Derivatives $\hat{L}^{(ff)}$, $\hat{L}^{(\zeta\zeta)}$, $\hat{L}^{(SS)}$, $\hat{L}^{(S_e S_e)}$ and $\hat{L}^{(\Phi\Phi)}$

From Table 16.1, $\hat{L}^{(ff)} \sim \sum_k D_k^2 (D_k^{-1})^{(f)2} (1 + e_k)^{-2} \sim \sum_k D_k^2 (D_k^{-1})^{(f)2}$ since $e_k \ll 1$. Substituting $(D_k^{-1})^{(f)} = 4f^{-1} \beta_k^2 (\beta_k^2 - 1 + 2\zeta^2)$ from (16.23), writing $\beta_k^2 - 1 = (\beta_k - 1)(\beta_k + 1)$ and expanding the square,

$$\hat{L}^{(ff)} \sim 16f^{-2} \left\{ \sum_k D_k^2 (\beta_k - 1)^2 (\beta_k + 1)^2 \beta_k^4 + 4\zeta^2 \sum_k D_k^2 (\beta_k - 1)(\beta_k + 1) \beta_k^4 + 4\zeta^4 \sum_k D_k^2 \beta_k^4 \right\} \quad (16.64)$$

Since $(\beta_k + 1)$ and β_k are $O(1)$, the first sum $\sum_k D_k^2 (\beta_k - 1)^2 (\beta_k + 1)^2 \beta_k^4$ has the same order as $\sum_k D_k^2 (\beta_k - 1)^2$, which is $O(N_c \zeta^{-1})$ from Table 16.6. Similarly, the second sum $\sum_k D_k^2 (\beta_k - 1)(\beta_k + 1) \beta_k^4$ has the same order as $\sum_k D_k^2 (\beta_k - 1) = O(N_c \zeta^{-1})$ and so the second term is $O(N_c \zeta)$. The third sum $\sum_k D_k^2 \beta_k^4$ has the same order as $\sum_k D_k^2 = O(N_c \zeta^{-3})$ and so the third term is $O(N_c \zeta)$. Thus, the first term in the brace dominates, giving

$$\hat{L}^{(ff)} \sim 16f^{-2} \sum_k D_k^2 (\beta_k - 1)^2 (\beta_k + 1)^2 \beta_k^4 \quad (16.65)$$

Since $(\beta_k - 1)^2$ has even power, using (16.63) we can take $\beta_k + 1 \sim 2$ and $\beta_k \sim 1$, and so $\hat{L}^{(ff)} \sim 64f^{-2} \sum_k D_k^2 (\beta_k - 1)^2$. Using Table 16.6 to evaluate the sum ($a = 2, b = 2$) gives the expression in Table 16.2.

For $\hat{L}^{(\zeta\zeta)}$, from Table 16.1 and using $(1 + e_k)^{-2} \sim 1$,

$$\hat{L}^{(\zeta\zeta)} \sim \sum_k D_k^2 (D_k^{-1})^{(\zeta)2} (1 + e_k)^{-2} \sim \sum_k D_k^2 (D_k^{-1})^{(\zeta)2} \quad (16.66)$$

Substituting $(D_k^{-1})^{(\zeta)} = 8\zeta \beta_k^2$ from (16.23), $\hat{L}^{(\zeta\zeta)} \sim 64\zeta^2 \sum_k D_k^2 \beta_k^4 \sim 64\zeta^2 \sum_k D_k^2$. Using Table 16.6 to evaluate the sum gives the expression in Table 16.2.

For $\hat{L}^{(SS)}$, from Table 16.1,

$$\hat{L}^{(SS)} \sim S^{-2} \sum_k (1 + e_k)^{-2} \sim S^{-2} \sum_k 1 = S^{-2} N_f \quad (16.67)$$

as in Table 16.2.

For $\hat{L}^{(S_e S_e)}$, from Table 16.1, $\hat{L}^{(S_e S_e)} \sim S_e^{-2} [(n-1)N_f + \sum_k e_k^2 (1 + e_k)^{-2}]$. The second term in the bracket is dominated by the first and so

$$\hat{L}^{(S_e S_e)} \sim S_e^{-2} (n-1)N_f \quad (16.68)$$

as in Table 16.2.

For $\hat{L}^{(\Phi\Phi)}$, from Table 16.1, using $1 + e_k \sim 1$ and $e_k^{-1} = SD_k/S_e$,

$$\hat{L}^{(\Phi\Phi)} \sim [2 \sum_k (1 + e_k)^{-1} e_k^{-1}] (\mathbf{I}_n - \bar{\Phi}\bar{\Phi}^T) \sim \left(\frac{2S}{S_e} \sum_k D_k\right) (\mathbf{I}_n - \bar{\Phi}\bar{\Phi}^T) \quad (16.69)$$

Using Table 16.6 to evaluate the sum gives the expression in Table 16.2.

Cross Derivatives $\hat{L}^{(f\zeta)}$, $\hat{L}^{(fS)}$ and $\hat{L}^{(\zeta S)}$

From Table 16.1,

$$\hat{L}^{(f\zeta)} \sim \sum_k D_k^2 (D_k^{-1})^{(f)} (D_k^{-1})^{(\zeta)} (1 + e_k)^{-2} \sim \sum_k D_k^2 (D_k^{-1})^{(f)} (D_k^{-1})^{(\zeta)} \quad (16.70)$$

Substituting $(D_k^{-1})^{(f)} = 4f^{-1}\beta_k^2(\beta_k^2 - 1 + 2\zeta^2)$ and $(D_k^{-1})^{(\zeta)} = 8\zeta\beta_k^2$, and expanding,

$$\hat{L}^{(f\zeta)} \sim 32f^{-1}\zeta \left[\sum_k D_k^2 (\beta_k - 1) \beta_k^5 + \sum_k D_k^2 (\beta_k - 1) \beta_k^4 + 2\zeta^2 \sum_k D_k^2 \beta_k^4 \right] \quad (16.71)$$

Evaluating the first two sums using (16.61) and the third using Table 16.6 gives the expression in Table 16.2.

For $\hat{L}^{(fS)}$, from Table 16.1,

$$\hat{L}^{(fS)} \sim -S^{-1} \sum_k D_k (D_k^{-1})^{(f)} (1 + e_k)^{-2} \sim -S^{-1} \sum_k D_k (D_k^{-1})^{(f)} \quad (16.72)$$

Substituting $(D_k^{-1})^{(f)} = 4f^{-1}\beta_k^2(\beta_k^2 - 1 + 2\zeta^2)$ and expanding,

$$\hat{L}^{(fS)} \sim -4f^{-1}S^{-1} \left[\sum_k D_k (\beta_k - 1) \beta_k^3 + \sum_k D_k (\beta_k - 1) \beta_k^2 + 2\zeta^2 \sum_k D_k \beta_k^2 \right] \quad (16.73)$$

Evaluating the first two sums using (16.61) and the third using Table 16.6 gives the expression in Table 16.2.

For $\hat{L}^{(\zeta S)}$, from Table 16.1,

$$\hat{L}^{(\zeta S)} \sim -S^{-1} \sum_k D_k (D_k^{-1})^{(\zeta)} (1 + e_k)^{-2} \sim -S^{-1} \sum_k D_k (D_k^{-1})^{(\zeta)} \quad (16.74)$$

Substituting $(D_k^{-1})^{(\zeta)} = 8\zeta\beta_k^2$, $\hat{L}^{(\zeta S)} \sim -8\zeta S^{-1} \sum_k D_k \beta_k^2 \sim -8\zeta S^{-1} \sum_k D_k$.
Evaluating the sum using Table 16.6 gives the expression in Table 16.2.

Cross Derivatives $\hat{L}^{(fS_e)}$, $\hat{L}^{(\zeta S_e)}$ and $\hat{L}^{(SS_e)}$

From Table 16.1,

$$\hat{L}^{(fS_e)} \sim -S^{-1} \sum_k (D_k^{-1})^{(f)} (1 + e_k)^{-2} \sim -S^{-1} \sum_k (D_k^{-1})^{(f)} \quad (16.75)$$

Substituting $(D_k^{-1})^{(f)} = 4f^{-1}\beta_k^2(\beta_k^2 - 1 + 2\zeta^2)$ and expanding gives

$$\hat{L}^{(fS_e)} \sim -4f^{-1}S^{-1} \left[\sum_k (\beta_k - 1)\beta_k^3 + \sum_k (\beta_k - 1)\beta_k^2 + 2\zeta^2 \sum_k \beta_k^2 \right] \quad (16.76)$$

Evaluating the first two sums using (16.61) and noting $\sum_k \beta_k^2 \sim N_f = 2\zeta\kappa N_c$ gives the expression in Table 16.2.

For $\hat{L}^{(\zeta S_e)}$, from Table 16.1,

$$\hat{L}^{(\zeta S_e)} \sim -S^{-1} \sum_k (D_k^{-1})^{(\zeta)} (1 + e_k)^{-2} \sim -S^{-1} \sum_k (D_k^{-1})^{(\zeta)} \quad (16.77)$$

Substituting $(D_k^{-1})^{(\zeta)} = 8\zeta\beta_k^2$ gives

$$\hat{L}^{(\zeta S_e)} \sim -8\zeta S^{-1} \sum_k \beta_k^2 \sim -8\zeta S^{-1} N_f = -\frac{16N_c\zeta^2\kappa}{S} \quad (16.78)$$

as in Table 16.2.

For $\hat{L}^{(SS_e)}$, from Table 16.1,

$$\hat{L}^{(SS_e)} \sim S^{-2} \sum_k D_k^{-1} (1 + e_k)^{-2} \sim S^{-2} \sum_k D_k^{-1} \quad (16.79)$$

Evaluating the sum using (16.60) gives the expression in Table 16.2.

Appendix C First Order of NLLF Derivatives w.r.t. f, ζ, S

In this appendix we derive expressions for $\hat{L}^{(ff)}$, $\hat{L}^{(\zeta\zeta)}$, $\hat{L}^{(SS)}$ and $\hat{L}^{(\zeta S)}$ up to the first order of the n/e ratio $v = S_e/S$. It leads to results in the form of (16.43).

Second Derivative $\hat{L}^{(ff)}$

From Table 16.1 and using $(1 + e_k)^{-2} \sim 1 - 2e_k$,

$$\hat{L}^{(ff)} = \sum_k D_k^2 (D_k^{-1})^{(f)2} (1 + e_k)^{-2} \sim \sum_k D_k^2 (D_k^{-1})^{(f)2} (1 - 2e_k) = \hat{L}_0^{(ff)} + \hat{L}_1^{(ff)} \quad (16.80)$$

where $\hat{L}_0^{(ff)} = \sum_k D_k^2 (D_k^{-1})^{(f)2}$ is the zeroth order whose expression is given in Table 16.2; and, using $D_k e_k = v$,

$$\hat{L}_1^{(ff)} = -2 \sum_k D_k^2 (D_k^{-1})^{(f)2} e_k = -2v \sum_k D_k (D_k^{-1})^{(f)2} \quad (16.81)$$

is the first order. Substituting $(D_k^{-1})^{(f)} = 4f^{-1}\beta_k^2(\beta_k^2 - 1 + 2\zeta^2)$ from (16.23), writing $\beta_k^2 - 1 = (\beta_k - 1)(\beta_k + 1)$ and expanding the square,

$$\begin{aligned} \sum_k D_k (D_k^{-1})^{(f)2} &= \frac{16}{f^2} \left[\sum_k D_k (\beta_k - 1)^2 (\beta_k + 1)^2 \beta_k^4 \right. \\ &\quad \left. + 4\zeta^2 \sum_k D_k (\beta_k - 1)(\beta_k + 1) \beta_k^4 + 4\zeta^4 \sum_k D_k \beta_k^4 \right] \end{aligned} \quad (16.82)$$

In the bracket, the first sum is of the same order as $\sum_k D_k (\beta_k - 1)^2 = O(N_c \zeta)$ from Table 16.6. Similarly, the second sum is of the same order as $\sum_k D_k (\beta_k - 1) = O(N_c \zeta)$ and so the second term is $O(N_c \zeta^3)$. The last sum is of the same order as $\sum_k D_k = O(N_c \zeta^{-1})$ and so the last term is also $O(N_c \zeta^3)$. Thus, the first term in the bracket dominates:

$$\sum_k D_k (D_k^{-1})^{(f)2} \sim 16f^{-2} \sum_k D_k (\beta_k - 1)^2 (\beta_k + 1)^2 \beta_k^4 \quad (16.83)$$

Since $(\beta_k - 1)^2$ has even power, using (16.63) we can take $\beta_k + 1 \sim 2$ and $\beta_k \sim 1$ to get $\sum_k D_k (D_k^{-1})^{(f)2} \sim 64f^{-2} \sum_k D_k (\beta_k - 1)^2$. Using Table 16.6 to evaluate the sum,

$$\sum_k D_k (D_k^{-1})^{(f)2} \sim \frac{32N_c \zeta}{f^2} (\kappa - \tan^{-1} \kappa) \quad (16.84)$$

Substituting into (16.81),

$$\hat{L}_1^{(ff)} = -\frac{64N_c v \zeta}{f^2} (\kappa - \tan^{-1} \kappa) \quad \frac{\hat{L}_1^{(ff)}}{\hat{L}_0^{(ff)}} = - \underbrace{\left[\frac{4(\kappa - \tan^{-1} \kappa)}{\tan^{-1} \kappa - \kappa/(\kappa^2 + 1)} \right]}_{c_{ff}} \underbrace{(4v\zeta^2)}_{\gamma^{-1}} \quad (16.85)$$

We can then write

$$\hat{L}^{(ff)} \sim \hat{L}_0^{(ff)} (1 - c_{ff} \gamma^{-1}) \quad c_{ff} = \frac{4(\kappa - \tan^{-1} \kappa)}{\tan^{-1} \kappa - \kappa/(\kappa^2 + 1)} \quad (16.86)$$

Second Derivative $\hat{L}^{(\zeta\zeta)}$

From Table 16.1 and using $(1 + e_k)^{-2} \sim 1 - 2e_k$,

$$\hat{L}^{(\zeta\zeta)} = \sum_k D_k^2 (D_k^{-1})^{(\zeta)2} (1 + e_k)^{-2} \sim \sum_k D_k^2 (D_k^{-1})^{(\zeta)2} (1 - 2e_k) = \hat{L}_0^{(\zeta\zeta)} + \hat{L}_1^{(\zeta\zeta)} \quad (16.87)$$

where $\hat{L}_0^{(\zeta\zeta)} = \sum_k D_k^2 (D_k^{-1})^{(\zeta)2}$ is the zeroth order in Table 16.2; and $\hat{L}_1^{(\zeta\zeta)} = -2v \sum_k D_k (D_k^{-1})^{(\zeta)2}$ is the first order. Substituting $(D_k^{-1})^{(\zeta)} = 8\zeta\beta_k^2$ from (16.23) gives $\hat{L}_1^{(\zeta\zeta)} = -128v\zeta^2 \sum_k D_k \beta_k^4 \sim -128v\zeta^2 \sum_k D_k$. Using Table 16.6 to evaluate the sum,

$$\hat{L}_1^{(\zeta\zeta)} \sim -64N_c v \zeta \tan^{-1} \kappa \quad \frac{\hat{L}_1^{(\zeta\zeta)}}{\hat{L}_0^{(\zeta\zeta)}} \sim - \underbrace{\left[\frac{4 \tan^{-1} \kappa}{\tan^{-1} \kappa + \kappa/(\kappa^2 + 1)} \right]}_{c_{\zeta\zeta}} \underbrace{(4v\zeta^2)}_{\gamma^{-1}} \quad (16.88)$$

We can then write

$$\hat{L}^{(\zeta\zeta)} \sim \hat{L}_0^{(\zeta\zeta)} (1 - c_{\zeta\zeta} \gamma^{-1}) \quad c_{\zeta\zeta} = \frac{4 \tan^{-1} \kappa}{\tan^{-1} \kappa + \kappa/(\kappa^2 + 1)} \quad (16.89)$$

Second Derivative $\hat{L}^{(SS)}$

From Table 16.1 and using $(1 + e_k)^{-2} \sim 1 - 2e_k$, $\hat{L}^{(SS)} \sim S^{-2} \sum_k (1 - 2e_k) = \hat{L}_0^{(SS)} + \hat{L}_1^{(SS)}$ where $\hat{L}_0^{(SS)} = S^{-2} N_f$ is the zeroth order; and $\hat{L}_1^{(SS)} = -2vS^{-2} \sum_k D_k^{-1}$ is the first order. Using (16.60) to evaluate the sum gives

$$\hat{L}_1^{(SS)} = -8S^{-2}N_f v \zeta^2 \left(1 + \frac{\kappa^2}{3}\right) \quad \frac{\hat{L}_1^{(SS)}}{\hat{L}_0^{(SS)}} = -2 \underbrace{\left(1 + \frac{\kappa^2}{3}\right)}_{c_{SS}} \underbrace{(4v\zeta^2)}_{\gamma^{-1}} \quad (16.90)$$

We can then write

$$\hat{L}^{(SS)} \sim \hat{L}_0^{(SS)}(1 - c_{SS}\gamma^{-1}) \quad c_{SS} = 2 \left(1 + \frac{\kappa^2}{3}\right) \quad (16.91)$$

Cross Derivative $\hat{L}^{(\zeta S)}$

From Table 16.1, using $(1 + e_k)^{-2} \sim 1 - 2e_k$ and $(D_k^{-1})^{(\zeta)} = 8\zeta\beta_k^2$,

$$\hat{L}^{(\zeta S)} \sim -S^{-1} \sum_k D_k (8\zeta\beta_k^2)(1 - 2e_k) = \hat{L}_0^{(\zeta S)} + \hat{L}_1^{(\zeta S)} \quad (16.92)$$

where $\hat{L}_0^{(\zeta S)} = -8S^{-1}\zeta \sum_k D_k \beta_k^2 \sim -4S^{-1}N_c \tan^{-1} \kappa$ is the zeroth order and

$$\hat{L}_1^{(\zeta S)} = 16S^{-1}v\zeta \sum_k \beta_k^2 \sim 32S^{-1}N_c v \zeta^2 \kappa \quad (16.93)$$

is the first order. Thus

$$\frac{\hat{L}_1^{(\zeta S)}}{\hat{L}_0^{(\zeta S)}} = - \underbrace{\left(\frac{2\kappa}{\tan^{-1} \kappa}\right)}_{c_{\zeta S}} \underbrace{(4v\zeta^2)}_{\gamma^{-1}} \quad (16.94)$$

and we can write

$$\hat{L}^{(\zeta S)} \sim \hat{L}_0^{(\zeta S)}(1 - c_{\zeta S}\gamma^{-1}) \quad c_{\zeta S} = \frac{2\kappa}{\tan^{-1} \kappa} \quad (16.95)$$

Reference

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Appendix A. Complex Gaussian and Wishart Distribution

When analyzing the spectral properties of stochastic processes, we often need to deal with their Fast Fourier Transform (FFT), which is a sequence of complex-valued vectors. Under a variety of situations the real and imaginary parts of these vectors are jointly Gaussian, by virtue of the Central Limit Theorem. The vectors are then said to have a ‘complex Gaussian distribution’. From first principle one can analyze the joint distribution of their real and imaginary parts but this is algebraically cumbersome. By working with complex vectors and matrices rather than real-valued ones, it turns out that the algebra can be significantly simplified. The probability density function (PDF) and other statistical results can be expressed in a compact manner.

This chapter gives a basic introduction to complex Gaussian vectors, which has been studied extensively in multivariate statistics. The theory provides an efficient tool for analyzing the distribution of the FFT of ambient data (Chap. 4) and is useful for general applications. Complex Wishart distribution is also discussed, which is related to the distribution of the averaged sample power spectral density matrix. This chapter is intended for readers with a theoretical pursuit. It may be skipped on first reading or for those primarily interested in applications.

A.1 Complex Gaussian Vectors

Without much loss of generality, assume in this chapter that all complex Gaussian variables or vectors have zero mean. A complex-valued scalar random variable Z is said to have a complex Gaussian distribution if its real and imaginary parts are jointly Gaussian. That is, if $X = \text{Re } Z$ and $Y = \text{Im } Z$, then they have a joint Gaussian PDF:

$$p_{XY}(x, y) = \frac{1}{2\pi\sqrt{|\mathbf{C}|}} \exp\left\{-\frac{1}{2} \begin{bmatrix} x \\ y \end{bmatrix}^T \mathbf{C}^{-1} \begin{bmatrix} x \\ y \end{bmatrix}\right\} \quad x, y \text{ real scalar} \quad (\text{A.1})$$

where

$$\mathbf{C} = \begin{bmatrix} E[X^2] & E[XY] \\ E[YX] & E[Y^2] \end{bmatrix} = \begin{bmatrix} E[\text{Re } Z \text{ Re } Z] & E[\text{Re } Z \text{ Im } Z] \\ E[\text{Im } Z \text{ Re } Z] & E[\text{Im } Z \text{ Im } Z] \end{bmatrix} \quad (\text{A.2})$$

is the covariance matrix of X and Y . Since two Gaussian variables are not necessarily jointly Gaussian, it is not sufficient to have just X and Y individually Gaussian distributed. They need to be jointly Gaussian, i.e., with a joint Gaussian PDF.

Clearly, there is an invertible transformation between Z and (X, Y) . The PDF of Z is the joint PDF of (X, Y) and is denoted by

$$p_Z(z) = p_{XY}(x, y) \quad \begin{array}{l} z = x + iy \\ x, y \text{ real scalar} \end{array} \quad (\text{A.3})$$

A complex-valued vector $\mathbf{Z}$ ($n \times 1$ complex) is called a complex Gaussian vector if its real and imaginary parts are jointly Gaussian. The PDF of $\mathbf{Z}$ is the joint PDF of $\mathbf{X} = \text{Re } \mathbf{Z}$ and $\mathbf{Y} = \text{Im } \mathbf{Z}$, denoted by

$$p_Z(\mathbf{z}) = p_{XY}(\mathbf{x}, \mathbf{y}) = \frac{1}{(2\pi)^n \sqrt{|\mathbf{C}|}} \exp\left\{-\frac{1}{2} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}^T \mathbf{C}^{-1} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}\right\} \quad \begin{array}{l} \mathbf{z} = \mathbf{x} + i\mathbf{y} \\ \mathbf{x}, \mathbf{y} \ n \times 1 \text{ real} \end{array} \quad (\text{A.4})$$

where

$$\mathbf{C} = \begin{bmatrix} E[\mathbf{X}\mathbf{X}^T] & E[\mathbf{X}\mathbf{Y}^T] \\ E[\mathbf{Y}\mathbf{X}^T] & E[\mathbf{Y}\mathbf{Y}^T] \end{bmatrix} = \begin{bmatrix} E[\text{Re } \mathbf{Z} \text{ Re } \mathbf{Z}^T] & E[\text{Re } \mathbf{Z} \text{ Im } \mathbf{Z}^T] \\ E[\text{Im } \mathbf{Z} \text{ Re } \mathbf{Z}^T] & E[\text{Im } \mathbf{Z} \text{ Im } \mathbf{Z}^T] \end{bmatrix} \quad (\text{A.5})$$

A group of $n \times 1$ complex-valued vectors $\{\mathbf{Z}_k\}_{k=1}^M$ are said to have a complex Gaussian distribution when their real and imaginary parts are jointly Gaussian. That is, if $\mathbf{X}_k = \text{Re } \mathbf{Z}_k$ and $\mathbf{Y}_k = \text{Im } \mathbf{Z}_k$, then $\mathbf{X}_p, \mathbf{Y}_p, \mathbf{X}_q$ and $\mathbf{Y}_q$ are jointly Gaussian for any $p, q = 1, \dots, M$. Let

$$\mathbf{Z} = \begin{bmatrix} \mathbf{Z}_1 \\ \vdots \\ \mathbf{Z}_M \end{bmatrix} \quad (\text{A.6})$$

Clearly, the joint PDF of $\{\mathbf{Z}_k\}_{k=1}^M$ is the PDF of $\mathbf{Z}$. This PDF is characterized by the following covariance matrices

$$E[\text{Re } \mathbf{Z}_p \text{Re } \mathbf{Z}_q^T], E[\text{Re } \mathbf{Z}_p \text{Im } \mathbf{Z}_q^T], E[\text{Im } \mathbf{Z}_p \text{Re } \mathbf{Z}_q^T], E[\text{Im } \mathbf{Z}_p \text{Im } \mathbf{Z}_q^T] \quad p, q = 1, \dots, M \quad (\text{A.7})$$

The PDF is still given by (A.4) but now with

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_1 \\ \vdots \\ \mathbf{z}_M \end{bmatrix} \quad \begin{matrix} \mathbf{z}_1, \dots, \mathbf{z}_M \\ n \times 1 \text{ complex} \end{matrix} \quad (\text{A.8})$$

The covariance matrix $\mathbf{C}$ in (A.4) is now a $2nM \times 2nM$ matrix containing in its partitions the matrices in (A.7).

A.2 Covariance and Pseudo-covariance Matrix

Characterizing the second order statistics of complex vectors in terms of the covariance matrices of their real and imaginary parts is cumbersome for both analysis and computation, involving operations that are similar or repetitive. Consider two complex vectors $\mathbf{Z}_1$ and $\mathbf{Z}_2$ (both $n \times 1$). The second order statistics between these two vectors involve the following matrices:

$$E[\text{Re } \mathbf{Z}_1 \text{Re } \mathbf{Z}_2^T], \quad E[\text{Re } \mathbf{Z}_1 \text{Im } \mathbf{Z}_2^T], \quad E[\text{Im } \mathbf{Z}_1 \text{Re } \mathbf{Z}_2^T], \quad E[\text{Im } \mathbf{Z}_1 \text{Im } \mathbf{Z}_2^T] \quad (\text{A.9})$$

Instead of working with these matrices, an equivalent but more elegant way is to work with the ‘(complex) covariance matrix’ and ‘pseudo-covariance matrix’ (also called ‘relation matrix’), defined respectively as

$$\mathbf{E}_{12} = E[\mathbf{Z}_1 \mathbf{Z}_2^*] \quad \mathbf{Q}_{12} = E[\mathbf{Z}_1 \mathbf{Z}_2^T] \quad (\text{A.10})$$

where ‘*’ denotes conjugate transpose. Clearly, $\mathbf{E}_{12}$ and $\mathbf{Q}_{12}$ can be determined by the matrices in (A.9). Conversely,

$$\begin{aligned} E[\text{Re } \mathbf{Z}_1 \text{Re } \mathbf{Z}_2^T] &= \frac{1}{2} (\text{Re } \mathbf{E}_{12} + \text{Re } \mathbf{Q}_{12}) & E[\text{Im } \mathbf{Z}_1 \text{Im } \mathbf{Z}_2^T] &= \frac{1}{2} (\text{Re } \mathbf{E}_{12} - \text{Re } \mathbf{Q}_{12}) \\ E[\text{Re } \mathbf{Z}_1 \text{Im } \mathbf{Z}_2^T] &= \frac{1}{2} (-\text{Im } \mathbf{E}_{12} + \text{Im } \mathbf{Q}_{12}) & E[\text{Im } \mathbf{Z}_1 \text{Re } \mathbf{Z}_2^T] &= \frac{1}{2} (\text{Im } \mathbf{E}_{12} + \text{Im } \mathbf{Q}_{12}) \end{aligned} \quad (\text{A.11})$$

These relationships can be verified by taking expectation of the following identities:

$$\begin{aligned} \text{Re } \mathbf{Z}_1 \text{Re } \mathbf{Z}_2^T &= \frac{1}{2} [\text{Re}(\mathbf{Z}_1 \mathbf{Z}_2^*) + \text{Re}(\mathbf{Z}_1 \mathbf{Z}_2^T)] & \text{Im } \mathbf{Z}_1 \text{Im } \mathbf{Z}_2^T &= \frac{1}{2} [\text{Re}(\mathbf{Z}_1 \mathbf{Z}_2^*) - \text{Re}(\mathbf{Z}_1 \mathbf{Z}_2^T)] \\ \text{Re } \mathbf{Z}_1 \text{Im } \mathbf{Z}_2^T &= \frac{1}{2} [-\text{Im}(\mathbf{Z}_1 \mathbf{Z}_2^*) + \text{Im}(\mathbf{Z}_1 \mathbf{Z}_2^T)] & \text{Im } \mathbf{Z}_1 \text{Re } \mathbf{Z}_2^T &= \frac{1}{2} [\text{Im}(\mathbf{Z}_1 \mathbf{Z}_2^*) + \text{Im}(\mathbf{Z}_1 \mathbf{Z}_2^T)] \end{aligned} \quad (\text{A.12})$$

The above in turn can be shown by direct algebra using the following identities:

$$\operatorname{Re} \mathbf{Z}_i = \frac{1}{2}(\mathbf{Z}_i + \overline{\mathbf{Z}_i}) \quad \operatorname{Im} \mathbf{Z}_i = -\frac{j}{2}(\mathbf{Z}_i - \overline{\mathbf{Z}_i}) \quad i = 1, 2 \quad (\text{A.13})$$

It can be verified that

$$\mathbf{E}_{12}^* = \mathbf{E}_{21} \quad \mathbf{Q}_{12}^T = \mathbf{Q}_{21} \quad (\text{A.14})$$

This also implies that $\mathbf{E}_{ii}^* = \mathbf{E}_{ii}$ (Hermitian) and $\mathbf{Q}_{ii}^T = \mathbf{Q}_{ii}$ (symmetric).

A.3 Complex Gaussian PDF

The PDF in (A.4) can be rewritten in a compact manner in terms of $\mathbf{E} = E[\mathbf{Z}\mathbf{Z}^*]$ and $\mathbf{Q} = E[\mathbf{Z}\mathbf{Z}^T]$:

$$p_{\mathbf{Z}}(\mathbf{z}) = \frac{1}{\pi^n \sqrt{|\mathbf{E}||\mathbf{P}|}} \exp \left\{ -\frac{1}{2} \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix}^* \begin{bmatrix} \mathbf{E} & \mathbf{Q} \\ \mathbf{Q}^* & \bar{\mathbf{E}} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix} \right\} \quad \mathbf{z} \, n \times 1 \text{ complex} \quad (\text{A.15})$$

$$\mathbf{P} = \bar{\mathbf{E}} - \mathbf{Q}^* \mathbf{E}^{-1} \mathbf{Q} \quad (n \times n \text{ Hermitian}) \quad (\text{A.16})$$

See the end for proof. Analogous to the standard Gaussian distribution, the ‘standard complex Gaussian distribution’ refers to the case with identity covariance matrix ($\mathbf{E} = \mathbf{I}_n$) and zero pseudo-covariance matrix ($\mathbf{Q} = \mathbf{0}$).

As a literature note, complex Gaussian distribution was introduced in Wooding (1956). Goodman (1963) is often cited for introduction of the subject with discussion on applications. The topic is discussed in textbooks, e.g., Miller (1974). Reviews can be found in Miller (1969) and Krishnaiah (1976). In common applications the pseudo-covariance matrix is zero. Generalization to non-zero pseudo-covariance matrix was considered by Van den Bos (1995).

Example A.1 (Complex Gaussian scalar) Consider $Z = X + jY$ where X and Y are real-valued jointly Gaussian variables with zero mean. The joint PDF of (X, Y) is determined by

$$c_{xx} = E[X^2] \quad c_{yy} = E[Y^2] \quad c_{xy} = E[XY] \quad (\text{A.17})$$

The PDF of Z is the PDF of (X, Y) and is given by (A.4):

$$p_Z(z) = \frac{1}{2\pi\sqrt{c_{xx}c_{yy} - c_{xy}^2}} \exp\left\{-\frac{1}{2} \begin{bmatrix} x \\ y \end{bmatrix}^T \begin{bmatrix} c_{xx} & c_{xy} \\ c_{xy} & c_{yy} \end{bmatrix}^{-1} \begin{bmatrix} x \\ y \end{bmatrix}\right\} \quad z = x + \mathbf{i}y \quad (\text{A.18})$$

Alternatively, the PDF of Z can be obtained via the PDF of a complex Gaussian variable. Using (A.15),

$$\mathbf{E} = E[\mathbf{Z}\mathbf{Z}^*] = E[X^2 + Y^2] = c_{xx} + c_{yy} \quad (\text{A.19})$$

$$\mathbf{Q} = E[\mathbf{Z}\mathbf{Z}^T] = E[X^2 - Y^2 + 2\mathbf{i}XY] = c_{xx} - c_{yy} + 2\mathbf{i}c_{xy} \quad (\text{A.20})$$

$$\mathbf{P} = \bar{\mathbf{E}} - \mathbf{Q}^* \mathbf{E}^{-1} \mathbf{Q} = c_{xx} + c_{yy} - \frac{(c_{xx} - c_{yy})^2 + 4c_{xy}^2}{c_{xx} + c_{yy}} = \frac{4(c_{xx}c_{yy} - c_{xy}^2)}{c_{xx} + c_{yy}} \quad (\text{A.21})$$

$$|\mathbf{E}||\mathbf{P}| = 4(c_{xx}c_{yy} - c_{xy}^2) \quad (\text{A.22})$$

The PDF is then given by

$$p_Z(z) = \frac{1}{\pi\sqrt{4(c_{xx}c_{yy} - c_{xy}^2)}} \times \exp\left\{-\frac{1}{2} \begin{bmatrix} x + \mathbf{i}y \\ x - \mathbf{i}y \end{bmatrix}^* \begin{bmatrix} c_{xx} + c_{yy} & c_{xx} - c_{yy} + 2c_{xy}\mathbf{i} \\ c_{xx} - c_{yy} - 2c_{xy}\mathbf{i} & c_{xx} + c_{yy} \end{bmatrix}^{-1} \begin{bmatrix} x + \mathbf{i}y \\ x - \mathbf{i}y \end{bmatrix}\right\} \quad (\text{A.23})$$

Equations (A.18) and (A.23) look different but they are in fact identical. First note that the constants in front of the exponential term are the same. It remains to verify that the quadratic terms in the exponent are the same. This can be done by writing

$$\begin{aligned} \begin{bmatrix} x + \mathbf{i}y \\ x - \mathbf{i}y \end{bmatrix} &= \begin{bmatrix} 1 & \mathbf{i} \\ 1 & -\mathbf{i} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \left\{ \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -\mathbf{i} & \mathbf{i} \end{bmatrix} \right\}^{-1} \begin{bmatrix} x \\ y \end{bmatrix} \\ \begin{bmatrix} x + \mathbf{i}y \\ x - \mathbf{i}y \end{bmatrix}^* &= \begin{bmatrix} x \\ y \end{bmatrix}^T \left\{ \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -\mathbf{i} & \mathbf{i} \end{bmatrix}^* \right\}^{-1} \end{aligned} \quad (\text{A.24})$$

Substituting into the quadratic form in (A.23) gives a quadratic form in $[x, y]$ with the middle matrix given by

$$\begin{aligned}
 & \left\{ \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -\mathbf{i} & \mathbf{i} \end{bmatrix}^* \right\}^{-1} \begin{bmatrix} c_{xx} + c_{yy} & c_{xx} - c_{yy} + 2c_{xy}\mathbf{i} \\ c_{xx} - c_{yy} - 2c_{xy}\mathbf{i} & c_{xx} + c_{yy} \end{bmatrix}^{-1} \left\{ \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -\mathbf{i} & \mathbf{i} \end{bmatrix} \right\}^{-1} \\
 &= \left\{ \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -\mathbf{i} & \mathbf{i} \end{bmatrix} \begin{bmatrix} c_{xx} + c_{yy} & c_{xx} - c_{yy} + 2c_{xy}\mathbf{i} \\ c_{xx} - c_{yy} - 2c_{xy}\mathbf{i} & c_{xx} + c_{yy} \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -\mathbf{i} & \mathbf{i} \end{bmatrix}^* \right\}^{-1} \\
 &= \begin{bmatrix} c_{xx} & c_{xy} \\ c_{xy} & c_{yy} \end{bmatrix}^{-1}
 \end{aligned} \tag{A.25}$$

This is the same matrix inside the inverse of the quadratic form in (A.18). ■

Proof of (A.15) (Complex Gaussian PDF)

The key to deriving (A.15) from (A.4) is to express $[\text{Re } \mathbf{z}; \text{Im } \mathbf{z}]$ in terms of $[\mathbf{z}; \bar{\mathbf{z}}]$; and $\mathbf{C}$ in (A.5) in terms of $\mathbf{E} = E[\mathbf{Z}\mathbf{Z}^*]$ and $\mathbf{Q} = E[\mathbf{Z}\mathbf{Z}^T]$. For the former, assembling $\text{Re } \mathbf{z} = (\mathbf{z} + \bar{\mathbf{z}})/2$ and $\text{Im } \mathbf{z} = -(\mathbf{z} - \bar{\mathbf{z}})\mathbf{i}/2$ in matrix form gives

$$\begin{bmatrix} \text{Re } \mathbf{z} \\ \text{Im } \mathbf{z} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \mathbf{I}_n & \mathbf{I}_n \\ -\mathbf{i}\mathbf{I}_n & \mathbf{i}\mathbf{I}_n \end{bmatrix} \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{I}_n & \mathbf{i}\mathbf{I}_n \\ \mathbf{I}_n & -\mathbf{i}\mathbf{I}_n \end{bmatrix}^{-1}}_{\mathbf{T}} \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix} = \mathbf{T}^{-1} \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix} \tag{A.26}$$

Since $[\text{Re } \mathbf{z}; \text{Im } \mathbf{z}]$ is a real vector,

$$\begin{bmatrix} \text{Re } \mathbf{z} \\ \text{Im } \mathbf{z} \end{bmatrix}^T = \begin{bmatrix} \text{Re } \mathbf{z} \\ \text{Im } \mathbf{z} \end{bmatrix}^* = \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix}^* (\mathbf{T}^{-1})^* = \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix}^* (\mathbf{T}^*)^{-1} \tag{A.27}$$

The quadratic form in the exponent of (A.4) can then be written as

$$\begin{bmatrix} \text{Re } \mathbf{z} \\ \text{Im } \mathbf{z} \end{bmatrix}^T \mathbf{C}^{-1} \begin{bmatrix} \text{Re } \mathbf{z} \\ \text{Im } \mathbf{z} \end{bmatrix} = \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix}^* (\mathbf{T}^*)^{-1} \mathbf{C}^{-1} \mathbf{T}^{-1} \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix} = \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix}^* (\mathbf{T}\mathbf{C}\mathbf{T}^*)^{-1} \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix} \tag{A.28}$$

We next write $\mathbf{T}\mathbf{C}\mathbf{T}^*$ in terms of $\mathbf{E}$ and $\mathbf{Q}$. Using (A.11) and (A.5),

$$\mathbf{C} = \frac{1}{2} \begin{bmatrix} \text{Re } \mathbf{E} + \text{Re } \mathbf{Q} & -\text{Im } \mathbf{E} + \text{Im } \mathbf{Q} \\ \text{Im } \mathbf{E} + \text{Im } \mathbf{Q} & \text{Re } \mathbf{E} - \text{Re } \mathbf{Q} \end{bmatrix} \tag{A.29}$$

Then

$$\mathbf{T}\mathbf{C}\mathbf{T}^* = \frac{1}{2} \begin{bmatrix} \mathbf{I}_n & \mathbf{i}\mathbf{I}_n \\ \mathbf{I}_n & -\mathbf{i}\mathbf{I}_n \end{bmatrix} \begin{bmatrix} \text{Re } \mathbf{E} + \text{Re } \mathbf{Q} & -\text{Im } \mathbf{E} + \text{Im } \mathbf{Q} \\ \text{Im } \mathbf{E} + \text{Im } \mathbf{Q} & \text{Re } \mathbf{E} - \text{Re } \mathbf{Q} \end{bmatrix} \begin{bmatrix} \mathbf{I}_n & \mathbf{i}\mathbf{I}_n \\ \mathbf{I}_n & -\mathbf{i}\mathbf{I}_n \end{bmatrix}^* \tag{A.30}$$

Multiplying the matrices and using $\bar{\mathbf{Q}} = \mathbf{Q}^*$ gives

$$\mathbf{TCT}^* = \begin{bmatrix} \mathbf{E} & \mathbf{Q} \\ \mathbf{Q}^* & \bar{\mathbf{E}} \end{bmatrix} \quad (\text{A.31})$$

Substituting into (A.28) gives

$$\begin{bmatrix} \text{Re } \mathbf{z} \\ \text{Im } \mathbf{z} \end{bmatrix}^T \mathbf{C}^{-1} \begin{bmatrix} \text{Re } \mathbf{z} \\ \text{Im } \mathbf{z} \end{bmatrix} = \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix}^* \begin{bmatrix} \mathbf{E} & \mathbf{Q} \\ \mathbf{Q}^* & \bar{\mathbf{E}} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{z} \\ \bar{\mathbf{z}} \end{bmatrix} \quad (\text{A.32})$$

which is the quadratic form in the exponent of (A.15).

It remains to write $|\mathbf{C}|$ in terms of $\mathbf{E}$ and $\mathbf{Q}$. Taking determinant on both sides of (A.31),

$$|\mathbf{T}||\mathbf{C}||\mathbf{T}^*| = \left| \begin{bmatrix} \mathbf{E} & \mathbf{Q} \\ \mathbf{Q}^* & \bar{\mathbf{E}} \end{bmatrix} \right| \quad (\text{A.33})$$

Using $|A \otimes B| = |A|^b |B|^a$ where $A \otimes B$ denotes the Kronecker product of two square matrices A and B respectively of size a and b ,

$$|\mathbf{T}| = \left| \begin{bmatrix} 1 & \mathbf{i} \\ 1 & -\mathbf{i} \end{bmatrix} \otimes \mathbf{I}_n \right| = \left| \begin{bmatrix} 1 & \mathbf{i} \\ 1 & -\mathbf{i} \end{bmatrix} \right|^n |\mathbf{I}_n|^2 = (-2\mathbf{i})^n \quad (\text{A.34})$$

On the other hand, using the block matrix determinant formula (Sect. C.5.2),

$$\left| \begin{bmatrix} \mathbf{E} & \mathbf{Q} \\ \mathbf{Q}^* & \bar{\mathbf{E}} \end{bmatrix} \right| = |\mathbf{E}| \underbrace{|\bar{\mathbf{E}} - \mathbf{Q}^* \mathbf{E}^{-1} \mathbf{Q}|}_{\mathbf{P}} = |\mathbf{E}| |\mathbf{P}| \quad (\text{A.35})$$

Substituting (A.34) and (A.35) into (A.33), noting $|\mathbf{T}||\mathbf{T}^*| = (-2\mathbf{i})^n (2\mathbf{i})^n = 2^{2n}$ and rearranging gives

$$|\mathbf{C}| = 2^{-2n} |\mathbf{E}| |\mathbf{P}| \quad (\text{A.36})$$

Substituting (A.32) and (A.36) into (A.4) gives (A.15). ■

A.4 Circular Symmetry

A complex Gaussian vector $\mathbf{Z}$ is said to be ‘circularly symmetric’ when its distribution does not change under an arbitrary rotation in the complex plane. That is, $\mathbf{Z}$ and $\mathbf{Z}' = e^{i\phi} \mathbf{Z}$ have the same distribution for any real ϕ . Circularly symmetric

Gaussian vectors appear frequently in applications and they are traditionally the central subject of interest in the study of complex Gaussian vectors.

Circular symmetry is directly associated with pseudo-covariance. It can be shown that $\mathbf{Z}$ is circularly symmetric *if and only if* it has zero mean and zero pseudo-covariance, i.e., $E[\mathbf{Z}] = \mathbf{0}$ and $E[\mathbf{Z}\mathbf{Z}^T] = \mathbf{0}$. A standard complex Gaussian vector is therefore circularly symmetric with identity covariance matrix. Circular symmetry is also related to linear transformation. It can be shown that $\mathbf{Z}$ is circularly symmetric *if and only if* $\mathbf{Z} = \mathbf{R}\boldsymbol{\xi}$ for some complex matrix $\mathbf{R}$ ($n \times n$) and standard complex Gaussian vector $\boldsymbol{\xi}$ ($n \times 1$). See the end for proof.

When $\mathbf{Z}$ is circularly symmetric, its distribution is fully determined by the covariance matrix $\mathbf{E} = E[\mathbf{Z}\mathbf{Z}^*]$. In this case the PDF in (A.15) reduces to

$$p_{\mathbf{Z}}(\mathbf{z}) = \frac{1}{\pi^n |\mathbf{E}|} \exp(-\mathbf{z}^* \mathbf{E}^{-1} \mathbf{z}) \quad (\text{A.37})$$

In terms of real-valued covariance matrices, circular symmetry means that

$$E[\text{Re } \mathbf{Z} \text{Re } \mathbf{Z}^T] = E[\text{Im } \mathbf{Z} \text{Im } \mathbf{Z}^T] \quad E[\text{Re } \mathbf{Z} \text{Im } \mathbf{Z}^T] = -E[\text{Im } \mathbf{Z} \text{Re } \mathbf{Z}^T] \quad (\text{A.38})$$

Proof (Circular Symmetry $\Leftrightarrow$ Zero Mean and Zero Pseudo-covariance)

If $\mathbf{Z}$ is circularly symmetric then it has the same distribution as $\mathbf{Z}' = e^{i\phi} \mathbf{Z}$ for any real ϕ . Then $\mathbf{Z}$ and $\mathbf{Z}'$ have the same mean, covariance and pseudo-covariance matrix. Since $E[\mathbf{Z}'] = e^{i\phi} E[\mathbf{Z}]$ and generally $e^{i\phi} \neq 1$, it is necessary that $E[\mathbf{Z}] = \mathbf{0}$. For covariance, $E[\mathbf{Z}' \mathbf{Z}'^*] = e^{i\phi} e^{-i\phi} E[\mathbf{Z} \mathbf{Z}^*] = E[\mathbf{Z} \mathbf{Z}^*]$ is automatically satisfied. For pseudo-covariance, $E[\mathbf{Z}' \mathbf{Z}'^T] = e^{2i\phi} E[\mathbf{Z} \mathbf{Z}^T]$. Again, $e^{2i\phi} \neq 1$ and so $E[\mathbf{Z} \mathbf{Z}^T] = \mathbf{0}$.

Conversely, suppose $\mathbf{Z}$ has zero mean and zero pseudo-covariance. Since $\mathbf{Z}' = e^{i\phi} \mathbf{Z}$ is complex Gaussian, its PDF is fully determined by the mean, covariance and pseudo-covariance. It is sufficient to show that these quantities are the same as those of $\mathbf{Z}$. This follows directly from the properties of $\mathbf{Z}$:

$$\begin{aligned} E[\mathbf{Z}'] &= e^{i\phi} \underbrace{E[\mathbf{Z}]}_{\mathbf{0}} = \mathbf{0} \\ E[\mathbf{Z}' \mathbf{Z}'^*] &= e^{i\phi} e^{-i\phi} E[\mathbf{Z} \mathbf{Z}^*] = E[\mathbf{Z} \mathbf{Z}^*] \\ E[\mathbf{Z}' \mathbf{Z}'^T] &= e^{i\phi} e^{i\phi} \underbrace{E[\mathbf{Z} \mathbf{Z}^T]}_{\mathbf{0}} = \mathbf{0} \end{aligned} \quad (\text{A.39})$$

■

Proof (Linear Transformation $\Leftrightarrow$ Zero Mean and Zero Pseudo-covariance)

Suppose $\mathbf{Z} = \mathbf{R}\xi$ where $\mathbf{R}$ ($n \times n$) is a constant complex matrix and ξ ($n \times 1$) is standard complex Gaussian. Then

$$\begin{aligned} E[\mathbf{Z}] &= \mathbf{R} \underbrace{E[\xi]}_{\mathbf{0}} = \mathbf{0} \\ E[\mathbf{Z}\mathbf{Z}^T] &= \mathbf{R} \underbrace{E[\xi\xi^T]}_{\mathbf{0}} \mathbf{R}^T = \mathbf{0} \end{aligned} \quad (\text{A.40})$$

Conversely, suppose $E[\mathbf{Z}] = \mathbf{0}$ and $E[\mathbf{Z}\mathbf{Z}^T] = \mathbf{0}$; and let $\mathbf{E}$ be the covariance matrix of $\mathbf{Z}$. Since $\mathbf{E}$ is Hermitian and positive definite, it has the representation $\mathbf{E} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^*$ where $\mathbf{\Lambda}$ ($n \times n$) is a diagonal matrix of (real positive) eigenvalues; and $\mathbf{U}$ ($n \times n$ complex) is a unitary matrix ($\mathbf{U}^*\mathbf{U} = \mathbf{I}_n$) consisting in its columns the eigenvectors of $\mathbf{E}$. Then $\mathbf{Z}$ can be represented as $\mathbf{Z} = \mathbf{U}\sqrt{\mathbf{\Lambda}}\xi$ where ξ is a $n \times 1$ standard complex Gaussian vector, because then

$$\begin{aligned} E[\mathbf{Z}] &= \mathbf{U}\sqrt{\mathbf{\Lambda}} \underbrace{E[\xi]}_{\mathbf{0}} = \mathbf{0} \\ E[\mathbf{Z}\mathbf{Z}^T] &= \mathbf{U}\sqrt{\mathbf{\Lambda}} \underbrace{E[\xi\xi^T]}_{\mathbf{0}} \sqrt{\mathbf{\Lambda}}\mathbf{U}^T = \mathbf{0} \\ E[\mathbf{Z}\mathbf{Z}^*] &= \mathbf{U}\sqrt{\mathbf{\Lambda}} \underbrace{E[\xi\xi^*]}_{\mathbf{I}_n} \sqrt{\mathbf{\Lambda}}\mathbf{U}^* = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^* = \mathbf{E} \end{aligned} \quad (\text{A.41})$$

■

A.5 Complex Wishart Distribution

Let $\{\mathbf{Z}_r\}_{r=1}^M$ ($n \times 1$) be an i.i.d. sequence of circularly symmetric complex Gaussian vectors, each with covariance matrix $\mathbf{E}$. An estimator for $\mathbf{E}$ based on simple averaging is

$$\hat{\mathbf{E}} = \frac{1}{M} \mathbf{W} \quad (\text{A.42})$$

where

$$\mathbf{W} = \sum_{r=1}^M \mathbf{Z}_r \mathbf{Z}_r^* \quad (\text{A.43})$$

Note that $E[\mathbf{W}] = M\mathbf{E}$ and $E[\hat{\mathbf{E}}] = \mathbf{E}$; and so $\hat{\mathbf{E}}$ is an unbiased estimator of $\mathbf{E}$. It can also be shown that $\hat{\mathbf{E}}$ is the maximum-likelihood estimator of $\mathbf{E}$ (Goodman 1963). The statistical properties of $\mathbf{W}$ (and hence $\hat{\mathbf{E}}$) have been studied extensively in multivariate statistics. Clearly, $\mathbf{W}$ is a sum of M independent matrices, each of rank 1. It is Hermitian, i.e., $\mathbf{W}^* = \mathbf{W}$. For $M < n$, $\mathbf{W}$ has rank M , less than its dimension and is therefore singular. The case of relevance in applications is $M \geq n$. In this case $\mathbf{W}$ has rank n and its PDF is given by

$$p_{\mathbf{W}}(\mathbf{w}) = \frac{|\mathbf{w}|^{M-n}}{I(\mathbf{E})} e^{-\text{tr}(\mathbf{E}^{-1}\mathbf{w})} \quad (M \geq n) \quad (\text{A.44})$$

$$I(\mathbf{E}) = \pi^{n(n-1)/2} |\mathbf{E}|^M \prod_{i=1}^n (M-i)! \quad (\text{A.45})$$

This is called the ‘complex Wishart distribution’ with M degrees of freedom. The PDF is valid for Hermitian positive semi-definite $n \times n$ matrix $\mathbf{w}$ (and is zero otherwise). It is in fact the joint PDF of the (real-valued) diagonal entries and the real and imaginary parts of the lower off-diagonal entries of $\mathbf{W}$.

The PDF of $\hat{\mathbf{E}} = \mathbf{W}/M$ can be obtained by a PDF transformation, which gives

$$p_{\hat{\mathbf{E}}}(\mathbf{s}) = M^{nM} \frac{|\mathbf{s}|^{M-n}}{I(\mathbf{E})} e^{-M \text{tr}(\mathbf{E}^{-1}\mathbf{s})} \quad (M \geq n) \quad (\text{A.46})$$

Again, this PDF is valid for $n \times n$ Hermitian positive semi-definite matrix $\mathbf{s}$ (and zero otherwise). It is the joint PDF of the (real-valued) diagonal entries and the real and imaginary parts of the lower off-diagonal entries of $\hat{\mathbf{E}}$.

As a literature note, Wishart distribution was first introduced by Wishart (1928) as a generalization of Chi-square distribution to multi-dimensions. Details can be found in texts on multivariate Gaussian distributions, e.g., Anderson (1984) and Tong (1990). Goodman (1963) gave an introduction to complex Wishart distribution with discussion on applications. Srivastava (1965) gave a simplified derivation of the distribution, though requiring some familiarity with PDF transformation with variables of complex matrices. See Maiwald & Kraus (2000) for statistical moments of complex Wishart distribution.

In the scalar case ($n = 1$), the complex Wishart PDF can be derived based on basic probability principles; see Sect. A.5.1. The general case ($n \geq 1$) is much more involved because $\mathbf{W}$ is a ‘structured matrix’, i.e., with constraints among its entries. The Jacobian of transformation between the entries of $\mathbf{W}$ and those of $\mathbf{Z}_r$ requires special consideration. See Srivastava (1965).

A.5.1 Scalar Variable

Consider the scalar case where $\{Z_r\}_{r=1}^M$ are i.i.d. scalar circularly symmetric Gaussian variables with covariance $E[Z_r Z_r^*] = S > 0$. Denote $\mathbf{W}$ in (A.43) by W_M :

$$W_M = \sum_{r=1}^M Z_r Z_r^* = \sum_{r=1}^M |Z_r|^2 \quad (\text{A.47})$$

Equations (A.44) and (A.46) reduce to

$$p_{W_M}(w) = \frac{w^{M-1}}{S^M (M-1)!} e^{-w/S} \quad w \geq 0 \quad (\text{A.48})$$

$$p_{\hat{E}}(s) = \frac{s^{M-1} M^M}{S^M (M-1)!} e^{-Ms/S} \quad s \geq 0 \quad (\text{A.49})$$

The PDF of W_M (and hence $\hat{E} = W_M/M$) can be derived as follow. Being circularly symmetric, Z_r can be represented as

$$Z_r = \sqrt{\frac{S}{2}}(X_r + iY_r) \quad (\text{A.50})$$

where X_r and Y_r are i.i.d. standard Gaussian. Check that

$$E[Z_r] = 0 \quad E[Z_r Z_r^T] = 0 \quad E[Z_r Z_r^*] = S \quad (\text{A.51})$$

Note that

$$W_M = \frac{S}{2} \sum_{r=1}^M (X_r^2 + Y_r^2) \quad (\text{A.52})$$

This implies that $2W_M/S$ is a sum of the squares of $2M$ i.i.d. standard Gaussian variables, and hence has a Chi-square distribution with $2M$ degrees of freedom. The PDF of W_M can then be obtained by a PDF transformation of the Chi-square PDF.

Alternatively, by viewing W_M as the sum of M i.i.d. variables distributed as W_1 , its PDF can be obtained by successive convolution of PDFs. For $M = 1$, $2W_1/S$ is Chi-square with 2 degrees of freedom, i.e., exponentially distributed with mean 2. This implies that W_1 is exponentially distributed with mean S . Its PDF is then given by

$$p_{W_1}(w) = \frac{1}{S} \exp\left(-\frac{w}{S}\right) \quad w \geq 0 \quad (\text{A.53})$$

For $M = 2$, since W_2 is the sum of two i.i.d. variables distributed as W_1 , its PDF can be obtained as the convolution of two PDFs of W_1 :

$$p_{W_2}(w) = \int_0^w p_{W_1}(x)p_{W_1}(w-x)dx = \frac{1}{S^2} \int_0^w e^{-x/S} e^{-(w-x)/S} dx = \frac{w}{S^2} e^{-w/S} \quad (\text{A.54})$$

Similarly,

$$p_{W_3}(w) = \int_0^w p_{W_2}(x)p_{W_1}(w-x)dx = \frac{1}{S^3} \int_0^w x e^{-x/S} e^{-(w-x)/S} dx = \frac{w^2}{2S^3} e^{-w/S} \quad (\text{A.55})$$

$$p_{W_4}(w) = \int_0^w p_{W_3}(x)p_{W_1}(w-x)dx = \frac{1}{2S^4} \int_0^w x^2 e^{-x/S} e^{-(w-x)/S} dx = \frac{w^3}{(2 \cdot 3)S^4} e^{-w/S} \quad (\text{A.56})$$

Continuing this procedure, we see that in general

$$p_{W_M}(w) = \frac{w^{M-1}}{[1 \cdot 2 \cdot 3 \cdots (M-1)]S^M} e^{-w/S} \quad (\text{A.57})$$

This agrees with (A.48).

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Appendix B. Hessian Under Constraints

This chapter discusses the Hessian of a scalar function under constraints. It supplements the theory on posterior covariance matrix in Sect. 11.2. Consider the minimum of a scalar function $L(\boldsymbol{\theta})$ w.r.t. $\boldsymbol{\theta}$ ($n_\theta \times 1$) subjected to n_c independent constraints:

$$G_i(\boldsymbol{\theta}) = 0 \quad i = 1, \dots, n_c \quad (\text{B.1})$$

Simply taking the Hessian of $L(\boldsymbol{\theta})$ w.r.t. $\boldsymbol{\theta}$ at the minimum does not account for constraints. Such Hessian is not even positive definite. One way to account for constraints is to consider the Hessian of the composite function

$$L_c(\boldsymbol{\theta}) = L(\mathbf{v}_c(\boldsymbol{\theta})) \quad (\text{B.2})$$

where $\mathbf{v}_c(\boldsymbol{\theta})$ ($n_\theta \times 1$) is a mapping that automatically satisfies constraints, i.e., $G_i(\mathbf{v}_c(\boldsymbol{\theta})) = 0$ ($i = 1, \dots, n_c$) for any $\boldsymbol{\theta}$. In this chapter we consider more generally the Hessian of $L_c(\mathbf{u}) = L(\mathbf{v}_c(\mathbf{u}))$ where $\mathbf{v}_c(\mathbf{u})$ maps a set of free parameters $\mathbf{u}$ ($p \times 1$) to $\boldsymbol{\theta}$ ($n_\theta \times 1$) that always satisfies constraints; p can be different from n_θ . Since there are n_θ parameters subjected to n_c constraints, it is necessary that $p \geq n_\theta - n_c$.

The Hessian of $L_c(\mathbf{u}) = L(\mathbf{v}_c(\mathbf{u}))$ w.r.t. $\mathbf{u}$ is more complicated than that of $L(\boldsymbol{\theta})$ w.r.t. $\boldsymbol{\theta}$. Formulas are presented that express the former in terms of the latter. Section B.1 presents a formula based on direct differentiation. It is applicable for any parameter value. Section B.2 presents a formula based on Lagrange multiplier concepts. It is only applicable at the optimum but is more insightful and involves much less computation.

Example B.1 (Quadratic form) This simple example demonstrates that incorrect conclusion results if the constraint is not taken into account when evaluating the Hessian. Let $\mathbf{A}$ be a $p \times p$ real symmetric and positive definite matrix. Consider maximizing $F(\mathbf{u}) = \mathbf{u}^T \mathbf{A} \mathbf{u}$ w.r.t. $\mathbf{u}$ ($p \times 1$) under norm constraint $\|\mathbf{u}\|^2 = \mathbf{u}^T \mathbf{u} = 1$. According to Lagrange multiplier method, the optimum is a stationary point of

$$J(\mathbf{u}, \lambda) = F(\mathbf{u}) + \lambda(1 - \mathbf{u}^T \mathbf{u}) = \mathbf{u}^T \mathbf{A} \mathbf{u} + \lambda(1 - \mathbf{u}^T \mathbf{u}) \quad (\text{B.3})$$

where λ is Lagrange multiplier.

Optimum

Direct differentiation of J and using symmetry of $\mathbf{A}$ yields

$$\nabla_{\mathbf{u}} J = 2\mathbf{u}^T \mathbf{A} - 2\lambda \mathbf{u}^T \quad \frac{\partial J}{\partial \lambda} = 1 - \mathbf{u}^T \mathbf{u} \quad (\text{B.4})$$

Solving $\nabla_{\mathbf{u}} J = \mathbf{0}$ shows that the optimum satisfies the eigenvalue equation:

$$\mathbf{A} \mathbf{u} = \lambda \mathbf{u} \quad (\text{B.5})$$

Note that $\partial J / \partial \lambda = 0$ simply gives the constraint equation. Since $\mathbf{A}$ is positive definite, all its eigenvalues are positive. Substituting (B.5), the value of F at the optimum is λ . As we want to maximize F , the optimum $\hat{\mathbf{u}}$ should be the eigenvector of $\mathbf{A}$ with the largest eigenvalue and scaled to have unit norm. Let $\{\mathbf{b}_i\}_{i=1}^p$ be the eigenvectors (with unit norm) of $\mathbf{A}$ with eigenvalues $\{\kappa_i\}_{i=1}^p$ (in ascending order of magnitude). Then $\hat{\mathbf{u}} = \mathbf{b}_p$ and the value of λ at the optimum is $\hat{\lambda} = \kappa_p$.

Hessian at Optimum

Consider now the Hessian of F at the optimum. Intuitively it should be negative definite because F is maximized there. This is only true when the constraint is correctly accounted for. Direct differentiation gives $\nabla^2 F(\hat{\mathbf{u}}) = 2\mathbf{A}$, which is positive definite (on the contrary). The Hessian of F that accounts for norm constraint can be obtained as the Hessian of the following composite function

$$F_c(\mathbf{u}) = F(\|\mathbf{u}\|^{-1} \mathbf{u}) = \left(\frac{\mathbf{u}}{\|\mathbf{u}\|}\right)^T \mathbf{A} \left(\frac{\mathbf{u}}{\|\mathbf{u}\|}\right) = \frac{\mathbf{u}^T \mathbf{A} \mathbf{u}}{\mathbf{u}^T \mathbf{u}} \quad (\text{B.6})$$

Direct differentiation gives

$$\nabla F_c = 2(\mathbf{u}^T \mathbf{u})^{-1}(\mathbf{u}^T \mathbf{A}) - 2(\mathbf{u}^T \mathbf{u})^{-2}(\mathbf{u}^T \mathbf{A} \mathbf{u}) \mathbf{u}^T \quad (\text{B.7})$$

$$\begin{aligned} \nabla^2 F_c &= 2(\mathbf{u}^T \mathbf{u})^{-1} \mathbf{A} - 4(\mathbf{u}^T \mathbf{u})^{-2}(\mathbf{u} \mathbf{u}^T \mathbf{A}) + 8(\mathbf{u}^T \mathbf{u})^{-3}(\mathbf{u}^T \mathbf{A} \mathbf{u}) \mathbf{u} \mathbf{u}^T \\ &\quad - 4(\mathbf{u}^T \mathbf{u})^{-2}(\mathbf{A} \mathbf{u} \mathbf{u}^T) - 2(\mathbf{u}^T \mathbf{u})^{-2}(\mathbf{u}^T \mathbf{A} \mathbf{u}) \mathbf{I}_p \end{aligned} \quad (\text{B.8})$$

where $\mathbf{I}_p$ denotes the $p \times p$ identity matrix. Substituting $\mathbf{u} = \hat{\mathbf{u}}$ into ∇F_c and using $\mathbf{A} \hat{\mathbf{u}} = \kappa_p \hat{\mathbf{u}}$ shows that $\nabla F_c(\hat{\mathbf{u}}) = \mathbf{0}$, as expected. Substituting $\mathbf{u} = \hat{\mathbf{u}}$ into $\nabla^2 F_c$ gives the Hessian at the optimum:

$$\nabla^2 \hat{F}_c = 2(\mathbf{A} - \kappa_p \mathbf{I}_p) \quad (\text{B.9})$$

where ‘^’ denotes that it is evaluated at optimum. Substituting $\mathbf{A} = \sum_{i=1}^p \kappa_i \mathbf{b}_i \mathbf{b}_i^T$ and $\mathbf{I}_p = \sum_{i=1}^p \mathbf{b}_i \mathbf{b}_i^T$ into (B.9) and rearranging gives the eigenvector representation:

$$\nabla^2 \hat{F}_c = \sum_{i=1}^p \underbrace{2(\kappa_i - \kappa_p)}_{\text{eigenvalue}} \mathbf{b}_i \mathbf{b}_i^T \quad (\text{B.10})$$

Since $\kappa_p \geq \kappa_i$, the largest eigenvalue of $\nabla^2 \hat{F}_c$ is zero and the remaining ones are all negative. This implies that $\nabla^2 \hat{F}_c$ is a negative semi-definite matrix. The zero eigenvalue is due to norm constraint. ■

Example B.2 (Quadratic form, geometric illustration) Suppose in Example B.1

$$\mathbf{A} = \begin{bmatrix} 1.2 & -0.4 \\ -0.4 & 1.8 \end{bmatrix}$$

Let $\mathbf{u} = [u_1, u_2]^T$. Figure B.1a shows the mesh of $F(\mathbf{u}) = \mathbf{u}^T \mathbf{A} \mathbf{u}$, whose unconstrained minimum is at the origin. Figure B.1b shows the contour of F . The thick line shows the unit circle $\mathbf{u}^T \mathbf{u} = 1$. Going along this circle and observing the value of F encounters the constrained minima and maxima of F subjected to $\mathbf{u}^T \mathbf{u} = 1$. The value of F along this circle is plotted in Fig. B.1c. The two minima are opposite to each other; the same applies to the maxima. The coordinates of the minima and maxima correspond to the eigenvectors of $\mathbf{A}$ with the smallest and largest eigenvalue, respectively.

Figure B.2 shows the mesh and contour of $F_c(\mathbf{u}) = F(\mathbf{v}_c(\mathbf{u}))$, where $\mathbf{v}_c(\mathbf{u}) = [u_1, u_2]^T / (u_1^2 + u_2^2)^{1/2}$ maps all points (u_1, u_2) to the unit circle and therefore

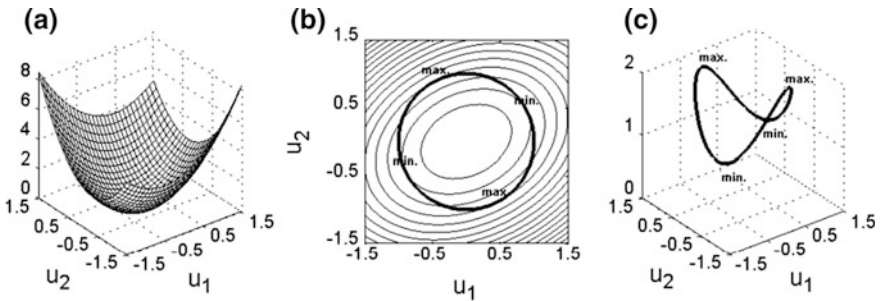


Fig. B.1 Plots of quadratic form. **a** Mesh; **b** contour; *thick solid line*: unit circle; **c** value of quadratic form along unit circle

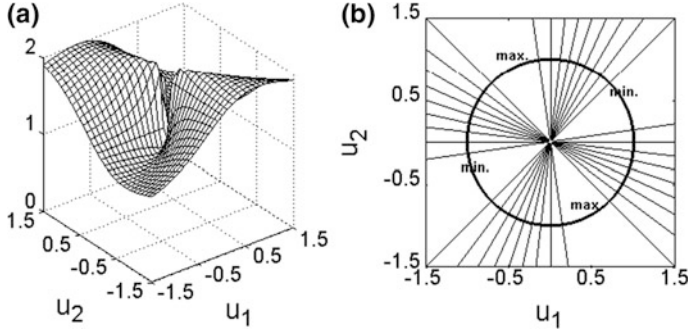


Fig. B.2 Plots of constrained quadratic form. **a** Mesh; **b** contour; *thick solid line*: unit circle

automatically satisfies unit norm constraint. The value of $\mathbf{v}_c(\mathbf{u})$ does not change along any ray stemming from the origin. The same is also true for $F_c(\mathbf{u})$. ■

B.1 Direct Formula

By direct differentiation and repeated application of chain rule, it can be shown that

$$\nabla^2 L_c = \underbrace{(\nabla \mathbf{v}_c)^T \nabla^2 L (\nabla \mathbf{v}_c)}_{p \times n_0 p} + \underbrace{(\mathbf{I}_p \otimes \nabla L)}_{p \times n_0 p} \nabla^2 \mathbf{v}_c \quad (\text{B.11})$$

where ‘ $\otimes$ ’ denotes the Kronecker product;

$$\nabla L = \nabla L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\mathbf{v}_c(\mathbf{u})} \quad \nabla^2 L = \nabla^2 L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\mathbf{v}_c(\mathbf{u})} \quad (\text{B.12})$$

Equation (B.11) is applicable for general $\mathbf{u}$, not just at the optimum. It is also applicable for any mapping $\mathbf{v}_c(\mathbf{u})$ that need not be related to constraints. See Sect. B.1.1 for proof.

Gradient and Hessian of Matrix Functions

Equation (B.11) involves the gradient and Hessian of the vector function $\mathbf{v}_c(\mathbf{u})$. For general matrix function $\mathbf{X}(\mathbf{u})$ ($m \times n$), ∇ and ∇^2 are interpreted via the Kronecker product:

$$\nabla \mathbf{X} = \nabla_{1 \times p} \otimes \mathbf{X}_{m \times n} = \left[\frac{\partial \mathbf{X}}{\partial u_1}, \dots, \frac{\partial \mathbf{X}}{\partial u_p} \right]_{m \times n \times m \times n} \quad (\text{B.13})$$

$$\nabla^2 \mathbf{X} = \nabla^2_{mp \times np} \otimes \mathbf{X}_{p \times p \times m \times n} = \begin{bmatrix} \frac{\partial^2 \mathbf{X}}{\partial u_1^2} & \frac{\partial^2 \mathbf{X}}{\partial u_1 \partial u_2} & \cdots & \frac{\partial^2 \mathbf{X}}{\partial u_1 \partial u_p} \\ & \ddots & & \frac{\partial^2 \mathbf{X}}{\partial u_2 \partial u_p} \\ & & \ddots & \vdots \\ \text{Block sym.} & & & \frac{\partial^2 \mathbf{X}}{\partial u_p^2} \end{bmatrix} \quad (\text{B.14})$$

This interpretation is consistent with the conventional definition of gradient and Hessian when $\mathbf{X}$ is a scalar. Thus, in (B.11),

$$\nabla \mathbf{v}_c = \left[\frac{\partial \mathbf{v}_c}{\partial u_1}, \dots, \frac{\partial \mathbf{v}_c}{\partial u_p} \right]_{n_\theta \times p} \quad \nabla^2 \mathbf{v}_c = \begin{bmatrix} \frac{\partial^2 \mathbf{v}_c}{\partial u_1^2} & \frac{\partial^2 \mathbf{v}_c}{\partial u_1 \partial u_2} & \cdots & \frac{\partial^2 \mathbf{v}_c}{\partial u_1 \partial u_p} \\ & \ddots & & \frac{\partial^2 \mathbf{v}_c}{\partial u_2 \partial u_p} \\ & & \ddots & \vdots \\ \text{Block sym.} & & & \frac{\partial^2 \mathbf{v}_c}{\partial u_p^2} \end{bmatrix} \quad (\text{B.15})$$

Example B.3 (Quadratic form, direct formula) Consider Example B.1 again to determine the constrained Hessian of $F(\mathbf{u}) = \mathbf{u}^T \mathbf{A} \mathbf{u}$ subjected to $\mathbf{u}^T \mathbf{u} = 1$, where $\mathbf{A}$ is a $p \times p$ real symmetric and positive definite matrix. In this example we obtain the Hessian of $F_c(\mathbf{u}) = F(\|\mathbf{u}\|^{-1} \mathbf{u})$ using (B.11). Although the expression is generally applicable for any $\mathbf{u}$, to simplify presentation we evaluate it at the optimum. The mapping satisfying constraint can be taken as

$$\mathbf{v}_c(\mathbf{u}) = \bar{\mathbf{u}} = \frac{\mathbf{u}}{\|\mathbf{u}\|} = \frac{\mathbf{u}}{(\mathbf{u}^T \mathbf{u})^{1/2}} \quad (\text{B.16})$$

Direct differentiation and algebra gives

$$\nabla F(\mathbf{u}) = 2\mathbf{u}^T \mathbf{A} \quad \nabla^2 F(\mathbf{u}) = 2\mathbf{A} \quad (\text{B.17})$$

$$\nabla \mathbf{v}_c(\mathbf{u}) = \|\mathbf{u}\|^{-1} (\mathbf{I}_p - \bar{\mathbf{u}} \bar{\mathbf{u}}^T) \quad (\text{B.18})$$

$$\nabla^2 \mathbf{v}_c(\mathbf{u}) = \|\mathbf{u}\|^{-2} [(3\bar{\mathbf{u}}\bar{\mathbf{u}}^T - \mathbf{I}_p) \otimes \bar{\mathbf{u}} - \bar{\mathbf{u}} \otimes \mathbf{I}_p - \text{vec}(\mathbf{I}_p)\bar{\mathbf{u}}^T] \quad (\text{B.19})$$

where $\text{vec}(\mathbf{I}_p)$ denotes the ‘vectorization’ of $\mathbf{I}_p$, i.e., a $n^2 \times 1$ vector obtained by stacking column-wise the columns of $\mathbf{I}_p$. Evaluating at $\mathbf{u} = \hat{\mathbf{u}}$ (optimum),

$$\nabla \hat{F} = 2\kappa_p \hat{\mathbf{u}}^T \quad \nabla^2 \hat{F} = 2\mathbf{A} \quad (\text{B.20})$$

$$\nabla \hat{\mathbf{v}}_c = \mathbf{I}_p - \hat{\mathbf{u}}\hat{\mathbf{u}}^T \quad \nabla^2 \hat{\mathbf{v}}_c = (3\hat{\mathbf{u}}\hat{\mathbf{u}}^T - \mathbf{I}_p) \otimes \hat{\mathbf{u}} - \hat{\mathbf{u}} \otimes \mathbf{I}_p - \text{vec}(\mathbf{I}_p)\hat{\mathbf{u}}^T \quad (\text{B.21})$$

where ‘^’ denotes that the quantity is evaluated at the optimum. Applying (B.11) at the optimum,

$$\nabla^2 \hat{F}_c = (\nabla \hat{\mathbf{v}}_c)^T (\nabla^2 \hat{F}) (\nabla \hat{\mathbf{v}}_c) + (\mathbf{I}_p \otimes \nabla \hat{F}) (\nabla^2 \hat{\mathbf{v}}_c) \quad (\text{B.22})$$

Direct substitution gives an expression that hardly looks like (B.9) in Example B.1. It can be shown to be equivalent after simplifications. For the first term in (B.22),

$$(\nabla \hat{\mathbf{v}}_c)^T (\nabla^2 \hat{F}_c) (\nabla \hat{\mathbf{v}}_c) = 2(\mathbf{I}_p - \hat{\mathbf{u}}\hat{\mathbf{u}}^T)^T \mathbf{A} (\mathbf{I}_p - \hat{\mathbf{u}}\hat{\mathbf{u}}^T) = 2(\mathbf{A} - \kappa_p \hat{\mathbf{u}}\hat{\mathbf{u}}^T) \quad (\text{B.23})$$

after expanding and making use of $\hat{\mathbf{u}}^T \hat{\mathbf{u}} = 1$ and $\mathbf{A}\hat{\mathbf{u}} = \kappa_p \hat{\mathbf{u}}$. For the second term in (B.22),

$$\begin{aligned} & (\mathbf{I}_p \otimes \nabla \hat{F}) (\nabla^2 \hat{\mathbf{v}}_c) \\ &= (\mathbf{I}_p \otimes 2\kappa_p \hat{\mathbf{u}}^T) [(3\hat{\mathbf{u}}\hat{\mathbf{u}}^T - \mathbf{I}_p) \otimes \hat{\mathbf{u}} - \hat{\mathbf{u}} \otimes \mathbf{I}_p - \text{vec}(\mathbf{I}_p)\hat{\mathbf{u}}^T] \\ &= 2\kappa_p \{ (\mathbf{I}_p \otimes \hat{\mathbf{u}}^T) [(3\hat{\mathbf{u}}\hat{\mathbf{u}}^T - \mathbf{I}_p) \otimes \hat{\mathbf{u}}] - (\mathbf{I}_p \otimes \hat{\mathbf{u}}^T) (\hat{\mathbf{u}} \otimes \mathbf{I}_p) - (\mathbf{I}_p \otimes \hat{\mathbf{u}}^T) \text{vec}(\mathbf{I}_p)\hat{\mathbf{u}}^T \} \end{aligned} \quad (\text{B.24})$$

The first two terms in the brace can be further simplified using the following identity $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ where A , B , C and D are matrices of appropriate size. Applying this to the first term,

$$(\mathbf{I}_p \otimes \hat{\mathbf{u}}^T) [(3\hat{\mathbf{u}}\hat{\mathbf{u}}^T - \mathbf{I}_p) \otimes \hat{\mathbf{u}}] = [\mathbf{I}_p (3\hat{\mathbf{u}}\hat{\mathbf{u}}^T - \mathbf{I}_p)] \otimes (\hat{\mathbf{u}}^T \hat{\mathbf{u}}) = 3\hat{\mathbf{u}}\hat{\mathbf{u}}^T - \mathbf{I}_p \quad (\text{B.25})$$

For the second term,

$$(\mathbf{I}_p \otimes \hat{\mathbf{u}}^T) (\hat{\mathbf{u}} \otimes \mathbf{I}_p) = (\mathbf{I}_p \hat{\mathbf{u}}) \otimes (\hat{\mathbf{u}}^T \mathbf{I}_p) = \hat{\mathbf{u}} \otimes \hat{\mathbf{u}}^T = \hat{\mathbf{u}}\hat{\mathbf{u}}^T \quad (\text{B.26})$$

The third term in (B.24) can be simplified using the identity $\text{vec}(AXB) = (B^T \otimes A)\text{vec}(X)$ for matrices A , B and X of appropriate size:

$$\underbrace{(\mathbf{I}_p \otimes \hat{\mathbf{u}}^T) \text{vec}(\mathbf{I}_p)}_{\text{vec}(\hat{\mathbf{u}}^T \mathbf{I}_p \mathbf{I}_p^T)} \hat{\mathbf{u}}^T = \underbrace{\text{vec}(\hat{\mathbf{u}}^T \mathbf{I}_p \mathbf{I}_p^T)}_{\hat{\mathbf{u}}} \hat{\mathbf{u}}^T = \hat{\mathbf{u}}\hat{\mathbf{u}}^T \quad (\text{B.27})$$

Substituting (B.25), (B.26) and (B.27) into (B.24) gives

$$(\mathbf{I}_p \otimes \nabla \hat{F})(\nabla^2 \hat{\mathbf{v}}_c) = 2\kappa_p(\hat{\mathbf{u}}\hat{\mathbf{u}}^T - \mathbf{I}_p) \quad (\text{B.28})$$

Substituting (B.23) and (B.28) into (B.22) gives the same result in (B.9). This example demonstrates that, although the direct formula in (B.11) can be implemented systematically, its calculation involves a large number of intermediate terms and so it may not be computationally efficient. ■

B.1.1 Derivation

Recall the context of (B.11). Let $L(\boldsymbol{\theta})$ be a scalar function of $\boldsymbol{\theta}$ ($n_\theta \times 1$); and $L_c(\mathbf{u}) = L(\mathbf{v}_c(\mathbf{u}))$, where $\mathbf{v}_c(\mathbf{u})$ is a $n_\theta \times 1$ vector function of $\mathbf{u}$ ($p \times 1$). We shall show that

$$\nabla L_c = \nabla L \nabla \mathbf{v}_c \quad (\text{B.29})$$

$1 \times p \quad 1 \times n_\theta \quad n_\theta \times p$

$$\nabla^2 L_c = (\nabla \mathbf{v}_c)^T \nabla^2 L (\nabla \mathbf{v}_c) + \underbrace{(\mathbf{I}_p \otimes \nabla L)}_{p \times n_\theta p} \nabla^2 \mathbf{v}_c \quad (\text{B.30})$$

$p \times p \quad p \times n_\theta \quad n_\theta \times n_\theta \quad n_\theta \times p \quad p \times p \quad 1 \times n_\theta \quad n_\theta p \times p$

Let $\mathbf{v}_c(\mathbf{u}) = [v_1(\mathbf{u}), \dots, v_{n_\theta}(\mathbf{u})]^T$ and $\mathbf{u} = [u_1, \dots, u_p]^T$. For simplicity in notation, omit dependence on variables. For $i = 1, \dots, p$,

$$\frac{\partial L_c}{\partial u_i} = \sum_{j=1}^{n_\theta} \frac{\partial L}{\partial \theta_j} \frac{\partial v_j}{\partial u_i} = \left[\frac{\partial L}{\partial \theta_1}, \dots, \frac{\partial L}{\partial \theta_{n_\theta}} \right] \left[\frac{\partial v_1}{\partial u_i}, \dots, \frac{\partial v_{n_\theta}}{\partial u_i} \right]^T = \nabla L \frac{\partial \mathbf{v}_c}{\partial u_i} \quad (\text{B.31})$$

Assembling this result row-wise for $i = 1, \dots, p$,

$$\nabla L_c = \left[\frac{\partial L_c}{\partial u_1}, \dots, \frac{\partial L_c}{\partial u_p} \right] = \left[\nabla L \frac{\partial \mathbf{v}_c}{\partial u_1}, \dots, \nabla L \frac{\partial \mathbf{v}_c}{\partial u_p} \right] = (\nabla L) \left[\frac{\partial \mathbf{v}_c}{\partial u_1}, \dots, \frac{\partial \mathbf{v}_c}{\partial u_p} \right] = \nabla L \nabla \mathbf{v}_c \quad (\text{B.32})$$

which proves (B.29). To show (B.30), note that

$$\nabla^2 L_c = \begin{bmatrix} \frac{\partial \nabla L_c}{\partial u_1} \\ \vdots \\ \frac{\partial \nabla L_c}{\partial u_p} \end{bmatrix} \quad (\text{B.33})$$

For $i = 1, \dots, p$, using (B.29) and chain rule,

$$\frac{\partial \nabla L_c}{\partial u_i} = \frac{\partial \nabla L}{\partial u_i} \nabla \mathbf{v}_c + \nabla L \frac{\partial \nabla \mathbf{v}_c}{\partial u_i} \quad (\text{B.34})$$

For the first term,

$$\begin{aligned} \frac{\partial \nabla L}{\partial u_i} &= \frac{\partial}{\partial u_i} \left[\frac{\partial L}{\partial \theta_1}, \dots, \frac{\partial L}{\partial \theta_{n_\theta}} \right] = \sum_{j=1}^{n_\theta} \left[\frac{\partial^2 L}{\partial \theta_j \partial \theta_1}, \dots, \frac{\partial^2 L}{\partial \theta_j \partial \theta_{n_\theta}} \right] \frac{\partial v_j}{\partial u_i} \\ &= \left[\frac{\partial v_1}{\partial u_i}, \dots, \frac{\partial v_{n_\theta}}{\partial u_i} \right] (\nabla^2 L) = \left(\frac{\partial \mathbf{v}_c}{\partial u_i} \right)^T (\nabla^2 L) \end{aligned} \quad (\text{B.35})$$

Thus, $(\partial \nabla L / \partial u_i) \nabla \mathbf{v}_c = (\partial \mathbf{v}_c / \partial u_i)^T (\nabla^2 L) \nabla \mathbf{v}_c$. Assembling column-wise for $i = 1, \dots, p$,

$$\begin{bmatrix} \frac{\partial \nabla L}{\partial u_1} \nabla \mathbf{v}_c \\ \vdots \\ \frac{\partial \nabla L}{\partial u_p} \nabla \mathbf{v}_c \end{bmatrix} = \begin{bmatrix} \left(\frac{\partial \mathbf{v}_c}{\partial u_1} \right)^T \\ \vdots \\ \left(\frac{\partial \mathbf{v}_c}{\partial u_p} \right)^T \end{bmatrix} (\nabla^2 L) \nabla \mathbf{v}_c = (\nabla \mathbf{v}_c)^T (\nabla^2 L) \nabla \mathbf{v}_c \quad (\text{B.36})$$

For the second term in (B.34),

$$(\nabla L) \frac{\partial \nabla \mathbf{v}_c}{\partial u_i} = (\nabla L) \left[\frac{\partial^2 \mathbf{v}_c}{\partial u_i \partial u_1}, \dots, \frac{\partial^2 \mathbf{v}_c}{\partial u_i \partial u_p} \right] \quad (\text{B.37})$$

If we divide $\nabla^2 \mathbf{v}_c$ ($n_\theta p \times p$) column-wise into p partitions of $n_\theta \times p$ matrices, then the i th partition, say $\mathbf{P}_i$ ($n_\theta \times p$), is the rightmost term in the above equation. That is,

$$\nabla^2 \mathbf{v}_c = \begin{bmatrix} \mathbf{P}_1 \\ \vdots \\ \mathbf{P}_p \end{bmatrix} \quad \mathbf{P}_i = \left[\frac{\partial^2 \mathbf{v}_c}{\partial u_i \partial u_1}, \dots, \frac{\partial^2 \mathbf{v}_c}{\partial u_i \partial u_p} \right] \quad (\text{B.38})$$

Assembling (B.37) column-wise for $i = 1, \dots, p$,

$$\begin{bmatrix} (\nabla L) \frac{\partial \nabla \mathbf{v}_c}{\partial u_1} \\ \vdots \\ (\nabla L) \frac{\partial \nabla \mathbf{v}_c}{\partial u_p} \end{bmatrix} = \begin{bmatrix} (\nabla L) \mathbf{P}_1 \\ \vdots \\ (\nabla L) \mathbf{P}_p \end{bmatrix} = \begin{bmatrix} \nabla L & & \\ & \ddots & \\ & & \nabla L \end{bmatrix} \begin{bmatrix} \mathbf{P}_1 \\ \vdots \\ \mathbf{P}_p \end{bmatrix} = (\mathbf{I}_p \otimes \nabla L) (\nabla^2 \mathbf{v}_c) \quad (\text{B.39})$$

Assembling (B.34) column-wise for $i = 1, \dots, p$ and using (B.36) and (B.39) gives (B.30).

B.2 Lagrange Multiplier Formula

The direct formula in (B.11) is generally applicable for any $\mathbf{u}$ but it involves multiplication of large matrices and hence may not be computationally efficient. It is applicable for any $\mathbf{v}_c(\mathbf{u})$ and so it has not used the fact that it does satisfy constraints. In this section we present a formula that takes advantage of this but is only applicable at the optimum. It is derived based on Lagrange multiplier concepts.

Assume that $\mathbf{v}_c(\mathbf{u})$ satisfies the constraints, i.e., $G_i(\mathbf{v}_c(\mathbf{u})) = 0$ ($i = 1, \dots, n_c$) for any $\mathbf{u}$. Then it can be shown that the Hessian of $L_c(\mathbf{u}) = L(\mathbf{v}_c(\mathbf{u}))$ w.r.t. $\mathbf{u}$ at the optimum $\hat{\mathbf{u}}$ (correspondingly $\hat{\boldsymbol{\theta}} = \mathbf{v}_c(\hat{\mathbf{u}})$) can be calculated as

$$\nabla^2 \hat{L}_c = (\nabla \hat{\mathbf{v}}_c)^T \left(\nabla^2 \hat{L} + \sum_{i=1}^{n_c} \hat{\lambda}_i \nabla^2 \hat{G}_i \right) \nabla \hat{\mathbf{v}}_c \quad (\text{B.40})$$

$\begin{matrix} p \times p & & p \times n_\theta & & n_\theta \times n_\theta & & n_c & & n_\theta \times n_\theta & & n_\theta \times p \end{matrix}$

where

$$\nabla \hat{\mathbf{v}}_c = \nabla \mathbf{v}_c(\mathbf{u})|_{\mathbf{u}=\hat{\mathbf{u}}} \quad \nabla \hat{G}_i = \nabla G_i(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \quad \nabla^2 \hat{G}_i = \nabla^2 G_i(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \quad (\text{B.41})$$

$$\hat{\boldsymbol{\lambda}} = \begin{bmatrix} \hat{\lambda}_1 \\ \vdots \\ \hat{\lambda}_{n_c} \end{bmatrix} = - \underbrace{[(\nabla \hat{\mathbf{G}}) (\nabla \hat{\mathbf{G}})^T]^{-1}}_{n_c \times n_c} \begin{matrix} n_c \times n_\theta & n_\theta \times n_c \\ (\nabla \hat{\mathbf{G}}) & (\nabla \hat{L})^T \end{matrix} \quad (\text{B.42})$$

$\begin{matrix} n_c \times n_\theta & n_\theta \times n_c \\ n_c \times n_\theta & n_\theta \times 1 \end{matrix}$

is a $n_c \times 1$ vector of Lagrange multipliers at the optimum; and

$$\nabla \hat{L} = \nabla L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \quad \nabla^2 \hat{L} = \nabla^2 L(\boldsymbol{\theta})|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}} \quad \nabla \hat{\mathbf{G}} = \begin{bmatrix} \nabla \hat{G}_1 \\ \vdots \\ \nabla \hat{G}_{n_c} \end{bmatrix} \quad (\text{B.43})$$

$\begin{matrix} n_c \times n_\theta \end{matrix}$

Equation (B.40) is invariant to the choice of $\{G_i(\boldsymbol{\theta})\}_{i=1}^{n_c}$ as long as they are consistent with $\mathbf{v}_c(\mathbf{u})$; see Sect. B.2.2.

Example B.4 (Quadratic form, Lagrange formula) Consider Example B.1 again to determine the constrained Hessian of $F(\mathbf{u}) = \mathbf{u}^T \mathbf{A} \mathbf{u}$ subjected to $\mathbf{u}^T \mathbf{u} = 1$, where $\mathbf{A}$ is a $p \times p$ real symmetric and positive definite matrix. In this example we obtain the Hessian of $F_c(\mathbf{u}) = F(\mathbf{v}_c(\mathbf{u}))$ using (B.40). There is only one constraint, $G(\mathbf{u}) = 1 - \mathbf{u}^T \mathbf{u} = 0$. As in Example B.3, the mapping satisfying constraint is taken as $\mathbf{v}_c(\mathbf{u}) = \mathbf{u}/\|\mathbf{u}\| = \mathbf{u}/(\mathbf{u}^T \mathbf{u})^{1/2}$ so that $G(\mathbf{v}_c(\mathbf{u})) \equiv 0$ for any $\mathbf{u}$. Recall the derivatives at the optimum from (B.20) and (B.21) in Example B.3:

$$\nabla \hat{F} = 2\kappa_p \hat{\mathbf{u}}^T \quad \nabla^2 \hat{F} = 2\mathbf{A} \quad \nabla \hat{\mathbf{v}}_c = \mathbf{I}_p - \hat{\mathbf{u}}\hat{\mathbf{u}}^T \quad (\text{B.44})$$

where $\hat{\mathbf{u}}$ is the eigenvector of $\mathbf{A}$ with the largest eigenvalue and scaled to have unit norm; ‘ $\wedge$ ’ denotes that the quantity is evaluated at the optimum. By direct differentiation and evaluating at the optimum,

$$\nabla \hat{G} = -2\hat{\mathbf{u}}^T \quad \nabla^2 \hat{G} = -2\mathbf{I}_p \quad (\text{B.45})$$

Applying (B.42) gives

$$\hat{\lambda} = -[(\nabla \hat{G})(\nabla \hat{G})^T]^{-1}(\nabla \hat{G})(\nabla \hat{F})^T = -[(-2\hat{\mathbf{u}}^T)(-2\hat{\mathbf{u}})]^{-1}(-2\hat{\mathbf{u}}^T)(2\kappa_p \hat{\mathbf{u}}) = \kappa_p \quad (\text{B.46})$$

Using (B.40),

$$\nabla^2 \hat{F}_c = 2(\mathbf{I}_p - \hat{\mathbf{u}}\hat{\mathbf{u}}^T)(\mathbf{A} - \kappa_p \mathbf{I}_p)(\mathbf{I}_p - \hat{\mathbf{u}}\hat{\mathbf{u}}^T) \quad (\text{B.47})$$

Recall from (B.9) that

$$\nabla^2 \hat{F}_c = 2(\mathbf{A} - \kappa_p \mathbf{I}_p) \quad (\text{B.48})$$

Equations (B.47) and (B.48) look different but they are in fact the same. To see this,

$$(\mathbf{A} - \kappa_p \mathbf{I}_p)(\mathbf{I}_p - \hat{\mathbf{u}}\hat{\mathbf{u}}^T) = \mathbf{A} - \underbrace{\mathbf{A}\hat{\mathbf{u}}}_{\kappa_p \hat{\mathbf{u}}} - \kappa_p \mathbf{I}_p + \kappa_p \hat{\mathbf{u}}\hat{\mathbf{u}}^T = \mathbf{A} - \kappa_p \mathbf{I}_p \quad (\text{B.49})$$

Similarly,

$$(\mathbf{I}_p - \hat{\mathbf{u}}\hat{\mathbf{u}}^T)(\mathbf{A} - \kappa_p \mathbf{I}_p)(\mathbf{I}_p - \hat{\mathbf{u}}\hat{\mathbf{u}}^T) = (\mathbf{I}_p - \hat{\mathbf{u}}\hat{\mathbf{u}}^T)(\mathbf{A} - \kappa_p \mathbf{I}_p) = \mathbf{A} - \kappa_p \mathbf{I}_p \quad (\text{B.50})$$

Equations (B.47) and (B.48) are therefore the same. ■

B.2.1 Derivation

Here we derive (B.40), which is recalled below:

$$\nabla_{p \times p}^2 \hat{L}_c = (\nabla_{p \times n_\theta} \hat{\mathbf{v}}_c)^T \left(\nabla_{n_\theta \times n_\theta}^2 \hat{L} + \sum_{i=1}^{n_c} \hat{\lambda}_i \nabla_{n_\theta \times n_\theta}^2 \hat{G}_i \right) \nabla_{n_\theta \times p} \hat{\mathbf{v}}_c \quad (\text{B.51})$$

$$\hat{\boldsymbol{\lambda}} = \begin{bmatrix} \hat{\lambda}_1 \\ \vdots \\ \hat{\lambda}_{n_c} \end{bmatrix} = - \underbrace{[(\nabla \hat{\mathbf{G}}) (\nabla \hat{\mathbf{G}})^T]^{-1}}_{n_c \times n_c} \underbrace{(\nabla \hat{\mathbf{G}}) (\nabla \hat{L})^T}_{n_c \times n_\theta \quad n_\theta \times 1} \quad (\text{B.52})$$

The key is to show that the second term in (B.11) at the optimum is given by

$$(\mathbf{I}_p \otimes \nabla \hat{L})(\nabla^2 \hat{\mathbf{v}}_c) = (\nabla \hat{\mathbf{v}}_c)^T \left(\sum_{i=1}^{n_c} \hat{\lambda}_i \nabla^2 \hat{G}_i \right) (\nabla \hat{\mathbf{v}}_c) \quad (\text{B.53})$$

First note that $\mathbf{v}_c(\mathbf{u})$ satisfies the constraints:

$$G_i(\mathbf{v}_c(\mathbf{u})) \equiv 0 \quad i = 1, \dots, n_c \quad \text{for any } \mathbf{u} \quad (\text{B.54})$$

Taking the Hessian of (B.54) w.r.t. $\mathbf{u}$ gives a LHS identical to the RHS of (B.11) except that L_c is replaced by G_i . This Hessian is always zero. Thus, evaluating at the optimum,

$$(\nabla \hat{\mathbf{v}}_c)^T (\nabla^2 \hat{G}_i) (\nabla \hat{\mathbf{v}}_c) + (\mathbf{I}_p \otimes \nabla \hat{G}_i) (\nabla^2 \hat{\mathbf{v}}_c) = \mathbf{0} \quad i = 1, \dots, n_c \quad (\text{B.55})$$

The second term on the LHS can be expressed in terms of a similar term associated with L . Consider the Lagrangian

$$J(\boldsymbol{\theta}, \boldsymbol{\lambda}) = L(\boldsymbol{\theta}) + \sum_{i=1}^{n_c} \lambda_i G_i(\boldsymbol{\theta}) \quad (\text{B.56})$$

where $\boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_{n_c}]^T$ contains the Lagrange multipliers. If $\hat{\boldsymbol{\theta}}$ minimizes L under constraints $G_i(\boldsymbol{\theta}) = 0$ ($i = 1, \dots, n_c$), then there is a $\hat{\boldsymbol{\lambda}} = [\hat{\lambda}_1, \dots, \hat{\lambda}_{n_c}]^T$ such that $(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\lambda}})$ is a stationary point of J , i.e.,

$$\nabla_{\boldsymbol{\theta}} J(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\lambda}}) = \nabla \hat{L} + \sum_{i=1}^{n_c} \hat{\lambda}_i \nabla \hat{G}_i = \mathbf{0} \quad (\text{B.57})$$

where $\nabla_{\boldsymbol{\theta}} J$ denotes the gradient of J w.r.t. $\boldsymbol{\theta}$. This can be written as a matrix equation for $\hat{\boldsymbol{\lambda}}$:

$$\underbrace{\begin{bmatrix} \hat{\lambda}_1 & \dots & \hat{\lambda}_{n_c} \end{bmatrix}}_{\hat{\boldsymbol{\lambda}}^T \ (1 \times n_c)} \underbrace{\begin{bmatrix} \nabla \hat{G}_1 \\ \vdots \\ \nabla \hat{G}_{n_c} \end{bmatrix}}_{\nabla \hat{\mathbf{G}} \ (n_c \times n_\theta)} = - \nabla \hat{L}_{1 \times n_\theta} \quad (\text{B.58})$$

Despite the fact that there can be more equations than unknowns ($n_c \leq n_\theta$), solution exists for $\hat{\lambda}$ because $\nabla \hat{L}$ is a linear combination of the rows in $\nabla \hat{\mathbf{G}}$. Right-multiplying both sides by $(\nabla \hat{\mathbf{G}})^T$ gives a full rank system of dimension n_c , inverting which and taking transpose gives the expression of $\hat{\lambda}$ in (B.52).

Taking Kronecker product of $\mathbf{I}_p$ with (B.57) and right-multiplying the resulting expression with $\nabla^2 \hat{\mathbf{v}}_c$ gives, after rearranging,

$$(\mathbf{I}_p \otimes \nabla \hat{L})(\nabla^2 \hat{\mathbf{v}}_c) = - \sum_{i=1}^{n_c} \hat{\lambda}_i (\mathbf{I}_p \otimes \nabla \hat{G}_i)(\nabla^2 \hat{\mathbf{v}}_c) \quad (\text{B.59})$$

Substituting the term $(\mathbf{I}_p \otimes \nabla \hat{G}_i)(\nabla^2 \hat{\mathbf{v}}_c)$ from (B.55) gives (B.53), which completes the proof.

B.2.2 Transformation Invariance

Equation (B.40) is invariant to the choice of constraint functions $\{G_i(\boldsymbol{\theta})\}_{i=1}^{n_c}$ as long as they are consistent with $\mathbf{v}_c(\mathbf{u})$. Specifically, the value computed in (B.40) remains the same if the constraint functions are replaced by $\{H_i(G_i(\boldsymbol{\theta}))\}_{i=1}^{n_c}$, where $H_i(\cdot)$ is any scalar function with a non-zero derivative at 0.

Proof (Transformation Invariance)

Let $K_i(\boldsymbol{\theta}) = H_i(G_i(\boldsymbol{\theta}))$, $i = 1, \dots, n_c$. Let $\hat{\lambda}' = [\hat{\lambda}'_1, \dots, \hat{\lambda}'_{n_c}]^T$ and $\hat{\lambda} = [\hat{\lambda}_1, \dots, \hat{\lambda}_{n_c}]^T$ be the vectors of Lagrange multipliers at the optimum when the constraint functions are $\{K_i(\boldsymbol{\theta})\}_{i=1}^{n_c}$ and $\{G_i(\boldsymbol{\theta})\}_{i=1}^{n_c}$, respectively. It is sufficient to show

$$(\nabla \hat{\mathbf{v}}_c)^T (\hat{\lambda}'_i \nabla^2 \hat{K}_i) \nabla \hat{\mathbf{v}}_c = (\nabla \hat{\mathbf{v}}_c)^T (\hat{\lambda}_i \nabla^2 \hat{G}_i) \nabla \hat{\mathbf{v}}_c \quad i = 1, \dots, n_c \quad (\text{B.60})$$

since these are the only terms in (B.40) affected by the choice of constraint functions.

For any $\boldsymbol{\theta}$,

$$\nabla K_i = H'_i \nabla G_i \quad \nabla^2 K_i = H''_i (\nabla G_i)^T \nabla G_i + H'_i \nabla^2 G_i \quad (\text{B.61})$$

where

$$H'_i = \left. \frac{dH_i(x)}{dx} \right|_{x=G_i(\boldsymbol{\theta})} \quad H''_i = \left. \frac{d^2 H_i(x)}{dx^2} \right|_{x=G_i(\boldsymbol{\theta})} \quad (\text{B.62})$$

Evaluating at the optimum and noting $G_i(\hat{\boldsymbol{\theta}}) = 0$,

$$\nabla \hat{K}_i = H'_i(0) \nabla \hat{G}_i \quad \nabla^2 \hat{K}_i = H''_i(0) (\nabla \hat{G}_i)^T \nabla \hat{G}_i + H'_i(0) \nabla^2 \hat{G}_i \quad (\text{B.63})$$

We now evaluate $\hat{\lambda}'$ based on (B.42) for constraint functions $\{K_i(\mathbf{0})\}_{i=1}^{n_c}$. Using (B.61), at the optimum,

$$\nabla \hat{\mathbf{K}} = \begin{bmatrix} \nabla \hat{K}_1 \\ \vdots \\ \nabla \hat{K}_{n_c} \end{bmatrix} = \underbrace{\begin{bmatrix} H'_1(0) & & \\ & \ddots & \\ & & H'_{n_c}(0) \end{bmatrix}}_{\mathbf{D}} \begin{bmatrix} \nabla \hat{G}_1 \\ \vdots \\ \nabla \hat{G}_{n_c} \end{bmatrix} = \mathbf{D} \nabla \hat{\mathbf{G}} \quad (\text{B.64})$$

Then

$$\begin{aligned} \hat{\lambda}' &= -[(\nabla \hat{\mathbf{K}})(\nabla \hat{\mathbf{K}})^T]^{-1} (\nabla \hat{\mathbf{K}})(\nabla \hat{L})^T \\ &= -[\mathbf{D}(\nabla \hat{\mathbf{G}})(\nabla \hat{\mathbf{G}})^T \mathbf{D}^T]^{-1} \mathbf{D}(\nabla \hat{\mathbf{G}})(\nabla \hat{L})^T \\ &= -\mathbf{D}^{-1} [(\nabla \hat{\mathbf{G}})(\nabla \hat{\mathbf{G}})^T]^{-1} \mathbf{D}^{-1} \mathbf{D}(\nabla \hat{\mathbf{G}})(\nabla \hat{L})^T = \mathbf{D}^{-1} \hat{\lambda} \end{aligned} \quad (\text{B.65})$$

This implies

$$\hat{\lambda}'_i = \frac{\hat{\lambda}_i}{H'_i(0)} \quad i = 1, \dots, n_c \quad (\text{B.66})$$

Using this and (B.63),

$$\hat{\lambda}'_i \nabla^2 \hat{K}_i = \hat{\lambda}_i \frac{H''_i(0)}{H'_i(0)} (\nabla \hat{G}_i)^T \nabla \hat{G}_i + \hat{\lambda}_i \nabla^2 \hat{G}_i \quad (\text{B.67})$$

and so

$$(\nabla \hat{\mathbf{v}}_c)^T (\hat{\lambda}'_i \nabla^2 \hat{K}_i) \nabla \hat{\mathbf{v}}_c = \hat{\lambda}_i \frac{H''_i(0)}{H'_i(0)} (\nabla \hat{G}_i \nabla \hat{\mathbf{v}}_c)^T \nabla \hat{G}_i \nabla \hat{\mathbf{v}}_c + (\nabla \hat{\mathbf{v}}_c)^T (\hat{\lambda}_i \nabla^2 \hat{G}_i) \nabla \hat{\mathbf{v}}_c \quad (\text{B.68})$$

Taking gradient w.r.t. $\mathbf{u}$ on $G_i(\mathbf{v}_c(\mathbf{u})) = 0$ and evaluating at the optimum gives $\nabla \hat{G}_i \nabla \hat{\mathbf{v}}_c = \mathbf{0}$, and so the first term in (B.68) is zero. Equation (B.60) then follows, which completes the proof. $\blacksquare$

Appendix C. Mathematical Tools

This appendix provides some mathematical background to supplement the main chapters.

C.1 Asymptotics

Let $F(x)$ and $G(x)$ be functions of x . Then $F(x)$ is ‘asymptotic to’ $G(x)$ as $x \rightarrow a$ if $F(x)/G(x) \rightarrow 1$ as $x \rightarrow a$. This is denoted by

$$F(x) \sim G(x) \quad x \rightarrow a \quad (\text{C.1})$$

Asymptotics and limit are related but different concepts. Limit is defined in terms of the absolute difference of two quantities; asymptotics is in terms of their ratio. When the limit of the subject quantity is non-zero and finite, asymptotics and limit are equivalent. Otherwise they are different and neither one implies the other. As a counter example, $x+1 \sim x$ as $x \rightarrow \infty$ because $(x+1)/x \rightarrow 1$; but x does not converge to $x+1$ because their difference is always 1. On the other hand, $x^2 \rightarrow x$ as $x \rightarrow 0$ but x^2 is not asymptotic to x as $x \rightarrow 0$ because $x^2/x = x$ tends to 0 instead of 1 as $x \rightarrow 0$.

Dominant Behavior

Asymptotics is a useful notion for investigating the limiting behavior of functions. For example, although we know that $x^2 + x + \ln x \rightarrow \infty$ as $x \rightarrow \infty$, it would be useful to know the ‘dominant behavior’ for large x . Since

$$\frac{x^2 + x + \ln x}{x^2} = 1 + \frac{1}{x} + \frac{\ln x}{x^2} \rightarrow 1 \quad x \rightarrow \infty \quad (\text{C.2})$$

it can be concluded that

$$x^2 + x + \ln x \sim x^2 \quad x \rightarrow \infty \quad (\text{C.3})$$

Thus, for large x the function $x^2 + x + \ln x$ ‘behaves like’ or is ‘dominated by’ x^2 . On a relative basis the ‘dominated’ terms x and $\ln x$ can be ignored. Since $x^2 + x + \ln x$ does not converge to x^2 as $x \rightarrow \infty$, the notion of limit does not help (in this case too stringent). Recognizing the dominant term x^2 is trivial here but it could be an ‘art’ in practice when dealing with expressions that are complicated or cannot even be explicitly written.

Non-uniqueness

Unlike limit, asymptotic expression is not unique. For example, $x^2 + x + \ln x$ is also asymptotic to $x^2 + x$ as $x \rightarrow \infty$. Both x^2 and $x^2 + x$ describe the asymptotic behavior of $x^2 + x + \ln x$. Which one to use is a matter of trade-off between the simplicity of expression and the quality of approximation under non-asymptotic situations.

Order of Magnitude

In analysis, it is useful to know the order of magnitude of a quantity and how it scales with other quantities. Two quantities $F(x)$ and $G(x)$ are asymptotically ‘of the same order’ as $x \rightarrow a$ if $\lim_{x \rightarrow a} F(x)/G(x) = C$ where $0 < C < \infty$. Equivalently, we may say that ‘ $F(x)$ is of the order of $G(x)$ ’ or simply ‘ $F(x) = O(G(x))$ ’ as $x \rightarrow a$.

The concept of asymptotics can be applied to obtain useful approximation of complicated solutions. See, e.g., Holmes (1995) for solving equations and Bleistein and Handelsman (1986) for integrals.

C.2 Linear Algebra

For convenience, we use R^n to denote the n -dimensional Euclidean space; $\mathbf{x} \in R^n$ to denote that $\mathbf{x}$ is a $n \times 1$ real vector; $\mathbf{x} \in R^{m \times n}$ to denote that $\mathbf{x}$ is a $m \times n$ real-valued matrix; and $\mathbf{x} \in C^{m \times n}$ to denote that $\mathbf{x}$ is a $m \times n$ complex-valued matrix.

C.2.1 Linear Independence, Span, Basis and Dimension

A set of vectors $\{\mathbf{x}_i \in R^n\}_{i=1}^m$ is said to be ‘linearly independent’ if any (non-trivial) linear combination of the vectors is a non-zero vector. That is, the only possibility for

$$a_1 \mathbf{x}_1 + a_2 \mathbf{x}_2 + \cdots + a_m \mathbf{x}_m = \mathbf{0} \quad (\text{C.4})$$

to hold is $a_1 = a_2 = \cdots = a_m = 0$. Otherwise $\{\mathbf{x}_i\}_{i=1}^m$ are said to be linearly dependent. A set of m vectors in R^n are always linearly dependent if $m > n$. Intuitively if one already has n linearly independent vectors in R^n then it is impossible to find an additional one that is linearly independent of all the n vectors.

Such additional vector can always be expressed as a linear combination of the n vectors.

A ‘linear space’ S is a collection of vectors such that $a\mathbf{x} + b\mathbf{y} \in S$ for any scalars a and b , and vectors $\mathbf{x}, \mathbf{y} \in S$. A set of vectors $\{\mathbf{x}_i\}_{i=1}^m$ is said to ‘span’ S if any member in S can be expressed as a linear combination of $\{\mathbf{x}_i\}_{i=1}^m$. That is, for any $\mathbf{y} \in S$, one can always find scalars $\{c_i\}_{i=1}^m$ such that $\mathbf{y} = c_1\mathbf{x}_1 + \cdots + c_m\mathbf{x}_m$. A set of vectors forms a ‘basis’ of S if the vectors are linearly independent and they span S . A basis is not unique but it always has the same number of vectors, which is referred as the ‘dimension’ of S .

C.2.2 Linear Transformation, Rank and Nullity

A ‘linear transformation’ L from a linear space X to another linear space Y maps a member in X to one in Y such that $L(a_1\mathbf{x}_1 + a_2\mathbf{x}_2) = a_1L(\mathbf{x}_1) + a_2L(\mathbf{x}_2)$ for any scalars a_1 and a_2 , and $\mathbf{x}_1, \mathbf{x}_2 \in X$. A matrix $\mathbf{A} \in R^{m \times n}$ can be viewed as a linear transformation from R^n to R^m , because $\mathbf{y} = \mathbf{A}\mathbf{x}$ maps $\mathbf{x} \in R^n$ to $\mathbf{y} \in R^m$; and $\mathbf{A}(a_1\mathbf{x}_1 + a_2\mathbf{x}_2) = a_1(\mathbf{A}\mathbf{x}_1) + a_2(\mathbf{A}\mathbf{x}_2)$. The ‘range space’ of $\mathbf{A}$ is the collection of vectors $\mathbf{y} = \mathbf{A}\mathbf{x}$ for all $\mathbf{x} \in R^n$. The ‘row rank’ of $\mathbf{A}$ is the number of linearly independent rows; the ‘column rank’ is the number of linearly independent columns. It can be shown that row rank and column rank are equal; and so they are simply referred as the ‘rank’ of $\mathbf{A}$. The ‘null space’ of $\mathbf{A}$ is the collection of all $\mathbf{z}$ in R^n for which $\mathbf{A}\mathbf{z} = \mathbf{0}$. The dimension of the null space is called ‘nullity’. The ‘rank-nullity theorem’ says that

$$\text{Rank} + \text{Nullity} = \text{Number of columns} \quad (\text{C.5})$$

It can be reasoned that range space and null space are linear space.

C.2.3 Euclidean Norm, Inner Product and Orthogonality

The ‘Euclidean norm’ of a real vector $\mathbf{x} = [x_1, \dots, x_n]^T$ is defined as

$$\|\mathbf{x}\| = (\mathbf{x}^T \mathbf{x})^{1/2} = \left(\sum_{i=1}^n x_i^2 \right)^{1/2} \quad (\text{C.6})$$

It is a measure of the ‘length’ of a vector. The ‘Euclidean inner product’ between two real vectors $\mathbf{x} = [x_1, \dots, x_n]^T$ and $\mathbf{y} = [y_1, \dots, y_n]^T$ is defined as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i \quad (\text{C.7})$$

When two real vectors have unit norm, their Euclidean inner product can be interpreted as the cosine of the hyper-angle between them, and is therefore a measure of proximity. Two real vectors $\mathbf{x}$ and $\mathbf{y}$ are ‘orthogonal’ (w.r.t. Euclidean inner product) if $\mathbf{x}^T \mathbf{y} = 0$. Two orthogonal vectors are linearly independent but the reverse is generally not true.

C.2.4 Eigenvalue Problem

Let $\mathbf{A}$ be a $n \times n$ real-valued square matrix. The ‘standard eigenvalue problem’ associated with $\mathbf{A}$, or simply ‘eigenvalue problem’ when understood, is to find ‘eigenvalue’ $\lambda \in \mathbb{C}$ and ‘eigenvector’ $\mathbf{q} \in \mathbb{C}^n$ (with $\mathbf{q} \neq \mathbf{0}$) such that

$$\mathbf{A}\mathbf{q} = \lambda\mathbf{q} \quad (\text{C.8})$$

By writing the equation as $(\mathbf{A} - \lambda\mathbf{I}_n)\mathbf{q} = \mathbf{0}$, for non-trivial solution ($\mathbf{q} \neq \mathbf{0}$) to exist, the eigenvalue must satisfy the ‘characteristic equation’:

$$|\mathbf{A} - \lambda\mathbf{I}_n| = 0 \quad (\text{C.9})$$

This is a n th order polynomial in λ and so there are n roots. A $n \times n$ matrix therefore always has n eigenvalues. These eigenvalues need not be real-valued or distinct. The ‘algebraic multiplicity’ of an eigenvalue is the number of times it is repeated among all eigenvalues. The n eigenvalues correspond to n eigenvectors but these eigenvectors need not be linearly independent. One example is

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

which has two eigenvalues both equal to 0 (hence algebraic multiplicity 2) but there is only one eigenvector $\mathbf{q} = [1 \ 0]^T$ such that $\mathbf{A}\mathbf{q} = 0 \cdot \mathbf{q}$. One can check that any other vector that satisfies this equation must be proportional to $[1 \ 0]^T$. The ‘geometric multiplicity’ of an eigenvalue is the number of linearly independent eigenvectors associated with it.

Sum and Product of Eigenvalues

It can be shown that the sum of eigenvalues of a matrix is equal to its ‘trace’, i.e., the sum of its diagonal entries; the product of eigenvalues of a matrix is equal to its determinant. These can be reasoned as follow.

The characteristic polynomial of $\mathbf{A}$ has the form

$$p(\lambda) = |\mathbf{A} - \lambda\mathbf{I}_n| = (-1)^n \lambda^n + (-1)^{n-1} \text{tr}(\mathbf{A}) \lambda^{n-1} + \cdots + |\mathbf{A}| \quad (\text{C.10})$$

where $\text{tr}(\mathbf{A})$ denotes the trace of $\mathbf{A}$. The λ^n and λ^{n-1} term can be shown by induction w.r.t. n . The constant term follows from the fact that it is equal to the value of $p(\lambda)$ when $\lambda = 0$, and noting that $p(0) = |\mathbf{A} - 0 \cdot \mathbf{I}_n| = |\mathbf{A}|$. On the other hand, if $\{\lambda_i\}_{i=1}^n$ are the eigenvalues of $\mathbf{A}$, then

$$(\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_n) = 0 \quad (\text{C.11})$$

Expanding and collecting terms gives

$$\lambda^n - \left(\sum_{i=1}^n \lambda_i\right) \lambda^{n-1} + \cdots + (-1)^n \prod_{i=1}^n \lambda_i = 0 \quad (\text{C.12})$$

Multiplying by $(-1)^n$ so that the λ^n term matches that of (C.10),

$$(-1)^n \lambda^n + (-1)^{n-1} \left(\sum_{i=1}^n \lambda_i\right) \lambda^{n-1} + \cdots + \prod_{i=1}^n \lambda_i = 0 \quad (\text{C.13})$$

Comparing the λ^{n-1} and constant term with their counterparts in (C.10) shows respectively

$$\sum_{i=1}^n \lambda_i = \text{tr}(\mathbf{A}) \quad \prod_{i=1}^n \lambda_i = |\mathbf{A}| \quad (\text{C.14})$$

C.2.5 Diagonalizable Matrices and Their Functions

Let $\{\mathbf{q}_i \in C^n\}_{i=1}^n$ be n linearly independent eigenvectors corresponding to eigenvalues $\{\lambda_i \in C\}_{i=1}^n$ of $\mathbf{A} \in C^{n \times n}$. Collecting the eigenvectors row-wise gives the $n \times n$ ‘eigenmatrix’:

$$\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_n] \quad (\text{C.15})$$

Assembling the eigenvalue equation $\mathbf{A}\mathbf{q}_i = \lambda_i \mathbf{q}_i$ row-wise for $i = 1, \dots, n$,

$$\mathbf{A} \underbrace{[\mathbf{q}_1, \dots, \mathbf{q}_n]}_{\mathbf{Q}} = \underbrace{[\mathbf{q}_1, \dots, \mathbf{q}_n]}_{\mathbf{Q}} \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}}_{\mathbf{\Lambda}} \quad \text{i.e., } \mathbf{A}\mathbf{Q} = \mathbf{Q}\mathbf{\Lambda} \quad (\text{C.16})$$

Since $\{\mathbf{q}_i\}_{i=1}^n$ are linearly independent, $\mathbf{Q}$ is invertible and so $\mathbf{Q}^{-1}\mathbf{A}\mathbf{Q} = \Lambda$ is a diagonal matrix. In this sense $\mathbf{A}$ is said to be ‘diagonalizable’ by $\mathbf{Q}$. Right-multiplying (C.16) by $\mathbf{Q}^{-1}$ gives the eigenmatrix representation of $\mathbf{A}$:

$$\mathbf{A} = \mathbf{Q}\Lambda\mathbf{Q}^{-1} \quad (\mathbf{A} \text{ diagonalizable}) \quad (\text{C.17})$$

Let $F(x)$ be a scalar function of scalar variable x . The function $F(\mathbf{A})$ of a diagonalizable matrix $\mathbf{A}$ is a $n \times n$ matrix defined as

$$F(\mathbf{A}) = \mathbf{Q}F(\Lambda)\mathbf{Q}^{-1} \quad (\text{C.18})$$

where $F(\Lambda)$ for diagonal matrix Λ is defined as a diagonal matrix with entries $\{F(\lambda_i)\}_{i=1}^n$. It follows from the definition that $F(\mathbf{A})$ has the same eigenvectors as $\mathbf{A}$ but its eigenvalues are $\{F(\lambda_i)\}_{i=1}^n$.

C.2.6 Real Symmetric and Hermitian Matrices

Let $\mathbf{A}$ be a $n \times n$ real symmetric matrix, i.e., all its entries are real-valued and $\mathbf{A} = \mathbf{A}^T$. Then it can be shown that all its eigenvalues and eigenvectors are real-valued. For each eigenvalue, the algebraic multiplicity is equal to the geometric multiplicity. There are then n linearly independent eigenvectors that span R^n . The eigenvectors of distinct eigenvalues are orthogonal to each other. For eigenvectors of the same eigenvalue, it is always possible to find a set of orthogonal eigenvectors whose number is equal to the algebraic multiplicity of the eigenvalue. Consequently, $\mathbf{A}$ has a set of orthogonal eigenvectors that spans R^n . If we scale these eigenvectors to have unit Euclidean norm, they are ‘orthonormal’. This orthonormal set of eigenvectors is a basis for R^n .

Eigenvector Representation

Let $\{\mathbf{q}_i\}_{i=1}^n$ be the orthonormal eigenvectors of $\mathbf{A}$ corresponding to eigenvalues $\{\lambda_i\}_{i=1}^n$; and $\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_n]$. Then

$$\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_n \quad (\text{C.19})$$

since the (i, j) -entry of $\mathbf{Q}^T\mathbf{Q}$ is $\mathbf{q}_i^T\mathbf{q}_j$, which is equal to 1 if $i = j$ and zero otherwise. Right-multiplying both sides of (C.19) by $\mathbf{Q}^{-1}$ gives $\mathbf{Q}^T = \mathbf{Q}^{-1}$ and so $\mathbf{Q}\mathbf{Q}^T = \mathbf{I}_n$ as well. Substituting $\mathbf{Q}^{-1} = \mathbf{Q}^T$ into (C.17) and expanding gives the eigenvector representation of $\mathbf{A}$:

$$\mathbf{A} = \mathbf{Q}\Lambda\mathbf{Q}^T = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^T \quad (\mathbf{A} \text{ real symmetric}) \quad (\text{C.20})$$

Positive Definite Matrix

A $n \times n$ real square matrix $\mathbf{A}$ is said to be ‘positive definite’ if

$$\mathbf{x}^T \mathbf{A} \mathbf{x} > 0 \quad \text{for any non-zero } \mathbf{x} \in \mathbb{R}^n \quad (\text{C.21})$$

It can be reasoned that a real symmetric matrix is positive definite if and only if all its eigenvalues are positive. A similar terminology and result hold if the eigenvalues are all negative (negative definite), non-negative (positive semi-definite), etc.

Hermitian Matrix

A $n \times n$ complex-valued square matrix $\mathbf{A}$ is ‘Hermitian’ if $\mathbf{A}^* = \mathbf{A}$, where ‘*’ denotes the conjugate transpose. Hermitian matrices are analogous to real symmetric matrices. A Hermitian matrix $\mathbf{A}$ has real eigenvalues $\{\lambda_i \in \mathbb{R}\}_{i=1}^n$ and an orthonormal set of eigenvectors $\{\mathbf{q}_i \in \mathbb{C}^n\}_{i=1}^n$, where orthogonality is now defined as $\mathbf{q}_i^* \mathbf{q}_j = 0$ ($i \neq j$). Analogous to (C.19) and (C.20),

$$\mathbf{Q}^* \mathbf{Q} = \mathbf{I}_n \quad (\text{C.22})$$

$$\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^* = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^* \quad (\mathbf{A} \text{ Hermitian}) \quad (\text{C.23})$$

A real symmetric matrix can be considered as a special case of a Hermitian matrix.

Proof (Real Eigenvalues and Orthogonal Eigenvectors)

Let $\mathbf{A}$ be a Hermitian matrix. Here we show that its eigenvalues are real-valued and its eigenvectors are orthogonal.

If λ is an eigenvalue with eigenvector $\mathbf{q}$ then $\mathbf{A} \mathbf{q} = \lambda \mathbf{q}$. Pre-multiplying both sides by $\mathbf{q}^*$ gives $\mathbf{q}^* \mathbf{A} \mathbf{q} = \lambda (\mathbf{q}^* \mathbf{q})$. Taking conjugate transpose and noting $\mathbf{A}^* = \mathbf{A}$ gives another equation $\mathbf{q}^* \mathbf{A} \mathbf{q} = \bar{\lambda} (\mathbf{q}^* \mathbf{q})$. Subtracting these two equations gives $(\lambda - \bar{\lambda})(\mathbf{q}^* \mathbf{q}) = 0$. Since $\mathbf{q}^* \mathbf{q} > 0$, this implies $\bar{\lambda} = \lambda$, i.e., λ is real-valued.

To show orthogonality of eigenvectors, first consider two distinct eigenvalues λ_1 and λ_2 with corresponding eigenvectors $\mathbf{q}_1$ and $\mathbf{q}_2$. Then

$$\mathbf{A} \mathbf{q}_1 = \lambda_1 \mathbf{q}_1 \quad \Rightarrow \quad \mathbf{q}_2^* \mathbf{A} \mathbf{q}_1 = \lambda_1 \mathbf{q}_2^* \mathbf{q}_1 \quad (\text{C.24})$$

$$\mathbf{A} \mathbf{q}_2 = \lambda_2 \mathbf{q}_2 \quad \Rightarrow \quad \mathbf{q}_1^* \mathbf{A} \mathbf{q}_2 = \lambda_2 \mathbf{q}_1^* \mathbf{q}_2 \quad \Rightarrow \quad \mathbf{q}_2^* \mathbf{A} \mathbf{q}_1 = \lambda_2 \mathbf{q}_2^* \mathbf{q}_1 \quad (\text{C.25})$$

conjugate
transpose,
 $\mathbf{A}^* = \mathbf{A}, \lambda_2 = \bar{\lambda}_2$

Subtracting (C.25) from (C.24) gives $(\lambda_1 - \lambda_2) \mathbf{q}_2^* \mathbf{q}_1 = 0$, which implies $\mathbf{q}_2^* \mathbf{q}_1 = 0$ since $\lambda_1 \neq \lambda_2$.

For repeated eigenvalues, let λ be an eigenvalue repeating m times, i.e., with algebraic multiplicity m . The eigenvectors associated with λ form a linear subspace.

To see this, if $\mathbf{x}_1$ and $\mathbf{x}_2$ are eigenvectors associated with λ , then so is $\mathbf{y} = a_1\mathbf{x}_1 + a_2\mathbf{x}_2$ for any scalars a_1 and a_2 :

$$\left. \begin{array}{l} \mathbf{A}\mathbf{x}_1 = \lambda\mathbf{x}_1 \\ \mathbf{A}\mathbf{x}_2 = \lambda\mathbf{x}_2 \end{array} \right\} \Rightarrow \mathbf{A} \underbrace{(a_1\mathbf{x}_1 + a_2\mathbf{x}_2)}_{\mathbf{y}} = \lambda \underbrace{(a_1\mathbf{x}_1 + a_2\mathbf{x}_2)}_{\mathbf{y}} \quad (\text{C.26})$$

Since $\mathbf{A}$ is Hermitian, there are always m linearly independent eigenvectors associated with λ (geometric = algebraic multiplicity), and so the subspace has a dimension of m . We can then always choose m mutually orthogonal vectors in the subspace as the eigenvectors associated with λ . These eigenvectors are also orthogonal to the eigenvectors of other eigenvalues different from λ . ■

C.3 Lagrange Multiplier Method

Let $F(\mathbf{x})$ and $G(\mathbf{x})$ be scalar functions of $\mathbf{x} = [x_1, \dots, x_n]^T \in \mathbb{R}^n$. Suppose we want to determine the optimal value $\hat{\mathbf{x}}$ that minimizes $F(\mathbf{x})$ subjected to constraint $G(\mathbf{x}) = 0$. The Lagrange multiplier method converts this constrained optimization problem into an unconstrained one by introducing an auxiliary scalar variable λ ('Lagrange multiplier') and considering the following objective function, called 'Lagrangian':

$$J(\mathbf{x}, \lambda) = F(\mathbf{x}) + \lambda G(\mathbf{x}) \quad (\text{C.27})$$

It can be shown that if there is a solution to the original constrained optimization problem then one can find $(\hat{\mathbf{x}}, \hat{\lambda})$ as a stationary point of J . That is, $\nabla_{\mathbf{x}}J$ (gradient of J w.r.t. $\mathbf{x}$) and $\partial J/\partial \lambda$ are zero at $(\hat{\mathbf{x}}, \hat{\lambda})$.

Using (C.27) and setting $\partial J/\partial \lambda = 0$ simply returns the constraint equation $G(\mathbf{x}) = 0$. On the other hand, post-multiplying $\nabla_{\mathbf{x}}J = \mathbf{0}$ by $(\nabla G)^T$ and rearranging gives the solution of λ :

$$\hat{\lambda} = - \underbrace{[(\nabla \hat{G})]_{1 \times n}}_{1 \times n} \underbrace{[(\nabla \hat{G})^T]_{n \times 1}}_{n \times 1}^{-1} \underbrace{(\nabla \hat{G})_{1 \times n}}_{1 \times n} \underbrace{(\nabla \hat{F})^T_{n \times 1}}_{n \times 1} \quad (\text{C.28})$$

where a hat '^' denotes the value at the optimum $\hat{\mathbf{x}}$.

Multiple Constraints in Vector Form

The above can be extended for multiple constraints. To minimize $F(\mathbf{x})$ subjected to constraints $\{G_i(\mathbf{x}) = 0\}_{i=1}^p$, one can consider the following Lagrangian:

$$J(\mathbf{x}, \boldsymbol{\lambda}) = F(\mathbf{x}) + \sum_{i=1}^p \lambda_i G_i(\mathbf{x}) = F(\mathbf{x}) + \underbrace{\boldsymbol{\lambda}^T}_{1 \times p} \underbrace{\mathbf{G}(\mathbf{x})}_{p \times 1} \quad (\text{C.29})$$

where λ_i is the Lagrange multiplier associated with $G_i(\mathbf{x})$; $\boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_p]^T$ and $\mathbf{G}(\mathbf{x}) = [G_1(\mathbf{x}), \dots, G_p(\mathbf{x})]^T$. The value of $\boldsymbol{\lambda}$ at the optimum is analogous to (C.28):

$$\hat{\boldsymbol{\lambda}} = \begin{bmatrix} \hat{\lambda}_1 \\ \vdots \\ \hat{\lambda}_p \end{bmatrix} = - \underbrace{[(\nabla \hat{\mathbf{G}})]_{p \times n}}_{p \times n} \underbrace{(\nabla \hat{\mathbf{G}})^T}_{n \times p}^{-1} \underbrace{(\nabla \hat{\mathbf{G}})}_{p \times n} \underbrace{(\nabla \hat{F})^T}_{n \times 1} \quad (\text{C.30})$$

where $\nabla \hat{\mathbf{G}} = [\partial \hat{\mathbf{G}} / \partial x_1, \dots, \partial \hat{\mathbf{G}} / \partial x_n]$ ($p \times n$) is the gradient of $\mathbf{G}(\mathbf{x})$ at the optimum.

Multiple Constraints in Matrix Form

In some problems the constraint functions may be structured by two indices, say, $\{G_{ij}(\mathbf{x}) : i = 1, \dots, p; j = 1, \dots, q\}$. One of course can re-structure these with a single index and use the vector form of constraints just discussed. In some cases, it may be possible to exploit the double-index structure to facilitate analysis. One may then consider the Lagrangian

$$J(\mathbf{x}, \Lambda) = F(\mathbf{x}) + \sum_{i=1}^p \sum_{j=1}^q \lambda_{ij} G_{ij}(\mathbf{x}) \quad (\text{C.31})$$

where λ_{ij} is the Lagrange multiplier associated with $G_{ij}(\mathbf{x})$; Λ is a $p \times q$ matrix of Lagrange multipliers whose (i, j) -entry is λ_{ij} . The Lagrangian can be further expressed in a compact form by writing the double sum as the trace (sum of diagonal entries) of a matrix:

$$J(\mathbf{x}, \Lambda) = F(\mathbf{x}) + \text{tr}[\Lambda^T \mathbf{G}(\mathbf{x})] \quad (\text{C.32})$$

where $\text{tr}(\cdot)$ denotes the trace of the argument matrix; $\mathbf{G}(\mathbf{x})$ is a $p \times q$ matrix whose (i, j) -entry is $G_{ij}(\mathbf{x})$. This form follows because the j th diagonal entry of $\Lambda^T \mathbf{G}(\mathbf{x})$ is $\sum_{i=1}^p \lambda_{ij} G_{ij}(\mathbf{x})$ and so summing it over $j = 1, \dots, q$ gives the trace equal to the double sum in (C.31).

C.4 Minimizing Quadratic Forms

Let $\mathbf{x} = [x_1, \dots, x_n]^T \in R^n$, $\mathbf{b} = [b_1, \dots, b_n]^T \in R^n$ and $\mathbf{A} \in R^{n \times n}$ whose (i, j) -entry is a_{ij} . Then

$$\mathbf{b}^T \mathbf{x} = \sum_{i=1}^n b_i x_i \quad \mathbf{x}^T \mathbf{A} \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j \quad (\text{C.33})$$

are scalar functions of $\mathbf{x}$, referred respectively as ‘linear form’ and ‘quadratic form’. Their gradient and Hessian w.r.t. $\mathbf{x}$ are

$$\begin{aligned} \nabla(\mathbf{b}^T \mathbf{x}) &= \mathbf{b}^T & \nabla^2(\mathbf{b}^T \mathbf{x}) &= \mathbf{0} \\ \nabla(\mathbf{x}^T \mathbf{A} \mathbf{x}) &= \mathbf{x}^T (\mathbf{A} + \mathbf{A}^T) & \nabla^2(\mathbf{x}^T \mathbf{A} \mathbf{x}) &= \mathbf{A} + \mathbf{A}^T \end{aligned} \quad (\text{C.34})$$

These can be derived from first principle by differentiating w.r.t. each x_i and assembling into matrix form, although it could be cumbersome. Alternatively, they can be derived by considering differentials w.r.t. $\mathbf{x}$, using chain rule and noting from Taylor series that the first and second order differentials of a scalar function $F(\mathbf{x})$ are $(\nabla F) \Delta \mathbf{x}$ and $\Delta \mathbf{x}^T (\nabla^2 F) \Delta \mathbf{x} / 2$, respectively; ‘ Δ ’ denotes a differential. For example,

$$\Delta(\mathbf{x}^T \mathbf{A} \mathbf{x}) = (\Delta \mathbf{x}^T) \mathbf{A} \mathbf{x} + \mathbf{x}^T \mathbf{A} \Delta \mathbf{x} = \mathbf{x}^T \mathbf{A}^T \Delta \mathbf{x} + \mathbf{x}^T \mathbf{A} \Delta \mathbf{x} = \underbrace{\mathbf{x}^T (\mathbf{A}^T + \mathbf{A})}_{\nabla(\mathbf{x}^T \mathbf{A} \mathbf{x})} \Delta \mathbf{x} \quad (\text{C.35})$$

In the following, let $\mathbf{A}$ be a $n \times n$ real symmetric and positive definite matrix. We consider minimizing some common quadratic forms with or without norm constraints. Let $F(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$. Clearly, $F(\mathbf{x}) \geq 0$ and it has a unique minimum value of 0 at $\mathbf{x} = \mathbf{0}$.

Consider the non-trivial problem of minimizing $F(\mathbf{x})$ w.r.t. $\mathbf{x}$ subjected to constraint $\mathbf{x}^T \mathbf{x} = 1$. Using Lagrange multiplier method, the solution is a stationary point of the Lagrangian $J(\mathbf{x}, \lambda) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \lambda(1 - \mathbf{x}^T \mathbf{x})$ where λ is a Lagrange multiplier. The gradient of J w.r.t. $\mathbf{x}$ is $\nabla_{\mathbf{x}} J = 2\mathbf{x}^T \mathbf{A} - 2\lambda \mathbf{x}^T = 2\mathbf{x}^T (\mathbf{A} - \lambda \mathbf{I}_n)$. Setting this to zero gives the standard eigenvalue problem $\mathbf{A} \mathbf{x} = \lambda \mathbf{x}$. Substituting into $F(\mathbf{x})$ and noting $\mathbf{x}^T \mathbf{x} = 1$ shows that its value at the optimum is λ . This implies that the minimizing solution of $\mathbf{x}$ is the eigenvector of $\mathbf{A}$ with the smallest eigenvalue and scaled to have unit norm.

Consider now minimizing $F(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + 2\mathbf{b}^T \mathbf{x}$. This function has gradient $\nabla F = 2(\mathbf{x}^T \mathbf{A} + \mathbf{b}^T)$ and Hessian $\nabla^2 F = 2\mathbf{A}$ (positive definite). Solving $\nabla F = \mathbf{0}$ gives $\mathbf{x} = -\mathbf{A}^{-1} \mathbf{b}$, at which $F(\mathbf{x})$ attains its minimum equal to $-\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}$.

Finally, consider minimizing $F(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + 2\mathbf{b}^T \mathbf{x}$ subjected to constraint $\mathbf{x}^T \mathbf{x} = 1$. Define the Lagrangian $J(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + 2\mathbf{b}^T \mathbf{x} + \lambda(1 - \mathbf{x}^T \mathbf{x})$. Solving for $\nabla_{\mathbf{x}} J = \mathbf{0}$ gives $\mathbf{A} \mathbf{x} + \mathbf{b} = \lambda \mathbf{x}$. This is a ‘constrained eigenvalue problem’ whose solution is generally different from the standard one. Details can be referred to Gander (1980) and Gander et al. (1989). The problem can be converted into a standard eigenvalue problem by defining an augmented vector. Let

Table C.1 Summary of minimizing solutions of quadratic forms

Objective function $F(\mathbf{x})$	Constraint	Solution $\hat{\mathbf{x}}$	Minimized value $F(\hat{\mathbf{x}})$
$\mathbf{x}^T \mathbf{A} \mathbf{x}$	–	$\mathbf{0}$	0
$\mathbf{x}^T \mathbf{A} \mathbf{x}$	$\mathbf{x}^T \mathbf{x} = 1$	Eigenvector of $\mathbf{A}$ (smallest eigenvalue) and scaled to unit norm	Smallest eigenvalue of $\mathbf{A}$
$\mathbf{x}^T \mathbf{A} \mathbf{x} + 2\mathbf{b}^T \mathbf{x}$	–	$-\mathbf{A}^{-1} \mathbf{b}$	$-\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}$
$\mathbf{x}^T \mathbf{A} \mathbf{x} + 2\mathbf{b}^T \mathbf{x}$	$\mathbf{x}^T \mathbf{x} = 1$	First half of eigenvector (smallest real eigenvalue) of $\mathbf{D} = \begin{bmatrix} \mathbf{A} & \mathbf{b} \mathbf{b}^T \\ \mathbf{I}_n & \mathbf{A} \end{bmatrix}$ and scaled to unit norm	$\lambda + \mathbf{b}^T \hat{\mathbf{x}}$ $\lambda =$ smallest real eigenvalue of $\mathbf{D}$

$$\mathbf{y} = -(\mathbf{A} - \lambda \mathbf{I}_n)^{-1} \mathbf{x} \quad \mathbf{z} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \quad (\text{C.36})$$

Then it can be shown that $\mathbf{z}$ satisfies the standard eigenvalue problem:

$$\underbrace{\begin{bmatrix} \mathbf{A} & \mathbf{b} \mathbf{b}^T \\ \mathbf{I}_n & \mathbf{A} \end{bmatrix}}_{\mathbf{D}} \underbrace{\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}}_{\mathbf{z}} = \lambda \underbrace{\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}}_{\mathbf{z}} \quad \text{i.e., } \mathbf{D} \mathbf{z} = \lambda \mathbf{z} \quad (\text{C.37})$$

Taking the first half of the eigenvector of $\mathbf{D}$ with the smallest real eigenvalue and scaling it to have unit norm gives the minimizing solution of $\mathbf{x}$.

Equation (C.37) can be shown as follow, although the process is somewhat tricky. Rearrange $\mathbf{A} \mathbf{x} + \mathbf{b} = \lambda \mathbf{x}$ to give $\mathbf{x} = -(\mathbf{A} - \lambda \mathbf{I})^{-1} \mathbf{b}$. Substituting into $\mathbf{y}$ in (C.36) gives

$$\mathbf{y} = (\mathbf{A} - \lambda \mathbf{I}_n)^{-2} \mathbf{b} \quad (\text{C.38})$$

$$\mathbf{b}^T \mathbf{y} = \mathbf{b}^T (\mathbf{A} - \lambda \mathbf{I})^{-2} \mathbf{b} = \mathbf{x}^T \mathbf{x} = 1 \quad (\text{C.39})$$

Thus $\mathbf{A} \mathbf{x} + \mathbf{b} = \lambda \mathbf{x}$ can be written as $\mathbf{A} \mathbf{x} + \mathbf{b} \mathbf{b}^T \mathbf{y} = \lambda \mathbf{x}$. On the other hand, pre-multiplying both sides of (C.36) by $(\mathbf{A} - \lambda \mathbf{I}_n)$ and rearranging gives $\mathbf{x} + \mathbf{A} \mathbf{y} = \lambda \mathbf{y}$. Assembling these two equations gives (C.37).

Table C.1 summarizes the solutions of the problems discussed.

C.5 Identities and Inequalities

Some identities and inequalities relevant to the main chapters are summarized in this section. Unless otherwise mentioned, all matrices and vectors are complex-valued. See Horn and Johnson (2013) and Brookes (2011) for further details on matrix algebra and identities.

C.5.1 Matrix Inverse Lemma and Determinant Theorem

For any matrices $\mathbf{A}$, $\mathbf{C}$, $\mathbf{U}$ and $\mathbf{V}$ of appropriate size, with $\mathbf{A}$ and $\mathbf{C}$ invertible,

$$(\mathbf{UCV} + \mathbf{A})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{VA}^{-1}\mathbf{U} + \mathbf{C}^{-1})^{-1}\mathbf{VA}^{-1} \quad (\text{C.40})$$

$$|\mathbf{UCV} + \mathbf{A}| = |\mathbf{A}||\mathbf{C}||\mathbf{VA}^{-1}\mathbf{U} + \mathbf{C}^{-1}| \quad (\text{C.41})$$

C.5.2 Block Matrix Determinant and Inverse

Let $\mathbf{A}_{11}$ and $\mathbf{A}_{22}$ be square matrices; $\mathbf{A}_{12}$ and $\mathbf{A}_{21}$ be matrices of appropriate size. Then

$$\left| \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} \right| = |\mathbf{A}_{11}| |\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}| \quad (\text{C.42})$$

$$\begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{B}_{11} & -\mathbf{A}_{11}^{-1}\mathbf{A}_{12}\mathbf{B}_{22} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{B}_{11} & \mathbf{B}_{22} \end{bmatrix} \quad (\text{C.43})$$

$$\begin{aligned} \mathbf{B}_{11} &= (\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})^{-1} = \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1}\mathbf{A}_{12}\mathbf{B}_{22}\mathbf{A}_{21}\mathbf{A}_{11}^{-1} \\ \mathbf{B}_{22} &= (\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1} = \mathbf{A}_{22}^{-1} + \mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{B}_{11}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \end{aligned} \quad (\text{C.44})$$

C.5.3 Derivatives of Log-Determinant and Inverse

Let a superscripted variable in parenthesis denote differentiation w.r.t. it. For any square invertible matrix $\mathbf{A}$ that depends on real variables x and y (possibly $x = y$),

$$(\ln |\mathbf{A}|)^{(x)} = \text{tr}[\mathbf{A}^{-1}\mathbf{A}^{(x)}] \quad (\text{C.45})$$

$$(\ln |\mathbf{A}|)^{(xy)} = \text{tr}[\mathbf{A}^{-1}(\mathbf{A}^{(xy)} - \mathbf{A}^{(x)}\mathbf{A}^{-1}\mathbf{A}^{(y)})] \quad (\text{C.46})$$

$$(\mathbf{A}^{-1})^{(x)} = -\mathbf{A}^{-1}\mathbf{A}^{(x)}\mathbf{A}^{-1} \quad (\text{C.47})$$

$$(\mathbf{A}^{-1})^{(xy)} = \mathbf{A}^{-1}[\mathbf{A}^{(x)}\mathbf{A}^{-1}\mathbf{A}^{(y)} + \mathbf{A}^{(y)}\mathbf{A}^{-1}\mathbf{A}^{(x)} - \mathbf{A}^{(xy)}]\mathbf{A}^{-1} \quad (\text{C.48})$$

where $\text{tr}(\cdot)$ denotes the trace (sum of diagonal entries) of the argument matrix.

C.5.4 Gradient and Hessian of Rayleigh Quotient

Let $\mathbf{K}$ and $\mathbf{M}$ be $n \times n$ real symmetric matrices and $\mathbf{u}$ be a $n \times 1$ real vector. The ‘Rayleigh quotient’ is defined as

$$R(\mathbf{u}) = \frac{\mathbf{u}^T \mathbf{K} \mathbf{u}}{\mathbf{u}^T \mathbf{M} \mathbf{u}} \quad (\text{C.49})$$

As a scalar function of $\mathbf{u}$, the gradient and Hessian of $R(\mathbf{u})$ are given respectively by

$$\nabla R(\mathbf{u}) = 2(\mathbf{u}^T \mathbf{M} \mathbf{u})^{-1} \mathbf{u}^T \mathbf{K} - 2(\mathbf{u}^T \mathbf{M} \mathbf{u})^{-2} (\mathbf{u}^T \mathbf{K} \mathbf{u}) \mathbf{u}^T \mathbf{M} \quad (\text{C.50})$$

$$\begin{aligned} \nabla^2 R(\mathbf{u}) = & 2(\mathbf{u}^T \mathbf{M} \mathbf{u})^{-1} \mathbf{K} - 4(\mathbf{u}^T \mathbf{M} \mathbf{u})^{-2} \mathbf{M} \mathbf{u} \mathbf{u}^T \mathbf{K} + 8(\mathbf{u}^T \mathbf{M} \mathbf{u})^{-3} (\mathbf{u}^T \mathbf{K} \mathbf{u}) \mathbf{M} \mathbf{u} \mathbf{u}^T \mathbf{M} \\ & - 4(\mathbf{u}^T \mathbf{M} \mathbf{u})^{-2} \mathbf{K} \mathbf{u} \mathbf{u}^T \mathbf{M} - 2(\mathbf{u}^T \mathbf{M} \mathbf{u})^{-2} (\mathbf{u}^T \mathbf{K} \mathbf{u}) \mathbf{M} \end{aligned} \quad (\text{C.51})$$

When $\mathbf{u}^T \mathbf{M} \mathbf{u} = 1$, (C.50) and (C.51) simplify to

$$\nabla R(\mathbf{u})|_{\mathbf{u}^T \mathbf{M} \mathbf{u} = 1} = 2\mathbf{u}^T \mathbf{K} - 2(\mathbf{u}^T \mathbf{K} \mathbf{u}) \mathbf{u}^T \mathbf{M} \quad (\text{C.52})$$

$$\nabla^2 R(\mathbf{u})|_{\mathbf{u}^T \mathbf{M} \mathbf{u} = 1} = 2\mathbf{K} - 4\mathbf{M} \mathbf{u} \mathbf{u}^T \mathbf{K} + 8(\mathbf{u}^T \mathbf{K} \mathbf{u}) \mathbf{M} \mathbf{u} \mathbf{u}^T \mathbf{M} - 4\mathbf{K} \mathbf{u} \mathbf{u}^T \mathbf{M} - 2(\mathbf{u}^T \mathbf{K} \mathbf{u}) \mathbf{M} \quad (\text{C.53})$$

When $\mathbf{u}$ is an eigenvector (with eigenvalue λ) of the generalized eigenvalue problem $\mathbf{K} \mathbf{u} = \lambda \mathbf{M} \mathbf{u}$, (C.50) and (C.51) simplify to

$$\nabla R(\mathbf{u})|_{\mathbf{K} \mathbf{u} = \lambda \mathbf{M} \mathbf{u}} = \mathbf{0} \quad (\text{C.54})$$

$$\nabla^2 R(\mathbf{u})|_{\mathbf{K} \mathbf{u} = \lambda \mathbf{M} \mathbf{u}} = 2(\mathbf{u}^T \mathbf{M} \mathbf{u})^{-1} (\mathbf{K} - \lambda \mathbf{M}) \quad (\text{C.55})$$

C.5.5 Gradient and Hessian of Unit Vector

Let $\mathbf{u} = [u_1, \dots, u_n]^T$ be a $n \times 1$ real vector. Define the unit vector

$$\bar{\mathbf{u}} = \|\mathbf{u}\|^{-1} \mathbf{u} = (\mathbf{u}^T \mathbf{u})^{-1/2} \mathbf{u} \quad (\text{C.56})$$

The gradient and Hessian of $\bar{\mathbf{u}}$ w.r.t. $\mathbf{u}$ are respectively

$$\nabla_{n \times n} \bar{\mathbf{u}} = \begin{bmatrix} \frac{\partial \bar{\mathbf{u}}}{\partial u_1} & \cdots & \frac{\partial \bar{\mathbf{u}}}{\partial u_n} \end{bmatrix} \quad (\text{C.57})$$

$$\nabla^2 \bar{\mathbf{u}} = \begin{matrix} n^2 \times n \end{matrix} \begin{bmatrix} \frac{\partial^2 \bar{\mathbf{u}}}{\partial^2 u_1} & \frac{\partial^2 \bar{\mathbf{u}}}{\partial u_1 \partial u_2} & \cdots & \frac{\partial^2 \bar{\mathbf{u}}}{\partial u_1 \partial u_n} \\ \frac{\partial^2 \bar{\mathbf{u}}}{\partial u_2 \partial u_1} & \frac{\partial^2 \bar{\mathbf{u}}}{\partial u_2 \partial u_2} & \cdots & \frac{\partial^2 \bar{\mathbf{u}}}{\partial u_2 \partial u_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 \bar{\mathbf{u}}}{\partial u_n \partial u_1} & \frac{\partial^2 \bar{\mathbf{u}}}{\partial u_n \partial u_2} & \cdots & \frac{\partial^2 \bar{\mathbf{u}}}{\partial u_n^2} \end{bmatrix} \quad (\text{C.58})$$

By direct differentiation and assembling in matrix form, it can be shown that

$$\nabla \bar{\mathbf{u}} = \|\mathbf{u}\|^{-1} (\mathbf{I}_n - \bar{\mathbf{u}} \bar{\mathbf{u}}^T) \quad (\text{C.59})$$

$$\nabla^2 \bar{\mathbf{u}} = \|\mathbf{u}\|^{-2} [(3\bar{\mathbf{u}} \bar{\mathbf{u}}^T - \mathbf{I}_n) \otimes \bar{\mathbf{u}} - \bar{\mathbf{u}} \otimes \mathbf{I}_n - \text{vec}(\mathbf{I}_n) \bar{\mathbf{u}}^T] \quad (\text{C.60})$$

where $\otimes$ denotes the Kronecker product and $\text{vec}(\mathbf{I}_n)$ denotes the ‘vectorization’ of $\mathbf{I}_n$, i.e., a $n^2 \times 1$ vector obtained by stacking the columns of $\mathbf{I}_n$.

C.5.6 Cauchy-Schwartz Inequality

Let $\mathbf{x} = [x_1, \dots, x_n]^T$ and $\mathbf{y} = [y_1, \dots, y_n]^T$ be two complex-valued vectors. The Cauchy-Schwartz inequality says that

$$|\mathbf{x}^* \mathbf{y}|^2 \leq (\mathbf{x}^* \mathbf{x})(\mathbf{y}^* \mathbf{y}) \quad (\text{C.61})$$

where $\mathbf{x}^*$ denotes the complex conjugate transpose of $\mathbf{x}$. In terms of the entries of the vectors,

$$|\sum_{i=1}^n \bar{x}_i y_i|^2 \leq (\sum_{i=1}^n |x_i|^2) (\sum_{j=1}^n |y_j|^2) \quad (\text{C.62})$$

where $\bar{x}_i$ denotes the complex conjugate of x_i . Replacing $\bar{x}_i$ by x_i and noting that they have the same modulus gives also

$$|\sum_{i=1}^n x_i y_i|^2 \leq (\sum_{i=1}^n |x_i|^2) (\sum_{j=1}^n |y_j|^2) \quad (\text{C.63})$$

Equation (C.61) can be shown as follow. When $\mathbf{y} = \mathbf{0}$ it holds trivially. In the general case for $\mathbf{y} \neq \mathbf{0}$, note that for any complex number s ,

$$(\mathbf{x} - s\mathbf{y})^* (\mathbf{x} - s\mathbf{y}) \geq 0 \quad (\text{C.64})$$

Expanding the LHS and setting $s = (\mathbf{y}^* \mathbf{x}) / (\mathbf{y}^* \mathbf{y})$ gives (C.61) after simplification.

The Cauchy-Schwartz inequality also holds for functions. For two complex-valued scalar functions $F(u)$ and $G(u)$ of real scalar variable u ,

$$\left| \int F(u)G(u)du \right|^2 \leq \left[\int |F(u)|^2 du \right] \left[\int |G(v)|^2 dv \right] \quad (\text{C.65})$$

C.5.7 Inequality for Partition of Inverse

Let a real symmetric positive definite matrix $\mathbf{A}$ and its inverse be partitioned as

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} \quad \mathbf{A}^{-1} = \begin{bmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{bmatrix} \quad (\text{C.66})$$

Then

$$\mathbf{B}_{11} \geq \mathbf{A}_{11}^{-1} \quad (\text{C.67})$$

in the sense that $(\mathbf{B}_{11} - \mathbf{A}_{11}^{-1})$ is a positive semi-definite matrix.

This inequality can be reasoned using the (1,1)-partition in (C.44) of the block matrix inverse formula:

$$\mathbf{B}_{11} - \mathbf{A}_{11}^{-1} = \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{B}_{22} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} = (\mathbf{A}_{21} \mathbf{A}_{11}^{-1})^T \mathbf{B}_{22} (\mathbf{A}_{21} \mathbf{A}_{11}^{-1}) \quad (\text{C.68})$$

since $(\mathbf{A}_{11}^{-1})^T = \mathbf{A}_{11}^{-1}$ and $\mathbf{A}_{21}^T = \mathbf{A}_{12}$. For any non-zero column vector $\mathbf{x}$ of appropriate size,

$$\mathbf{x}^T (\mathbf{B}_{11} - \mathbf{A}_{11}^{-1}) \mathbf{x} = \underbrace{(\mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{x})^T}_{\mathbf{y}^T} \mathbf{B}_{22} \underbrace{(\mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{x})}_{\mathbf{y}} = \mathbf{y}^T \mathbf{B}_{22} \mathbf{y} \geq 0 \quad (\text{C.69})$$

since $\mathbf{B}_{22}$ is positive definite. The equality cannot be ruled out because it is possible that $\mathbf{y} = \mathbf{0}$.

C.5.8 Inequality for Covariance of Two Vectors

For any two random real vectors $\mathbf{X}$ ($m \times 1$) and $\mathbf{Y}$ ($n \times 1$) with positive definite covariance matrices,

$$E[\mathbf{Y}\mathbf{Y}^T] \geq E[\mathbf{Y}\mathbf{X}^T] E[\mathbf{X}\mathbf{X}^T]^{-1} E[\mathbf{X}\mathbf{Y}^T] \quad (\text{C.70})$$

in the sense that the LHS minus RHS is a positive semi-definite matrix. Equality holds if and only if $\mathbf{Y} = E[\mathbf{Y}\mathbf{X}^T] E[\mathbf{X}\mathbf{X}^T]^{-1} \mathbf{X}$.

This inequality can be reasoned as follow. Let $\mathbf{A}$ be a $n \times m$ constant matrix. Then $\mathbf{Y} - \mathbf{AX}$ is a $n \times 1$ random vector. Clearly,

$$E[(\mathbf{Y} - \mathbf{AX})(\mathbf{Y} - \mathbf{AX})^T] \geq \mathbf{0} \quad (\text{C.71})$$

in the sense that the LHS is a positive semi-definite matrix. Expanding gives

$$E[\mathbf{YY}^T] - \mathbf{AE}[\mathbf{XY}^T] - E[\mathbf{YX}^T]\mathbf{A}^T + \mathbf{AE}[\mathbf{XX}^T]\mathbf{A}^T \geq \mathbf{0} \quad (\text{C.72})$$

Taking $\mathbf{A} = E[\mathbf{YX}^T]E[\mathbf{XX}^T]^{-1}$ and simplifying gives (C.70). It is clear from (C.71) that the equality holds if and only if $\mathbf{Y} = \mathbf{AX}$, i.e., $\mathbf{Y} = E[\mathbf{YX}^T]E[\mathbf{XX}^T]^{-1}\mathbf{X}$.

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